M503-E404H
Apr. 2024

Mobile X-Ray System

MobileDaRt/Art Evolution (MX8 Version)

Installation Manual

Common Section

This manual is for professional service personnel.
Do NOT use this manual unless you have received proper training.

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Revision History

Table 1

Revision	Date	Changes
First edition	Jul. 2021	------
A	Sep. 2021	• Added the description in "4.3.1 Assembly of FPD Box". • Modified the image in "4.23 Attaching Screw Cap". • Added the description in "4.15 Installing Access Point (Only for MobileDaRt Evolution)" • Corrected typos. • Deleted Expendable Parts List from "11 [Appendix] Maintenance and Inspection".
B	Apr. 2022	• Added the description for the model combined with VIVIX-S series in "1.2 Document List". • Added the description for the model combined with VIVIX-S series in "2.1 Conditions for Installation (Usage)". • Added the description for the model combined with FXRD-2530VAW / FXRD-2530VAW PLUS in "4.22 Attaching Cushion for 10x12 FPD". • Added the description for the model combined with VIVIX-S series in "6.1 XCONT-2018 Board". • Modified the description for the model combined with VIVIX-S series in "7.21 DR Setting Mode". • Added the description for High Voltage Transformer D150MH-40 in "9.1 Preparation". • Added the description for High Voltage Transformer D150MH-40 in "9.4.3 Adjusting inverter control circuit of XCONT-2018 board" and "9.4.5 Adjusting VR9 and VR10 on XCONT-2018 Board". • Added the description for High Voltage Transformer D150MH-40 in "9.6.4 Fine adjustment of tube voltage waveform". • Added the error code in "10.1 Error message list".
C	Jul. 2022	• Changed the title of "5.7 Dose Area Product Meter Option". • Added "Dose Area Product Meter VACUDAP" and "Dose Area Product Meter Producted by Shimadzu" in "5.7.1 Component of Dose Area Product Meter Option". • Changed the title of "5.7.2 Installing Dose Area Product Meter VACUDAP Option". • Added "5.7.3 Installing Dose Area Product Meter Shimadzu Option".
D	Sep.2022	• Changed the component of "5.6.1 Component of Infrared Remote Controller Option".
E	Nov.2022	• Added the description for the model with Multi Box MB-02 and without Multi Box in "6.1 XCONT-2018 Board". • Added the description for CXDI (Combined with Multi Box MB-4A) in "7.21 DR Setting Mode". • Added the error code in "10.1 Error message list".

Revision	Date	Changes
F	May 2023	• Changed the list of "4.1 Installation Procedures". • Added "5.10 Assembly of FPD Docking Mechanism (Only Konica Minolta DR-equipped System)". • Added "4.14 Installing GIU SZ (Only for MobileDaRt Evolution)".
G	Jan. 2024	• Added a note in "11.9 PC Maintenance Function".
H	Mar. 2024	• Changed the list of "4.1 Installation Procedures". • Added "4.4 Installing Cushion ASSY(When using FPD Docking Mechanism with 14x17 or 10x12 size FPD)" • Addeds"4.5 Assembly of FPD Docking Mechanism (For CXDI)". • Added "4.6 Assembly of FPD Docking Mechanism (For VXvue)". • Added "4.7 Assembly of FPD Docking Mechanism (For AeroDR)". • Added "4.8 Installing cover for concealing docking mechanism (When the docking mechanism is not used)".

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1 Introduction

Fully understand the contents of this Installation Manual before installing the unit. Install the unit so that the system will be able to deliver its full performance and functions.

In addition, read the Operation Manual and Service Manual thoroughly.

For MobileDaRt Evolution, refer to the following document for installation instructions on the DR system.

Reference M503-E410 MobileDaRt Evolution （MX8 Version）Installation Manual (DR Section)

1.1 Directions for using the unit

Directions about safety are described in this Section. Read this before installing the unit.

The meanings of the following precaution and prohibition terms used in this manual are defined as below:

Table 1-1

⚠ WARNING	Indicates a potentially hazardous situation which, if not avoided, could result in serious injury or possibly deated.
⚠ CAUTION	Indicates a potentially hazardous situation which, if not avoided, may result in minor to moderate injury or equipment damage.
NOTE	Emphasizes additional information that is provided to ensure the proper use of this product.

⚠ WARNING

For 32kW type (MobileDaRt Evolution/MobileArt Evolution high-power type), large-capacity capacitors are connected to the battery.
Electrical charge remains in the capacitor even after turning OFF the key switch or the main breaker. Be sure to discharge the capacitor when working on this system.

Reference "3.4 Discharging the Internal Capacitor"

1.2 Document List

If necessary, the following documents can be obtained.

Table 1-2

	Document No.					
	MobileDaRt Evolution				MobileArt Evolution	
	Equipped with CXDI	Equipped with DR-ID800	Equipped with AeroDR(CS-7)	Equipped with VIVIX-S	high-power type	standard type
Operation Manual	M503-E068				M503-E091	
Installation Manual (Common Section)	M503-E404 (This manual)					
Installation Manual (DR Section)	M503-E410				-	
Reference Guide	M503-E070	M503-E071	M503-E080	M503-E096	M503-E092	
Parts List	M503-4025					
Connection Diagram	M503-2070					
X-ray tube Operation Manual	M534-E318					M535-E218

2 Outline

This chapter describes the outline and features of the equipment.
Read this chapter before installing the system.

2.1 Conditions for Installation (Usage)

■ Use environment

The use environment varies depending on the type of device. Refer to the table below.

Table 2-1

	MobileDaRt Evolution				MobileArt Evolution
	Equipped with CXDI	Equipped with DR-ID800	Equipped with AeroDR (CS-7)	Equipped with VIVIX-S	
Atmosphere	No explosive or corrosive gases				
Atmospheric pressure	800 to 1060 hPa				
Environment luminosity	150 to 500 lx				
Ambient temperature	10 to 30℃	15 to 30℃	10 to 30℃	10 to 30℃	10 to 40℃
Relative humidity	30 to 80% (no dew condensation)				30 to 85% (no dew condensation)

NOTE The battery must be charged in the environment of usage.
Do NOT charge the battery in the storage environment.

■ Power supply

AC power
System : Single phase AC
Frequency : 50/60 Hz
Standard voltages : 100, 110, 120, 200, 220, 230, 240 V
Voltage variation range : ±10 % of standard voltages
Supply capacity : 1kVA
Supply Impedance : 100, 110, 120 V : 1.0 Ω or less
200, 220, 230, 240 V : 4.0 Ω or less

■ Grounding

Ground Terminal : Max. ground resistance: 100 Ω
Additional Ground Terminal : Max. ground resistance: 100 Ω

2.2 Conditions for Transportation and Storage

NOTE This condition can be applied only for a packing state of transportation and storage.

■ Transportation and Storage environment

Ambient temperature: -20 °C to 50 °C

Relative humidity: 20 % to 90 % (no dew condensation)

Atmospheric pressure: 700 hPa to 1060 hPa

NOTE
- In case the unit is kept in storage for a long time before its installation, it is recommended to charge the battery within the following period in order to keep the performance of the battery:
  Storage temperature 40 °C : Once every 2 months
  30 °C : Once every 4 months
  25 °C : Once every 6 months
- When the unit is kept in storage for a long time until its installation, it is recommended to travel the motor once every 3 month in order to keep the performance of the motor.

⚠ WARNING

DO NOT CHARGE THE BATTERY WHEN PACKED.

Place the system in a well-ventilated area to charge the battery.

2.3 Name of Each Part of the Unit

2.3.1 Appearance

MobileDaRt Evolution

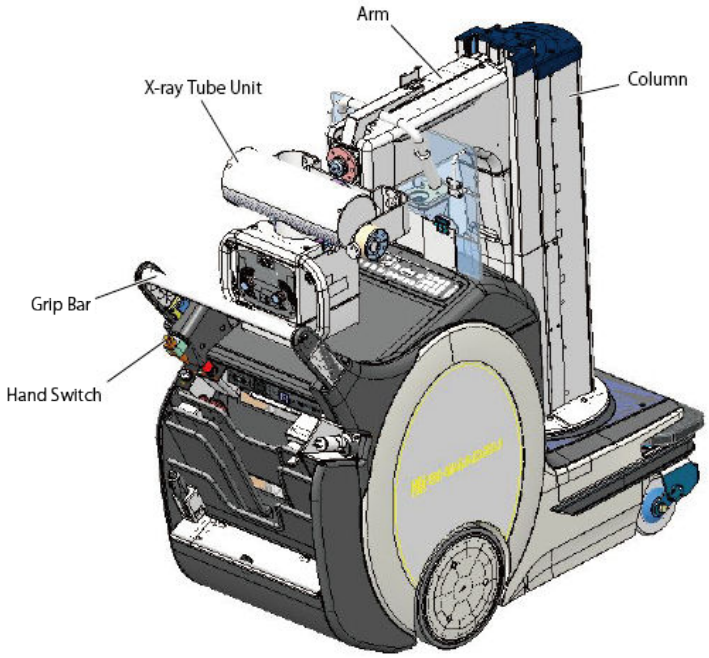


Fig. 2-1 MobileDaRt Evolution (Appearance)

MobileArt Evolution

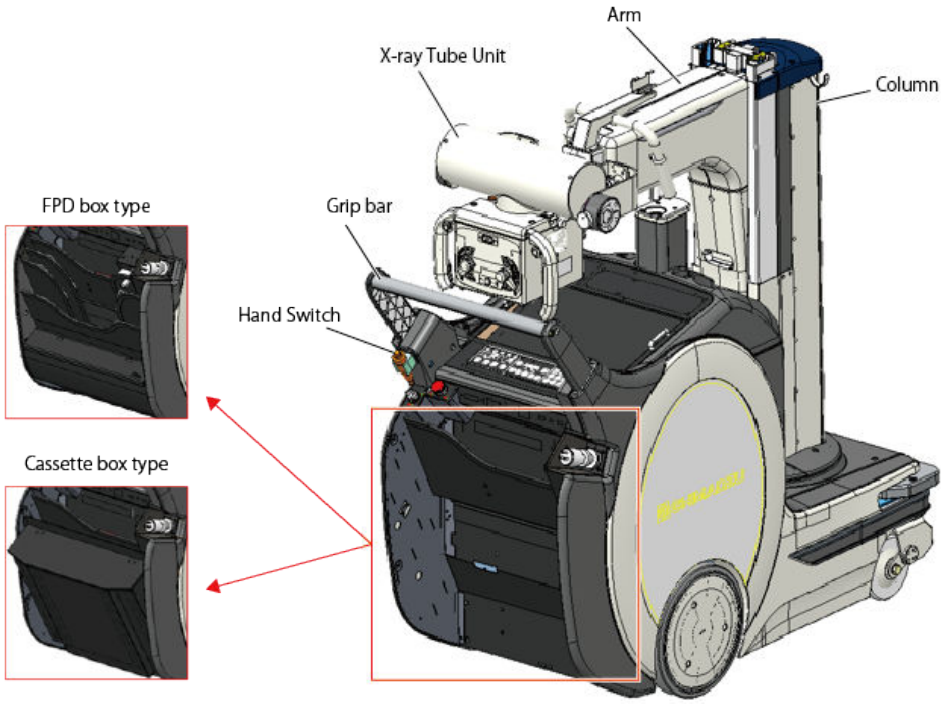


Fig. 2-2 MobileArt Evolution (Appearance)

2.3.2 Operating Section

MobileDaRt Evolution

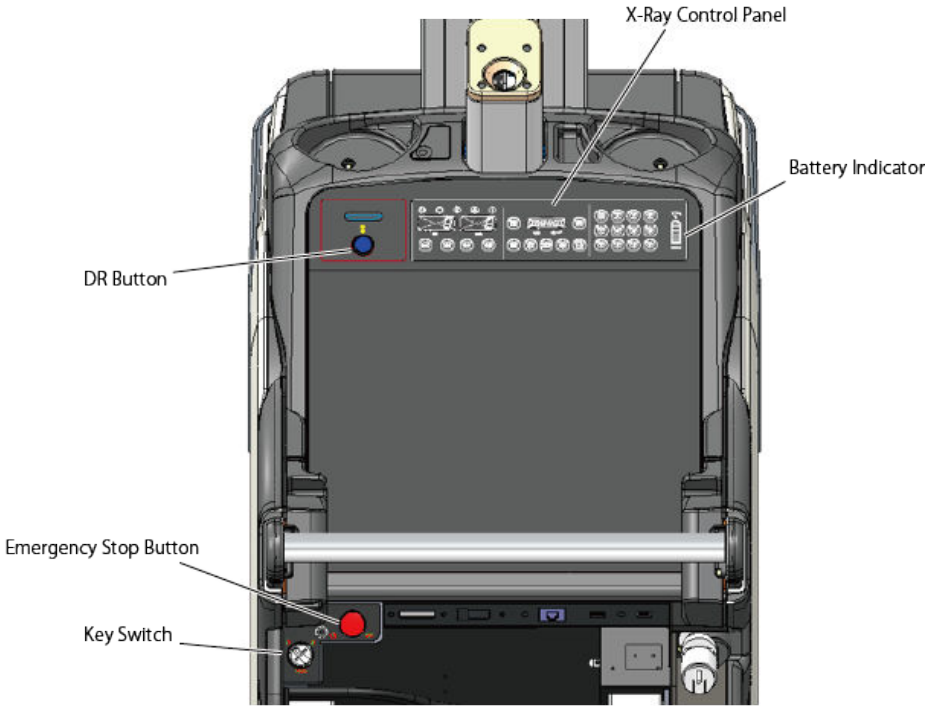


Fig. 2-3 MobileDaRt Evolution (Operating section)

MobileArt Evolution

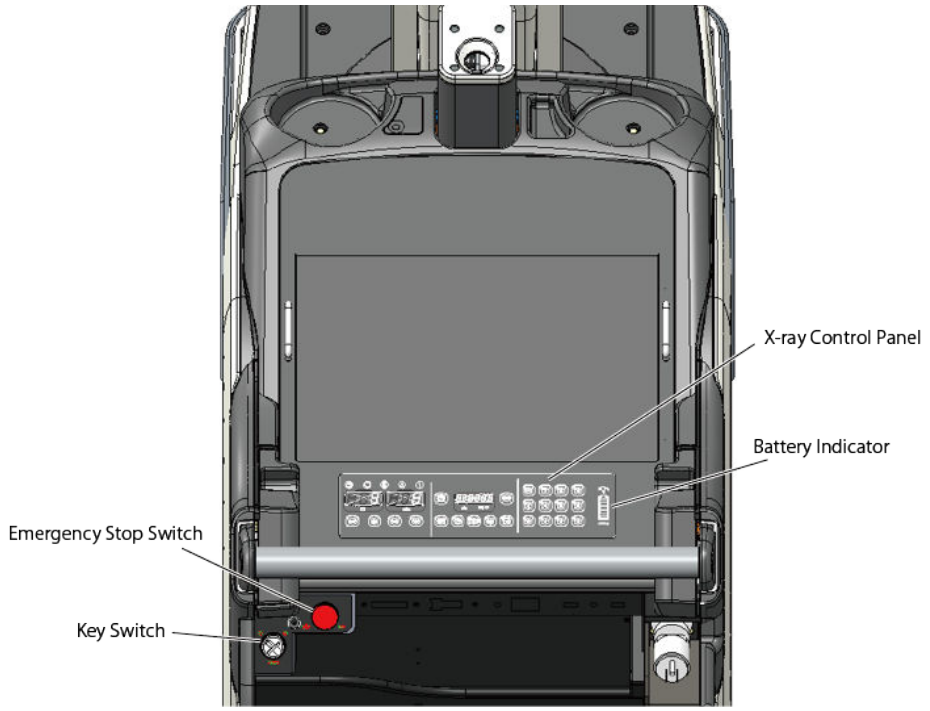


Fig. 2-4 MobileArt Evolution (Operating sectio)

2.3.3 Power Plug and Main Circuit Breaker

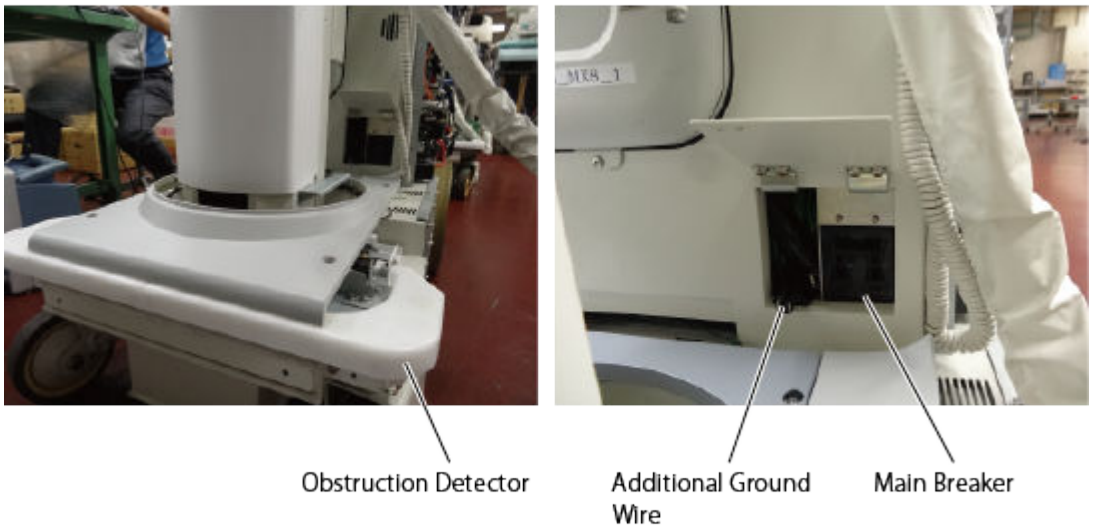
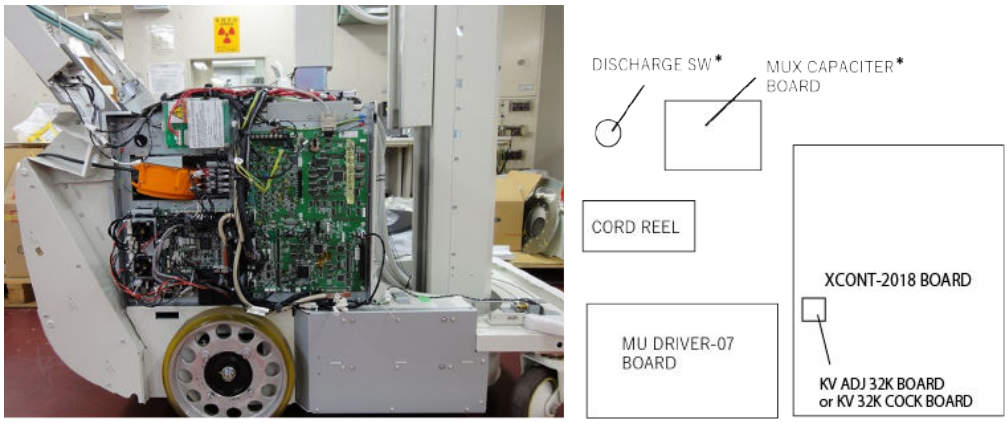


Fig. 2-5

NOTE If the main breaker stops at the intermediate position between "ON" and "OFF", return the breaker to the "OFF" side and move it to the "ON" position before reinserting it.

2.4 Name and Layout of Each Circuit Board

2.4.1 Right Side of Unit



* Available only for 32kW type (MobileDaRt Evolution/MobileArt Evolution high-power type)

Fig. 2-6

2.4.2 Left Side of Unit

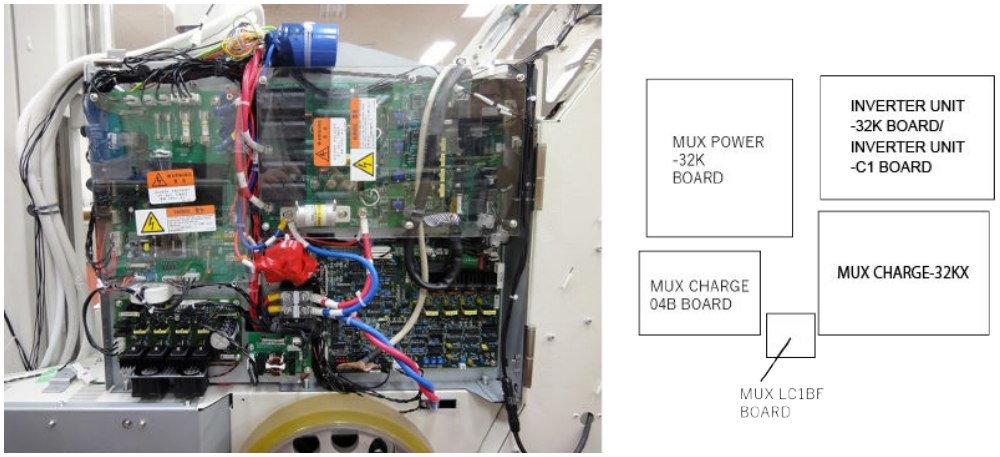


Fig. 2-7

2.4.3 Rear Side of Unit

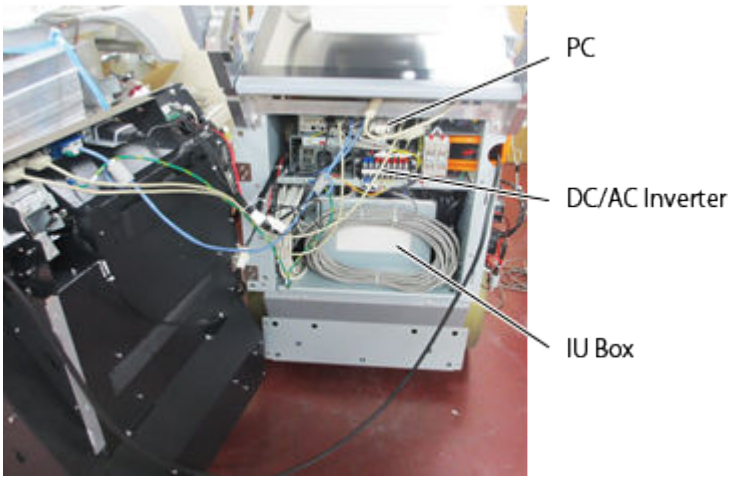


Fig. 2-8 Rear Side of Unit (For CXDI)

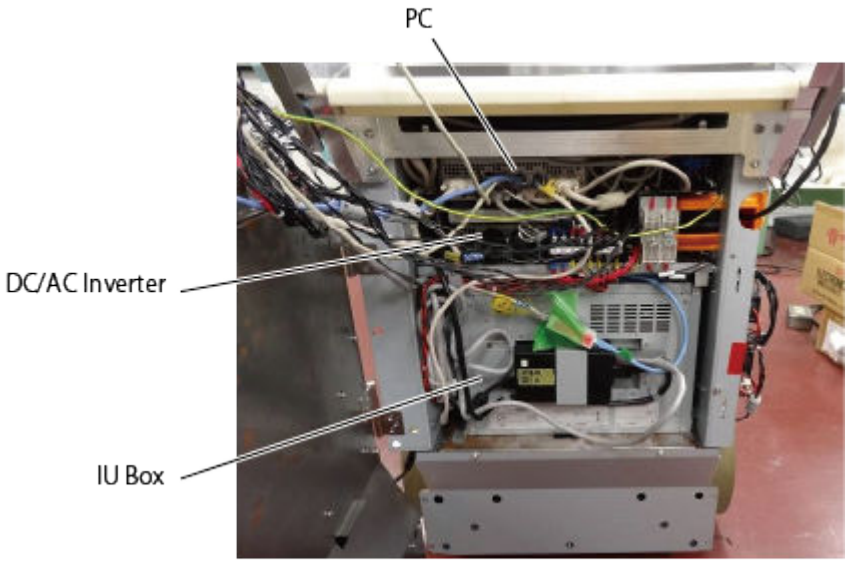


Fig. 2-9 Rear Side of Unit (For DR-ID 800)

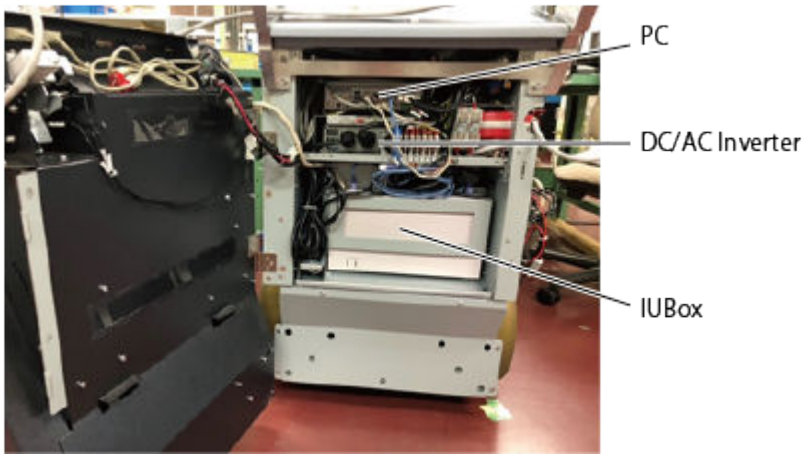


Fig. 2-10 Rear Side of Unit (For AeroDR (CS-7))

2.4.4 Inside of Collimator

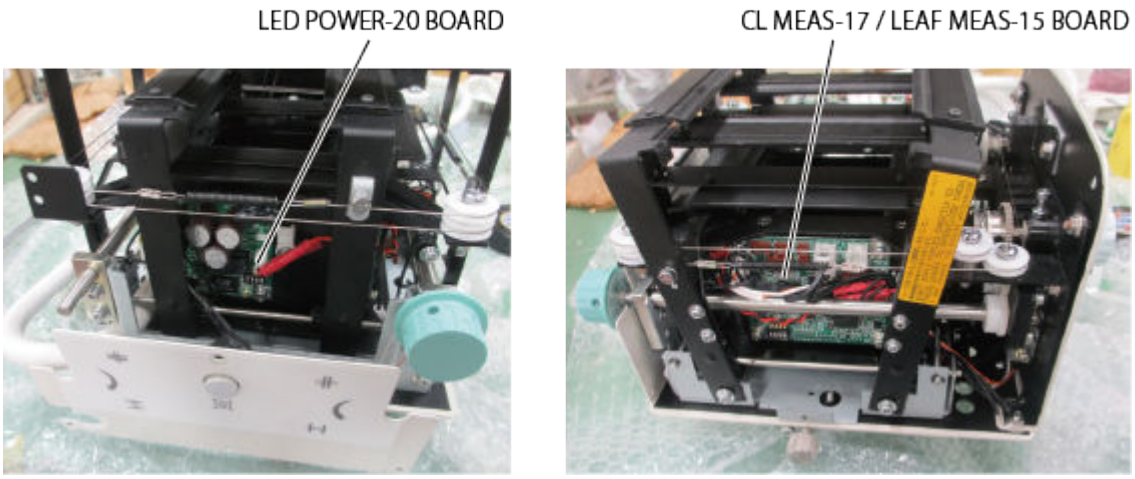


Fig. 2-11

3 Preparations for Installation

Before installing the unit, thoroughly read this chapter and make necessary preparations.

3.1 Tools Required for Installation

When installing the unit, prepare the following tools:

- Screw drivers (cross slot and straight slot)
- Hexagon wrench: No. 2.5 to10 (mm)
- Nipper
- Pliers
- Tester
- Cable tie
- Grease
- Cutter
- Waste cloth
- Alligator clip
- Alcohol for cleaning
- Oscilloscope
- Hexagon socket head cap screwt M8x16: (Required when exchanging X-ray tube assembly)
- Pliers for FU Lock nut AW04: (Required when exchanging X-ray tube assembly)
- Spring gauge: (Required when adjusting X-ray tube assembly or adjustment of handle)

> NOTE For MobileDaRt Evolution, also prepare the following tools to set up the DR system:

- USB port hub: 3 ports or more
- Keyboard (USB type)
- Mouse (USB type)
- DVD Drive (USB type) *1
- Power cable for DVD Drive *1*2
- External hard disk drive (USB type) or USB flash drive
- LAN cable: (Required when adjusting XCONT-2018 board)

> NOTE For MobileDaRt Evolution equipped with CXDI type or AeroDR (CS-7), also prepare the following tools to set up the DR system:

- Personal computer (PC) (Windows 7 OS or higher. RS232C serial port is installed or USB/RS-232C conversion cable is needed separately. Required for upgrading software.)

- RS232C serial cable (9Pin female to female, cross cable type. Required for upgrading software.)

*1: Packaged with the equipment

*2: Used only if the DVD drive does not have USB bus power

3.2 Unpacking

Table 3-1

			Quantity	Description
Main unit	MobileDaRt Evolution／MobileArt Evolution		1	
	Components	Arm cover	1	Screws are attached on the tap hole on the arm.
		Fuse, 313.500P	1	Spare part
		Fuse, 313001P	1	Spare part
		Fuse, 313002P	1	Spare part
		Fuse, 313005P	1	Spare part
		Fuse, 3136.25	1	Spare part
		Fuse, 313004P	1	Spare part
		Fuse, 326010XP	1	Spare part
		Fuse, 660CF-10ULTC	1	Spare part (Only for 32kW type*1)
		Fuse, 660FH-100UL	1	Spare part (Only for 12.5kW type*2)
		Fuse, FLM10	1	Spare part
		Label, Input Voltage	1	Spare part
		Cap	2	White. Spare part
		Cap	2	Gray. Spare part
		Cushion for 14X17 FPD	2	(Only for MobileDaRt Evolution)
		Recovery CD FD, MUX	1	Spare part (Only for MobileDaRt Evolution)
		Collimator Rotation-Lock Knob	1	
		OPERATION MANUAL	1	
		Reference Guide	1	
		Operation Guide	1	
		Document CD	1	
		Attachment Document	1	

*1 32kW type: MobileDaRt Evolution/MobileArt Evolution high-power type

*2 12.5kW type: MobileArt Evolution standard type

NOTE For the DR kit of MobileDaRt Evolution, refer to M503-E410 MobileDaRt Evolution (MX8 Version) Installation Manual (DR Section)

3.3 Removing Fixing bracket for device shipment

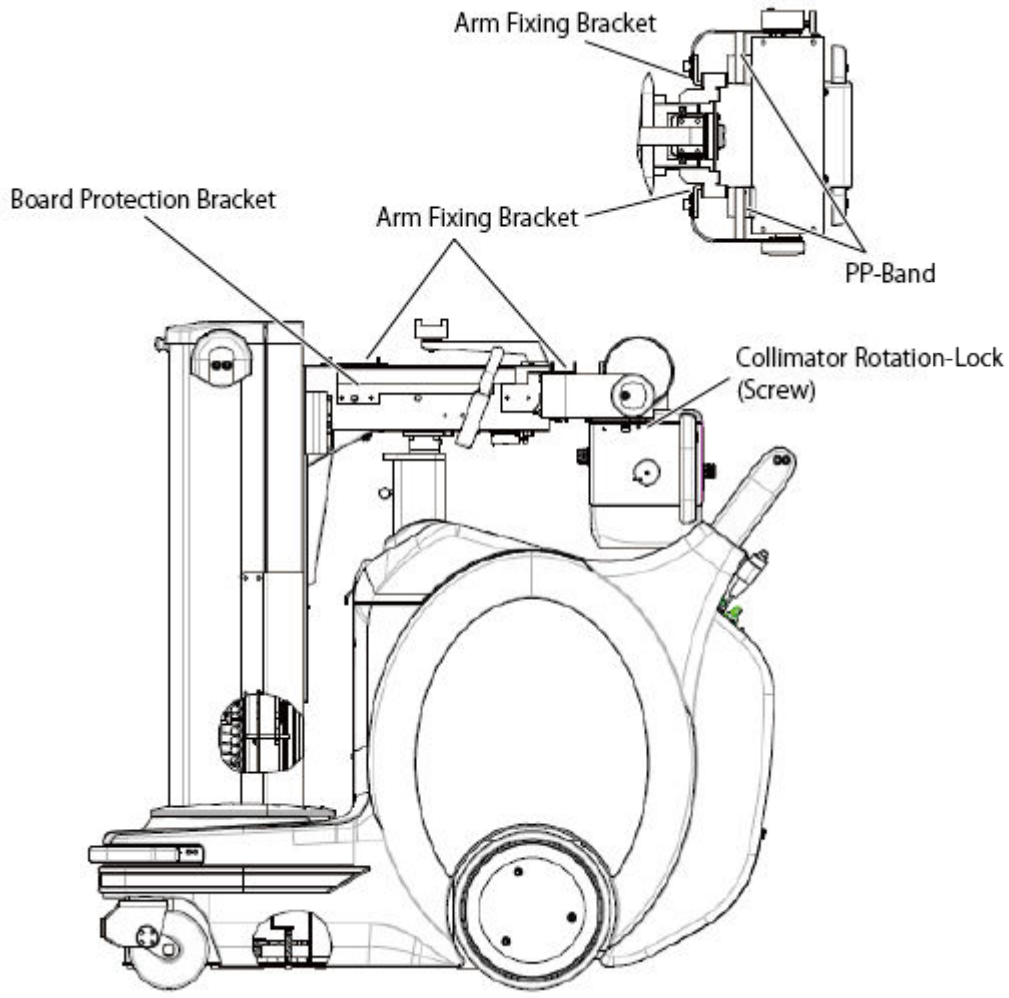


Fig. 3-1

NOTE Remove the collimator rotation-stop screws at checking movement of the collimator unit.

1 **Remove the arm fixing brackets and the board protection brackets**

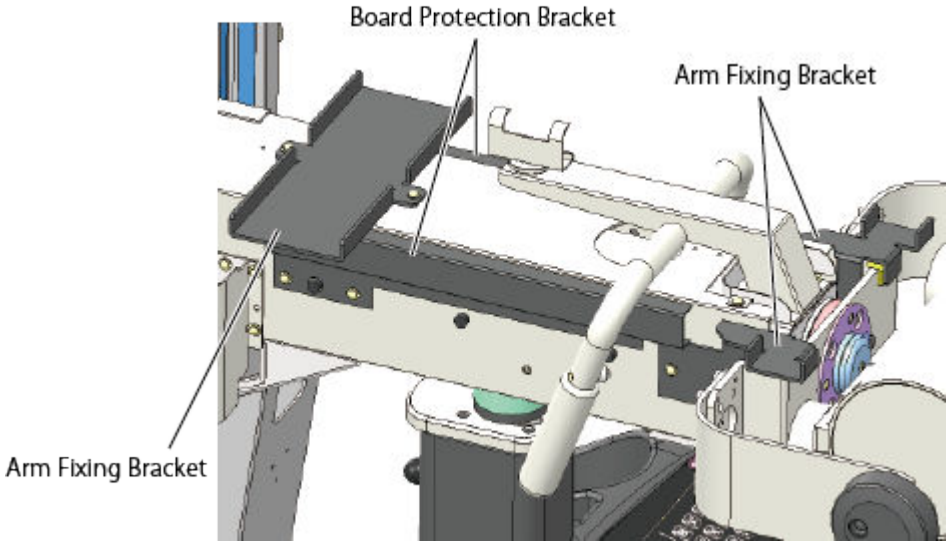


Fig. 3-2

2 **Remove two PP-bands fastening the X-ray tube and the fixing bracket.**

3 **Turn ON the main breaker and key switch, and then move the device to place where the X-ray radiography is permitted and the power outlet is available.**

Hint The main breaker is located at the front side of the system.

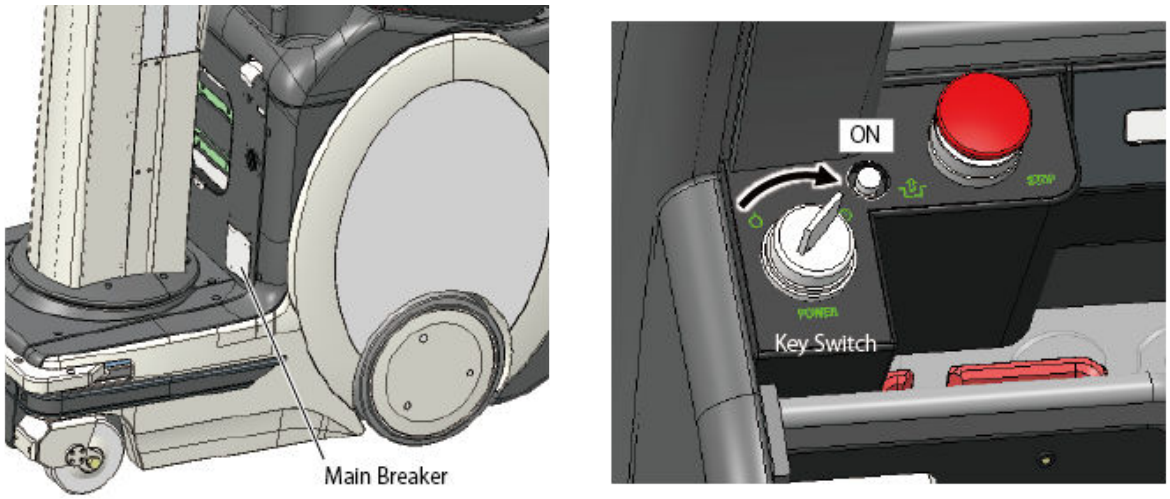


Fig. 3-3

NOTE

- The wheel may be flattened due to leaving the unit in one place for a long time. The wheel will return to its normal shape by driving the unit.
- Use the fixing brackets when transporting the unit by vehicle.
- Keep the fixing brackets for transportation without disposing.

3.4 Discharging the Internal Capacitor

⚠ WARNING

In the system for 32kW type (MobileDaRt Evolution/MobileArt Evolution high-power type) , large-capacity capacitors are connected to the battery.
Electrical charge remains in the capacitor even after turning OFF the key switch or the main breaker.
Be sure to discharge the capacitor when working on this system.

NOTE The system for 12.5kW type (MobileArt Evolution standard type) does not have the internal capacitor.

1 **Confirm that both the key switch and the main breaker are turned OFF.**

2 **Remove the right inside cover and the right-side cover.**

Reference "3.5 How to Open/Close Covers"

3 **Confirm that the discharge switch is set to "NORMAL" (bottom side).**

4 **Disconnect the CN1/CN2 cable from the CN2 connector "NORMAL" on the MUX CAPACITOR board and connect it to the CN1 connector "DISCHARGE".**

If electrical charge is remaining in the capacitor, the LED on the MUX CAPACITOR board will be lit.

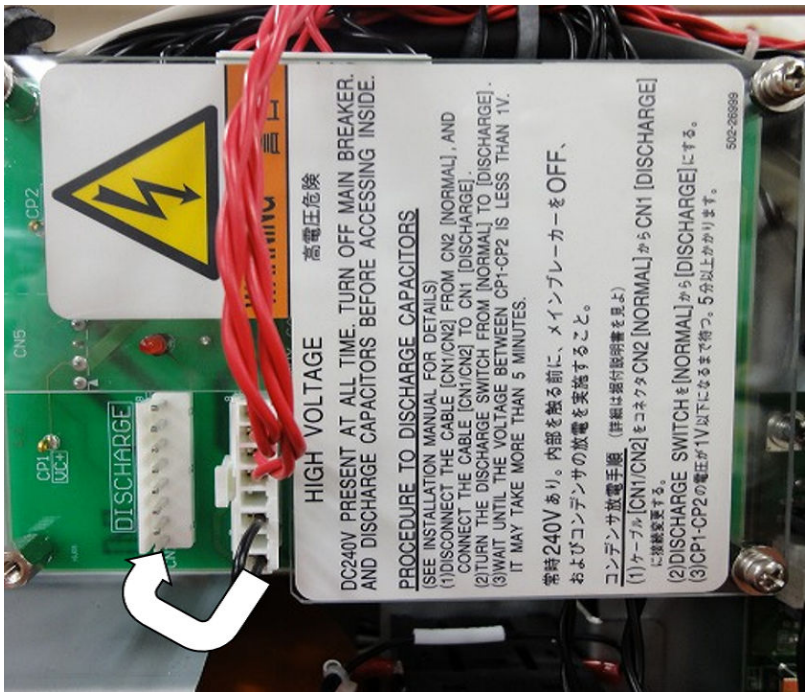


Fig. 3-4

5 **Set the discharge switch to "DISCHARGE".**

Discharge of the internal capacitor will start.

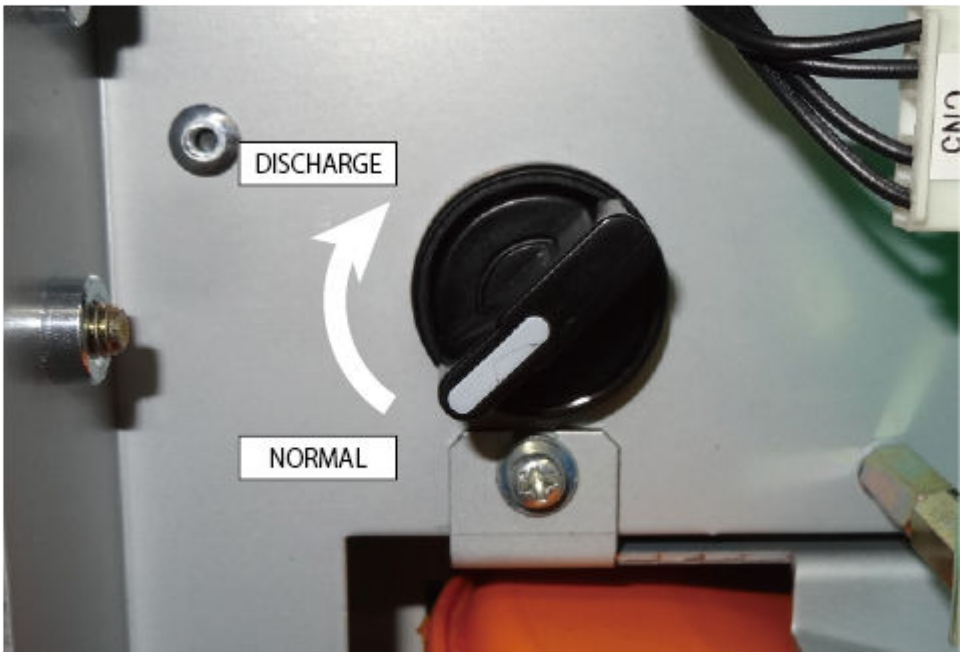


Fig. 3-5

6 **Wait until the voltage of the capacitor drops below 1.0V. (May take 5 minutes or more.)**

The voltage of the capacitor can be measured between test points, CP1 and CP2, on the MUX CAPACITOR board.

7 **After the capacitor is discharged, follow the procedures below to return the system to normal operation status.**

1. Confirm that both the key switch and the main breaker are turned OFF
2. Set the discharge switch to "NORMAL" (bottom side)
3. Reconnect the CN1/CN2 cable to the CN2 connector "NORMAL" on the MUX CAPACITOR board.

3.5 How to Open/Close Covers

Follow the instuction appropriate to each cover to open/close the cover.

⚠ WARNING

Be sure to turn off the main breaker before opening the covers. There is a danger of receiving electric shock.

3.5.1 Name and Layout of the Covers

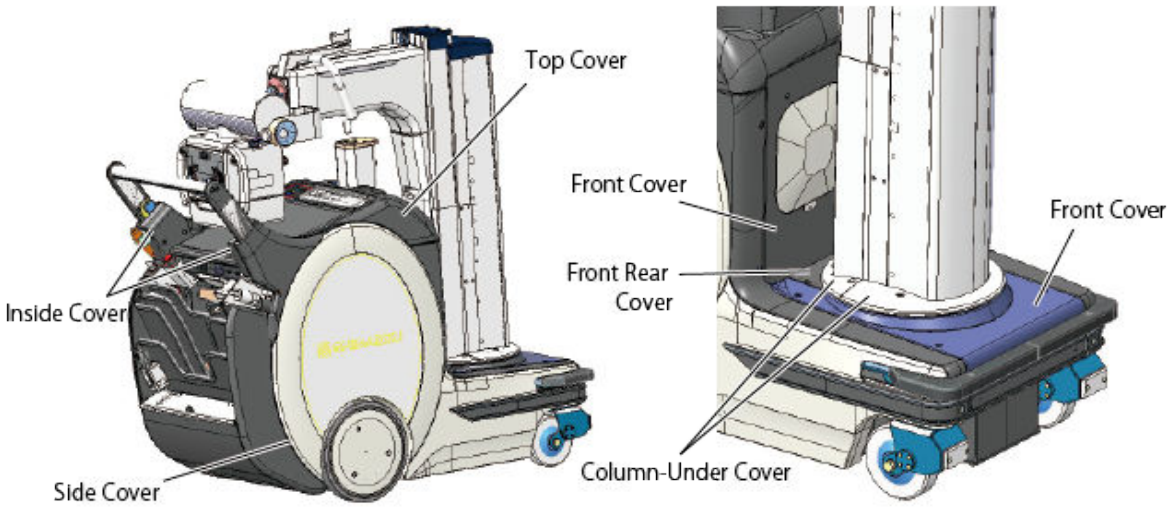


Fig. 3-6 Name and layout of the covers

⚠ CAUTION

Do NOT pull the imaging unit or its cable when the cable is caught on something. Otherwise, it may cause a damage to the equipment, malfunctions, fire, or electric shock.

3.5.2 How to Open/Close Inside Cover

1 **Unscrew two screws as shown in the figure below.**

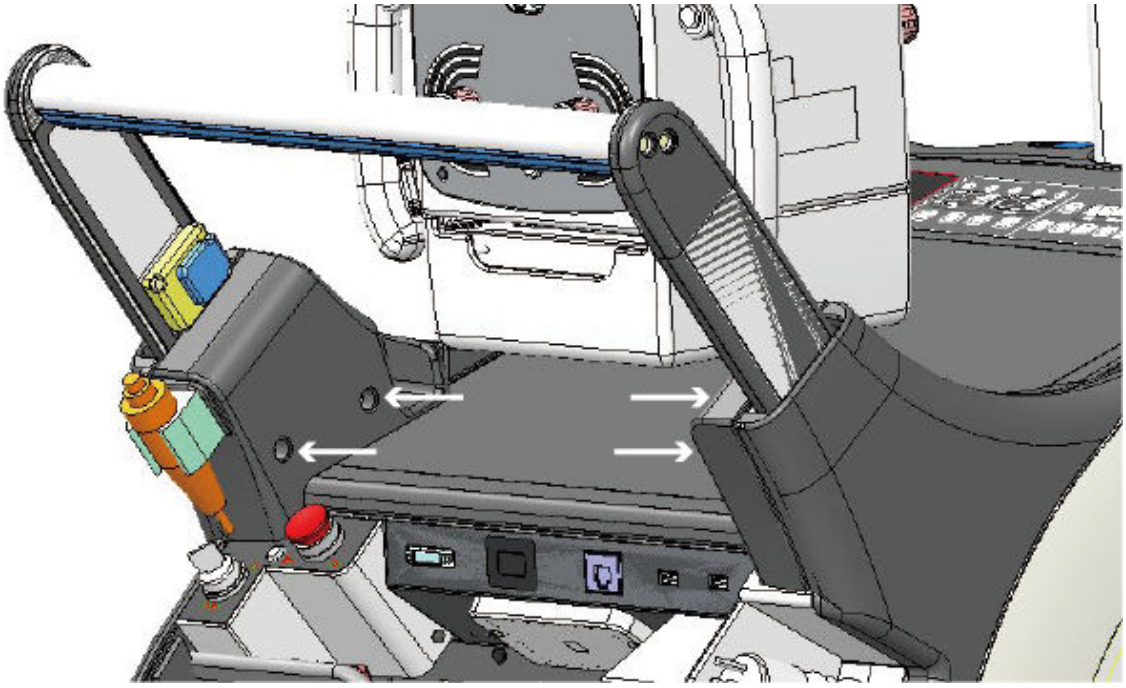


Fig. 3-7

2 **Pull out the inside cover laterally to remove it.**

3.5.3 How to Open/Close Side Cover

1 **Remove the inside cover.**

2 **Unscrew screws as shown in the figure (a).**

3 **Unscrew screws as shown in the figure (b), and remove the side cover.**

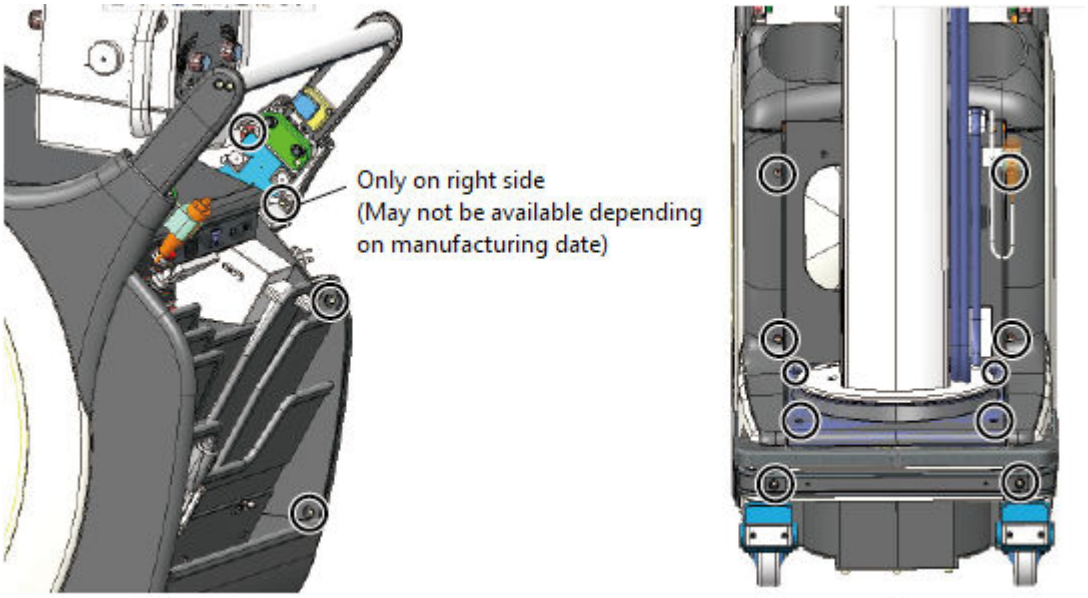


Fig. 3-8

3.5.4 How to Open/Close Top Cover

1 **Remove the inside cover and side cover.**

2 **Unscrew screws as shown in the figure (a), and open the FPD box.**

3 **Remove screw caps, and unscrew screws as shown in the figure (b).**

4 **Unscrew screws as shown in the figure (c).**

5 **Turn the arm lock release lever as shown in the figure (d) in counterclockwise direction to remove it.**

6 **Lift up the top cover to remove it.**

⚠ CAUTION

The top cover and the main unit are connected with cables. Be careful when removing the top cover.

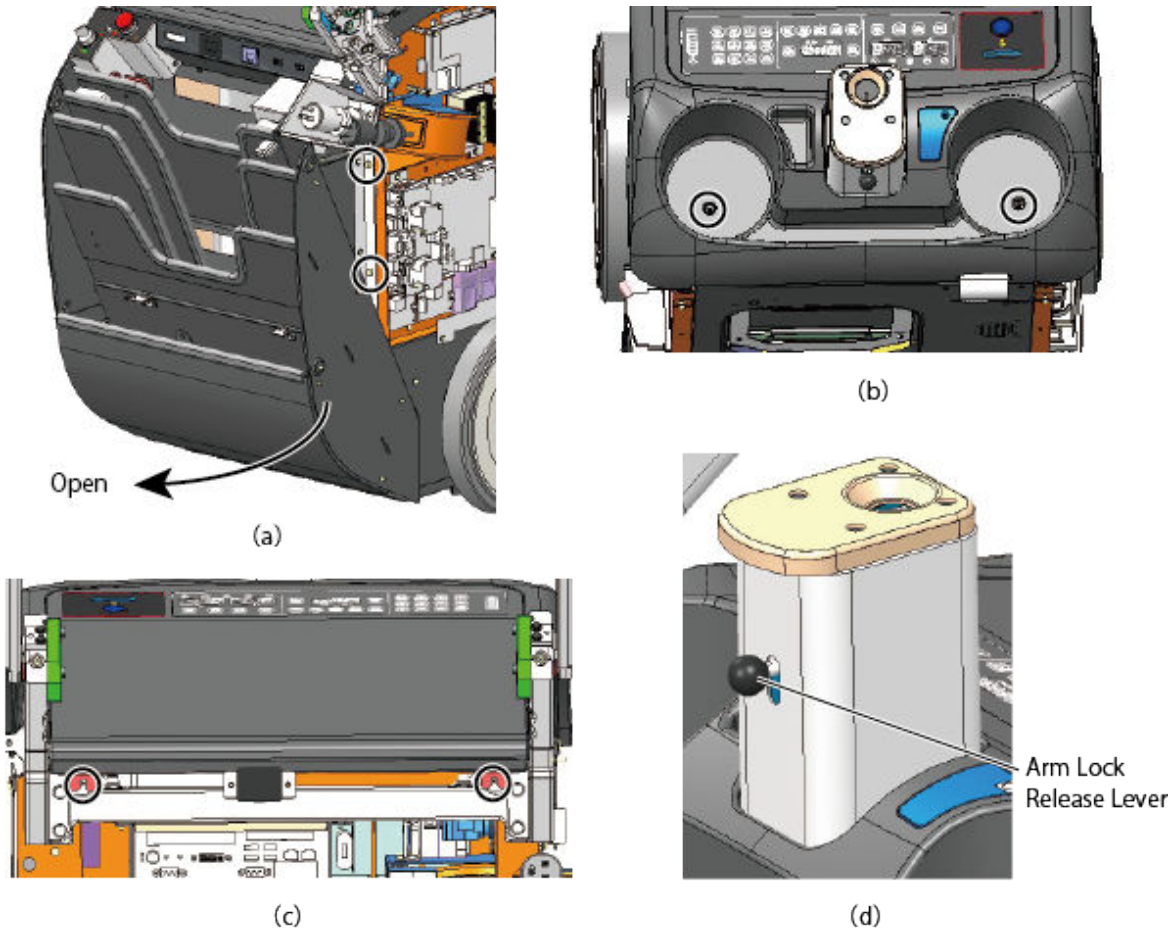


Fig. 3-9

3.5.5 How to Open/Close Front Cover

1 **Unscrew ten screws as shown in the figure.**

2 **Remove the column-under cover and front rear cover, and then pull out the front cover forward to remove it.**

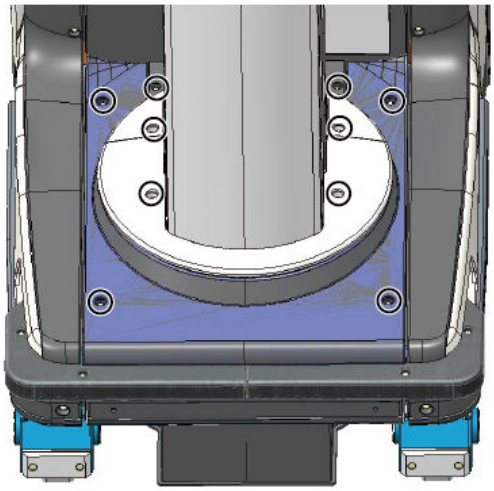


Fig. 3-10

3.5.6 How to Open/Close Front Rear Cover

1 **Unscrew four screws as shown in the figure.**

2 **Remove the column-under cover, and remove the front rear-cover.**

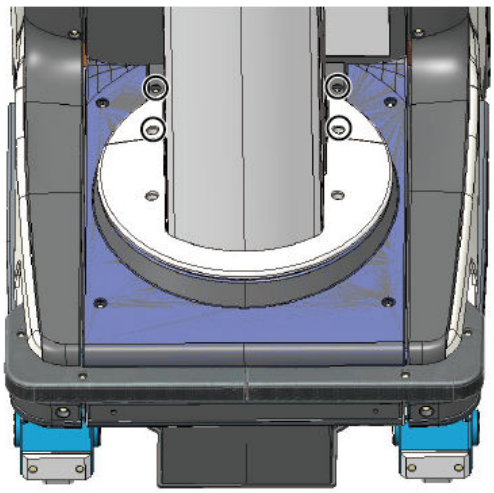


Fig. 3-11

3.5.7 How to Open/Close Front Cover

1 **Remove the inside cover, side cover, top cover, front cover, and front rear cover.**

2 **Unscrew 4 screws as shown below and remove the cable cover.**

⚠ CAUTION

The access point, the hand switch (if added), and the main unit are connected with cables. Be careful when removing the front cover.

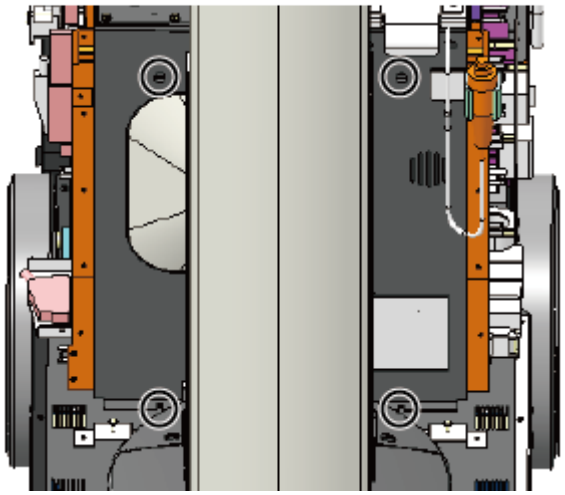


Fig. 3-12

3.5.8 How to Open/Close the Arm Joint

1 **Remove four hexagon socket head screw as shown in the figure.**

2 **Rotate the arm lock release lever in a counterclockwise to remove it.**

3 **Pull out the top plate and arm joint main cover.**

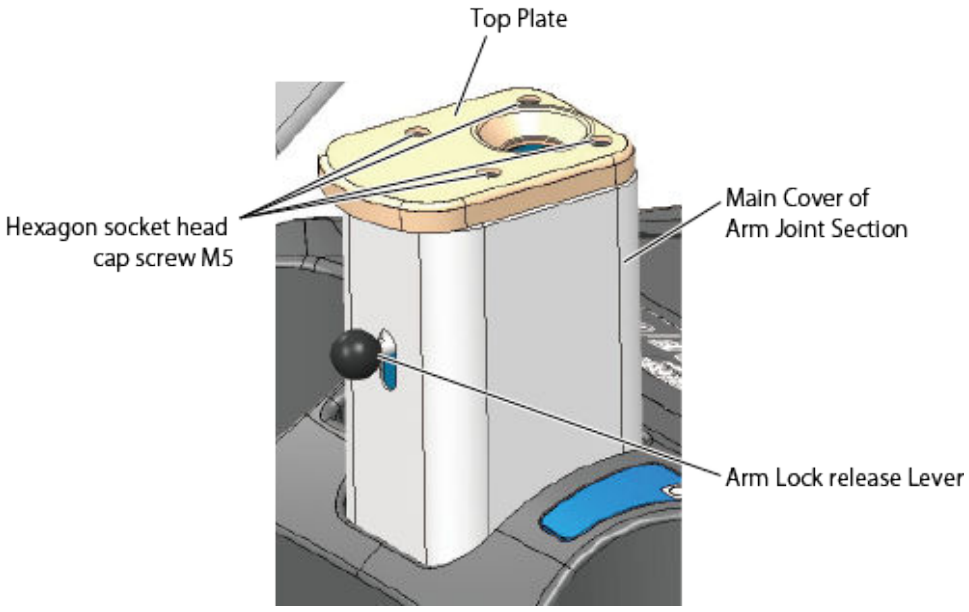


Fig. 3-13

4 Installation

This Chapter describes the installation procedures for MobileDaRt Evolution.

⚠ WARNING

In the system for 32kW type (MobileDaRt Evolution/MobileArt Evolution high-power type) , large-capacity capacitors are connected to the battery.
Even after turning OFF the key switch or the main breaker, electrical charge remains in the capacitor.
Be sure to discharge the capacitor when working on this system.

4.1 Installation Procedures

Install the system according to the procedures below.

Table 4-1 Installation Procedures

Step	Procedure
1	"4.1 Installation Procedures"
2	"4.2 Checking Safety Function"
3	"4.3 Assembly of FPD Box/Cassette Box (Only for MobileArt Evolution) "
4	"4.4 Installing Cushion ASSY(When using FPD Docking Mechanism with 14x17 or 10x12 size FPD)"
	"4.5 Assembly of FPD Docking Mechanism (For CXDI)"
	"4.6 Assembly of FPD Docking Mechanism (For VXvue)"
	"4.7 Assembly of FPD Docking Mechanism (For AeroDR)"
	"4.8 Installing cover for concealing docking mechanism (When the docking mechanism is not used)"
5	"4.9 Connection Change According to Power-Supply Voltage"
6	"4.10 Checking Performance of Each Part"
7	"4.11 Checking Dose Area Product Calculation Unit"
8	"4.12 Assembly of Multi Box (Only for MobileDaRt Evolution)"
9	"4.13 Installing AeroDR Portable RF Unit 2 (Only for MobileDaRt Evolution)"
10	"4.14 Installing GIU SZ (Only for MobileDaRt Evolution)"
11	"4.15 Installing Access Point (Only for MobileDaRt Evolution)"
12	"4.16 DR System Settings (Only for MobileDaRt Evolution) "
13	"4.17 Option Installations"
14	"4.18 Adjustment of Image Quality (Only for MobileDaRt Evolution) "
15	"4.19 Attaching Arm Cover"
16	"4.20 Attaching Knob for Collimator Rotation-Lock"
17	"4.21 Attaching Cushion for 14X17 FPD"
18	"4.22 Attaching Cushion for 10x12 FPD"
19	"4.23 Attaching Screw Cap"
20	"4.24 Resetting All Parts Usage Information"
21	"4.25 Charging Battery After Installation"
22	"4.26 Assembler Test"
23	"4.27 Set up of McAfee Embedded Control (MEC) (Only for MobileDaRt Evolution) "
24	"4.28 Backup After Installation (Only for MobileDaRt Evolution) "

4.2 Checking Safety Function

■ Check the function of Key switch

Confirm that the key switch can be turned ON/OFF.

■ Check the function of brake

1. Grip the brake release bar to see the brake can be released.
2. Release the brake bar to see that the brake is ON.

■ Check the function of emergency brake release

1. Turn off the key switch.
2. Press the emergency stop switch.
3. Press the emergency brake release switch to release the brake.

■ Check the function of emergency stop

1. Press the emergency stop switch.
2. It bans the running and x-ray radiography.

4.3 Assembly of FPD Box/Cassette Box (Only for MobileArt Evolution)

NOTE MobileArt Evolution allows users to select the FPD box/cassette box, and to install it locally.
Refer to the corresponding procedure for installation.

Reference
- "4.3.1 Assembly of FPD Box"
- "4.3.2 Assembly of Cassette Box"

4.3.1 Assembly of FPD Box

The FPD box kit consists of the following parts. Make sure that all these parts are contained in the package.

Table 4-2 Component of FPD box kit

No.	Component	Amount	Remarks
1	Cushion 1	1	
2	Cushion 2	1	
3	Frame 4 ASSY	1	
4	Frame 3 ASSY	1	
5	Frame 5 ASSY	1	
6	Frame 6 ASSY	1	
7	Frame 7 ASSY	1	
8	Frame 8 ASSY	1	
9	FPD LOCK ASSY	1	
10	Belt	1	
11	Velcro fastener	0.3 m	
12	Cushion for 14x17 FPD	2	Used in "4.21 Attaching Cushion for 14X17 FPD"
13	Cushion for 10x12 FPD	1	Used in "4.22 Attaching Cushion for 10x12 FPD"

Cushion

Frame 4 ASSY

Frame 3 ASSY

Frame 5,6 ASSY

Frame 7,8 ASSY

FPD LOCK ASSY

Belt

Velcro fastener

Fig. 4-1

1 **Turn the main breaker OFF.**

2 **Remove the inside cover and both side covers from the system.**

Reference "3.5 How to Open/Close Covers"

3 **With the slit in the figure widened, attach the Cushion 1 and Cushion 2.**

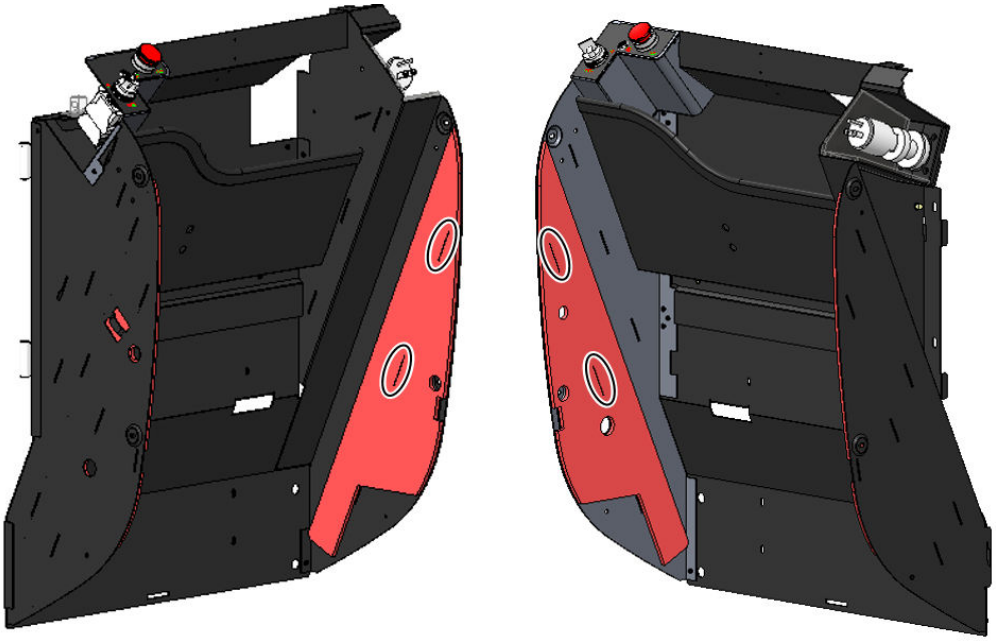


Fig. 4-2

4 **Remove two bolts for fixing the FPD box/Cassette box, and open the FPD box.**

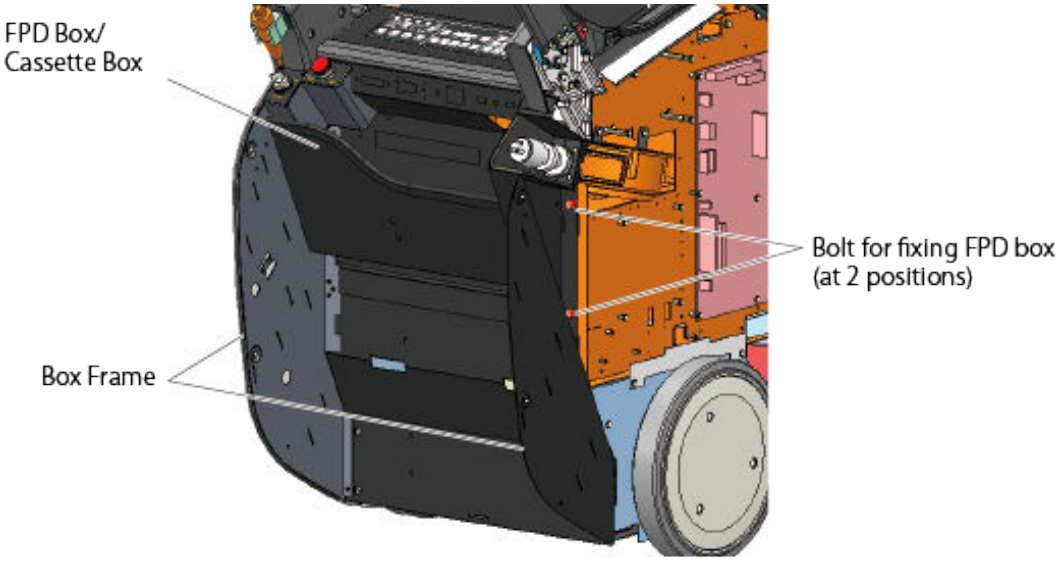


Fig. 4-3

5 **Loosen the screw in (1) and remove the screw in (2) to widen the space between the right and left box frames.**

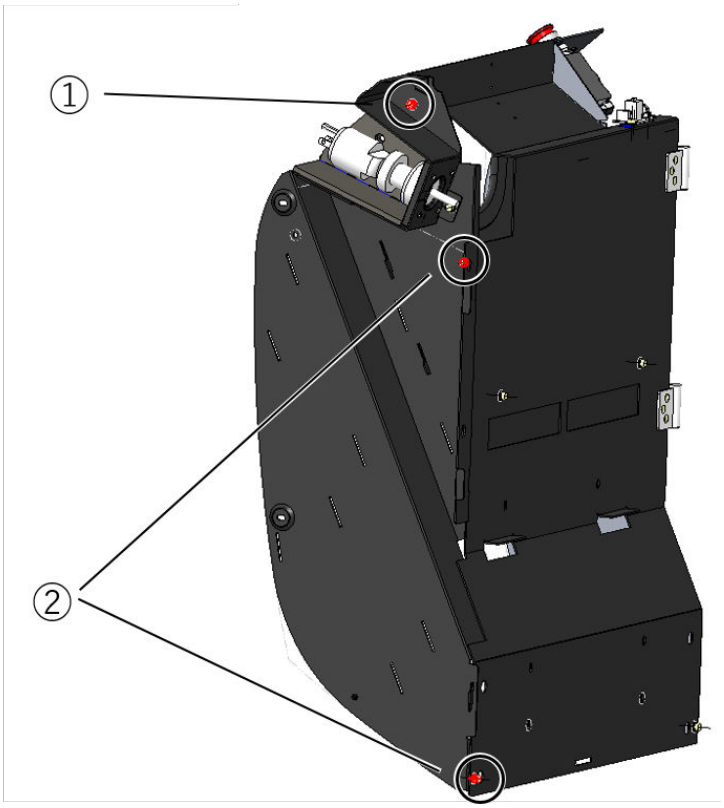


Fig. 4-4

6 **Insert the Frame 4.**

Hook and attach the pawl of the Frame 4 to the square hole of the box frame.

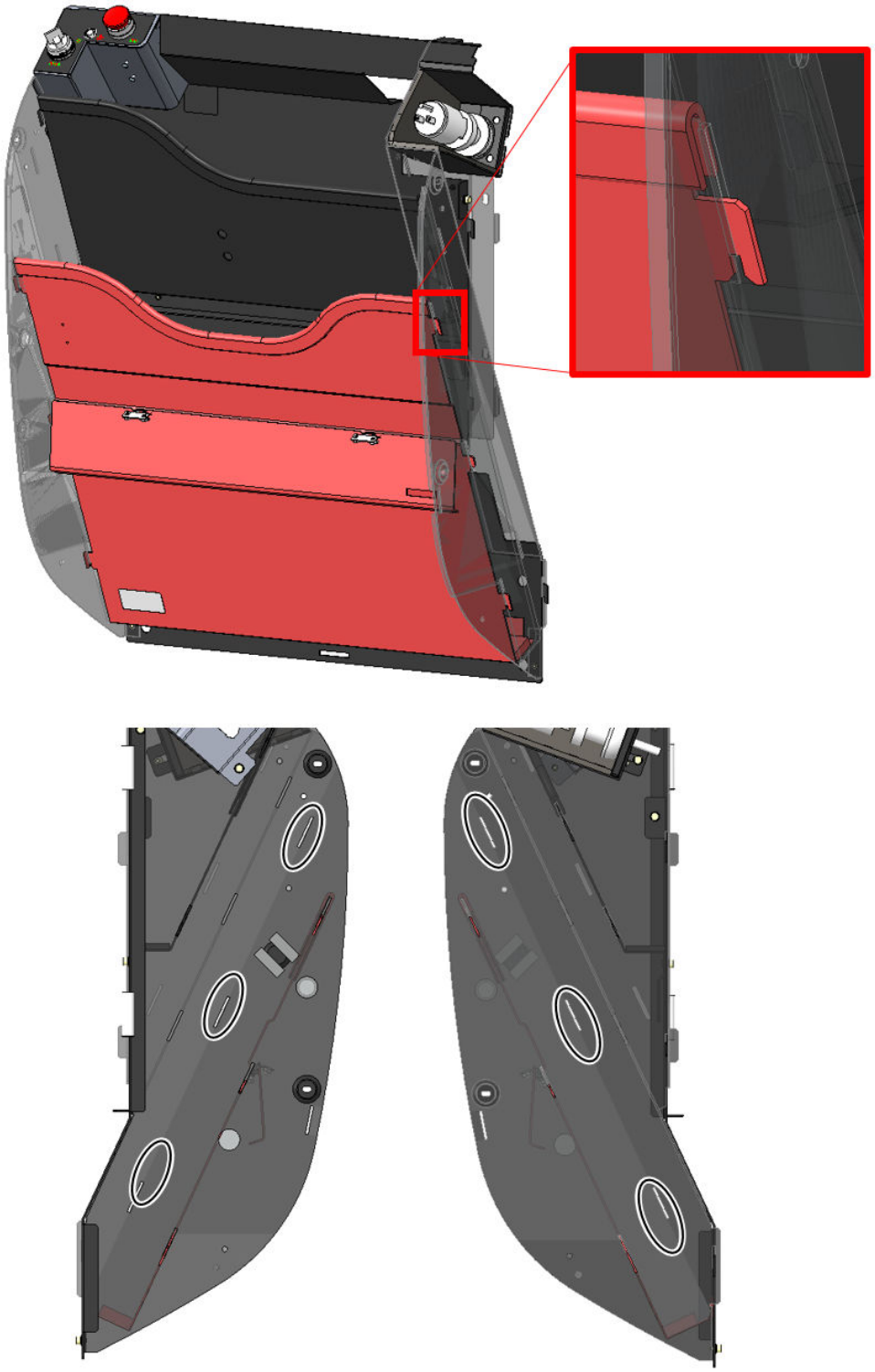


Fig. 4-5

7 **Insert the Frame 3.**

Lift the lid of the Frame 4 and insert the pawl of the Frame 3 into the square hole of the box frame, and then fasten the screw at two position teporarily.

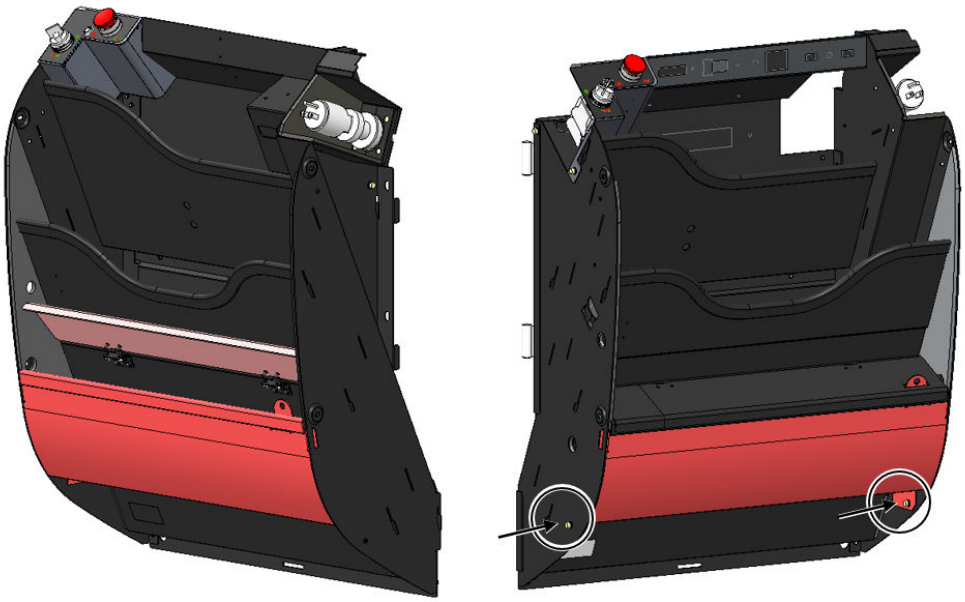


Fig. 4-6

8 **Attach the Frame 5 and the Frame 6.**

Hook each click of the Frame 5 and the Frame 6 to the square hole of the box frame, and fasten temporarily screws at 4 positions in each side and back side

Fig. 4-7

9 **Attach the Frame 7 and the Frame 8.**

Hook each click of the Frame 7 and the Frame 8 to the square hole of the box frame, and fasten temporarily screws at 4 positions in each side and back side.

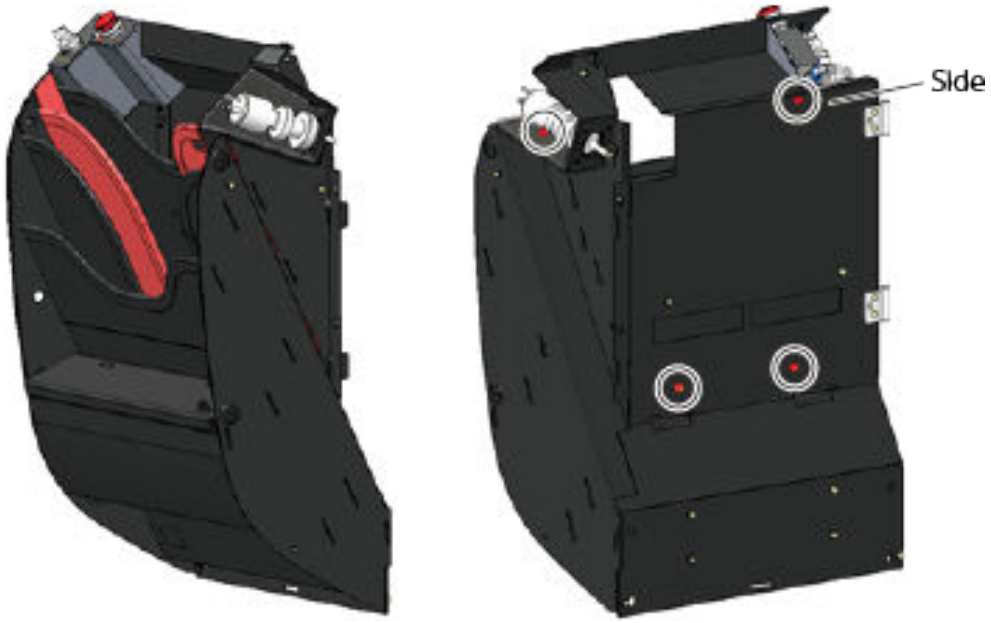


Fig. 4-8

10 Attach FPD LOCK ASSY.

1 Put FPD LOCK ASSY from the back side, and insert the lock block into the square hole of the Frame 8.

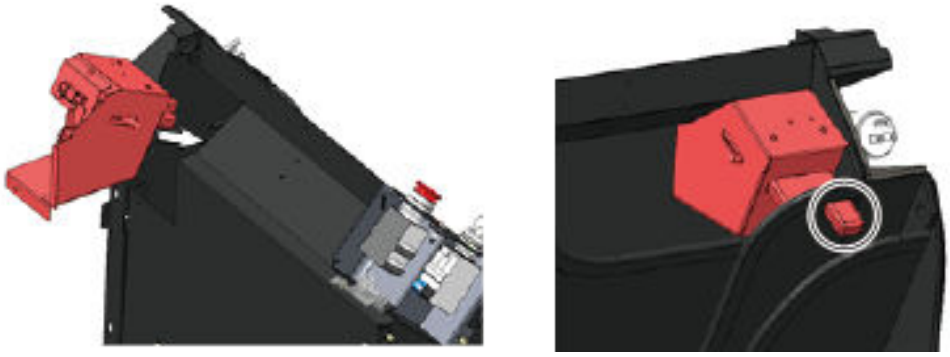


Fig. 4-9

2 Fasten temporarily two screws to secure FPD LOCK.

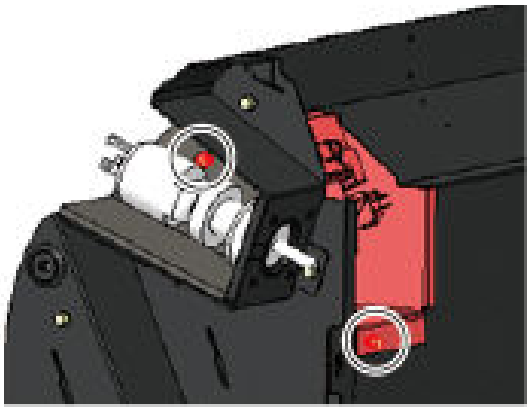


Fig. 4-10

11 Secure the frames.

1 Fastern the screws, which were loosen in Step 5, to fix the box frame.

2 Tighten the screws at 7 positions on the lateral side and 7 positions on the back to secure the Frame 3 to 8 and FPD LOCK ASSY.

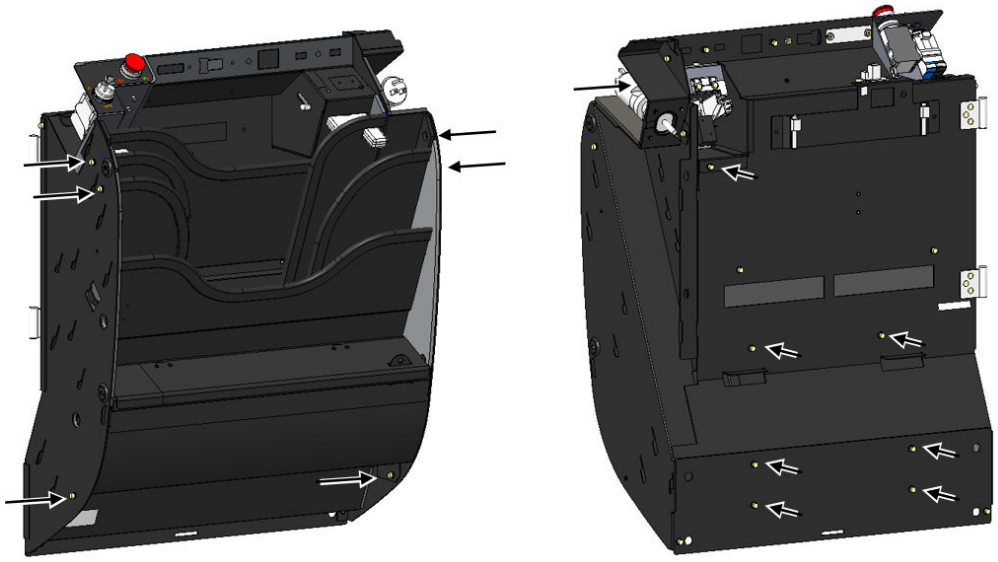


Fig. 4-11

12 Attach the belt through the belt loop.

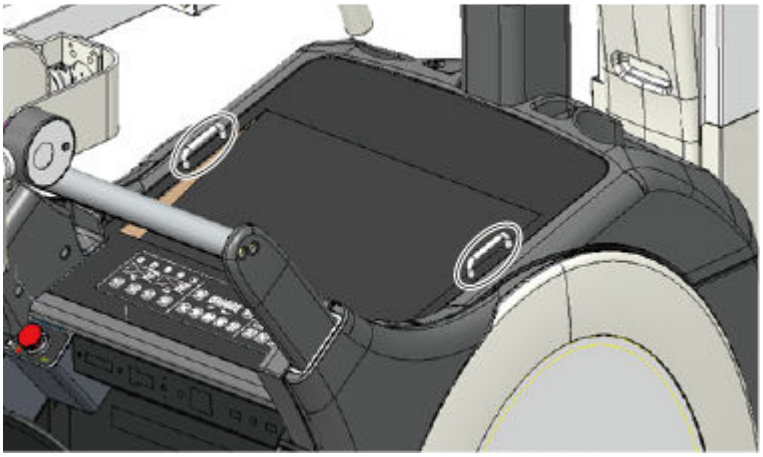


Fig. 4-12

NOTE The belt is used for fixing a laptop so that it doesn't fall while driving. refer to the following document for how to use the belt.

Reference
- M503-E070 MobilDaRt Evolution MX8 type CXDI Reference Guide
- M503-E071 MobilDaRt Evolution MX8 type DR-ID800 Reference Guide
- M503-E080 MobileDaRt Evolution MX8 type type AeroDR(CS-7) Reference Guide
- M503-E092 MobileArt Evolution MX8 Reference Guide

13 **When installing a connection device such as an access point, attach the velcro fastener to the area in the figure below and also the connection device.**

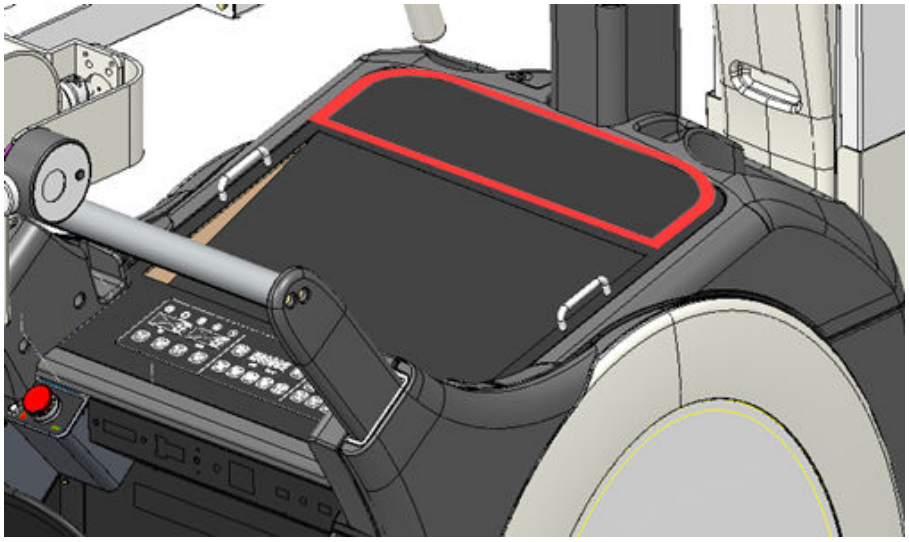


Fig. 4-13

NOTE Ensure that there is sufficient distance between the connection device and the collimator.
Store the arm in the catch section, and fix the connecting device in a position where a space of 30 mm or more can be made even if the tube or collimator is rotated.
When combining the folding screen option or infrared remote controller option, install the option parts and then secure the connecting device in a non-interfering position.

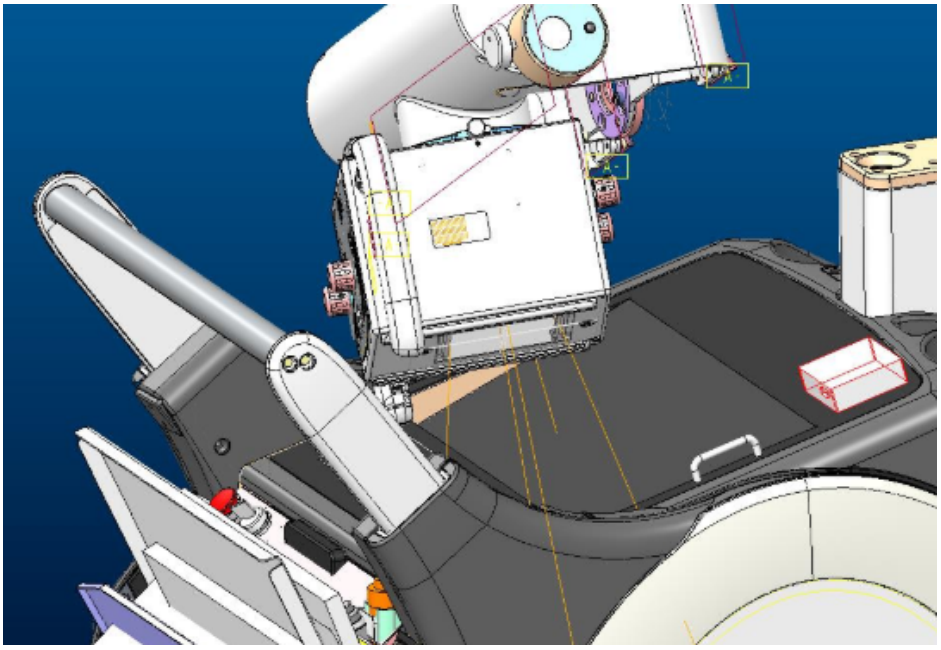


Fig. 4-14

14 **Turn ON DIP SW 3-4 on the XCONT-2018 board.**

Reference "6 [Appendix] DIP Switches and Jumpers Setting on Each Board"

15 **Attach the removed covers in reverse order of Step 2 and 4.**

4.3.2 Assembly of Cassette Box

The cassette box kit consists of the following parts. Make sure that all these parts are contained in the package.

Table 4-3 Component of cassette box

No.	Component	Amount
1	Frame A	1
2	Frame B	1
3	Cushion A	1
4	Cushion rubber	1
5	Frame cover	1
6	Hexagon socket head SEMS screw M4x6	2
7	Bind screw M4x6	2

Fig. 4-15

1 **Turn the main breaker OFF.**

2 **Remove the inside cover and both side covers from the system.**

Reference "3.5 How to Open/Close Covers"

3 **Remove two bolts for fixing the FPD box/Cassette box, and open the FPD box.**

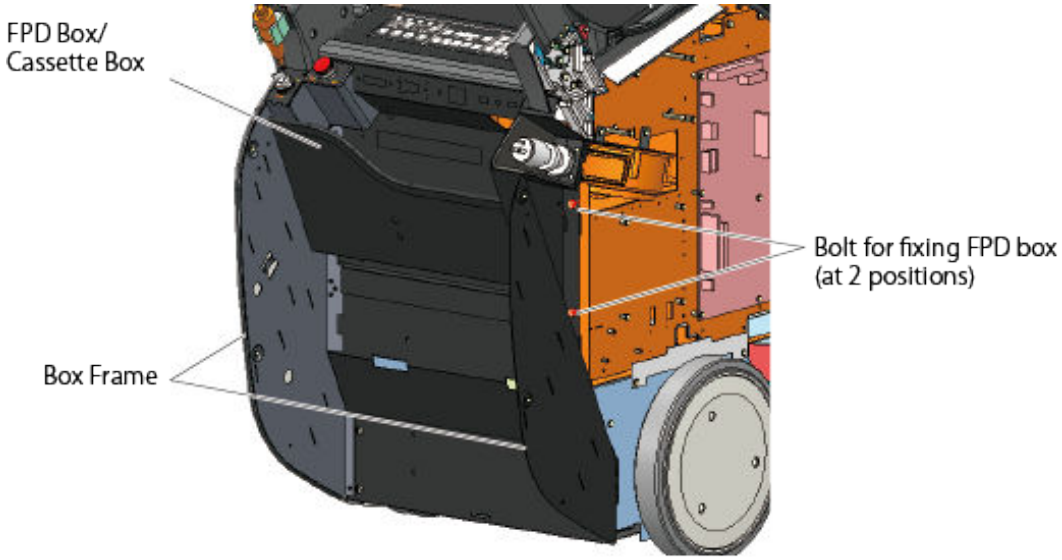


Fig. 4-16

4 **Loosen the screw (1) and remove the screw (2), and widen the space between the right and left box frames.**

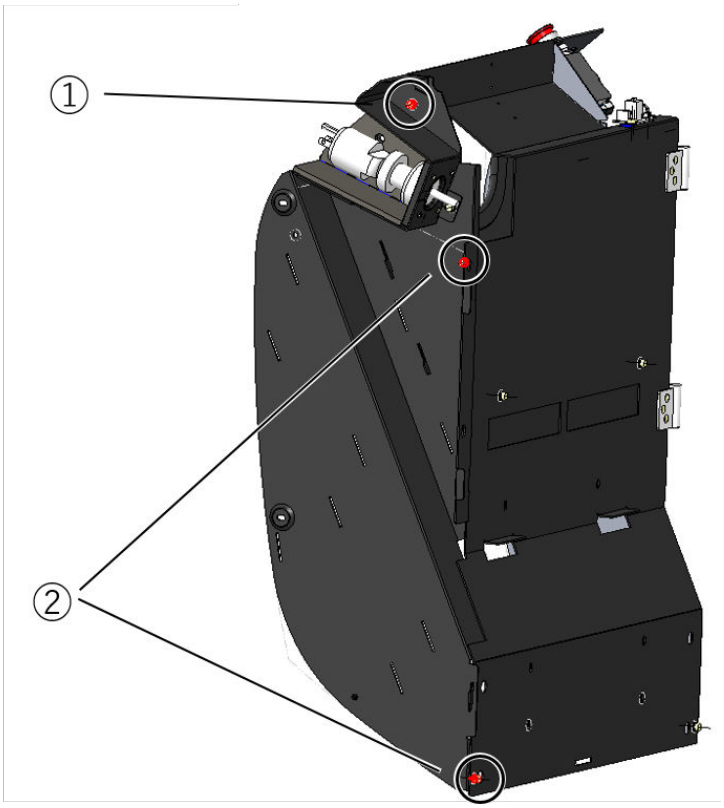


Fig. 4-17

5 **Insert the Frame A.**

Insert the click of the Frame A into the square hole of the box frame.

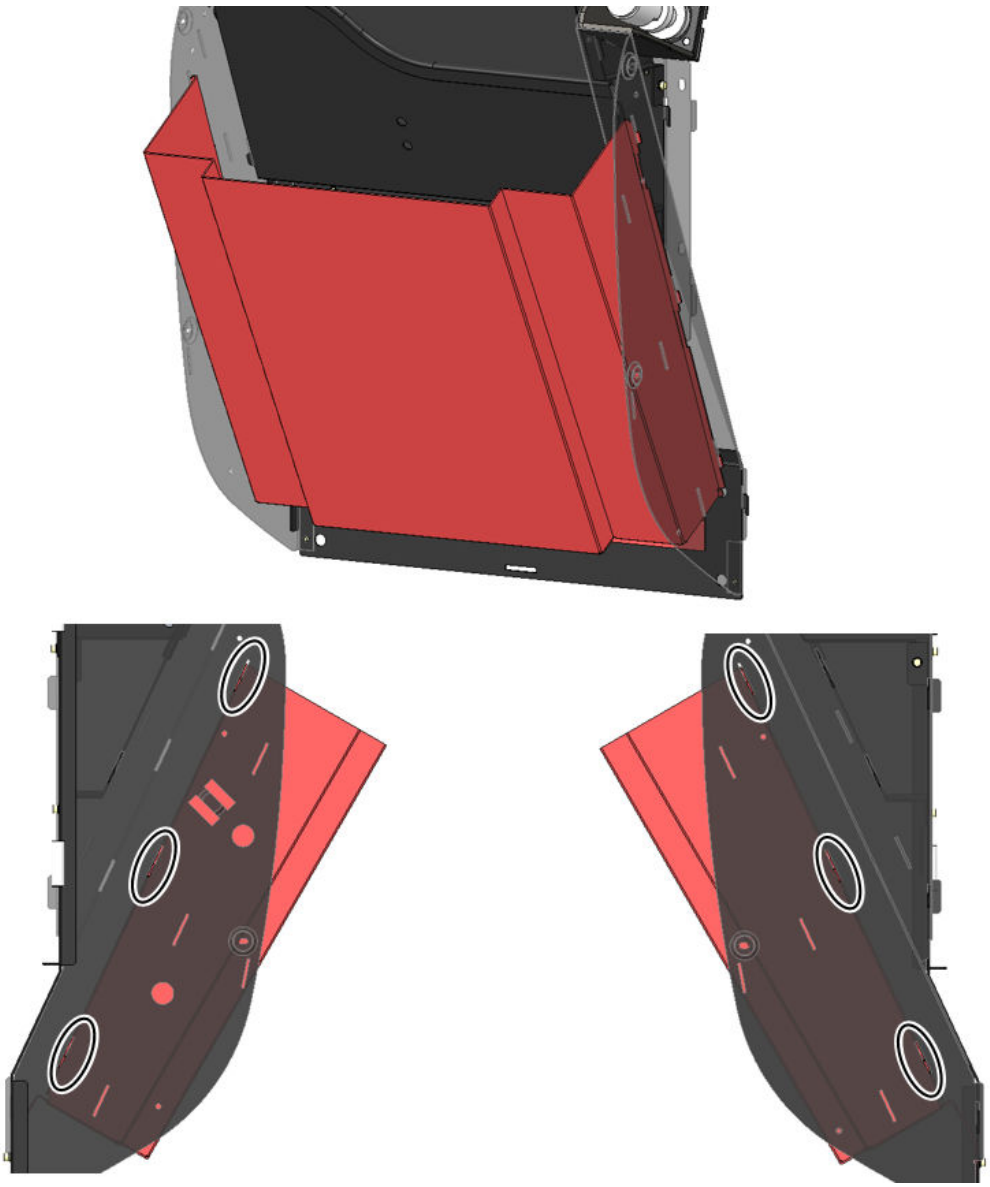


Fig. 4-18

6 **Insert the Frame B.**

Hook and attach the pawl of the Frame A to the square hole of the box frame.

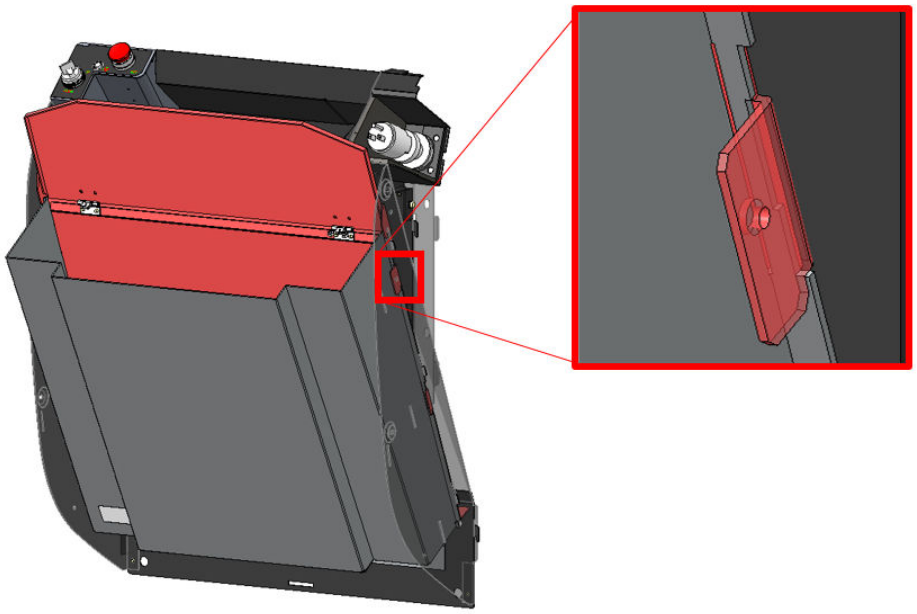


Fig. 4-19

7 Secure the frames.

1 Fastern the screws, which were loosen in Step 4, to fix the box frame.

2 Fasten screws at 4 positions in the back side and 2 positions in the side to secure the Frame A and the Frame B.

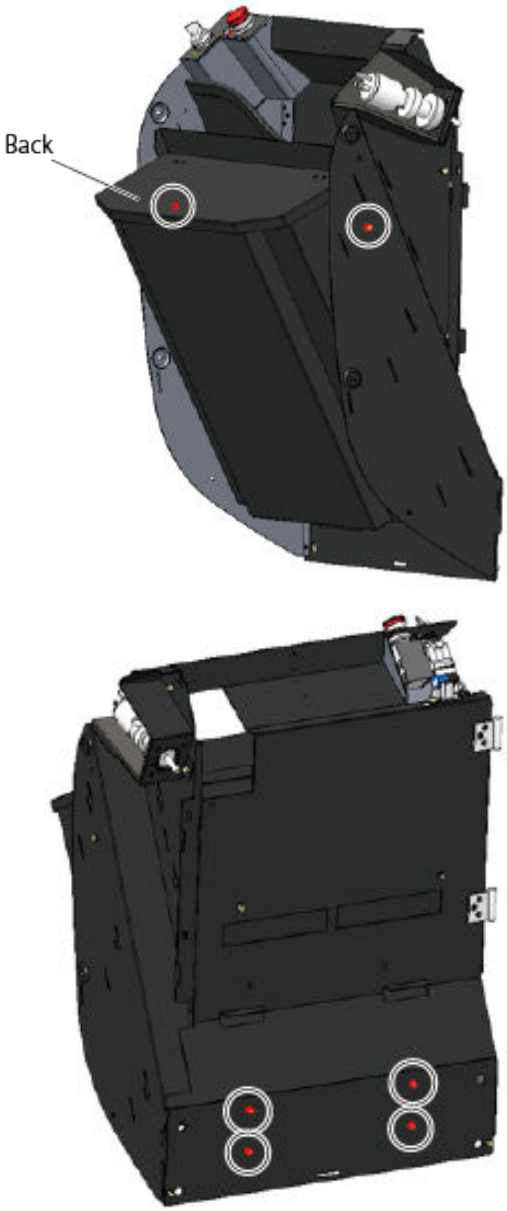


Fig. 4-20

8 **Attach the cushion.**

Place the Cushion A and Cushion rubber on the bottom of the Frame A.

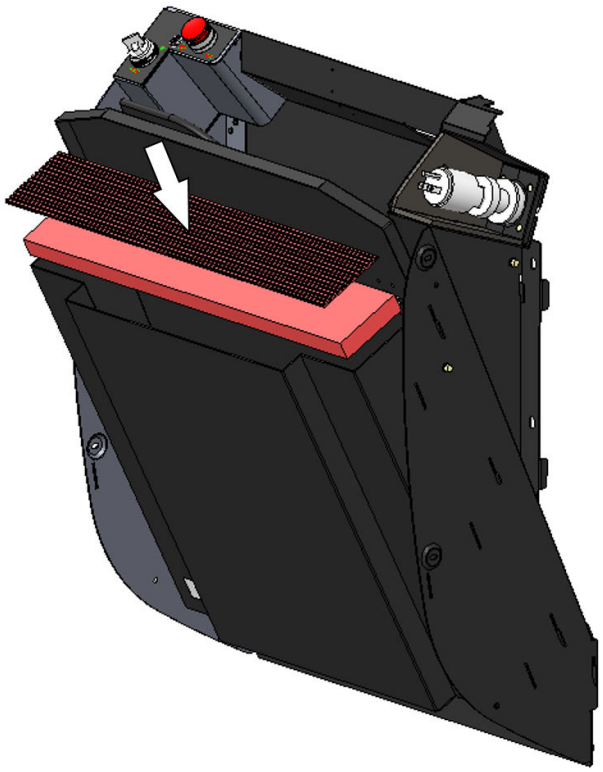


Fig. 4-21

9 Attach the frame cover.

1 Insert the frame cover from the back, and insert the pawl into the square hole of the frame box..

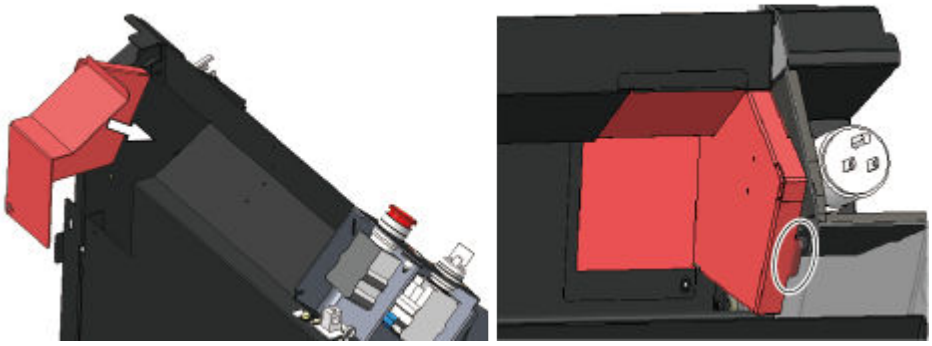


Fig. 4-22

2 Fasten two screws to secure the frame cover.

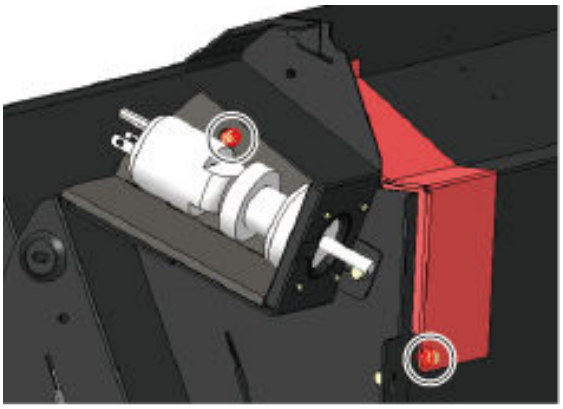


Fig. 4-23

10 **Attach the removed covers in reverse order of Step 2 and 3.**

In the case of the cassette box, it is necessary to change the fastener components for workability. Replace the side cover fixing Sems screws M4x10 2PC in the figure below with (1) Bind screw M4x8 and (2) hexagon socket head SEMS screw M4x6 to attach the cover.

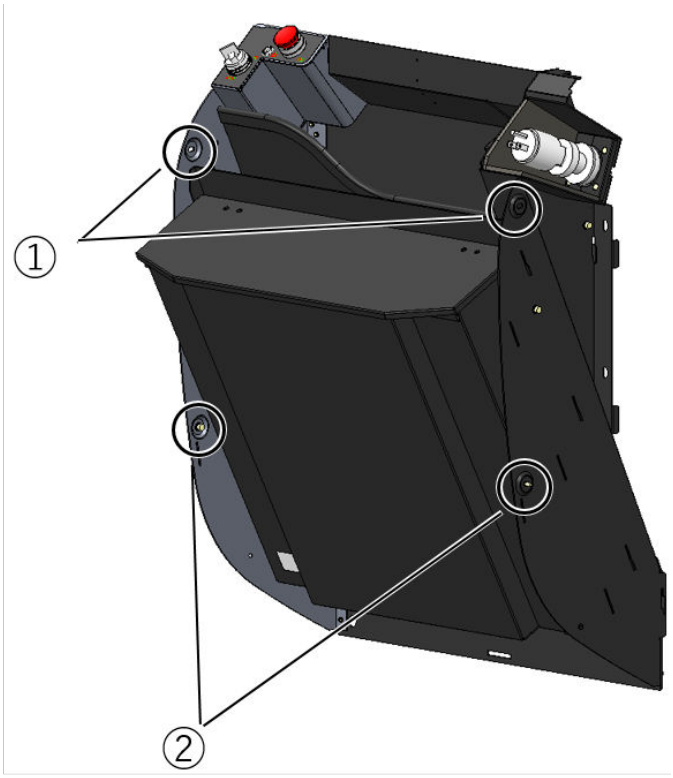


Fig. 4-24

4.4 Installing Cushion ASSY(When using FPD Docking Mechanism with 14x17 or 10x12 size FPD)

Before installation, please prepare the items listed in the table below from cushion ASSY.

Table 4-4

No.	Component	Amount
1	Cushion for Half A	1
2	Cushion for Half B	2
3	Cushion support base ASSY	1

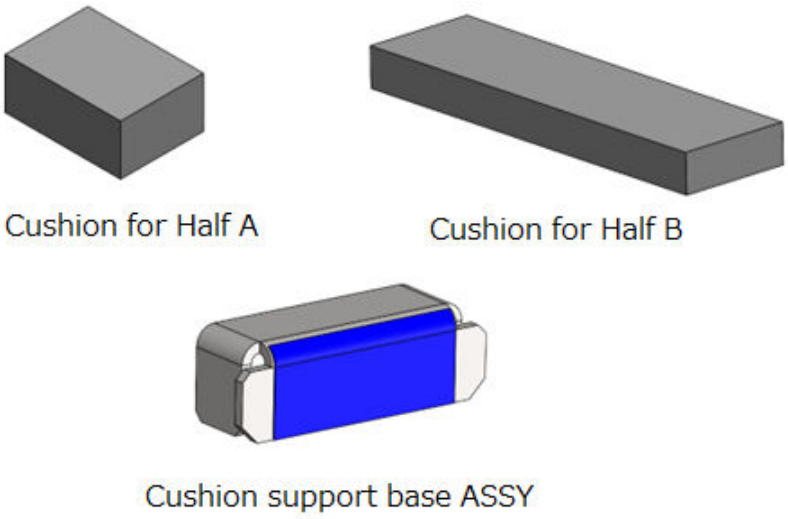


Fig. 4-25

1 **Turn OFF the main breaker.**

2 **Remove the inside cover and right and left side covers.**

Reference "3.5 How to Open/Close Covers"

3 **Discharge the internal capacitor.**

Reference "3.4 Discharging the Internal Capacitor"

4 **Remove the FPD box fixing bolt, and open the FPD box.**

5 **Remove two screws, and remove frame 7.**

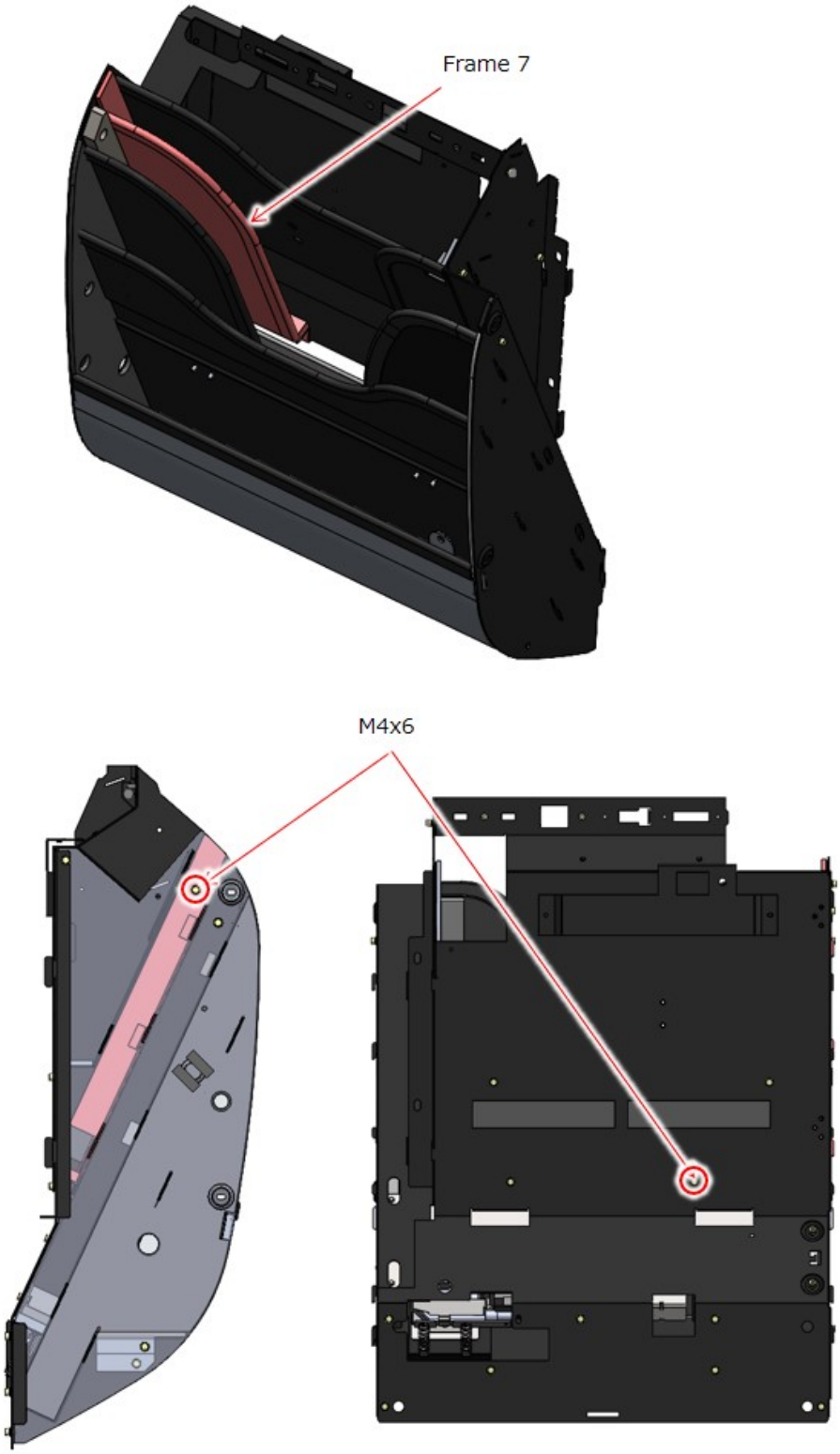


Fig. 4-26

6 **Remove 7 screws shown in the figure below.**

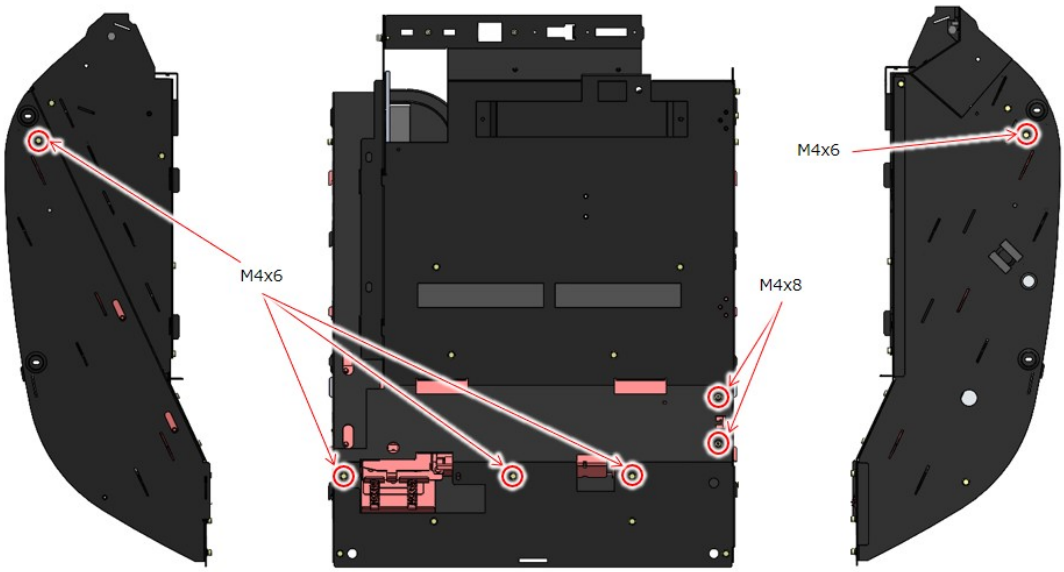


Fig. 4-27

7 Remove Left frame ASSY and Right frame ASSY.

Hint

- The frames can be removed by removing the claws that are caught in the square holes of the box frame.
- Remove the left frame ASSY first while lifting the right frame ASSY.

NOTE Do not disassemble the FPD-guided right frame ASSY because the dimensions are adjusted when the parts are assembled at the factory.

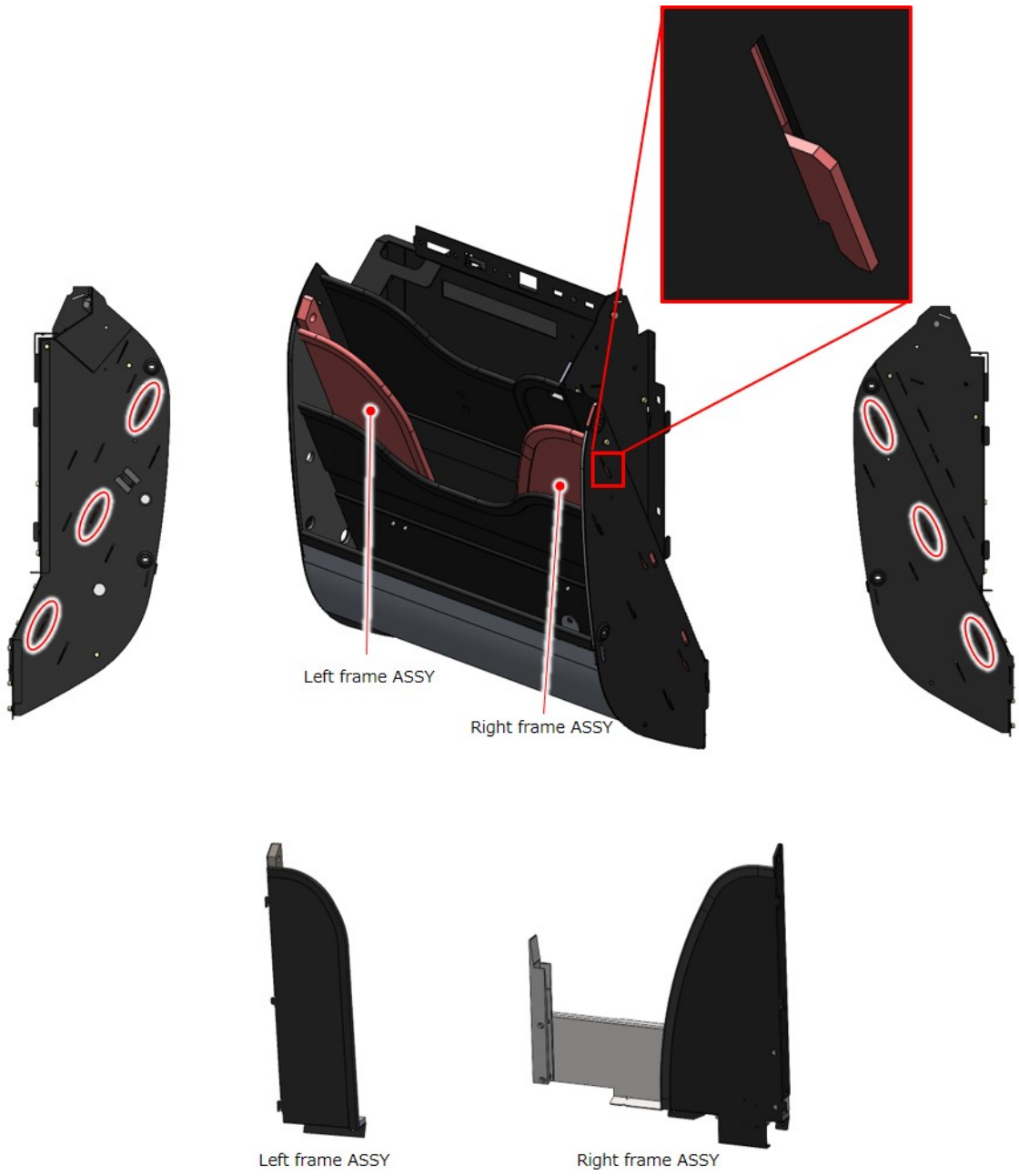


Fig. 4-28

8 **Remove two springs from the right frame ASSY.**

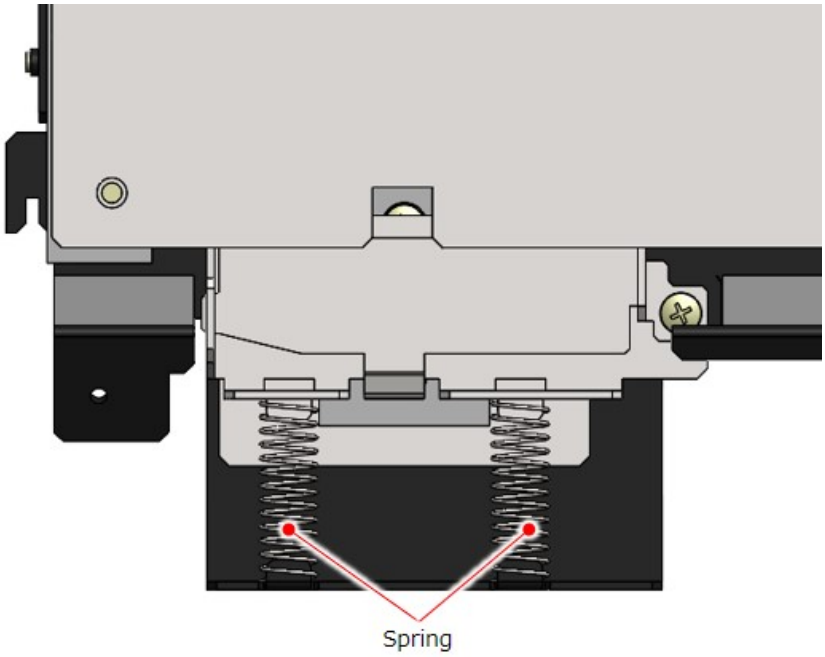


Fig. 4-29

9 **Remove the release coated paper of the cushion for half A and attach it to the gap in the POM guide.**

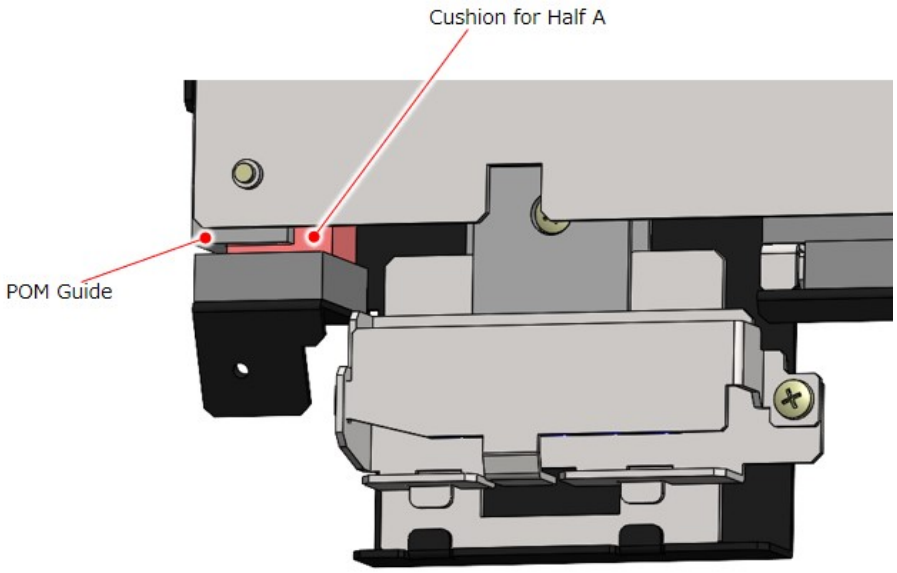


Fig. 4-30

10 **Remove the release coated paper of the cushion support base ASSY and attach it to the side of the cushion R while placing the end face of the cushion support base ASSY to the plate BR.**

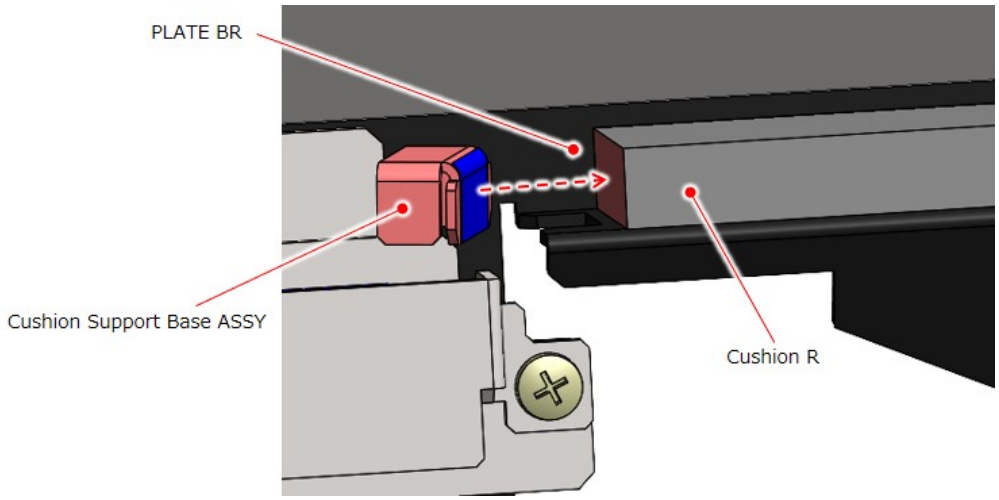


Fig. 4-31

11 **Remove the release coated paper of the cushion for half B and attach it to the top of cushion support base ASSY and cushion R.**

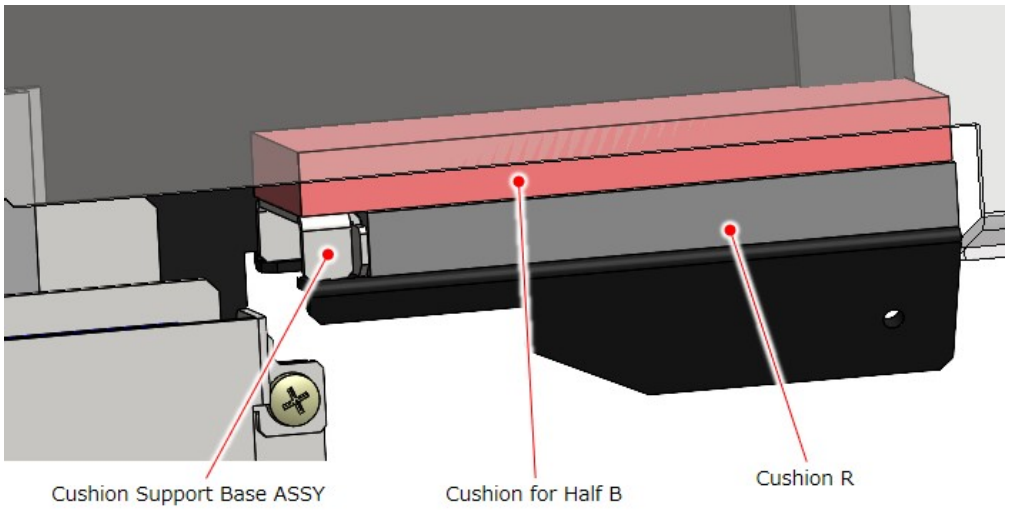


Fig. 4-32

12 Attach the two springs removed in step 8.

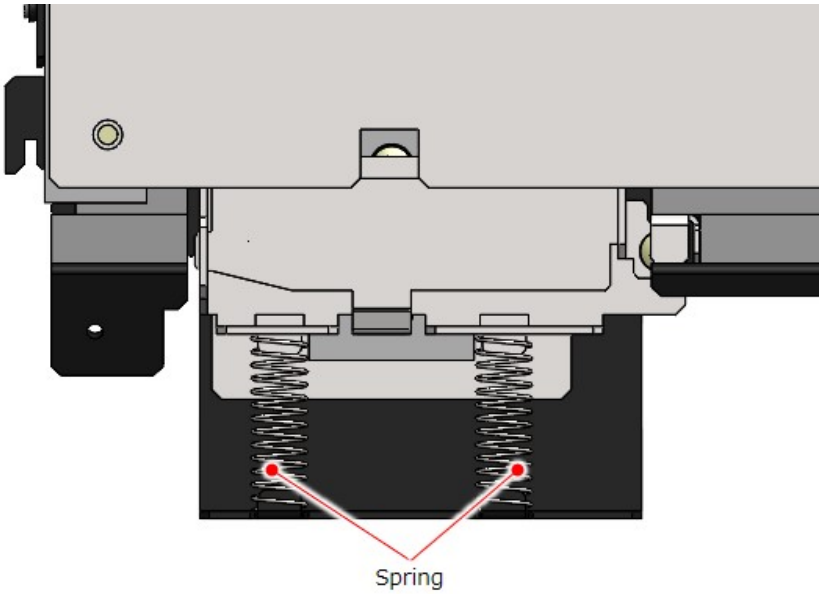


Fig. 4-33

13 Remove the release coated paper of cushion for half B and attach it to the top of cushion 8 of the left frame ASSY.

> **NOTE** Note that the end faces shown in the figure should be aligned and that the cushion for half B should be installed while pressing it to the left frame ASSY.

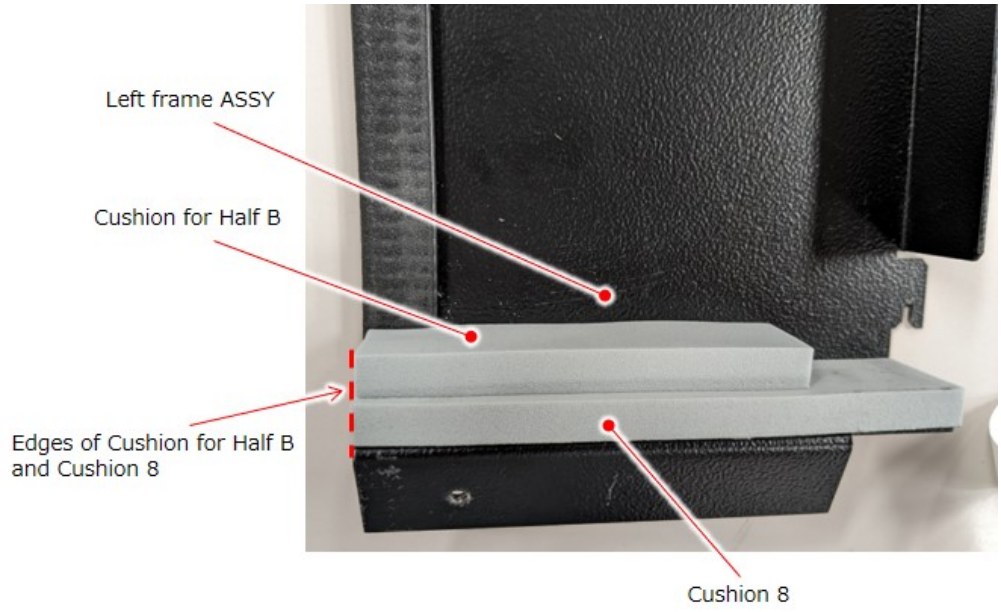


Fig. 4-34

14 Attach the left frame ASSY, right frame ASSY and frame 7 in reverse order of step 5-7.

15 **Reset the internal capacity discharge to “NORMAL”.(Both discharge switch and connector should be reset.)**

16 **Attach the FPD box and the covers in reverse order of step 2 and 4.**

4.5 Assembly of FPD Docking Mechanism (For CXDI)

Before installation, please prepare the items listed in the table below.

Table 4-5

No.	Component	Amount
1	Wiring Cable	1
2	Cable tie (Accessory)	5

NOTE Before assembling the FPD docking mechanism to the FPD box, install the Power Box Option or Wired Connection Option.
Refer to the corresponding procedure for installation.

Reference M503-E410 MobileDaRt Evolution (MX8 Version) installation Manual (DR Section)
- " Installing Power Box Option"
- "Installing Wired Connection Option (MB-02) "

If using 14x14 or 10x12 FPD, install cushion ASSY first.

Reference "4.4 Installing Cushion ASSY(When using FPD Docking Mechanism with 14x17 or 10x12 size FPD)"

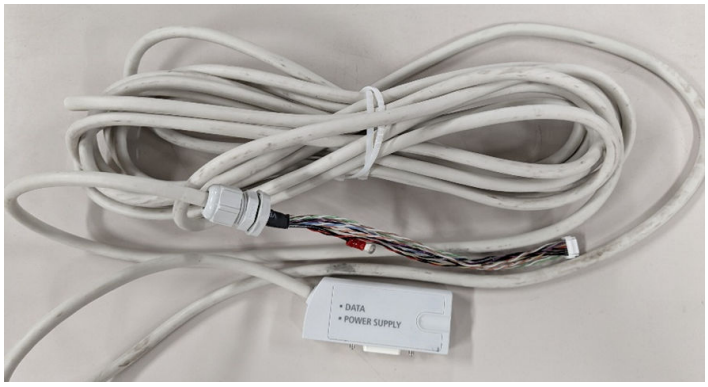


Fig. 4-35 Wiring Cable

1 **Turn OFF the main breaker.**

2 **Remove the inside cover and right and left side covers.**

Reference "3.5 How to Open/Close Covers"

3 **Discharge the internal capacitor.**

Reference "3.4 Discharging the Internal Capacitor"

4 **Remove the FPD box fixing bolt, and open the FPD box.**

5 **Remove two springs.**

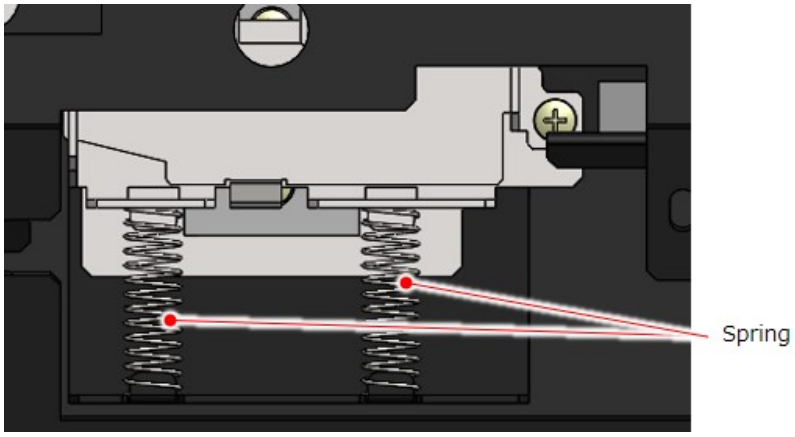


Fig. 4-36

6 **Remove the screw (sems screw and two washers), and remove the cover ASSY.**

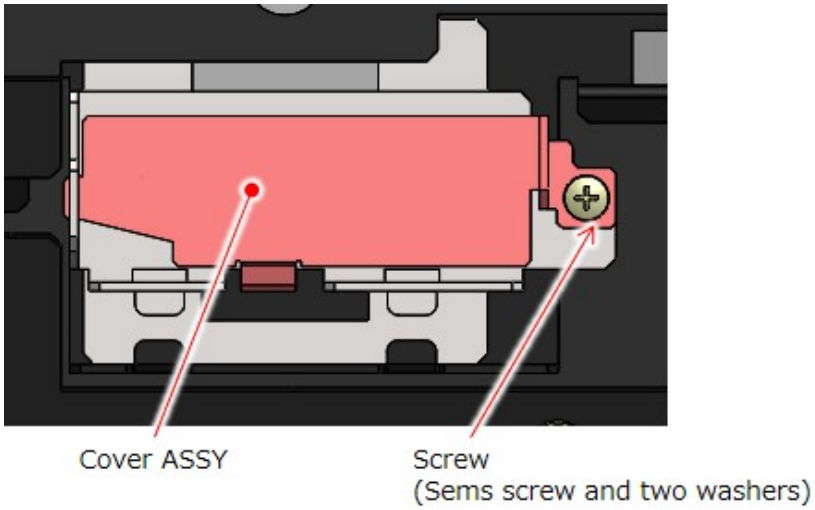


Fig. 4-37

7 **Remove the release coated paper of the double-sided tape of cover ASSY.**

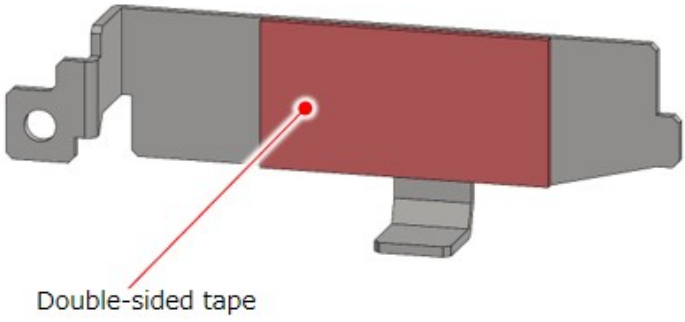


Fig. 4-38

8 **Attach double-sided tape to the wiring cable connector with the sheet metal of the cover ASSY in contact with the wiring cable connector as shown in the figure below. Ensure that the top surface of the sheet metal is parallel to the terminal surface of the connector.**

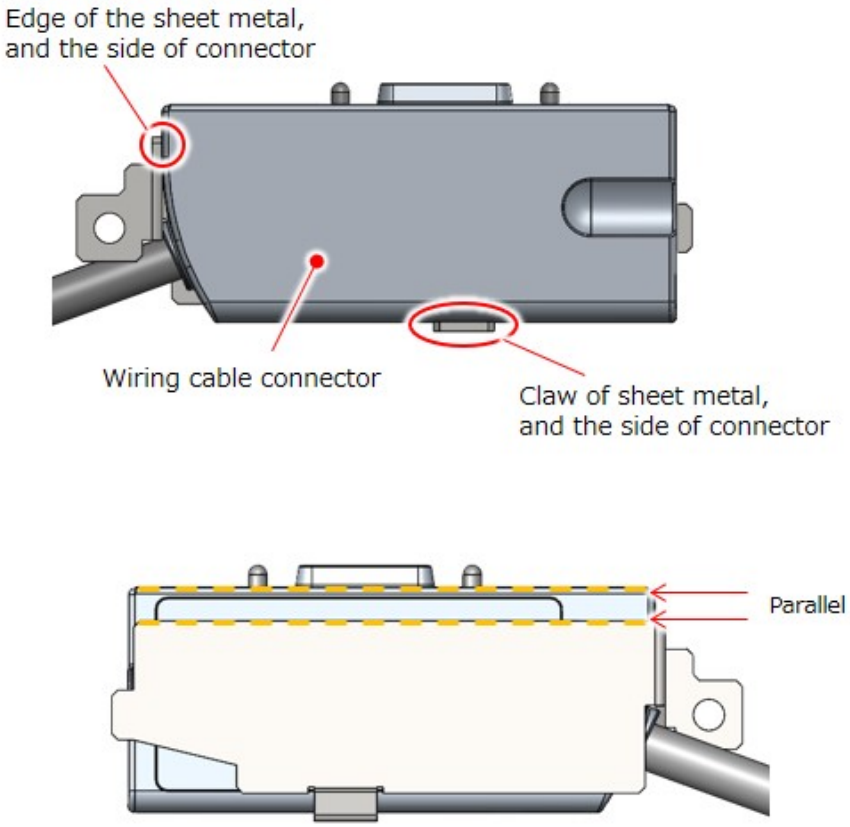


Fig. 4-39

9 Wire the wiring cable as shown in the figure below.

Hint If it is difficult to pass the cable through, pass the cable through while lifting the base plate upward.

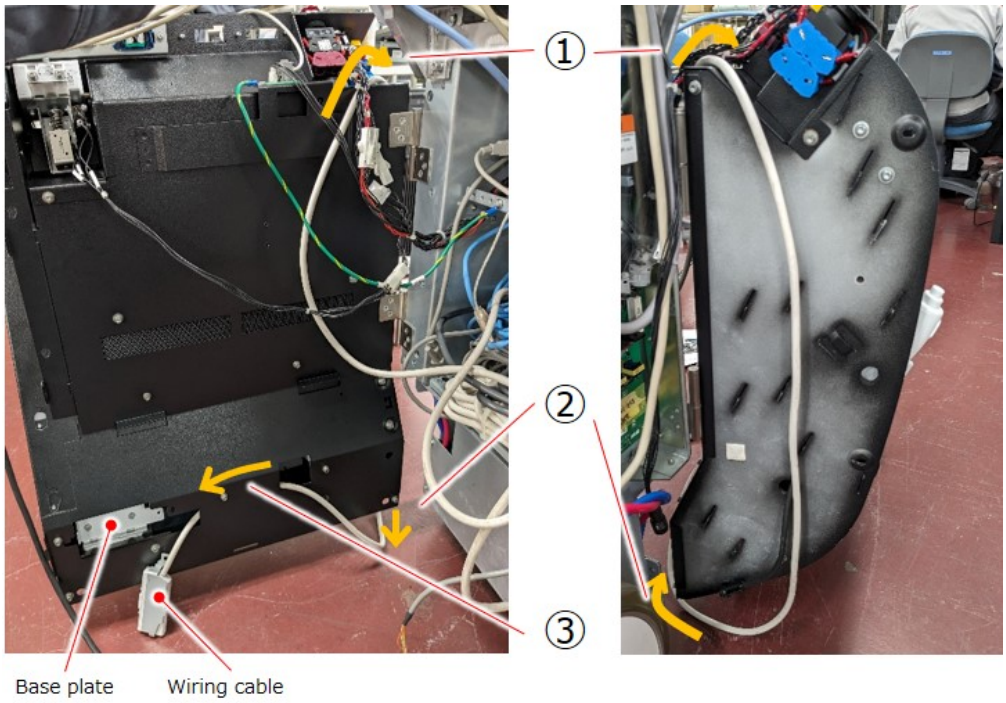


Fig. 4-40

10 Using the sems screw and two washers removed in step 6, attach the cover ASSY with the wiring cable connector to the base plate. When attaching, press and hold the wiring cable connector against the indicated position of the base plate in the figure, and screw it.

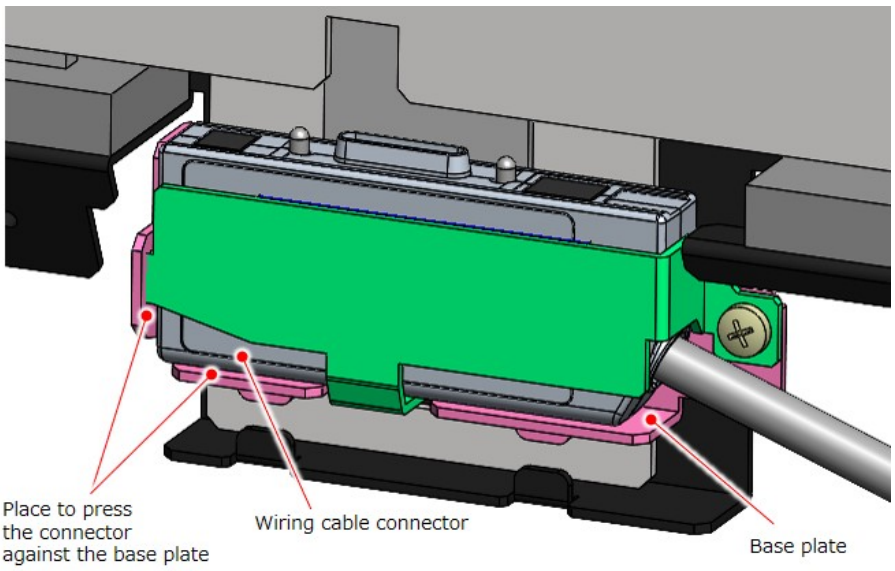


Fig. 4-41

11 **Attach the two springs removed in step5.**

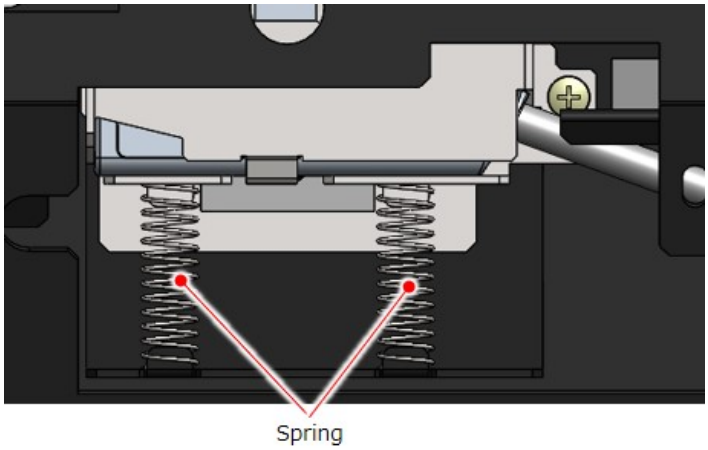


Fig. 4-42

12 **Wire the wiring cable along the rear frame of the FPD box as shown in the figure below, and fix the wiring cable with cable ties.**

NOTE
- Fix the cable in the center of the left-right movement allowance of the wiring cable connector.
- Fix the cable in the state of natural wiring to avoid excessive force.
- Be especially careful not to twist the cable at fixing position (1).

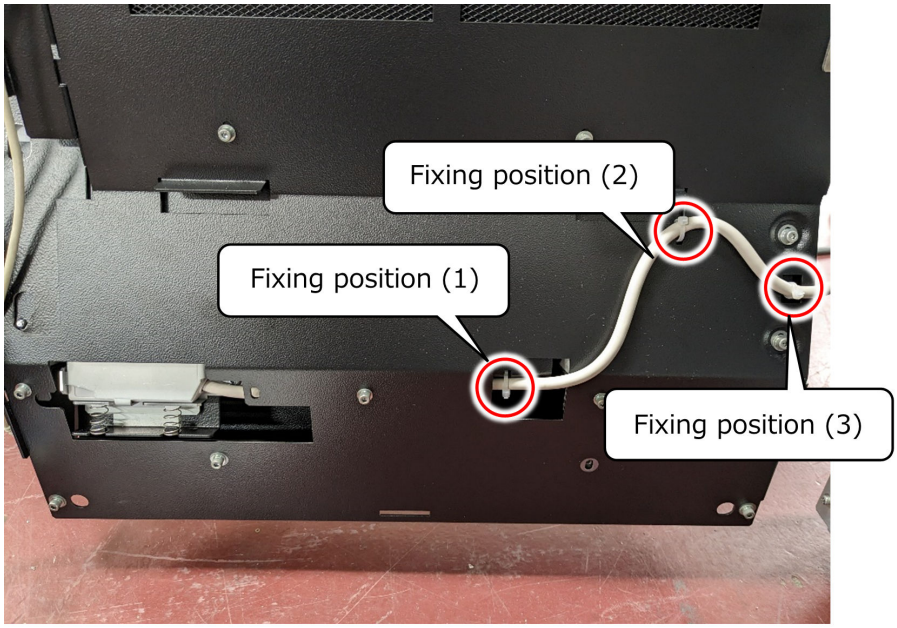


Fig. 4-43

13 **Insert the FPD to be used at the delivery destination into the FPD box and confirm that there is no insertion resistance and that the wiring cable connector mates with the FPD connector.**

14 **Wire the cable from FPD box side to the reverse side of the emergency stop button, as shown in the figure below.**

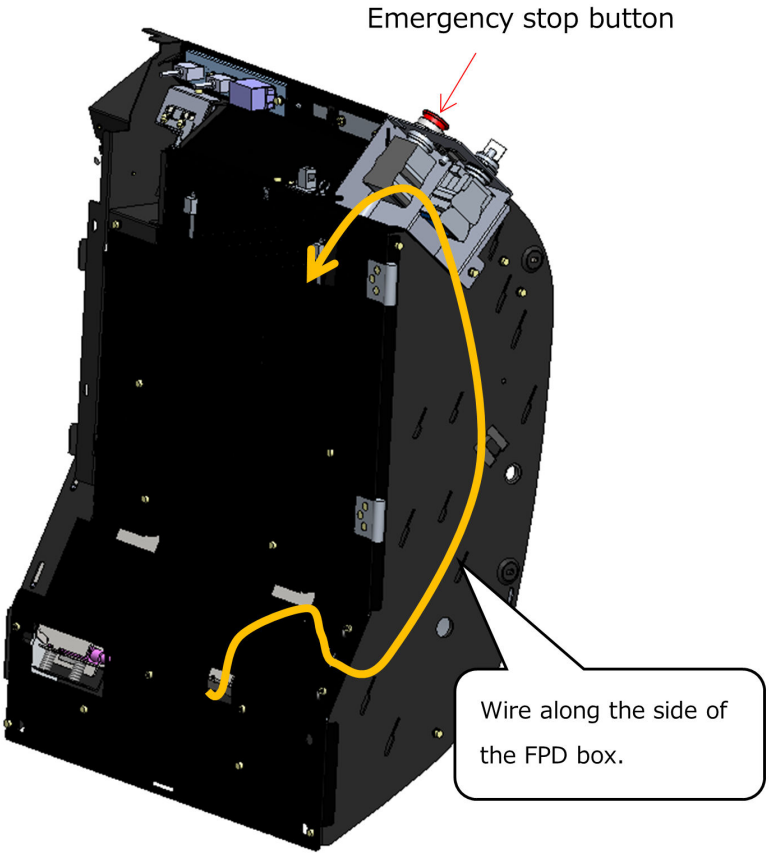


Fig. 4-44

15 **Wire the wiring cable to the side of the FPD box for natural deflection, and fix it along with the cable extending from the emergency stop button with a cable tie.**

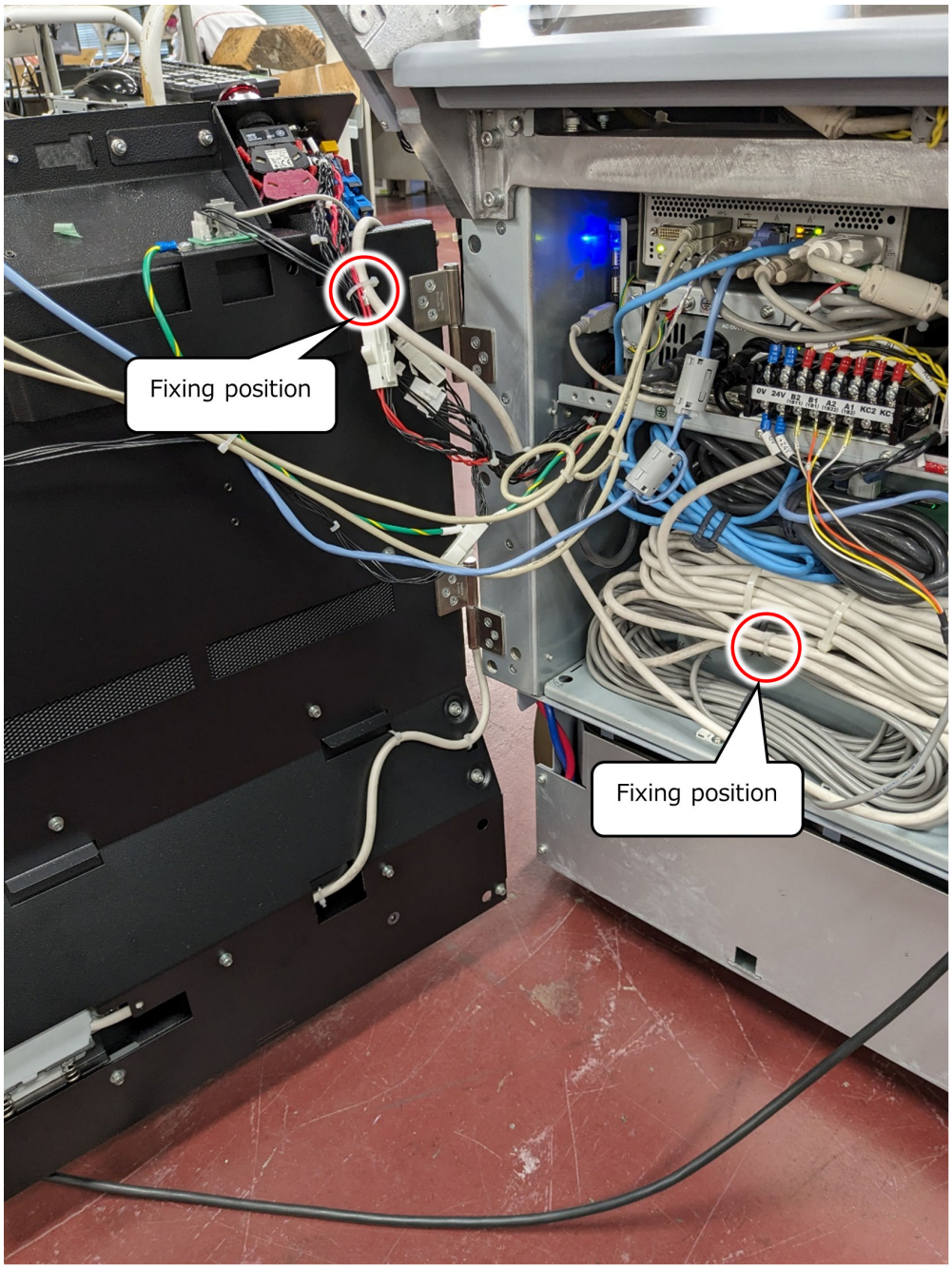


Fig. 4-45

16 **Reset the internal capacity discharge to "NORMAL". (Both discharge switch and connector should be reset.)**

17 **Attach the FPD box and the covers in reverse order of step 2 and 4.**

4.6 Assembly of FPD Docking Mechanism (For VXvue)

Before installation, please prepare the items listed in the table below.

Table 4-6

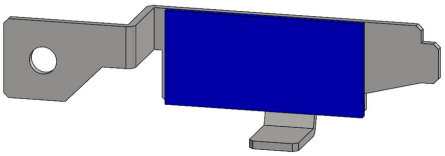
No.	Component	Amount
1	Thether I/F cable	1
2	Cover ASSY (8C-8V U/G ASSY)	1
3	Base plate (8C-8V U/G ASSY)	1
4	Cable tie (8C-8V U/G ASSY)	4

NOTE Before assembling the FPD docking mechanism to the FPD box, install the Wired Charging Kit Option.
Refer to the corresponding procedure for installation.

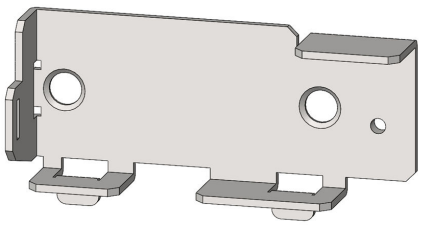
Reference M503-E410 MobileDaRt Evolution (MX8 Version) installation Manual (DR Section)
- "Wired Charging Kit Option."

If using 14x14 or 10x12 FPD, install cushion ASSY first.

Reference "4.4 Installing Cushion ASSY(When using FPD Docking Mechanism with 14x17 or 10x12 size FPD)"



Cover ASSY



Base plate

Thether I/F cable

Fig. 4-46

1 **Turn OFF the main breaker.**

2 **Remove the inside cover and right and left side covers.**

Reference "3.5 How to Open/Close Covers"

3 **Discharge the internal capacitor.**

Reference "3.4 Discharging the Internal Capacitor"

4 **Remove the FPD box fixing bolt, and open the FPD box.**

5 **Remove two springs.**

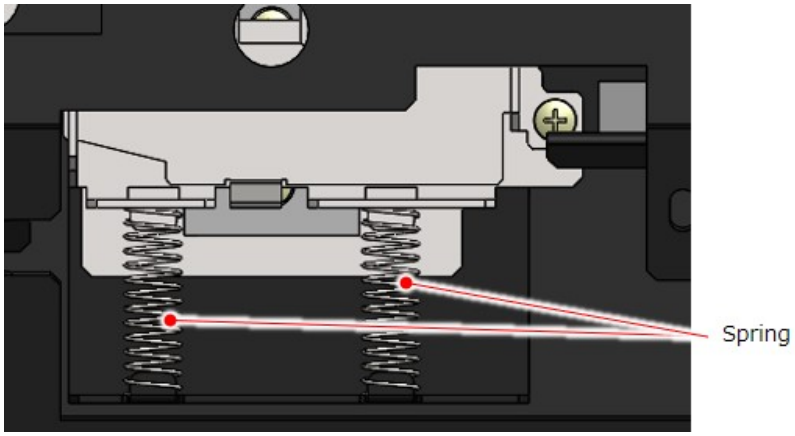


Fig. 4-47

6 **Remove the bind screw, and remove the cover ASSY.**

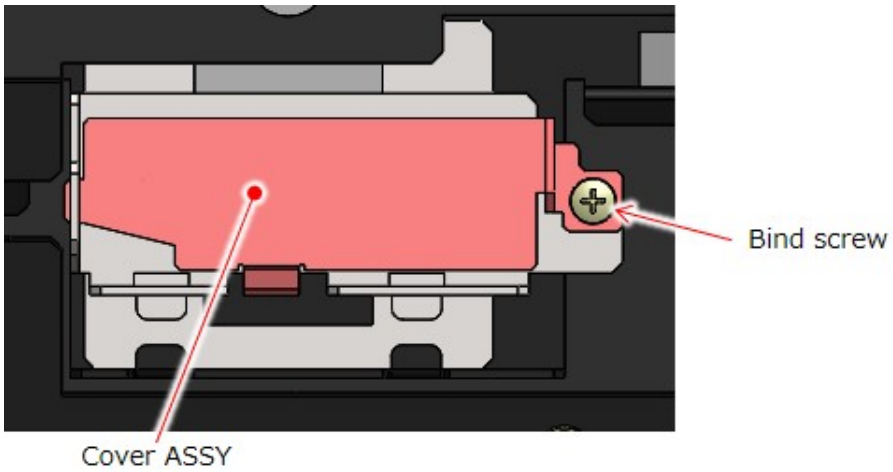


Fig. 4-48

7 Remove two flat screws, and replace the base plate for CXDI with one for VXvue.

NOTE
- Ensure that the spacer is inserted into the protrusion of the barring.
- Cover ASSY and base plate for CXDI will no longer be used.

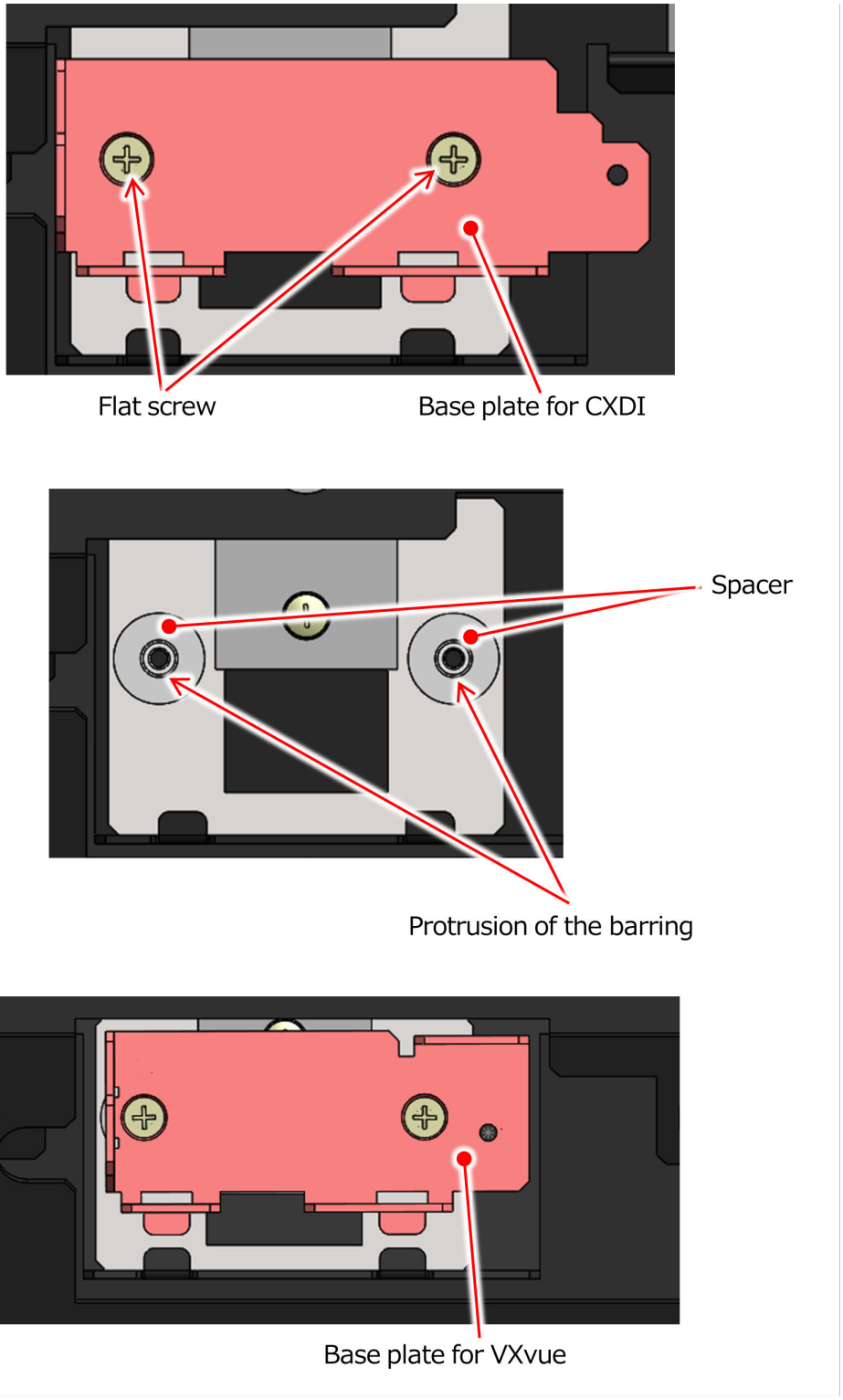


Fig. 4-49

8 **Remove the release coated paper of the double-sided tape of cover ASSY.**

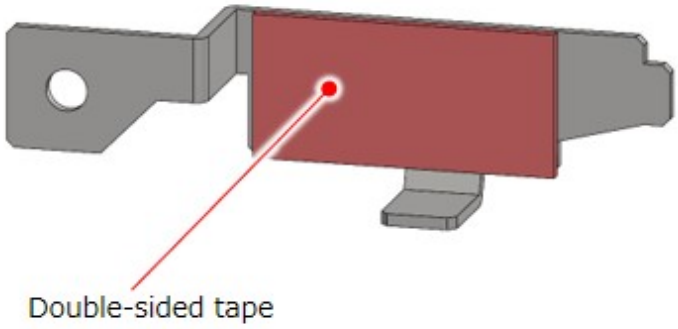


Fig. 4-50

9 **Attach double-sided tape to the thether I/F cable connector with the sheet metal of the cover ASSY in contact with the thether I/F cable connector as shown in the figure below. Ensure that the top surface of the sheet metal is parallel to the terminal surface of the connector.**

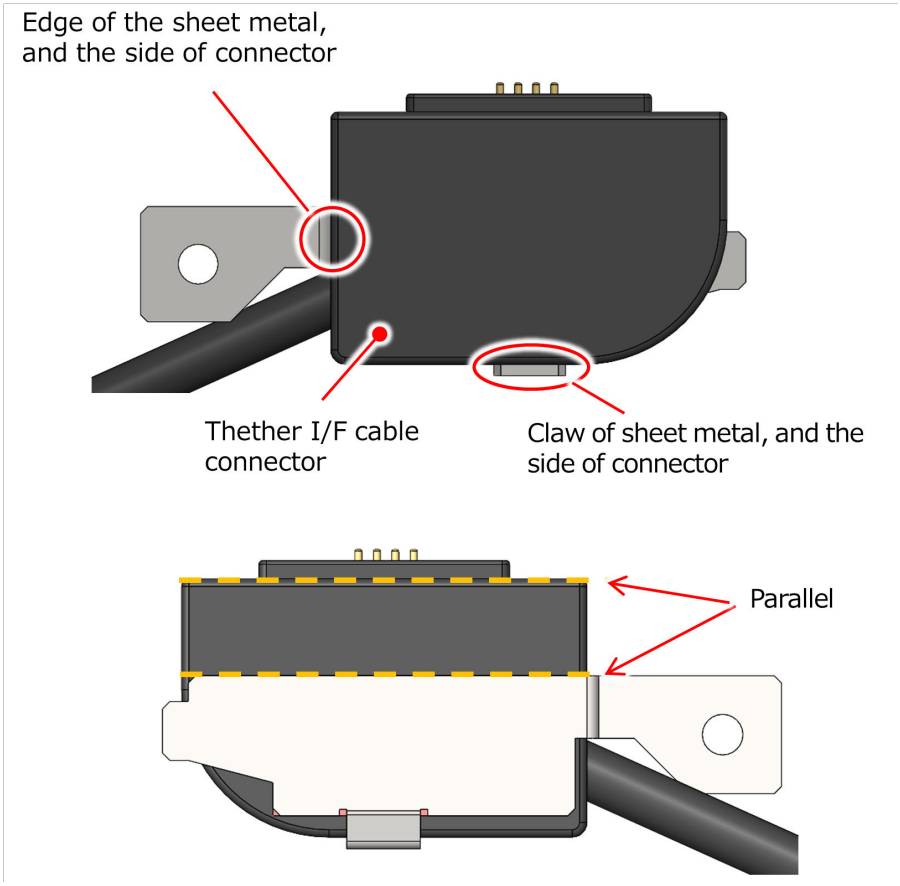


Fig. 4-51

10 Wire the thether I/F cable as shown in the figure below.

Hint If it is difficult to pass the cable through, pass the cable through while lifting the base plate upward.

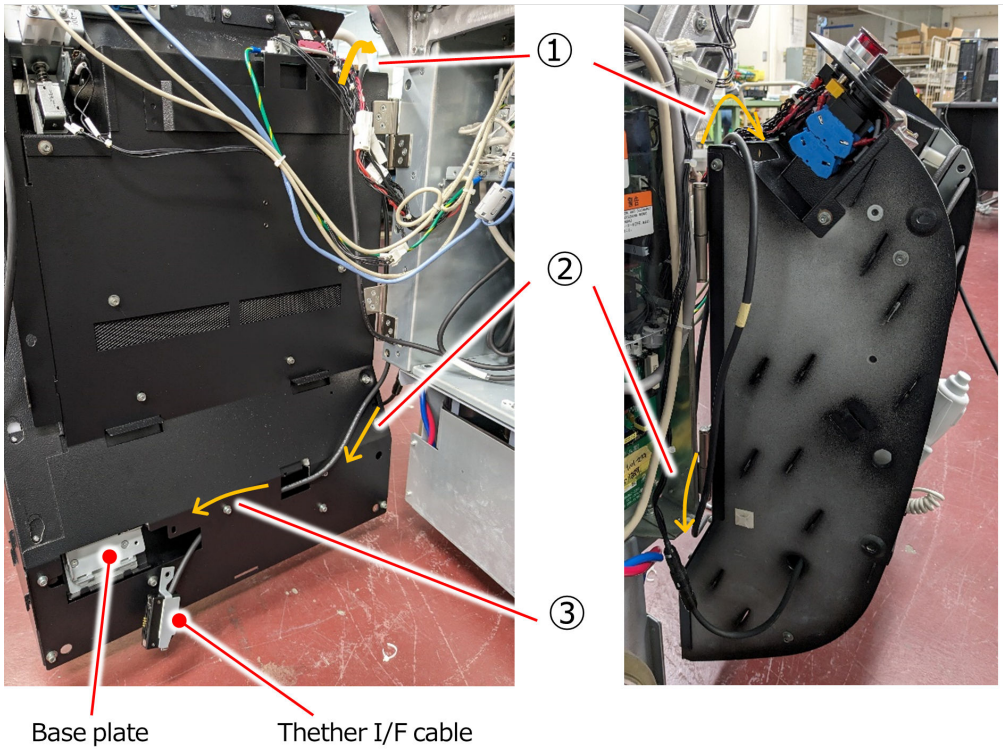


Fig. 4-52

11 **Using the bind screw removed in step 6, attach the cover ASSY with the thether I/F cable connector to the base plate. When attaching, press and hold the thether I/F cable connector against the indicated position of the base plate in the figure, and screw it.**

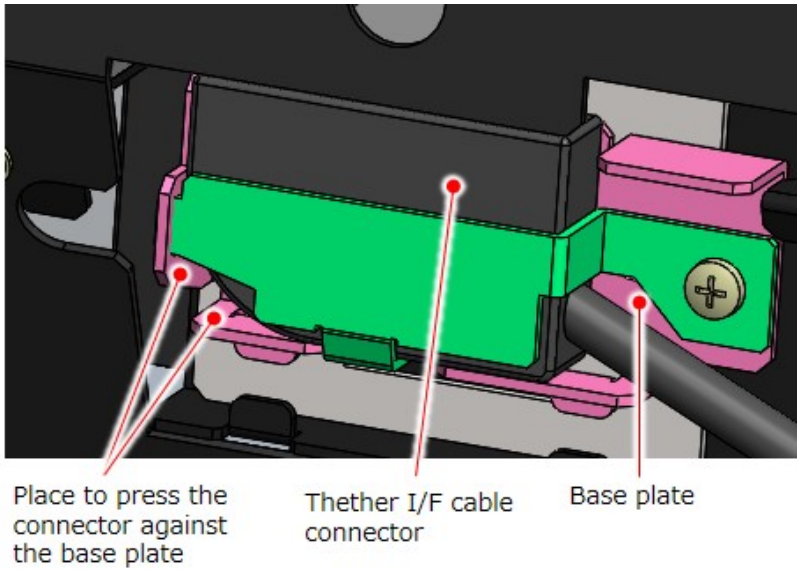


Fig. 4-53

12 **Attach the two springs removed in step 5.**

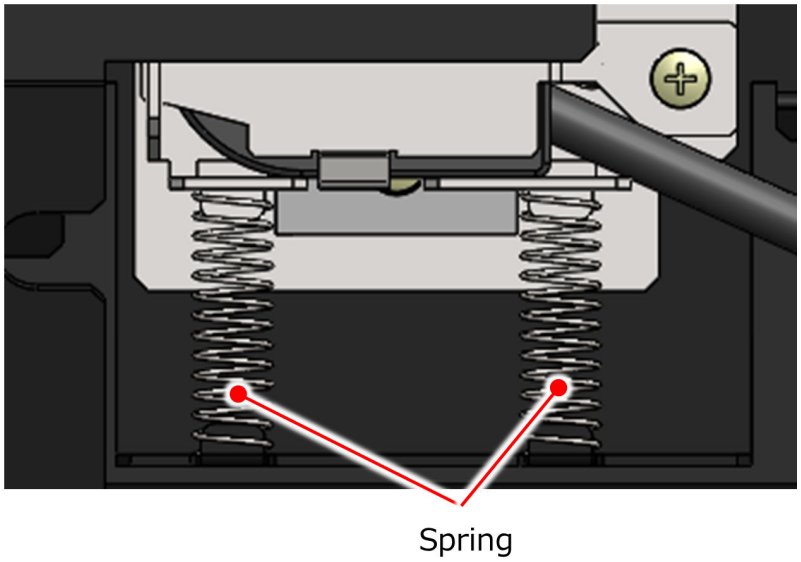


Fig. 4-54

13 Wire the thether I/F cable along the rear frame of the FPD box as shown in the figure below, and fix the thether I/F cable with cable ties.

NOTE
- Fix the cable in the center of the left-right movement allowance of the thether I/F cable connector.
- Fix the cable in the state of natural wiring to avoid excessive force.
- Be especially careful not to twist the cable at fixing position (1).

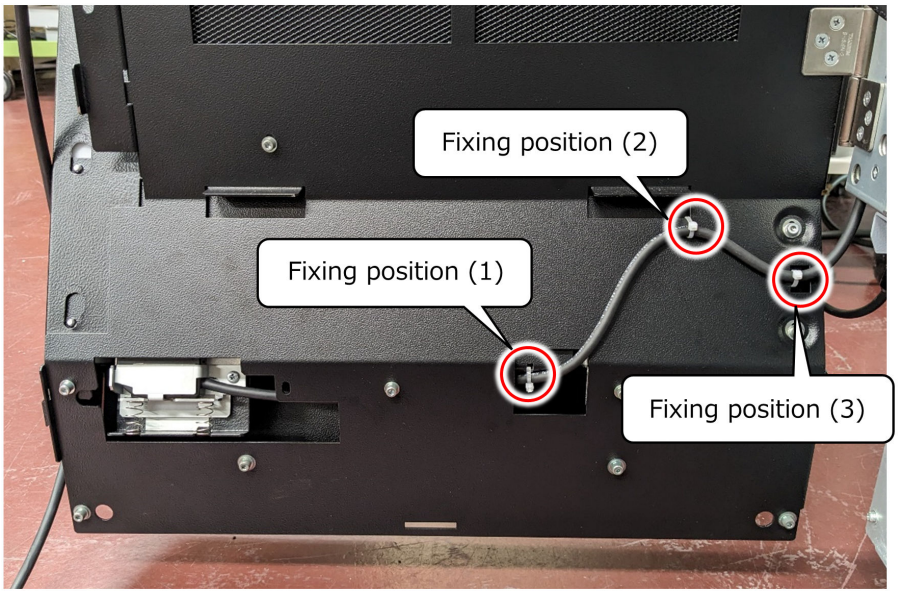


Fig. 4-55

14 Insert the FPD to be used at the delivery destination into the FPD box and confirm that there is no insertion resistance and that the thether I/F cable connector mates with the FPD connector.

15 **Wire the cable from FPD box side to the reverse side of the emergency stop button, as shown in the figure below.**

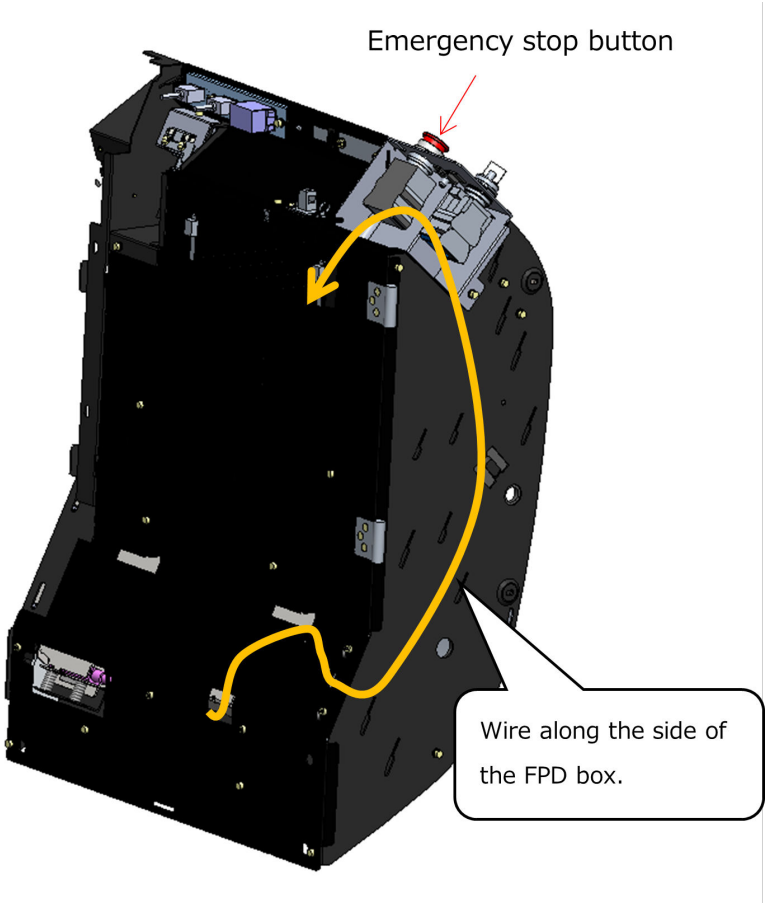


Fig. 4-56

16 Use a screwdriver to remove two screws and remove the cover of the controller GIU SZ.

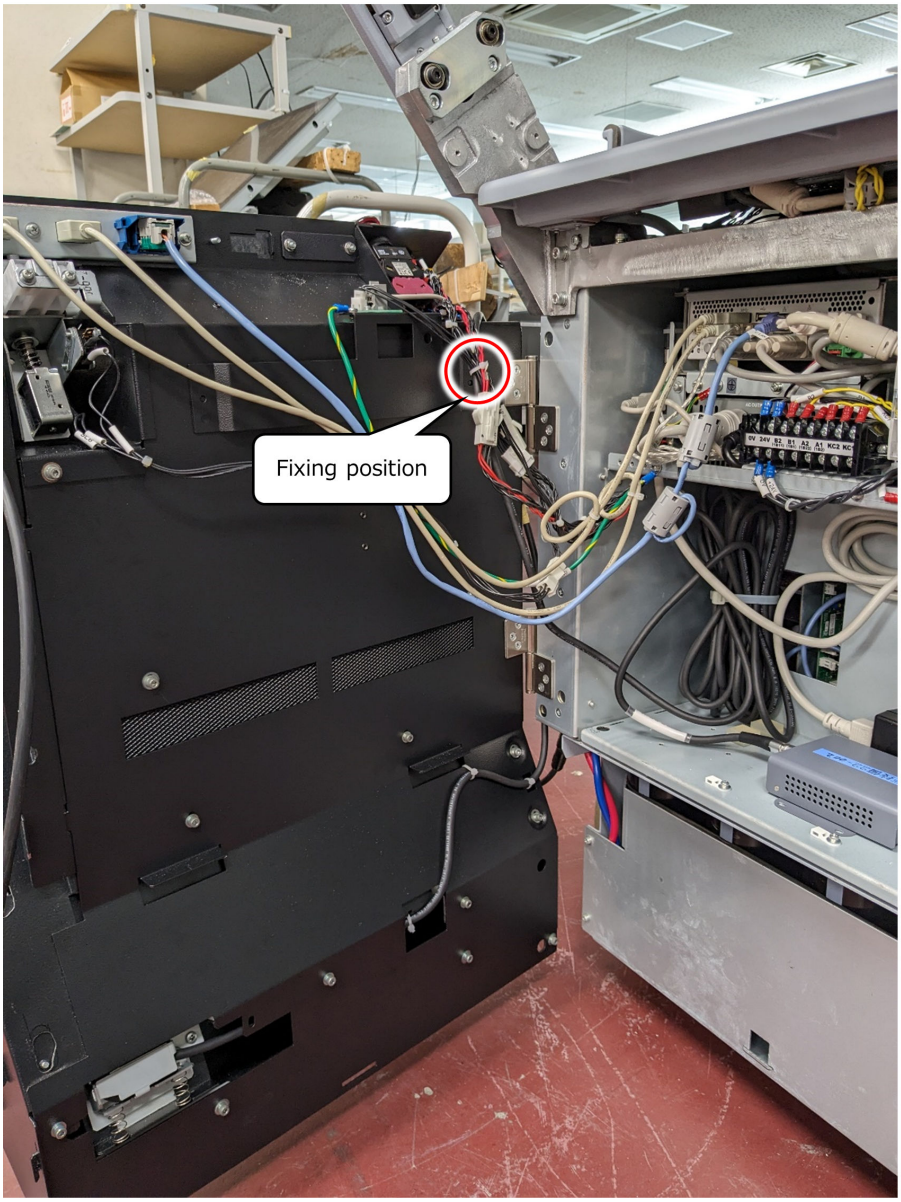


Fig. 4-57

17 Reset the internal capacity discharge to "NORMAL". (Both discharge switch and connector should be reset.)

18 Attach the FPD box and the covers in reverse order of step 2 and 4.

4.7 Assembly of FPD Docking Mechanism (For AeroDR)

NOTE If using a FPD docking mechanism kit, refer to corresponding procedure.
Reference "5.10 Assembly of FPD Docking Mechanism (Only Konica Minolta DR-equipped System)"

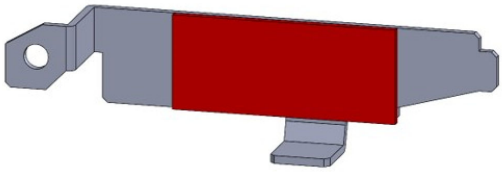
Before installation, please prepare the items listed in the table below.

Table 4-7

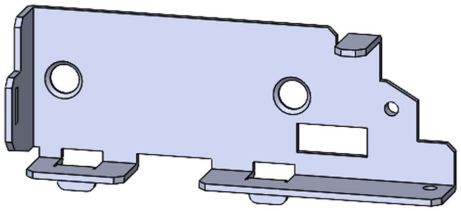
No.	Component	Amount
1	I/F cable	1
2	Cover ASSY (FPD UG kit 8C-8K)	1
3	Base plate (FPD UG kit 8C-8K)	1
4	Cable tie (FPD UG kit 8C-8K)	7

NOTE Before assembling the FPD docking mechanism to the FPD box, install the controller GIU SZ.
Refer to the corresponding procedure for installation.
Reference "4.14 Installing GIU SZ (Only for MobileDaRt Evolution)"

If using 14x14 or 10x12 FPD, install cushion ASSY first.
Reference "4.4 Installing Cushion ASSY(When using FPD Docking Mechanism with 14x17 or 10x12 size FPD)"



Cover ASSY



Base plate

I/F cable

Fig. 4-58

1 **Turn OFF the main breaker.**

2 **Remove the inside cover and right and left side covers.**

Reference "3.5 How to Open/Close Covers"

3 **Discharge the internal capacitor.**

Reference "3.4 Discharging the Internal Capacitor"

4 **Remove the FPD box fixing bolt, and open the FPD box.**

5 **Remove two springs.**

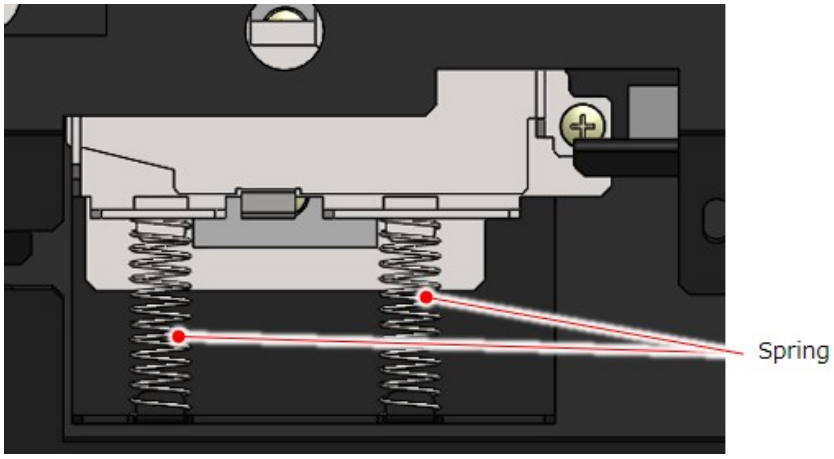


Fig. 4-59

6 **Remove the bind screw, and remove the cover ASSY.**

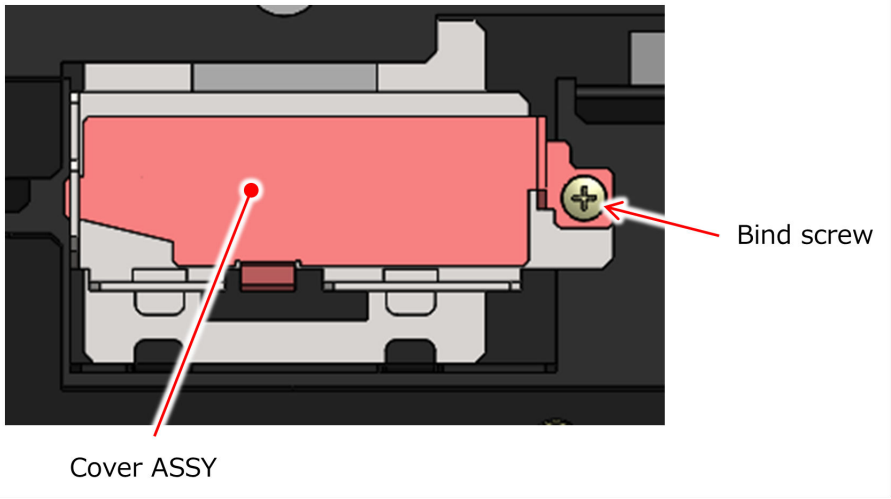


Fig. 4-60

7 Remove two flat screws, and replace the base plate for CXDI with one for AeroDR.

NOTE
- Ensure that the spacer is inserted into the protrusion of the barring.
- Cover ASSY and base plate for CXDI will no longer be used.

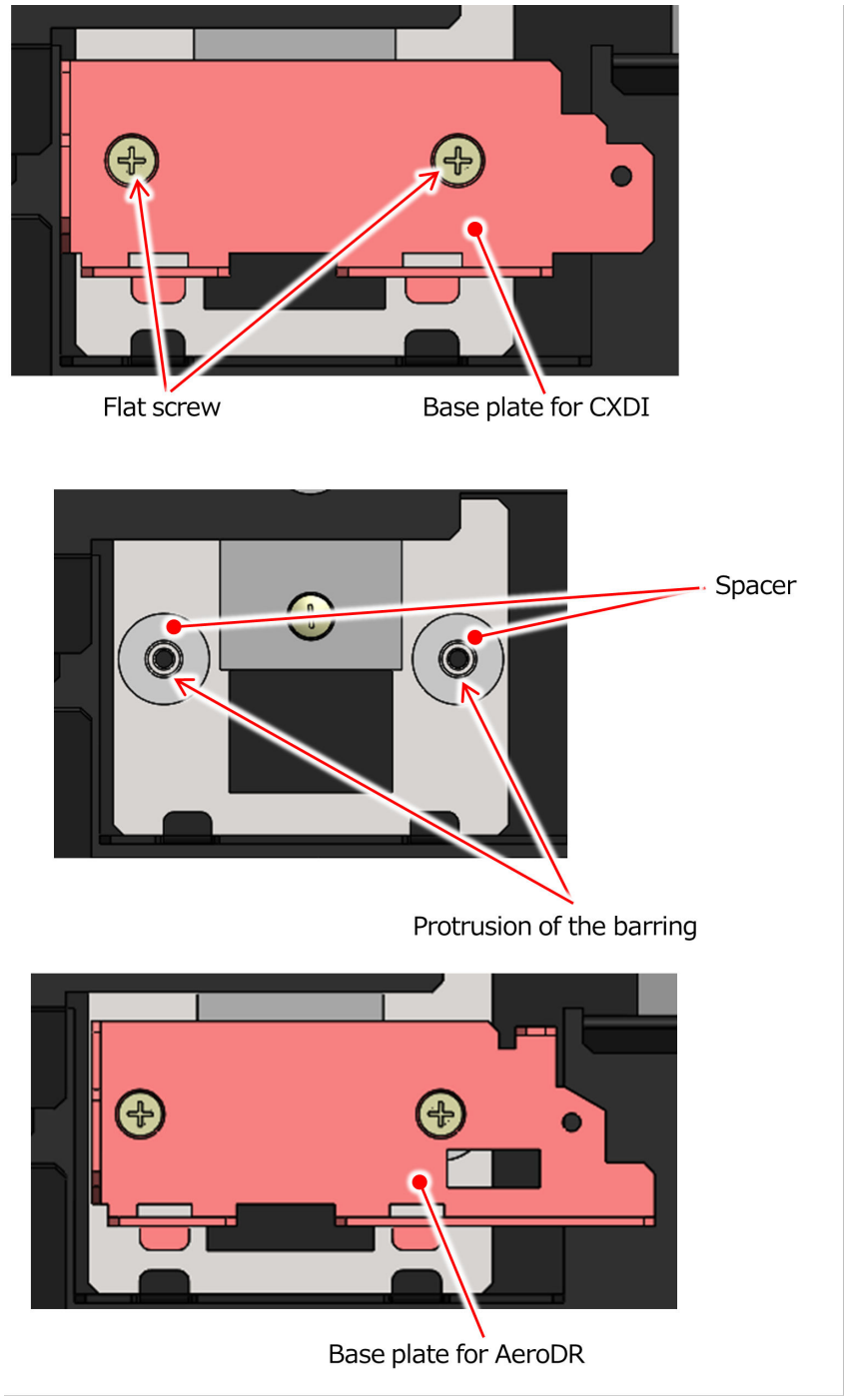


Fig. 4-61

8 **Remove the release coated paper of the double-sided tape of cover ASSY.**

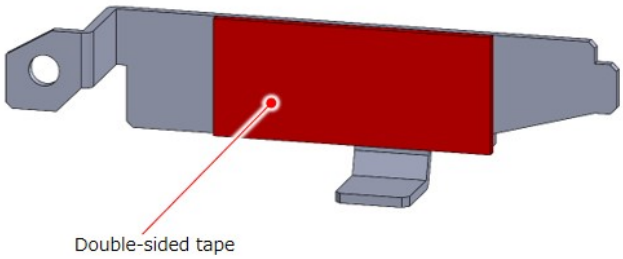


Fig. 4-62

9 **Attach double-sided tape to the I/F cable connector with the sheet metal of the cover ASSY in contact with the I/F cable connector as shown in the figure below. Ensure that the top surface of the sheet metal is parallel to the terminal surface of the connector.**

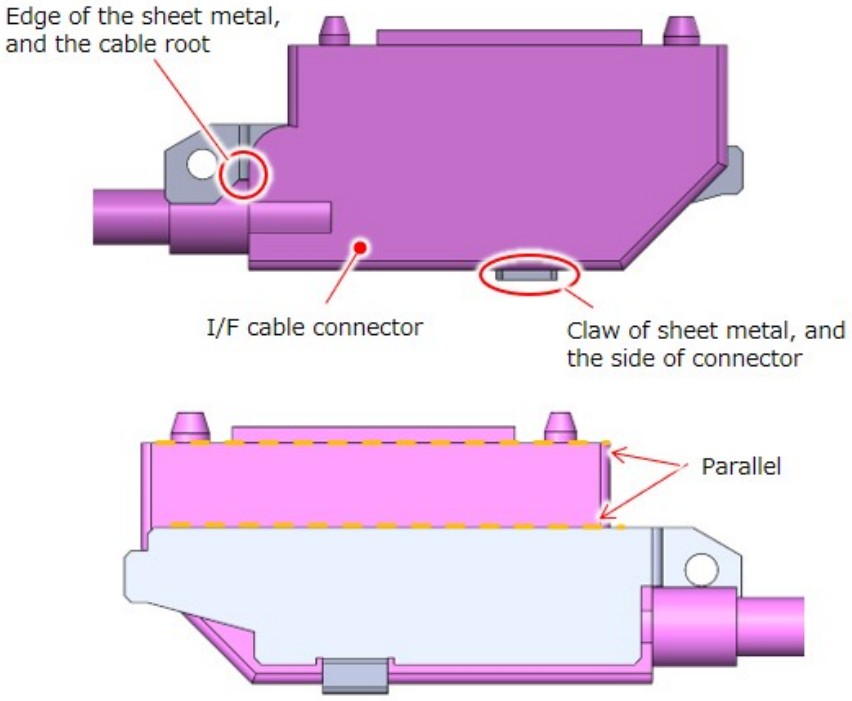


Fig. 4-63

10 Wire the I/F cable as shown in the figure below.

Hint If it is difficult to pass the cable through, pass the cable through while lifting the base plate upward.

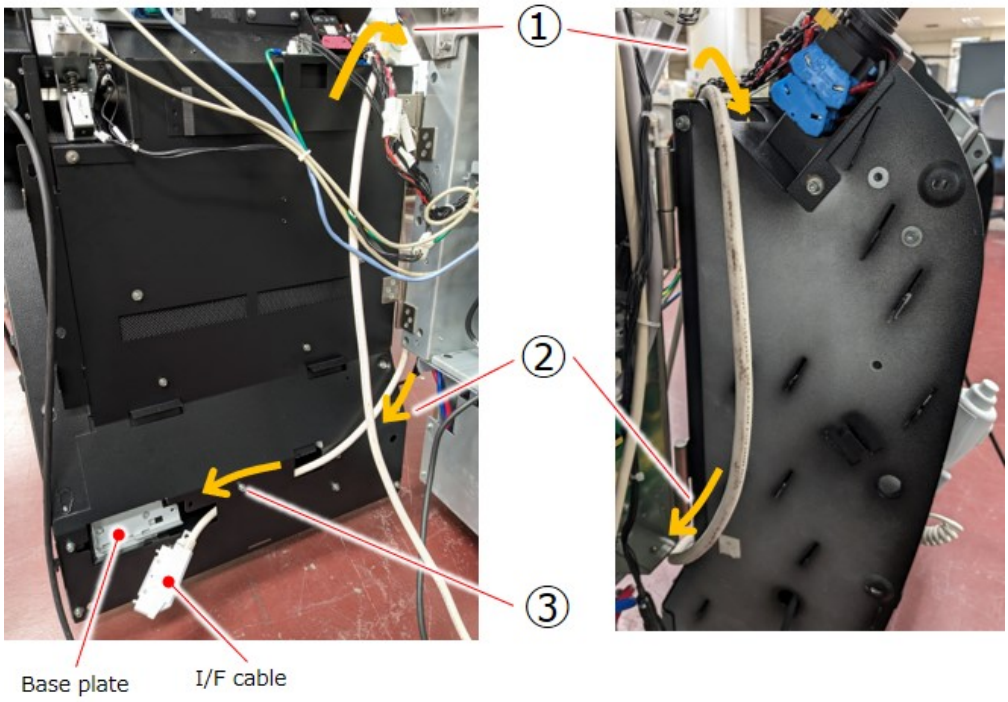


Fig. 4-64

11 **Using the bind screw removed in step 6, attach the cover ASSY with the charging cable connector to the base plate. When attaching, press and hold the I/F cable connector against the indicated position of the base plate in the figure, and screw it.**

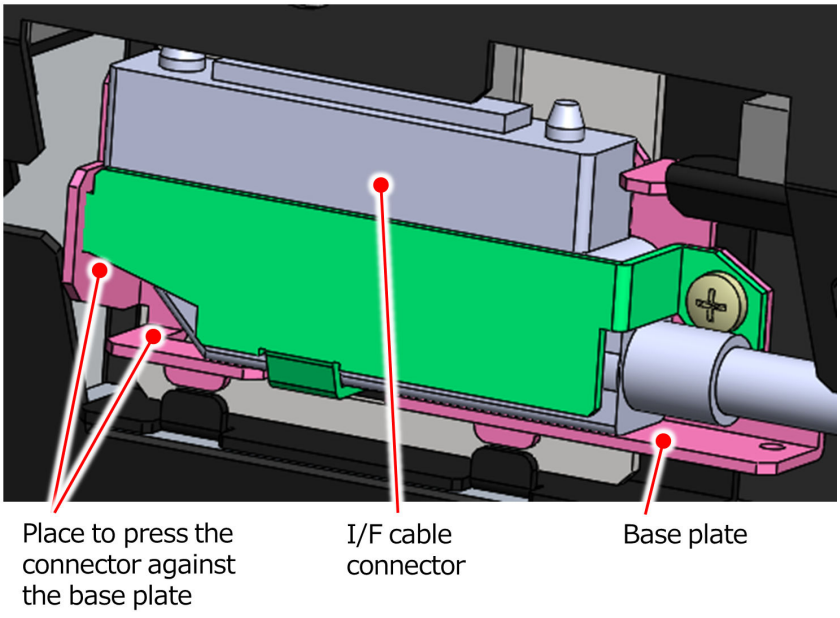


Fig. 4-65

12 **Press the I/F cable against the indicated position of the base plate in the figure, and fix it with a cable tie.**

> NOTE Be careful not to place the knot of the cable tie above the cable.

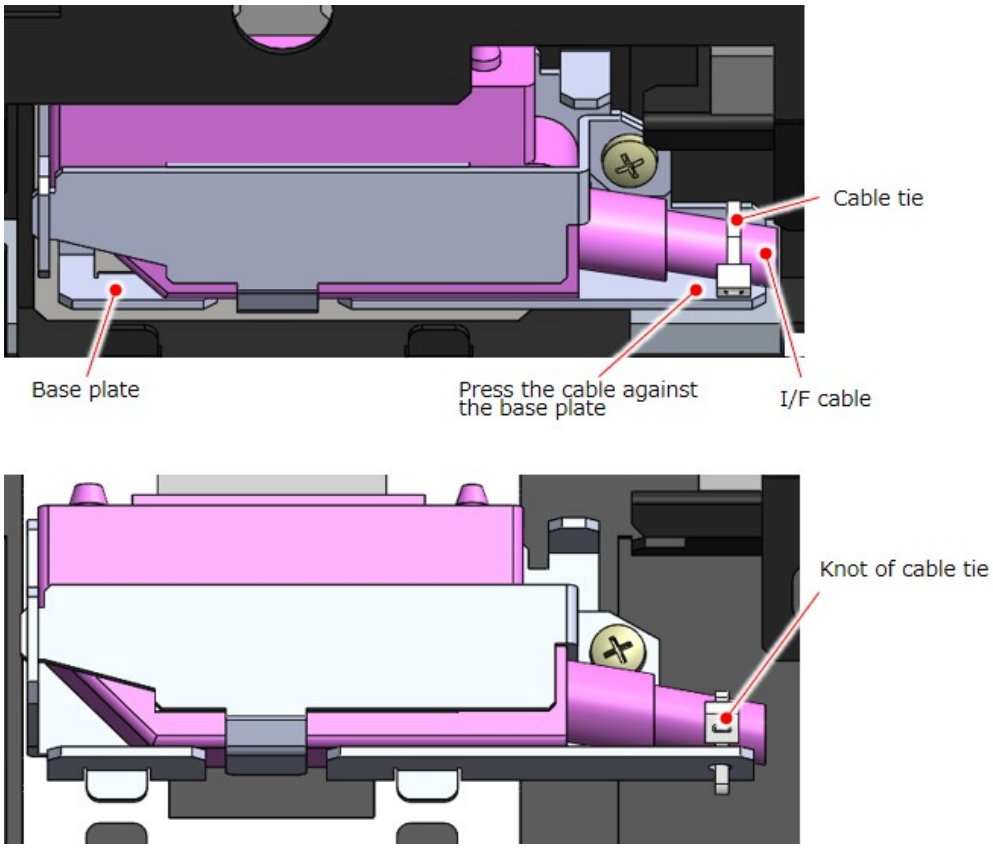


Fig. 4-66

13 Attach the two springs removed in step5.

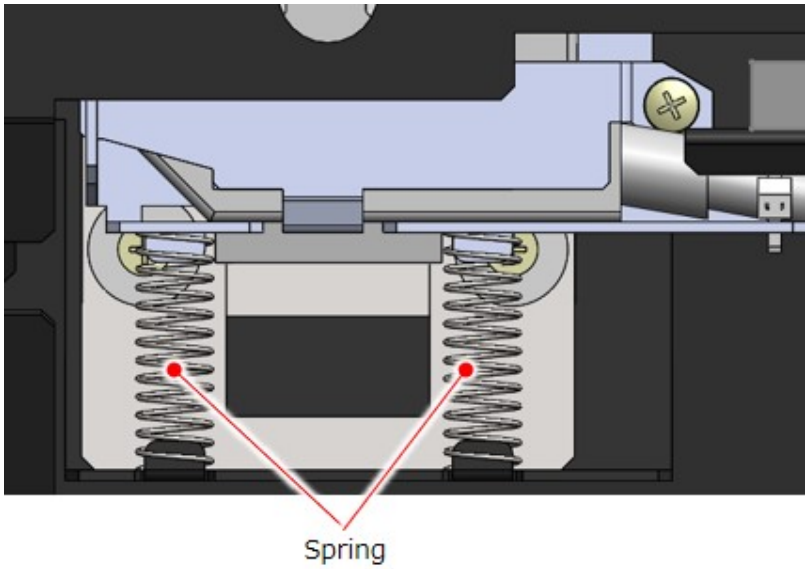


Fig. 4-67

14 Wire the I/F cable along the rear frame of the FPD box as shown in the figure below, and fix the I/F cable with cable ties.

NOTE

- Fix the cable in the center of the left-right movement allowance of the IF cable connector.
- Fix the cable in the state of natural wiring to avoid excessive force.
- Keep the I/F cable vertical at the red dotted line in the figure below.
- Be especially careful not to twist the cable at fixing position (1).

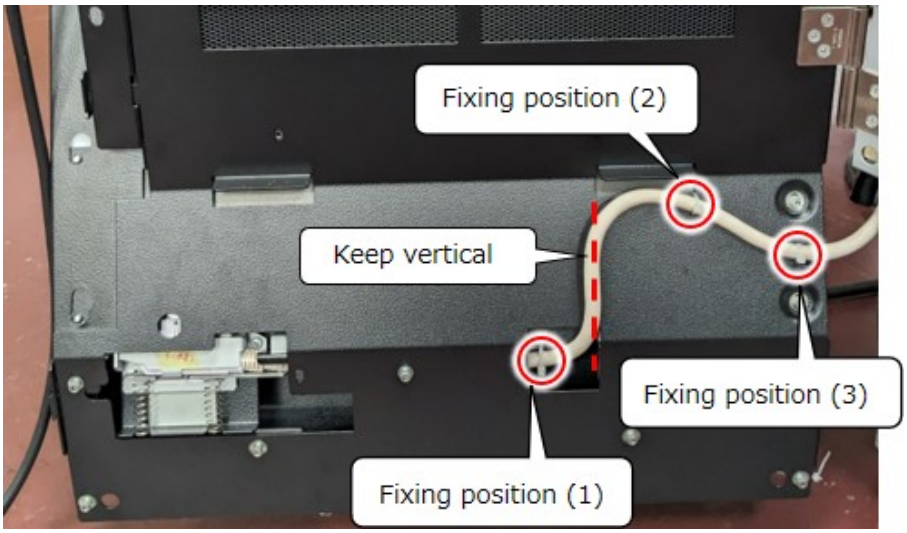


Fig. 4-68

15 **Insert the FPD to be used at the delivery destination into the FPD box and confirm that there is no insertion resistance and that the I/F cable connector mates with the FPD connector.**

16 **Wire the cable from FPD box side to the reverse side of the emergency stop button, as shown in the figure below.**

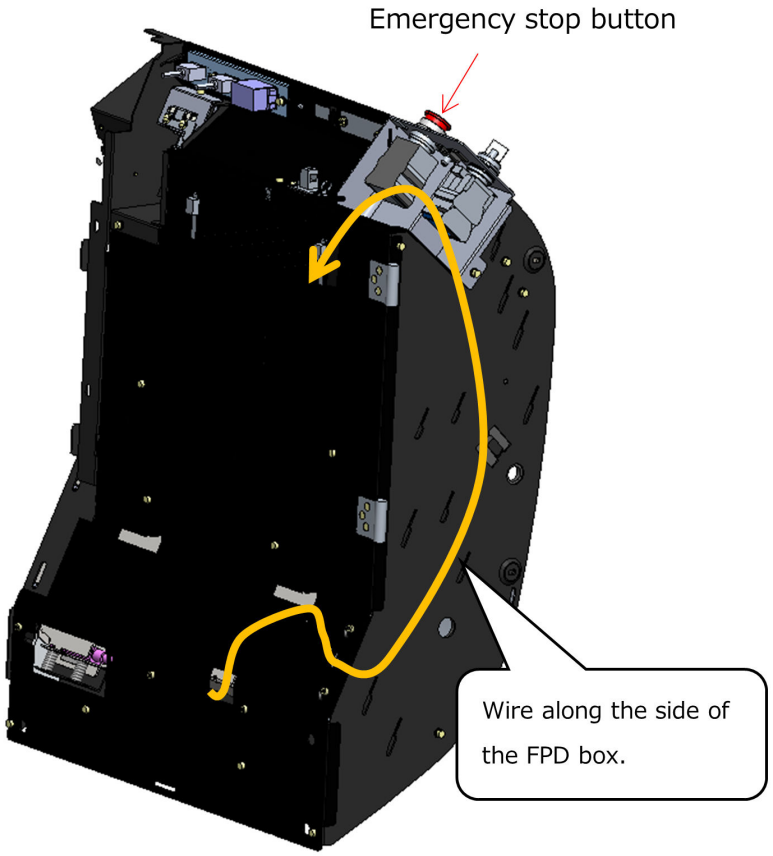


Fig. 4-69

17 **Use a screwdriver to remove two screws and remove the cover of the controller GIU SZ.**

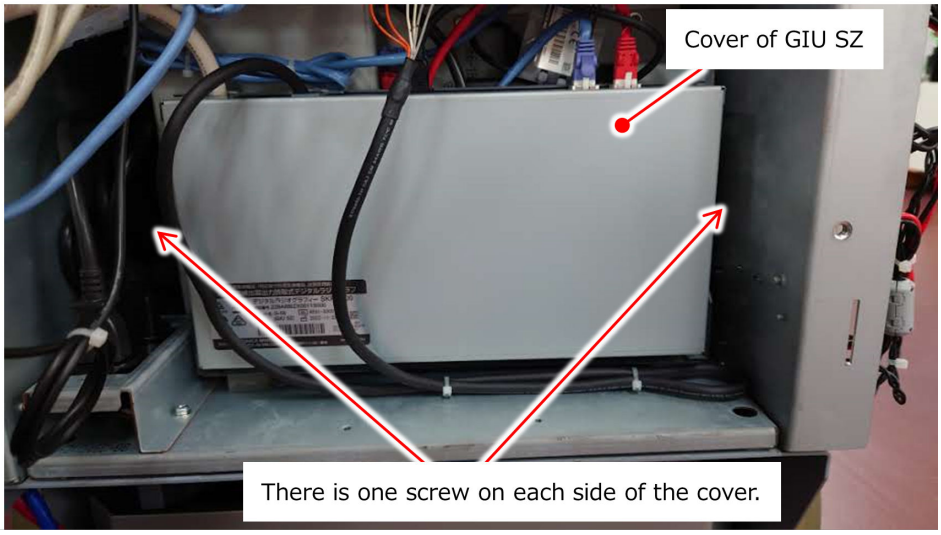


Fig. 4-70

18 **Plug the IF cable connector into the board and fix the IF cable fixing fittings into the GIU SZ with a screw.**

> **NOTE** Fix the fixing fittings in a position where they do not touch the black heat-shrinkable tubing of the IF cable.

Hint The fixing fittings and a screw are included with the IF cable

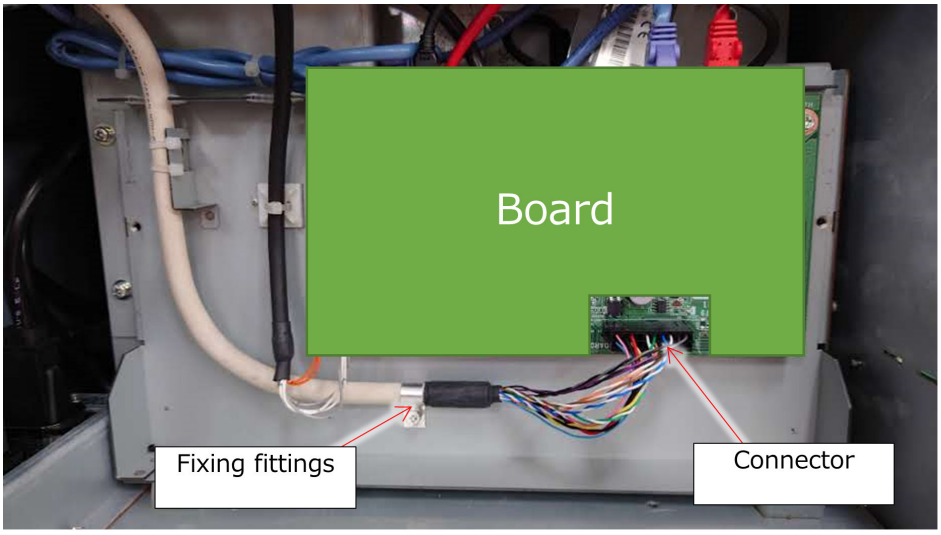


Fig. 4-71

19 **Fix the IF cable using cable ties at the indicated positions in the figure below inside GIU SZ.**

Hint Fit cable ties into their respective notches of the fixing plate and tie them off.

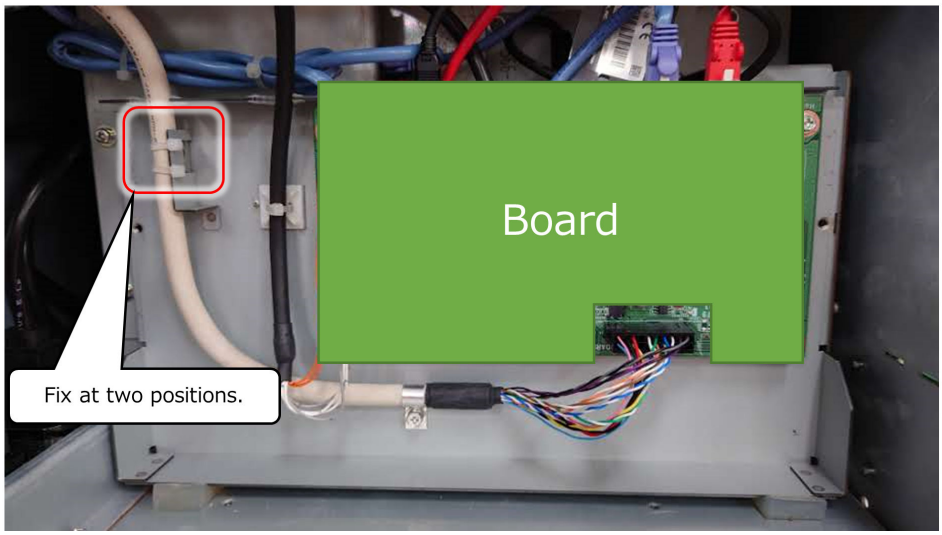


Fig. 4-72

20 **Re-fix the cover of GIU SZ with two screws removed in step 12.**

21 **Wire the IF cable to the side of the FPD box for natural deflection, and fix it along with the cable extending from the emergency stop button with a cable tie.**

> NOTE Slacken the remaining cable (approx. 70 cm) as shown in the figure below and store it inside the unit.

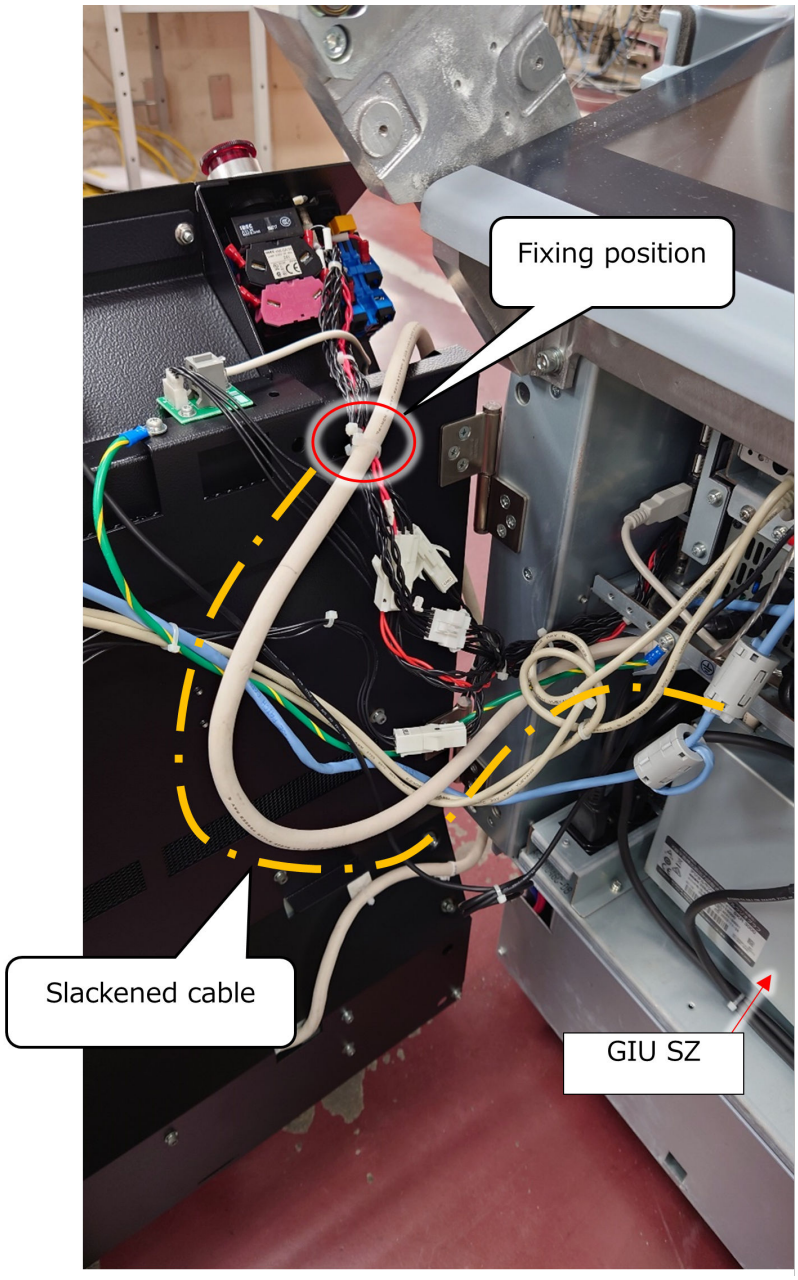


Fig. 4-73

22 **Reset the internal capacity discharge to "NORMAL". (Both discharge switch and connector should be reset.)**

23 **Attach the FPD box and the covers in reverse order of step 2 and 4.**

4.8 Installing cover for concealing docking mechanism (When the docking mechanism is not used)

If the docking mechanism is not used, install the cover sheet metal. Prepare the following parts from the accessories.

Table 4-8

No.	Component	Amount
1	Cover	1
2	Sems screw M4×8	1

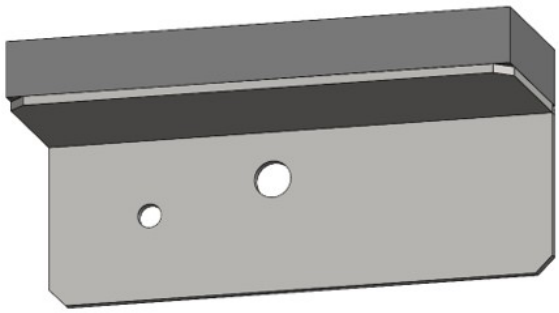


Fig. 4-74 Cover

1 **Turn OFF the main breaker.**

2 **Remove the inside cover and right and left side covers.**

Reference "3.5 How to Open/Close Covers"

3 **Discharge the internal capacitor.**

Reference "3.4 Discharging the Internal Capacitor"

4 **Remove the FPD box fixing bolt, and open the FPD box.**

5 **Remove two springs.**

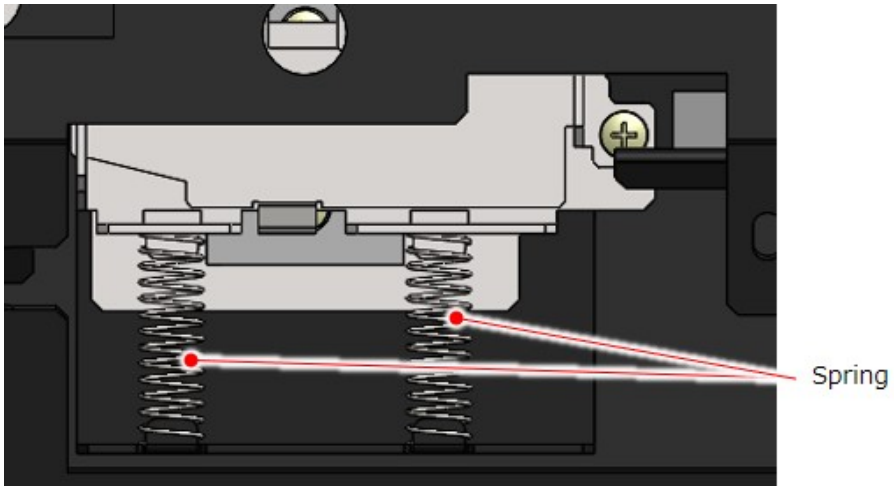


Fig. 4-75

6 **Remove the bind screw, and remove the cover ASSY.**

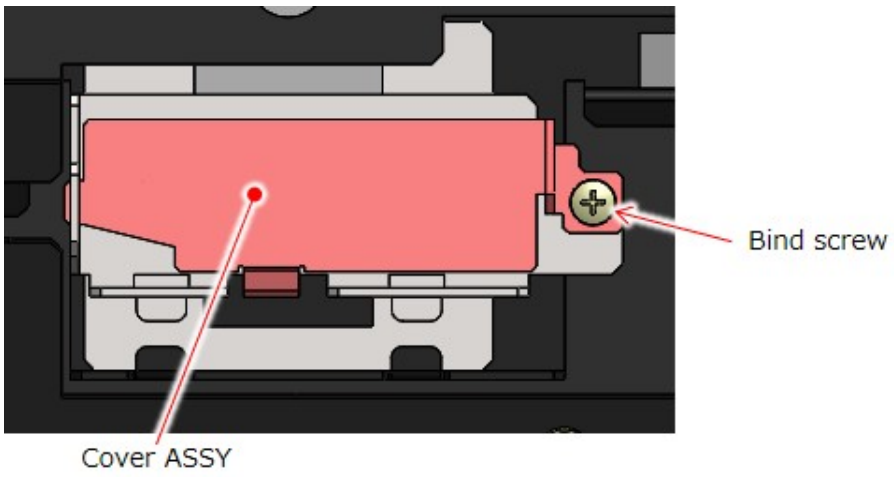


Fig. 4-76

7 **Remove two flat screws, and remove the base plate and spacers.**

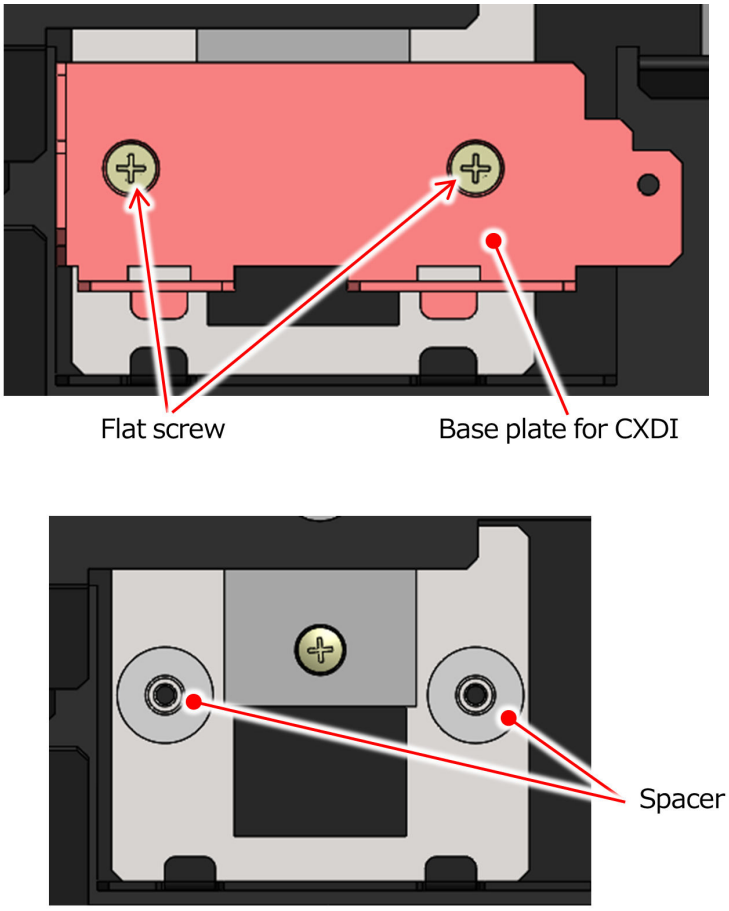


Fig. 4-77

8 **Remove the bind screws, and remove the POM rail and slide plate.**

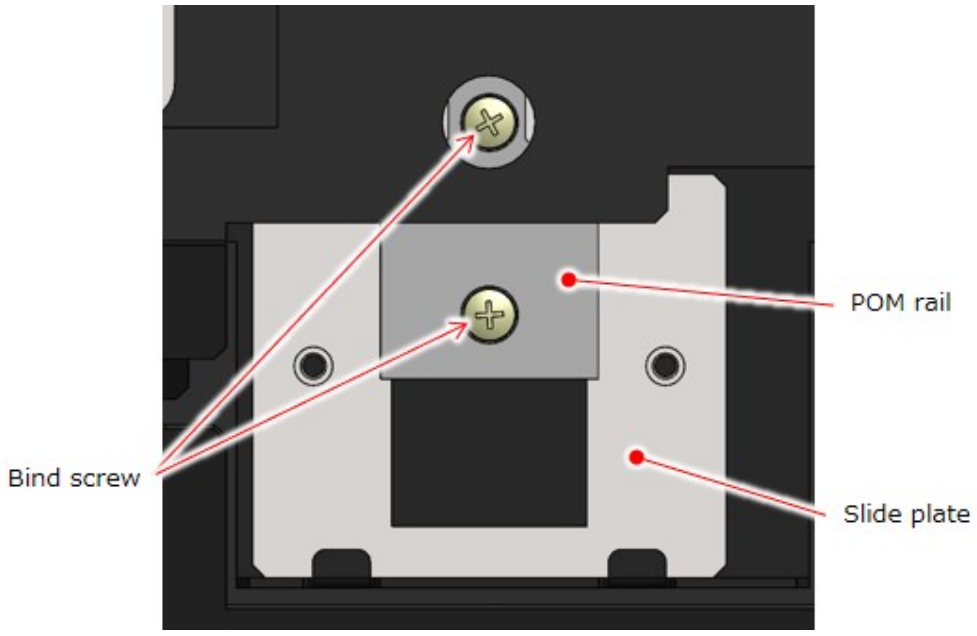


Fig. 4-78

9 **Attach the cover with sems screw.**

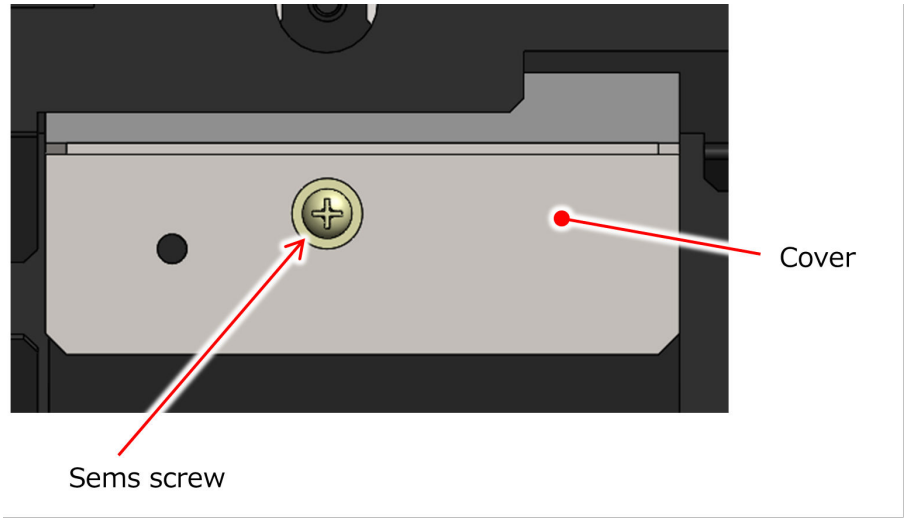


Fig. 4-79

10 **Reset the internal capacity discharge to "NORMAL". (Both discharge switch and connector should be reset.)**

11 **Attach the FPD box and the covers in reverse order of step 2 and 4.**

4.9 Connection Change According to Power-Supply Voltage

■ **Change the connection settings of P1, P2 connectors in the unit, according to the electrical power requirements of the installation site.**

1 **Turn OFF the main circuit breaker.**

Reference "2.3 Name of Each Part of the Unit"

2 **Open the right side cover.**

Reference "3.5 How to Open/Close Covers"

3 **Change the connections based on "Table 4-9 Changing the connection according to supply voltage".**

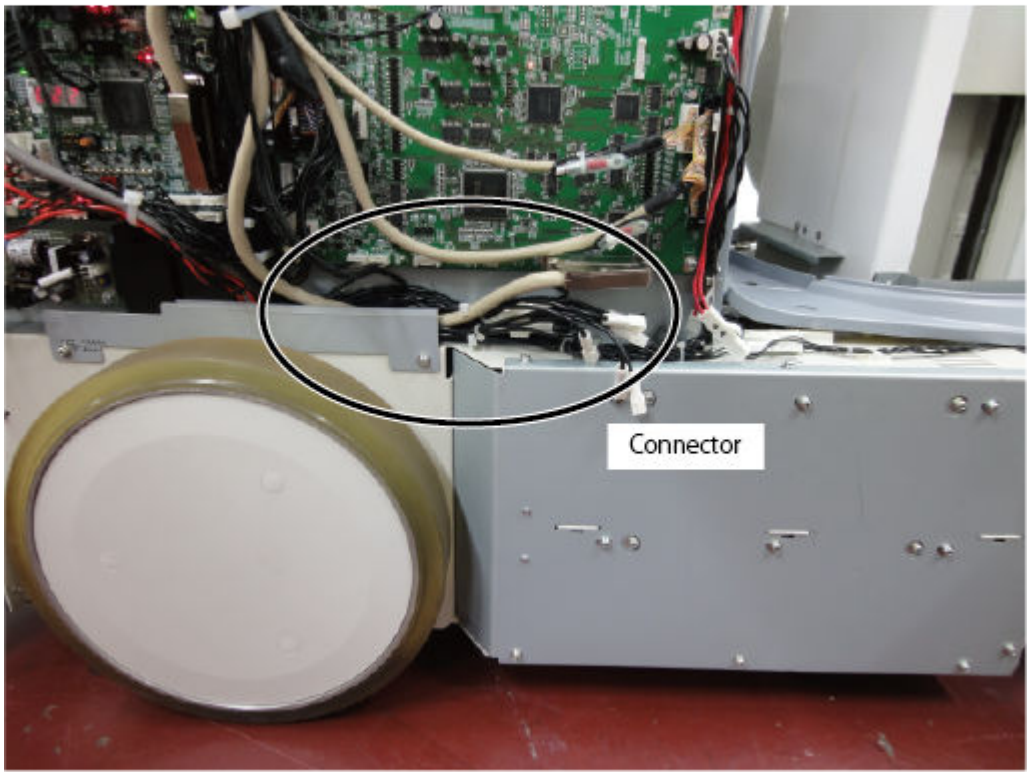


Fig. 4-80

4 **Stick the supply voltage indication label on the outlet of the power plug.**

The label is included in the maintenance parts.

5 **Turn ON the main circuit breaker.**

⚠ WARNING

The connection settings must be changed according to the local mains electrical power supply requirements. Attempting to charge the batteries with incorrectly set connections may cause burnout or fire of electrical parts. The factory default settings are for an electrical power supply of 100 VAC.

Table 4-9 Changing the connection according to supply voltage

Supply voltage	Connector P1	Connector P2
240V±10%	A0	A240
230V±10%	A+10	A220
220V±10%	A0	A220
200V±10%	A0	A200
120V±10%	A0	A120
110V±10%	A+10	A100
100V±10%	A0	A100

4.10 Checking Performance of Each Part

⚠ WARNING

Make sure that the connections for the supply voltage matches the electrical power requirements of the installation site before the performance check.

4.10.1 System Warm-Up

Perform the warm-up operations as shown in the following table. Take exposures with different settings, in the correct order, from step 1 to step 13.

If the system behaves abnormally or becomes unstable, go back two steps and try again.

When measuring the voltage waveform, use an oscilloscope to measure CP9 TKV, tube voltage feedback signals (1V = 20kV), on the XCONT-2018 board.

■ For MobileDaRt Evolution

1 **Press and hold the cassette radiography switch for about 3 seconds.**

The LED of the switch lights up, and the system enters the cassette radiography mode.

2 **Press the key <0> until "tUbeUP" appears on the dose display.**

The system enters the warm-up mode.

3 **Specify the tube voltage and the number of times for Step 1 by using kV[+] and kV[-], or mAs[+] and mAs[-]. The tube voltage should be 50kV, and the number of times should be two, which is indicated as [n2].**

- Tube voltage: The kV display
- The number of times: The mAs display (1 is indicated as [n1], and 2 as [n2]. If nothing is set, it is indicated as [n-].)

4 **According to the "Table 4-10 Warm-Up Steps (For MobileDaRt Evolution)", perform each radiography for Step 1 to Step 13.**

5 **After completing the radiography for all steps, turn the system power OFF.**

Table 4-10 Warm-Up Steps (For MobileDaRt Evolution)

Step	Tube voltage (kV)	Number of exposures	Pause time (Sec.)
1	60	2	60
2	60	2	60
3	70	2	60
4	80	2	60
5	90	2	60
6	100	2	60
7	105	2	60
8	110	2	60
9	115	2	60
10	120	2	60
11	125	2	60
12	130	2	60
13	133	2	60

⚠ WARNING

Fully close the collimator leaves and ensure adequate X-ray protection measures are used before commencing the warm-up procedure.

⚠ WARNING

The warm-up during the system installation should be performed according to the step order in "Table 4-10 Warm-Up Steps (For MobileDaRt Evolution)".

■ For MobileArt Evolution

1 According to the "Table 4-11 Warm-Up Steps (For MobileArt Evolution)", perform each radiography for Step 1 to Step 11.

2 After completing the radiography for all steps, turn the system power OFF.

Table 4-11 Warm-Up Steps (For MobileArt Evolution)

Step	Tube voltage (kV)	Tube current - time product (mAs)	Number of exposures	Pause time (Sec.)	
1	50	10	2	60	
2	60	10	2	60	
3	70	10	2	60	
4	80	10	2	60	
5	90	10	2	60	
6	100	10	2	60	
7	105	10	2	60	
8	110	10	2	60	
9	115	10	2	60	
10	120	10	2	60	
11	125	10	2	60	
12	130	10	2	60	onny for 32kW type
13	133	10	2	60	onny for 32kW type

⚠ WARNING

Fully close the collimator leaves and ensure adequate X-ray protection measures are used before commencing the warm-up procedure.

⚠ WARNING

The warm-up during the system installation should be performed according to the step order in "Table 4-11 Warm-Up Steps (For MobileArt Evolution)".

4.10.2 Checking Tube Current

■ **Check the tube current with the following steps.**

1 **Enter the Adjustment mode.**

Reference
- For MobileDaRt Evolution: "7.1 How to Start Adjustment Mode"
- For MobileArt Evolution: "8.1 How to Start Adjustment Mode"

2 **Enter the tube current two-point adjustment mode.**

Reference
- For MobileDaRt Evolution: "7.4 Adjustment of Tube Current"
- For MobileArt Evolution: "8.4 Adjustment of Tube Current"

3 **Check that tube current value displayed on panel is within the range below. Adjust tube current when the tube current is not within the range.**

Reference
- For MobileDaRt Evolution: "7.4 Adjustment of Tube Current"
- For MobileArt Evolution: "8.4 Adjustment of Tube Current"

For 32kW type (MobileDaRt Evolution/MobileArt Evolution high-power type)

Large focus

- 80 kV, 50 mA point: 45–50 mA
- 80 kV, 400 mA point: 370–400mA

Small focus

- 80 kV, 50 mA point: 45–50 mA
- 80 kV, 200 mA point: 185–200mA

For 12.5kW type (MobileArt Evolution standard type)

- 80 kV, 50 mA point: 45-50 mA
- 80 kV,160 mA point: 148-160 mA

4.10.3 Operation Check of Collimator Unit

■ **Make sure that the collimator is correctly mounted.**

1 **Loose the rotation-lock screw for the collimator.**

2 **Check that there is no large play on the rotation of the collimator.**

■ **Check the light field**

1 **Turn the Irradiation field adjusting knob completely open.**

2 **Push the collimator lamp switch and confirm the light field.**

3 **Turn the Irradiation field adjusting knob completely close.**

4 **Push the collimator lamp switch and confirm the light field is completely shaded.**

Reference Refer to "11.5 Centering the Collimator and X-Ray Focus" to adjust the centers of the X-ray focus to the collimator's light focus.

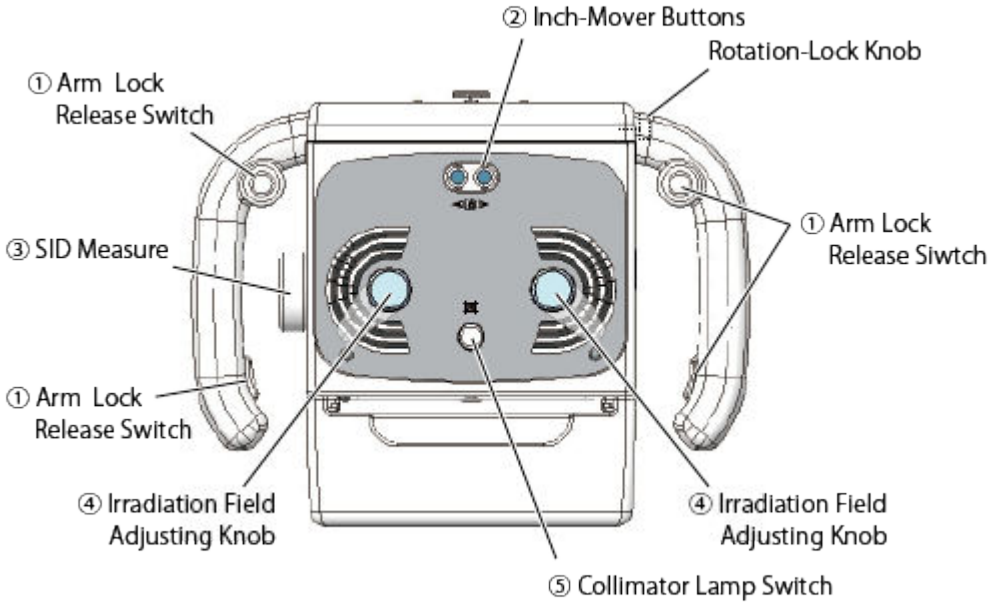


Fig. 4-81 Name of parts of the collimator

4.10.4 Operation Check of Unit

■ **Check functions of the driving handle and the Inch-Mover Buttons. The Inch-Mover Button is at the front of the collimator.**

1 **Turn on the key switch.**

Reference "2.3 Name of Each Part of the Unit")

2 **Confirm the system runs forward when pushing the driving handle.**

3 **Confirm the system runs backward when pulling the driving handle.**

4 **Push the arm lock release switch and lift up the X-ray tube assembly.**

5 **Confirm the system moves to correct direction when pushing the Inch-Mover Buttons.**

6 **Confirm the system stops when pushing the both Inch-Mover Buttons at the same time.**

> NOTE When the driving handle is operated, the Inch-Mover Buttons are disabled for safety reasons.

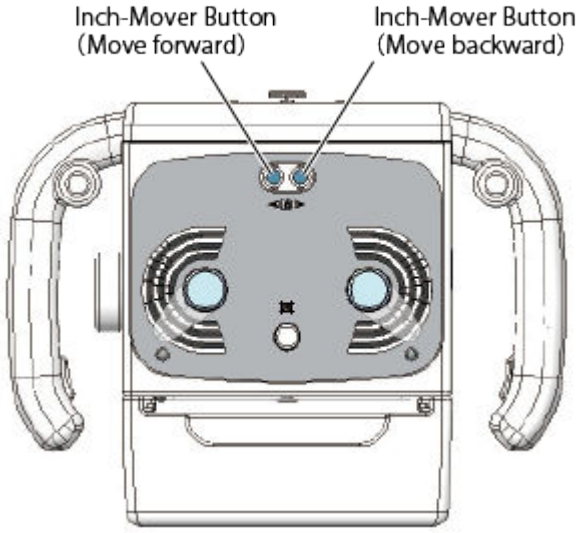


Fig. 4-82 Inch-Mover Button

4.10.5 Checking Each Parts Operation and Lock Function

■ **Check the arm stretch, up/down and rotation.**

1 **Move the stretch, up/down and rotation of the arm with pressing the "Arm Lock Release Button". Confirm whether there is neither abnormal noise nor stroke shortage.**

2 **Release the "Arm Lock Release Button" and confirm that the arm is locked.**

4.11 Checking Dose Area Product Calculation Unit

Check a calculated error of the dose area product before the operation check.

If the calculated error exceeds the error margin, make readjustment of the collimator leaf opening amount and ckV.

4.11.1 Adjustment of Dose Area Product Calculation Unit Setting

Dose Area Product Calculation Unit has been adjusted without an additional filter at the factory.

To use an additional filter, please adjust the coefficients of each filter.

Reference • For MobileDaRt Evolution: "7.20 Dose Area Product Calculation Unit Setting Mode"
• For MobileArt Evolution: "8.20 Dose Area Product Calculation Unit Setting Mode"

4.11.2 Checking Dose Area Product Calculation Value

After the adjustment of the dose are product calculation unit, check the calculated error of the dose area product.

If the calculated error exceeds the error margin, make readjustment of the collimator leaf opening amount and ckV.

■ Setting of Dose Area Product Calculation Unit

Use the calibrated dose area product calculation unit to set the measurement system as shown below.

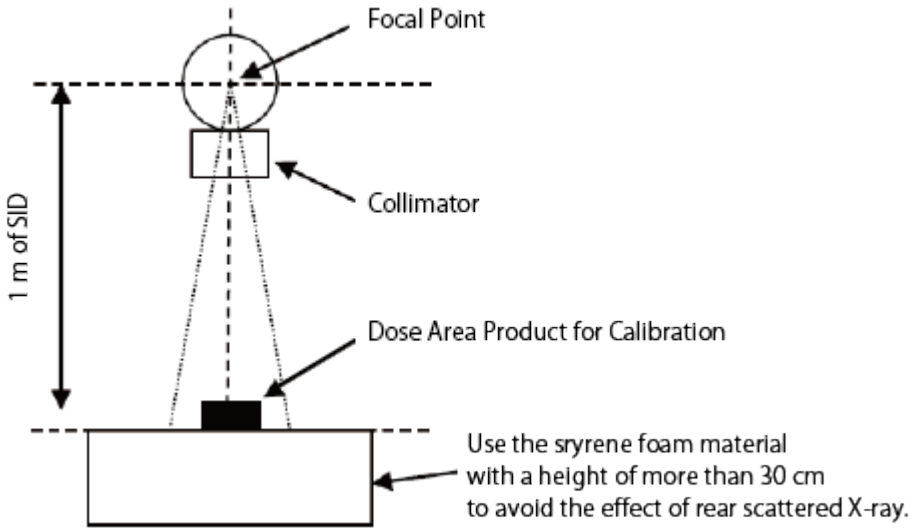


Fig. 4-83 Measurement System

■ Check of Measurement Value and Calculated Error

- ❶Irradiation dose measurement value by the dosimeter
  Write a measurement value [μGy] on the dosimeter to the table below.
  If the dosimeter shows a value by mGy, convert the value into μGy by multiplying it by 1000.
- ❷Dose area product measurement value
  Calculate a value from the following formula, and write it to the table below.
  (❷Dose area product measurement value) = (❶Irradiation dose measurement value) × (irradiation field size)
- ❸Dose area product calculation value
  Write a value that is displayed on the control panel to the table below.
- ❹Error
  Calculate an error from the following formula, and write it to the table below.
  (❹Error) = ((❸Dose area product calculation value) - (❷Dose area product measurement value)) / (❷Dose area product measurement value) × 100

NOTE Set the tube voltage according to the power type.
- 32 kW type: MobileDaRt Evolution/MobileArt Evolution high-power type
- 12.5kw type: MobileArt Evolution standard type

Table 4-12 When Additional Filter is not used.

Irradiation field per 1 m of SID [m^2]		0.04m^2=0.2m×0.2m			0.16m^2=0.4m×0.4m		
Tube voltage [kV]	32kW type	40	80	133	40	80	133
	12.5kW type			125			125
Tube current-time product [mAs]		10			10		
❶Irradiation dose measurement value by the dosimeter [μGy]							
❷ Dose area product measurement value [μGym^2]							
❸Dose area product calculation value [μGym^2]							
❹Error [%]							
Error margin [%]		±30			±30		

Table 4-13 When Additional Filter 1 is used.

Irradiation field per 1 m of SID [m^2]		$0.04m^2$=0.2m×0.2m			$0.16m^2$=0.4m×0.4m		
Tube voltage [kV]	32kW type	40	80	133	40	80	133
	12.5kW type			125			125
Tube current-time product [mAs]		10			10		
❶Irradiation dose measurement value by the dosimeter [μGy]							
❷ Dose area product measurement value [μGym^2]							
❸Dose area product calculation value [μGym^2]							
❹Error [%]							
Error margin [%]		±30			±30		

Table 4-14 When Additional Filter 2 is used.

Irradiation field per 1 m of SID [m^2]		$0.04m^2$=0.2m×0.2m			$0.16m^2$=0.4m×0.4m		
Tube voltage [kV]	32kW type	40	80	133	40	80	133
	12.5kW type			125			125
Tube current-time product [mAs]		10			10		
❶Irradiation dose measurement value by the dosimeter [μGy]							
❷ Dose area product measurement value [μGym^2]							
❸Dose area product calculation value [μGym^2]							
❹Error [%]							
Error margin [%]		±30			±30		

4.12 Assembly of Multi Box (Only for MobileDaRt Evolution)

NOTE This section is applicable only to the following system.
- MobileDaRt Evolution Type CXDI

4.12.1 Preparations

1 **Turn off the main circuit breaker.**

2 **Remove the inside cover and both side covers, and open the FPD Box.**

3 **Discharge the internal capacitor.**

Reference "3.4 Discharging the Internal Capacitor"

⚠ WARNING

In this system, large-capacity capacitors are connected to the battery.
Even after turning OFF the key switch or the main breaker, electrical charge remains in the capacitor.
Be sure to discharge the capacitor when working on this system.

4.12.2 Assembly of Multi Box

1 **Remove two screws on each side panels (a total of 4 screws) and two screws on the bottom panel of the multi box.**

NOTE Take care not to strip the screw heads of the side panel and bottom panel.

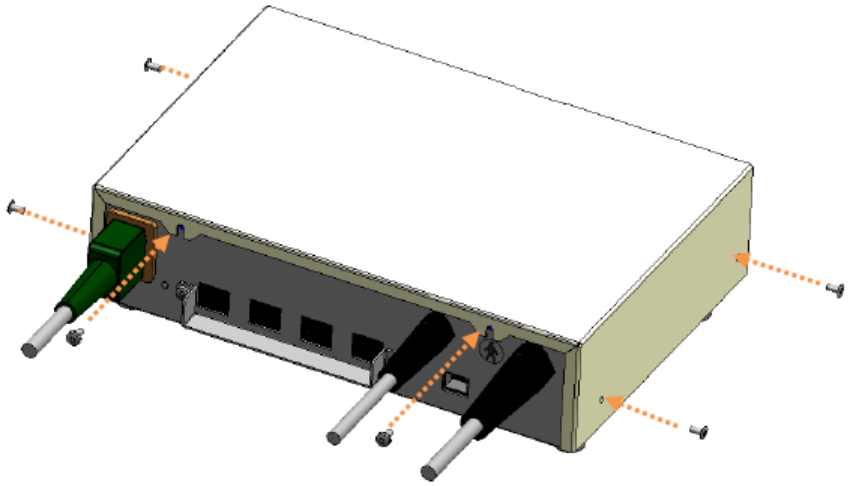


Fig. 4-84

2 **Lift up and remove the top cover from the multi box.**

NOTE Too much tilting of the top cover may cause a damage to the inside of the multi box.

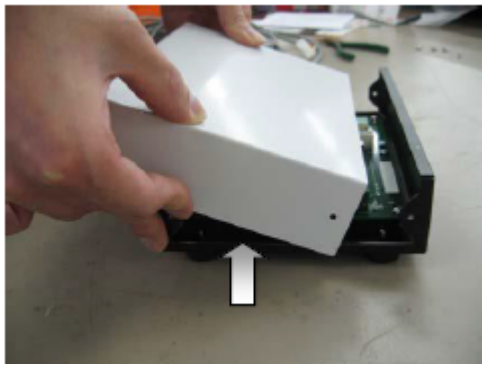


Fig. 4-85

3 Wire the multi box cable as shown below.

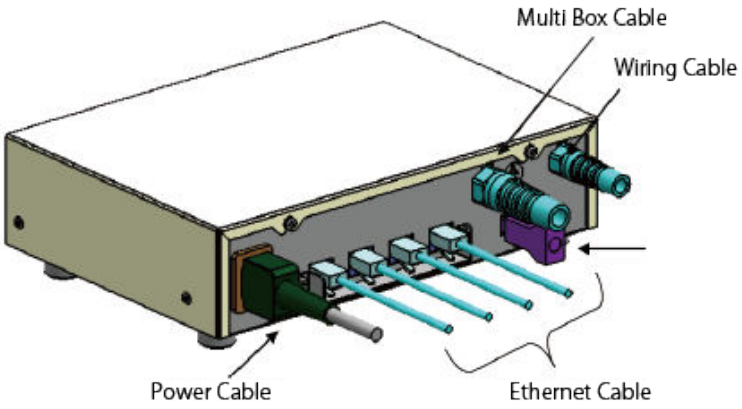


Fig. 4-86

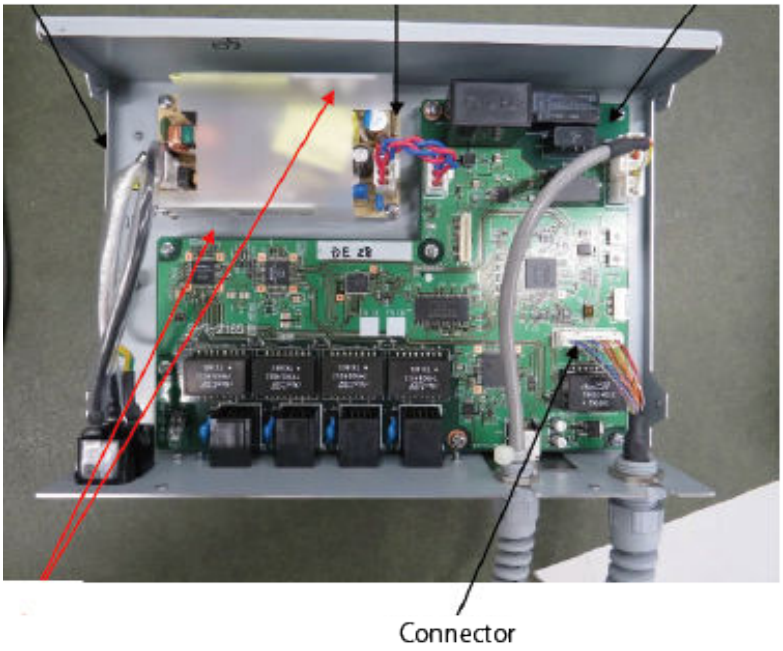


Fig. 4-87

4 After wiring the cable, attach the covers of the multi box.

NOTE Be careful not to strip the screw heads of the side panel and bottom panel.
The screws used for mounting must be tightened up in the following sequence: bottom panel (screws with external teeth) → both side panels (brazier head screws).

5 Connect the power cable and Ethernet cable to the multi box

4.12.3 Attachment of Multi Box

1 **Remove the bracket.**

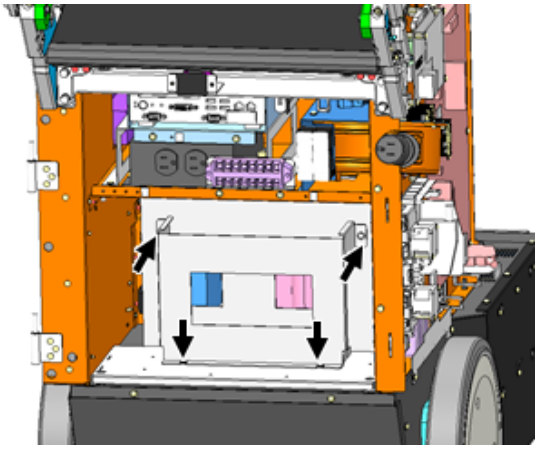


Fig. 4-88

2 **Attach the multi box.**

1 Place the multi box inside the DR-ASSY. The connectors should be located at the upper side.

2 Fix the multi box to the DR-ASSY with the bracket.

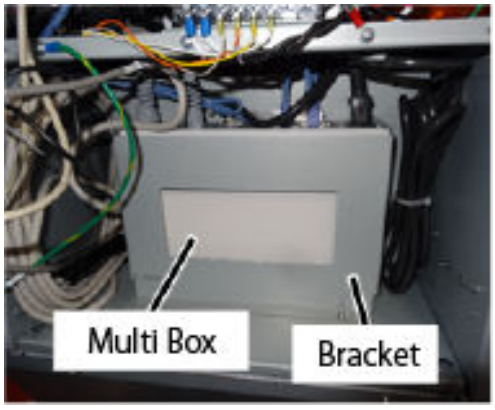


Fig. 4-89

3 Connect the multi box

1 Connect the power cable to the power outlet.

2 Connect the LAN cable connecting from Access Point to the multi box.

3 Connect the other LAN cable between multi box and PC.

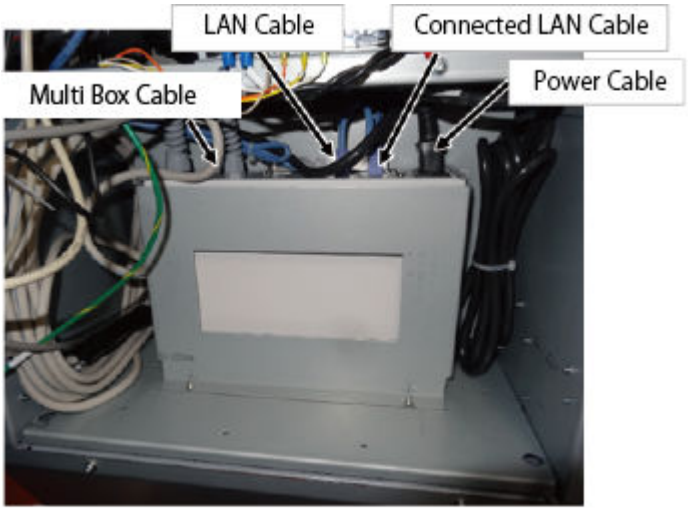


Fig. 4-90

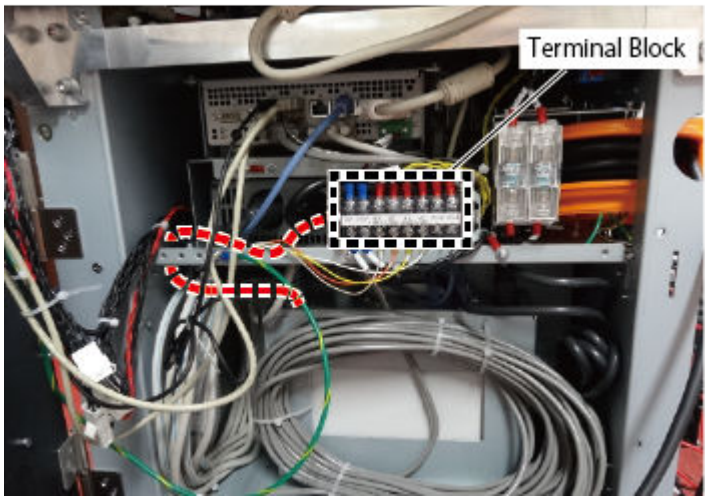


Fig. 4-91 Wiring route of multi cable

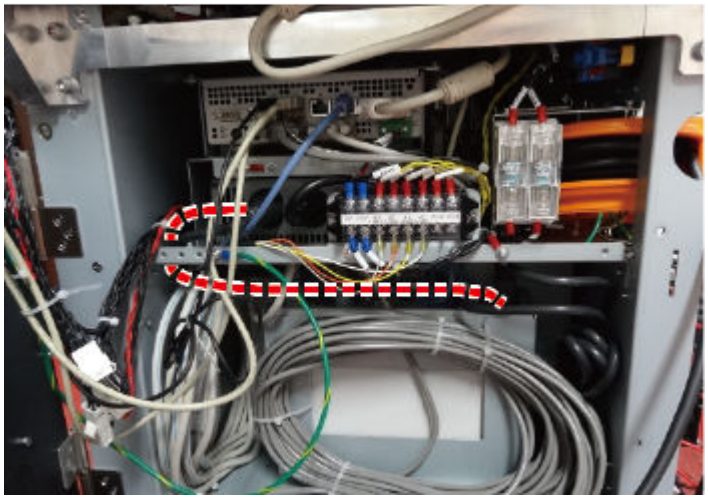


Fig. 4-92 Wiring route of Power Cable

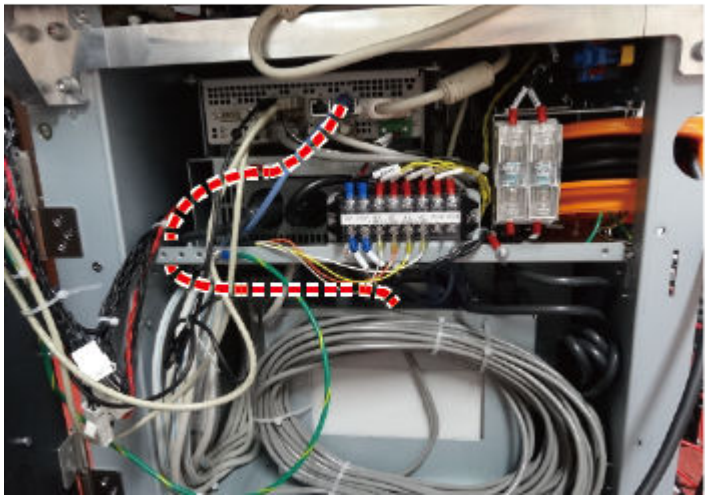


Fig. 4-93 Wiring route of LAN Cable

4 **Connect the multi box cable to the terminal block (terminals A1, A2, B1, and B2).**
Make sure that each cable markers matches the marker on the terminal block.

5 **Attach the removed FPD box and covers in reverse order.**

6 **Turn the main breaker on.**

7 **Reset the internal capacitor setting to "NORMAL".**

8 **Upload the APR default data for the installed CXDI from the attached CD to the XCONT-2018 board.**

▶▶ Reference M503-E410 MobileDaRt Evolution (MX8Version) Installation Manual (DR Section)

4.13 Installing AeroDR Portable RF Unit 2 (Only for MobileDaRt Evolution)

NOTE This section is applicable only to the following system.
- MobileDaRt Evolution Type AeroDR (CS-7)

4.13.1 Installation Procedures for AeroDR Portable RF Unit 2

If AeroDR Potable RF Unit 2 is not mounted, install AeroDR Potable RF Unit 2 according to the flowchart below.

Table 4-15

Start

↓

Installation Procedures for AeroDR Portable RF Unit 2

↓

Check of bundled items

↓

Assembly of AeroDR Portable RF Unit 2

↓

Installation of AeroDR Portable RF Unit 2

↓

FPD Box for AeroDR 1012HQ

↓

End

4.13.2 Check of bundled items

■ Component of AeroDR portable RF unit 2

Open the device packaging and check the bundled items to confirm that there are no scratches, cracks or other external damage.

The configuration of the AeroDR portable RF unit 2 is shown below.

Table 4-16

Item	Parts Name	Quantity
AeroDR I/F Cable2	AeroDR I/F Cable2	1
	Bundled items list	1
AeroDR Portable RF Unit2	AeroDR Portable RF Unit2	1
	LAN cable (*Not used)	1
	Cable tie	3
	Bushing SRB-R-1	1

Item	Parts Name	Quantity
	Plug cushion rubbers	2
	Bushing	1
	Bush	1
	Bundled items list	1
AeroDR Portable AC adapter	AC adapter	1
	Bundled items list	1
AeroDR Battery Charging KIT	PSW board	1
	FLED cable	1
	Power cable	1
	Cable bush	1
	Cable tie	7
	Small bind screws M3-5	4
	Bundled items list	1

NOTE
- Use the identification labels (seals) affixed on the packing box to perform the activation of device registrations. Keep the packing box until installation has been completed.
- Bundled items differ depending on the product or destination. For details, refer to the bundled items list.

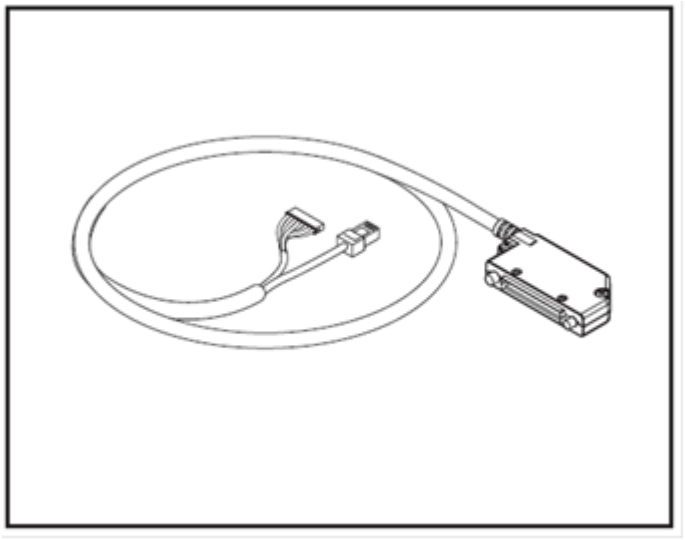


Fig. 4-94 Fig. 4-14 AeroDR I/F Cable2

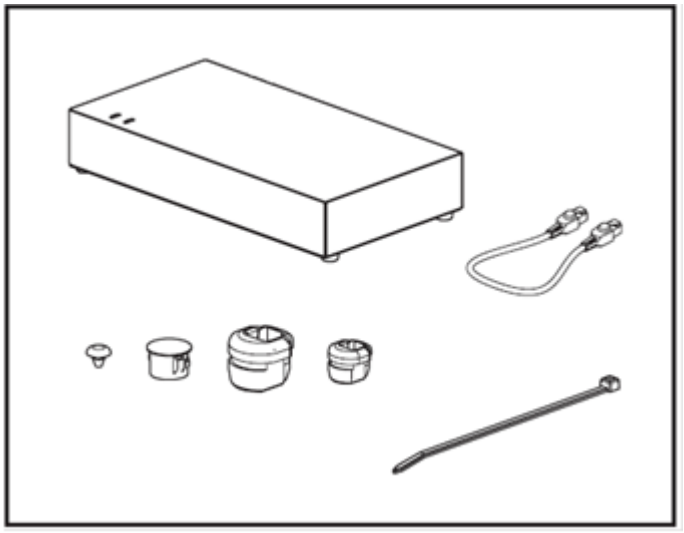


Fig. 4-95 Fig. 4-15 AeroDR Portable RF Unit 2

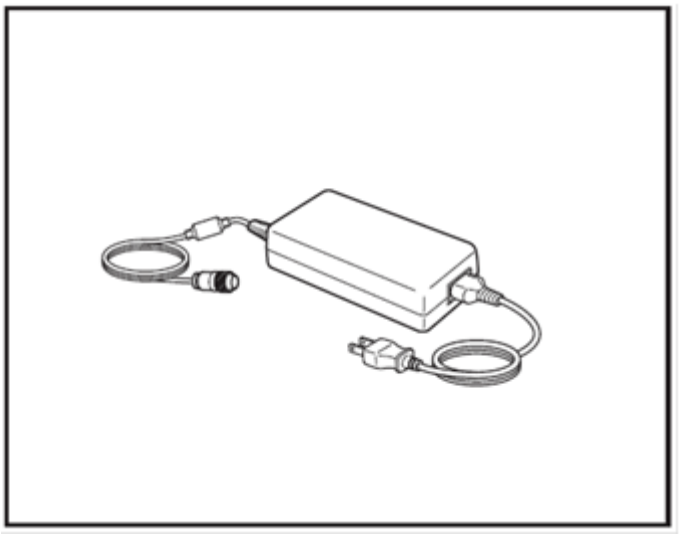


Fig. 4-96 Fig. 4-16 AeroDR AC Adapter

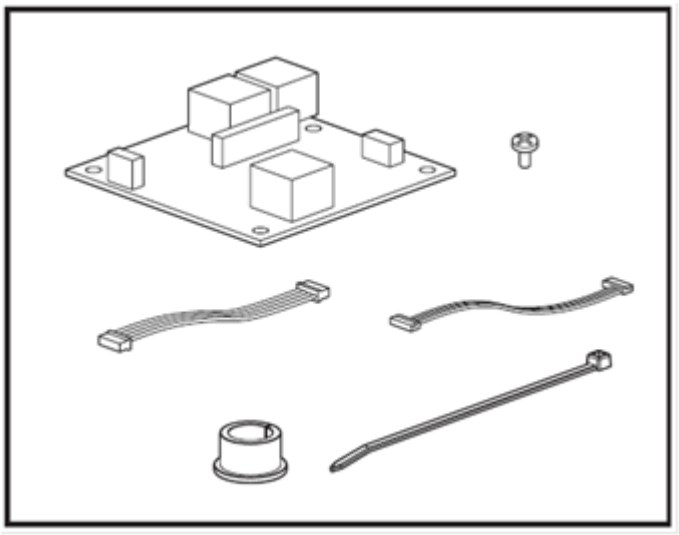


Fig. 4-97 Fig. 4-17 AeroDR Battery Charging KIT

■ Component of FPD Upgrade Kit KM

Use the FPD upgrade kit KM, install AeroDR portable RF unit 2.

Open the device packaging and check the bundled items to confirm that there are no scratches, cracks or other external damage.

The component of the FPD Upgrade Kit KM is shown below.

Table 4-17

Item	Parts Name	Quantity
IF BOX ASSY,KM	Plate for AeroDR portable RF unit 2	1
Socket	Socket for AeroDR I/F cable 2 connector	1
CS-7 OS M430J	Three DVD-media	1
DART KM USA Manual		1
Cushion15	Cushion for AeroDR 1012HQ	1
XRAY IF CABLE(KM)	X-ray interlock cable	1
Cable, LD-CTT/BU1/RS	LAN cable	2
Bolt, S Hole M4X8 ZN3W	Bolt for fixed Socket	2
Seal,Hidden Link	Seal	1

4.13.3 Assembly of AeroDR Portable RF Unit 2

■ Removal of the cover assy.

1 **Remove the screws (4).**

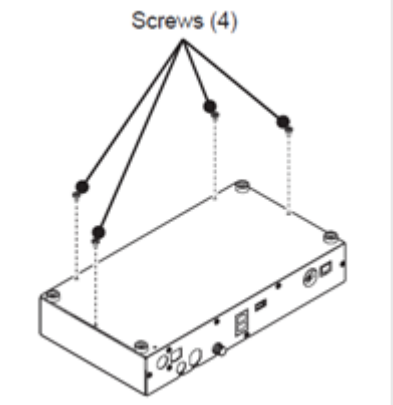


Fig. 4-98

2 **Remove the screws (3), and remove the cover assy.**

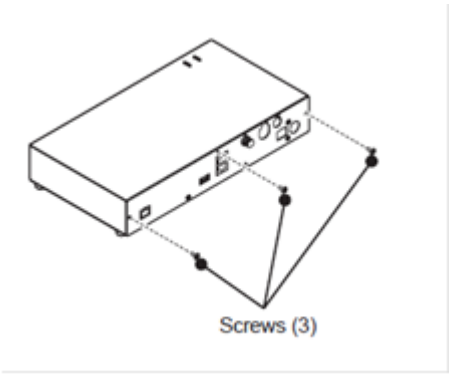


Fig. 4-99

NOTE
- When removing, pay attention to the FLED Assy cable.
- When attaching, exercise care not to pinch the cables.

■ Attachment of AeroDR I/F Cable2

1 **Temporarily fix the cable bush, to AeroDR I/F Cable2.**

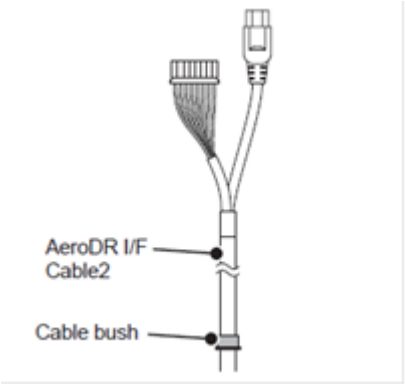


Fig. 4-100

1 Attach the cable bush to the thinner part of AeroDR I/F Cable2 connector first, and then move it to the cable side.

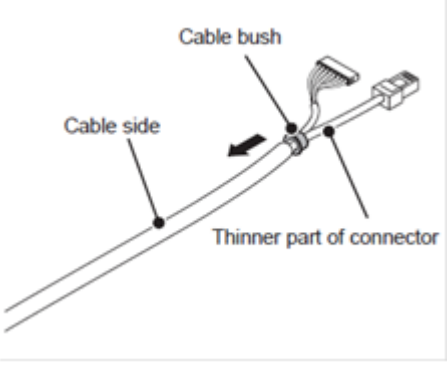


Fig. 4-101

2 **Remove the screws (2), and detach the opening plate from the connector mounting plate.**

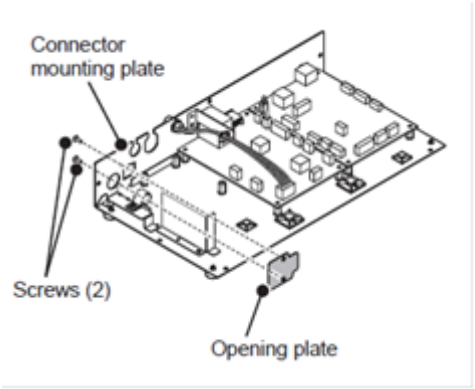


Fig. 4-102

3 **Remove the screw (1), and remove the ground clamp from the ground mounting plate of the base assy.**

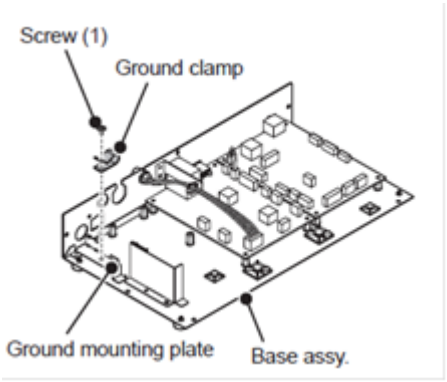


Fig. 4-103

4 **Pass AeroDR I/F Cable2 through the outlet for I/F Cable2 of the connector mounting plate.**

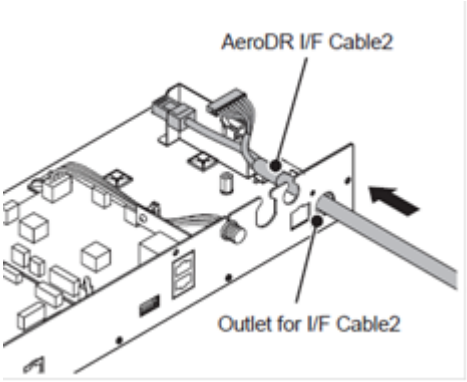


Fig. 4-104

5 **Attach the ground clamp to AeroDR I/F Cable2, and secure it to the ground mounting plate using the screw (1).**

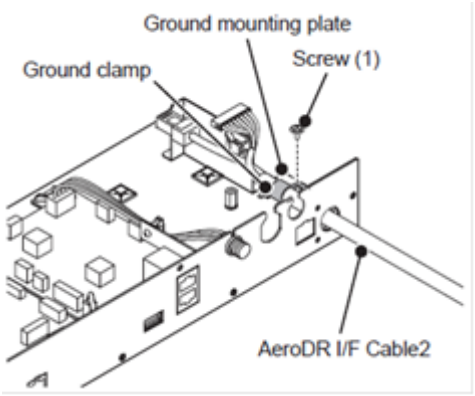


Fig. 4-105

6 **Pass the binding bands (2) through the ground mounting plate and secure AeroDR I/F Cable2.**

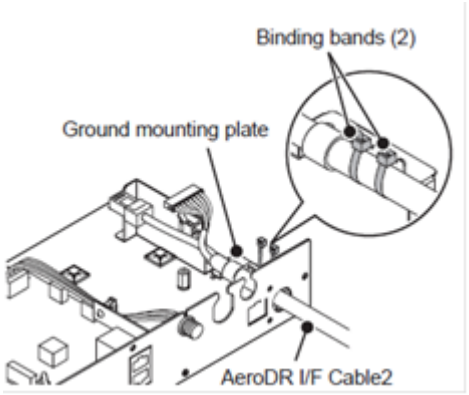


Fig. 4-106

7 **Pass the binding bands (2) through the grooves of the ground mounting plate and secure them.**

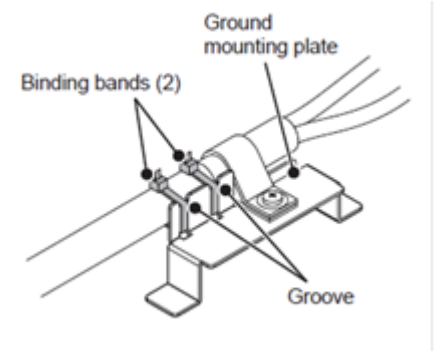


Fig. 4-107

8 **Secure AeroDR I/F Cable2 to the connector mounting plate. Secure the cable bush to the outlet for AeroDR I/F Cable2 of the connector mounting plate.**

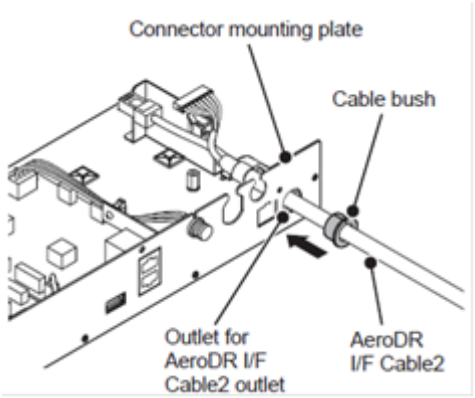


Fig. 4-108

■ Addition of the PSW board

1 **1 Attach PSW board to the base assy. Using the screws (4).**

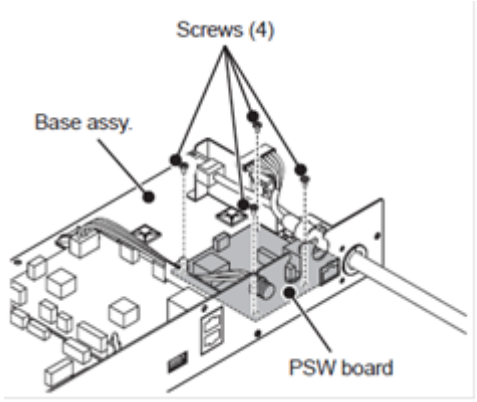


Fig. 4-109

1 1 Attach the PSW board so that CN1 on the PSW board is on the connector mounting plate side.

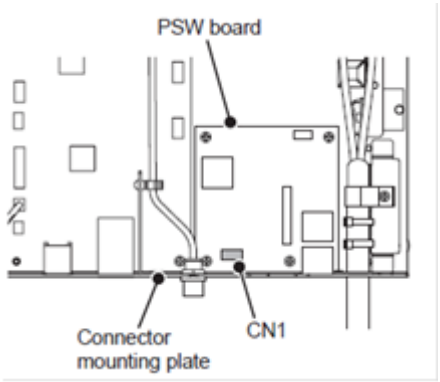


Fig. 4-110

2 2 Press CN2 on the PSW board into the LAN port on the connector mounting plate to attach the PSW board.

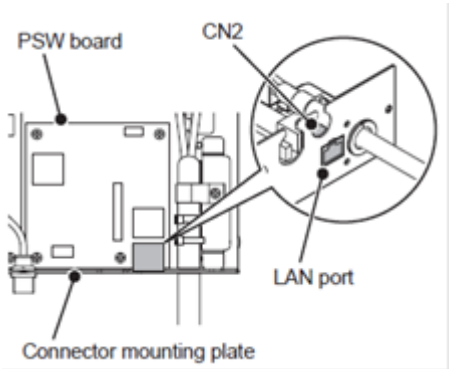


Fig. 4-111

2 **Connect the Ethernet cable connector of AeroDR I/F Cable2 to CN3 on the PSW board.**

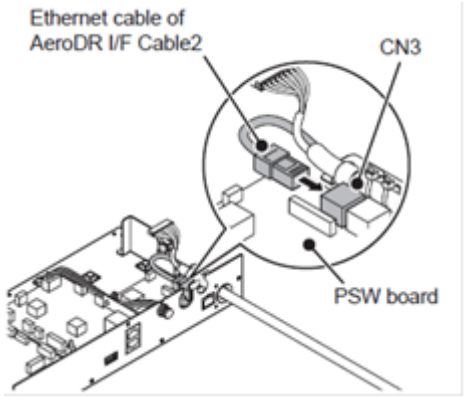


Fig. 4-112

3 **Connect the power cable of AeroDR I/F Cable2 to CN5 on the PSW board.**

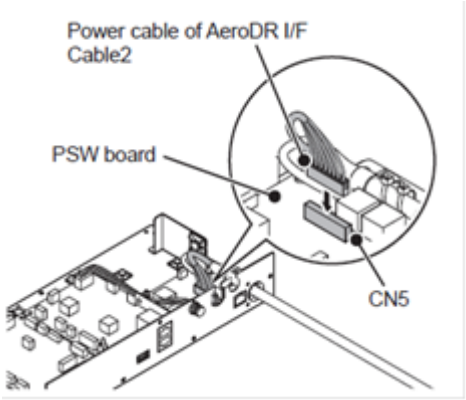


Fig. 4-113

4 **Connect the PSCN1 connector of power cable to CN1 on the PSW board.**

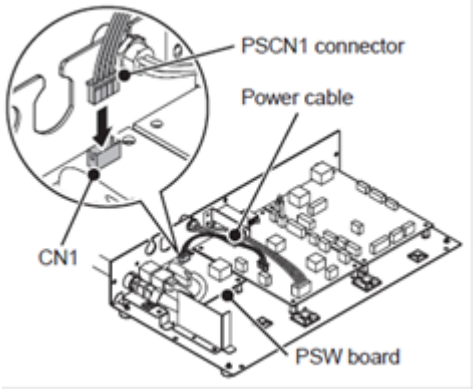


Fig. 4-114

5 **Connect the MCN4 connector of power cable to CN4 on the MPD Assy.**

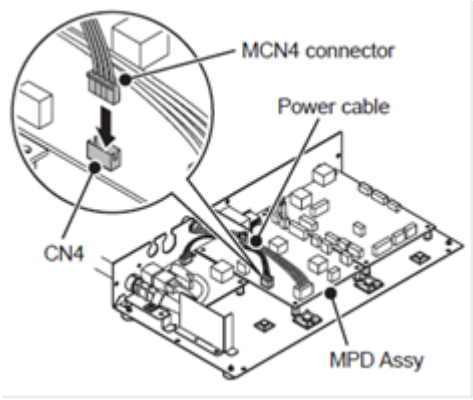


Fig. 4-115

6 **Bind the MPD Power Cable and power cable together using the locking wire saddle on the harness binding plate.**

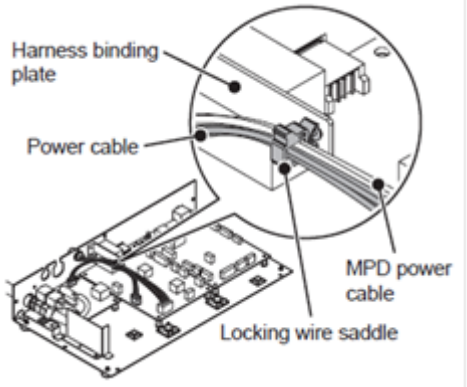


Fig. 4-116

7 **Bind MPD Power Cable and power cable together using the binding bands. Bind them together just above CN4 of the MPD Assy.**

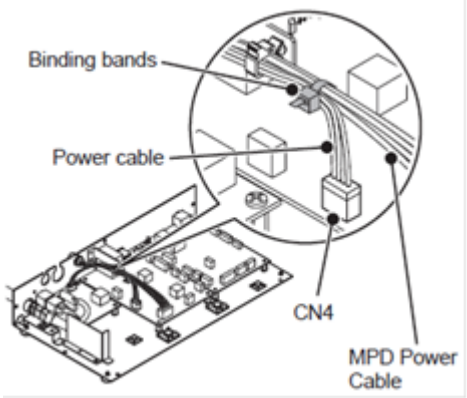


Fig. 4-117

■ Removal of the FLED Assy

1 **Remove the U FLED cable from the binding band of the base assy.**

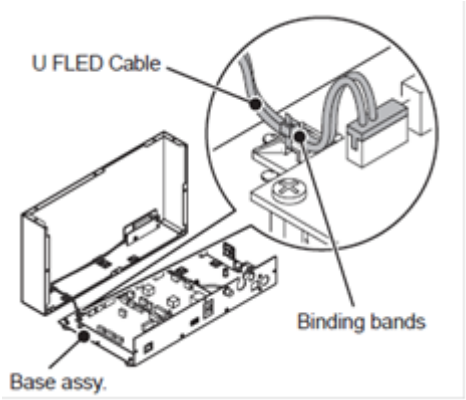


Fig. 4-118

2 **Disconnect the UCN12 connector of U FLED Cable from CN12 on the UEC board.**

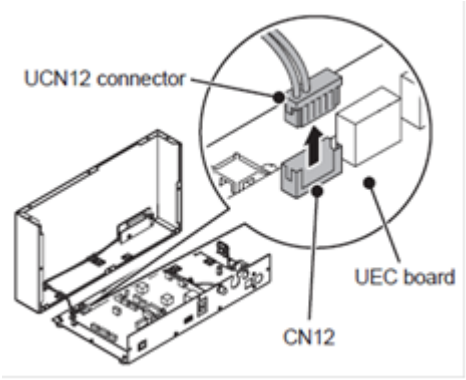


Fig. 4-119

3 **Remove U FLED Cable from the binding bands (3) of the base assy.**

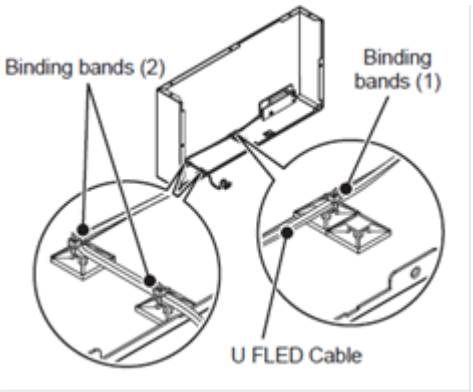


Fig. 4-120

4 **Remove the screws (2), and remove the FLED Assy.**

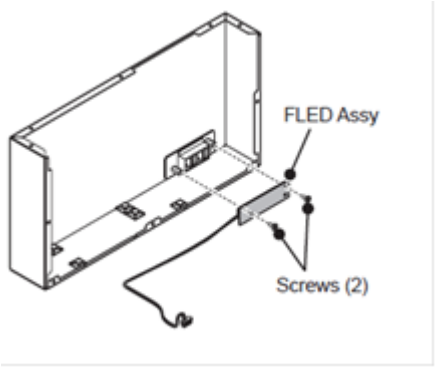


Fig. 4-121

5 **Disconnect the FLCN1 connector from CN1 on the FLED Assy.**

NOTE
- Do not disconnect the UCN12 connector of U FLED Cable from CN12 on the UEC board.
- Do not remove U FLED Cable from the base assy. And cover assy.

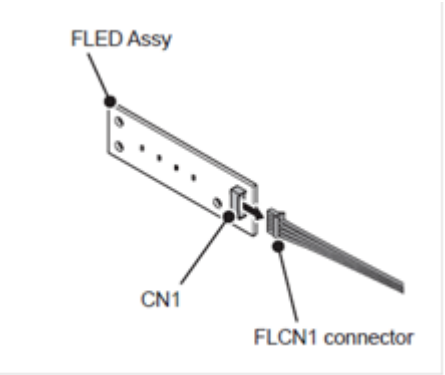


Fig. 4-122

6 **Pass the binding band through the mount base of the cover assy., and secure the U FLED cable.**

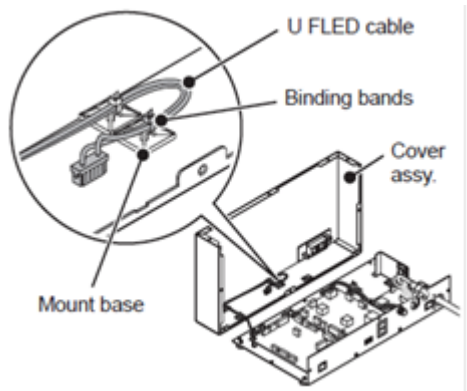


Fig. 4-123

■ Attachment of the FLED Assy

1 **Connect the FLCN1 connector of FLED cable to CN1 on the FLED Assy.**

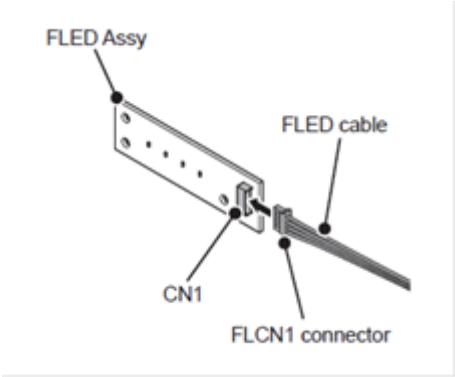


Fig. 4-124

2 **Attach the FLED Assy to the cover assy. using the screws (2).**

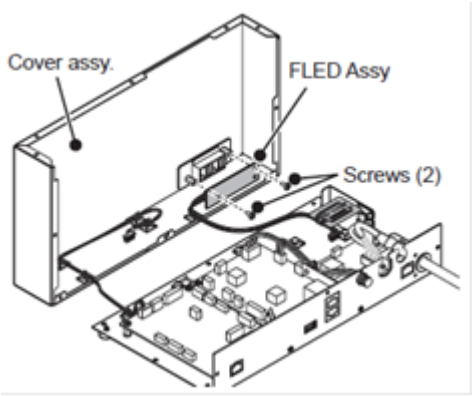


Fig. 4-125

3 **Pass the binding band through the mount base of the cover assy., and secure FLED cable.**

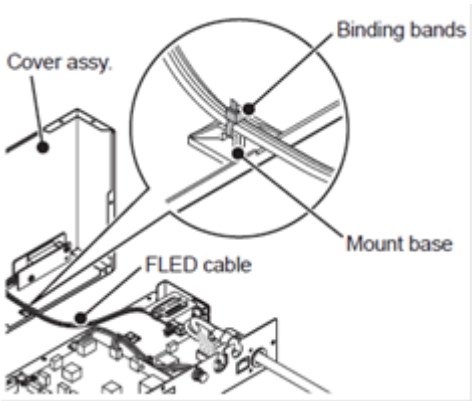


Fig. 4-126

4 **Connect the PSCN4 connector of FLED cable to CN4 on the PSW board.**

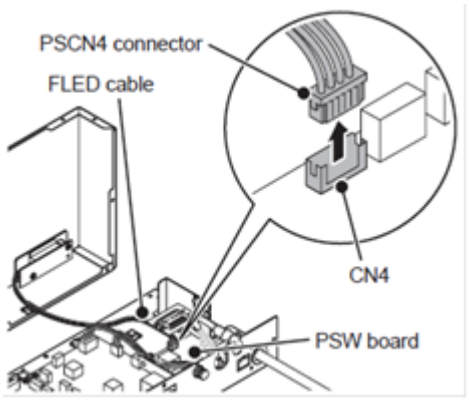


Fig. 4-127

5 **Pass the binding band through the mount base of the base assy., and secure FLED cable.**

- Secure it approximately 60mm away from the edge of the connector.
- The FLED cable fixing position is specified to prevent the nameplate from coming in contact with the board.

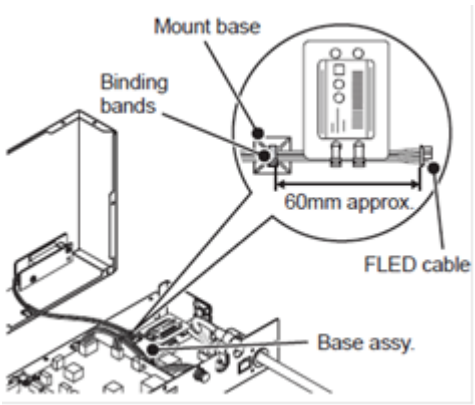


Fig. 4-128

6 **Pass a binding band through the mount base of the base assy, and secure the nameplate of the FLED cable.**

• Secure the nameplate to prevent it from coming in contact with the board.

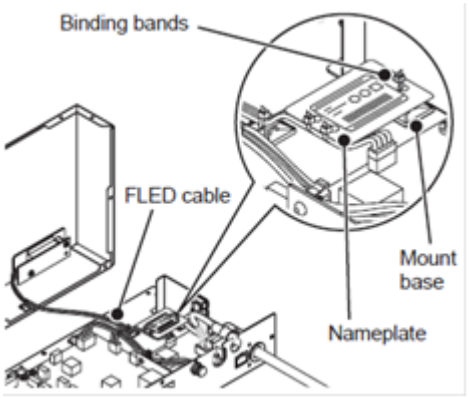


Fig. 4-129

7 **Attach the plug cushion rubbers (2) to the connector mounting plate.**

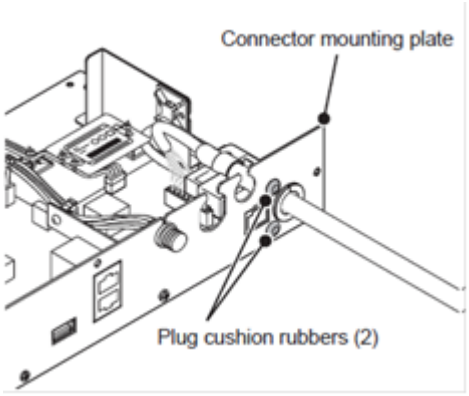


Fig. 4-130

■ Attachment of XRAY IF CABLE(KM)

1 **Connect the UCN8 connector of XRAY IF CABLE(KM) to CN8 on the UEC board.**

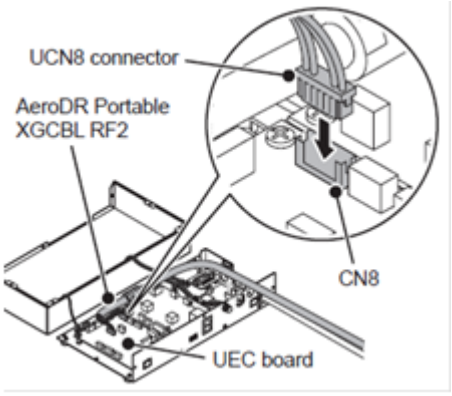


Fig. 4-131

2 **Connect the UCN10 connector of XRAY IF CABLE(KM) to CN10 on the UEC board.**

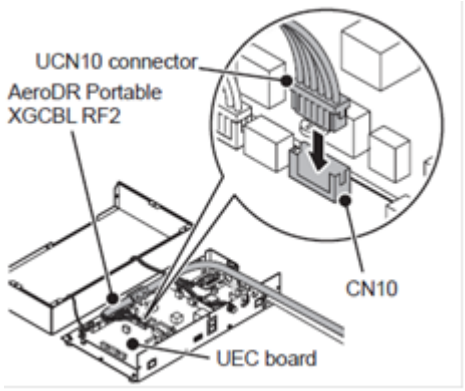


Fig. 4-132

3 **Secure XRAY IF CABLE(KM) to the connector mounting plate.**

- Temporarily fix the bushing to the front of the binding band.
- Secure the bushing to the outlet for X-ray linkage cable of the connector mounting plate.

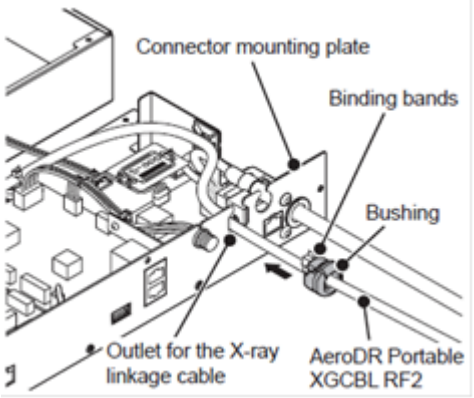


Fig. 4-133

4 **Pass the binding band through the mount base of the harness binding plate, and secure XRAY IF CABLE(KM).**

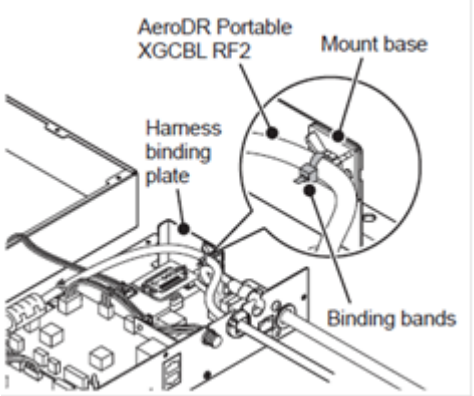


Fig. 4-134

5 **Attach the bush to the outlet for the hand switch cable of the connector mounting plate.**

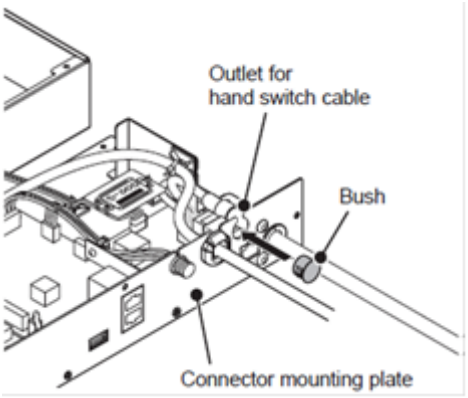


Fig. 4-135

6 **Attach the cover assy.**

4.13.4 Installation of AeroDR Portable RF Unit 2

1 **Turn OFF the key switch.**

2 **Turn OFF the main breaker.**

3 **Remove the inside cover and both side covers, and open FPD Box.**

Reference "3.5 How to Open/Close Covers"

4 **Discharge the internal capacitor.**

Reference "3.4 Discharging the Internal Capacitor"

5 **Remove four screws, and remove the plate.**

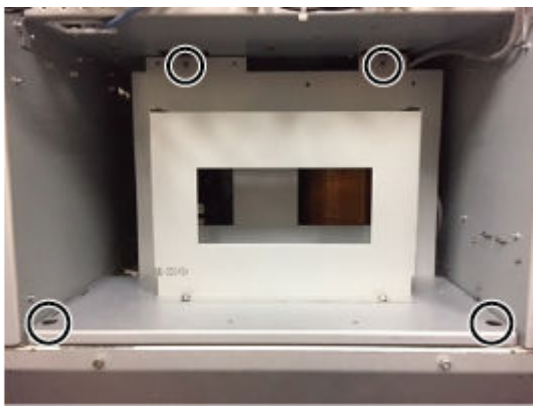


Fig. 4-136

6 **Remove the infrared unit, and attach a seal to hide "LINK".**

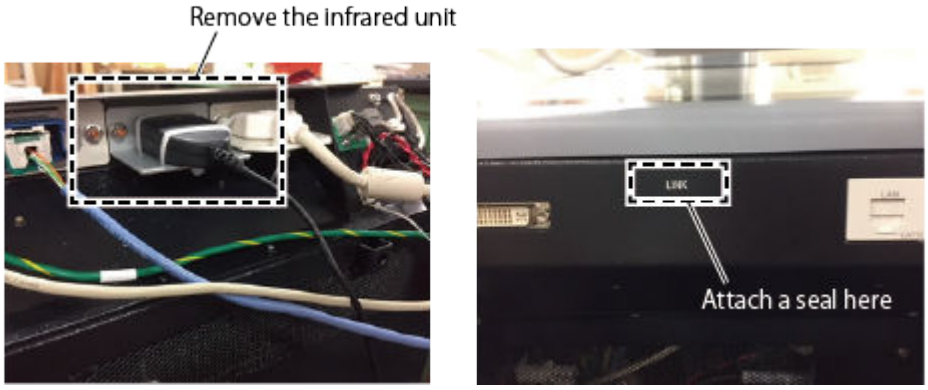


Fig. 4-137

7 **Remove the outlet and the hold bracket from the base plate of IF BOX ASSY,KM.**

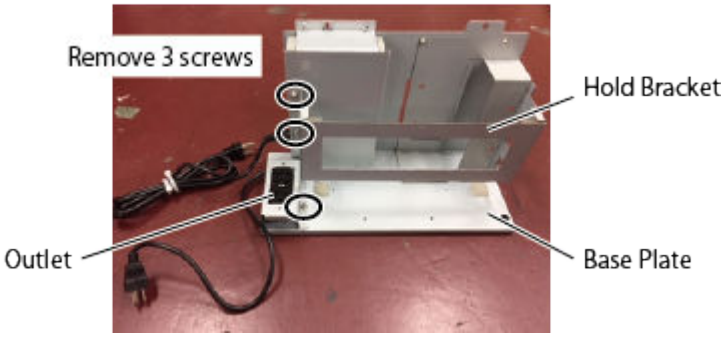


Fig. 4-138

8 **Remove the adapter bracket from the base plate.**

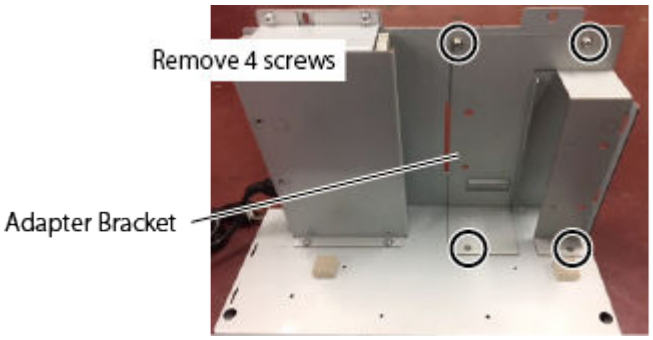


Fig. 4-139

9 **Fix the AC adapter to the adapter bracket, and attach the bracket to the base plate again.**

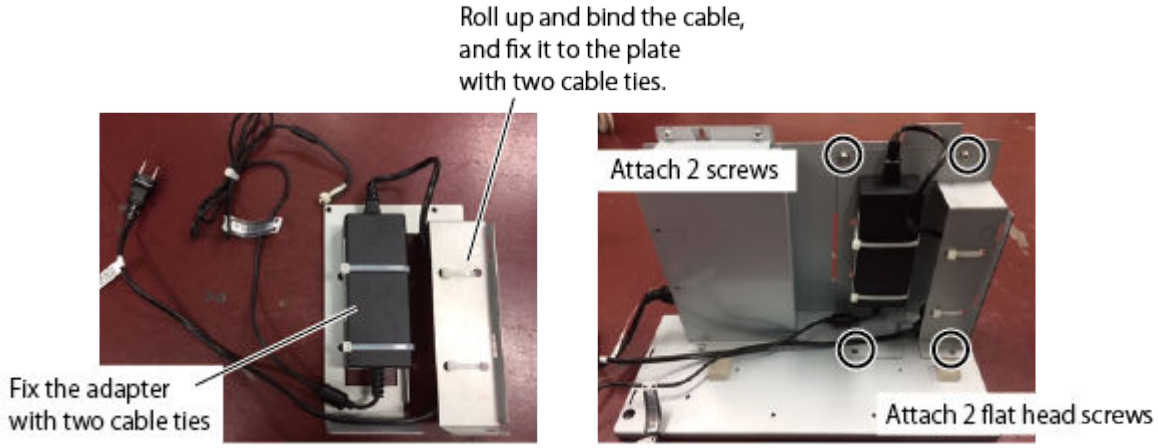


Fig. 4-140

10 Wire a cable of AeroDR portable RF unit 2.

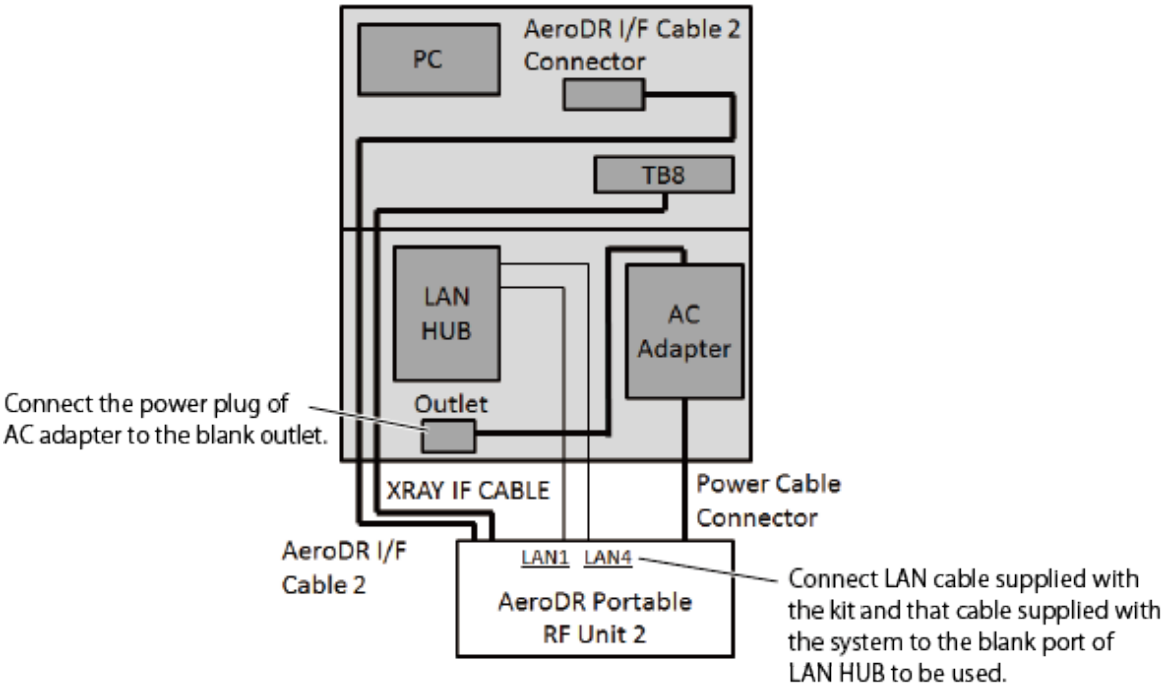


Fig. 4-141

1 Connect LAN cable to the second top port on LAN HUB.

> **NOTE** The connected position is recommended. Any LAN port can be used for connection.

Fig. 4-142

11 Attach AeroDR Portable RF Unit2 covered with the hold bracket to the base plate.

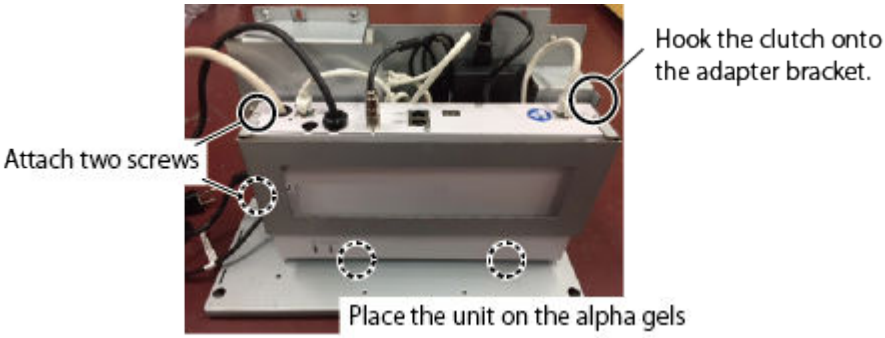


Fig. 4-143

12 Connect each LAN cable for Access point and for PC to the top port on LAN HUB.

> **NOTE** The connected position is recommended. Any LAN port can be used for connection.

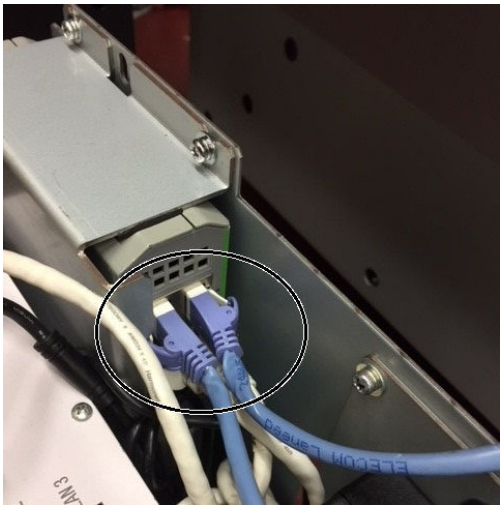


Fig. 4-144

13 Mount the base plate to the mobile X-ray system.

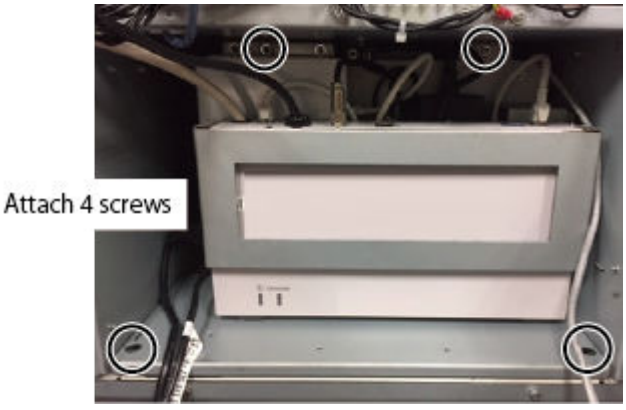


Fig. 4-145

14 Wire the XRAY IF cables.

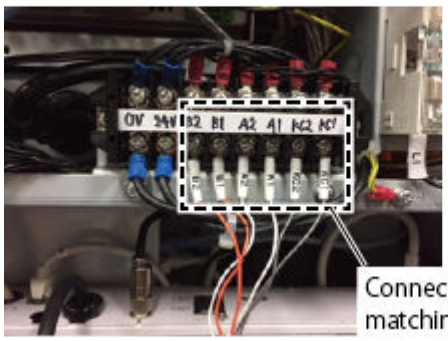


Fig. 4-146

15 Wire the power cable of the outlet and fix it to the base plate.

Fig. 4-147

16 Connect each power cable for LAN HUB and for AeroDR Portable RF Unit2 to the outlet.

Fig. 4-148

17 **Unscrew three fixing screws on SW BOX and one fixing screw on the back-side cover, and pull out SW BOX upward.**

Fig. 4-149

18 **Pass the AeroDR I/F cable under SW BOX.**

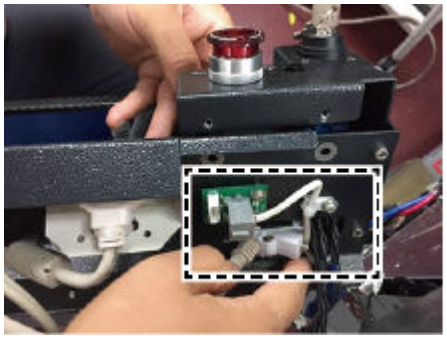


Fig. 4-150

19 **Pull out the AeroDR I/F cable about 45 cm length, and fix SW BOX.**

Fig. 4-151

20 **Fix the AeroDR I/F cable with a cable tie.**

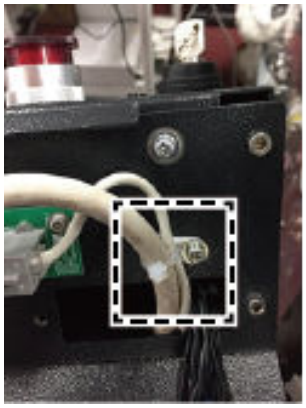


Fig. 4-152

21 **Fix the connector holder to FPD BOX.**

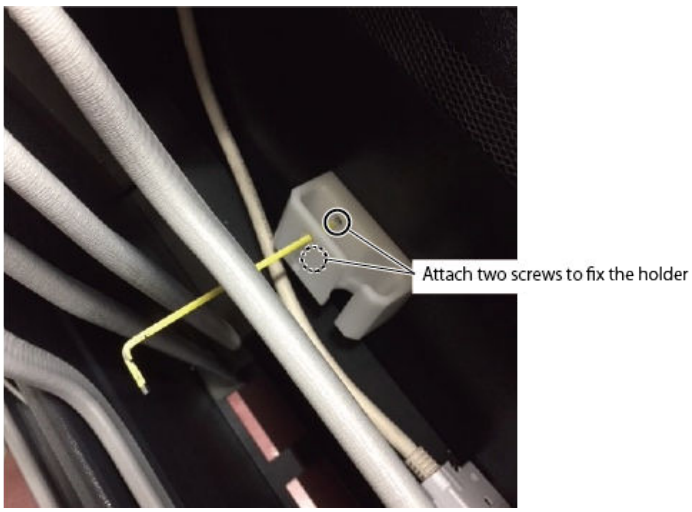


Fig. 4-153

4.13.5 FPD Box for AeroDR 1012HQ

When using AeroDR 1012HQ, attach the accessory cushion sheet to the left side inside the slot.

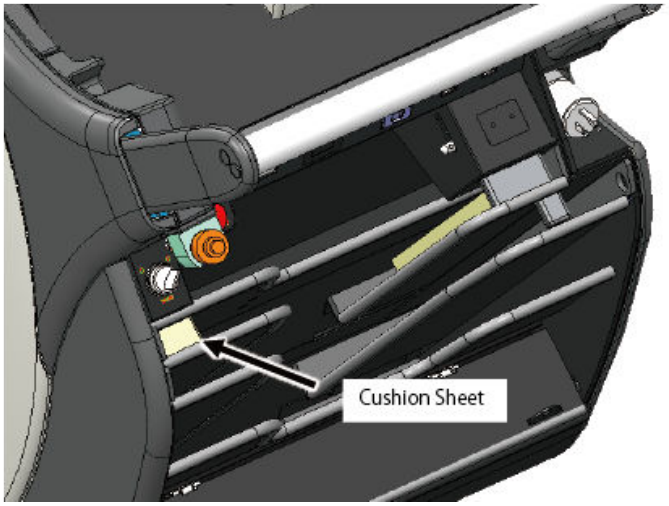


Fig. 4-154

4.14 Installing GIU SZ (Only for MobileDaRt Evolution)

NOTE This section is applicable only to the following system.
- MobileDaRt Evolution Type AeroDR (CS-7 with serial radiography)

4.14.1 Installation Procedures for GIU SZ

If GIU SZ is not mounted, install GIU SZ according to the flowchart below.

Table 4-18

Start
↓
Installation Procedures for GIU SZ
↓
Check of bundled items
↓
FPD Box for AeroDR 1012HQ
↓
Installation of GIU SZ
↓
"5.10 Assembly of FPD Docking Mechanism (Only Konica Minolta DR-equipped System)"
↓
End

4.14.2 Check of bundled items

■ Component of GIU SZ

Open the device packaging and check the bundled items to confirm that there are no scratches, cracks or other external damage.

The configuration of the GIU SZ is shown below.

Table 4-19

Item	Parts Name	Quantity
I/F cable4 2mU	I/F cable4 2mU	1
	Device identity lists	1
	Inspection Certificate	1
	Bundled items list	1
GIU SZ	Generator Interface Unit	1
	Earth Clamp	1
	Screw M4x8	1

Item	Parts Name	Quantity
	Cable Tie	3
	Inspection Certificate	1
	Packing insertion	1
	Bundled items list	1
GIU SZ AC Adapter	AC adapter	1
	Power Cable	1
	Inspection Certificate	1
	Bundled items list	1

NOTE
- Use the identification labels (seals) affixed on the packing box to perform the activation of device registrations. Keep the packing box until installation has been completed.
- Bundled items differ depending on the product or destination. For details, refer to the bundled items list.

4.14.3 FPD Box for AeroDR 1012HQ

When using AeroDR 1012HQ, attach the accessory cushion sheet to the left side inside the slot.

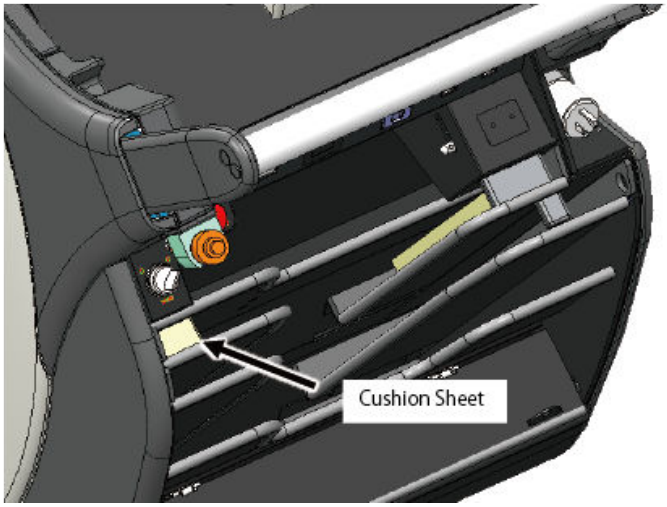


Fig. 4-155

4.14.4 Installation of GIU SZ

1 **Turn OFF the key switch.**

2 **Turn OFF the main breaker.**

3 **Remove the inside cover and both side covers, and open FPD Box.**

Reference "3.5 How to Open/Close Covers"

4 **Discharge the internal capacitor.**

Reference "3.4 Discharging the Internal Capacitor"

5 **Remove four screws, and remove the plate.**

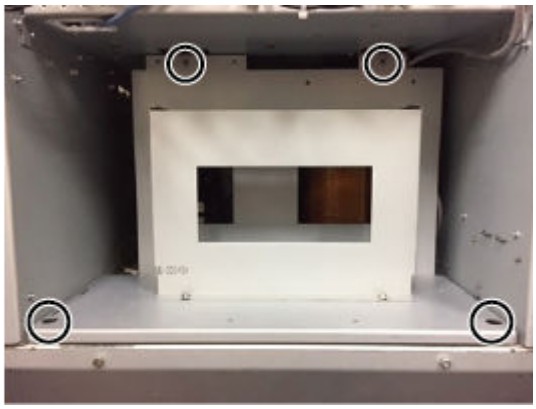


Fig. 4-156

6 **Remove the infrared unit, and attach a seal to hide "LINK".**

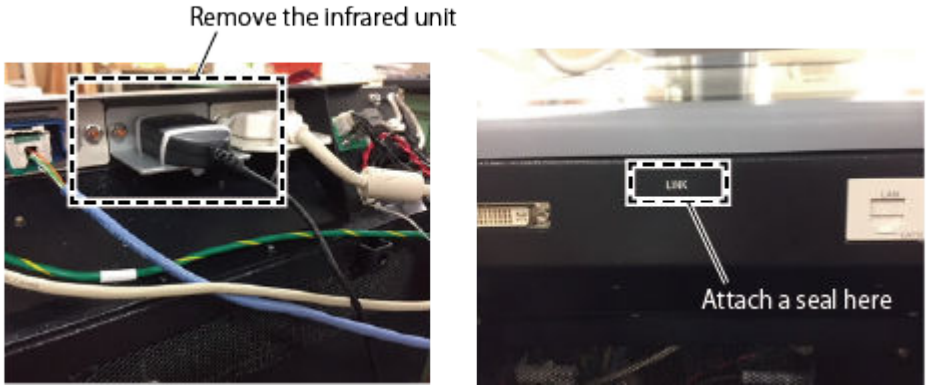


Fig. 4-157

7 **Remove the outlet and the hold bracket from the base plate of IF BOX ASSY,KM.**

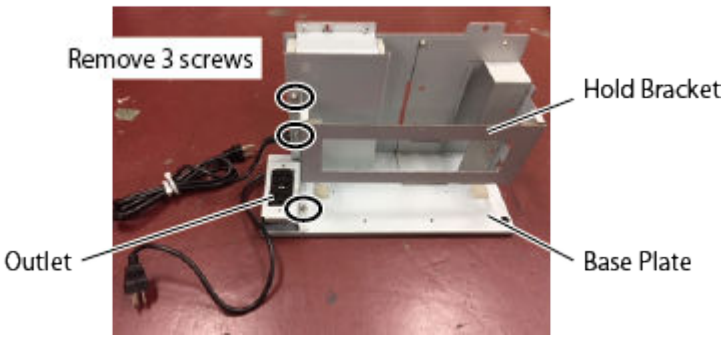


Fig. 4-158

NOTE Hold Bracket is not used in this procedure.

8 **Remove the adapter bracket from the base plate.**

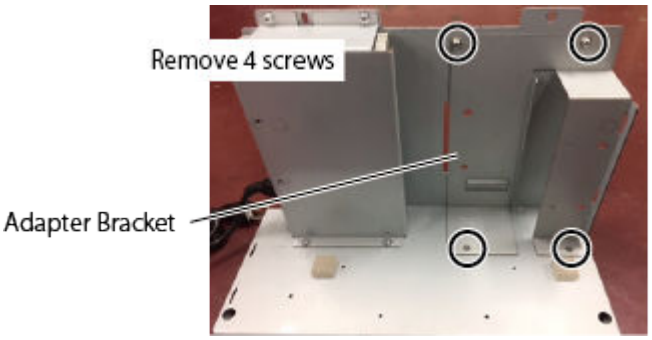


Fig. 4-159

9 **Fix the AC adapter to the adapter bracket, and attach the bracket to the base plate again.**

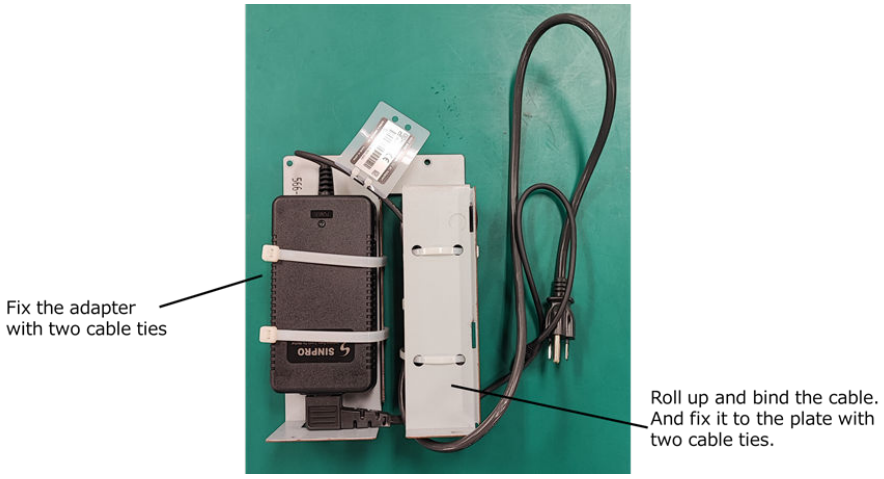


Fig. 4-160

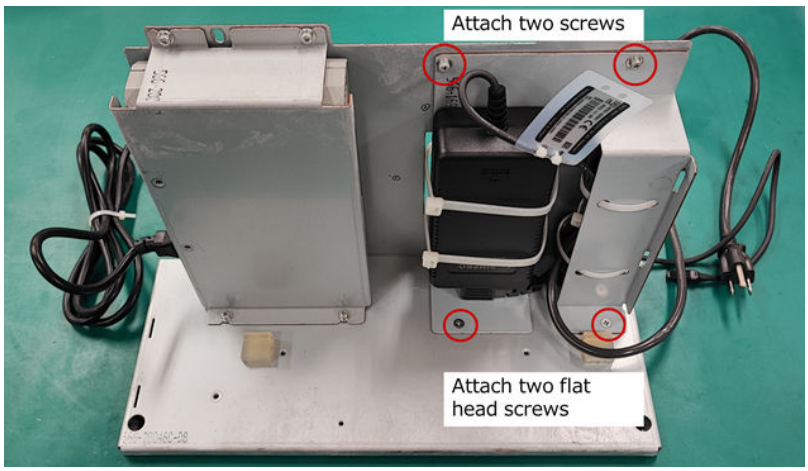


Fig. 4-161

NOTE The AC adapter should be installed with the L-shaped connector facing down.

10 **Mount the base plate to the mobile X-ray system.**

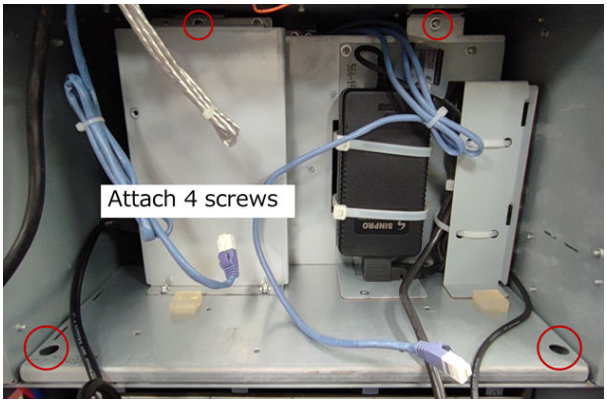


Fig. 4-162

11 Mount the GIU SZ to the base plate, and wire a cable of GIU SZ.

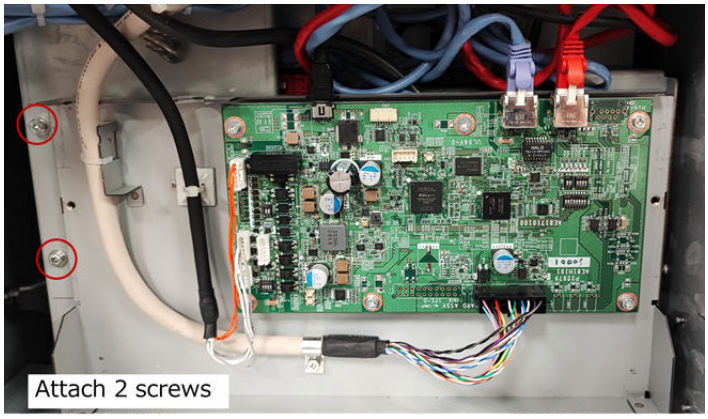


Fig. 4-163

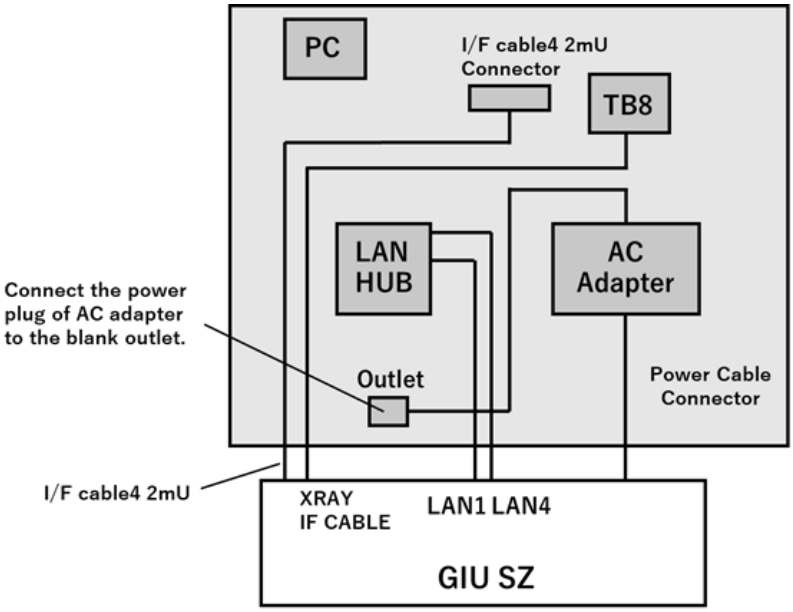


Fig. 4-164

1 Refer to "5.10 Assembly of FPD Docking Mechanism (Only Konica Minolta DR-equipped System)" to connect I/F cable4 2mU.

2 Connect LAN cable to the second top port on LAN HUB.

NOTE The connected position is recommended. Any LAN port can be used for connection.

Fig. 4-165

12 Connect each LAN cable for Access point and for PC to the top port on LAN HUB.

NOTE The connected position is recommended. Any LAN port can be used for connection.

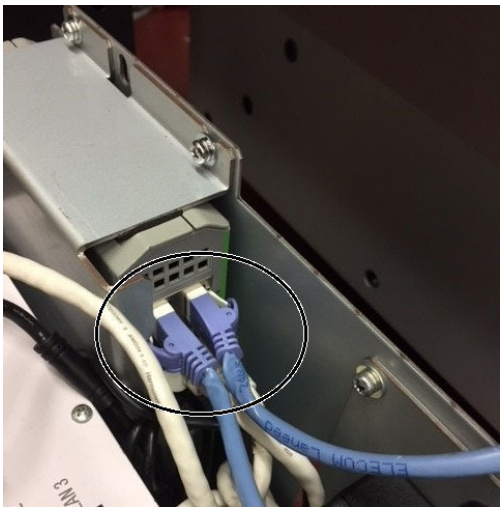


Fig. 4-166

13 Attach GIU SZ covere to the base plate.

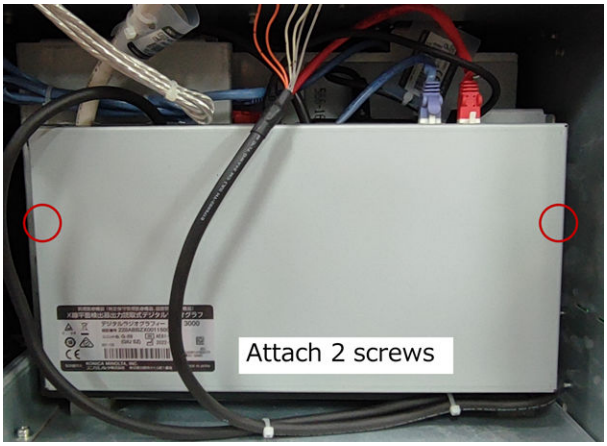


Fig. 4-167

14 Wire the XRAY IF cables.

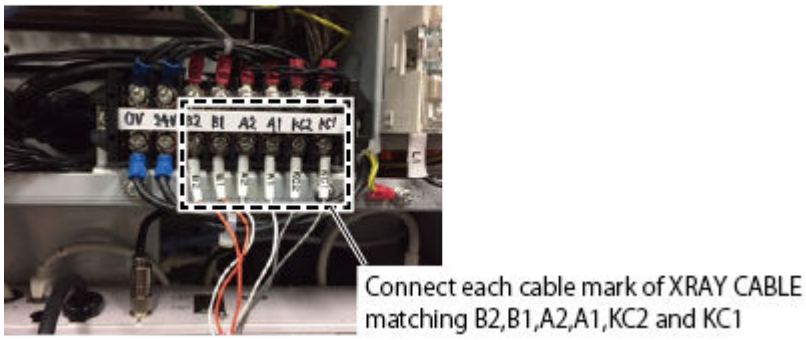


Fig. 4-168

15 Wire the power cable of the outlet and fix it to the base plate.

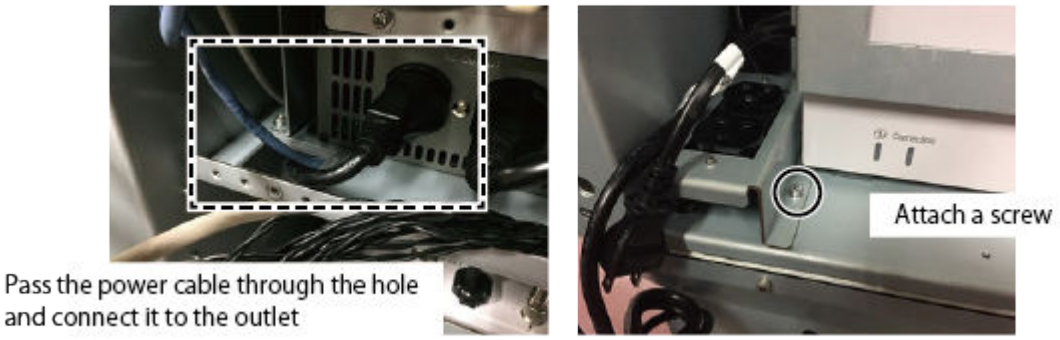


Fig. 4-169

16 **Connect each power cable for LAN HUB and for GIU SZ to the outlet.**

Fig. 4-170

4.14.5 Assembly of FPD Docking Mechanism

After installation of GIU SZ, refer to "5.10 Assembly of FPD Docking Mechanism (Only Konica Minolta DR-equipped System)" to assemble the FPD docking mechanism.

4.15 Installing Access Point (Only for MobileDaRt Evolution)

NOTE This system is not equipped with Access Point, which needs to be purchased separately.

- Manufacturer: Aruba Networks
- Model: AP-303

For setting of Aruba access point, refer to the document below.

Reference
- For CXDI: Aruba Access Point Setup Manual (MQRU-18D012)
- For DR-ID800: Release Notes For DR-ID800 (M517-E865)
- For AeroDR(CS-7): MobileDaRt Evolution (MX8 Version) Installation Manual (DR Section)（M503-E410）

1 Set the Access point as below.

Fig. 4-171

2 Attach the fixing clip and the bracket B while twisting as below.

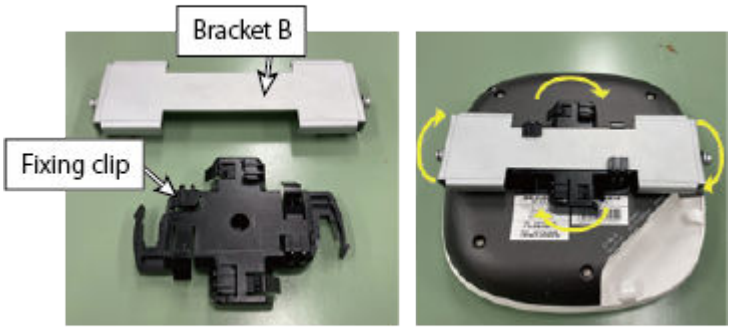


Fig. 4-172

3 **Attach the bracket A with 2 sems screws M4x8 as below.**

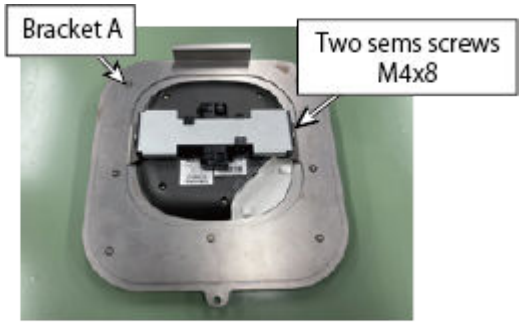


Fig. 4-173

4 **Connect LAN cable and AC power cable to the Access point. These cable have already wired in the system.**

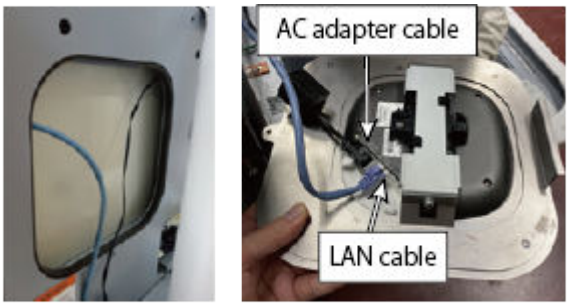


Fig. 4-174

5 **Fix the Access Point to the front cover with the toothed washer M4 and the hexagon socket button bolt M4x8.**

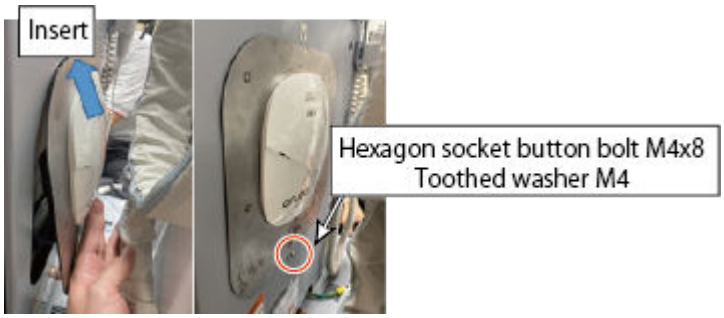


Fig. 4-175

4.16 DR System Settings (Only for MobileDaRt Evolution)

For MobileDaRt Evolution, refer to the document below to configure the DR system.

Reference M503-E410 MobileDaRt Evolution (MX8 Version) Installation Manual (DR Section)

4.17 Option Installations

Proceed to the following for option installations.

Reference "5 Installing the Options"

⚠ WARNING

In the system for 32kW type (MobileDaRt Evolution/MobileArt Evolution high-power type), large-capacity capacitors are connected to the battery.
Even after turning OFF the key switch or the main breaker, electrical charge remains in the capacitor.
Be sure to discharge the capacitor when opening the cover of this system.

Reference "3.4 Discharging the Internal Capacitor"

4.18 Adjustment of Image Quality (Only for MobileDaRt Evolution)

For MobileDaRt Evolution, refer to the document below to adjust the image quality.

Reference M503-E410 MobileDaRt Evolution (MX8 Version) Installation Manual (DR Section)

4.19 Attaching Arm Cover

1 **Expand the arm.**

Fig. 4-176

2 **Loosen screws fixing the cover at the bottom of the arm. (M4 hex bolt, M4 flat washer at 2 positios)**

Fig. 4-177

3 **Attach the left-side arm cover to the arm. (M4 sems screw at 2 positions, M4 bind screw at 1 position)**

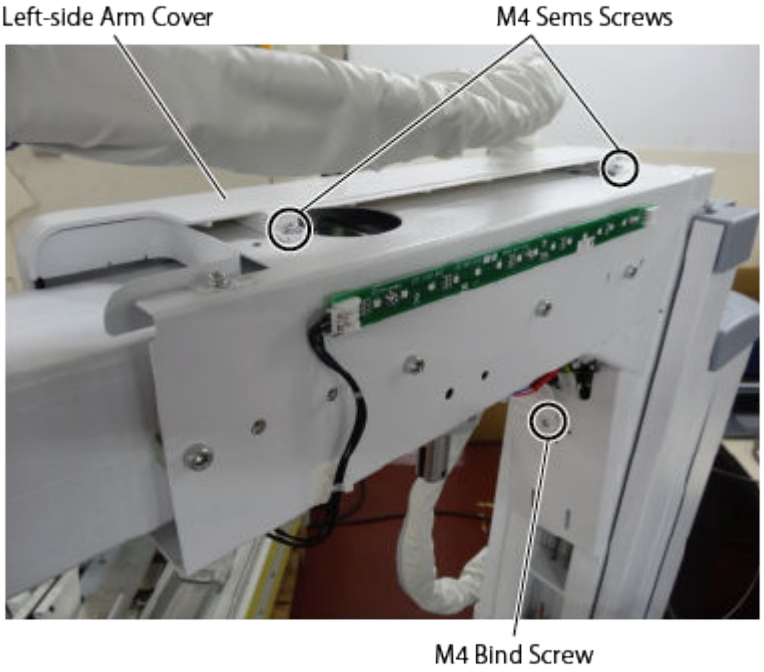


Fig. 4-178

4 **Attach the right-side arm cover to the arm. (M4 sems screw at 2 positions, M4 bind screw at 1 position)**

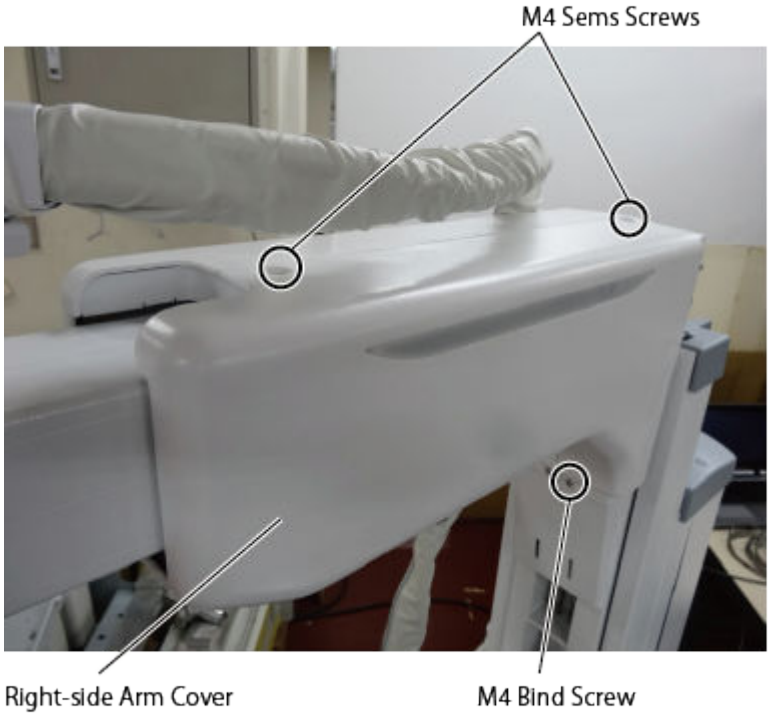


Fig. 4-179

5 **Fasten screws that was loosened in the step 2 from the screw holes.**

Fig. 4-180

6 **Attach the bottom cover. (M4 sems screw at 2 postions)**

Fig. 4-181

4.20 Attaching Knob for Collimator Rotation-Lock

Attach a knob for locking collimator rotation as shown below. The knob is packed together with screws for adjustment of X-ray axis.

Fig. 4-182

4.21 Attaching Cushion for 14X17 FPD

To use 14X17 FPD, attach the push-up cushion to the bottom of middle-slot as shown below.

> **NOTE** This procedure applies to MobileDaRt Evolution, or MobileArt Evolution combining with FPD Box OP.

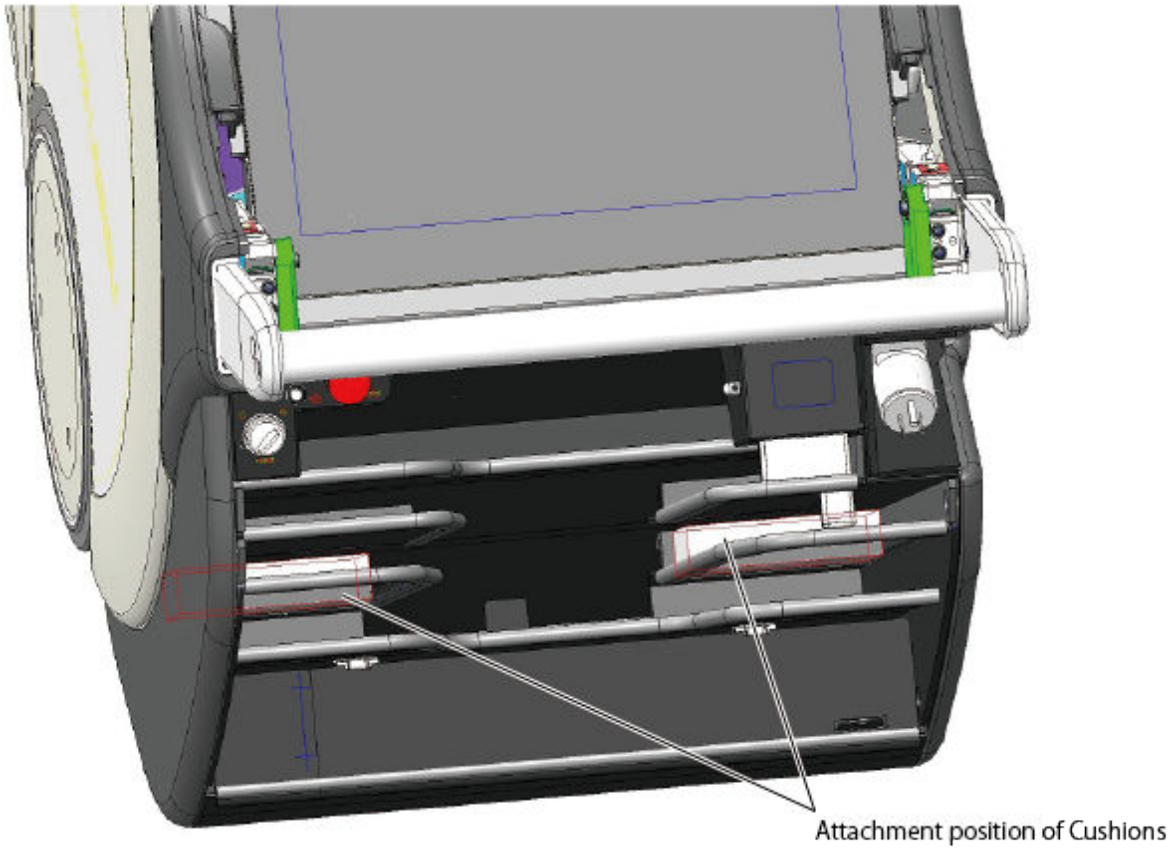


Fig. 4-183

4.22 Attaching Cushion for 10x12 FPD

When using 10x12 FPD, attach the accessory cushion on the side of the slot as shown below.

> **NOTE** This step applies to MobileDaRt Evolution combined with FXRD-2530VAW / FXRD-2530VAW PLUS, or MobileArt Evolution combined with FPD Box OP.

Fig. 4-184

4.23 Attaching Screw Cap

Attach the screw cap to the position in the figure below. The screw cap is included in the repair parts.

Side Cover (2 positions on each side, white)
Rear Wheel (3 positions on each wheel, white)

Column-under Cover
(5 positions on each side, gray)
Obstruction Detector
(3positions on each side, white 4PC, gray 2PC)

Front Cover (2 spositions, gray)

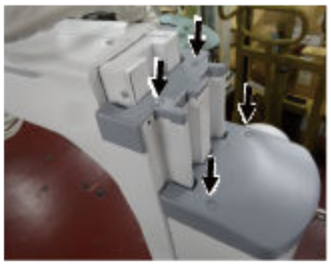
Top Cover (2 positions, gray)

Driving Hangle (2 positions on each side, gray)

Inside Cover (2 positions on each side, gray)

Collimator (2 positions, white)
X-ray tube support (1 position, gray)

Arm (6 postions, white)



Column (4 positions, gray)

Column Hook Cover (2 positions, white)

Side of Column (2 positions, white)

Fig. 4-185

NOTE Do NOT attach the screw cap to the double-hole on the lateral side of the column (circled in the figure).

4.24 Resetting All Parts Usage Information

Clear all parts usage information from the system with Usage Information Collective Setting Mode.

Reference • For MobileDaRt Evolution: "7.19 Usage Information Collective Setting Mode"
• For MobileArt Evolution: "8.19 Usage Information Collective Setting Mode"

4.25 Charging Battery After Installation

Make sure that the LEDs on the Battery Indicator are performing normally.

The rechargeable batteries may have little power due to self-discharging during transportation and installation. Be sure to inform users that the rechargeable batteries need to be charged before the first use, otherwise the batteries will lose efficiency earlier than usual.

4.26 Assembler Test

In USA, execute the assembler test according to the ASSEMBLER TEST MANUAL (Document No. M503-E330).

4.27 Set up of McAfee Embedded Control (MEC) (Only for MobileDaRt Evolution)

For MobileDaRt Evolution, refer to the document below to configure MEC.

Reference M503-E410 MobileDaRt Evolution (MX8 Version) Installation Manual (DR Section)

4.28 Backup After Installation (Only for MobileDaRt Evolution)

For MobileDaRt Evolution, refer to the document below to back up after installation.

Reference M503-E410 MobileDaRt Evolution (MX8 Version) Installation Manual (DR Section)

5 Installing the Options

How to install the options is described in this Chapter.

5.1 Wireless Hand Switch Option

5.1.1 Component of Wireless Hand Switch Option

The wireless hand switch option consists of the following parts. Make sure that all these parts are contained in the package.

> **NOTE** This option does not include a battery for the wireless handswitch. Prepare it separately beforehand.

Table 5-1 Component of Wireless Hans Switch Option

No	Parts Name	Quantity
1	Wireless Hand Switch	1
2	Wireless Holder	1
3	Harness	1
4	Cable tie	1
5	Mount base	1
6	Philipps-head screw M3X16	2
7	Cover A, FPD BOX	1

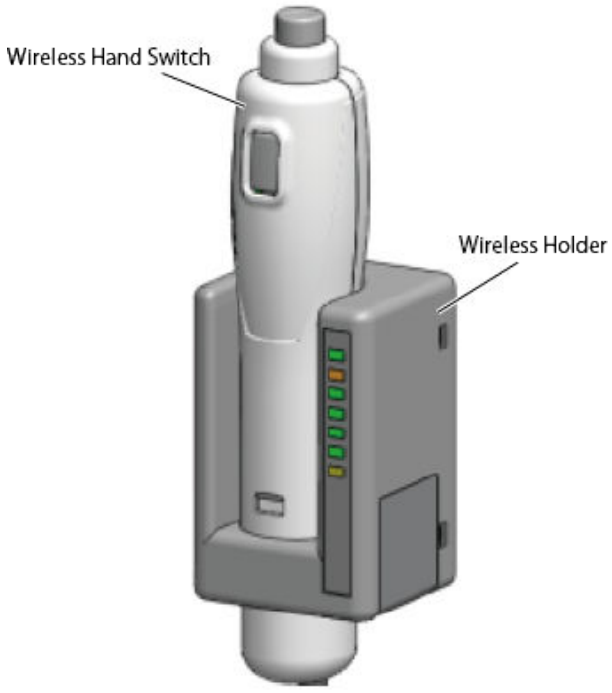


Fig. 5-1

5.1.2 Parts Details of Wireless Hand Switch Option

■ Wireless Hans Switch

Each function name of Wireless hand switch is shown below.

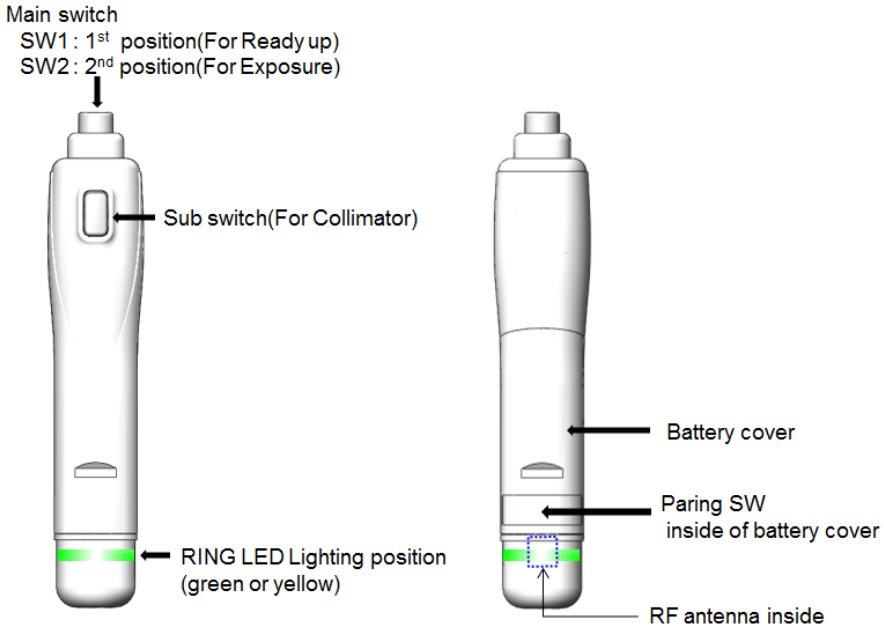


Fig. 5-2

■ Wireless Holder

Each function of Wireless Holder is shown below.

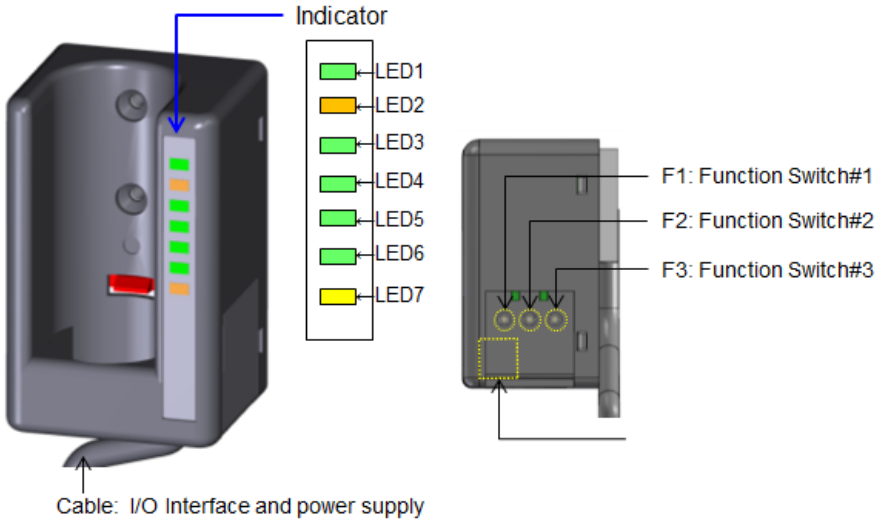


Fig. 5-3

Table 5-2

LED1 (Green)	POWER ON	Turns ON when power is supplied and the circuit is correctly activated.
LED2 (Orange)	LOW BATTERY	Indication for battery condition of HAND SWITCH Blinks when the battery voltage of HAND SWITCH is less than 2.5V. Turns ON when battery voltage of HAND SWITCH is 2.1V or less (change the battery immediately).
LED3 (Green)	LINK	Turns ON when Wireless communication has been established. Turns OFF when the HAND SWITCH is in following conditions. (1) Bluetooth pairing not established. (2) Power is OFF (3) Occurrence of wireless disconnection by out of range. (4) Entered in SLEEP MODE.
LED4 (Green)	TOUCH SENSOR setting	Turns ON when the TOUCH SENSOR function set to enabled.
LED5 (Green)	BUZZER setting	Turns ON when OPERATION LIMIT TIMER set to enabled.
LED6 (Green)	INDICATION of SW1	Turns ON when the SW1 of the HAND SWITCH is operated. And turns OFF when SW2 operated.
LED7 (Yellow)	INDICATION of SW2	Turns ON when the SW2 of the HAND SWITCH is operated.

5.1.3 Temporary Wiring of Wireless Hand Switch

1 **Turn OFF the main Breaker.**

2 **Remove the inside cover, both side covers, and top cover from the system.**

Reference "3.5 How to Open/Close Covers"

3 **Connect the harness to the J9W on the XCONT-2018 board and to the wireless holder cable.**

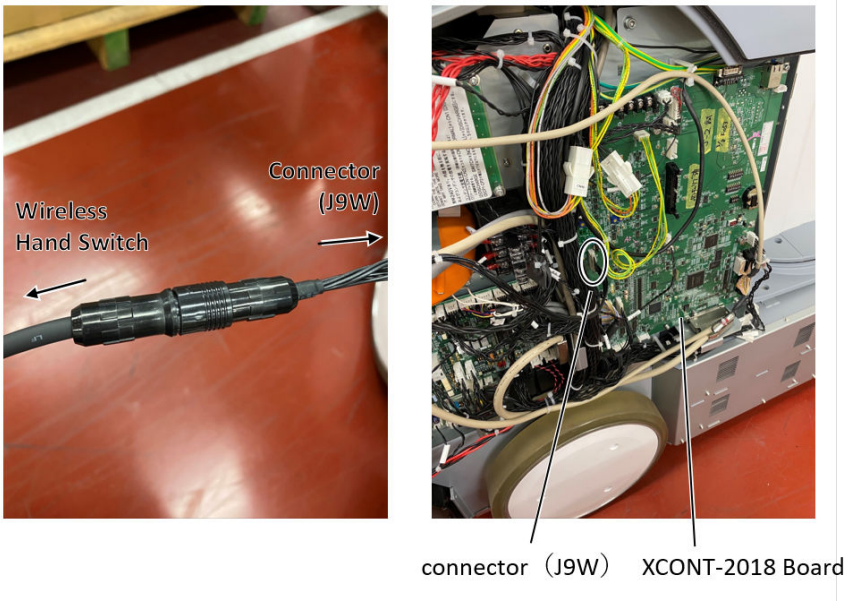


Fig. 5-4

4 **Turn ON the main breaker.**

5 **Turn ON the key switch.**

6 **Check that the LED1 of wireless holder turns ON in green.**

5.1.4 Setting Mode

Change each setting mode for the wireless holder.

Table 5-3

No.	Mode	Function	Recommend value
1	Pairing Setting	Configure the pairing setting of wireless hand switch and wireless holder.	-
2	Touch Sensor Setting	Modify the touch sensor setting. The setting is unnecessary for the wireless hand switch in Rev. B or later revisions.	Invalid
3	Operation Limit Timer Setting	Set migration time to operation limit.	30 mins
4	Buzzer Setting	When reached to operation limit timer, switch ON/OFF buzzer setting.	Invalid
5	Handover Setting	Switch ON/OFF of handover mode setting when using two wireless hand switches. (The handover has been set to be invalid as a factory setting. The setting change is not needed.)	Invalid

By pressing function switches, the setting can be changed as shown below.

NOTE
- Setting mode is canceled after 3 minutes passes after entering to this mode. Changed value will be discarded if the power of HOLDER turns off during setting mode.
- The touch sensor function setting is unnecessary for the wireless hand switch in Rev. B or later revisions.

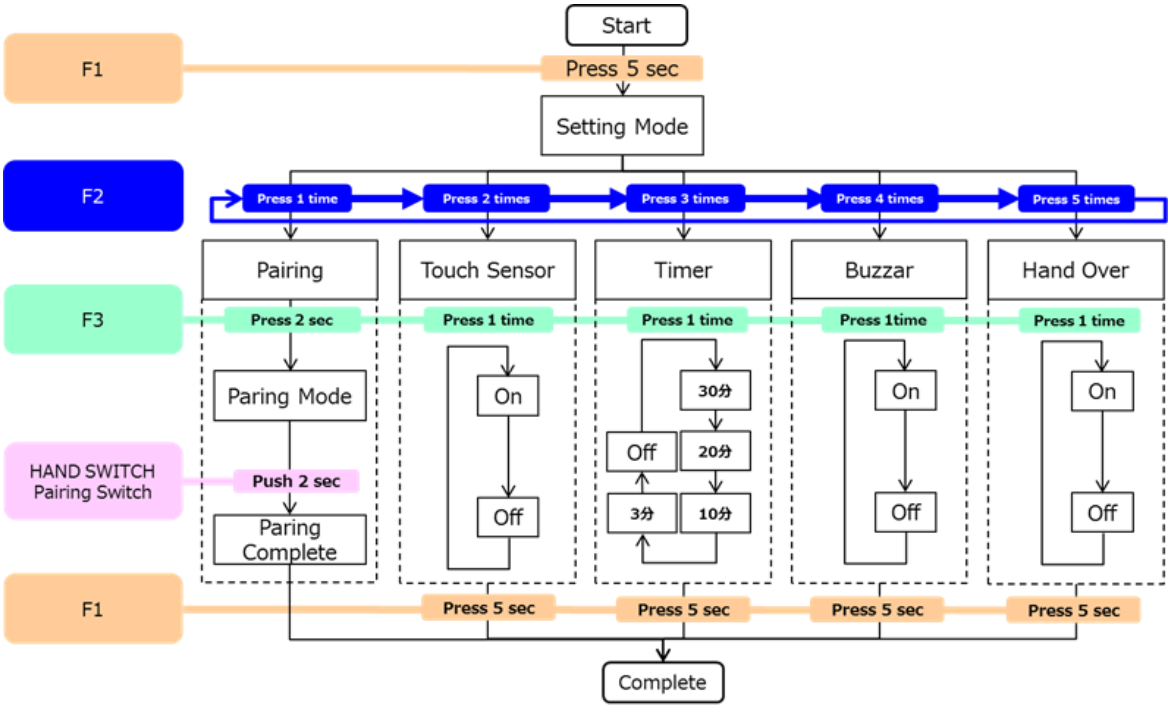


Fig. 5-5

Function switches are located below.

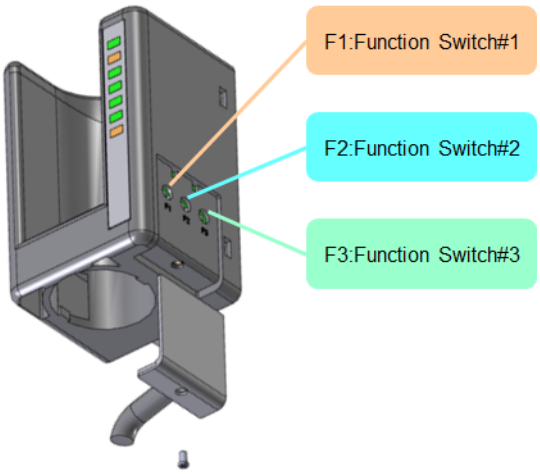


Fig. 5-6

■ Preparation before Setting

A cover of the wireless holder must be removed before access to function switch. Remove the cover according to the following instructions.

How to Open/Close the Cover

1 **Remove the cover screw. (1)**

2 **Slide the switch cover and remove it. (2)**

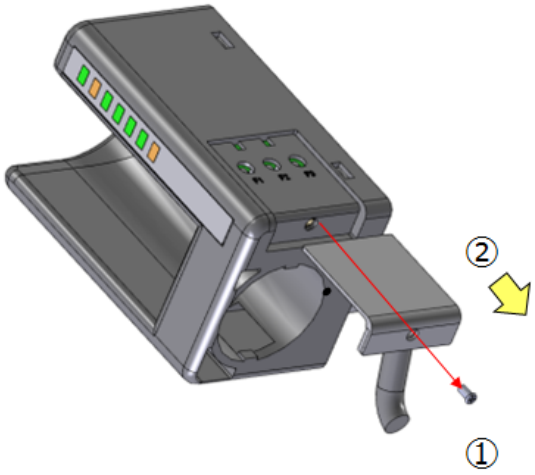


Fig. 5-7

3 **Reverse the above procedure to close the cover.**

■ Pairing Setting

Configure the pairing setting of wireless hand switch and wireless holder.

1 **Set wireless hand switch to the wireless holder. LED1 turns ON when the power is applied to the holder.**

2 **Keep pressing function switch1 (F1) of the holder until LEDs from LED2 to LED6 start blinking. (More than 5 seconds)**

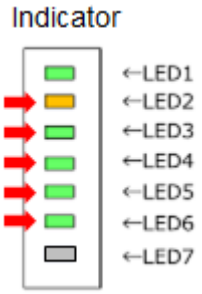


Fig. 5-8

3 **Press function switch2 (F2) once to light LED2.**

4 **Press function switch3 (F3) for more than 2 seconds.**

5 **LED2 starts blinking. Wireless holder starts searching for a wireless hand switch that is in a state of pairing mode.**

6 **Take out wireless hand switch from the holder, and press and hold the paring switch of wireless hand switch for more than 2 seconds within 1 minute since the holder is set to the pairing mode.**

Fig. 5-9

7 **LED3 turns ON when the pairing is accomplished. LED4 and LED5 show touch sensor status and operation limit timer status. (They will not turn ON if their settings are not available.)**

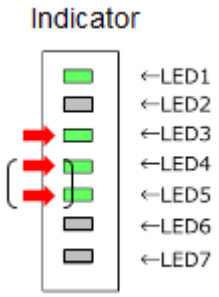


Fig. 5-10

NOTE

- If the LED blinks as green->yellow->green->yellow after pressing pairing switch of the wireless hand switch, more than two wireless holders (maximum number) are already paired to the hand switch. In this case, it is necessary to delete all the pairing information and try the pairing again. To delete the pairing information, see "5.1.8 How to Delete Pairing Information for Wireless Hand Switch". Then press the pairing switch for 2 seconds for pairing.
  *If the pairing information is deleted, already paired two holder information will be both deleted.
- For the wireless hand switch in Rev. B or later revisions, LED 4 will not turn ON even after the paring is accomplished since the touch sensor function has been disabled as default.

■ Touch Sensor Setting

Modify the touch sensor setting.

> NOTE
> - The touch sensor is not used for this system. It should be set as invalid.
> - The setting is unnecessary for the wireless hand switch in Rev. B or later revisions.

1 **Set wireless hand switch to the wireless holder. LED1 turns ON when the power is applied to the holder.**

2 **Press function switch1 (F1) of the holder until LEDs from LED2 to LED6 start blinking. (More than 5 seconds)**

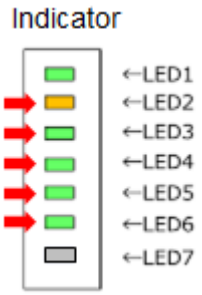


Fig. 5-11

3 **Press function switch2 (F2) twice.**

4 **LED3 and LED6 turn ON. (Touch sensor function at the factory setting is valid)**

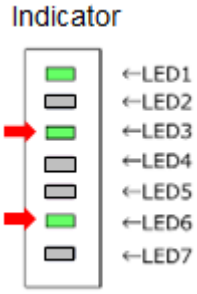


Fig. 5-12 Touch Sensor: Valid

5 **Each time F3 is pressed, the touch sensor switch valid/invalid.**

6 **LED3 and LED7 are lighted.**

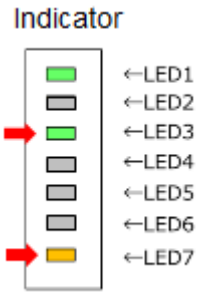


Fig. 5-13 Touch Sensor: Invalid

7 **Press function switch1 (F1) for more than 5 seconds.**

8 **Only LED1 is lighted and setting completed.**

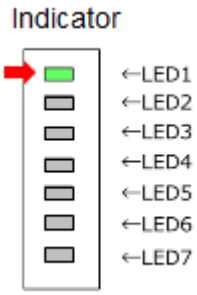


Fig. 5-14

■ Operation Limit Timer Setting

Set migration time to operation limit.

> NOTE The operation limit timer has been set to 30 minutes as a factory setting. To change the limit timer, follow the instruction below.

1 **Set wireless hand switch to the wireless holder. LED1 is lighted when the power is applied to the holder.**

2 **Keep pressing function switch1 (F1) of the holder until LEDs from LED2 to LED6 start blinking. (More than 5 seconds)**

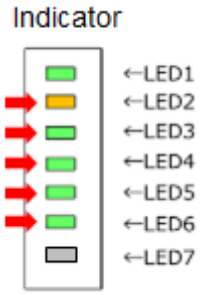


Fig. 5-15

3 **Press function switch2 (F2) for three times.**

4 **LEDs from LED4 to LED7 indicate current setting. (The setting is set to 30 minutes at factory setting)**

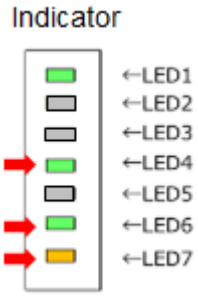


Fig. 5-16

5 **Each time F3 is pressed, the activation time changes 30 minutes->20 minutes->10 minutes->OFF->30 minutes.**
Refer to the following figure for details.

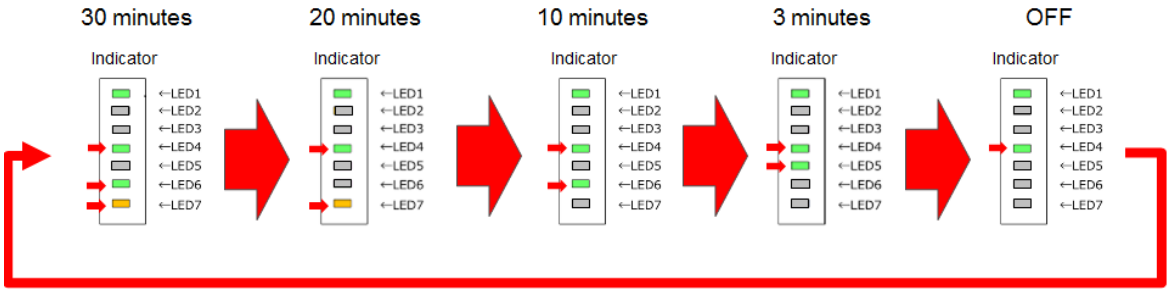


Fig. 5-17

6 **Press function switch1 (F1) for more than 5 seconds.**

7 **Only LED1 is lighted and setting completed.**

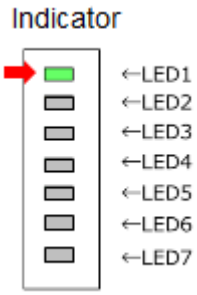


Fig. 5-18

NOTE To return from the operation limit timer setting, set the wireless hand switch to the wireless holder, and take out the wireless hand switch from the holder again.

■ Buzzer Setting

When reached to operation limit, switch ON/OFF buzzer setting.

> NOTE The buzzer is not used for this system. It should be set as invalid.

1 **Set wireless hand switch to the wireless holder. LED1 is lighted when the power is applied to the holder.**

2 **Press function switch1 (F1) of the holder until LEDs from LED2 to LED6 start blinking. (More than 5 seconds)**

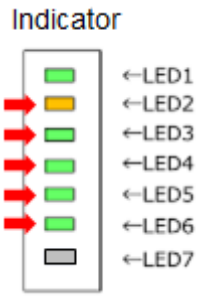


Fig. 5-19

3 **Press function switch2 (F2) for four times.**

4 **LED5 and LED6 are lighted. (Buzzar function at the factory setting is valid)**

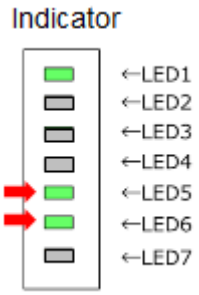


Fig. 5-20 Buzzar: Valid

> NOTE The adjustment of sound volume is not available.

5 **Each time F3 is pressed, valid/invalid of buzzer function is switched.**

6 **LED5 and LED7 are lighted.**

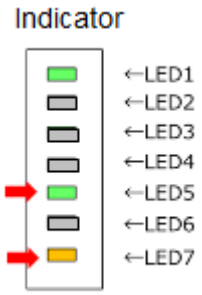


Fig. 5-21 Buzzar: Invalid

7 **Press function switch1 (F1) for more than 5 seconds.**

8 **Only LED1 is lighted and setting competed.**

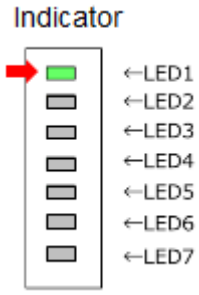


Fig. 5-22

5.1.5 Installing Wireless Hand Switch Option

NOTE The mounting position of the wireless hand switch depends on the FPD box/cassette box. Refer to the corresponding procedure for mounting.

Reference
- "Mounting Wireless Hand Switch for FPD BOX" P.169
- "Mounting Wireless Hand Switch for Cassette Box" P.172

■ Mounting Wireless Hand Switch for FPD BOX

1 **Turn OFF the main Breaker.**

2 **Remove the inside cover, both side covers, and top cover from the system.**

Reference "3.5 How to Open/Close Covers"

3 **Replace the accessory case cover.**

1 Remove 4 hexagon cap screws on the back to replace with optional short cover A, FPD BOX.

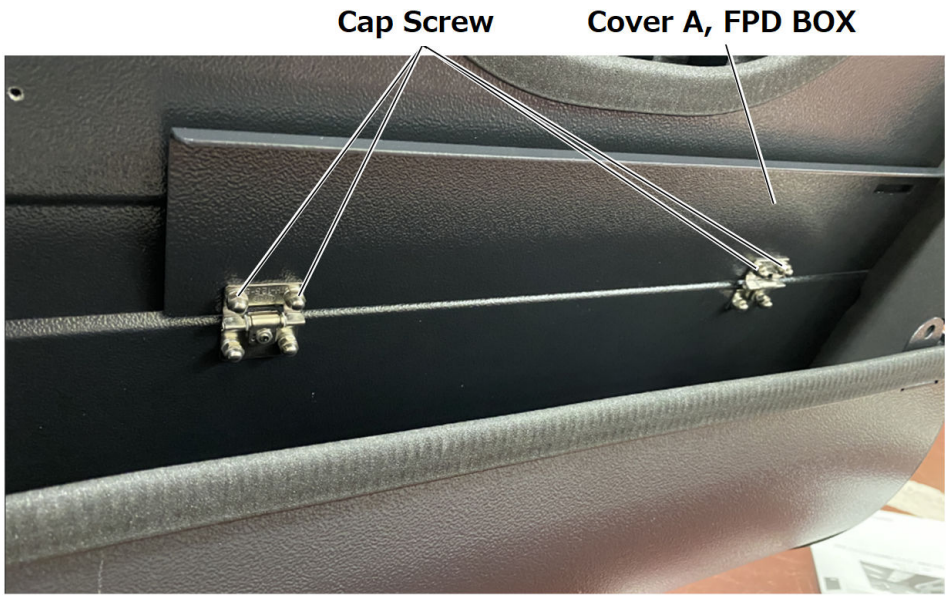


Fig. 5-23

4 **Mount the wireless holder with two phillips-flat-head screws M3x16 and wire the cable through a hole on the side of FPD box.**

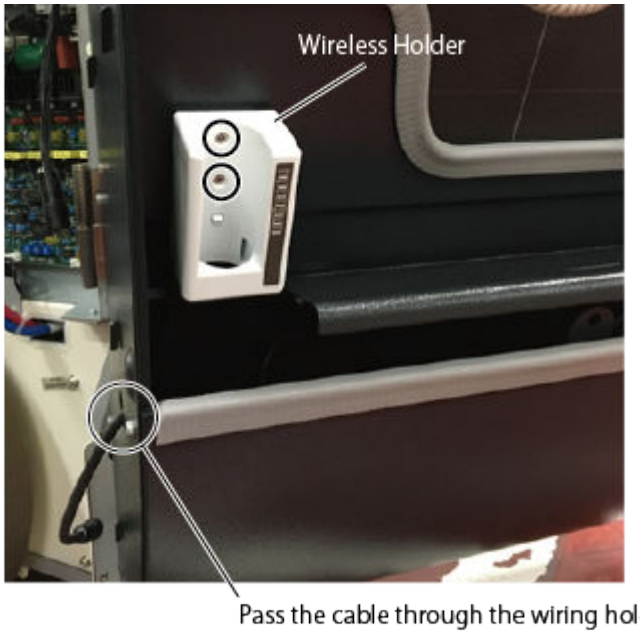


Fig. 5-24

5 **Connect and wire the harness to J9W on the XCONT-2018 board.**

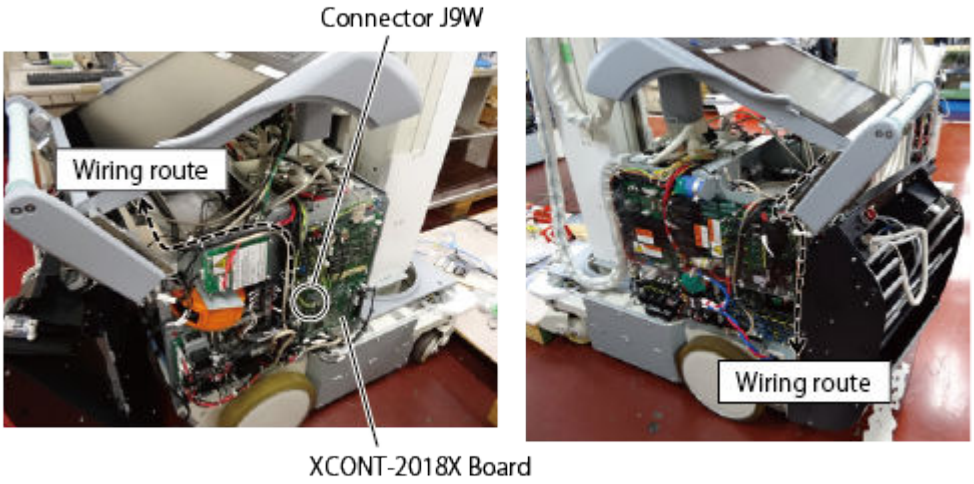


Fig. 5-25

6 **Connect the wireless hand switch cable and harness.**
The connected cable should be fixed with a cable tie and mount base, which is attached on the side of the FPD box.

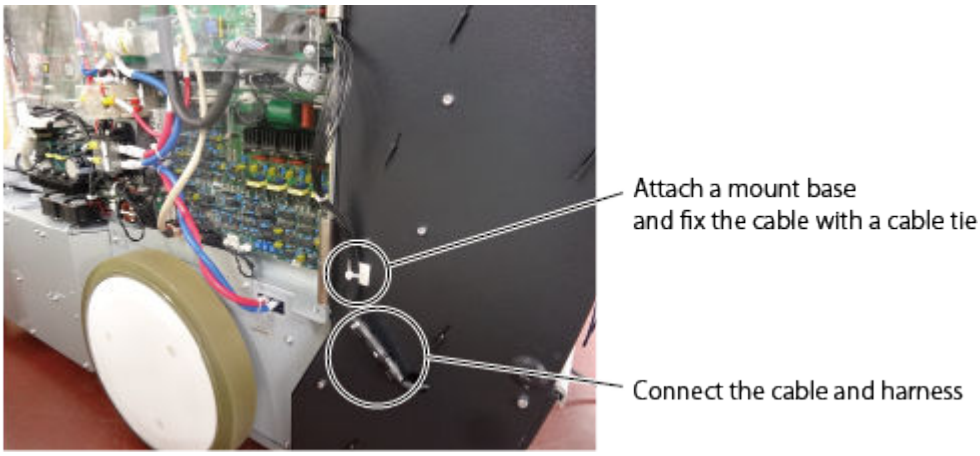


Fig. 5-26

7 **Attach the removed cover again.**

8 **Turn ON the main breaker.**

9 **Turn ON the key switch.**

10 **Check that the LED1 of wireless holder turns ON in green.**
Installation of the the wireless hand is complete. Go to the next step.

Reference "5.1.6 Operation Check"

■ Mounting Wireless Hand Switch for Cassette Box

1 **Turn OFF the main Breaker.**

2 **Remove the inside cover, both side covers, and top cover from the system.**

Reference "3.5 How to Open/Close Covers"

3 **Remove the frame cover.**

"4.3.2 Assembly of Cassette Box"

4 **Mount the wireless holder to the frame cover with two phillips-flat-head screws M3x16.**

Fig. 5-27

5 **Mount the frame cover so that the cable passes through the gap shown in the figure below.**

Fig. 5-28

6 **Connect and wire the harness to J9W on the XCONT-2018 board.**

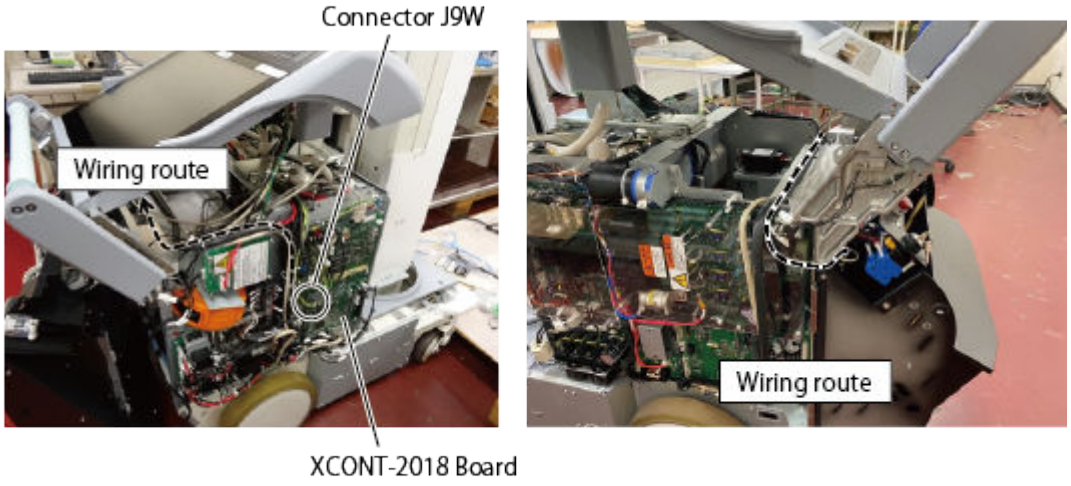


Fig. 5-29

7 **Connect the wireless hand switch cable and harness.**

Attach the mount base to the back of the cassette box and secure the connected cable with the cable tie.

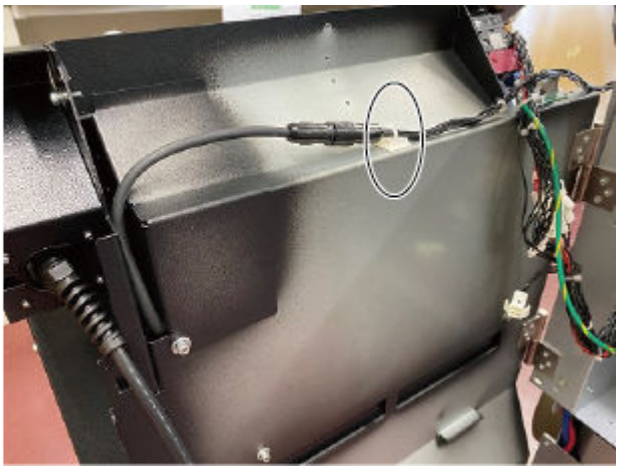


Fig. 5-30

8 **Attach the removed cover again.**

9 **Turn ON the main breaker.**

10 **Turn ON the key switch.**

11 **Check that the LED1 of wireless holder turns ON in green.**

Installation of the wireless hand switch is complete. Go to the next step.

Reference "5.1.6 Operation Check"

5.1.6 Operation Check

1. **Turn OFF and then ON the key switch and restart the system.**
2. **Check that only LED1 is lighted when the hand switch is set to the holder, and LED1, LED3, and LED5 are lighted when the hand switch is removed from the holder.**
3. **Check that LED6 is lighted after READY operation and LED7 is lighted after X-RAY operation.**
4. **Check that the collimator lamp is lighted by pressing SW3 on the wireless hand switch.**

5.1.7 How to Replace the Wireless Hand Switch Battery

■ How to Open/Close the Battery Cover

1. **Rotate the end cap 90 degree in a counterclockwise direction. (1)**
2. **Pull out the end cap. (2)**
3. **Slide battery cover to the end cap direction. (3)**
4. **Pull up the cover and remove. (4)**

5 **Reverse the above procedure to close the cover.**

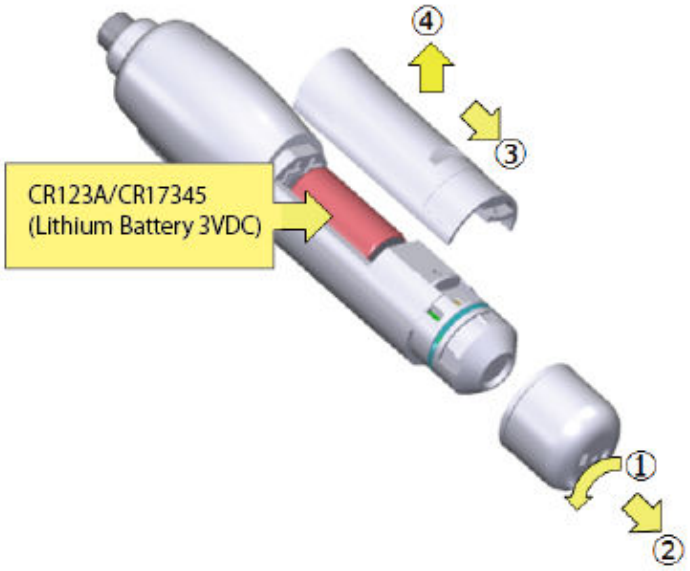


Fig. 5-31

NOTE

- Do not place the battery in wrong direction.
- The specified lithium primary battery, CR123A/CR17345 (nominal voltage: 3VDC), must be used.

5.1.8 How to Delete Pairing Information for Wireless Hand Switch

Delete the pairing setting of wireless hand switch and wireless holder.

1 **Set wireless hand switch to the wireless holder. LED1 turns ON when the power is applied to the holder.**

2 **Keep pressing function switch1 (F1) of the holder until LEDs from LED2 to LED6 start blinking. (More than 5 seconds)**

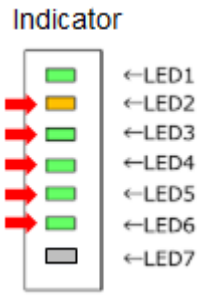


Fig. 5-32

3 **Take out wireless hand switch from the holder, and press and hold the paring switch of wireless hand switch for more than 10 seconds.**

After 2 seconds since the pressing of the paring switch, RING LED of the wireless hand switch blinks in yellow.
After 10 seconds, the deletion of paring setting is done, and the LED blinks in green.

Fig. 5-33

4 **Press function switch1 (F1) for more than 5 seconds.**

5 **Only LED1 is lighted and setting completed.**

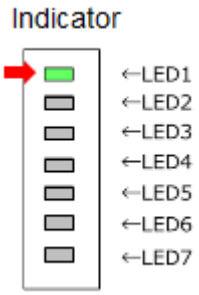


Fig. 5-34

5.2 Additional Hand Switch

5.2.1 Component of Additional Hand Switch

Table 5-4 Component of Additional Hand Switch

No.	Item	Qty.
1	Illuminated Hand Switch, C2U-23	1
2	Hand SW I/F board ASSY	1
3	Holder mount	1
4	Hand SW cable, J9B	1
5	Spacer	1
6	Cabletie	1
7	Flat head screw M3×8	2
8	Flange nut M4	2

5.2.2 IInstallation of Additional Hand Switch

1 **Turn OFF the main breaker.**

2 **Remove the inside cover, both side covers, top cover, and front cover from the system.**

Reference "3.5 How to Open/Close Covers"

3 **Remove the cover.**

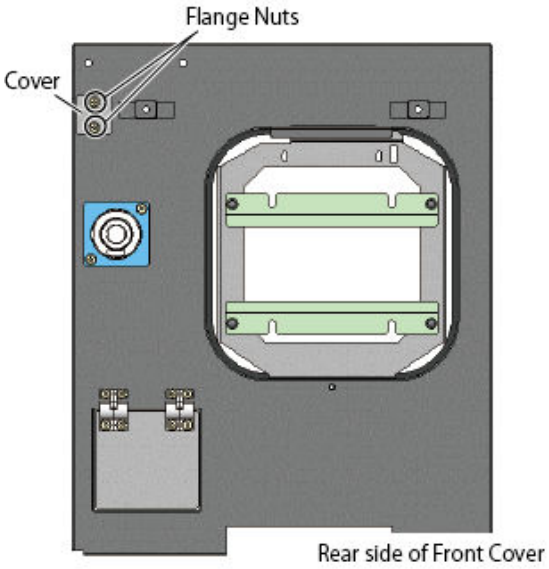


Fig. 5-35

4 **Attach HAND SW I/F Board ASSY, spacer, and holder attachment plate to the front cover with flange nuts removed in step 3 and accessory flange nuts. The hand switch cable should be passed through a hole of attachment plate.**

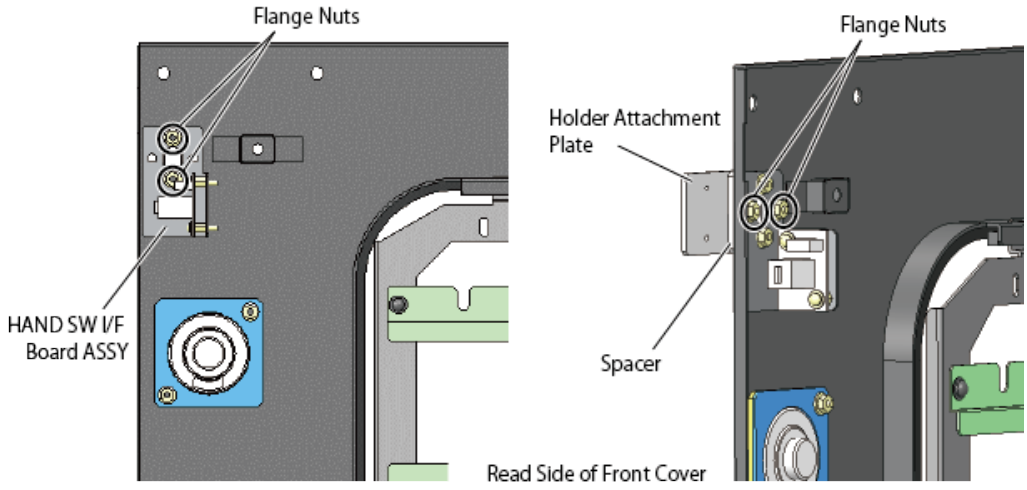


Fig. 5-36

5 **Fix the hand switch holder to attachment plate with attached Phillips-head screw.**

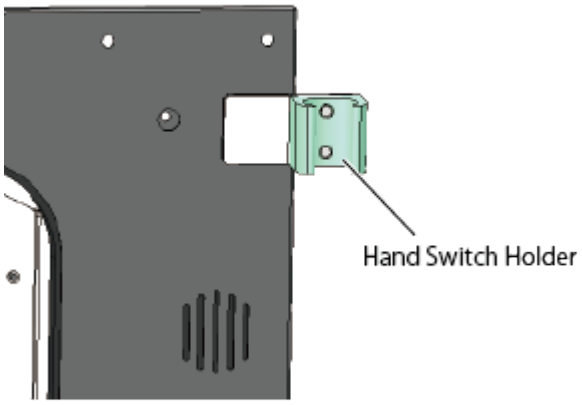


Fig. 5-37

6 **Insert the hand switch connector to HAND SW I/F board and fix it with a cable tie.**

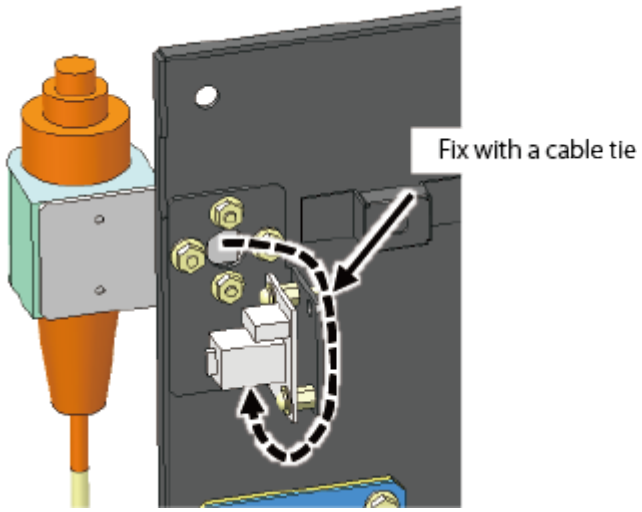


Fig. 5-38

7 **Connect to wire the hand SW cable J9B between J9AB connector on Hand SW I/F board and J9B connector on the XCONT-2018 board.**

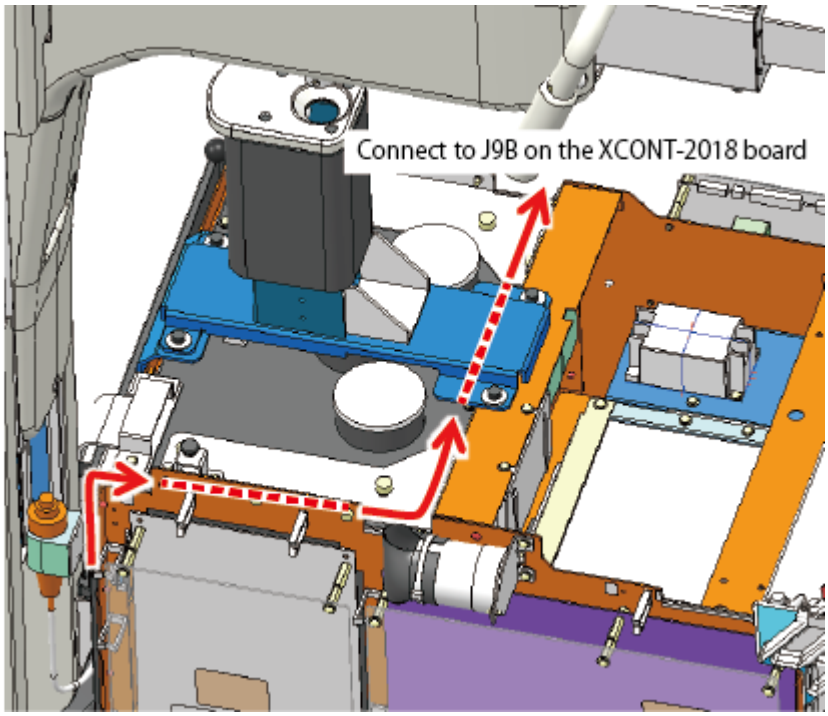


Fig. 5-39

8 **Attach the removed covers in reverse order. Turn ON the main breaker.**

9 **Check the performance.**

5.3 Luminous Hand Switch

5.3.1 Component of Luminous Hand Switch

Table 5-5 Component of Luminous Hand Switch

No.	Item	Qty.
1	Luminous Hand Switch	1

5.3.2 Installing the Luminous Hand Switch

NOTE The mounting position of the luminous hand switch depends on the FPD box/cassette box. Refer to the corresponding procedure for mounting.

Reference
- "Mounting Luminous Hand Switch for FPD BOX" P.183
- "Mounting Luminous Hand Switch for Cassette BOX" P.186

■ Mounting Luminous Hand Switch for FPD BOX

1 **Turn OFF the breaker. Remove the inside cover and both side covers from the system, and open the FPD box.**

2 **Cut the cable tie fixing the cable, and unplug the hand switch connector and disconnect the cable.**

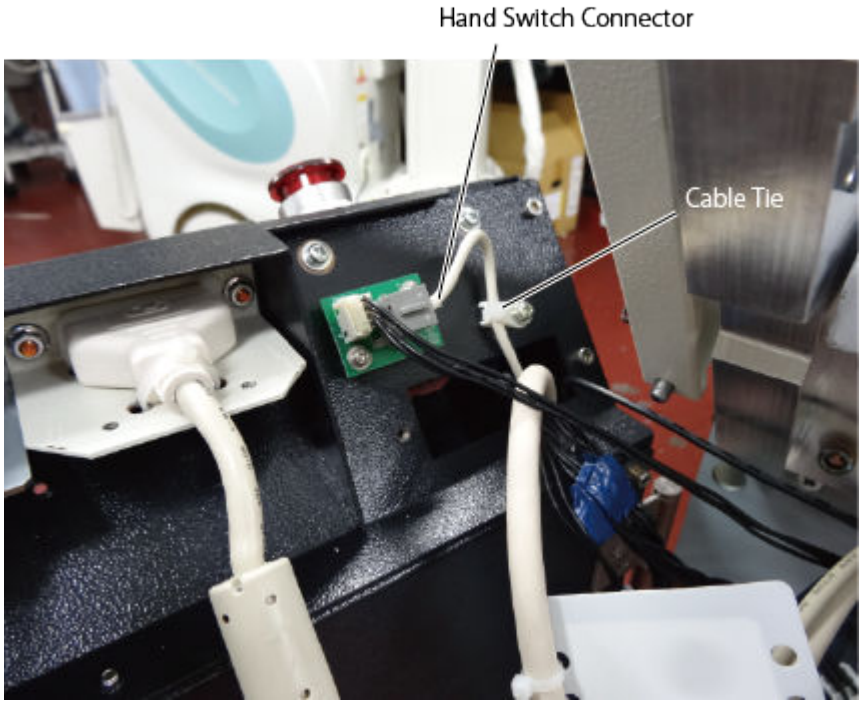


Fig. 5-40

3 **Route the cable through the holes on the side of the FPD box and connect it to the connectors as shown in the figure. And secure the hand switch cable with a cable tie to prevent the connector from coming off.**

Fig. 5-41

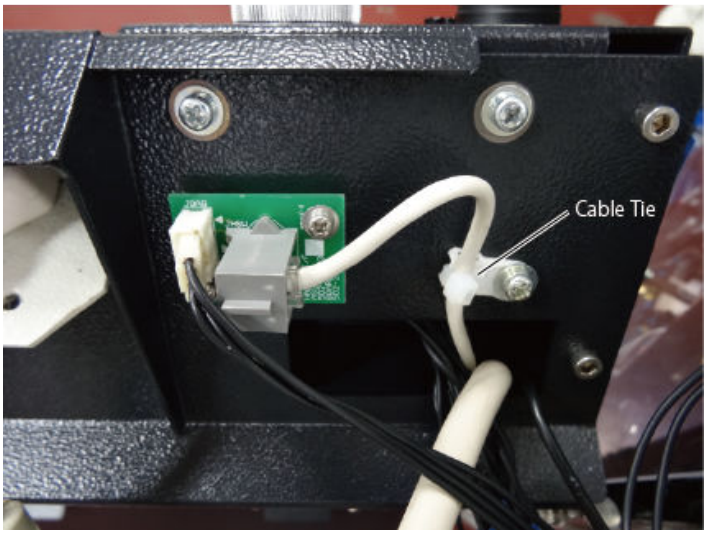


Fig. 5-42

4 **Adjust so that the length of hand switch cable coming out from the hole is about 600 mm as shown in the figure, and fix it with a cable tie.**

Fig. 5-43

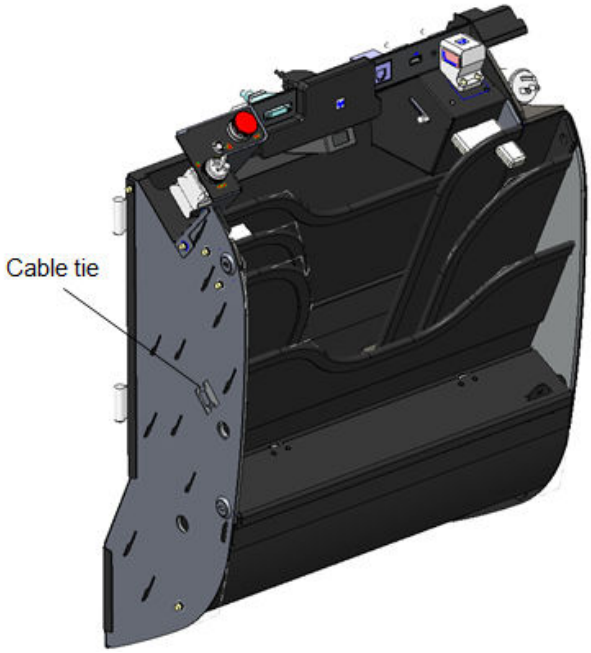


Fig. 5-44

5 **Check the operation.**

Reference For details about the illumination, see the Operation Manual.

■ Mounting Luminous Hand Switch for Cassette BOX

1 Turn OFF the breaker. Remove the inside cover and both side covers from the system, and open the Cassette box.

2 Cut the cable tie fixing the cable, and unplug the hand switch connector and disconnect the cable.

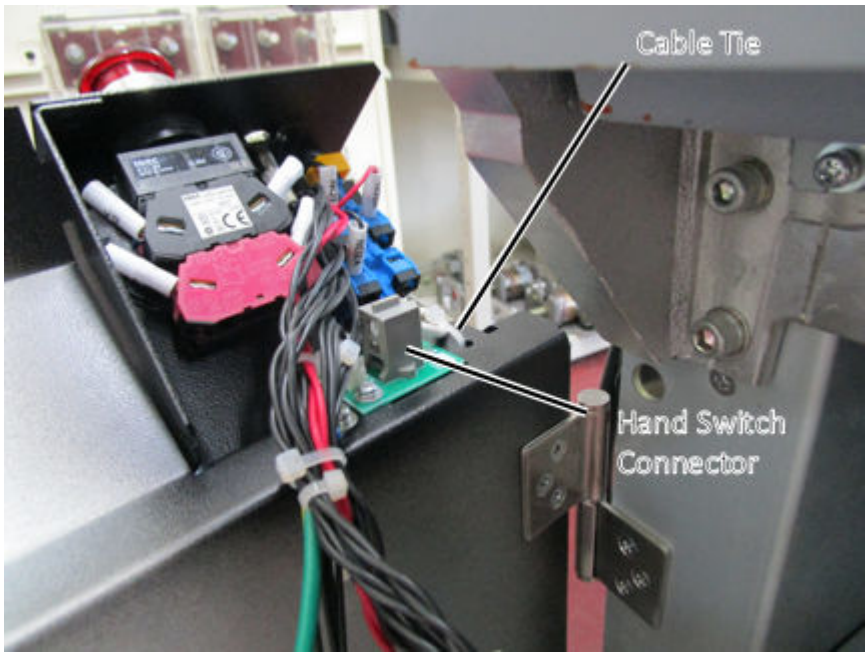


Fig. 5-45

3 **Route the luminous hand switch cable in the original path and connect it to the hand switch connector. And secure the cable with a cable tie to prevent the connector from coming off.**

Fig. 5-46

4 **Check the operation.**

Reference For details about the illumination, see the Operation Manual.

5.4 Keyless Entry

5.4.1 Keyless entry configuration

Keyless entry is composed of the following parts. Check when unpacking.

Table 5-6 Component of Keyless entry

No.	Item	Qty.
1	Keyless switch	1
2	SW bracket	1

NOTE The SW bracket is not used for this system.

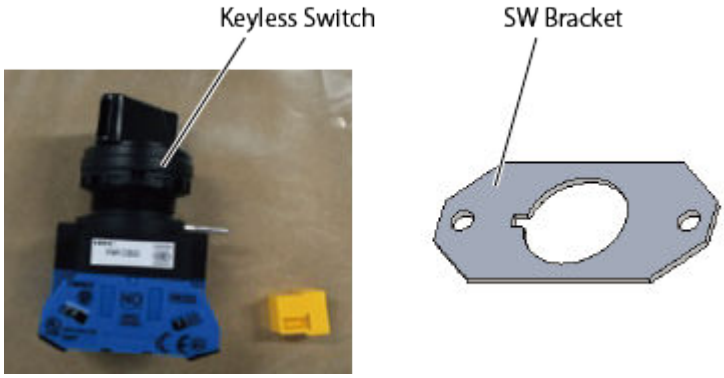


Fig. 5-47

5.4.2 Mount Keyless Entry Option

1 **Turn OFF the main breaker.**

2 **Remove the inside cover and both side covers from the system.**

Reference "3.5 How to Open/Close Covers"

3 **Remove two fixing bolts to open the FPD/Cassette box.**

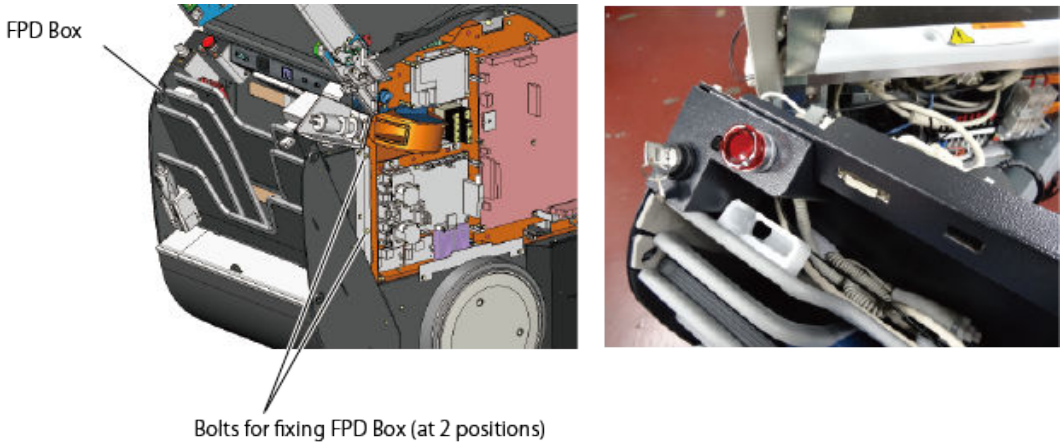


Fig. 5-48

4 **Disconnect all connectors from the SW BOX.**

5 **Unscrew a fixing screw on SW BOX, and pull out SW BOX upward.**

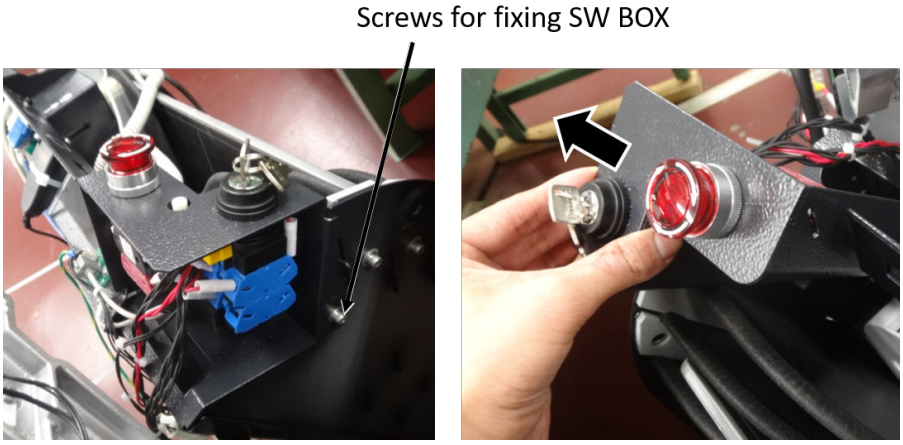


Fig. 5-49

6 **Remove the rotation-lock cover of the key switch, rotate the stopper lever to remove the contact unit of the key switch (at the connecting position).**

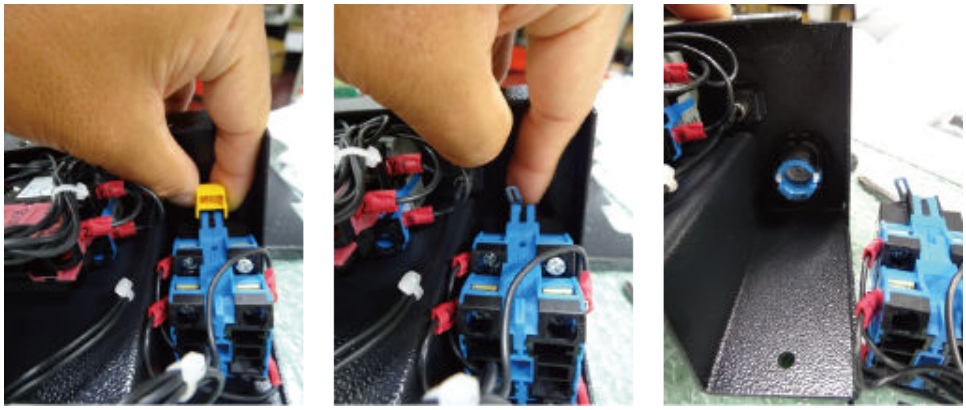


Fig. 5-50

7 **Loosen the nut on the back side, and remove the key switch from SW BOX. Also disassemble the rotation-lock ring from the nut.**

(The rotation-lock ring is reused for reassembly.)

Fig. 5-51

8 **Insert the keyless switch from the upper-side of SW BOX, and attach it with the rotation-lock ring and a resin nut.**

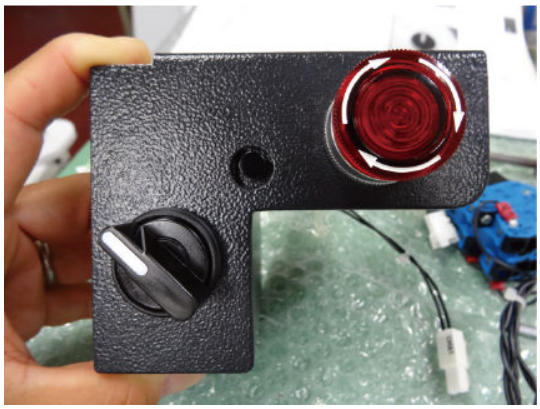


Fig. 5-52

NOTE Make sure the convex side of the rotation-lock ring is the direction of the fixing plate.

Fix the rotation-lock ring so that the convex part is firmly locked into the notch in the fixing plate, or otherwise the key switch may become loose.

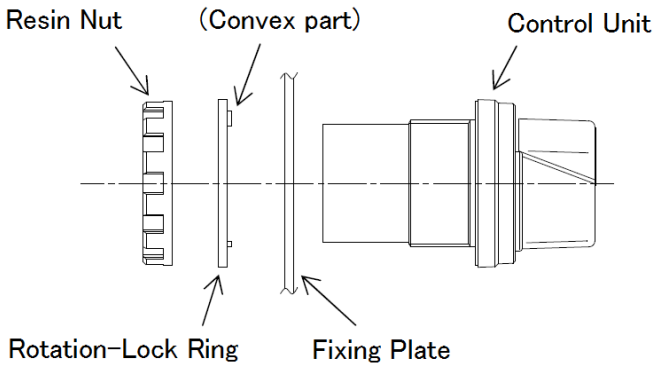


Fig. 5-53

9 **Reassemble the keyless switch in the reverse order of the step 8.**

Fig. 5-54

10 **Reassemble SW BOX to the FPD box.**

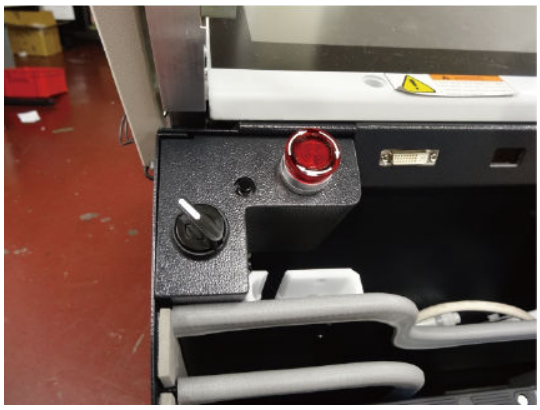


Fig. 5-55

11 **Restore the system in reverse order to the step 4.**

5.4.3 Setting of DIP switch

Turn ON the DIP switch SW3-2 on the XCONT-2018 board.

Reference "6 [Appendix] DIP Switches and Jumpers Setting on Each Board"

5.5 Changing the Grip Bar Height Kit

5.5.1 Component of Changing the Grip Bar Height Kit

The changing the grip bar height kit consists of the following parts. Make sure that all these parts are contained in the package.

Table 5-7 Component of Changing the Grip Bar Height Kit

番号	品名	数量
1	Handle arm L	1
2	Handle arm R	1
3	Cable guide	1
4	Harness bracket	1
5	Extension harness (CN11)	1
6	Ground cable (a) with a length of 120 mm	2
7	Ground cable (b) with a length of 230 mm	1
8	Cushion tape	1
9	Cable tie, Fasteners	1 set

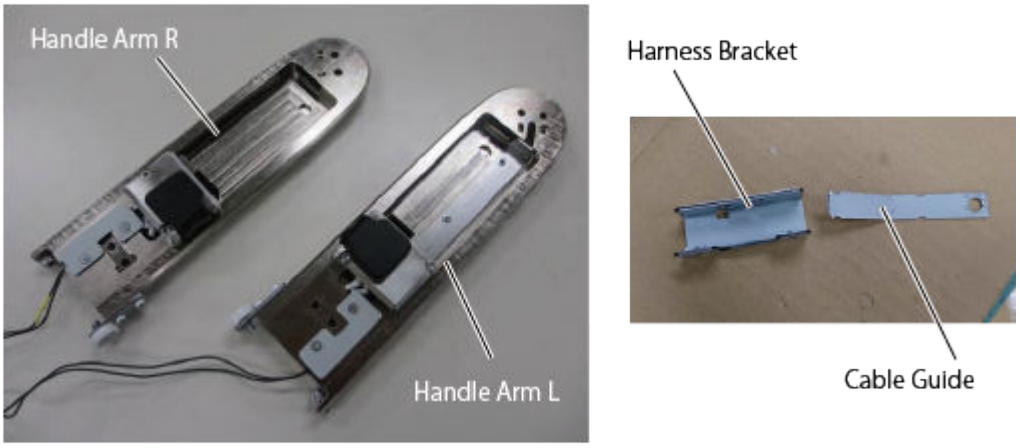


Fig. 5-56

5.5.2 Removing Normal Handle

1 **Turn OFF the power and the main breaker.**

2 **Remove the inside covers and side covers (right and left) from the system.**

Reference "3.5 How to Open/Close Covers"

3 **Remove the connector (CN11) and the ground cable from the right side of the handle, and also remove the cable tie.**

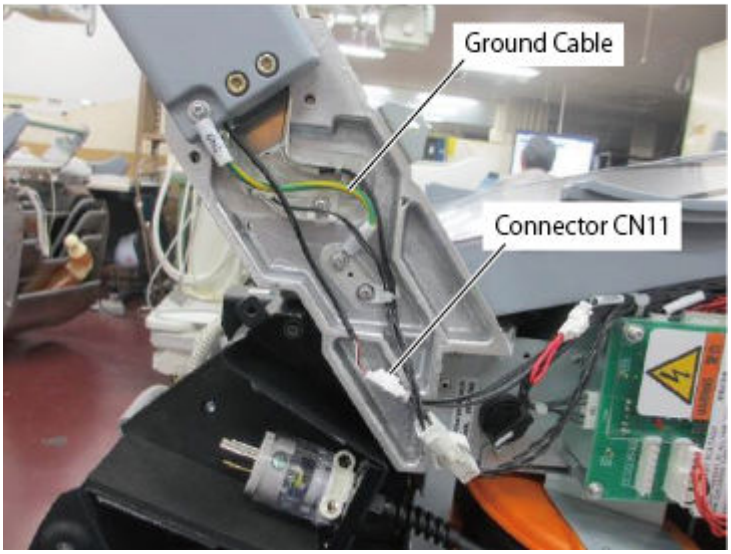
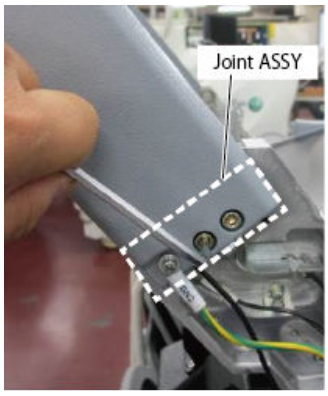
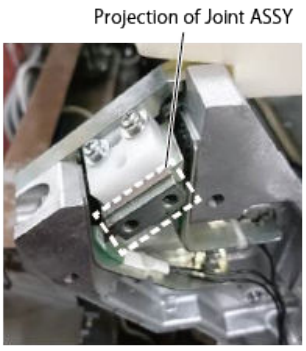


Fig. 5-57

4 **Remove two attachment bolts from each side of Joint ASSY fixing the handle arm, and remove the handle arm unit.**

> NOTE The handle arm is set in the projection of Joint ASSY. Pull both ends of the handle arm away from the projection to remove it.





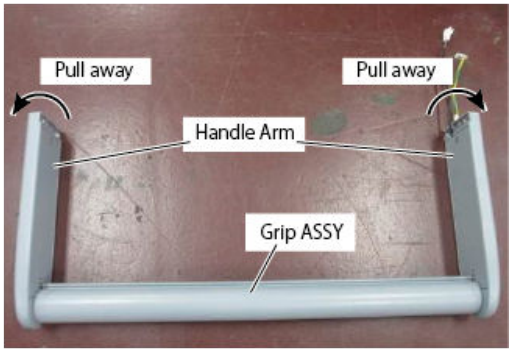


Fig. 5-58

5.5.3 Relocation of Grip ASSY

1 **Secure Grip ASSY with masking tapes as shown below.**

NOTE This step must be performed. If Grip ASSY turns inside out, the internal parts may spring out.

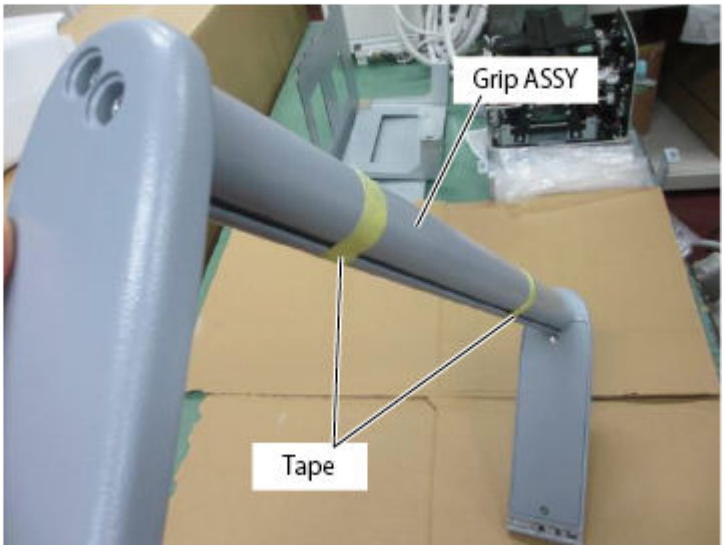


Fig. 5-59

2 **Remove the handle cover from the cable-wired side (right side) of the handle cover.**

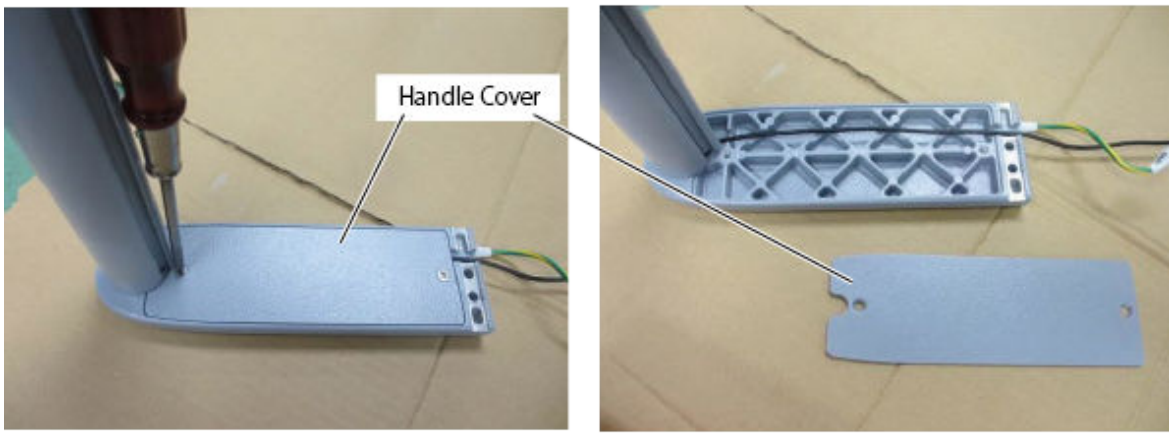


Fig. 5-60

3 **Remove two fixing bolt on each side of Grip ASSY, and disassemble the handle arm from Grip ASSY.**

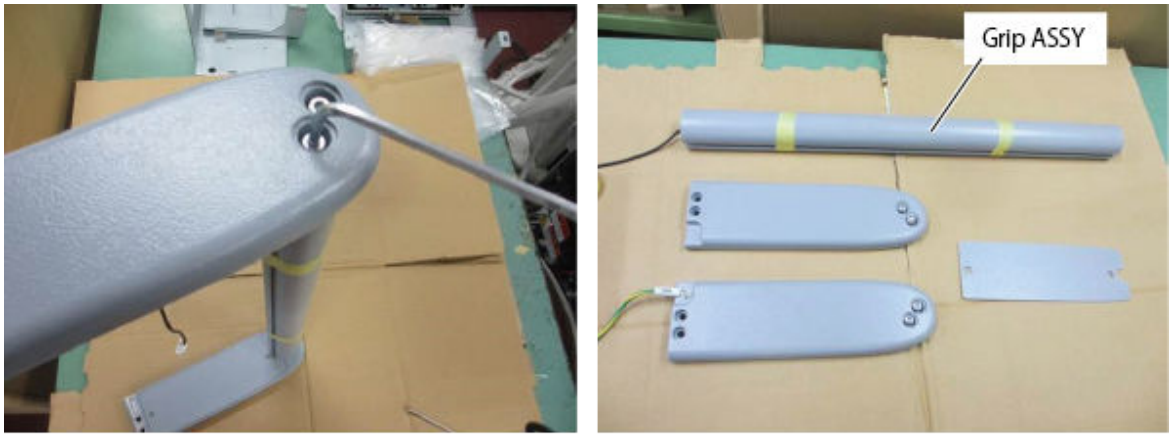


Fig. 5-61

4 **Remove the stopper bolt from the Handle arm R included in the option and loosen the bolt on the bearing attachment plate, and pull out Adjuster ASSY by holding down the black button.**

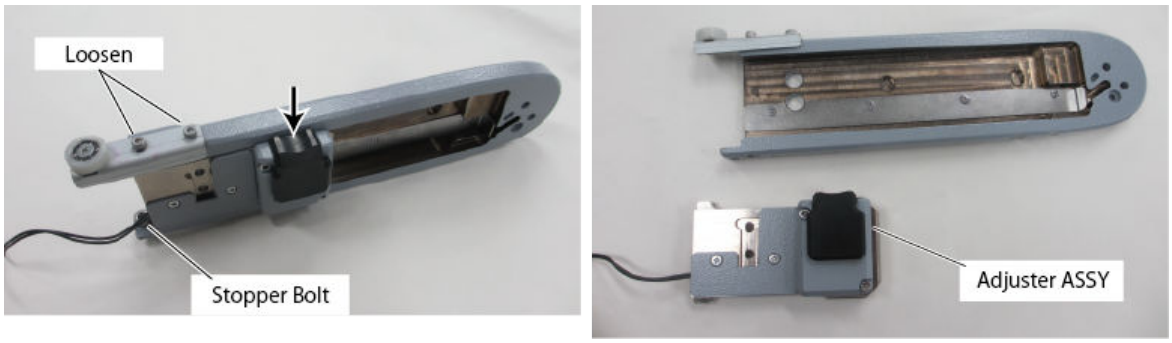


Fig. 5-62

5 **Remove the cable cover.**

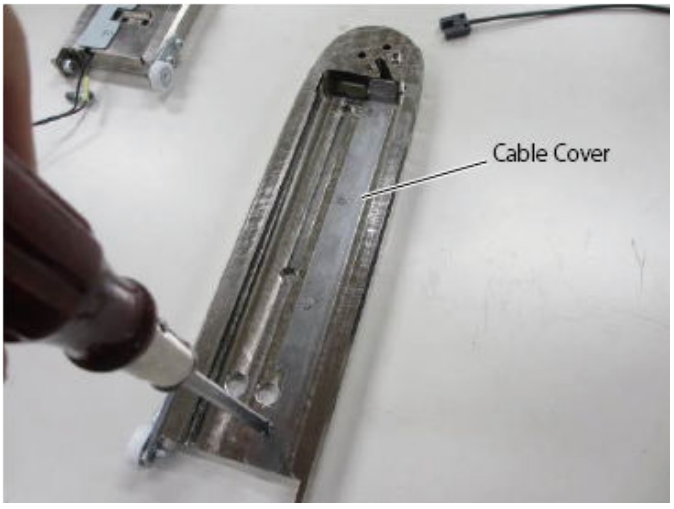
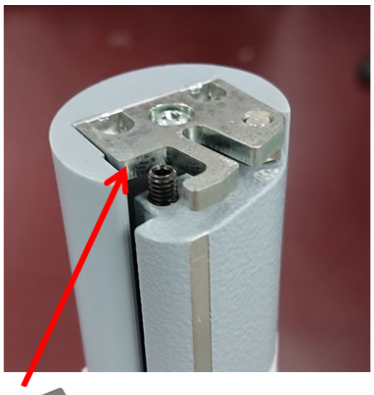


Fig. 5-63

6 Check the end of handle grip ASSY.

1 Use the bolts M5x20, which is provided with the grip bar height kit option.



Type with a supporting plate

Fig. 5-64

7 Fix the handle arm and Grip ASSY (with loctite 242 blue at two positios on both handle arms). After fixing, remove masking tapes andmake sure the brake bar can work smoothly.

NOTE

- If the bolt to fix the handle arm and the handle grip ASSY is mistaken, the bolt may loose. Be sure to check the type of the handle grip ASSY.
- Prevent the cable from being wedged between the handle arm and Grip ASSY.
- Fix the bolts at a level place such as a table to prevent the handle arms from being twisted.
- Turn the Grip ASSY in the direction shown below to expand the lever opening.

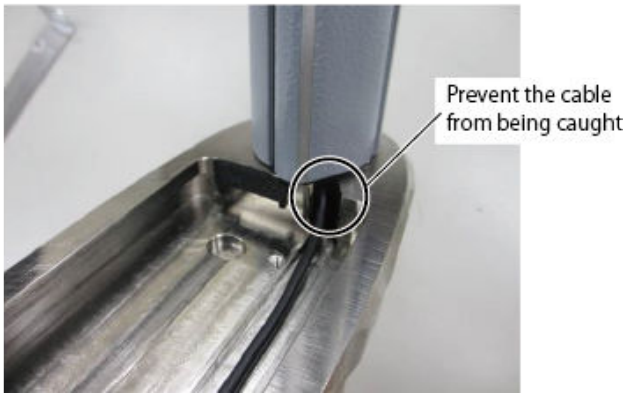


Fig. 5-65

8 **Pass the cable (CN11) from Grip ASSY through the groove of the handle arm, and attach the cable cover again carefully not to pinch the cable.**

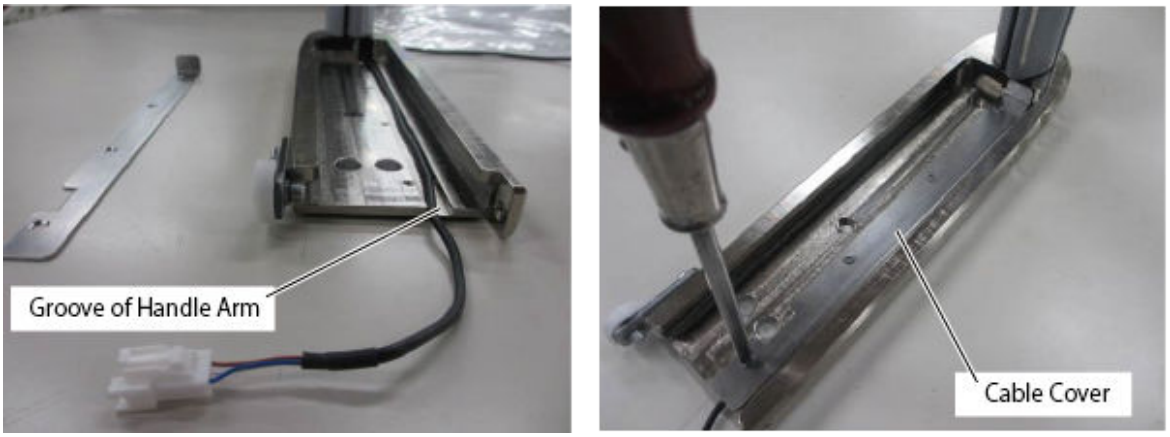


Fig. 5-66

9 **Insert Adjuster ASSY removed in Step 4 to the lock position.**

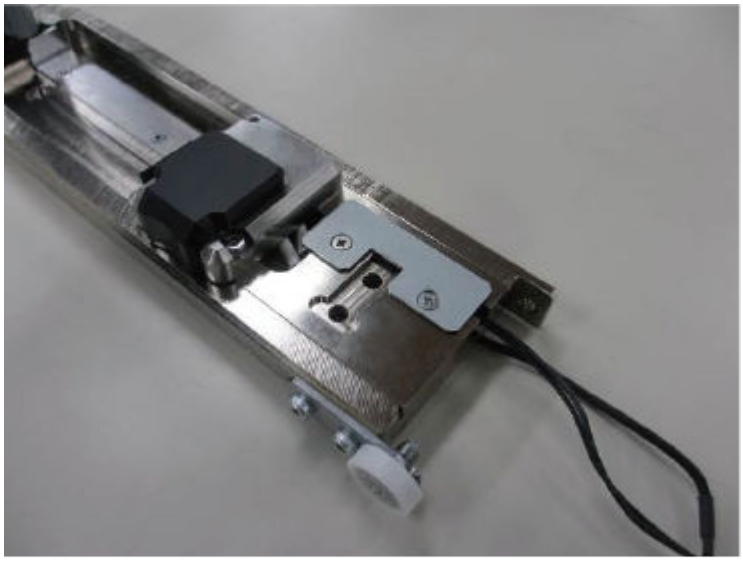


Fig. 5-67

10 **Jointly fasten the stopper bolt removed in Step 4 and the accessory cable guide. And attach the ground cable as shown below.**

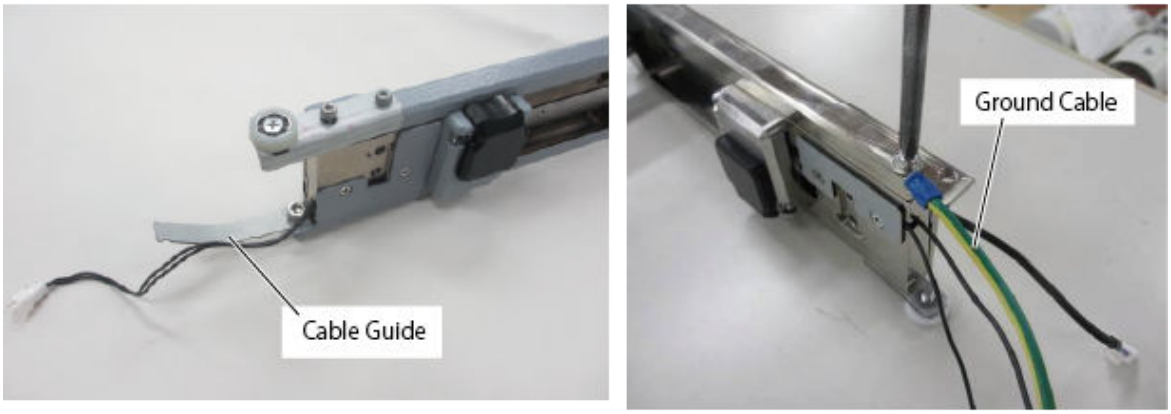


Fig. 5-68

11 **Remove the masking tape to complete the assembly of handle arm unit.**

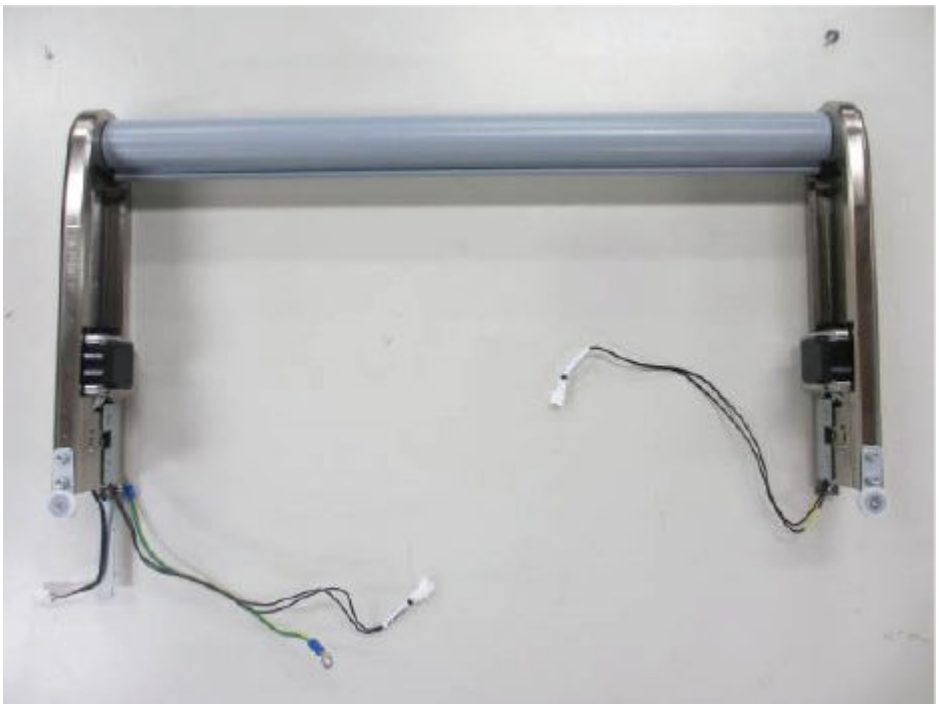


Fig. 5-69

5.5.4 Mounting Handle Arm Unit to Mobile X-ray System

1 **Attach the accessory ground cable (a) and the ground icon label to the bottom of Joint ASSY on each side of the system (with SEMS M3x6). The mounting direction of the ground cable (a) terminal should be as shown below.**

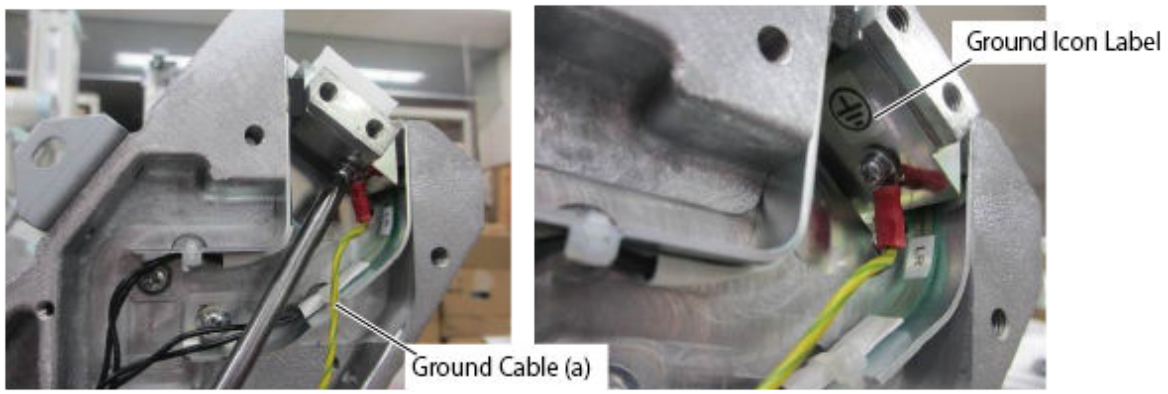


Fig. 5-70 Left side of Joint ASSY

2 **Mount the handle arm unit in Joint ASSY of the system, and attach two low head hexagon socket head cap screws M5X12 (included in the option) with loctite 242 (blue) to both side of the handle arm.**

> NOTE As same as the removal of the standard handle arm, pull both ends of the handle arm away to mount the groove to the projection.
>
> Reference "5.5.2 Removing Normal Handle", Step 4

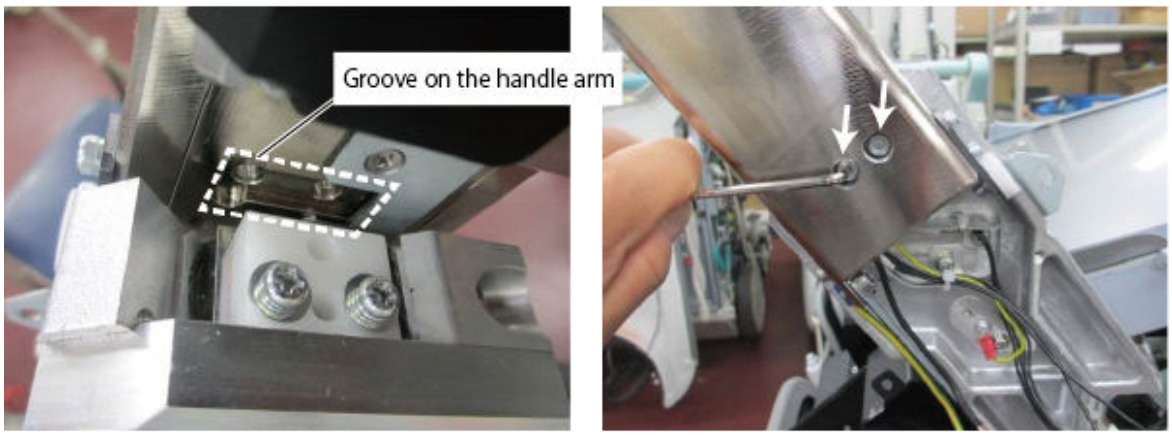


Fig. 5-71

3 **Connect the connector cable and ground cable on each side.**
Remove the existing short-circuit connector from both sides, and then connect each connector (CN38A/B),

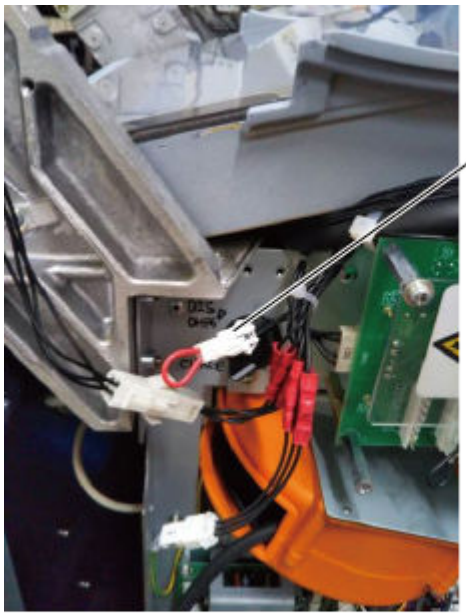


Fig. 5-72

4 **Fix the harness bracket (with SEMS M4x8, 2 places).**

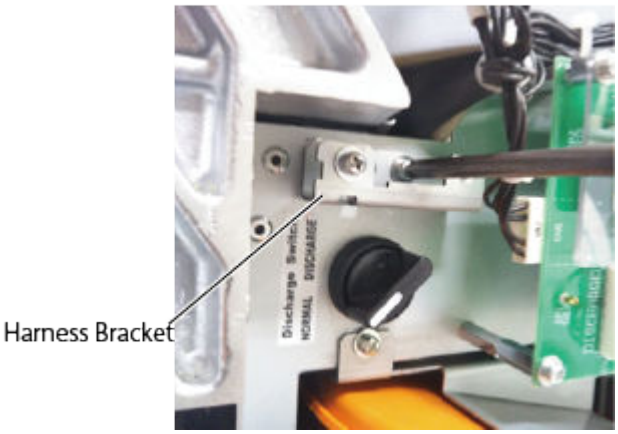


Fig. 5-73

5 **Connect the extention harness between connectors (CN11) of the handle arm and the system.**

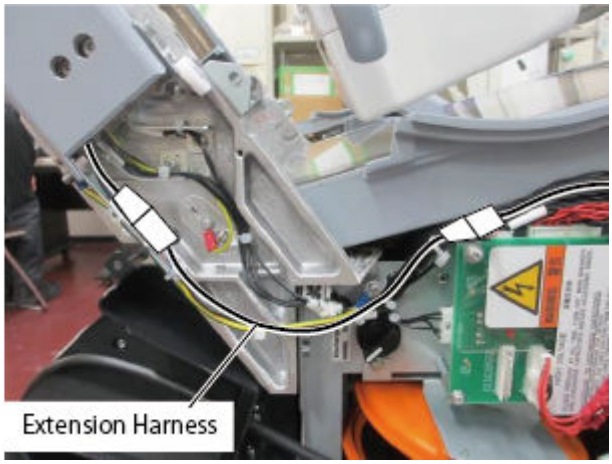


Fig. 5-74

5.5.5 Wiring

1 **Fix the groud cable (a), and attach the grounding symbol label.**

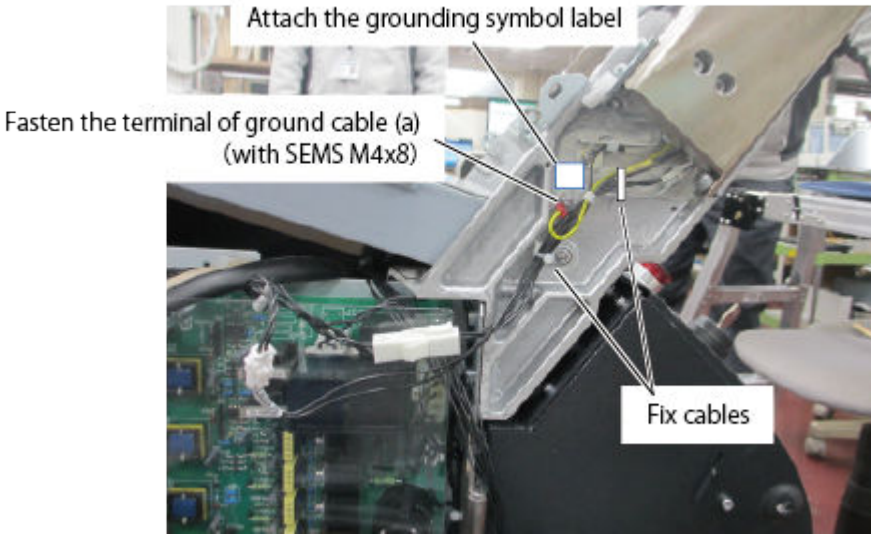


Fig. 5-75

2 **Bundle CN10 and CN38A cables and the ground cable (a), and fix them with cable ties. The connector of CN10 and CN38A is placed in the position as shown below.**

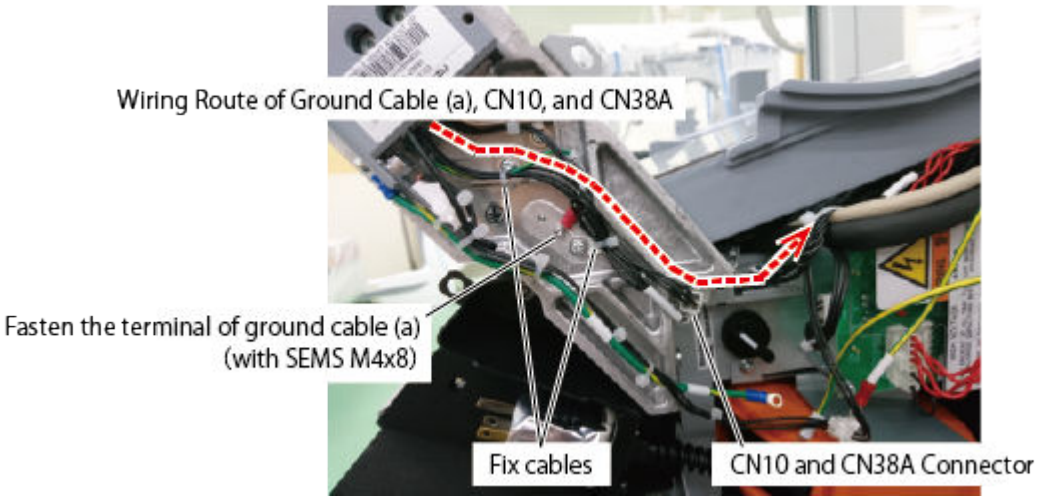


Fig. 5-76

3 **Fix the ground cable (b) and extension cable to the back and front of the cable guide. The connector CN11 of the extension harness is fixed onto the cable guide.**

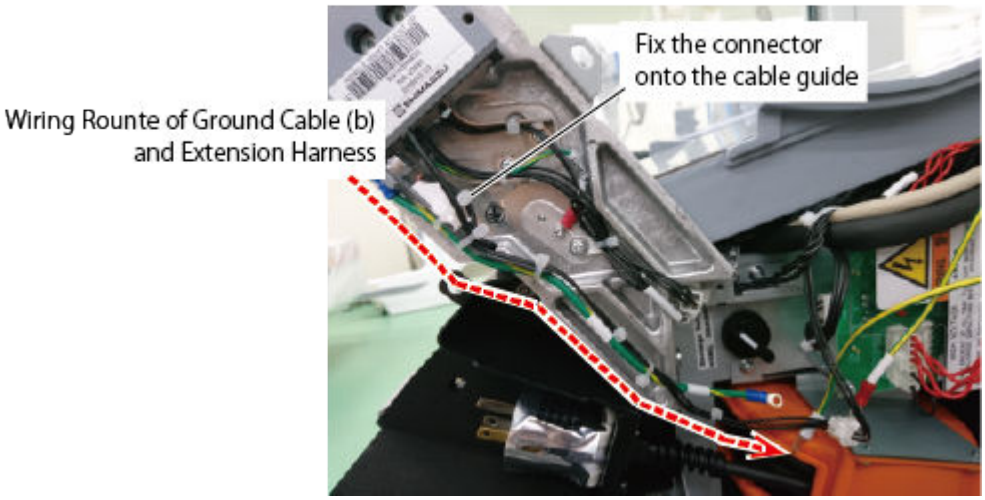


Fig. 5-77

4 **Fasten the terminal of the ground cable (b) to the harness bracket (with SEMS M4x8, 1 place), and fix the cable and the extension harness to the harness bracket.**

> NOTE Make sure that the ground cable is not pulled strongly. This may cause the device to turn left when releasing the brake.

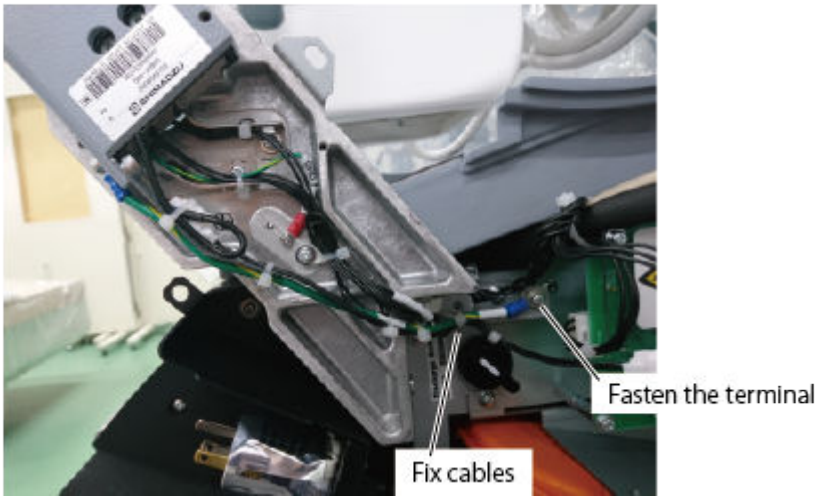


Fig. 5-78

5 **Bundle the extra length of the extension harness with a cable tie, and fasten the terminal of the ground cable to the harness brakcet.**

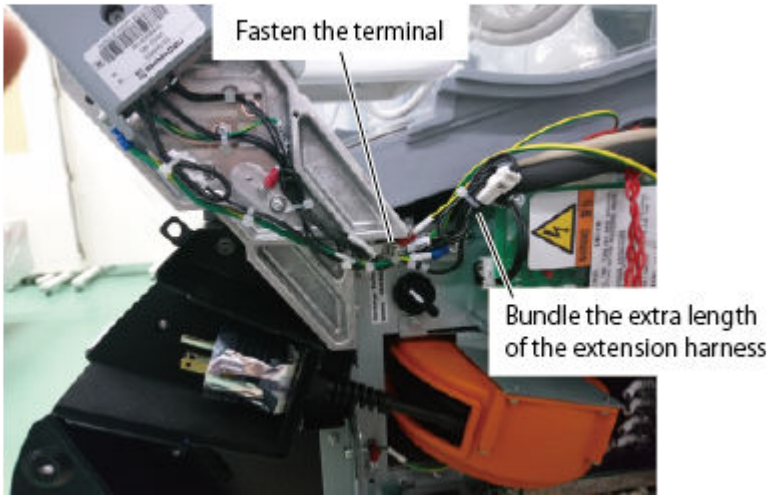


Fig. 5-79

6 **Move the handle to the lower position to make sure the ground cable (b) and extension harness do not contact the discharge switch and cord reel, and complete the wiring.**

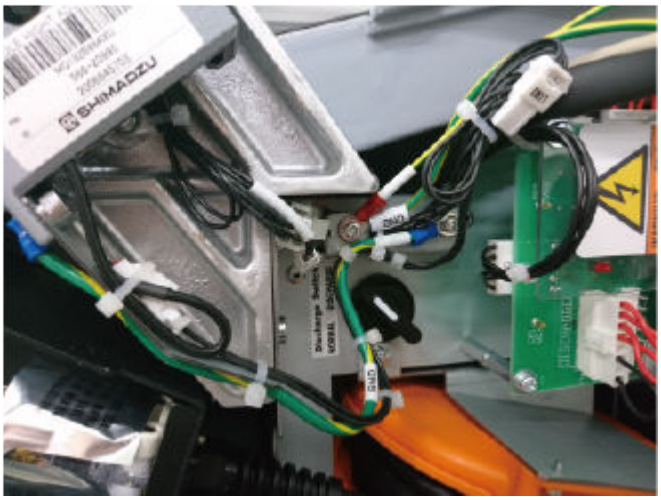


Fig. 5-80

5.5.6 Performance Check and Attachment of Cover

1 **While locking the handle at the high position, fix the fixing plate with a hex socket head cap so that the bearing lightly contacts the traveling rail.**

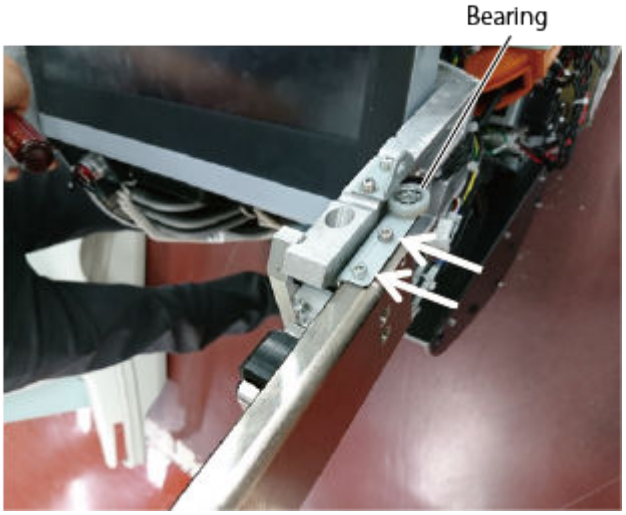


Fig. 5-81

2 **There are two set screws inside the screw hole. Contact the first screw toward the fixing plate, and insert the other screw into the screw hole and tighten it.**

Fig. 5-82

3 **Remove Handle lock ASSY on both sides with removing two hex head cap screws.**

> NOTE Be careful not to cut off the harness of Handle lock ASSY.

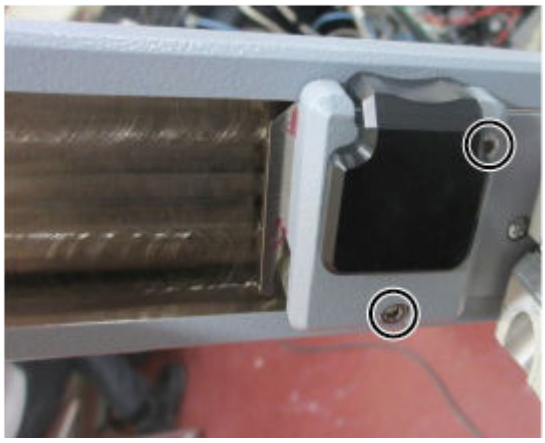


Fig. 5-83

4 **Check the gap on each side of the handle. The gap should be set to 0.2 mm to 0.5 mm on each side by adjusting the resin guide.**

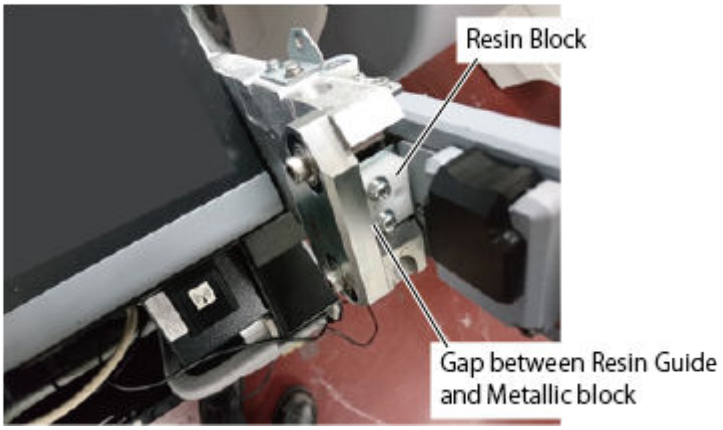


Fig. 5-84

5 **Move the arm up and down to check the following performances.**

- The arm can be locked at the default position and the low position.
- When the arm is unlocked at the low position, the arm is held at the same position and does not fall suddenly.
- The arm can work smoothly without catching a cable.

6 **Turn ON the main braker and the power supply, and release the brake while unlocking the arm to make sure D2 error is displayed.**

7 **Check if the version of the software is later than ver.1.50. Perform upgrade if neccessary.**

Reference • How to check the version: "7.1.2 Method 2"
• How to upgrade: "11.8 Updating Software"

8 **Turn OFF the main braker and the power supply.**

9 **Peel off the cushion tape, which is attched to both side covers, and attach the accessory cushion tape cut into 25 mm length as shown below.**

Fig. 5-85

10 **Attach the side covers. Check the gap between the side cover and the arm handle so that it does not scratch. Make sure the right side has more than 2 mm gap at the default handle position.**

> NOTE If the gap is small, adjust the position of the cover attachment stand.

Fig. 5-86

11 **Attach two caps on each side, and attach the handle inside cover.**

Fig. 5-87

12 **Adjust the FSR value at the default handle position.**

Reference "7.5 Adjustment of the Handle"

5.5.7 How to Adjust the Gripping Force of the Handle Arm

Whent the handle is unlocked at the default position, it is held at the same position by Roller ASSY, which is located on the leftside of the handle unit. The gripping force of the handle is already adjusted during the manufacturing process. However, if the handle arm descends too fast due to lack of force, or the lowering force becomes too large over time, follow the instruction below to adjust the gripping force.

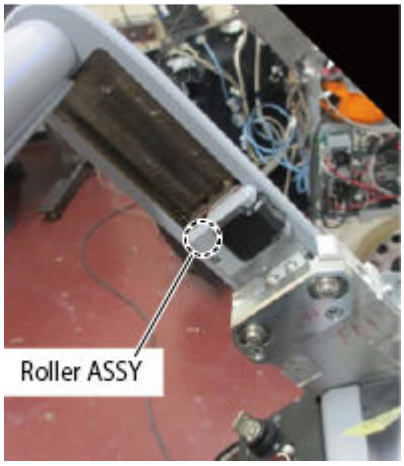


Fig. 5-88

1 **Lock the handle at the default position.**

2 **Remove Handle lock ASSY on the left side with removing two hex head cap screws.**

NOTE Be careful not to cut off the harness of Handle lock ASSY.

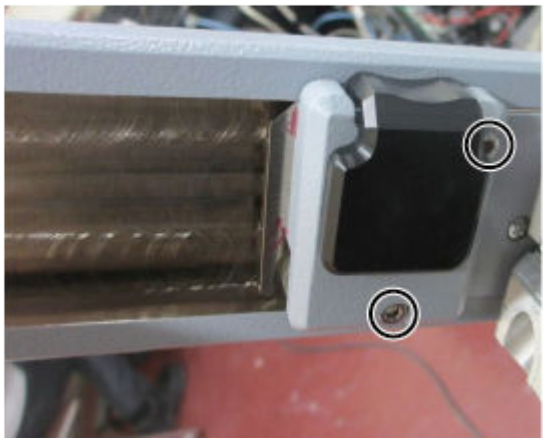


Fig. 5-89

3 **Adjust the U-nut mounted to Roller ASSY. The gripping force is increased by tighting the U-nut, and decreased by loosening it.**

NOTE
- After adjustment, make sure the roller rotates when moving the handle down. If the roller does not rotate, loosen the U-nut to decrease the holding force.
- The standard force to lower the handle arm when the Roller ASSY is rotating is approximately 10N.
- If the U-nut is too loose, the disc spring may be misaligned as shown below. In this case, align and hold the disc spring to the center by using a needle nose plier or an equivalent tool, and tighten the U-nut.

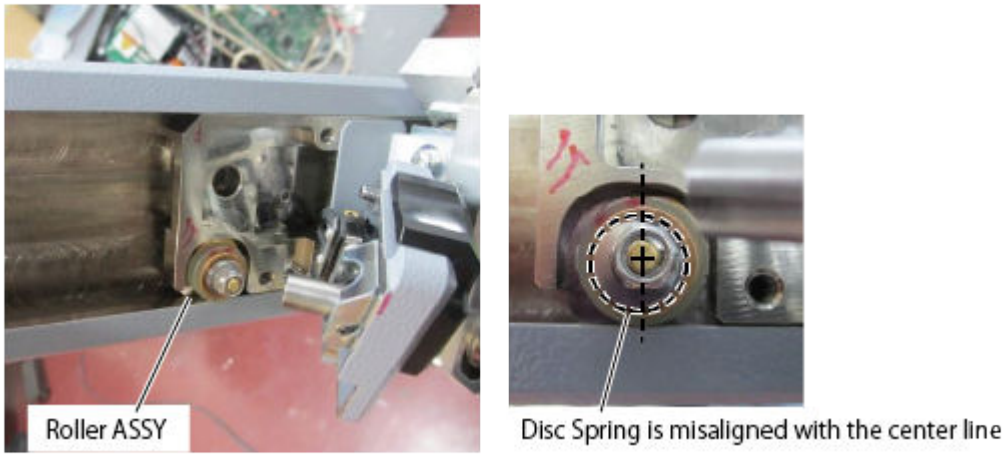


Fig. 5-90

5.5.8 Other Maintenance

Be sure to prevent grease and oil from adhering to the traveling rail for Roller ASSY. Otherwise, it will totally lose its grip force. In the case a grease adheres to the rail, dismantle and clean both Roller ASSY and travering rail.

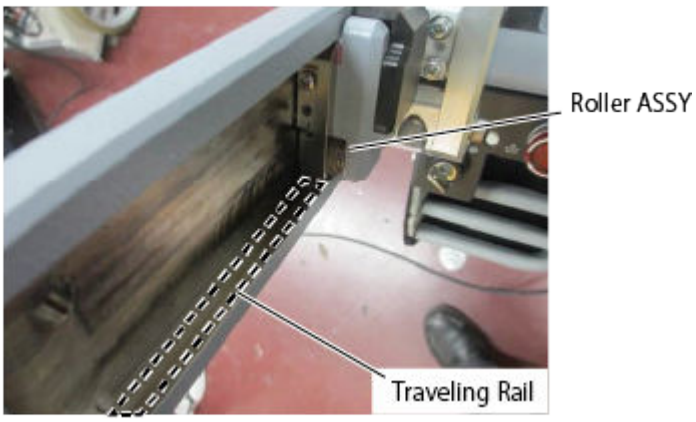


Fig. 5-91

5.6 Infrared Remote Controller Option

5.6.1 Component of Infrared Remote Controller Option

The infrared remote controller option consists of the following parts. Make sure that all these parts are contained in the package.

NOTE
- This option does not include a battery for the RC transmitter. Prepare it separately beforehand.
- The AA battery must be used.

Table 5-8 Component of Infrared Remote Controller Option

No	Parts Name	Quantity
1	RC transmitter	1
2	RC cable	1
3	Receiver ASSY	1
4	Catch cover ASSY	1

5.6.2 Attachment of Infrared Remote Controller Option

1 **Check if the version of the software is later than ver.1.50. Perform upgrade if neccessary.**

Reference How to check the version
- For MobileDaRt Evolution: "7.1.2 Method 2"
- For MobileArt Evolution: "8.1.2 Method 2"

How to upgrade: "11.8 Updating Software"

2 **Remove the top plate and the arm lock release lever, then remove the joint section cover and the spacer.**

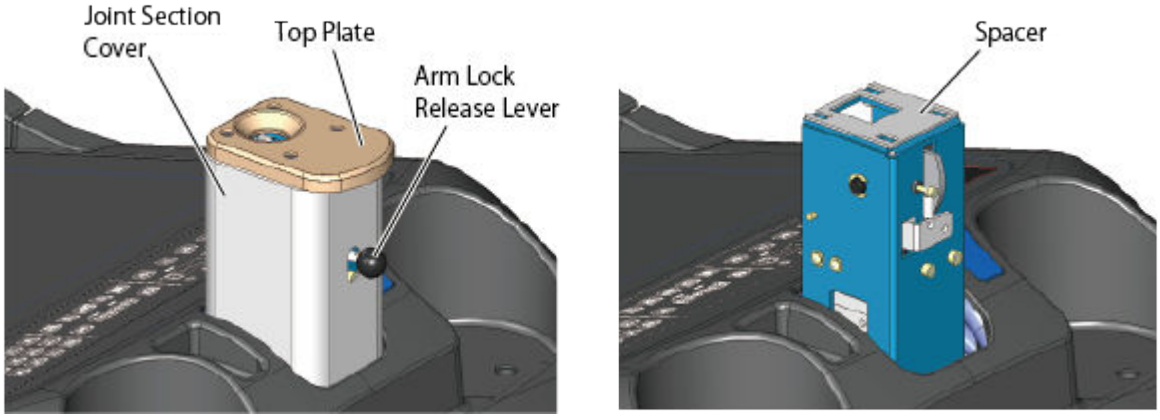


Fig. 5-92

3 Remove the inside cover, both side covers, and top cover from the system.

Reference "3.5 How to Open/Close Covers"

4 Connect the RC cable to JREM on the XCONT-2018 board, and then route it along other cables to the joint part.

Fig. 5-93

5 Turn ON DIP SW 3-3 on the XCONT-2018 board.

6 Attach the removed covers in reverse order of Step 2.

7 **Put the receiver ASSY on the joint part. Insert the RC cable into the connector on the receiver board and fix it with a cable tie.**

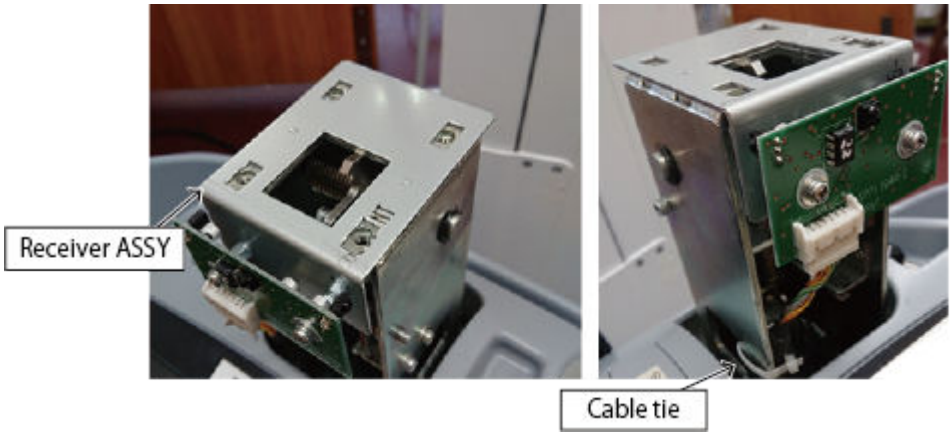


Fig. 5-94

8 **Attach the catch cover ASSY. Using a dowel of the receiver ASSY, move the position of the catch cover ASSY.**

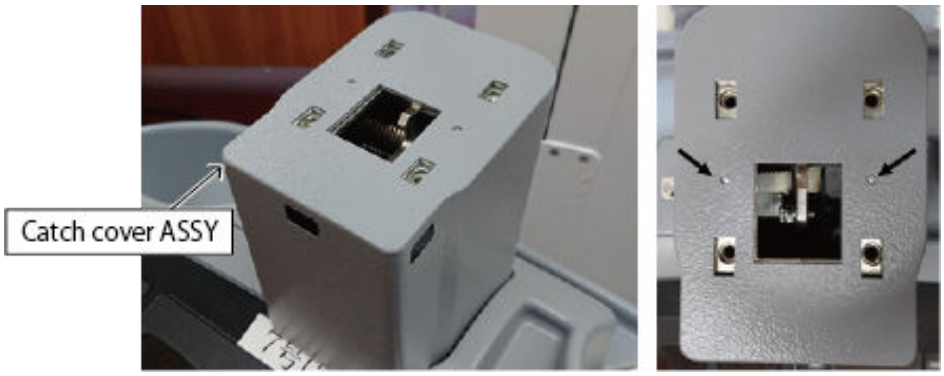


Fig. 5-95

9 **Attach the removed top plate and armlock release lever in reverse order of Step 2.**

10 **Put an AA battery in RC transmitter.**

NOTE • Do not place the battery in wrong direction.

11 **Perform operation check.**

Reference Refer to the operation manual for usage.

5.7 Dose Area Product Meter Option

5.7.1 Component of Dose Area Product Meter Option

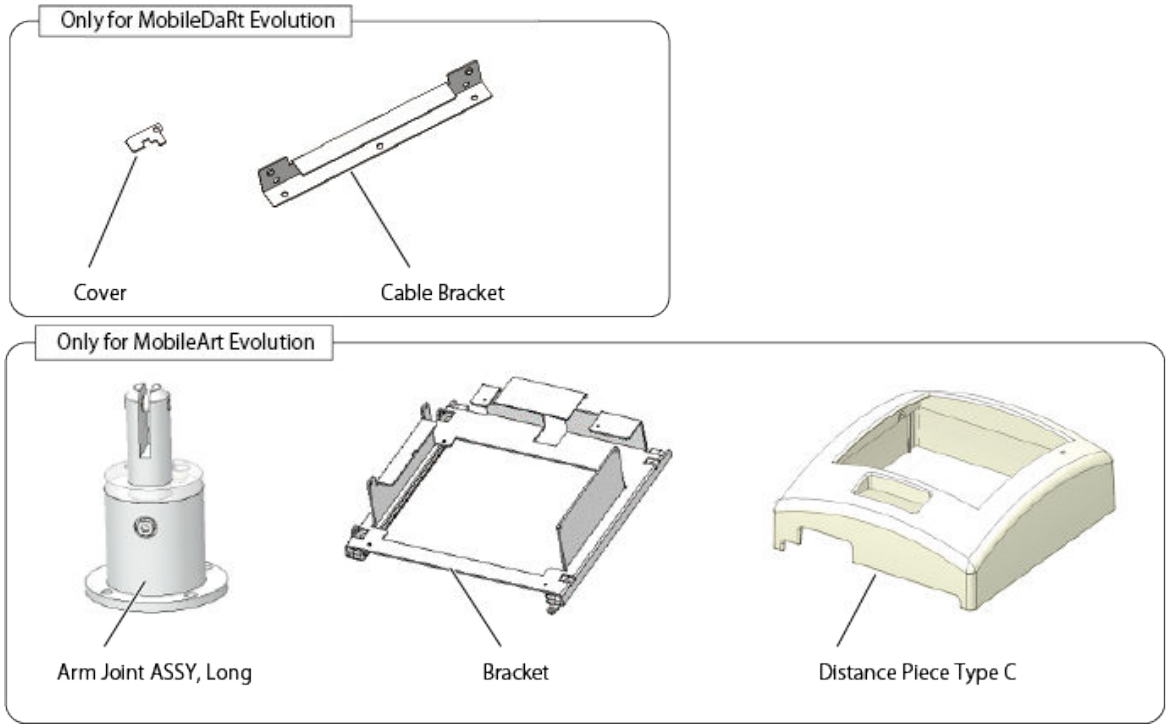
The Dose Area Product Meter Option consists of the following parts. Make sure that all these parts are contained in the package.

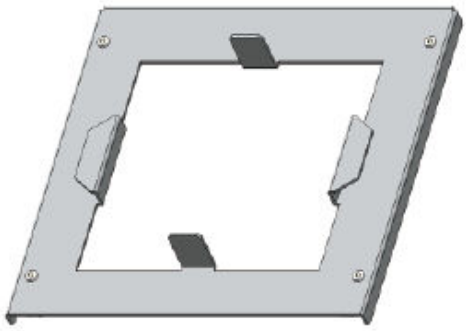
■ Dose Area Product Meter VACUDAP

Table 5-9 Component of Dose Area Product Meter VACUDAP

No	Parts Name		Quantity		Remarks
			MobileDaRt Evolution	MobileArt Evolution	
1	Cover (for DAP)		1	-	
2	Cable bracket		1	-	
3	DAP cable MUX A		1	1	
4	DAP cable MUX B		1	1	
5	Screw, sems M4X8		2	2	
6	Arm joint ASSY, Long		-	1	
7	Distance piece ASSY				
	8-1	Bracket	-	1	
	8-2	Distance piece	-	1	
	8-3	Hex socket head cap screw M3x5	-	1	
	8-4	Hex socket head cap screw M3x6	-	4	
8	Additional accessory rail		1	1	It depends on the shipping time. Not included in the transition period.

* The following items is not included. Please purchase these items separate from this option.
- Dose Area Product Meter VacuDAP 1580015





Additional Accessory Rail

Fig. 5-96

■ Dose Area Product Meter Producted by Shimadzu

The Dose Area Product Meter Option consists of the following parts. Make sure that all these parts are contained in the package.

Table 5-10 Component of Dose Area Product Meter Producted by Shimadzu

No	Parts Name		Quantity	
			MobileDaRt Evolution	MobileArt Evolution
1	DAP METER		1	1
2	DAP cable MUX B		1	1
3	Screw, sems M4X8		2	2
4	TAPPING Screw, SST FLAT 2X6		4	4
5	Arm joint ASSY, Long		-	1
6	Distance piece ASSY			
	6-1	Bracket	-	1
	6-2	Distance piece	-	1
	6-3	Hex socket head cap screw M3x5	-	1
	6-4	Hex socket head cap screw M3x6	-	4

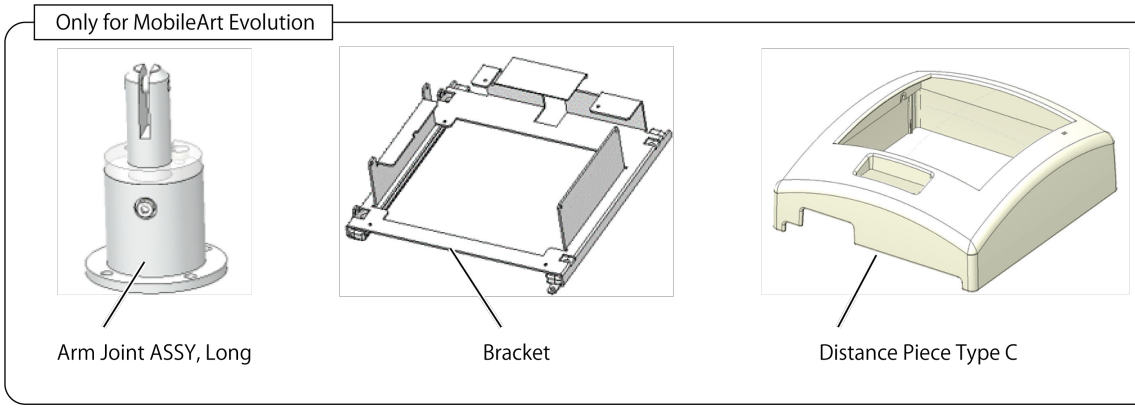


Fig. 5-97

5.7.2 Installing Dose Area Product Meter VACUDAP Option

NOTE The installation of the dose area product meter depends on the MobileDaRt Evolution/MobileArt Evolution. Refer to the corresponding procedure for installation.

Reference
- "Installing Dose Area Product Meter Option for MobileDaRt Evolution" P.219
- "Installing Dose Area Product Meter Option for MobileArt Evolution" P.223

■ Installing Dose Area Product Meter Option for MobileDaRt Evolution

1 **Turn OFF the main breaker.**

2 **Rmove the ditance piece.**

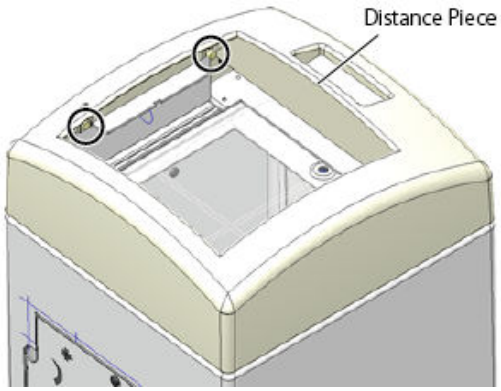


Fig. 5-98

3 **Remove the bracket and the rail.**

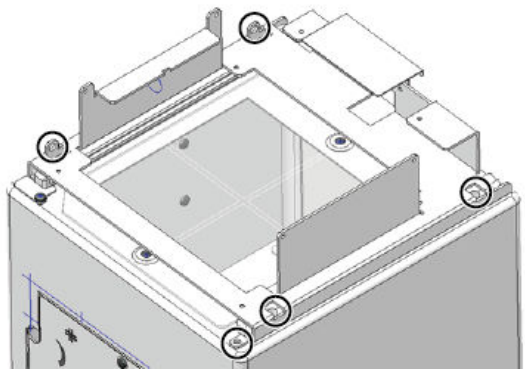


Fig. 5-99

4 **Remove the collimator cover, the rear cover, and the cover.**

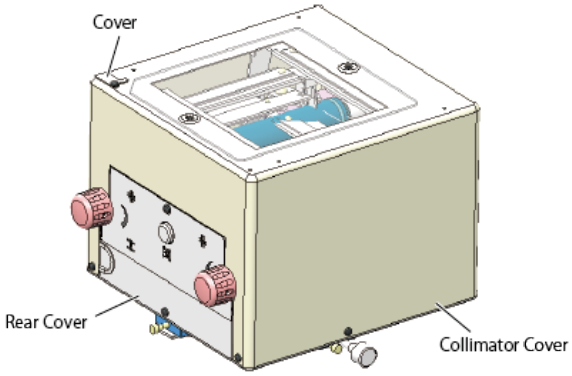


Fig. 5-100

5 **Connect and wire DAP cable MUX B to J2 connector on CL MEAS-17 board.**

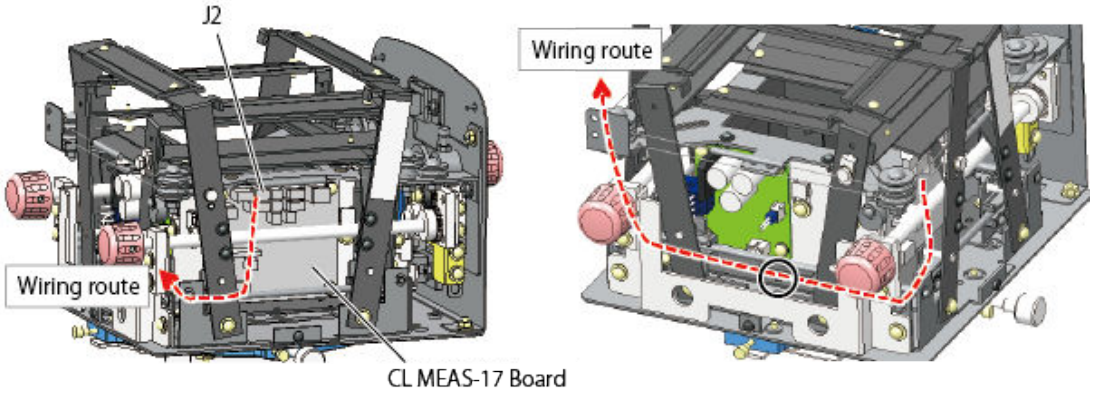


Fig. 5-101

6 **Attach the rear cover, pass DAP cable MUX B through the notch on the collimator cover, and attach the collimator cover.**

Do not attach the cover and the cover (for DAP). They are not needed.

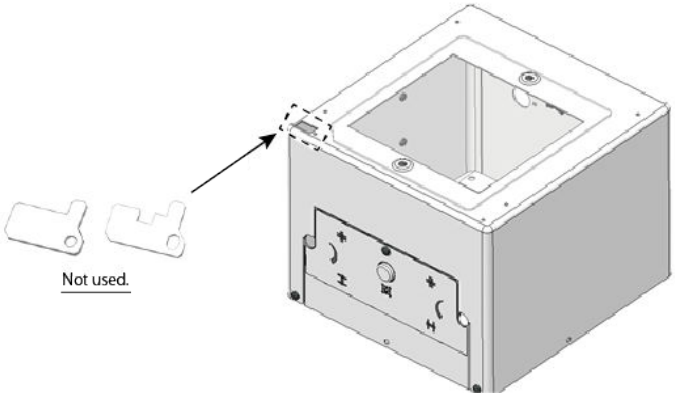


Fig. 5-102

7 **Remove two plates fixed in Dose Area Product Meter.**

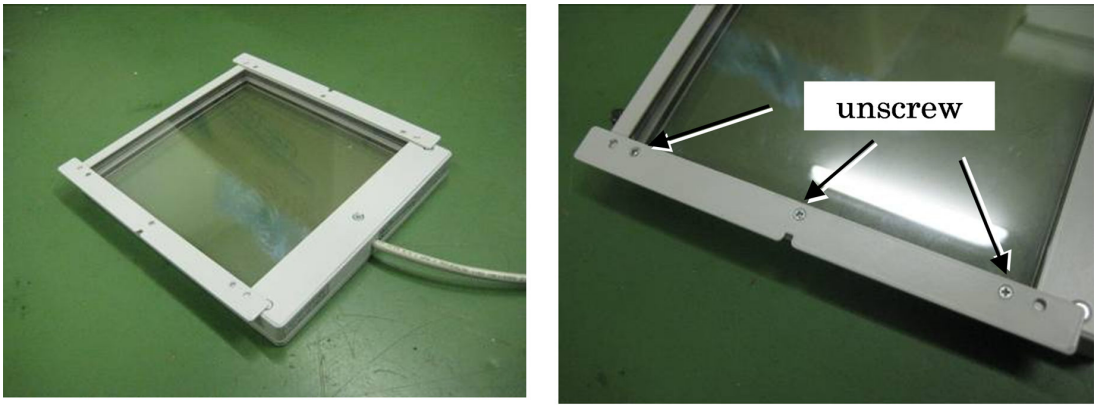


Fig. 5-103

8 **Fix the dose area product meter to the bracket with four screws removed in the step 7.**

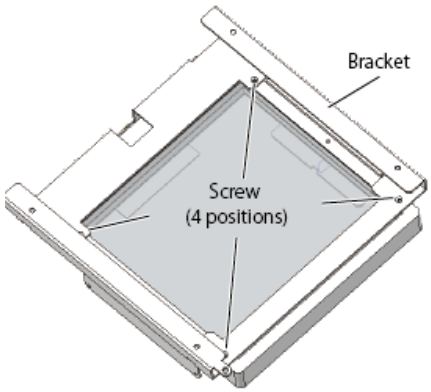


Fig. 5-104

9 **Attach the bracket with the dose area product meter.**

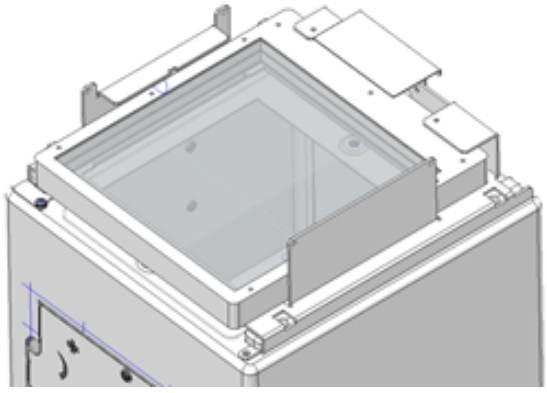


Fig. 5-105

10 **Connect DAP cable MUX A to the dose area product meter and DAP cable MUX B.**

11 Attach the bracket and secure the cable connector to this bracket with cable ties.

NOTE Do not attach the cable bracket. It is not needed.

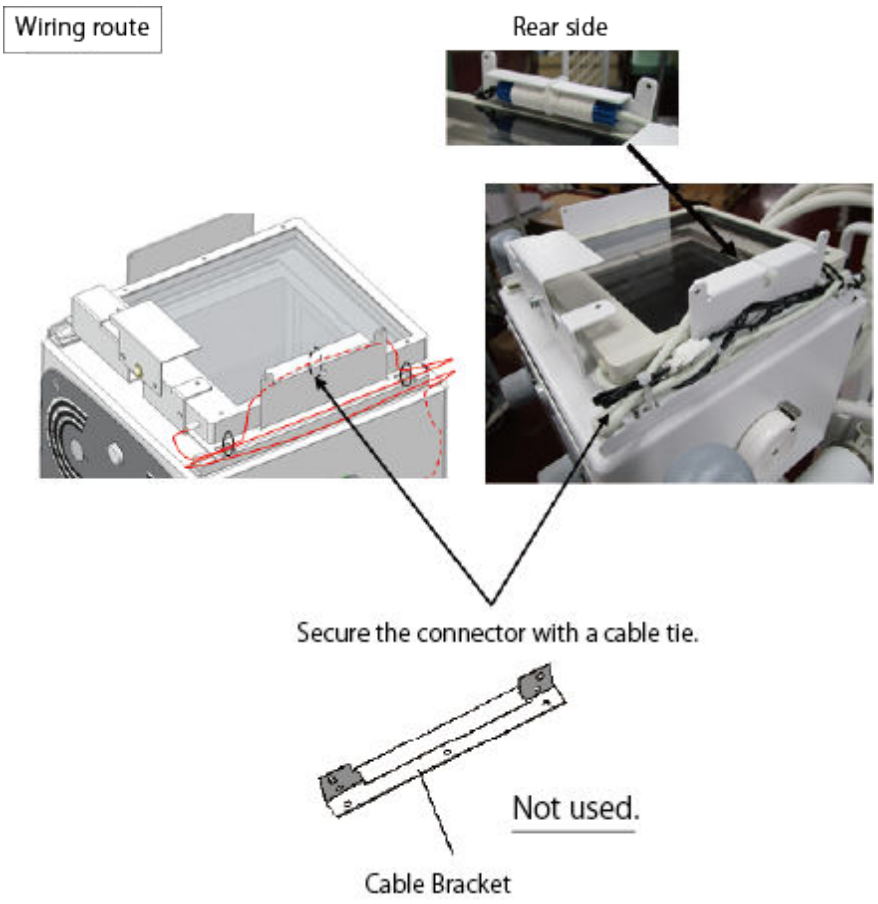


Fig. 5-106

12 Attach the Distance Piece.

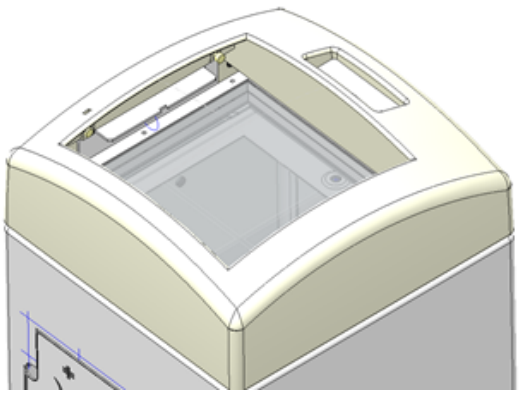


Fig. 5-107

■ Installing Dose Area Product Meter Option for MobileArt Evolution

Replacement of Arm Joint

> NOTE When MobileArt Evolution is combined with the dose area product meter option, it is necessary to change a height of height of the arm joint to maintain the distance between the top cover and the collimator.

1 **Remove the arm cover**

Reference "4.19 Attaching Arm Cover"

2 **Remove the Arm joint ASSY from the unit.**

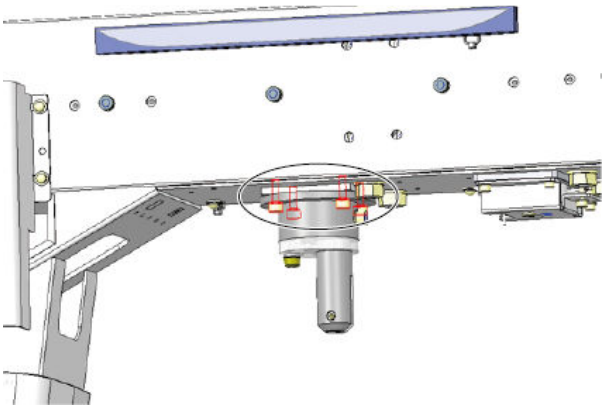


Fig. 5-108

3 **Attach the Arm joint ASSY, Long.**

> NOTE Make sure the M6 bolt in the figure below is firmly secured before attachment.

Fig. 5-109

4 **Pull the Arm joint ASSY, Long closer to the column support and fix it with bolts.**

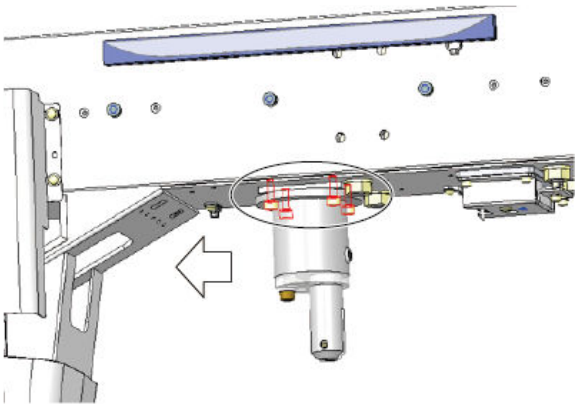


Fig. 5-110

Installing Dose Area Product Meter Option

1 **Turn OFF the main breaker.**

2 **Remove the rail of the collimator.**

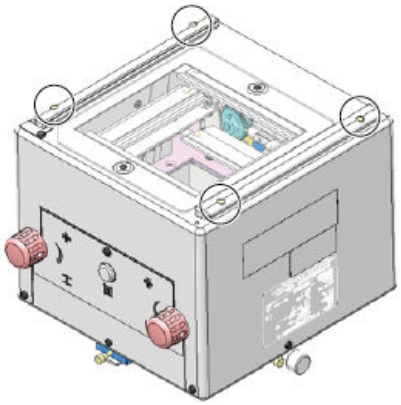


Fig. 5-111

3 **Remove the collimator cover, the rear cover, and the cover.**

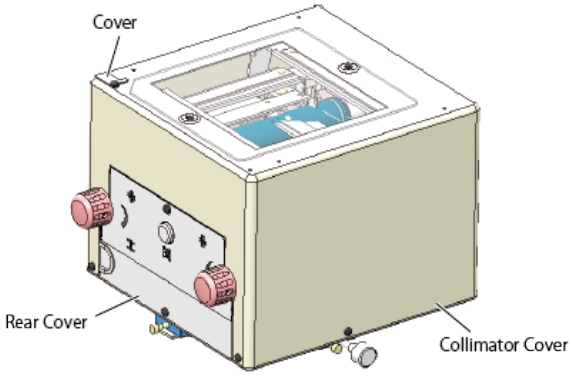


Fig. 5-112

4 **Connect and wire DAP cable MUX B to J2 connector on LEAF MEAS-15 board.**

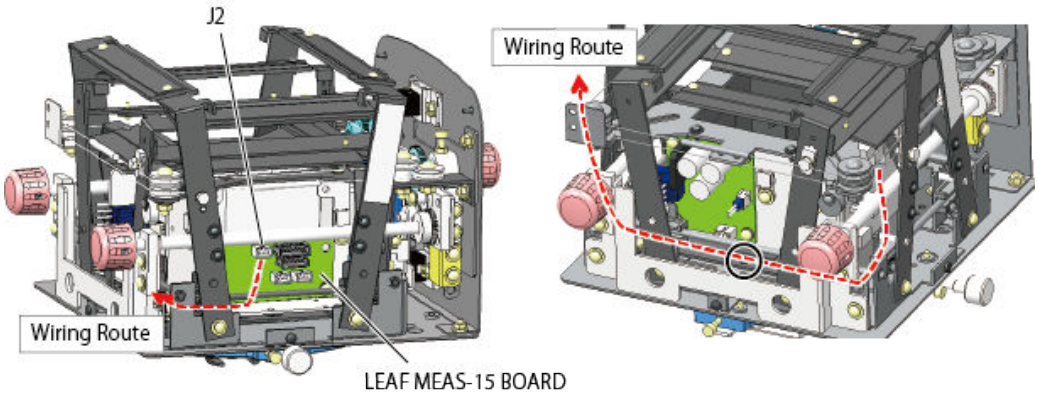


Fig. 5-113

5 **Attach the rear cover, pass DAP cable MUX B through the notch on the collimator cover, and attach the collimator cover.**

6 **Remove two plates fixed in Dose Area Product Meter.**

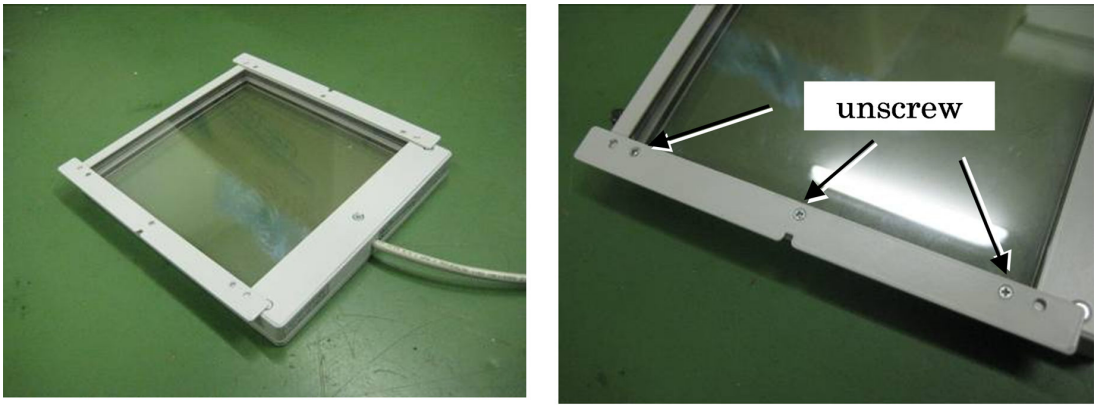


Fig. 5-114

7 **Fix the dose area product meter to the bracket with four screws removed in the step 6.**

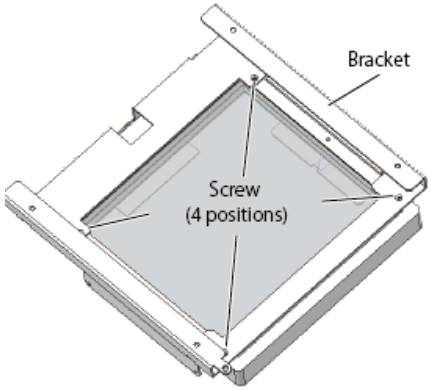


Fig. 5-115

8 **Using the rail removed in step 2, attach the bracket with the dose area product meter.**

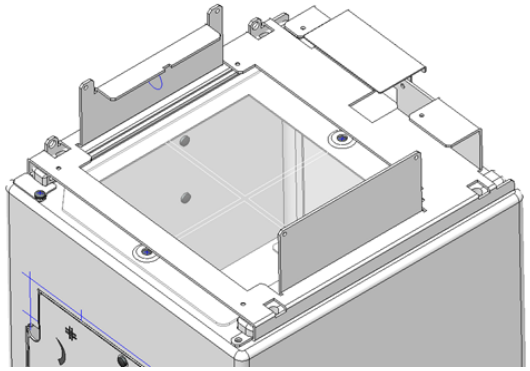


Fig. 5-116

9 **Connect DAP cable MUX A to the dose area product meter and DAP cable MUX B.**

10 **Attach the bracket and secure the cable connector to this bracket with cable ties.**

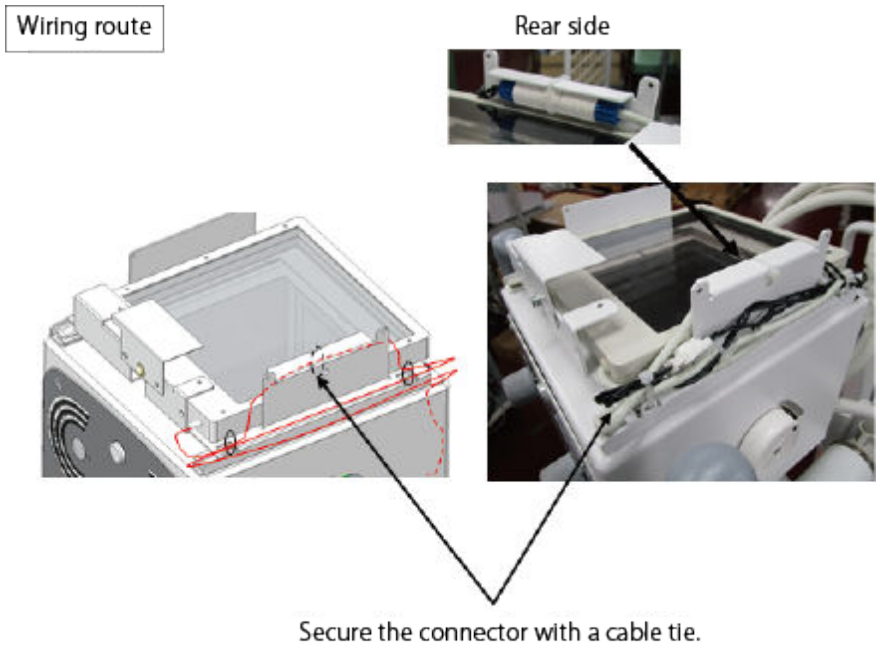


Fig. 5-117

11 **Attach the Distance Piece.**

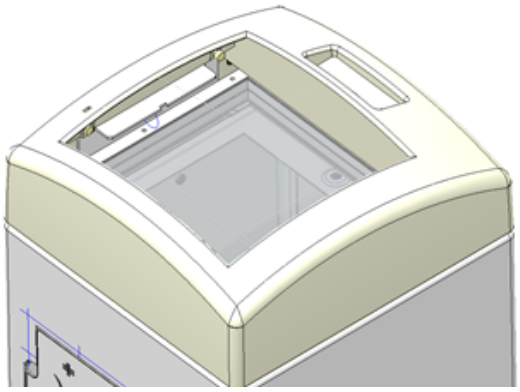


Fig. 5-118

5.7.3 Installing Dose Area Product Meter Shimadzu Option

NOTE The installation of the dose area product meter depends on the MobileDaRt Evolution/MobileArt Evolution. Refer to the corresponding procedure for installation.

Reference
- "Installing Dose Area Product Meter Option for MobileDaRt Evolution" P.227
- "Installing Dose Area Product Meter Option for MobileArt Evolution" P.230

■ Installing Dose Area Product Meter Option for MobileDaRt Evolution

1 **Turn OFF the main breaker.**

2 **Rmove the ditance piece.**

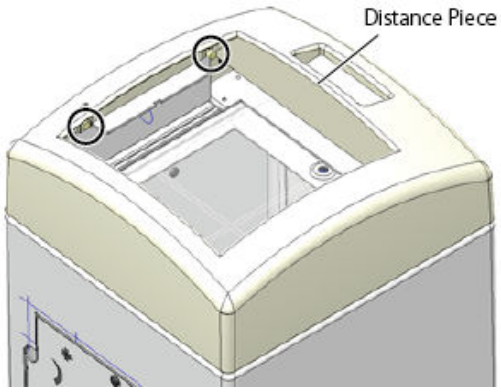


Fig. 5-119

3 Remove the bracket and the rail.

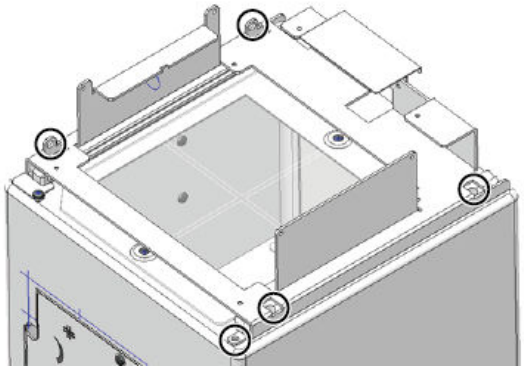


Fig. 5-120

4 Remove the collimator cover, the rear cover, and the cover.

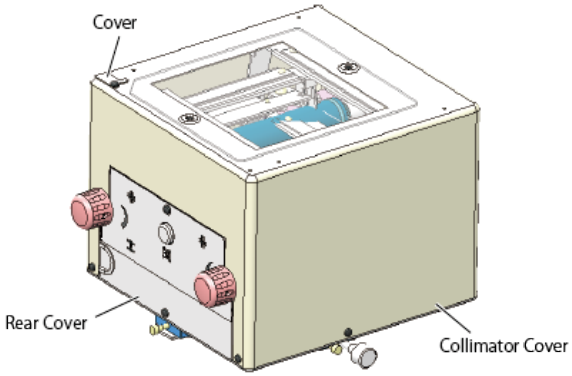


Fig. 5-121

5 Connect and wire DAP cable MUX B to J2 connector on CL MEAS-17 board.

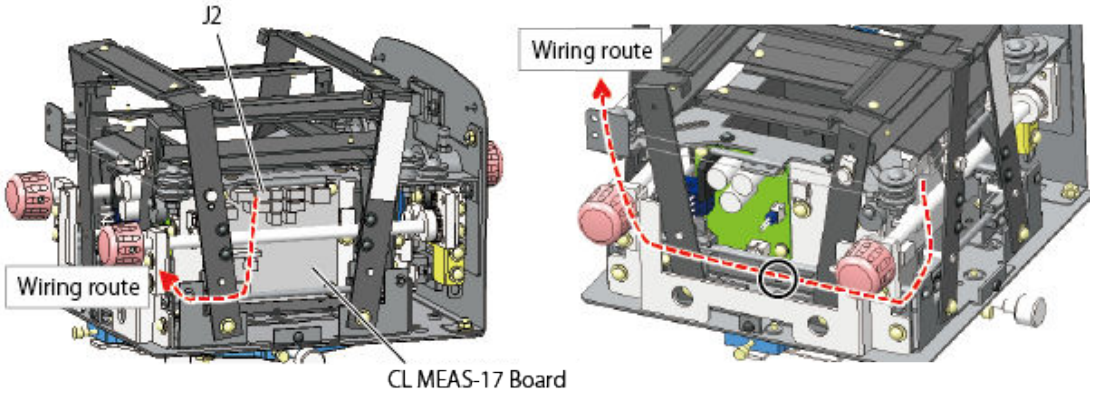


Fig. 5-122

6 **Attach the rear cover, pass DAP cable MUX B through the notch on the collimator cover, and attach the collimator cover.**

Do not attach the cover and the cover (for DAP). They are not needed.

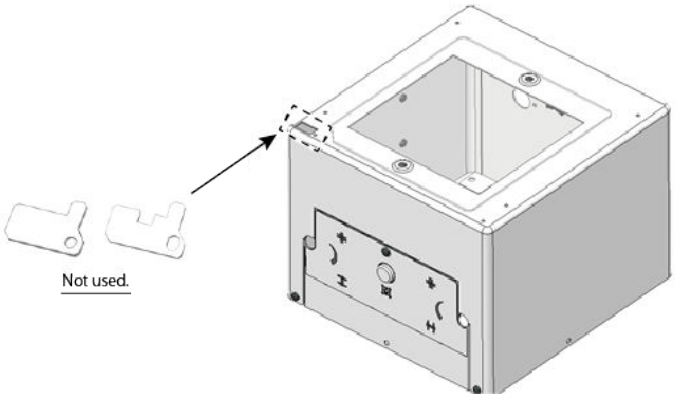


Fig. 5-123

7 **Attach the dose area product meter to the bracket with four tapping screws.**

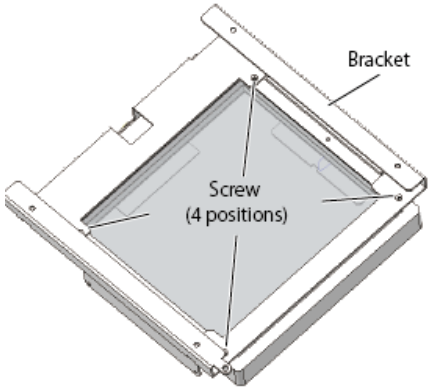


Fig. 5-124

8 **Attach the bracket with the dose area product meter.**

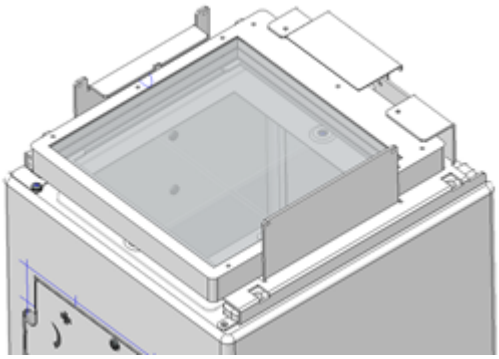
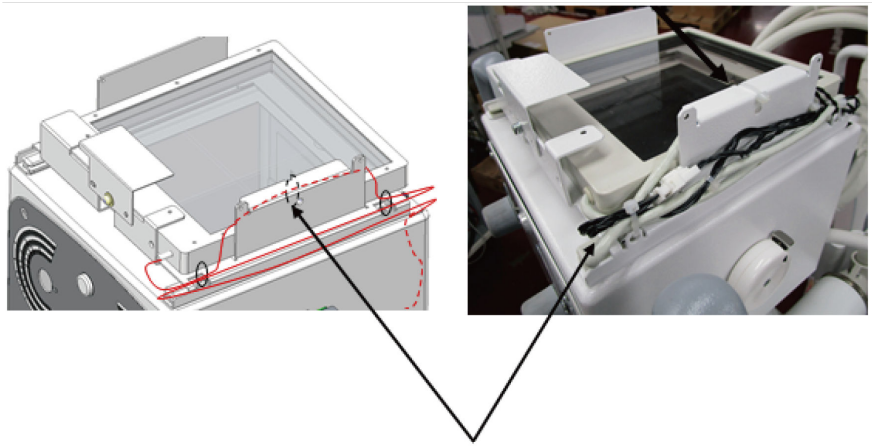


Fig. 5-125

9 Attach the bracket and secure the cable connector to this bracket with cable ties.

Wiring route



Secure the connector with a cable tie.

Fig. 5-126

10 Attach the Distance Piece.

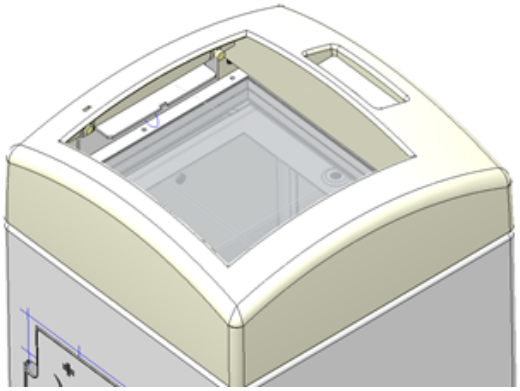


Fig. 5-127

■ Installing Dose Area Product Meter Option for MobileArt Evolution

Replacement of Arm Joint

NOTE When MobileArt Evolution is combined with the dose area product meter option, it is necessary to change a height of height of the arm joint to maintain the distance between the top cover and the collimator.

1 Remove the arm cover

Reference "4.19 Attaching Arm Cover"

2 **Remove the Arm joint ASSY from the unit.**

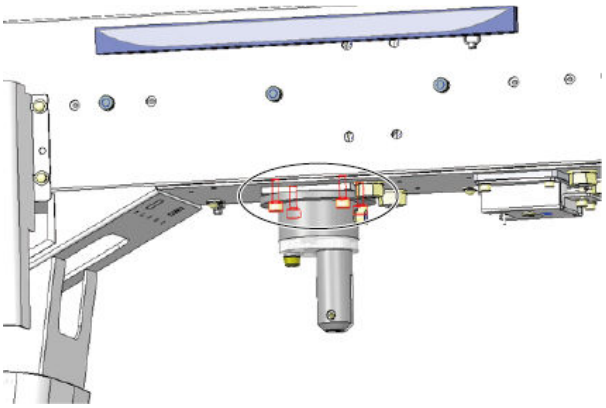


Fig. 5-128

3 **Attach the Arm joint ASSY, Long.**

NOTE Make sure the M6 bolt in the figure below is firmly secured before attachment.

Fig. 5-129

4 **Pull the Arm joint ASSY, Long closer to the column support and fix it with bolts.**

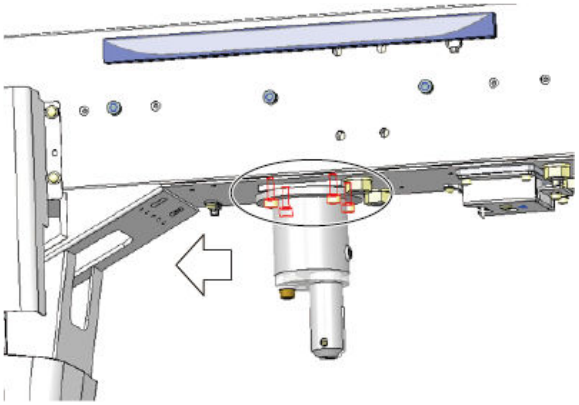


Fig. 5-130

Installing Dose Area Product Meter Option

1 Turn OFF the main breaker.

2 Remove the rail of the collimator.

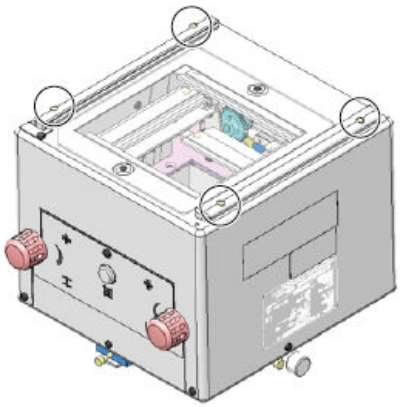


Fig. 5-131

3 Remove the collimator cover, the rear cover, and the cover.

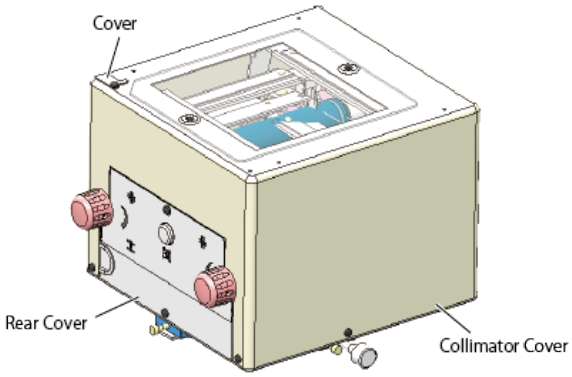


Fig. 5-132

4 Connect and wire DAP cable MUX B to J2 connector on LEAF MEAS-15 board.

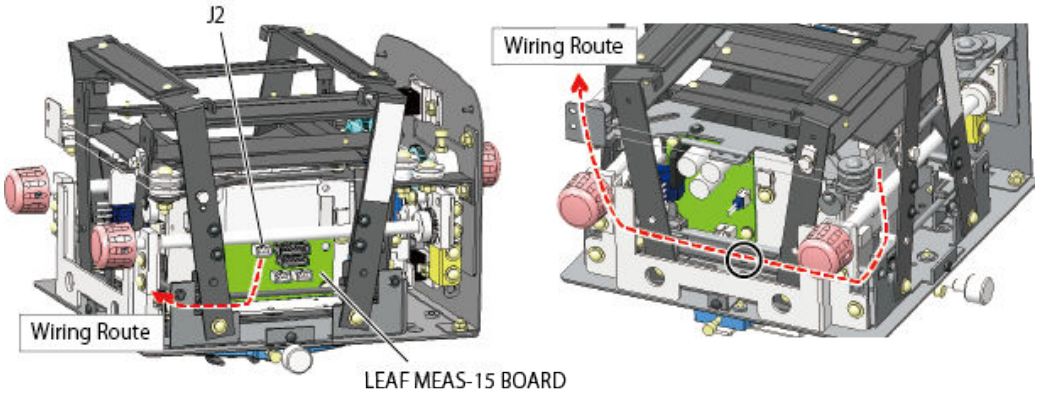


Fig. 5-133

5 **Attach the rear cover, pass DAP cable MUX B through the notch on the collimator cover, and attach the collimator cover.**

6 **Attach the dose area product meter to the bracket with four tapping screws.**

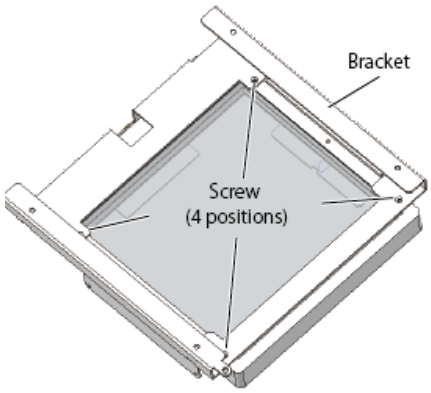


Fig. 5-134

7 **Using the rail removed in step 2, attach the bracket with the dose area product meter.**

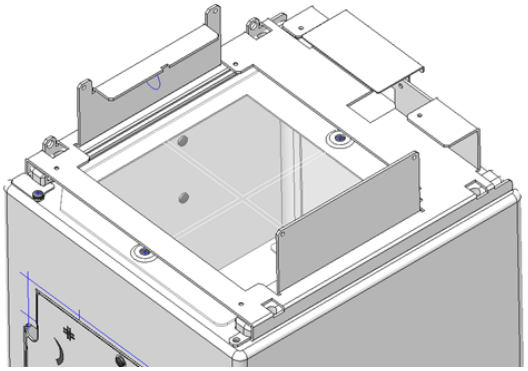


Fig. 5-135

8 **Attach the bracket and secure the cable connector to this bracket with cable ties.**

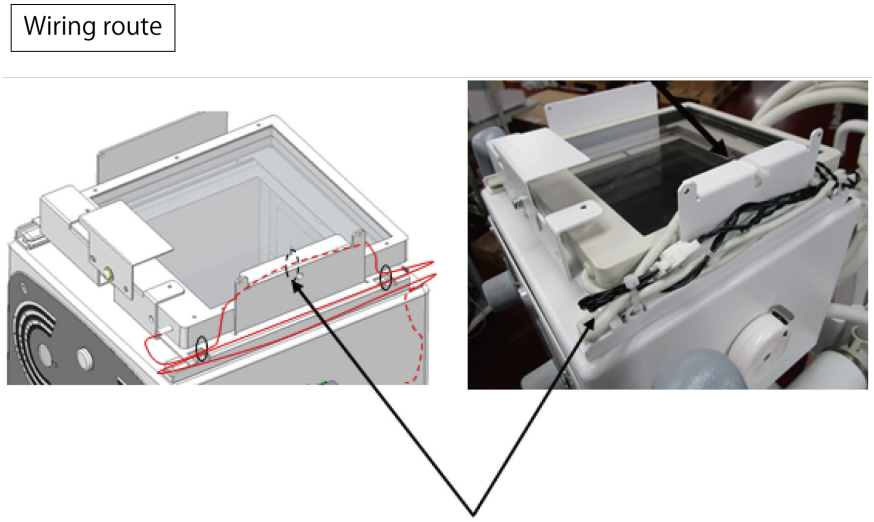


Fig. 5-136

9 **Attach the Distance Piece.**

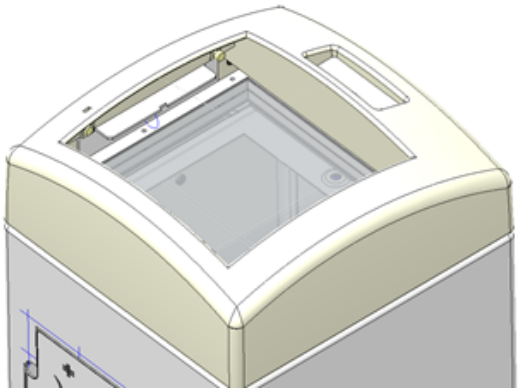


Fig. 5-137

5.7.4 Setting of DIP Switch

1 **Turn ON SW3-5 (enables Dose Area Product Meter Option) on the XCONT-2018 board.**

5.7.5 Operation Check of Dose Area Product Meter

1 **Turn ON the power.**

2 Adjust the Dose Area Product Meter.

Reference
- For MobileDaRt Evolution: "7.12 Dose Area Product Meter Adjustment Mode"
- For MobileArt Evolution:"8.12 Dose Area Product Meter Adjustment Mode"

3 After the adjustment is complete, operate the Dose Area Product Meter and check that it functions correctly.

NOTE The Dose Area Product Meter must be adjusted properly using its adjustment method. DO NOT use the Dose Area Product Meter without adjustment, or otherwise it may display incorrect DAP measurements.

Dose Area Product Meter values are displayed in mGy·cm^2. The Dose Area Product value in the DICOM image header is displayed in dGy·cm^2.

4 For CXDI-NE, change the annotation settings to show Total Area Dose (Display Annotation) and Area Dose (Image Box Annotation).

Reference CXDI NE Service Manual

NOTE This step is only applicable to MobileDaRt Evolution equipped with CXDI.

5 Check the operating force for the horizontal-shaft and back-and-forth rotations of the tube.

NOTE If the following problem occurs, adjust the tube rotation operating force according to "11.4.5 Adjustment of Tube Rotation Operating Force".
- Operating force is heavy.
- The collimator droops because it cannot be held at any angle.

5.7.6 Installing Additional Accessory Rail

The additional accessory rail is provided for installing AL phantom to the front face of DAP. Use this rail if required for dosimetry and calibration.

⚠ CAUTION

Prohibition

Do not operate the system with the additional accessory rail attached, or place the terescopic column in the catch unit.

The additional accessory rail is available only for dosimetry and calibration.

NOTE The irradiation range in the state with an additional accessory rail attached is a maximum of about 26 cm X 26 cm (SID 1000).

1 **When attaching the additional accessory rail to the distance piece type B, attach the supplied sponge tape to the position shown in the figure.**

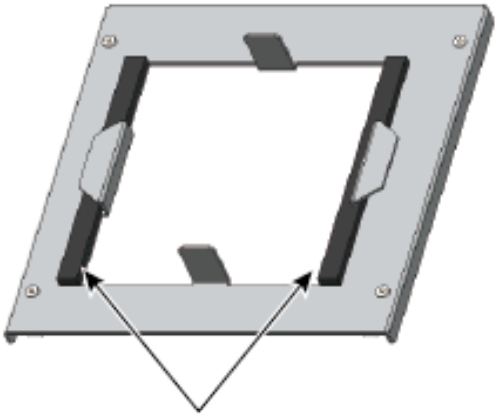


Fig. 5-138

2 **With the collimator horizontal, insert the additional accessory rail into the distance piece opening at a 45-degree rotation as shown.**

> **NOTE** Both types of distance piece Types A and B of means of attachment of a distance piece are the same.

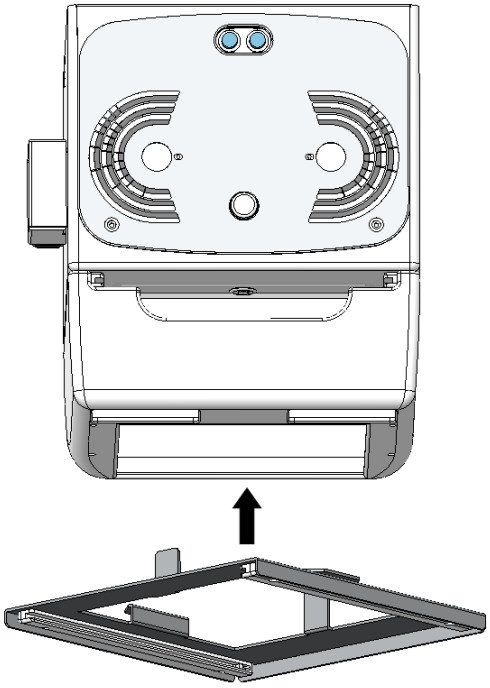


Fig. 5-139

3 Rotate the additional accessory rail 45 degrees as shown in the figure below and hook it on the edge of the distance piece.

NOTE Make sure the rail is parallel to the collimator.

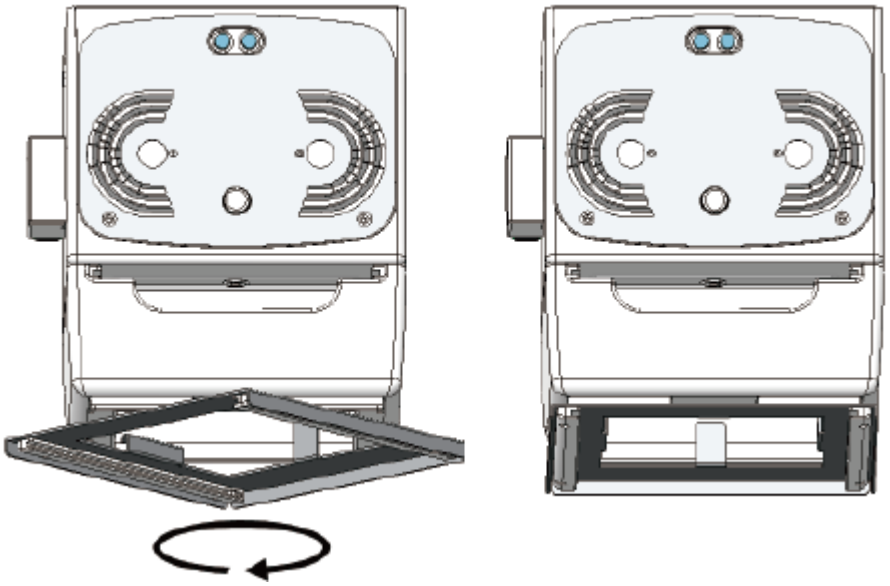


Fig. 5-140

4 Insert an AL phantom into the rail、and use it.

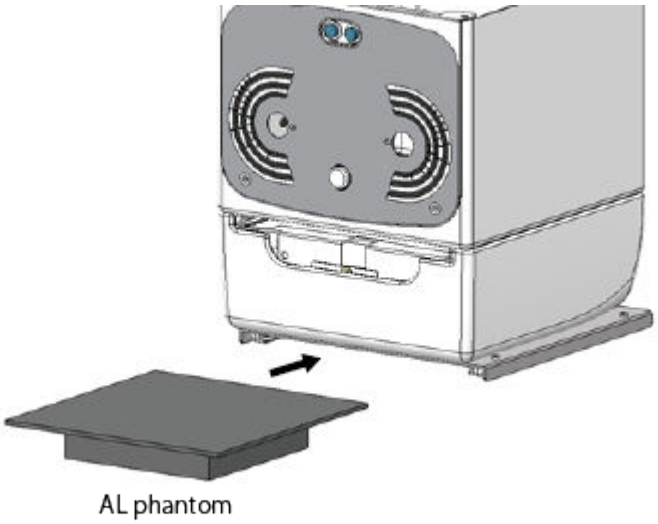


Fig. 5-141

⚠ CAUTION

Prohibition

- The AL phantom must be installed or uninstalled with the rail level.
- Do not move the unit with the additional accessory rail attached. The AL phantom or the additional accessory rail may fall.

5.8 Folding-type Protective Screen Option

5.8.1 Component of the Folding-type Protective Screen Option

The Protective screen option consists of the following parts. Make sure that all these parts are contained in the package.

Table 5-11 Component of folding-type Protective Screen Option (for 15 inch Monitor)

No	Parts Name	Quantity
1	Protective screen assy	1 set
2	SEMS SCREW M5x12 (accessory)	2

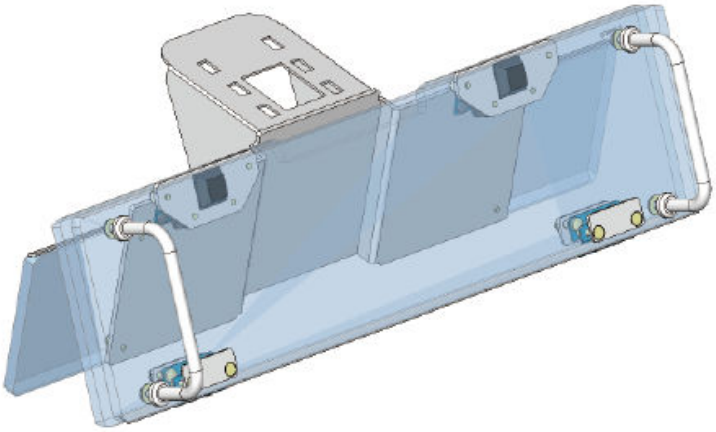


Fig. 5-142

5.8.2 Installation of the Protective Screen Option

1 **Remove the top plate and the arm lock release lever, then remove the joint section cover and the spacer.**

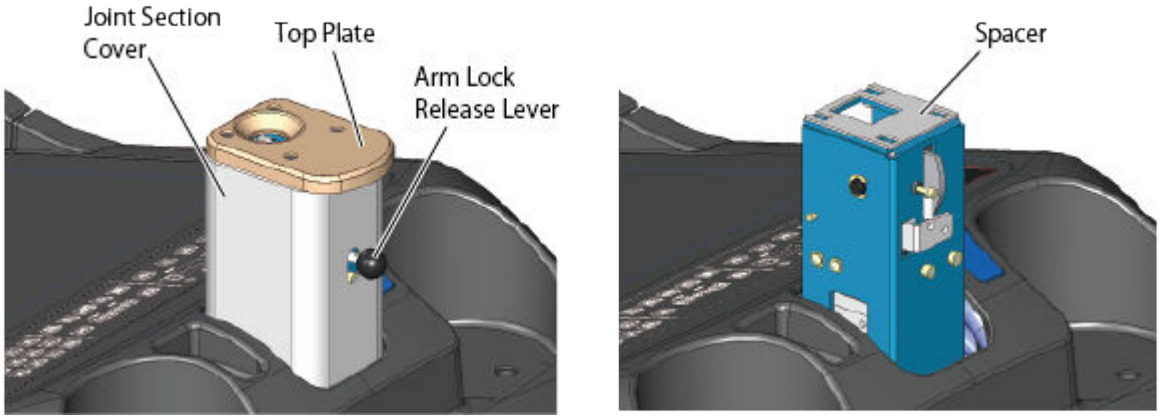


Fig. 5-143

2 **Place the joint section cover on the joint section. And attach the arm lock release lever, place the Protective Screen ASSY on the joint section cover, and fix it with SEMS M5. Use Loctite (blue) for fixing bolts.**

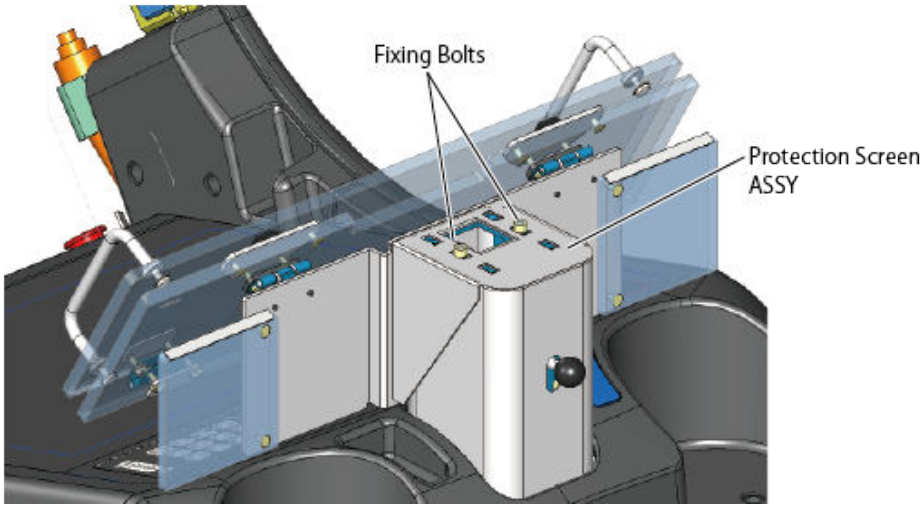


Fig. 5-144

3 **Place the top plate on the Protective screen and fix the protective screen.**

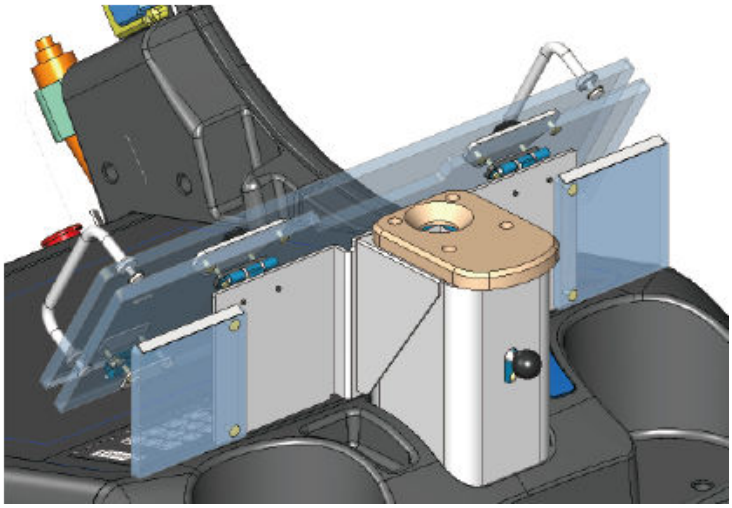


Fig. 5-145

5.8.3 Operation Check of the Folding-type protective screen option

Make sure that the unit can be operated successfully by raising (adjusting) and lowering the optional fixed protective screen several times.

Make sure that the bumper attached to the top cover sheet panel correctly makes contact with the protective screen in order to prevent any interference with the flat panel display.

5.9 Decoration Label

5.9.1 Component of Decoration Label Option

Decoration Label Option consists of items listed below. Please check while unpacking.

Table 5-12 Component of Decoration Label Option

No	Parts Name	Quantity
1	Label set	1

Fig. 5-146

5.9.2 Attaching Decoration Label

Put each label on MobileDaRt Evolution as shown below.

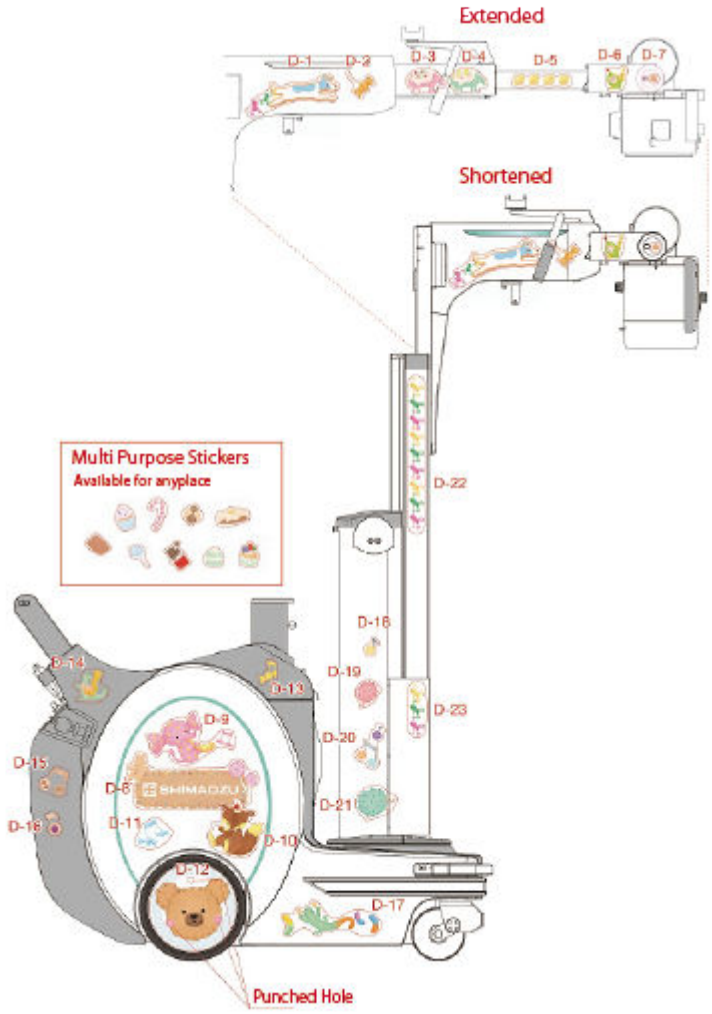


Fig. 5-147 Right Side

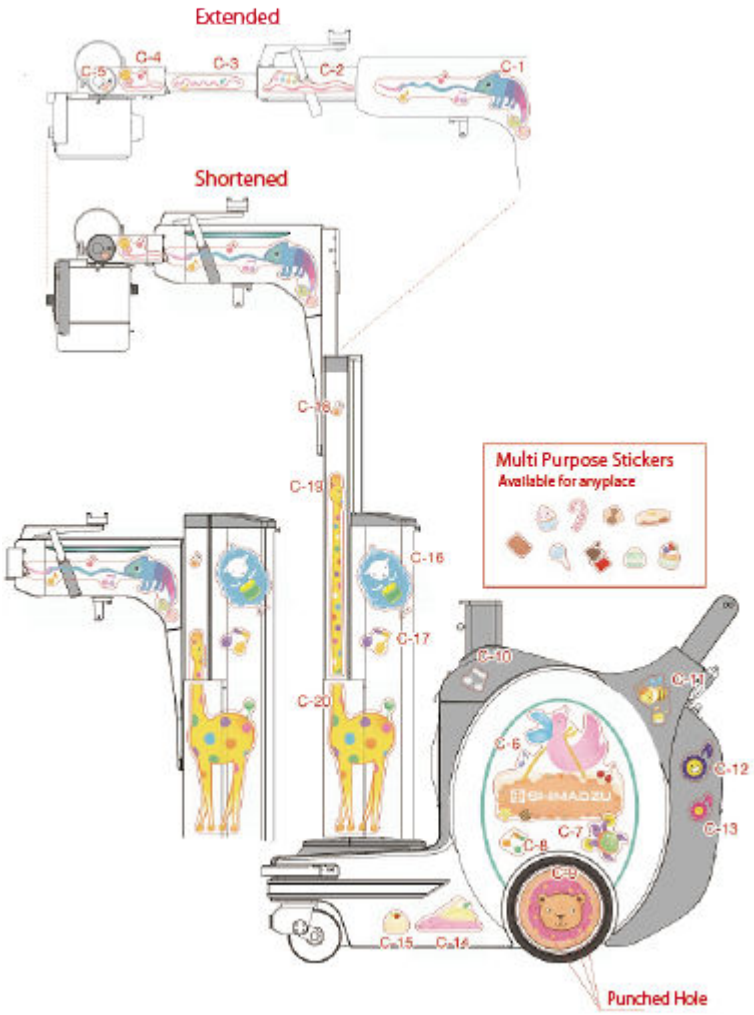


Fig. 5-148 Left Side

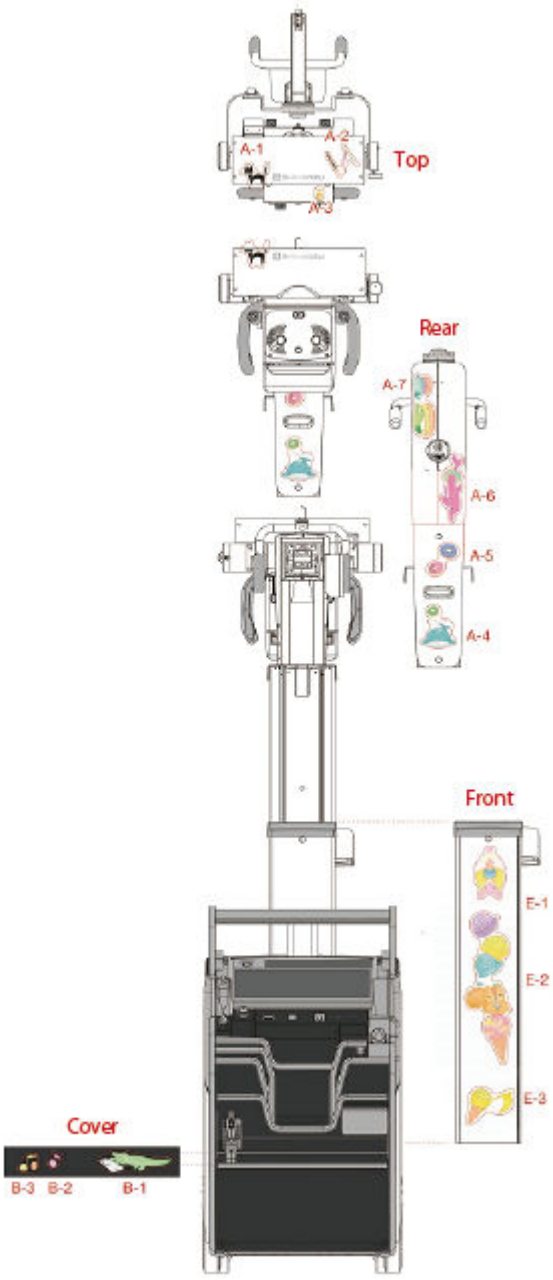


Fig. 5-149 Front Side

5.10 Assembly of FPD Docking Mechanism (Only Konica Minolta DR-equipped System)

The FPD docking mechanism kit is composed by the following parts. Check them when unpacking. Also, prepare I/F cables in addition to the items listed below.

Table 5-13 Table 4-4 Component of FPD docking mechanism kit

No.	Component	Amount
1	FPD-guided right frame ASSY	1
2	Left frame ASSY	1
3	Cable tie	6
4	Hexagon socket head SEMS screw M4x8	3

NOTE Do not disassemble the FPD-guided right frame ASSY because the dimensions are adjusted when the parts are assembled at the factory.

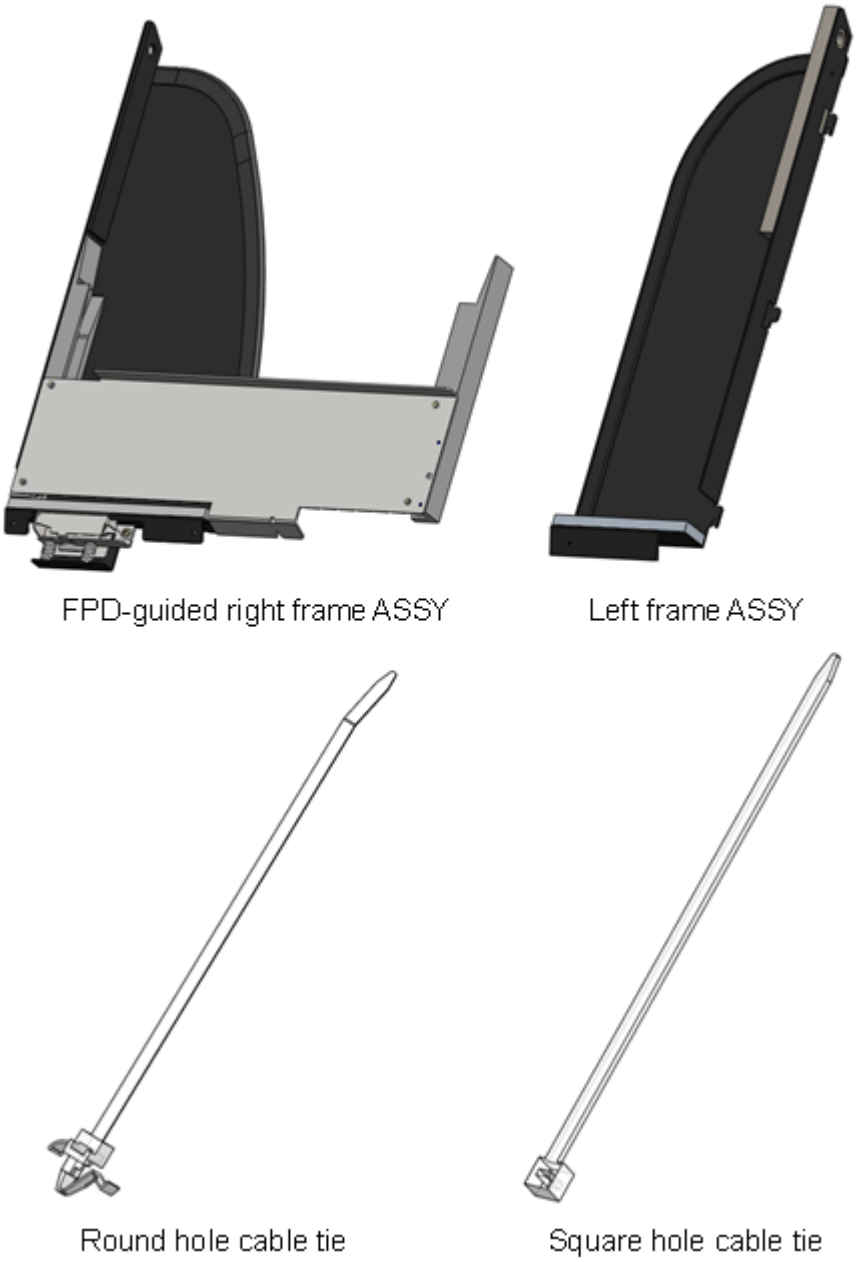

FPD-guided right frame ASSY
Left frame ASSY
Round hole cable tie
Square hole cable tie

Fig. 5-150

Fig. 5-151 IF Cable

5.10.1 Assembly of IF Cable to Right Frame ASSY with FPD Guide

1 **Remove the bind screw, and remove the cover ASSY.**

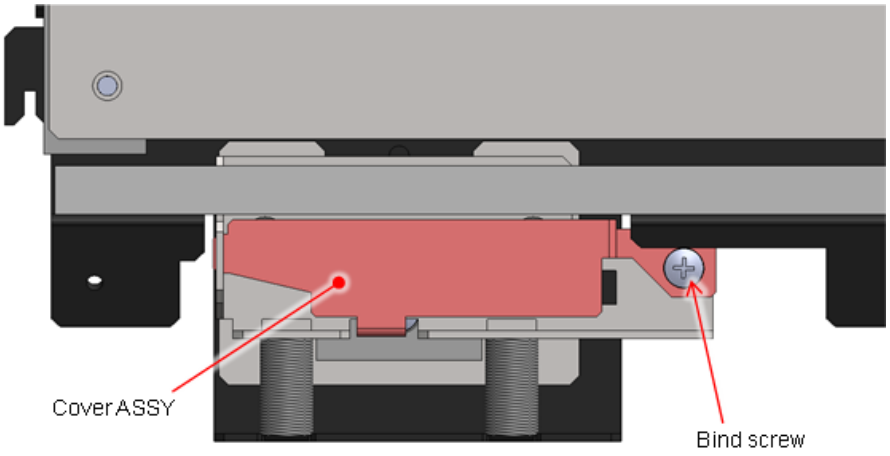


Fig. 5-152

2 **Remove the release coated paper of the double-sided tape of cover ASSY.**

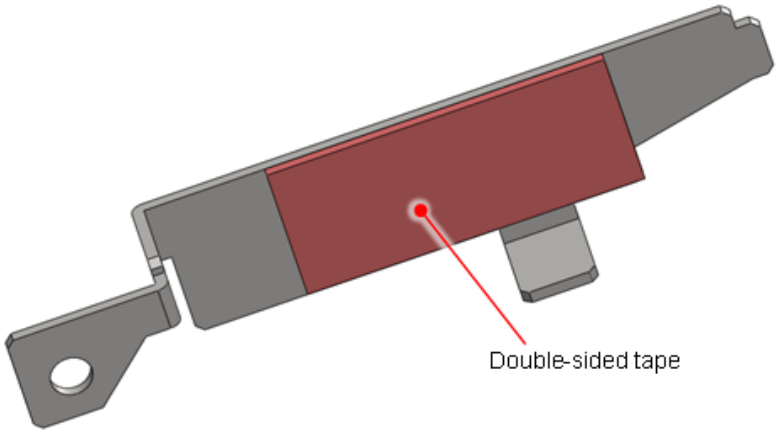


Fig. 5-153

3 **Attach double-sided tape to the IF cable connector with the sheet metal of the cover ASSY in contact with the IF cable connector as shown in the figure below.**

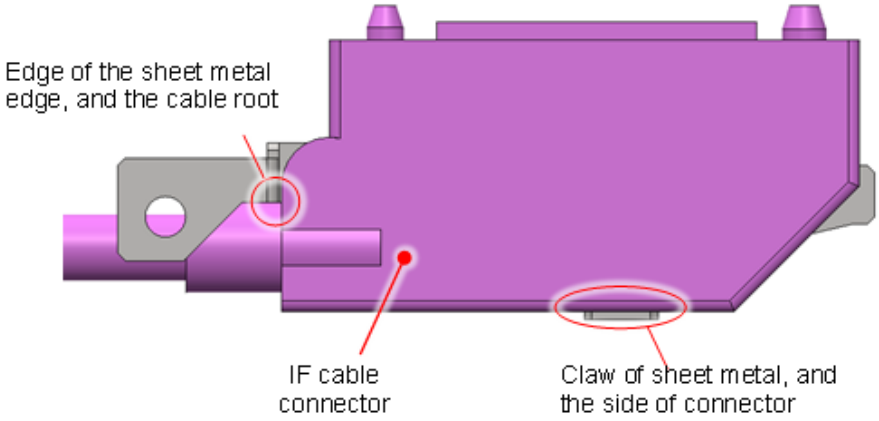


Fig. 5-154

4 **Using the bind screw removed in step 1, attach the cover ASSY with the IF cable connector to the base plate. When attaching, press and hold the IF cable connector against the indicated position of the base plate in the figure, and screw it.**

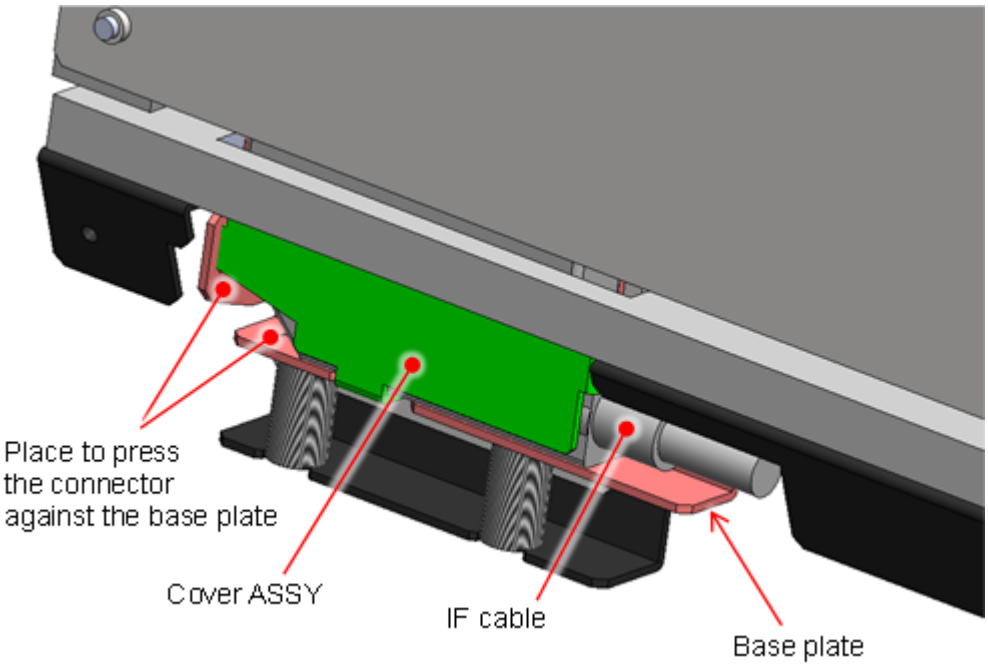


Fig. 5-155

5 Pass the IF cable through the square hole and a slit of the bridge plate and fix it with a cable tie.

NOTE
- Fix the cable in the center of the left-right movement allowance of the IF cable connector.
- Fix the cable in the state of natural wiring to avoid excessive force.

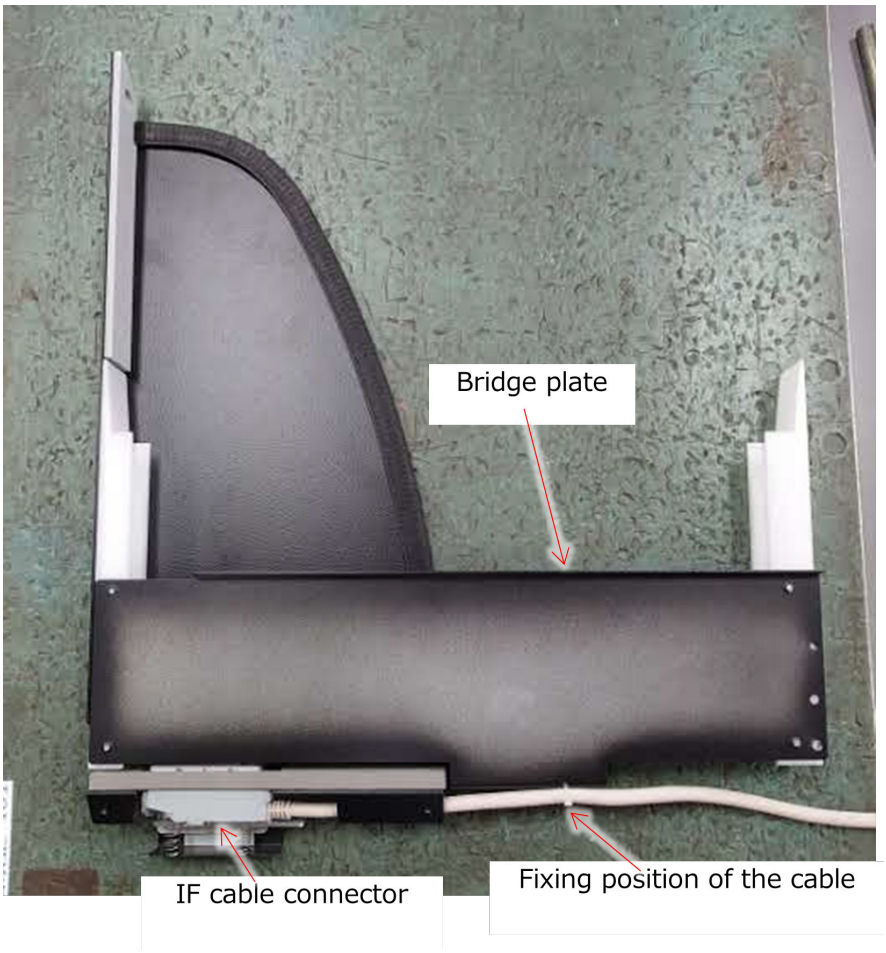


Fig. 5-156

6 Insert the FPD to be used at the delivery destination into the FPD-guided right frame ASSY and confirm that there is no insertion resistance and that the IF cable connector mates with the FPD connector.

5.10.2 Assembly of FPD Docking Mechanism to FPD Box

> **NOTE** Before assembling the FPD docking mechanism to the FPD box, install the controller GIU SZ.
> Refer to the corresponding procedure for installation.

Reference "4.14 Installing GIU SZ (Only for MobileDaRt Evolution)"

1 **Turn OFF the main breaker.**

2 **Remove the inside cover and right and left side covers.**

3 **Remove the FPD box fixing bolt, and open the FPD box.**

4 Remove the frames 5, 6, and 7.

NOTE Only frame 7 is reattached to the FPD box in a later step.

Hint The frames can be removed by unscrewing three screws on the rear and three screws on the sides and removing the claws of frames 5, 6 and 7 that are caught in the square holes of the box frame.

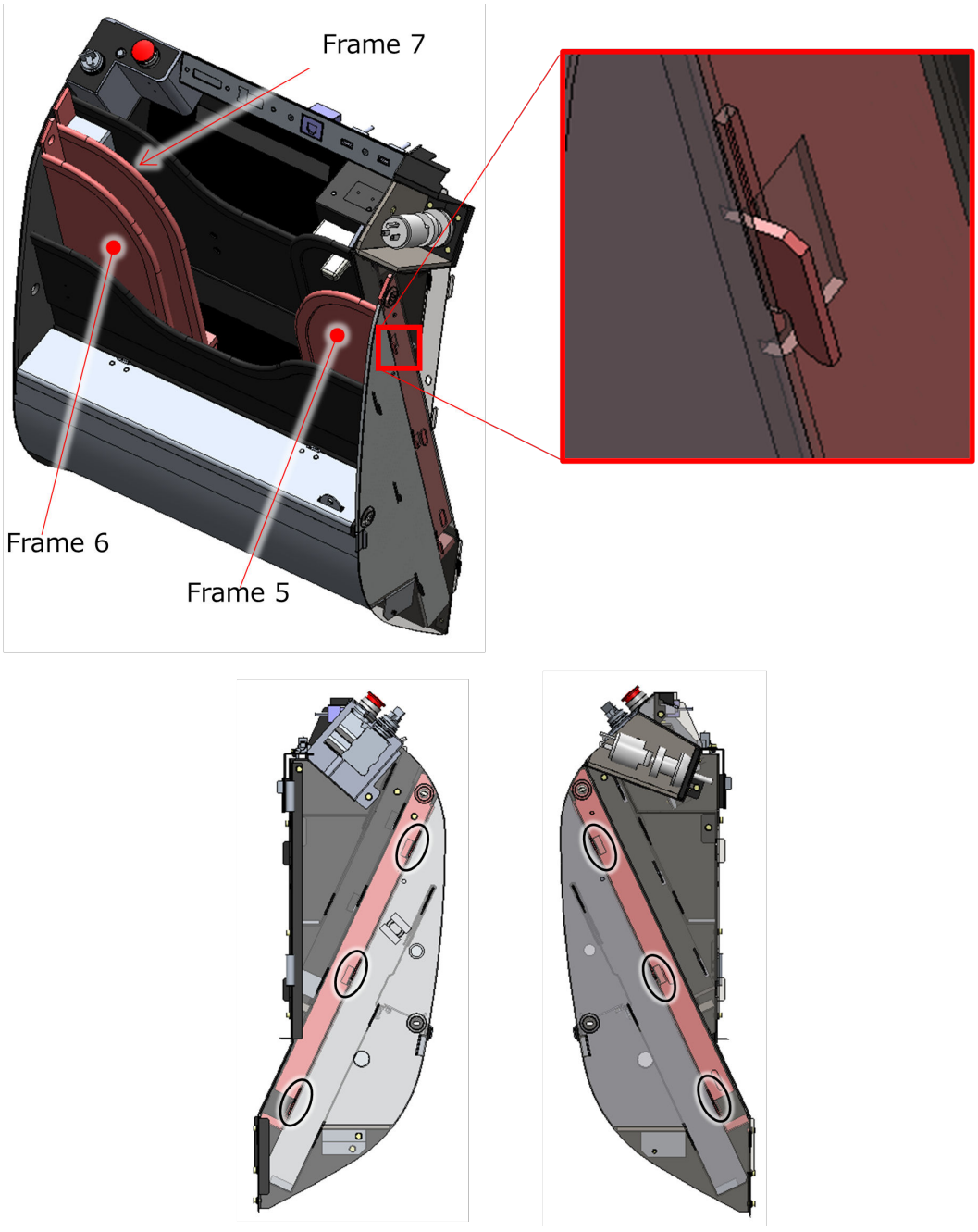


Fig. 5-157

5 **Pass the IF cable attached to the FPD-guided right frame ASSY through the hole on the rear frame of the FPD box, and pull out it from the inside to the outside of the FPD box.**

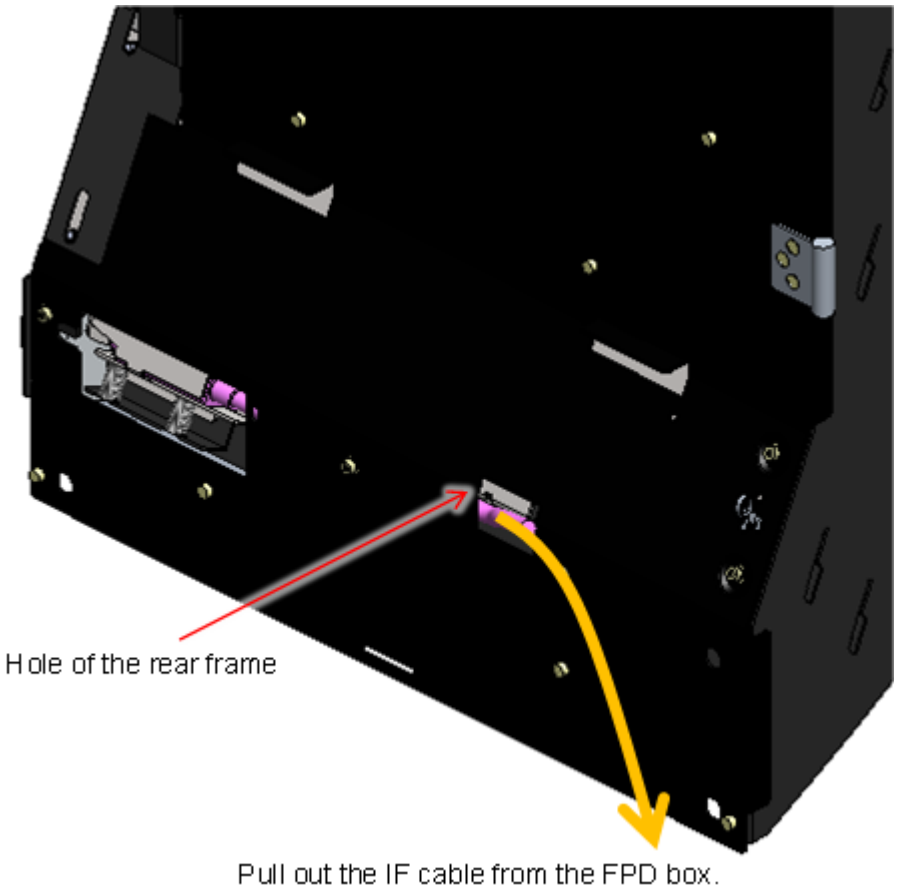


Fig. 5-158

6 Attach the FPD-guided right frame ASSY to the FPD box.

1 Insert the claw of the FPD-guided right frame ASSY into the square hole of the box frame, and fasten four screws on the rear side and one screw on the right side temporarily.

> **NOTE**
> - Be careful not to crush the IF cable by inserting the FPD-guided right frame ASSY into the FPD box while simultaneously pulling the IF cable out of the FPD box.
> - When inserting the FPD-guided right frame ASSY into the FPD box, be careful that the claw of the FPD-guided right frame ASSY does not damage the cushion inside the right frame of the FPD box.

Hint Use the M4x6 screws removed in step 4.

Fig. 5-159

7 Attach the left frame ASSY to the FPD box.

1 Insert the claw of the left frame ASSY into the square hole of the box frame, and fasten one screw on the rear side and one screw on the left side temporarily.

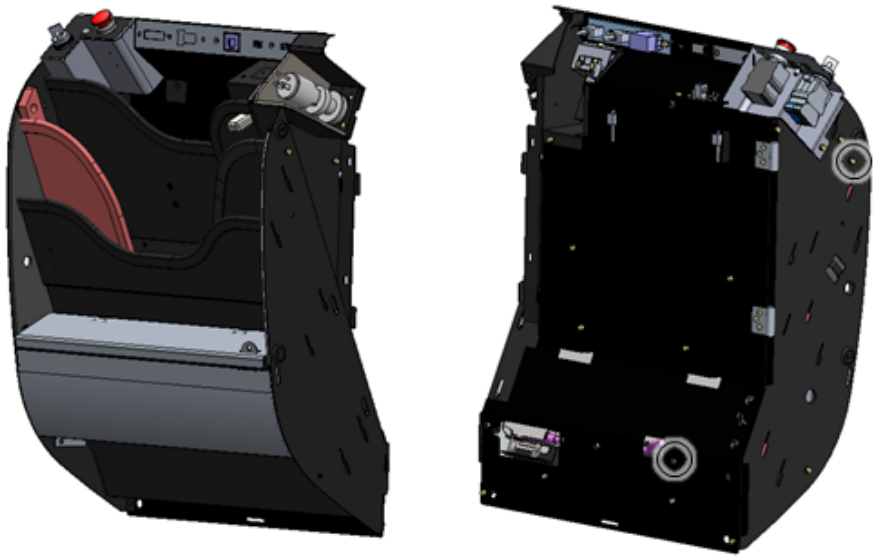


Fig. 5-160

8 Attach the frame 7 to the FPD box.

1 Insert the claw of the frame 7 into the square hole of the box frame, and fasten one screw on the rear side and one screw on the left side temporarily.

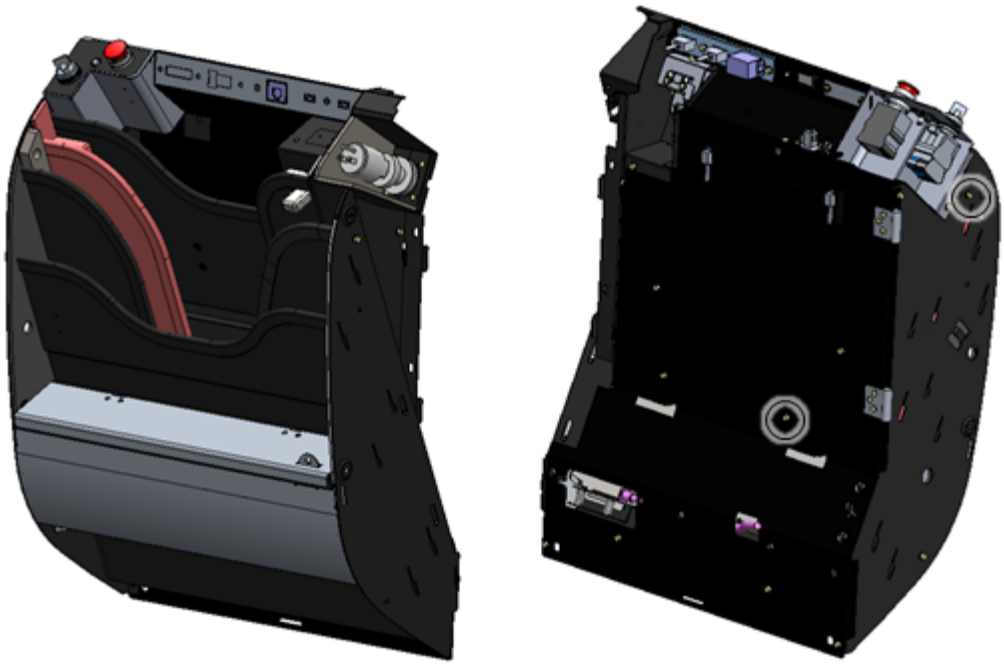


Fig. 5-161

9 **Fasten the screws temporarily fastened in step 6, 7, and 8 in any order to fix the FPD-guided right frame ASSY, the left frame ASSY, and the frame 7 to the FPD box.**

10 **Wire the IF cable along the rear frame of the FPD box as shown in the figure below, and fix the IF cable with cable ties.**

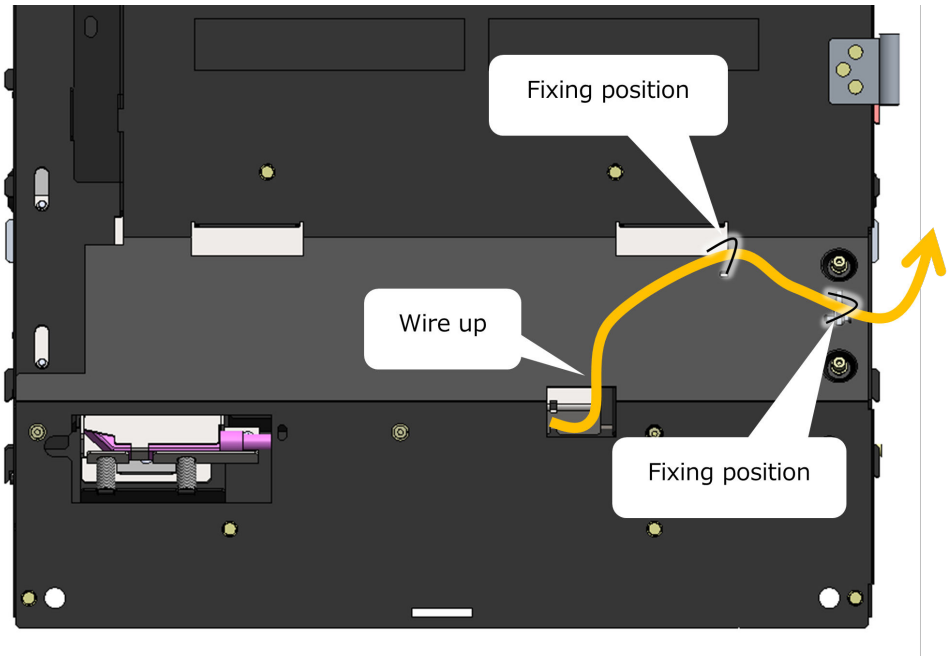


Fig. 5-162

11 **Wire the cable from FPD box side to the reverse side of the emergency stop button, as shown in the figure below.**

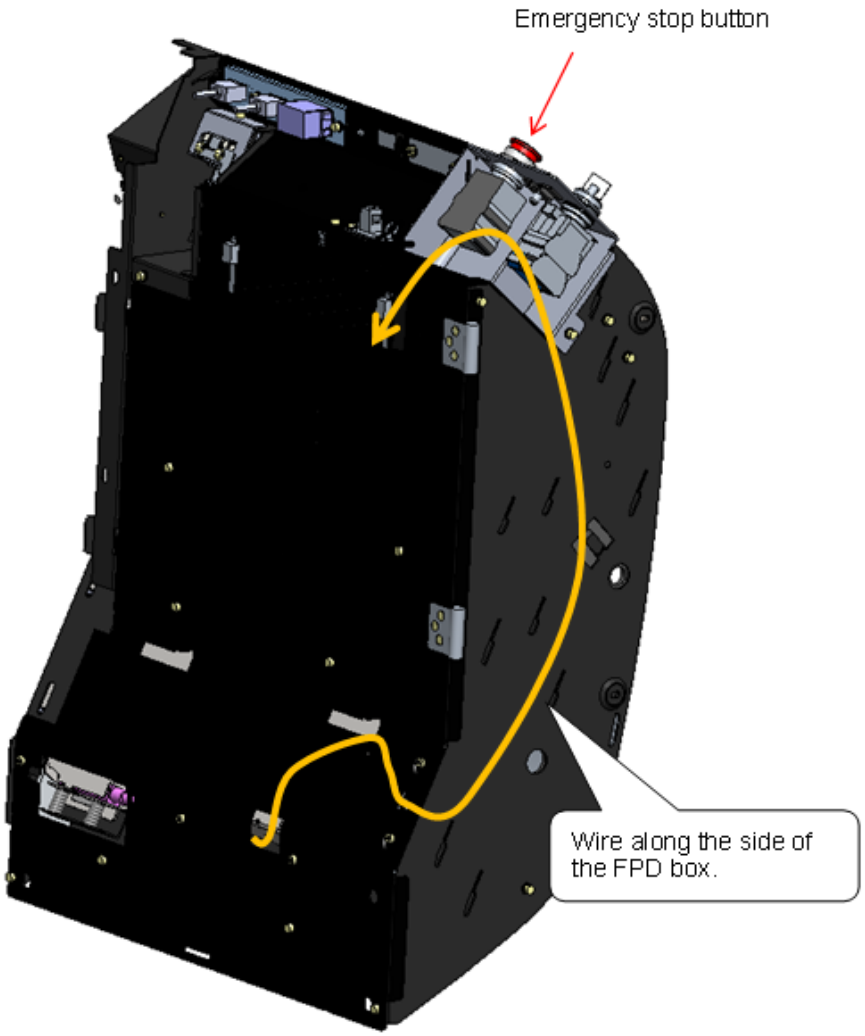


Fig. 5-163

12 Use a screwdriver to remove two screws and remove the cover of the controller GIU SZ.

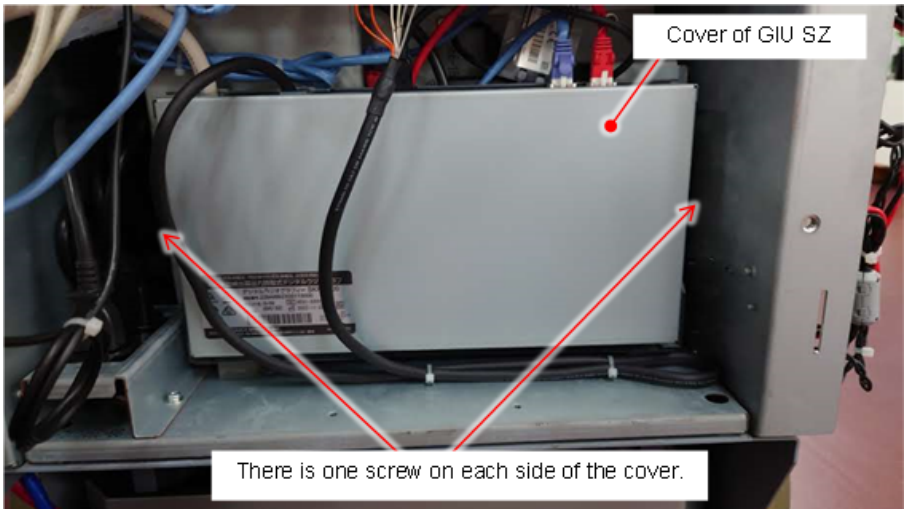


Fig. 5-164

13 Plug the IF cable connector into the board and fix the IF cable fixing fittings into the GIU SZ with a screw.

> **NOTE** Fix the fixing fittings in a position where they do not touch the black heat-shrinkable tubing of the IF cable.

Hint The fixing fittings and a screw are included with the IF cable.

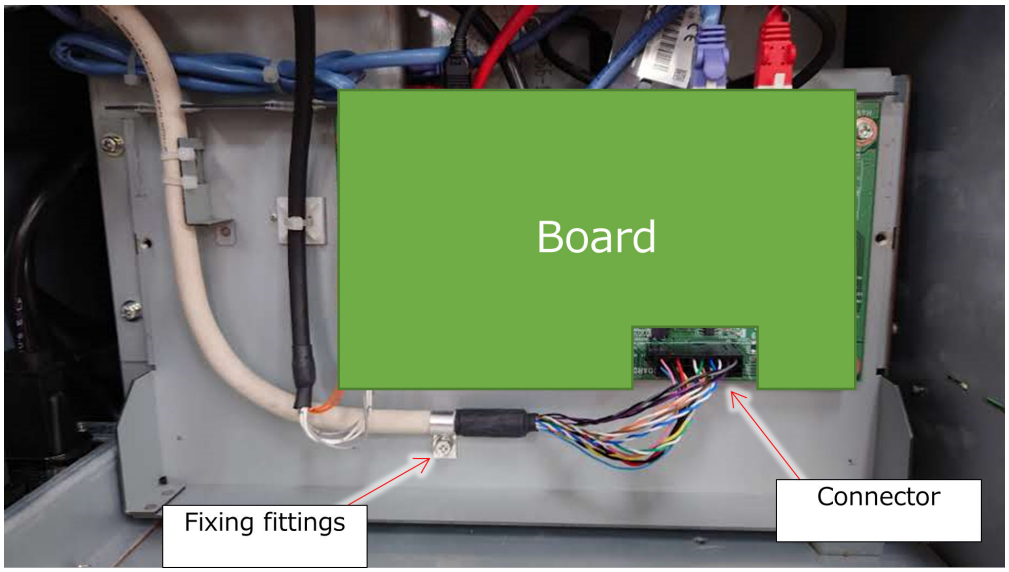


Fig. 5-165

14 **Fix the IF cable using cable ties at the indicated positions in the figure below inside GIU SZ.**

Hint Fit cable ties into their respective notches of the fixing plate and tie them off.

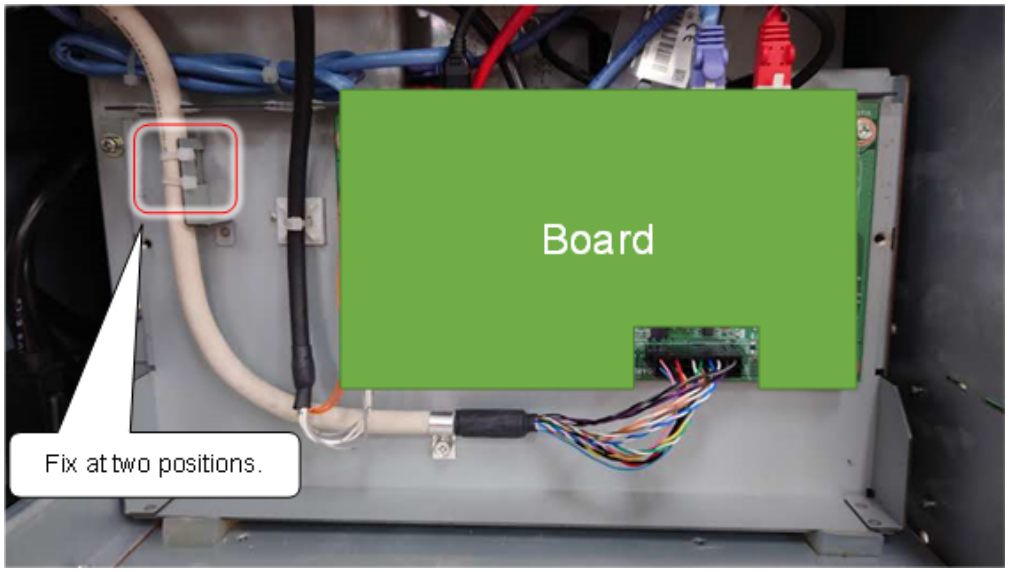


Fig. 5-166

15 **Re-fix the cover of GIU SZ with two screws removed in step 12.**

16 **Wire the IF cable to the side of the FPD box for natural deflection, and fix it along with the cable extending from the emergency stop button with a cable tie.**

> NOTE Slacken the remaining cable (approx. 70 cm) as shown in the figure below and store it inside the unit.

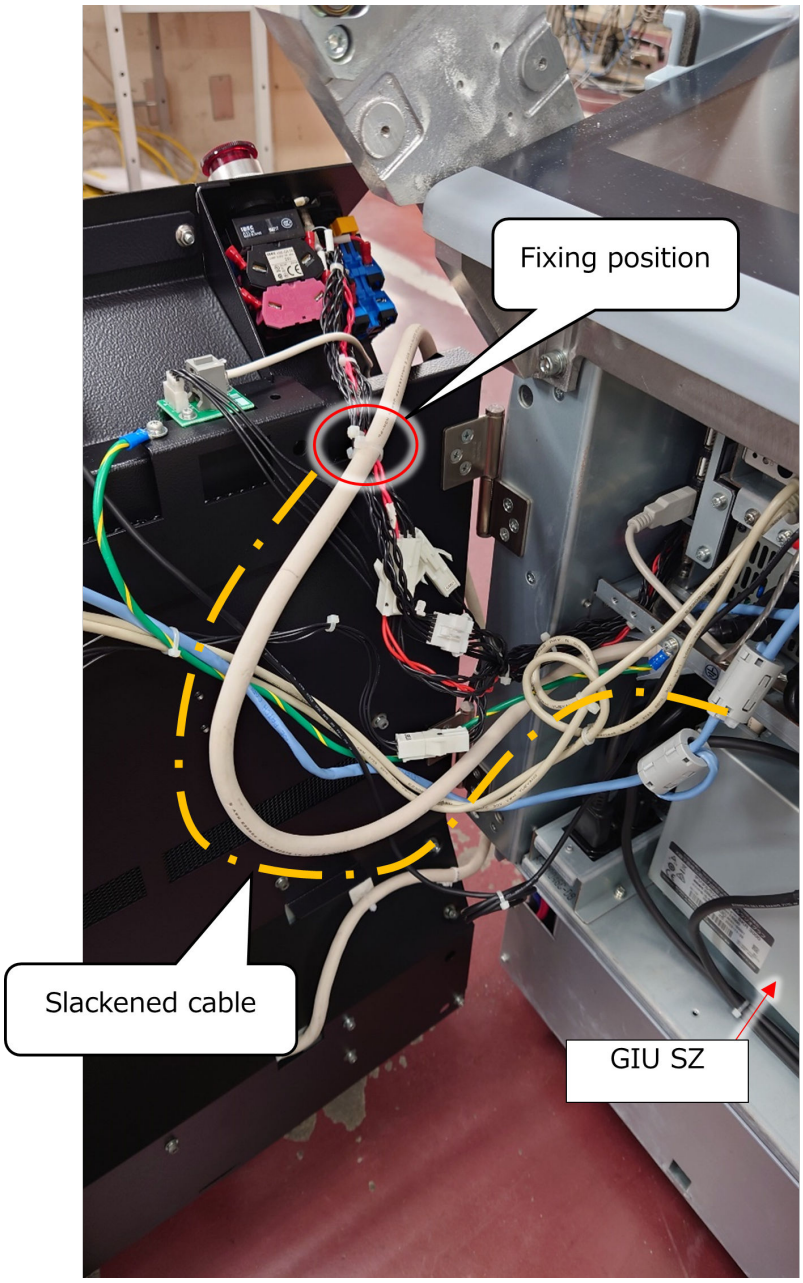


Fig. 5-167

17 **Attach the FPD box and the covers in reverse order of step 2 and 3.**

5.11 Options for MobileDaRt Evolution

For MobileDaRt Evolution, refer to the document below to install options that are applicable only to MobileDaRt Evolution.

Reference M503-E410 MobileDaRt Evolution （MX8Version） Installation Manual (DR Section)

6 [Appendix] DIP Switches and Jumpers Setting on Each Board

6.1 XCONT-2018 Board

NOTE For jumper settings, refer to "6.2 XCONT-2018 Board Jumper Setting".

Table 6-1 XCONT-2018 DIP Switches

Number	Function	Switch Status & Function			Default		
					MobileDaRt Evolution	MobileArt Evolution	
						High-power type	Standard type
SW2-1	System configuration	ON: MobileDaRt Evolution OFF: MobileArt Evolution			ON	OFF	OFF
SW2-2	12.5kW type	OFF: 32kW type ON: 12.5kW type			OFF	OFF	ON
SW2-3	Debug mode	ON: Debug mode OFF: Normal mode			OFF	OFF	OFF
SW2-4	DR communication*1	MobileDaRt Evolution	DR-200m	ON	*2	OFF	OFF
			CS-7 for Serial Radiography	ON			
			System other than DR-200m and CS-7 for Serial Radigraphy	OFF			
		MobileArt Evolution		OFF			
SW2-5	Measured value display mode	ON: Measured value display mode OFF: Normal mode			OFF	OFF	OFF
SW2-6	High voltage transformer	MobileArt Evolution	High-power type	ON	ON	ON	OFF
			Standard type	OFF			
		MobileDaRt Evolution	D150MH-40 (P/N: 562-25900)	OFF			
			D150MH-40 (P/N: 565-16863)	ON			

Number	Function	Switch Status & Function			Default		
					MobileDaRt Evolution	MobileArt Evolution	
						High-power type	Standard type
SW2-7	DR communication*1	MobileDaRt Evolution	DR-ID800	ON	*2	OFF	OFF
			DR-200m	ON			
			CS-7	OFF			
			CS-7 for Serial Radiography	ON			
			VIVIX-S	OFF			
			CXDI	OFF			
		MobileArt Evolution		OFF			
SW2-8	Initial setting mode	ON: Initial setting mode OFF: Normal mode			OFF	OFF	OFF
SW3-1	EEPROM initialization	ON: with SW2-8 ON to initialize CPU board			OFF	OFF	OFF
SW3-2	Keyless Entry Option	ON: Enable OFF: Disable			OFF	OFF	OFF
SW3-3	Infrared Remote Controller Option	ON: Enable OFF: Disable			OFF	OFF	OFF

Number	Function	Switch Status & Function				Default		
						MobileDaRt Evolution	MobileArt Evolution	
							High-power type	Standard type
SW3-4	DR communication or FPD enclosure	MobileDaRt Evolution	DR-I800		OFF	*2	OFF	OFF
			DR-200m		ON			
			CS-7		OFF			
			CS-7 for Serial Radiography		OFF			
			VIVIX-S		ON			
			CXDI	Multi Box MB-4A	OFF			
				Without Multi Box	OFF			
				Multi Box MB-02	ON			
		MobileArt Evolution	FPD box type		ON			
			Cassette box type		OFF			
SW3-5	Dose Area Product Meter Option	ON: Enable OFF: Disable				OFF	OFF	OFF

Number	Function	Switch Status & Function				Default		
						MobileDaRt Evolution	MobileArt Evolution	
							High-power type	Standard type
SW3-6	DR communication	MobileDaRt Evolution	DR-ID800		ON	*2	OFF	OFF
			DR-200m		ON			
			CS-7		ON			
			CS-7 for Serial Radiography		OFF			
			VIVIX-S		OFF			
			CXDI	Multi Box MB-4A	ON			
				Without Multi Box	OFF			
				Multi Box MB-02	ON			
		MobileArt Evolution			OFF			
SW3-7	Serial Radiography	ON: Enable OFF: Disable				OFF	OFF	OFF
SW3-8	Adjustment mode	ON: Adjustment mode OFF: Normal mode				OFF	OFF	OFF
SW4-1	Unused	Normally OFF				OFF	OFF	OFF
SW4-2	Software upgrade	ON: Enable OFF: Normal mode				OFF	OFF	OFF
SW4-3	Unused	Normally OFF				OFF	OFF	OFF
SW4-4	Unused	Normally OFF				OFF	OFF	OFF

*1 To use the PC maintenance function on Tera Term, set SW2-4 to ON, and SW2-7 to OFF.

*2 The default settings for SW2-4, SW2-7, SW3-4, and SW3-6 of MobileDaRt Evolution depend on the type of digital radiography equipment installed in the system..

6.2 XCONT-2018 Board Jumper Setting

NOTE MobileArt Evolution has an XCONT-2018 board. XCONT-2008X board cannot be installed.

Table 6-2 XCONT-2018 Board Jumpers

Number	Pin Number	Silk	Setting	Remarks
JP1	1-2	IN	Shorted	-
	2-3	OUT	Open	-

6.3 INVERTER UNIT-32K Board

NOTE The INVERTER UNIT-32 K board is installed on 32kW type (MobileDaRt Evolution/MobileArt Evolution high-power type).

Table 6-3 INVERTER UNIT 32K Board Dip Switches

Number	Function	Switch Status: Function	Default
SW1-1	Unit Setting	Normally OFF	OFF
SW1-2	Not used	Normally OFF	OFF

6.4 INVERTER UNIT-C1 Board

NOTE The INVERTER UNIT-C1 K board is installed on 12.5kW type (MobileArt Evolution standard type)

Table 6-4 INVERTER UNIT-C1 Board Dip Switches

Number	Function	Switch Status/Function	Default
SW1-1	Unit Setting	Normally OFF	OFF
SW1-2	Not used	Normally OFF	OFF

6.5 POWER-200D Board

NOTE The POWER-200D board is only mounted on MobileDaRt Evolution.

Table 6-5 POWER-200D Board Dip Switches

Number	Function	Switch Status: Function	Default
SW1-1	Confirm Operation	ON: DR power can be turned ON without permission signal from XCONT	OFF

Number	Function	Switch Status: Function	Default
SW1-2	Not used	Always OFF	OFF

6.6 MU DRIVER 07 Board

Table 6-6 MU DRIVER 07 Board Jumpers

Number	Pin Number	Silk	Setting	Remarks
JP2	1-2	FAN 24V	Shorted	-
	2-3	FAN 5V	Open	-
JP3	1-2	PS	Shorted	-
	2-3	AS	Open	-
JP4	1-2	PS	Shorted	-
	2-3	AS	Open	-
JP5	1-2	PS	Shorted	-
	2-3	AS	Open	-

Table 6-7 MU DRIVER 07 Board DIP switches

Number	Function	Switch Status: Function	Default
SW1-1	Not in use	Always OFF	OFF
SW1-2	Not in use	Always OFF	OFF
SW1-3	Not in use	Always OFF	OFF
SW1-4	Not in use	Always OFF	OFF
SW3	FSR ASSY	NEW FSR ASSY: For connecting FSR L ASSY/566-20907, FSR R ASSY/566-20908	ON
		OLD FSR ASSY For connecting FSR L ASSY/563-74355, FSR R ASSY/563-74356	

Table 6-8 MU DRIVER 07 Board Rotary switches

Number	Information	Range*1	Remarks
0	Latest Error	Error code	Note 1
1	Left FSR*2	0 to 1023	Note 2
2	Right FSR*2	0 to 1023	Note 2
3	Left motor current	0 to 1023	Note 2
4	Right motor current	0 to 1023	Note 2
5	Left velocity	-511 to 511	Note 3
6	Left output	-127 to 127	Note 4
7	Right velocity	-511 to 511	Note 3
8	Right output	-127 to 127	Note 4
9	AD conversion value VOUT1	0 to 1023	Note 2
A	AD conversion value VOUT2	0 to 1023	Note 2
B	AD conversion value VOUT3	0 to 1023	Note 2
C	AD conversion value VOUT4	0 to 1023	Note 2
D	Not in use	-	-
E	SEG LED is OFF	-	-
F	Version	0 to 999	Note 5

*1 Values are displayed on 7 segments LED of MU Driver board.

*2 Force sensing resistor, used in the handle for control.

■ Note 1: "E00" is displayed if no error was recorded. If any error was recorded "E" and the sum of the error codes (in hexadecimal) from the following table will be displayed .

Table 6-9

Error Code	Content of Error
01	Left side FET is short-circuited.
02	Right side FET was short-circuited.
04	Not used
08	FET error.
10	FET thermal switch is ON.
20	Dry cell battery error.
40	Contactors (K11) contact is melt
80	XCONT-Communication error

■ Note 2: Described in unsigned 10 bit Hexadecimal.

Table 6-10

Display	Value
000	1
001	2
・ ・ ・	・ ・ ・
00A	10
・ ・ ・	・ ・ ・
3FF	1023

■ Note 3: Described in signed 10 bit Hexadecimal.

Table 6-11

Display	Value
000	1
001	2
・ ・ ・	・ ・ ・
1FF	511
200	Not in use
201	-511
・ ・ ・	・ ・ ・
3FE	-2
3FF	-1

■ Note 4: Described in signed Hexadecimal.

Table 6-12

Display	Value
000	1
001	2
・ ・ ・	・ ・ ・
07F	127
080	Not in use
081	-127
・ ・ ・	・ ・ ・
0FE	-2
0FF	-1

■ Note 5: Described in unsigned decimal.

Table 6-13

Display	Value
000	0
001	1
・ ・ ・	・ ・ ・
999	999

6.7 LED POWER-20 Board

NOTE Refer to the appropriate table according to P/N of LED LIGHT ASSY to be used.

Table 6-14 LED POWER-20 Board DIP switches (LED LIGHT ASSY P/N: 565-12375-02)

Number	LED Name			
	PX	PY	PZ	QX
SW1-1	OFF	OFF	ON	ON
SW1-2	ON	ON	OFF	ON
SW1-3	OFF	ON	ON	OFF

Table 6-15 LED POWER-20 Board DIP switches (LED LIGHT ASSY P/N: 565-14690-02)

Number	
SW1-1	ON
SW1-2	ON
SW1-3	OFF

*1 These settings are not required in the usual installation of the unit.

6.8 CL MEAS-17 Board

NOTE CL MEAS-17 board is only mounted on MobileDaRt Evolution.

Table 6-16 CL MEAS-17 Board DIP switches

Number	Function	Switch Status: Function	Default
SW1-1	AD_CH1	The switch is available only in selfcheck mode (SW1-4 to ON). ON: AD_CH1 is attached to V leaf OFF: AD_CH1 is attached to H leaf	OFF
SW1-2	Not in use	OFF	OFF
SW1-3	Not in use	OFF	OFF
SW1-4	Mode change	ON: self check mode OFF: normal mode	OFF

■ Self check mode

Table 6-17

Position to Check	7 Seg Display	Check Item
Serial communication	[J1]	Verify serial communication is working
SPI port	[J3]	For future use.
Analog 1	[J4]	AD value is in the normal range
Analog 2	[J5]	AD value is in the normal range
Aanalog 3	[J6]	AD value is in the normal range
Analog 4	[J7]	AD value is in the normal range
Vleaf position	[Cn2]	AD value is in the normal range
Hleaf position	[Cn3]	AD value is in the normal range
Digitail SW	[Cn4]	Verify that 2 switches are closed
Acceleration sensor	[ACC]	Execute selftest about acceleration sensor itself.
Analog power	[AnA]	Verify if analog power is stable

■ LED indication according to check status

Table 6-18

Check Status	LD1	LD2	7 Seg LED
Check in progress (For 7 secnds after SW1-4 to ON)	Blinks alternatively with LD2	Blinks alternatively with LD1	Displays [- - -]
Passed	Lights-up	Lights-up	Displays [PA.S]
Failed	Lights-up	Lights-out	After [FAL] blinking, displays "Position to check" with lighting if OK, blinking if failed*

* * E.g. [J1] to [J7] and [Cn3] to [AnA] are "Passed", and [Cn2] is "Failed": [FAL] blinks → [J1] lights → [J3] lights → [J4] lights → [J5] lights → [J6] lights →[J7] lights → [Cn2] blinks → [Cn3] lights → [Cn4] lights → [ACC] lights → [AnA] lights

6.9 MUX CHARGE-32KX Board

Table 6-19 MUX CHARGE-32KX Board DIP switches

Number	Function	Switch Status/Function	Default 32kW type	Default 12.5kW type
SW1-1	Unit Setting	OFF	OFF	ON
SW1-2	Unused	OFF	OFF	OFF
SW1-3	Unused	OFF	OFF	OFF
SW1-4	Unused	OFF	OFF	OFF

7 [Appendix] Adjustment Mode (For MobileDaRt Evolution)

7.1 How to Start Adjustment Mode

There are two methods to start the Adjustment Mode.

7.1.1 Method 1

1 **Remove the power plug from the outlet, and turn OFF the main system.**

2 **Remove the right-inside cover and right-side cover.**

Reference "3.5 How to Open/Close Covers"

3 **Turn ON the DIP switch SW3-8 on the XCONT-2018 board.**

4 **Turn ON the main power.**

"TECI" is displayed immediately on the kV/mAs display fields of the X-ray control panel.

Fig. 7-1 Initial display on adjustment mode

By pressing the Registration key for each setting described in this chapter, the setting value is registered and "record" is displayed on the kV/mAs display fields.

Fig. 7-2 "record" Display when registering

■ **How to finish adjustment mode**

1 **Turn OFF the main system.**

2 **Turn OFF the DIP switch SW3-8 on the XCONT-2018 board.**

3 **Attach the removed covers in reverse order.**

Reference "3.5 How to Open/Close Covers"

4 **Restart the main system.**

7.1.2 Method 2

This method allows to enter initialization and adjustment mode without setting DIP switches on the XCONT-2018 board.

Hint This method allows to perform some settings below without opening the covers.

- Adjustment of X-ray tube current
- Initialization
- Check the irradiation counter

X-ray tube current can easily be checked because the unit is automatically brought into the measured value display mode.

1 **Remove the power plug from the outlet, and turn OFF the power to te system.**

2 **Turn ON the power to the system.**

3 **Press and hold the Setting Change key for more than 3 seconds.**

The Setting Change Mode will start.

4 **Wait until the display field changes from** **to** **(Personal ID Setting). When the display is changed, press <7> and then the collimator Lamp key within 3 seconds.**

5 **SUP OPE appears for approx. 1 second, and then TEC appears on the dose indicator.**

Hint At this time, the version of S/W "X.XX"is displayed on mAs display field. If the Collimator Lamp Key is pressed,"8" will be displayed as the type of software on the mAs display.

⚠ CAUTION

Be sure to confirm software compatibility after the replacement of the XCONT-2018 board.
The installation of software incompatible with the unit output may cause damage to the unit.

Fig. 7-3 Display of software version (example of Ver.1.00)

Fig. 7-4 Display of software type

6 **In this state, the display will switch between TEC (adjustment mode) and IN I (initialization) by pressing the INFO switch.**

7 **Press Program Register key with the chosen mode. The chosen mode will start.**

8 **Refer to the following for details of each mode.**

Reference
- "7.2 Initialization of XCONT-2018 Board".
- "7.3 Adjustment Mode List"

■ How to finish the mode

1 **Turn OFF the power to te system.**

2 **Turn ON the power to the system.**
The system returns to normal mode.

7.2 Initialization of XCONT-2018 Board

NOTE When replacing the XCONT-2018 board, it must be initialized.

7.2.1 Procedure of Initialization

1 **Remove the power plug from the outlet, and turn OFF the power to te system.**

2 **Remove the right-inside cover and right-side cover.**

Reference "3.5 How to Open/Close Covers"

3 **Set the DIP switch and the Jumpers on the XCONT-2018 board depending on the configuration of the device.**

Reference
- "6.1 XCONT-2018 Board"
- "6.2 XCONT-2018 Board Jumper Setting"

4 **Set the DIP switch SW3-1 on the XCONT-2018 board as shown below.**

- DIPSW2-8: ON
- DIPSW3-1: ON

5 **Turn ON the main power of the system.**

6 **"ins" is displayed on X-ray control panel for about 40 seconds.**

7 **After "ins" disappears, press and hold the Registration key until "CAL end" appears.**

8 **Set the DIP switch SW2-8 and SW3-1 on the XCONT-2018 board to original position.**

- DIPSW2-8: OFF
- DIPSW3-1: OFF

NOTE Be sure to reset both DIP switches to the original position once the initialization is complete.

9 **Attach the removed covers in reverse order.**

Reference "3.5 How to Open/Close Covers"

7.3 Adjustment Mode List

Mode list in Adjustment Mode is shown below.

■ Adjustment mode Page 1 (Key <1> blinks, and INFO switch goes off)

Table 7-1

Key	Name of mode	Notes
4	Adjustment mode of the battery detection circuit	Adjustment by unit. ▶▶ Reference "7.6 Adjustment of Battery Voltage Detection Circuit"
5	Setting Date and Time	▶▶ Reference "7.7 Setting Date and Time"
6	Adjustment mode of the Handle	Adjustment by unit. ▶▶ Reference "7.5 Adjustment of the Handle"
7	Tube current two-point adjustment mode	The standard adjustment of tube current. ▶▶ Reference "7.4 Adjustment of Tube Current"
8	Tube current each-point adjustment mode Setting Date and Time	The detailed check and adjustment of tube current. (Not usually performed.) ▶▶ Reference "7.4 Adjustment of Tube Current"
9	Battery voltage drop check mode	This mode checks the condition of the battery under load (ex. radiography). ▶▶ Reference "7.10 Battery Voltage Drop Check Mode"

■ Adjustment mode Page 2 (Key <1> blinks, and INFO switch blinks)

Table 7-2

Key	Name of mode	Reference
4	Battery charging test mode	Perform the operation of charging for a minute with charging current setting 1A. This mode is used to confirm the operation of charging circuit. ▶▶ Reference "9.7 Checking battery charging"
5	Displaying running speed	▶▶ Reference "7.9 Display of Running Speed"
6	Battery code setting mode	▶▶ Reference "7.11 Battery Code Setting Mode"
7	DR setting mode	▶▶ Reference "7.21 DR Setting Mode"

Key	Name of mode	Reference
8	Dose area product meter adjustment mode (SW3-5ON) Dose calculation adjustment mode or Dose Area Product Calculation Unit Setting (SW3-5OFF)	Reference • "7.12 Dose Area Product Meter Adjustment Mode" • "7.20 Dose Area Product Calculation Unit Setting Mode"
9	Error Log mode	Reference "7.8 Display and Reset of Error Log"

■ Adjustment mode Page 3 (INFO switch goes off, and key <2> blinks)

Table 7-3

Key	Name of mode	Reference
4	-	Not used
5	-	Not used
6	-	Not used
7	-	Not used
8	-	Not used
9	-	Not used

■ Adjustment mode Page 4 (INFO switch goes off, and key <3> blinks)

Table 7-4

Key	Name of mode	Reference
4	Motor usage information setting mode	Reference "7.17 Motor Usage Information Setting Mode"
5	X-ray tube unit usage information setting mode	Reference "7.16 X-ray Tube Unit Usage Information Setting Mode"
6	Usage information collective setting mode	Reference "7.19 Usage Information Collective Setting Mode"
7	Battery usage setting mode	Reference "7.13 Battery Usage Setting Mode"
8	CPU board battery usage information setting mode	Reference "7.14 CPU Board Battery Usage Information Setting Mode"
9	Emergency brake release battery usage Information setting mode	Reference "7.15 Emergency Brake Release Battery Usage Information Setting Mode"

■ Adjustment mode Page 5 (INFO switch blinks, and key <3> blinks)

Table 7-5

Key	Name of mode	Reference
4	Mechanical performance usage information setting mode	Reference "7.24 Mechanical Performance Usage Information Setting Mode"
5	-	Not used
6	-	Not used
7	Arm usage information setting mode	Reference "7.18 Arm Usage Information Setting Mode"
8	Cord Reel usage information setting mode	Reference "7.22 Cord Reel Usage Information Setting Mode"
9	FSR usage information setting mode	Reference "7.23 FSR Usage Information Setting Mode"

7.4 Adjustment of Tube Current

7.4.1 Tube Current Adjustment Mode

The tube current of this unit is adjusted by changing the setting of the filament current (called FVR value).
As shown below, there are 6 points of adjustable FVR values; 40, 60, 80, 100, 125 and 133 kV for each tube current. FVR values between tube voltage points are calculated using linear interpolation.

■ Small focus

Table 7-6

	50 mA	63 mA	80 mA	100 mA	125 mA	160 mA	200 mA
133 kV	○	○	○	○	-	-	-
125 k V	○	○	○	○	○	-	-
100 k V	○	○	○	○	○	○	-
80 k V	○◎	○	○	○	○	○	○◎
60 k V	○	○	○	○	○	○	○
40 k V	○	○	○	○	○	○	○

* ○ : Adjustment point in Tube Current Each-point Adjustment Mode
◎ : Adjustment point in Tube Current Two-point Adjustment Mode

■ Large focus

Table 7-7

	50 mA	63 mA	80 mA	100 mA	125 mA	160 mA	200 mA	250 mA	320 mA	400 mA
133 kV	○	○	○	○	○	○	○	-	-	-
125 k V	○	○	○	○	○	○	○	○	-	-
100 k V	○	○	○	○	○	○	○	○	○	-
80kV	○◎	○	○	○	○	○	○	○	○	○◎
60kV	○	○	○	○	○	○	○	○	○	○
40 k V	○	○	○	○	○	○	○	○	○	-

* ○ : Adjustment point in Tube Current Each-point Adjustment Mode
◎ : Adjustment point in Tube Current Two-point Adjustment Mode

There are two types of tube current adjustment mode. Normally, the tube current is adjusted with (1) Tube Current Two-point Adjustment Mode. When necessary, the tube current at each adjustment point is checked and adjusted with (2) Tube Current Each-point Adjustment Mode

- (1) Tube Current Two-point Adjustment Mode
  FVR value is adjusted with two points; minimum and maximum tube current of 80kV. The FVR values of other points are calculated based on the two values.

- (2) Tube Current Each-point Adjustment Mode
  FVR values are adjusted at each adjustment point.

> **NOTE** The Tube Current Two-point Adjustment Mode is to correct the currently-set FVR value table based on the adjustment results of the two points. Therefore, if any currently-set FVR value is invalid, the correction results could be wrong. Be sure to initialize FVR value before the tube current two-point adjustment.

7.4.2 To Initialize FVR Values

> **NOTE** The initialization is not required in the standard installation procedure of the unit. Perform the initializing methods in case such as when the circuit boards or other parts are replaced and readjustment is required.

1 **Press the focus key you choose to initialize the FVR value.**

2 **Press the key <0> for about 3 seconds until "c A n c e L" appears on the display.**
FVR value of selected focus will be initialized only.

> **NOTE** FVR value can also be initialized by initializing the XCONT-2018 board. However, other important data will also be initialized.

7.4.3 Others

- When the Tube Current Two-point or Each-point Adjustment Mode is selected, the tube current feedback circuit is automatically disabled. Radiography time is fixed at 20msec.
- To adjust the tube current, use the secondary tube current meter or neutral point tube current meter, or enter the actual measurement mode on the system. (During the tube current adjustment mode, the system automatically enters the actual measurement display mode.)

> **NOTE** The tube current of the same FVR value may differ due to the presence of the tube voltmeter connection. Make sure that the tube voltmeter is disconnected when the tube current adjustment is performed.

7.4.4 Tube Current Two-point Adjustment Mode

> **NOTE** This mode is to calculate the currently-set FVR value table based on the adjustment results of the two points. Therefore, if any currently-set FVR value is invalid, the correction results could be wrong. Be sure to initialize FVR value before the tube current two-point adjustment.

1 **Enter the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

2 **Press the key <1> to enter the Tube Current Two-point Adjustment Mode.**

LED of the switch and keys are indicated as follows.

- INFO switch goes off
- Key <1> blinks
- Key <7> blinks

3 **Tube voltage is displayed on kV display field and tube current is displayed on mAs display field when enter the Tube Current Two-point Adjustment Mode.**

4 **Press focus key and select small focus.**

Focus key will light.

5 **Set the tube voltage and tube current by pressing each key.**

The tube voltage is fixed at 80kV. The tube current can be set at two points, minimum and maximum. Set the minimum tube current first.

Table 7-8

Focus type	Adjustment point of tube current	Tube current value
Small focus (Focus key lighting)	Minimum	50mA
	Maximum	200mA
Large focus (Focus key light out)	Minimum	50mA
	Maximum	400mA

Fig. 7-5 Tube current two-point adjustment mode (the tube current is set to 50 mA)

6 Execute X-ray exposure.

Execute the exposure by pressing the hand switch.

7 Adjust the FVR value.

1 Press the Additional Filter key to show the FVR value on the tube voltage display. (The Additional Filter key blinks when the FVR value is displayed.)

2 Use the the Tube Voltage key to change the FVR value.

Hint

- To decrease the tube current, set the FVR value to a smaller value.
- To increase the tube current, set the FVR value to a larger value.

8 Repeat the steps 6 and 7 until the measured tube current value is within the recommended range.

Table 7-9

Focus type	Adjustment point	Recommended range
Small focus (Focus key lighting)	80kV 50mA	45mA to 50mA
	80kV 200mA	185mA to 200mA
Large focus (Focus key light out)	80kV 50mA	45mA to 50mA
	80kV 400mA	370mA to 400mA

Fig. 7-6 Tube current two-point adjustment mode-FVR value setting (the tube current is set to 50mA, the FVR ajusted to FVR = 3.92)

9 Register FVR values.

1 Register the FVR value after the tube current adjustment is complete.

2 To register the FVR value, press the Registration switch. Characters "r e c o r d" will show on the display.

NOTE Be sure to press the Registration key when the FVR value is changed. Otherwise, the changed FVR value will become invalid if the other point or mode is moved, or the switch is turned off. FVR value can also be registered on the display of the exposure condition (instead of the FVR value).

10 **Set the maximum tube current, and then repeat the steps 6 to 9.**

> NOTE However, the same adjustment results can be obtained in reverse order.
> If this mode is terminated with only one point setting being performed, the automatic adjustment on this mode is not executed.

11 **Press the Focus key and select Large focus. Repeat the steps 5 to 10.**

12 **Finish the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

7.4.5 Tube Current Each Point Adjustment Mode

■ How to Adjust FVR value

1 **Enter the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

2 **Press the key <8> to enter the Tube Current Each Point Adjustment Mode.**

LED of the switch and keys are indicated as follows.

- INFO switch goes off
- Key <1> blinks
- Key <8> blinks

3 **Tube voltage is displayed on kV display field and tube current is displayed on mAs display field when enter the Tube Current Each Point Adjustment Mode.**

4 **Select focus.**

Hint

- Small focus : focus key LED ON
- Large focus : focus key LED OFF

5 **Set the tube voltage and tube current by pressing each key.**

Fig. 7-7 Tube current each point adjustment mode – Exposure condition setting (In case of 160mA)

6 **Execute X-ray exposure.**

Execute the exposure by pressing the hand switch.

7 Adjust the FVR value.

1 Press the Filter key to show the FVR value on the tube voltage display. The Filter key blinks when the FVR value is displayed.

2 Use the Tube Voltage key to change the FVR value.

Hint
- To decrease the tube current, set the FVR value to a smaller value.
- To increase the tube current, set the FVR value to a larger value.

8 Repeat the steps 5 and 6 until the measured tube current value is within -15% to +5% of the set value.

Fig. 7-8 Tube current each point adjustment mode: FVR value adjusted (this figure shows an example of the display: the FVR is setting to FVR = 3.87)

9 Register FVR values.

Register the FVR value after the tube current adjustment is complete. To register the FVR value, press the Registration key. Characters "r e c o r d" will show on the display.

NOTE Be sure to press the Registration key when the FVR value is changed. Otherwise, the changed FVR value will become invalid if the other point or mode is moved, or the switch is turned off. FVR value can also be registered on the display of the exposure condition (instead of the FVR value).

10 Repeat the steps 5 to 8 for other points, as required.

11 Press focus key and select focus. Repeat the steps 4 to 9 for the other focus.

Hint
- Small focus : focus key LED ON
- Large focus : focus key LED OFF

12 Finish the adjustment mode.

Reference "7.1 How to Start Adjustment Mode"

7.5 Adjustment of the Handle

In this mode, the handle is adjusted to control the driving performance of the unit. Perform the following procedures if the handle or the XCONT-2018 board is exchanged.

1 **Enter the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

2 **Pess the key <6> to enter the Adjustment of the Handle Mode.**

LED of the switch and keys are indicated as follows.

- INFO switch goes off
- Key <1> blinks
- Key <6> blinks

3 **Set a spring gauge.**

As the adjustment position number is indicated on the tube voltage display, apply the force in the indicated direction with a spring gauge

Reference "Table 7-10 Relation between handle adjustment position number and pull power"
"Fig. 7-9 Adjustment of the Handle"

Tie a piece of string or the like round the end of the handle, then hook the spring gauge to the string, and apply the force. When driving handle pull forward, the adjustment value increase. When driving handle pull backward, the adjustment value decrease.

Fig. 7-9 Adjustment of the Handle

4 **Register the adjustment values (12 positions)**

Press the Registration swich. The next adjustment position number will appear. Repeat steps 3 and 4 with all the adjustment positions shown in the following table.

NOTE If the system is powered OFF without pressing the Registration key, the adjusted value will not be registered.

Table 7-10 Relation between handle adjustment position number and pull power

Adjustment position number	Direction and weight of pulling the spring gauge
1*1	Forward left direction (1 kg)*3
2	Forward left direction (1.5 kg)
3	Forward left direction (3 kg)
4	Backward left direction (1 kg)*3
5	Backward left direction (1.5 kg)
6	Backward left direction (3 kg)
7*2	Forward right direction (1 kg)*3
8	Forward right direction (1.5kg)
9	Forward right direction (3 kg)
10	Backward right direction (1 kg)*3
11	Backward right direction (1.5 kg)
12	Backward right direction (3 kg)

*1 When the adjustment position number indicates 1, pull the spring gauge backward left direction with the force of 3 kg, and then forward left direction with 1 kg.

*2 When the adjustment position number indicates 7, pull the spring gauge backward right direction with the force of 3 kg, and then forward right direction with 1 kg.

*3 To pull the spring gauge forward left, backward left, forward right, or backward right direction with the force of 1 kg, pull it gradually within 1kg.

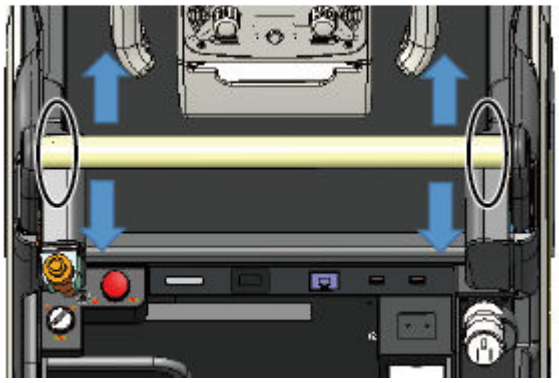


Fig. 7-10 Positions and directions for setting the spring gauge

5 **Press the Filter key to display the adjustment value on the X-ray control panel.**

"r" is displayed in front of the adjustment position number.
After adjustment, press ±kV keys to choose the adjustment position number, and make sure the adjustment values for each point are correctly registered.

NOTE If relation between the adjustment values are as described below, normal driving operability cannot be obtained. Be sure to adjust the values correctly.

- The values between 2 adjustment positions in the same direction are the same.
- The values are not sequential with the magnitude of the pull force.

6 **Finish the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

7.6 Adjustment of Battery Voltage Detection Circuit

This mode adjusts battery voltage detection circuit. Perform the following procedures when exchanging the MUX POWER board, the MUX CHARGE-32KX board or the XCONT-2018 board.

1 **Enter the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

2 **Press the key <4> to enter the Adjustment of the Battery Voltage Detection Circuit Mode. The detected battery voltage is displayed with four digits and 0.1V unit. Press the tube voltage key to select [+] or [-]. If [-] is selected, "-" is displayed at the second digits of kV display field.**

LED of the switch and keys are indicated as follows.

- INFO switch goes off
- Key <1> blinks
- Key <4> blinks

3 **Press [mAs +] and [mAs -] key to adjust the displayed value. Press the Registration key if the displayed value is within ±0.2V of the measured battery voltage.**

Hint

- Measure the actual battery voltage on + side with TER3 [B120+O] - TER5 [B0] of the MUX POWER-32K board.
- Measure the actual battery voltage on - side with TER5 [B0] -TER4 [B120-O] of the MUX POWER-32K board.

Fig. 7-11 Adjustment of the battery voltage detection circuit –Positive voltage (in case of 119.7V)

Fig. 7-12 Adjustment of the battery voltage detection circuit –Negative voltage (in case of 119.1V)

4 **Finish the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

7.7 Setting Date and Time

Set the present year, month, day and time for the unit with this mode.

1 Enter the adjustment mode.

Reference "7.1 How to Start Adjustment Mode"

2 Press the key <5> to enter the Date and Time Setting Mode.

LED of the switch and keys are indicated as follows.

- INFO switch goes off
- Key <1> blinks
- Key <5> blinks

Fig. 7-13 Setting Date (In case of February 3, 2018)

3 Change the date.

1. Press the Additional Filter key to enter the Year/Month/Day Setting Mode. The Additional Filter key LED turns OFF.
2. Press Tube Voltage keys to set the lower two digits of the year.
3. Press Current-Time Product (-) key to change the month.
4. Press Current-Time Product (+) key to change the date.

4 Change the time.

1 Press the Additional Filter key to enter the Time Setting Mode.
The Additional Filter key LED blinks.

2 Press the Tube Voltage keys to change the hour (displayed in 24-hour mode).

3 Press the Current-Time Product keys to change the minute.

Fig. 7-14 Setting Time (In case of PM 1:08)

5 Finish the adjustment mode.

Reference "7.1 How to Start Adjustment Mode"

7.8 Display and Reset of Error Log

In this mode error codes of the past is shown (maximum 100 times, in chronological order). If 100 errors are already recorded, the next new data will overwrite the oldest data.

7.8.1 To Check Error Log

1 **Enter the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

2 **Press the INFO switch to blink, and then the key <9> to enter the Error Log Mode.**

LED of the switch and keys are indicated as follows.

- INFO switch blinks
- Key <1> blinks
- Key <9> blinks

Fig. 7-15 Display of Error Log (In case of the latest error code was F03)

Fig. 7-16 Display of Error Log (In case of the latest error code was F68)

3 **Check the error codes.**

Error code is displayed on mAs display field and log number is displayed on kV display field.
"Fault" is displayed on the dose display field for failure, and "Alert" is displayed for warning.
The number "000" means the latest error and the number "099" means the oldest error. Press [kV-] to increase a log number and press [kV+] to decrease a log number. Change the Failure/Warning display with the Tube Current log number with the Current-Time Product keys.

Reference "10.1 Error message list"

4 **Check the date and time of the error.**

Press the Additional Filter key to switch between the "date and time" page and the "log number and error code" page.

7.8.2 To Reset Error Log

1 **Press the Registration key for more than 3 seconds.**

The error log is reset.

7.8.3 To Finish Error Log Display Mode

1 **Finish the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

7.9 Display of Running Speed

The running speed is calculated from the number of rotation detected by the encoder in the motor. Use this for checking failure of the encoder.

1 **Enter the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

2 **Press the INFO switch to blink, and then the keys in order of <1> and <5> to enter the Running Speed Display Mode.**

LED of the switch and keys are indicated as follows.

- INFO switch blinks
- Key <1> blinks
- Key <5> blinks

3 **Check the displayed running speed.**

The running speed is displayed on the kV and mAs display fields.

Hint The character "-" is displayed when driving backward.

Fig. 7-17 An example of running speed display (where the speed of both of the wheels is 0 km/hour)

4 Show the speed of each wheel.

Press the Additional Filter Key to show the following running speed.

- Running speed of both wheels (km/h)
- Running speed of the left wheel (km/h)
- Running speed of the right wheel (km/h)

Fig. 7-18 An example of running speed display (left wheel speed)

Fig. 7-19 An example of running speed display (right wheel speed)

5 Finish the adjustment mode.

Reference "7.1 How to Start Adjustment Mode"

7.10 Battery Voltage Drop Check Mode

Use this mode to check the condition of the battery and the inverter operation (VIN) when load of the X-ray exposure is applied.

> **NOTE** This mode cannot be entered during the low battery mode.

1 Enter the adjustment mode.

Reference "7.1 How to Start Adjustment Mode"

2 Press the key <9> to enter the Battery Voltage Drop Check Mode.

LED of the switch and keys are indicated as follows.

- INFO switch goes off
- Key <1> blinks
- Key <9> blinks

The exposure condition for inspection is shown on the mAs and kV display fields (80kV, 200mA, 100 msec).

Fig. 7-20 Example of the display of the battery voltage drop check mode (exposure condition is displayed)

3 Check the battery voltage drop.

1. Perform the X-ray exposure by pressing the hand switch.
   Values of tube voltage and tube current are measured after exposing for 80 msec.
2. The measured values are shown on the display.

> **Hint** In other modes, the tube voltage and tube current are measured after exposing for 20 msec. These values may not match the measured value of the Battery Voltage Drop Check Mode.

4 Show the measured values.

Press the Additional Filter key to show the following measured value.

- Exposure condition
- Battery voltage on the plus side
- Battery voltage on the minus side
- Measured tube voltage
- Measured tube current

Fig. 7-21 Example of the display of the battery voltage drop check mode (112.8V of the battery voltage on the plus side is displayed)

Fig. 7-22 Example of the display of the battery voltage drop check mode (110.1V of the battery voltage on the minus side is displayed)

Fig. 7-23 Example of the display of the battery voltage drop check mode (80kV of the measured tube voltage is displayed)

Fig. 7-24 Example of the display of the battery voltage drop check mode (192mA of the measured tube current is displayed)

Hint Battery voltage (plus and minus side) are measured 80 msec after the exposure starts in this mode. They do not match the value measured with the following section.

- "7.5 Adjustment of the Handle"

NOTE The measured value provided in this mode does not reflect the performance of the unit and the battery.

5 Finish the adjustment mode.

Reference "7.1 How to Start Adjustment Mode"

7.11 Battery Code Setting Mode

This mode allows to set the battery code corresponding to each battery type.

7.11.1 To Enter Battery Code Setting Mode

1 **Enter the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

2 **Press the INFO switch to blink, and then press the key <6> to enter the battery code setting mode.**

LED of the switch and keys are indicated as follows.

- INFO switch blinks
- Key <1> blinks
- Key <6> blinks

7.11.2 To Check Battery code

1 **The Setting Item Number "0" is shown at the third digit (from the right) of kV display.**

The "0" indicates the Battery Code Display Mode.

2 **The battery code is shown at the first digit of mAs display. The Battery code5, GS YUASA PXL12090 is registered as default data.**

Table 7-11

Battery code	Battery name
0	Reserved
1	Reserved
2	Reserved
3	Reserved
4	Reserved
5	GS Yuasa PX12090 or GS Yuasa PXL12072H FR
6	GS Yuasa PXL12090
E	New battery other than 0 - 6

7.11.3 To Change Battery Code

1 **Press [mAs±] key and select the battery code corresponding to the battery type, and then the Registration key to register.**

Fig. 7-25 Battery code setting (when Battery code6 is selected)

7.11.4 Setting of Battery Charge Restriction

For overcharge prevention of a battery, this system controls not to charge the battery if the battery remains enough.

Follow the instruction below to make sure the battery charge restriction is enabled. If not, set this setting to be enabled.

> **NOTE** Repetition of charging the battery even though the battery remains enough may cause the deterioration of the battery life and performance. The battery charge restriction must be enabled for overcharge prevention of the battery.

1 **On the battery code setting mode, press the Additional Filter key, then start the battery charge restriction setting mode.**

2 **At the third digit of kV display, the setting item number 9 (Battery charge restriction setting) is displayed.**

3 **The battery charge restriction ON/OFF setting is displayed at mAs display.**

Fig. 7-26 Battery charge restriction setting mode (Battery charge restriction is enabled)

Fig. 7-27 Battery charge restriction setting mode (Battery charge restriction is disabled)

4 **Press [mAs+] key, set the battery charge restriction ON.**

5 **When the battery charge restriction is set completely, press the Registration key to register.**

6 **Press the Added filter key to enter the setting mode of status indicator lamp during battery charge.**

7.11.5 Setting of Status Indicator Lamp During Battery Charge

The status indicator lamp blinks in blue while the battery is charging.
Follow the instruction below to make sure the status indicator lamp blinking during the battery charge is enabled. If being disabled unintentionally, set this setting to be enabled.

> NOTE The setting is supported by software version "1.50" or later.

1 **Press the Added filter button on the battery charge restriction setting mode, to enter the Status indicator lamp during battery charge setting mode.**

2 **The setting item number 10 (Status indicator lamp during battery charge setting) is displayed at the third digit of kV display.**

3 **The battery charge ON/OFF setting is displayed at mAs display.**

Fig. 7-28 Status indicator lamp during battery charge setting mode is ON

Fig. 7-29 Status indicator lamp during battery charge setting mode is OFF

4 **Press the Current-time product (+) button to set the status indicator during the battery charge.**

5 **When thesetting is complited, press the Registration button to register.**

6 **Press the Added filter button to return to the battery code setting mode.**

7.11.6 To Finish Battery Code Setting Mode

1 **Finish the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

7.12 Dose Area Product Meter Adjustment Mode

7.12.1 To Enter Dose Area Product Meter Adjustment Mode

1 **Remove the power plug from the outlet, and turn OFF the power to the system.**

2 **Remove the right-inside cover and right-side cover.**

Reference "3.5 How to Open/Close Covers"

3 **Set the DIP switches of the XCONT-2018 board as follows. Turn ON the main system power.**

- SW3-5(Dose area product meter) : ON
- SW3-8 (Adjustment mode) : ON

NOTE The SW3-5 must be ON to enter the dose area product meter adjustment mode.

4 **Press the INFO switch to blink, and then the key <8> to enter the dose area product meter adjustment mode.**

LED of the switch and keys are indicated as follows.

- INFO switch blinks
- Key <1> blinks
- Key <8> blinks

5 **"dAP 2" is displayed on the kV/mAs display fields. Press the Additional Filter key and go on to the radiography condition display.**

6 **The kV/mAs display fields on X-ray control panel shows the radiography conditions. Radiography conditions are 70kV and 2.0mAs.**

At this point, nothing is displayed at the mGy display because radiography has not been executed with the dose area product meter.

Fig. 7-30 Dose area product meter adjustment mode (Radiography condition display)

7.12.2 Preparation for Calibration of Dose Area Product Meter

1 **Attach the dose area product meter to the collimator and place the Reference Dose meter as shown in the following figure.**

1 Adjust SOD (source object distance) to 100 cm, and the exposure field to 35 cm (approx. 14 inch) square.

2 Place the Reference Dose meter, in the center of the exposure field apart from the floor surface to eliminate the effect of the backward scattered ray (with styrol box, for example).

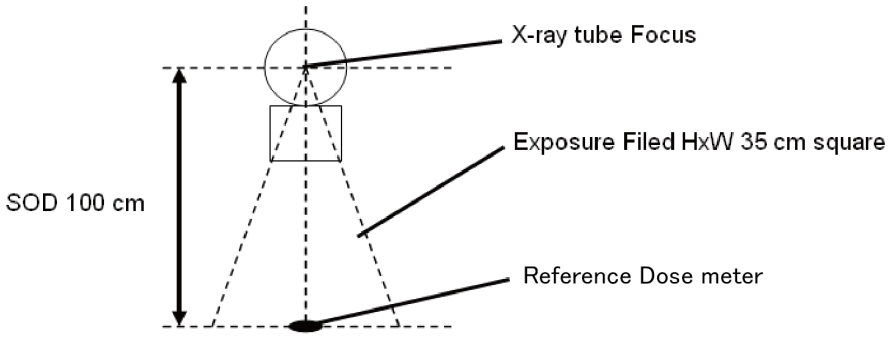


Fig. 7-31 Setting Dose Area Product Meter for calibration

7.12.3 Calibration of Dose Area Product Meter

1 **Press the Additional Filter key to enter the calibration of dose area product meter mode.**

2 **In the calibration of dose area product meter mode, the adjustment value shows on the kV and mAs display fields of X-ray control panel. (LED of Additional Filter key is blinking). The initial adjustment value is 100.**

At this point, nothing is displayed at the mGy display because radiography has not been executed with the dose Area Product Meter.

3 **Take the radiography and measure the exposure dose.**

After the radiography is completed, the exposure dose (Unit : mGy) calculated from the Dose Area Product is displayed.

Fig. 7-32 Dose area product meter adjustment mode (Adjustment value, Exposure dose display)

4 **Press the Current-Time Product key, to change the Adjustment Value.**

The exposure dose amount (Unit:mGy) display changes with the Adjustment Value. Adjust the Value so that this exposure dose and the measured dose (Reference Dose meter) are equal.

5 **Press the Registration key to register the Adjustment Value.**

6 **Press the Additional Filter key.**

X-ray control panel display is changed, then the radiography conditions will appear on the kV and mAs display on the panel.

7 **Check that the calibrated Dose Area Product is displayed at the μGm^2 display.**

Fig. 7-33 Dose area product meter adjustment mode (Radiography condition, dose area product display)

7.12.4 To Finish Dose Area Product Meter Adjustment Mode

1 **Turn OFF the power.**

2 **Set the DIP switches of the XCONT-2018 board as follows.**

- DIPSW3-5 (Dose area product mete): ON
- DIPSW3-8 (Adjustment mode): OFF

3 **Attach the removed covers in reverse order.**

Reference "3.5 How to Open/Close Covers"

7.13 Battery Usage Setting Mode

Battery usage setting mode is used to check the starting date of battery usage and the charging count, and to set the battery usage information when the battery ,or the XCONT-2018 board is replaced. Adjust the setting with the following procedure when the battery ,or the XCONT-2018 board is replaced.

7.13.1 To Enter Battery Usage Setting Mode

1 **Enter the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

2 **Press keys in order of <3> and <7> to enter the battery usage setting mode.**

LED of the switch and keys are indicated as follows.

- INFO switch goes off
- Key <3> blinks
- Key <7> blinks

7.13.2 To Check Battery Usage Information

The starting date of the battery usage information is displayed at kV and mAs displays. Battery usag and the charge count are displayed at the first to fifth digits of mGy display. Press the Collimator Lamp key to switch the usage information on the mGy display.

Table 7-12

6th Digits of mGy	Item
0	Charge count (by single count)
1	Full charge count (by single count)
2	Charge limit count (by single count)
3	Low battery occurence count (by single count)
4	Error bbb occurence count (by single count)
5	Error bb0 occurence count (by single count)
6	Error bb1 occurence count (by single count)

Fig. 7-34 EXAMPLE: Battery Starting date of use and battery Charge count display (Feb.1.2018, 58 counts)

7.13.3 To Reset Battery Usage Information

It is required to reset the starting date of the battery usage when the battery is replaced.

1 **While the usage information is displayed, press and hold the Registration key for 3 seconds.**

The starting date of the battery usage is reset to current date, and the charge count is reset to 0.

7.13.4 To Finish Battery Usage Setting Mode

1 **Finish the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

7.14 CPU Board Battery Usage Information Setting Mode

This mode is used to check the starting date of CPU board battery usage, and to set the usage information of CPU board battery when the CPU board battery or, the XCONT-2018 board is replaced. Adjust the setting by the following procedure when the CPU board battery, or the XCONT-2018 board is replaced.

7.14.1 To Enter CPU Board Battery Usage Information Setting

1 **Enter the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

2 **Press keys in order of <3> and <8> to enter the CPU board battery usage information setting mode.**

LED of the switch and keys are indicated as follows.

- INFO switch goes off
- Key <3> blinks
- Key <8> blinks

7.14.2 To Check CPU Board Battery Usage Information

The usage information is displayed at kV and mAs displays.

Fig. 7-35 EXAMPLE: CPU board battery's starting date of use (Feb.3.2018)

7.14.3 To Reset CPU Board Battery Usage Information

It is required to reset the starting date of use when CPU board battery is replaced.

1 **While the usage information is displayed, press and hold the Registration key for 3 seconds.**

The starting date of CPU board battery usage is set to the current date.

7.14.4 To Finish CPU Board Battery Usage Information Setting

1 **Finish the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

7.15 Emergency Brake Release Battery Usage Information Setting Mode

This mode is used to check the starting date of use of the emergency brake release battery, and to set its starting date of use when the emergency brake release battery, or the XCONT-2018 board is replaced. Adjust the setting by the following procedure when the emergency brake release battery, or the XCONT-2018 board is replaced.

7.15.1 To Enter Emergency Brake Release Battery Usage Information Setting Mode

1 **Enter the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

2 **Press the keys in order of <3> and <9> to enter the emergency brake release battery usage information setting mode.**

LED of the switch and keys are indicated as follows.

- INFO switch goes off
- Key <3> blinks
- Key <9> blinks

7.15.2 To Check Emergency brake release battery usage information

The usage information is displayed at kV and mAs displays.

Fig. 7-36 EXAMPLE: Emergency brake release battery's starting date of use (Feb.3.2018)

7.15.3 To Reset Emergency Brake Release Battery Usage Information

It is required to reset the starting date of use after replacing the emergency brake release battery.

1 **While the usage information is displayed, press and hold the Registration key for 3 seconds.**

The starting date of Emergency brake release battery is set to the current date.

7.15.4 To Finish Emergency Brake Release Battery Usage Information Setting Mode

1 **Finish the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

7.16 X-ray Tube Unit Usage Information Setting Mode

This mode is used to check the X-ray tube unit's starting date of use and the count of radiography, and to set the X-ray tube unit usage information when the X-ray tube unit, or the XCONT-2018 board is replaced. Adjust the setting by the following procedure when the X-ray tube unit, or the XCONT-2018 board is replaced.

7.16.1 To Enter X-ray Tube Unit Usage Information Setting Mode

1 **Enter the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

2 **Press the keys in order of <3> and <5> to enter the X-ray tube unit usage information setting mode.**

LED of the switch and keys are indicated as follows.

- INFO switch goes off
- Key <3> blinks
- Key <5> blinks

7.16.2 To Check X-ray Tube Unit Usage Information

The usage information is displayed at kV and mAs displays.

Battery usage and the charge count are displayed at the first to fifth digits of mGy display. Press the Collimator Lamp key to switch the usage information on the mGy display.

Table 7-13

6th Digits of mGy	Item
0	Radiography count
1	Accumulated mAs
2	Tube discharge count
3	Accumulated filament heating time
4	Accumulated starter rotation time

Fig. 7-37 EXAMPLE: X-ray tube unit's starting date of use and count of radiography (Feb.1.2018, 26 counts)

7.16.3 To Reset X-ray Tube Unit Usage Information

It is required to reset the starting date of use after the X-ray tube unit is replaced.

1 **While the usage information is displayed, press and hold the Registration key for 3 seconds.**

The starting date of X-ray tube unit usage is set to the current date, and the count of radiography is set to 0.

7.16.4 To Finish X-ray Tube Unit Usage Information Setting Mode

1 **Finish the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

7.17 Motor Usage Information Setting Mode

This mode is used to set the usage information, such as the starting date of use of motor and the accumulated running distance, when the motor is replaced. Adjust with the following procedure.

7.17.1 To Enter Motor Usage Information Setting Mode

1 **Enter the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

2 **Press keys in order of <3> and <4> to enter the motor usage information setting mode.**

LED of the switch and keys are indicated as follows.

- INFO switch goes off
- Key <3> blinks
- Key <4> blinks

7.17.2 To Check Motor Usage Information

The usage information is displayed at kV and mAs displays.

The accumulated running distance and the starting date of use of motor is displayed at mGy display. Press the Collimator Lamp key to switch the usage information on the mGy display.

Fig. 7-38 EXAMPLE: Starting date of use of motor and accumulated running distance of motor (Feb.1.2018, 16km)

7.17.3 To Reset Motor Usage Information

It is required to reset the starting date of use after replacement of the motor.

1 **While the usage information is displayed, press and hold the Registration key for 3 seconds.**

The starting date of the motor usage is set to the current date, and the accumulated running distance is set to 0.

7.17.4 To FinishX-ray Tube Unit Usage Information Setting Mode

1 **Finish the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

7.18 Arm Usage Information Setting Mode

This mode is used set the usage information, such as wire's starting date of use and the arm moving distance, when the wire is replaced. Adjust with the following procedure.

7.18.1 To Enter Arm Usage Information Setting Mode

1 **Enter the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

2 **Press the INFO switch to blink, and then press the keys in order of <3> and <7> to enter the arm usage information setting mode.**

LED of the switch and keys are indicated as follows.

- INFO switch blinks
- Key <3> blinks
- Key <7> blinks

7.18.2 To Check Arm Usage Information

The usage information is displayed at kV and mAs displays.

The accumulated running distance and the starting date of use of motor. Press the Collimator Lamp key to switch the usage information on the mGy display.

Table 7-14

6th Digits of mGy	Item
0	Arm upward moving distance
1	Arm downward moving distance
2	Encoder usage setting

Fig. 7-39 EXAMPLE: Starting date of use of wire and accumulated moving distance of arm (Feb.1.2018, 3km)

7.18.3 To Reset Arm Usage Information

It is required to reset the starting date of use after replacement of the wire.

1 **While the usage information is displayed, press and hold the Registration key for 3 seconds.**

The starting date of the wire usage is set to the current date, and the arm moving distance is set to 0.

7.18.4 To Change Encoder Usage Setting

When the encoder is installed in the system without encoder, the encoder usage setting is required to be "on".

NOTE

- The encoder usage setting is available for the software version 1.10 or later.
- To update the software version from less than 1.10 to 1.10 or later, the encoder usage setting should be defaulted to "oFF".
- If the encoder usage setting is set to "on" on the system without encoder, "d62" error is detected by locking the X-ray tube unit to the arm interlocking unit.

1 **Show the encoder usage setting values on the control panel., and press and hold the Additional Filter key for three seconds .**

The encoder usage setting is set to "on".

Fig. 7-40 EXAMPLE: Display of the encoder usage setting ("on" is set)

7.18.5 To Finish Arm Usage Information Setting Mode

1 **Finish the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

7.19 Usage Information Collective Setting Mode

This mode is used to reset the starting date of use of components collectively, such as the battery, the CPU board battery, the emergency brake release battery, the X-ray tube unit, the motor, and the charging count, radiography count and the accumulated running distance.

7.19.1 To Enter Usage Information Collective Setting Mode

1 **Enter the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

2 **Press the INFO switch to turn off the light, and then the keys in order of <3> and <6> to enter the Usage Information Collective Setting Mode.**

LED of the switch and keys are indicated as follows.

- INFO switch goes off
- Key <3> blinks
- Key <6> blinks

7.19.2 To Reset Usage Information of All Components

When provide the system or replace all components, it is required to reset the usage information of all components.

1 **In this mode, the current date is displayed at the kV and mAs displays.**

2 **Press and hold the Registration key for 5 seconds.**

The starting dates of use of all components are set to the current date. The count of charging, the count of radiography, and the accumulated running distance are set to 0.

Fig. 7-41 EXAMPLE: Usage information collective setting mode (When the current date is Feb.3.2018)

7.19.3 To Finish Usage Information Collective Setting Mode

1 **Finish the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

7.20 Dose Area Product Calculation Unit Setting Mode

> **NOTE** When the registered information is modified, that information must be saved in the memory by pressing the Registration keyon the initial screen of the control panel.
>
> The registered information in each adjustment mode is the interim registration. If the adjustment mode is finished without saving in the memory, the modified information cannot be refleted to the system.

7.20.1 To Remove Collimator Cover

1 **Remove the power plug from the outlet, and turn OFF the power to the system.**

2 **Remove the collimator cover, and remove screws at eight positions as shown below.**

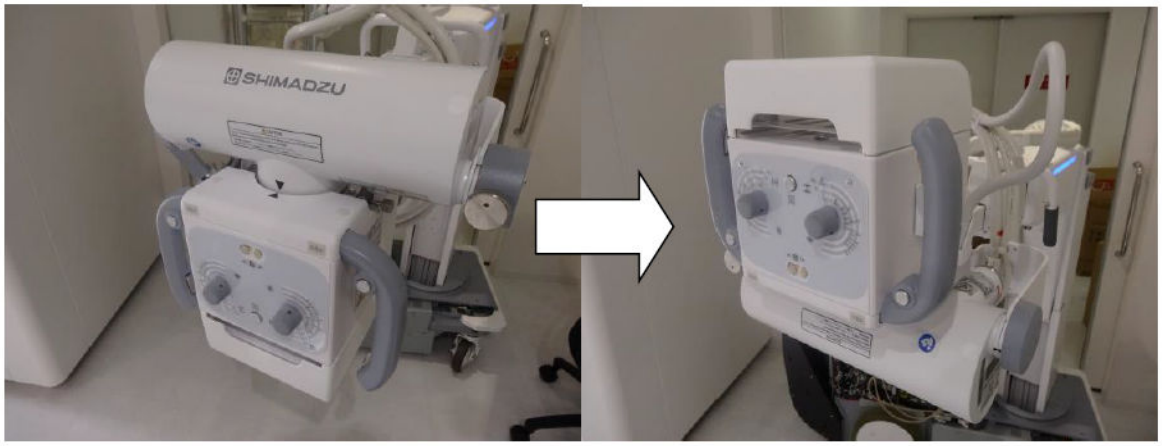


Fig. 7-42

Fig. 7-43 Front side of Collimator

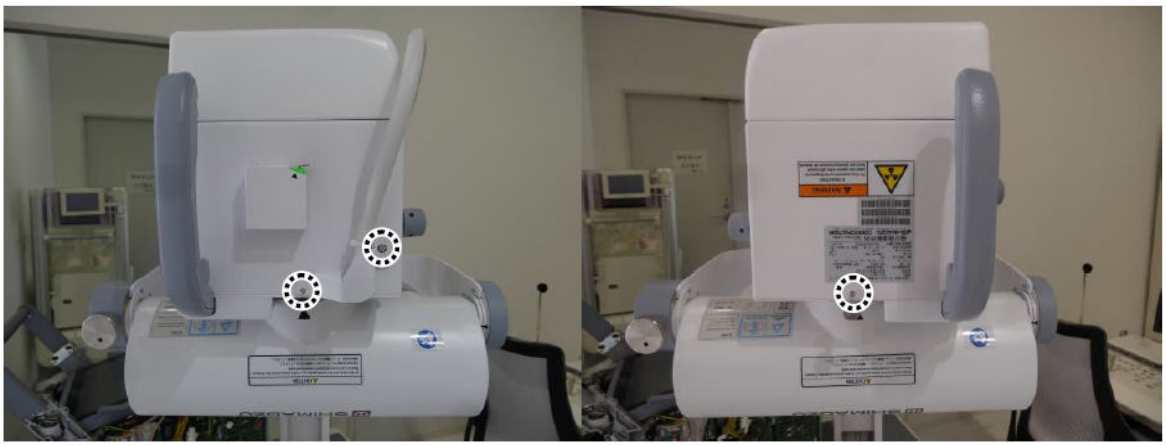


Fig. 7-44 Lateral side of Collimator

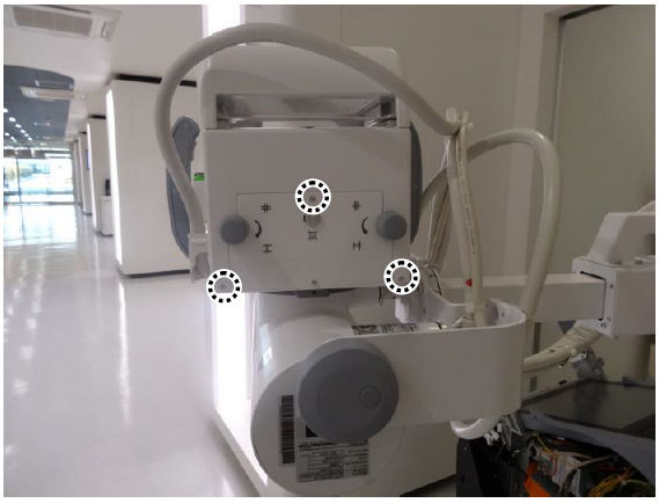


Fig. 7-45 Rear side of Collimator

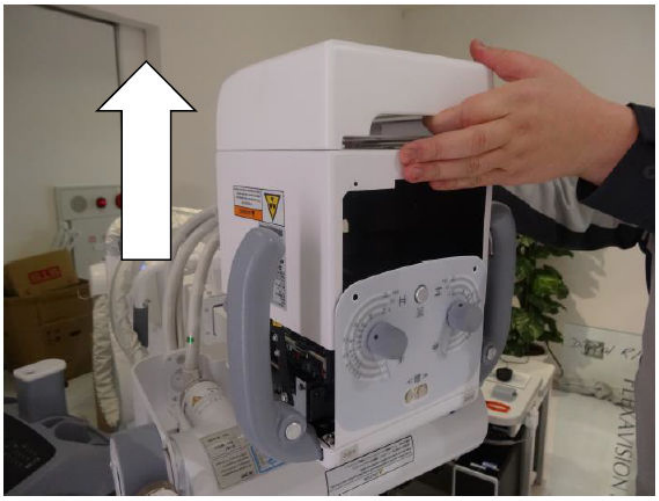


Fig. 7-46

7.20.2 To Enter Dose Aarea Product Calculation Unit Setting Mode

1 **Remove the power plug from the outlet, and turn OFF the power to the system.**

2 **Remove the right-inside cover and right-side cover.**

Reference "3.5 How to Open/Close Covers"

3 **Set each DIP SW on the XCONT-2018 board as shown below, and turn ON the power.**

- DIPSW3-5: OFF
- DIPSW3-8 (for execution of setting mode): ON

4 **Press the INFO switch to blink, and then the keys in order of <1> and <8> to enter the dose area product calculation unit setting mode.**

LED of the switch and keys are indicated as follows.

- INFO switch blinks
- Key <1> blinks
- Key <8> blinks

5 **When enter the dose area product calculation unit setting mode, the exposure condition is displayed in the control panel.**

6 **Make sure the default condition is displayed as 40kV, 10mAs.**

7 **Press the Additional Filter key to switch the screen to the ON/OFF setting of the dose area product calculation unit.**

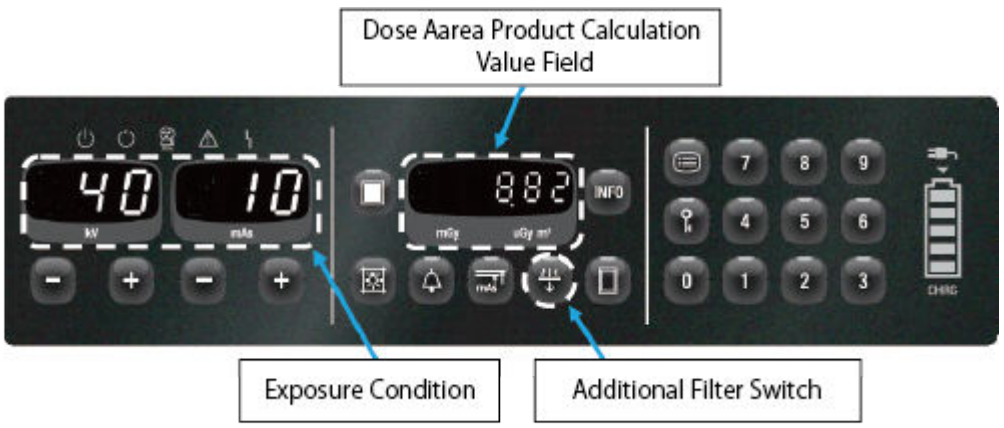


Fig. 7-47 Initial screen of the dose area product calculation unit

7.20.3 ON/OFF Setting of Dose Area Product Calculation Unit

1 **The ON/OFF setting of the dose area product calculation unit is displayed in the kV display.**

2 **Make sure the setting is ON. If OFF, press the Tube Voltage key to change the setting to ON, and then the Registration key.**

3 **Press the Additional Filter key to switch the screen to the setting the collimator leaf opening amount.**

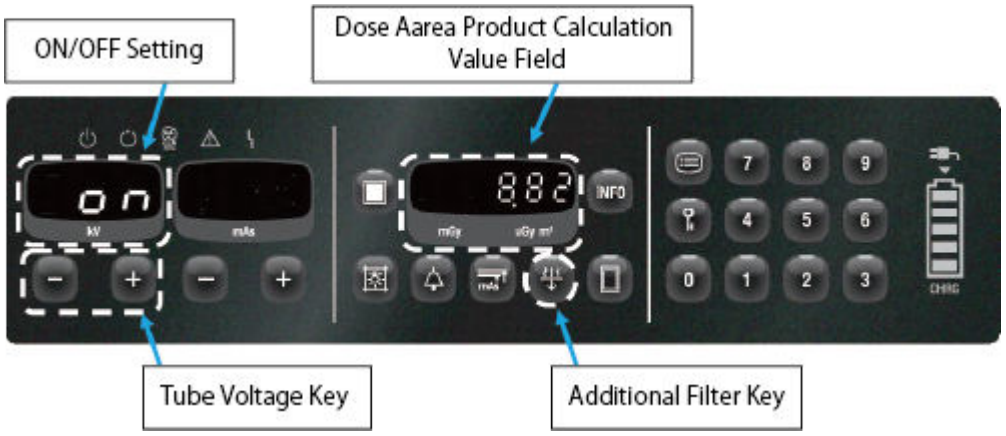


Fig. 7-48 ON/OFF setting screen of the dose area product calculation unit

7.20.4 Setting of collimator leaf opening amount

The collimator leaf opening amount has been specified at the factory. This setting is needed only for the re-adjustment of opening amount when the collimator is replaced. Press the Additional Filter key to enter the adjustment of whole number coefficients.

1 **The following information is displayed in the kV display field and mAs display field.**

Table 7-15

kV display field	mAs display field
The registration point from "h0", "h35", "v0", "v35"	The registered detection voltage (0.00to 5.00V) of the collimator leaf opening amount at the registration point displayed in kV display field.

* *Press the tube voltage switch to switch the registration point in kV display field.

Fig. 7-49 Setting screen of the collimator leaf opening amount

2 Remove the collimator cover.

Register the collimator leaf opening amount at each registration point, as shown in the table below. After adjustment, the opening amount at the registration point can be updated by pressing the Registration key at the registration point.

Table 7-16

Registration point (kV display field)	Collimator leaf state	How to adjust the collimator
h0	The horizontal leaf level when the horizontal radiation field becomes 0cm.	Adjust by the lighting field.
h35	The horizontal leaf level when the horizontal radiation field becomes 35cm at 1m of SID.	Adjust using the adjustment jig (*).
v0	The horizontal leaf level when the vertical leaf becomes 0cm.	Adjust by the lighting field.
v35	The vertical leaf level when the vertical radiation field becomes 35cm at 1m of SID.	Adjust using the adjustment jig (*).

* *(*)Protective tool supplied with the collimator

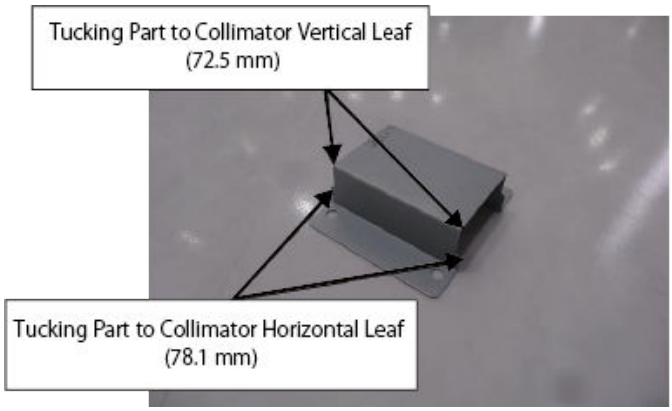


Fig. 7-50 Ajustment jig

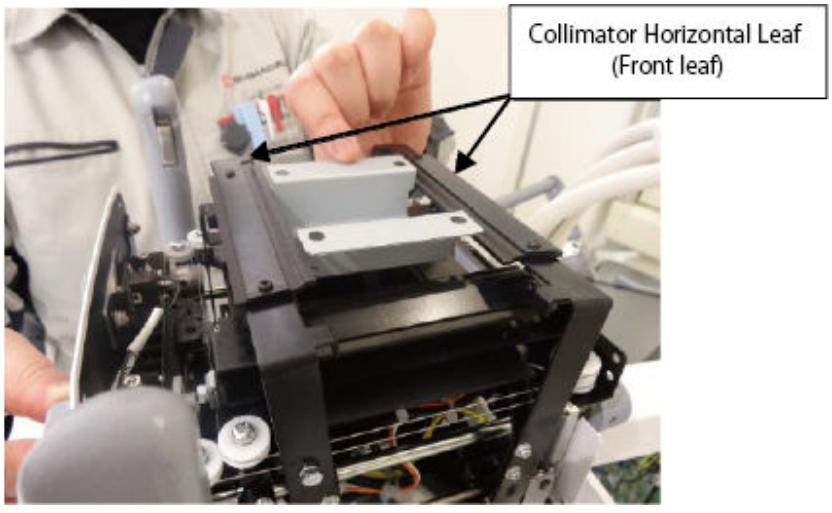


Fig. 7-51 Adjustment of horizontal leaf level

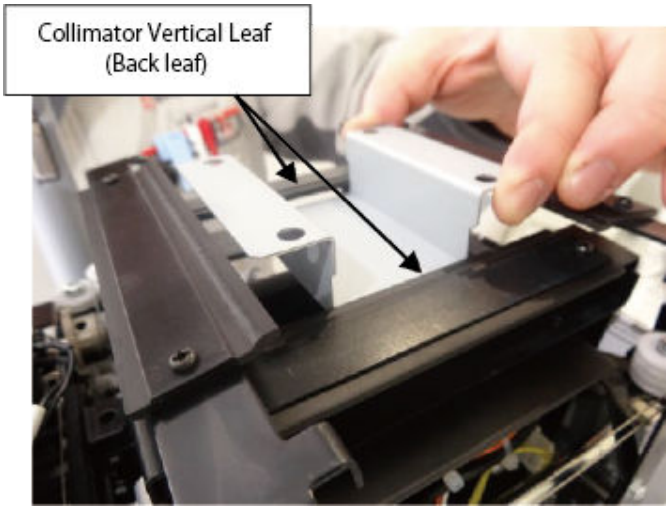


Fig. 7-52 Adjustment of vertical leaf level

3 **Press the Additional Filter key for two times to enter the dose area product calculation unit setting mode.**

4 **When the control panel shows the initial screen for the dose area product calculation unit setting mode, press the Registration key.**

All data registered on each adjustment screen is stored to the memory.

Fig. 7-53

Table 7-17 Registration of all data

NOTE When the registered information is modified, that information must be saved in the memory by pressing the Registration keyon the initial screen of the control panel.

The registered information in each adjustment mode is the interim registration. If the adjustment mode is finished without saving in the memory, the modified information cannot be refleted to the system.

5 **Turn OFF th power to the system.**

6 **Attach the removed collimator cover.**

7 **Turn ON the power to the system.**

8 **Press the INFO switch to blink, and then key <8> to enter the dose area product calculation unit setting mode.**

LED of the switch and keys are indicated as follows.

- INFO switch blinks
- Key <1> blinks
- Key <8> blinks

9 **Press the Additional Filter key for three times to switch the screen to set the whole number coefficients and select the additional filter.**

7.20.5 To Adjust Whole Coefficient and Select Additional Filter

1 **ALL Comp value (whole adjustment coefficient) of the dose area product meter is displayed in the kV display field, and the following additional filter is displayed in the mAs display field.**

Table 7-18

mAs display field	Filter information
0	No filter
1	Additional filter 1 (Arbitrary additional filter)
2	Additional filter 2 (Arbitrary additional filter)

Fig. 7-54 Screen to adjust the whole coefficient and to select the additional filter

2 **Make sure the All Comp value is 100. The modification of All Comp value is not needed. If modify, change the value by the Tube Voltage key, and press the Registration key to save that value.**

3 **Press the tube current time integration key to select a filter type.**

4 **Press the Additional Filter key to switch the screen to the calibrating the dose area product meter for the displayed filter type.**

7.20.6 To Calibrate Dose Area Product Calculation Unit

Calibrate the dose area product calculation unit for the filter type selected in the previous section, "To adjust the whole coefficients and to select the additional filter".

When "0" is selected in the filter type, this setting is not required because the calibration of the dose area product calculation unit has been made at the factory.

When Additional Filter 1 and 2 are used, the dose area product calculation unit must be calibrated.

Use the calibrated dose area product calculation unit to set the measurement system as shown below.

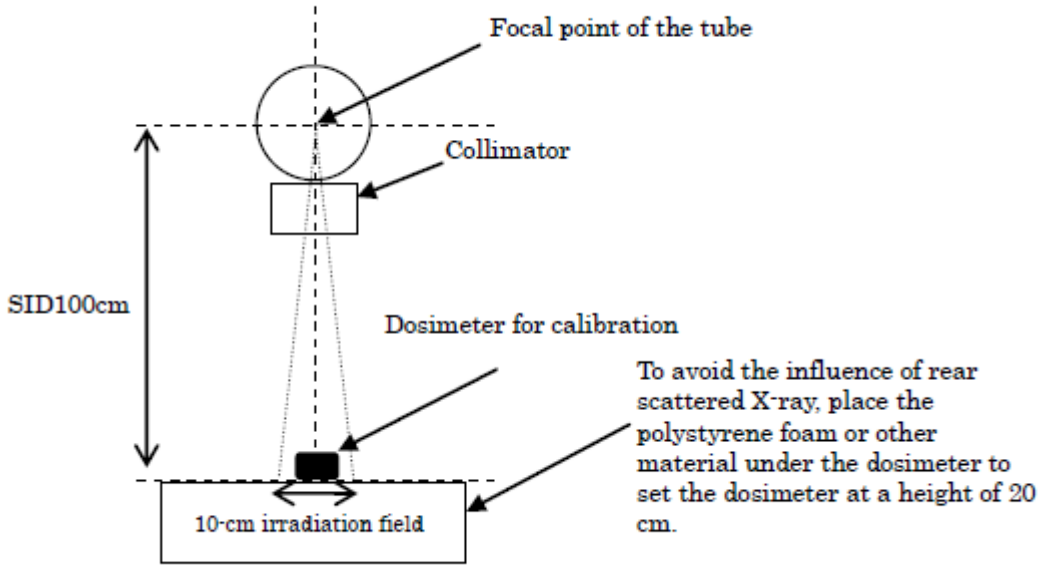


Fig. 7-55 Measurement system

1 The following information is displayed in each display field.

Table 7-19

kV display field	mAs display field	Ares dose display field
Tube voltage to be adjusted. The value is changed by step below: 40kV, 50kV, 60kV, 70kV, 80kV, 90kV, 100kV, 110kV, 120kV, 130kV, 133kV	Filter type selected in the adjustment of whole coefficients: 0: No filter 1: Additional filter 1 2: Additional filter 2	ckV at kV displayed in the kV display field. (Tube voltage total filtering coefficient basic data)

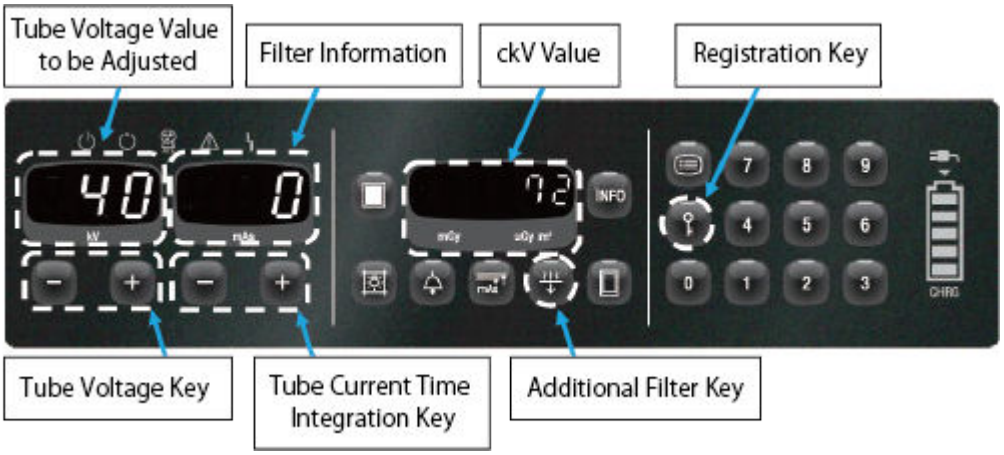


Fig. 7-56

Table 7-20 Calibration screen of dose area product calculation unit

2 Press the tube voltage key to change kV to be set.

NOTE The mAs value is not displayed. The value (10mAs) that was set on the initial screen is displayed on the mAs display.

3 **By changing kV, set ckV at each kV in the table below. Select the filter type, and perform an exposure by the measurement system at each kV, then write down the measurement value of the dosimeter in the table below. And also, calculate ckV at each kV using the following calculation formula, and write down the value in the table below. (These values must be memorized for the later confirmation.)**

- For the dosimeter in mGy unit ckV = ESD (measured value) x 1000
- For the dosimeter in μGy unit ckV = ESD (measured value)
- ESD = Measured dose value [mGy] or [μGy]

Table 7-21

kV	Measured value (mGy)	ckV
40		
50		
60		
70		
80		
90		
100		
110		
120		
130		
133		

* (Reference) 115mR = 1mGy, 1mGy = 1000μGy

* (Reference) The software of dose area product calculation unit calculates the following information.

ESD＝A/100×C/10000×R×1/D

ESD[mGy]: Irradiation dose
A: All Comp value (whole adjustment coefficient)
C: ckV value (Tube voltage total filtering coefficient basic data)
R[mAs]: Tube current time integration
D[m]: SID (=1m)

DAP=ESD×H×V

DAP[mGycm2]: Value of dose area product calculation unit (0.1mGycm2)
H[cm]: Horizontal opening amount of the collimator leaf at 1m of SID (0.1cm)
V[cm]: Vertical opening amount of the collimator leaf at 1m of SID (0.1cm)

4 **Press the tube current time integration key to change kV and set ckV at the displayed kV.**

5 **Press the Registration key to save ckV at each kV.**

6 **Press the Additional Filter key to go back to the initial screen of the dose area product calculation unit setting mode.**

7 **In the initial screen of the dose area product calculation unit setting mode, press the Registration key to save all data regitered in each adjustment mode.**

Fig. 7-57 Registration of All Data

NOTE When the registered information is modified, that information must be saved in the memory by pressing the Registration keyon the initial screen of the control panel.

The registered information in each adjustment mode is the interim registration. If the adjustment mode is finished without saving in the memory, the modified information cannot be refleted to the system.

8 **When you adjust the dose area product calculation unit for other filter types, press the Additional Fitler key for several times to display the screen of adjusting the whole coefficients and selecting the additional filter, and make adjustment.**

7.20.7 Initial Screen of Dose Area Product Calculation Unit (To Save Registered Information in Memory)

1 **Turn OFF the power to the system.**

2 **Set the DIP switch SW3-5 and SW3-8 on the XCONT-2018 board as shown below.**

- DIPSW3-5 5(Dose area product meter): OFF
- DIPSW3-8 (Adjustment mode): OFF

3 **Attach the removed covers in reverse order.**

Reference "3.5 How to Open/Close Covers"

7.21 DR Setting Mode

When equipped with DR-ID800 or CXDI or CS-7, this mode is aready set at the factory setting. Therefore, modification of the following setting is not necessary except when it is necessary specifically.
When equipped with VIVIX-S, perform setting exposure time restriction with DR radiography mode and APR initialization.

7.21.1 To Enter DR Setting Mode

1 **Enter the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

2 **Press the INFO switch to blink, and then the key <7> to enter the DR setting mode.**

LED of the switch and keys are indicated as follows.

- INFO switch blinks
- Key <1> blinks
- Key <7> blinks

7.21.2 Setting of DR

1 **When enter the setting mode, the setting item is displayed on kV display field and the setting value is displayed on mAs display field. Press the Tube Voltage key to switch the setting item on kV display field and the setting values on mAs display field in synchronization.**
However if keyless entry is disabled, "PaS" setting screen cannot be entered.

Table 7-22

kV display field	Description
EtS	Setting exposure time restriction with DR radiography mode
EdL	Setting exposure ime restriction with DR long-time radiography mode (for DR-ID800, CXDI, CS-7, CS-7 for Serial Radiography)
PaS	Setting necessity of password entry
Prc	Setting of communicatino protocol (for CXDI (Combined with Multi Box MB-4A), CS-7)
Apr	Setting APR initialization (for VIVIX-S)

2 **If "EtS" is displayed on kV display field, the setting number of an exposure time limit during DR radiography mode is displayed on mAs display field. When equipped with VIVIX-S, set the setting number of an exposure time limit during DR radiography mode according to Exposure Time of VXvue. When Exposure Time of VXvue is 800 msec, the setting number is 58. When Exposure Time of VXvue is 2800 msec, the setting number is 69.**

Press the tube current time integration key to change the setting number of a time limit by 1 step on mAs display field.
Press the Registration key to display "rec" "ord" on kV/mAs display fields and store the setting.

Fig. 7-58 Time limit during DR radiography mode ("60" is set)

⚠ WARNING

When equipped with DR-ID800 or CXDI or CS-7 or CS-7 for Serial Radiography, do not change the setting of a time limit during DR radiography mode.

3 **If "EtL" is displayed on kV display field, the setting number of an exposure time limt during DR long-time radiography mode is displayed on mAs display field.**

Press the tube current time integration key to change the setting number of a time limit by 1 step on mAs display field.
Press the Registration key to display "rec" "ord" on kV/mAs display fields and store the settings.

Fig. 7-59 Time limit during DR long-time radiography mode ("69" is set)

⚠ WARNING

Do not change the setting of a time limit during DR long-time radiography mode.

4 If "PaS" is displayed on kV display field, necessity of password entry is displayed on mAs display field.

Press the tube current time integration key to switch the mAs display field.
The default setting is "on".
Press the Registration key to display "rec" "ord" on kV/mAs display fields and store the settings.

Fig. 7-60 Requirement of Password entry ("on" is displayed)

Table 7-23

mAs display field	Description
on	Password entry is requested when return from auto power off
off	Password entry is not requested when return from auto power off

⚠ WARNING

Necessity of password entry must be set to "on".
Do not change the setting without user's request and agreement.

5 **If "Prc" is displayed on kV display field, the communication version is displayed on mAs display field.**

Press the tube current time integration key to switch the mAs display field.
The dafault setting is "11".
Press the Registration key to display kV/mAs display fields and store the settings.

Fig. 7-61 Communication version ("11" is displayed)

Table 7-24

mAs display field	Description
11	Normal
00	Unused

⚠ WARNING

Do NOT change the setting of the communication protocol.

6 **When "Apr" is displaied on kV display field, press the Registration key for approx. 3 seconds to start initializing the APR and display "ini" on mAs display field. If the APR is initialized, "rec" "ord" on kV/mAs display fields are displaied.**

7.21.3 To Finish DR Setting Mode

1 **Finish the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

7.22 Cord Reel Usage Information Setting Mode

This mode is used to check the starting date of the cord reel usage and the number of connection to the outlet, and to set the usage information of the Cord Reel.

Adjust the setting by the following procedure when the cord reel is replaced.

7.22.1 To Enter Cord Reel Usage Information Setting Mode

1 **Enter the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

2 **Press the INFO switch to blink, and then the keys in order of <3> and <8> to enter the cord reel usage information setting mode.**

LED of the switch and keys are indicated as follows.

- INFO switch blinks
- Key <3> blinks
- Key <8> blinks

7.22.2 To Check Cord Reel Usage Information

When the cord reel usage information setting mode is entered, the starting date of the cord reel usage is displayed in the kV and mAs display fields.

The number of connection to the outlet is displayed in the dose display field. The number of connection to the outlet is displayed by single count.

Fig. 7-62 EXAMPLE: Starting date of use of cord reel and the number of connection to the outlet (March 2, 2018, 562 times)

7.22.3 To Reset Cord Reel Usage Information

It is required to reset the usage information when the cord reel is replaced.

1 **Enter the cord reel usage information setting mode, and press the Registration key for 3 seconds.**

The starting date of the cord reel is reset to the current date, and the number of connection to the outlet is reset to 0.

7.22.4 To Finish Cord Reel Usage Information Setting Mode

1 **Finish the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

7.23 FSR Usage Information Setting Mode

This mode is used to check the starting date of the FSR usage and the number of FSR enhancement, and to set the usage information of the FSR.

Adjust the setting by the following procedure when FSR is replaced.

7.23.1 To Enter FSR Usage Information Setting Mode

1 **Enter the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

2 **Press the INFO switch to blink, and then the keys in order of <3> and <9> to enter the FSR usage information setting mode.**

LED of the switch and keys are indicated as follows.

- INFO switch blinks
- Key <3> blinks
- Key <9> blinks

7.23.2 To Check FSR Usage Information

When the FSR usage information setting mode is entered, the starting date of FSR usage is displayed in the kV and mAs display fields.

The number of FSR enhancement is displayed in the dose display field. The information in the dose display field can be switched by pressing the collimator lamp switch.

Table 7-25

6th Digits of mGy	Item
0	The number of left FSR enhancement (by single count)
1	The number of right FSR enhancement (by single count)

Fig. 7-63 EXAMPLE: Starting date of FSR usage and the number of SR enhancement (Feb. 28, 2018, 365 times of left FSR enhancement)

7.23.3 To Reset FSR Usage Information

It is required to reset the usage information when FSR is replaced.

1 **Enter the FSR usage information setting mode, and press the Registration key for 3 seconds.**

The starting date of the cord reel is reset to the current date., and the number of enhancement of FSR is reset to 0.

7.23.4 To Finish FSR Usage Information Setting Mode

1 **Finish the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

7.24 Mechanical Performance Usage Information Setting Mode

7.24.1 To Enter Mechanical Performance Usage Information Setting Mode

1 **Enter the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

2 **Press the INFO switch to blink, and then the keys in order of <3> and <4> to enter the mechanical performance usage information setting mode.**

LED of the switch and keys are indicated as follows.

- INFO switch blinks
- Key <3> blinks
- Key <4> blinks

7.24.2 To Check Mechanical Performance Usage Information

When the mechanical performance usage information setting mode is entered, the starting date of mechanical performance usage is displayed in the kV and mAs display fields.

The usage information of each item is displayed in the dose display field. The information on kV/mAs, and dose display fields can be switched by pressing the collimator lamp switch.

Table 7-26

kV/mAs display field	Description
Obstr	The number of obstacle detection (by single count)
Catch	The number of arm lock (by single count)

Fig. 7-64 EXAMPLE: The number of obstacle detection (96 times)

Fig. 7-65 EXAMPLE: The number of arm lock (758 times)

7.24.3 To Reset Mechanical Performance Usage Information

1 **Enter the mechanical performance usage information setting mode, and press the Registration key for 3 seconds.**

The number of obstacle detection and the number of arm lock are reset to 0.

7.24.4 To Finish Mechanical Performance Usage Information Setting Mode

1 **Finish the adjustment mode.**

Reference "7.1 How to Start Adjustment Mode"

8 [Appendix] Adjustment Mode (For MobileArt Evolution)

8.1 How to Start Adjustment Mode

There are two methods to start the Adjustment Mode.

8.1.1 Method 1

1 **Remove the power plug from the outlet, and turn OFF the main system.**

2 **Remove the right-inside cover and right-side cover.**

3 **Turn ON the DIP switch SW3-8 on the XCONT-2018 board.**

4 **Turn ON the main power.**
"TECI" is displayed immediately on the kV/mAs display fields of the X-ray control panel.

Fig. 8-1 Initial display on adjustment mode

By pressing the Registration button for each setting described in this chapter, the setting value is registered and "record" is displayed on the kV/mAs display fields.

Fig. 8-2 "record" Display when registering

■ **How to finish adjustment mode**

1 **Turn OFF the main system.**

2 **Turn OFF the DIP switch SW3-8 on the XCONT-2018 board.**

3 **Attach the removed covers in reverse order.**

4 **Restart the main system.**

8.1.2 Method 2

This method allows to enter initialization and adjustment mode without setting DIP switches on the XCONT-2018 board.

Hint This method allows to perform some settings below without opening the covers.

- Adjustment of X-ray tube current
- Initialization
- Check the irradiation counter

X-ray tube current can easily be checked because the unit is automatically brought into the measured value display mode.

1 **Remove the power plug from the outlet, and turn OFF the power to te system.**

2 **Turn ON the power to the system.**

3 **Press and hold the User settings button for more than 3 seconds.**
The mode for changing various settings is entered.

4 **Wait until the display field changes from** **to**
(Personal ID Setting). When the display is changed, press the chest radiography button and then the Collimator lamp button within 3 seconds.

5 **SUP OPE appears for approx. 1 second, and then 7EC appears on the dose indicator.**

Hint At this time, the version of S/W "X.XX"is displayed on mAs display field. If the Collimator lamp button is pressed,"8" will be displayed as the type of software on the mAs display.

⚠ CAUTION

Be sure to confirm software compatibility after the replacement of the XCONT-2018 board. The installation of software incompatible with the unit output may cause damage to the unit.

Fig. 8-3 Display of software version (example of Ver.1.50)

Fig. 8-4 Display of software type

6 **In this state, the display will switch between 7EC (adjustment mode) and INI (initialization) by pressing the Body thickness change button.**

7 **Press the Registration button in the mode to be executed to enter the selected mode.**

8 **Refer to the following on each adjustment mode and initialization.**

Reference
- "8.3 Adjustment Mode List".
- "8.2 Initialization of XCONT-2018 Board"

■ How to finish the mode

1 Turn OFF the power to te system.

2 Turn ON the power to the system.

8.2 Initialization of XCONT-2018 Board

NOTE When replacing the XCONT-2018 board, it must be initialized.

8.2.1 Procedure of Initialization

1 **Remove the power plug from the outlet, and turn OFF the power to te system.**

2 **Remove the right-inside cover and right-side cover.**

Reference "3.5 How to Open/Close Covers"

3 **Set the DIP switch and the Jumpers on the XCONT-2018 board depending on the configuration of the device.**

Reference
- "6.1 XCONT-2018 Board"
- "6.2 XCONT-2018 Board Jumper Setting"

4 **Set the DIP switch on the XCONT-2018 board as shown below.**

- DIPSW2-8: ON
- DIPSW3-1: ON

5 **Turn ON the main power of the system.**

6 **"ins" is displayed on the X-ray control panel for about 40 seconds.**

7 **After "ins" disappears, press and hold the Registration button until "CAL end" appears.**

8 **Set the following DIP switches on the XCONT-2018 board to original position.**

- DIPSW2-8: OFF
- DIPSW3-1: OFF

NOTE Be sure to reset both DIP switches to the original position once the initialization is complete.

9 **Attach the removed covers in reverse order.**

8.3 Adjustment Mode List

Mode list in Adjustment Mode is shown below.

■ Adjustment mode Page 1 (Front-view button blinks, and Body thickness change button goes off.)

Table 8-1

Button	Name of mode	Notes
Foot radiography button	Battery detection circuit adjustment mode	Used for adjustment of each device. Reference "8.6 Adjustment of Battery Voltage Detection"
Arm radiography button	Date and Time setting mode	Reference "8.7 Setting Date and Time"
Hand radiography button	Handle adjustment mode	Used for adjustment of each device. Reference "8.5 Adjustment of the Handle"
Chest radiography button	Tube current two-point adjustment mode	Used for normal tube current adjustment. Reference "8.4 Adjustment of Tube Current"
Abdominal radiography button	Tube current each-point adjustment mode	Used for the detailed check and adjustment of tube current. (Not normally used). Reference "8.4 Adjustment of Tube Current"
Lumbar radiography button	Battery voltage drop check mode	Used for checking the battery condition when performing the radiography then applying load. Reference "8.10 Battery Voltage Drop Check Mode"

■ Adjustment mode Page 2 (Front-view button blinks, and Body thickness change button blinks.)

Table 8-2

Button	Name of mode	Reference
Foot radiography button	Battery charging test mode	Perform the operation of charging for a minute with charging current setting 1A. This mode is used to confirm the operation of charging circuit. Reference "9.7 Checking battery charging"
Arm radiography button	Running speed display mode	Reference "8.9 Display of Running Speed"
Hand radiography button	Battery code setting mode	Reference "8.11 Battery Code Setting Mode"
Chest radiography button	-	Not used.
Abdominal radiography button	Dose area product meter adjustment mode (SW3-5ON) Dose area product calculation unit adjustment mode (SW3-5OFF)	Reference • "8.12 Dose Area Product Meter Adjustment Mode" • "8.20 Dose Area Product Calculation Unit Setting Mode"
Lumbar radiography button	Error Log mode	Reference "8.8 Display and Reset of Error Log"

■ Adjustment mode Page 3 (Body thickness change button goes off, and Lateral-view button blinks.)

Table 8-3

Button	Name of mode	Reference
Foot radiography button	-	Not used.
Arm radiography button	-	Not used.
Hand radiography button	-	Not used.
Chest radiography button	-	Not used.
Abdominal radiography button	-	Not used.
Lumbar radiography button	-	Not used.

■ Adjustment mode Page 4 (Body thickness change button goes off, and Oblique-view button blinks.)

Table 8-4

Button	Name of mode	Reference
Foot radiography button	Motor usage information setting mode	Reference "8.17 Motor Usage Information Setting Mode"
Arm radiography button	X-ray tube unit usage information setting mode	Reference "8.16 X-ray Tube Unit Usage Information Setting Mode"
Hand radiography button	Usage information collective setting mode	Reference "8.19 Usage Information Collective Setting Mode"
Chest radiography button	Battery usage setting mode	Reference "8.13 Battery Usage Setting Mode"
Abdominal radiography button	CPU board battery usage information setting mode	Reference "8.14 CPU Board Battery Usage Information Setting Mode"
Lumbar radiography button	Emergency brake release battery usage Information setting mode	Reference "8.15 Emergency Brake Release Battery Usage Information Setting Mode"

■ Adjustment mode Page 5 Body thickness change button blinks, and Oblique-view button blinks)

Table 8-5

Button	Name of mode	Reference
Foot radiography button	Mechanical performance usage information setting mode	Reference "8.23 Mechanical Performance Usage Information Setting Mode"
Arm radiography button	-	Not used.
Hand radiography button	-	Not used.
Chest radiography button	Arm usage information setting mode	Reference "8.18 Arm Usage Information Setting Mode"
Abdominal radiography button	Cord Reel usage information setting mode	Reference "8.21 Cord Reel Usage Information Setting Mode"
Lumbar radiography button	FSR usage information setting mode	Reference "8.22 FSR Usage Information Setting Mode"

8.4 Adjustment of Tube Current

8.4.1 Tube Current Adjustment Mode

The tube current of this unit is adjusted by changing the setting of the filament current (called FVR value).
As shown below, there are 6 points of adjustable FVR values; 40, 60, 80, 100, 125 and 133 kV for each tube current. FVR values between tube voltage points are calculated using linear interpolation.

■ 32 kW type (MobileArt Evolution high-power type)

Table 8-6 Small focus

	50 mA	63 mA	80 mA	100 mA	125 mA	160 mA	200 mA
133 kV	○	○	○	○	-	-	-
125 k V	○	○	○	○	○	-	-
100 k V	○	○	○	○	○	○	-
80 k V	○◎	○	○	○	○	○	○◎
60 k V	○	○	○	○	○	○	○
40 k V	○	○	○	○	○	○	○

* ○ : Adjustment point in Tube Current Each-point Adjustment Mode
◎ : Adjustment point in Tube Current Two-point Adjustment Mode

Table 8-7 Large focus

	50 mA	63 mA	80 mA	100 mA	125 mA	160 mA	200 mA	250 mA	320 mA	400 mA
133 kV	○	○	○	○	○	○	○	-	-	-
125 k V	○	○	○	○	○	○	○	○	-	-
100 k V	○	○	○	○	○	○	○	○	○	-
80kV	○◎	○	○	○	○	○	○	○	○	○◎
60kV	○	○	○	○	○	○	○	○	○	○
40 k V	○	○	○	○	○	○	○	○	○	-

* ○ : Adjustment point in Tube Current Each-point Adjustment Mode
◎ : Adjustment point in Tube Current Two-point Adjustment Mode

■ 12.5 kW type (MobileArt Evolution standard type)

Table 8-8 Small focus

	50 mA	63 mA	80 mA	100 mA	125 mA	160 mA
125 k V	○	○	○	○	-	-
100 k V	○	○	○	○	○	-
80 k V	○◎	○	○	○	○	○◎
60 k V	○	○	○	○	○	○
40 k V	○	○	○	○	○	○

* ○ : Adjustment point in Tube Current Each-point Adjustment Mode
◎ : Adjustment point in Tube Current Two-point Adjustment Mode

There are two types of tube current adjustment mode. Normally, the tube current is adjusted with (1) Tube Current Two-point Adjustment Mode. When necessary, the tube current at each adjustment point is checked and adjusted with (2) Tube Current Each-point Adjustment Mode

- (1) Tube Current Two-point Adjustment Mode
  FVR value is adjusted with two points; minimum and maximum tube current of 80kV. The FVR values of other points are calculated based on the two values.
- (2) Tube Current Each-point Adjustment Mode
  FVR values are adjusted at each adjustment point.

> **NOTE** The Tube Current Two-point Adjustment Mode is to correct the currently-set FVR value table based on the adjustment results of the two points. Therefore, if any currently-set FVR value is invalid, the correction results could be wrong. Be sure to initialize FVR value before the tube current two-point adjustment.

8.4.2 To Initialize FVR Values

> **NOTE** The initialization is not required in the standard installation procedure of the unit. Perform the initializing methods in case such as when the circuit boards or other parts are replaced and readjustment is required.

1 **Press the focus mode button you choose to initialize the FVR value.**

2 **Press the Pediatric radiography button for about 3 seconds until "c A n c e L" appears on the display.**

FVR value of selected focus will be initialized only.

> **NOTE** FVR value can also be initialized by initializing the XCONT-2018 board. However, other important data will also be initialized.

8.4.3 Others

- When the Tube Current Two-point or Each-point Adjustment Mode is selected, the tube current feedback circuit is automatically disabled. Radiography time is fixed at 20msec.
- To adjust the tube current, use the secondary tube current meter or neutral point tube current meter, or enter the actual measurement mode on the system. (During the tube current adjustment mode, the system automatically enters the actual measurement display mode.)

> **NOTE** The tube current of the same FVR value may differ due to the presence of the tube voltmeter connection. Make sure that the tube voltmeter is disconnected when the tube current adjustment is performed.

8.4.4 Tube Current Two-point Adjustment Mode

NOTE This mode is to calculate the currently-set FVR value table based on the adjustment results of the two points. Therefore, if any currently-set FVR value is invalid, the correction results could be wrong. Be sure to initialize FVR value before the tube current two-point adjustment.

1 **Enter the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

2 **Press the Front-view button to enter the Tube Current Two-point Adjustment Mode.**

The LEDs on the following buttons are:.

- The Body thickness change buttin goes off.
- The Front-view button blinks.
- The chest radiography button blinks

3 **Tube voltage is displayed on kV display field and tube current is displayed on mAs display field when enter the Tube Current Two-point Adjustment Mode.**

4 **Press the Focus mode button and select a focus.**

The LED on the Focus mode button is:.

- The LED lights when the small focus is selected.
- The LED goes off when the large focus is selected.

NOTE For 12.5 kW, only the small focus can be selected.

5 **Set the tube voltage by pressing the tube voltage buttons, and the tube current by pressing the current-time product buttons.**

The tube voltage is fixed at 80kV. The tube current can be set at two points, minimum and maximum. Set the minimum tube current first.

32 kW type (MobileArt Evolution high-power type)

Table 8-9 Small focus

Focus type	Adjustment point of tube current	Tube current value
Small focus The Focus mode button LED lights.	Minimum	50 mA
	Maximum	200 mA

Table 8-10 Large focus

Focus type	Adjustment point of tube current	Tube current value
Large focus The Focus mode button LED goes off.	Minimum	50 mA
	Maximum	400 mA

12.5 kW type (MobileArt Evolution standard type)

Table 8-11 Small focus

Focus type	Adjustment point of tube current	Tube current value
Small focus The Focus mode button LED lights.	Minimum	50 mA
	Maximum	160 mA

Fig. 8-5 Tube current two-point adjustment mode (the tube current is set to 50 mA)

6 **Press the handswitch to execute X-ray exposure.**

The measured tube current value is displayed on the mAs display field.

7 Adjust the FVR value.

1 Press the Added filter button to show the FVR value on the tube voltage display. (The Added filter button blinks when the FVR value is displayed.)

2 Use the the Tube voltage button to change the FVR value.

Hint
- To decrease the tube current, set the FVR value to a smaller value.
- To increase the tube current, set the FVR value to a larger value.

8 Repeat the steps 6 and 7 until the measured tube current value is within the recommended range.

32 kW type (MobileArt Evolution high-power type)

Table 8-12 Small focus

Focus type	Adjustment point	Recommended range
Small focus The Focus mode button LED lights.	80 kV 50 mA	45 to 50 mA
	80 kV 200 mA	185 to 200 mA

Table 8-13 Large focus

Focus type	Adjustment point	Recommended range
Large focus The Focus mode button LED goes off.	80 kV 50 mA	45 to 50 mA
	80 kV 400 mA	370 to 400 mA

12.5 kW type (MobileArt Evolution standard type)

Table 8-14 Small focus

Focus type	Adjustment point	Recommended range
Small focus The Focus mode button LED lights.	80 kV 50 mA	45 to 50 mA
	80 kV 160 mA	148 to 160 mA

Fig. 8-6 Tube current two-point adjustment mode-FVR value setting
(The tube current is set to 50mA, the FVR ajusted to FVR = 3.92)

9 Register FVR values.

1 Register the FVR value after the tube current adjustment is complete.

2 To register the FVR value, press the Registration button. Characters "r e c o r d" will show on the display.

> **NOTE** Be sure to press the Registration button when the FVR value is changed. Otherwise, the changed FVR value will become invalid if the other point or mode is moved, or the switch is turned off. FVR value can also be registered on the display of the exposure condition (instead of the FVR value).

10 Set the maximum tube current, and then repeat Step 6 to 9.

> **NOTE**
> - The result are same whether the adjustment is performed from the minimum tube current point or the maximum tube current point, but it is preferable to perform the adjustment from the minimum tube current point.
> - If this mode is terminated with only one point setting being registered, the other point are considered as having unchanged.

11 Select each focus, and repeat Step 5 to 10.

12 Finish the adjustment mode.

Reference "8.1 How to Start Adjustment Mode"

8.4.5 Tube Current Each Point Adjustment Mode

■ How to Adjust FVR value

1 **Enter the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

2 **Press the Abdominal radiography button to enter the Tube current each point adjustment mode.**

The LEDs on the following buttons are:.

- The Body thickness change buttin goes off.
- The Front-view button blinks.
- The Abdominal radiography button blinks.

3 **When the Tube current each point adjustment mode is entered, the tube voltage is displayed on kV display field and tube current is displayed on mAs display field.**

4 **Press the Focus mode button and select a focus.**

Hint

- The LED lights when the small focus is selected.
- The LED goes off when the large focus is selected.

5 **Set the tube voltage by pressing the tube voltage buttons, and the tube current by pressing the current-time product buttons.**

Fig. 8-7 Tube current each point adjustment mode – Exposure condition setting (In case of 80 kV and 160 mA)

6 **Press the handswitch to execute X-ray exposure.**

The measured tube current value is displayed on the mAs display field.

7 Adjust the FVR value.

1 Press the Added filter button to show the FVR value on the tube voltage display. The added filter button blinks when the FVR value is displayed.

2 Use the Tube Voltage button to change the FVR value.

Hint
- To decrease the tube current, set the FVR value to a smaller value.
- To increase the tube current, set the FVR value to a larger value.

8 Repeat Step 5 and 6 until the measured tube current value is within -15% to +5% of the set value.

Fig. 8-8 Tube current each point adjustment mode: FVR value adjusted
(This figure shows an example of the display: the FVR is setting to FVR = 3.87)

9 Register FVR values.

Register the FVR value after the tube current adjustment is complete. To register the FVR value, press the Registration button. Characters “r e c o r d” will show on the display.

NOTE Be sure to press the Registration button when the FVR value is changed. Otherwise, the changed FVR value will become invalid if the other point or mode is moved, or the switch is turned off. FVR value can also be registered on the display of the exposure condition (instead of the FVR value).

10 Repeat Step 5 to 8 for other points, as required.

11 Press the Focus mode button and select focus. And then repeat Step 4 to 9 for the other focus.

Hint
- The LED lights when the small focus is selected.
- The LED goes off when the large focus is selected.

12 Finish the adjustment mode.

Reference "8.1 How to Start Adjustment Mode"

8.5 Adjustment of the Handle

In this mode, the handle is adjusted to control the driving performance of the unit. Perform the following procedures if the handle or the XCONT-2018 board is exchanged.

1 **Enter the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

2 **Pess the Hand radiography button to enter the Handle adjustment mode.**

The LEDs on the following buttons are:.

- The Body thickness change buttin goes off.
- The Front-view button blinks.
- The Hand radiography button blinks.

3 **Pull the handle by a spring gauge.**

As the adjustment position number is indicated on the tube voltage display, apply the force in the indicated direction with a spring gauge

Reference "Table 8-15 Relation between handle adjustment position number and pull power"
"Fig. 8-9 Adjustment of the Handle"

Wrap a string around the end of the handle, and hook the spring gauge to the string, then apply the force. By pulling the handle forward, the adjustment value increases. By pulling the handle backward, the adjustment value decreases.

Fig. 8-9 Adjustment of the Handle

4 Register the correction values (12 positions)

Press the Registration button. The next adjustment position number will appear. Repeat Step 3 and 4 with all the adjustment positions shown in the following table.

> **NOTE** If the system is powered OFF without pressing the Registration button, the correction value will not be registered.

Table 8-15 Relation between handle adjustment position number and pull power

Adjustment position number	Direction and weight of pulling the spring gauge
1*1	Forward left direction (1 kg)*3
2	Forward left direction (1.5 kg)
3	Forward left direction (3 kg)
4	Backward left direction (1 kg)*3
5	Backward left direction (1.5 kg)
6	Backward left direction (3 kg)
7*2	Forward right direction (1 kg)*3
8	Forward right direction (1.5kg)
9	Forward right direction (3 kg)
10	Backward right direction (1 kg)*3
11	Backward right direction (1.5 kg)
12	Backward right direction (3 kg)

*1 When the adjustment position number indicates 1, pull the spring gauge backward left direction with the force of 3 kg, and then forward left direction with 1 kg.

*2 When the adjustment position number indicates 7, pull the spring gauge backward right direction with the force of 3 kg, and then forward right direction with 1 kg.

*3 To pull the spring gauge forward left, backward left, forward right, or backward right direction with the force of 1 kg, pull it gradually within 1kg.

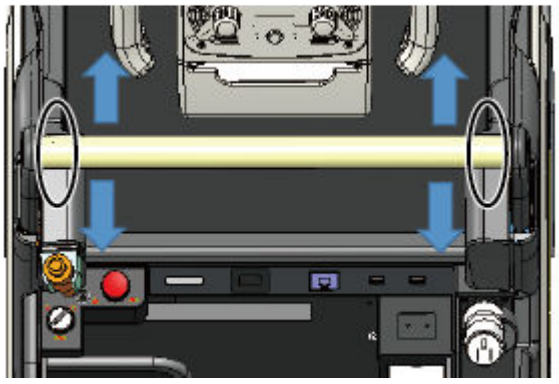


Fig. 8-10 Positions and directions for setting the spring gauge

5 Press the Added filter button to display the registered handle correction value.

"r" is displayed in front of the adjustment position number.
After adjustment, press ±kV keys to choose the adjustment position number, and make sure the adjustment values for each point are correctly registered.

> NOTE If the correction value is as described below, normal driving operability cannot be obtained. In this case, adjust the value correctly.
>
> - The correction values of two adjustment positions in the same direction are same.
> - The magnitude of the pulling force does not match the magnitude of the correction value.

6 Finish the adjustment mode.

Reference "8.1 How to Start Adjustment Mode"

8.6 Adjustment of Battery Voltage Detection

This mode adjusts battery voltage detection circuit. Perform the following procedures when exchanging the MUX POWER board, the MUX CHARGE-32KX board, or XCONT-2018 board.

1 **Enter the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

2 **Press the Foot radiography button to enter the Battery voltage detection adjustment mode. The detected battery voltage is displayed with four digits and 0.1V unit. Press the tube voltage buttons to select [+] or [-]. If [-] is selected, "-" is displayed at the second digits of kV display field.**

The LEDs on the following buttons are:.

- The Body thickness change buttin goes off.
- The Front-view button blinks.
- The Foot radiography button blinks.

3 **Press [mAs +] and [mAs -] key to adjust the displayed value. Press the Registration buttonif the displayed value is within ±0.2V of the measured battery voltage.**

Hint

- Measure the actual battery voltage on + side with TER3 [B120+O] - TER5 [B0] of the MUX POWER-32K board.
- Measure the actual battery voltage on - side with TER5 [B0] -TER4 [B120-O] of the MUX POWER-32K board.

Fig. 8-11 Adjustment of the battery voltage detection – Negative voltage (in case of 125.6 V)

Fig. 8-12 Adjustment of the battery voltage detection – Positive voltage (in case of 125.8 V)

4 **Finish the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

8.7 Setting Date and Time

Set the present year, month, day and time for the unit with this mode.

1 Enter the adjustment mode.

Reference "8.1 How to Start Adjustment Mode"

2 Press the Arm radiography button to enter the Date and Time setting mode.

The LEDs on the following buttons are:.

- The Body thickness change buttin goes off.
- The Front-view button blinks.
- The Arm radiography button blinks.

Fig. 8-13 Setting Date (In case of August 6, 2021)

3 Change the date.

1. Press the Added filter button to enter the Year/Month/Day setting mode.
   The Added filter button LED goes off.
2. Press the Tube voltage button to change a year (last two digits of the year).
3. Press the Current-time product (-) button to change a month.
4. Press the Current-time product (+) button to change a date.

4 Change the time.

1 Press the Added filter button to enter the Time setting mode.
The Added filter button LED blinks.

2 Press the Tube voltage button to change an hour (24-hour display).

3 Press the Current-time product buttons to change a minute.

Fig. 8-14 Setting Time (In case of PM 1:08)

5 Finish the adjustment mode.

Reference "8.1 How to Start Adjustment Mode"

8.8 Display and Reset of Error Log

In this mode error codes of the past is shown (maximum 100 times, in chronological order). If 100 errors are already recorded, the next new data will overwrite the oldest data.

8.8.1 To Check Error Log

1 **Enter the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

2 **Press the Body thickness change button to blink, and then The Lumbar radiography button to enter the Error log display mode.**

The LEDs on the following buttons are:.

- The Body thickness change button blinks
- The Front-view button blinks.
- The Lumbar radiography buttonblinks.

Fig. 8-15 Display of Error Log (In case of the latest error code was F03)

Fig. 8-16 Display of Error Log (In case of the latest error code was F6d)

3 **Check the error codes.**

Error code is displayed on mAs display field and log number is displayed on kV display field.
"Fault" is displayed on the dose display field for failure, and "Alert" is displayed for warning.
The number "000" means the latest error and the number "099" means the oldest error.
Press [kV-] to increase a log number and press [kV+] to decrease a log number. Change the Failure/Warning display with the Tube Current log number with the Current-time product buttons.

Reference "10.1 Error message list"

4 **Check the date and time of the error.**

Press the Added filter button to switch between the "date and time" page and the "log number and error code" page.

8.8.2 To Reset Error Log

1 **Press the Registration button for more than 3 seconds.**

The error log is reset.

8.8.3 To Finish Error Log Display Mode

1 **Finish the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

8.9 Display of Running Speed

The running speed is calculated from the number of rotation detected by the encoder in the motor. Use this for checking failure of the encoder.

1 **Enter the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

2 **Press the Body thickness change button to blink, and press buttons in order of the Front-view and Arm radiography to enter the Running speed display mode.**

The LEDs on the following buttons are:.

- Body thickness change button blinks
- The Front-view button blinks.
- The arm radiography button blinks

3 **Check the displayed running speed.**

The running speed is displayed on the kV and mAs display fields.

Hint The character "-" is displayed when driving backward.

Fig. 8-17 An example of running speed display (where the speed of both of the wheels is 0 km/hour)

4 Show the speed of each wheel.

Press the Added filter button to show the following running speed.

- Running speed of both wheels (km/h)
- Running speed of the left wheel (km/h)
- Running speed of the right wheel (km/h)

Fig. 8-18 An example of running speed display (left wheel speed)

Fig. 8-19 An example of running speed display (right wheel speed)

5 Finish the adjustment mode.

Reference "8.1 How to Start Adjustment Mode"

8.10 Battery Voltage Drop Check Mode

Use this mode to check the the amount of battery voltage drop when load of the X-ray exposure is applied.

NOTE This mode is not available in low battery mode.

1 Enter the adjustment mode.

Reference "8.1 How to Start Adjustment Mode"

2 Press the Lumbar radiography button to enter the Battery Voltage Drop Check Mode.

The LEDs on the following buttons are:.

- The Body thickness change button goes off
- The Front-view button blinks.
- The Lumbar radiography button blinks.

The exposure condition for inspection is shown on the mAs and kV display fields.

Table 8-16 The exposure condition for inspection

	kV display fields	mAs display fields
32kW type (MobileDaRt Evolution / MobileArt Evolution high-power type)	80	200
12.5kW type (MobileArt Evolution standard type)	80	160

Fig. 8-20 Example of the display of the battery voltage drop check mode (Inspection exposure condition display, 32 kW)

3 Check the battery voltage drop.

1 Perform the X-ray exposure by pressing the hand switch.
Values of tube voltage and tube current are measured after exposing for 80 msec.

2 The measured values are displayed.

Hint In other modes, the tube voltage and tube current are measured after exposing for 20 msec. These values may not match the measured value of the Battery Voltage Drop Check Mode.

4 Show the measured values.

Press the Added filter button to show the following measured value.

- Exposure condition
- Battery voltage on the plus side
- Battery voltage on the minus side
- Measured tube voltage
- Measured tube current

Fig. 8-21 Example of the display of the battery voltage drop check mode
(110.2 V of the battery voltage on the plus side is displayed)

Fig. 8-22 Example of the display of the battery voltage drop check mode
(110.8 V of the battery voltage on the minus side is displayed)

Fig. 8-23 Example of the display of the battery voltage drop check mode
(80kV of the measured tube voltage is displayed)

Fig. 8-24 Example of the display of the battery voltage drop check mode
(196 mA of the measured tube current is displayed)

Hint Battery voltage (plus and minus side) are measured 80 msec after the exposure starts in this mode. They do not match the value measured with the following section.

- "8.5 Adjustment of the Handle"

NOTE Currently, each value is a reference and the status of the device and battery from the values obtained in this mode cannot be determined.

5 Finish the adjustment mode.

Reference "8.1 How to Start Adjustment Mode"

8.11 Battery Code Setting Mode

This mode allows to set the battery code corresponding to each battery type.

8.11.1 To Enter Battery Code Setting Mode

1 **Enter the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

2 **Press the Body thickness change button to blink, and then press the hand radiography button to enter the Battery code setting mode.**

The LEDs on the following buttons are:.

- Body thickness change button blinks
- The Front-view button blinks.
- The hand radiography button blinks

8.11.2 To Check Battery code

1 **The Setting Item Number "0" is displayed at the third digit (from the right) of kV display.**

The "0" indicates the Battery Code Display Mode.

2 **The battery code is shown at the first digit of mAs display. The Battery code6, GS YUASA PXL12090 is registered as default.**

Table 8-17

Battery code	Battery name
0	Reserved
1	Reserved
2	Reserved
3	Reserved
4	Reserved
5	GS Yuasa PX12090
6	GS Yuasa PXL12090
E	New battery other than 0 - 6

8.11.3 To Change Battery Code

1 **Press the Current-time product buttons and select the battery code corresponding to the battery type, and then press the Registration button to register it.**

Fig. 8-25 Battery code setting (when Battery code6 is selected)

8.11.4 Setting of Battery Charge Restriction

For overcharge prevention of a battery, this system controls not to charge the battery if the battery remains enough.
Follow the instruction below to make sure the battery charge restriction is enabled. If disabled, set this setting to be enabled.

> **NOTE** Repetition of charging the battery even though the battery remains enough may cause the deterioration of the battery life and performance. The battery charge restriction must be enabled for overcharge prevention of the battery.

1 **Press the Added filter button on the battery code setting mode, to enter the Battery charge restriction setting mode.**

2 **The setting item number 9 (Battery charge restriction setting) is displayed at the third digit of kV display.**

3 **The battery charge restriction ON/OFF setting is displayed at mAs display.**

Fig. 8-26 Battery charge restriction setting mode (Battery charge restriction is enabled)

Fig. 8-27 Battery charge restriction setting mode (Battery charge restriction is disabled)

4 **Press the Current-time product (+) button to set the battery charge restriction to ON.**

5 **When the battery charge restriction is set completely, press the Registration button to register.**

6 **Press the Added filter key to enter the setting mode of status indicator lamp during battery charge.**

8.11.5 Setting of Status Indicator Lamp During Battery Charge

The status indicator lamp blinks in blue while the battery is charging.
Follow the instruction below to make sure the status indicator lamp blinking during the battery charge is enabled. If being disabled unintentionally, set this setting to be enabled.

1 **Press the Added filter button on the battery charge restriction setting mode, to enter the Status indicator lamp during battery charge setting mode.**

2 **The setting item number 10 (Status indicator lamp during battery charge setting) is displayed at the third digit of kV display.**

3 **The battery charge ON/OFF setting is displayed at mAs display.**

Fig. 8-28 ☒ 8-26 Status indicator lamp during battery charge setting mode is ON

Fig. 8-29 ☒ 8-27 Status indicator lamp during battery charge setting mode is OFF

4 **Press the Current-time product (+) button to set the status indicator during the battery charge.**

5 **When thesetting is complited, press the Registration button to register.**

6 **Press the Added filter button to return to the battery code setting mode.**

8.11.6 To Finish Battery Code Setting Mode

1 **Finish the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

8.12 Dose Area Product Meter Adjustment Mode

8.12.1 To Enter Dose Area Product Meter Adjustment Mode

1 **Remove the power plug from the outlet, and turn OFF the power to the system.**

2 **Remove the right-inside cover and right-side cover.**

Reference "3.5 How to Open/Close Covers"

3 **Set the DIP switches of the XCONT-2018 board as follows. Turn ON the main system power.**

- SW3-5 (Dose area product meter) : ON
- SW3-8 (Adjustment mode) : ON

NOTE The SW3-5 must be ON to enter the dose area product meter adjustment mode.

4 **Press the Body thickness change button to blink, and then the Abdominal radiography button to enter the dose area product meter adjustment mode.**

The LEDs on the following buttons are:.

- The Body thickness change button blinks
- The Front-view button blinks.
- The Abdominal radiography button blinks

5 **"dAP 2" is displayed on the kV/mAs display fields. Press the Added filter button and go on to the radiography condition display.**

6 **The kV/mAs display fields on X-ray control panel shows the radiography conditions. Radiography conditions are 70kV and 2.0mAs.**

At this point, nothing is displayed at the mGy display because radiography has not been executed with the dose area product meter.

Fig. 8-30 Dose area product meter adjustment mode (Radiography condition display)

8.12.2 Preparation for Calibration of Dose Area Product Meter

1 **Attach the dose area product meter to the collimator and place the Reference Dose meter as shown in the following figure.**

1 Adjust SOD (source object distance) to 100 cm, and the exposure field to 35 cm (approx. 14 inch) square.

2 Place the Reference Dose meter, in the center of the exposure field apart from the floor surface to eliminate the effect of the backward scattered ray (with styrol box, for example).

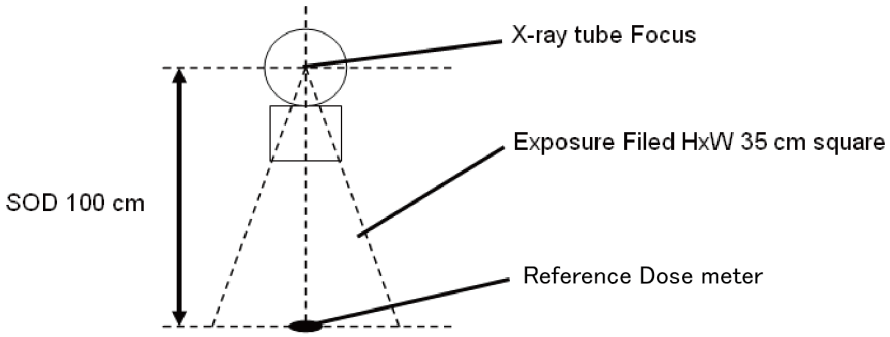


Fig. 8-31 Setting Dose Area Product Meter for calibration

8.12.3 Calibration of Dose Area Product Meter

1 **Press the Added filter button to enter the calibration screen of the dose area product meter.**

2 **In the calibration of dose area product meter mode, the adjustment value shows on the kV and mAs display fields of X-ray control panel. (LED of Added filter button is blinking). The initial adjustment value is 100.**

At this point, nothing is displayed at the mGy display because radiography has not been executed with the dose Area Product Meter.

3 **Take the radiography and measure the exposure dose.**

After the radiography is completed, the exposure dose (Unit : mGy) calculated from the Dose Area Product is displayed.

Fig. 8-32 Dose area product meter adjustment mode (Adjustment value, Exposure dose display)

4 **Press the Current-time product button to change the Adjustment Value.**

The exposure dose amount (Unit:mGy) display changes with the Adjustment Value. Adjust the Value so that this exposure dose and the measured dose (Reference Dose meter) are equal.

5 **Press the Registration button to register the Adjustment Value.**

6 **Press the Added filter button.**

X-ray control panel display is changed, then the radiography conditions will appear on the kV and mAs display on the panel.

7 **Check that the calibrated Dose Area Product is displayed in the dose display.**

Fig. 8-33 Dose area product meter adjustment mode (Radiography condition, dose area product display)

8.12.4 To Finish Dose Area Product Meter Adjustment Mode

1 **Turn OFF the power.**

2 **Set the DIP switches of the XCONT-2018 board as follows.**

- DIPSW3-5 (Dose area product mete): ON
- DIPSW3-8 (Adjustment mode): OFF

3 **Attach the removed covers in reverse order.**

8.13 Battery Usage Setting Mode

Battery usage setting mode is used to check the starting date of battery usage and the charging count, and to set the battery usage information when the battery or the XCONT-2018 board is replaced. Adjust the setting with the following procedure when the battery or the XCONT-2018 board is replaced.

8.13.1 To Enter Battery Usage Setting Mode

1 **Enter the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

2 **Press keys in order of the Oblique-view button and the Chest radiography button to enter the battery usage setting mode.**

The LEDs on the following buttons are:.

- The Body thickness change button goes off
- The Oblique-view button blinks
- The Chest radiography blinks

8.13.2 To Check Battery Usage Information

The starting date of the battery usage information is displayed at kV and mAs displays. Battery usag and the charge count are displayed at the first to fifth digits of mGy display. Press the Collimator Lamp key to switch the usage information on the mGy display.

Table 8-18

6th Digits of mGy	Item
0	Charge count (by single count)
1	Full charge count (by single count)
2	Charge limit count (by single count)
3	Low battery occurence count (by single count)
4	Error bbb occurence count (by single count)
5	Error bb0 occurence count (by single count)
6	Error bb1 occurence count (by single count)

Fig. 8-34 EXAMPLE: Battery Starting date of use and battery Charge count display (Aug. 6, 2021, 68 counts)

8.13.3 To Reset Battery Usage Information

It is required to reset the starting date of the battery usage when the battery is replaced.

1 **While the usage information is displayed, press and hold the Registration button for 3 seconds.**

The starting date of the battery usage is reset to current date, and the charge count is reset to 0.

8.13.4 To Finish Battery Usage Setting Mode

1 **Finish the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

8.14 CPU Board Battery Usage Information Setting Mode

This mode is used to check the starting date of CPU board battery usage, and to set the usage information of CPU board battery when the CPU board battery or the XCONT-2018 board is replaced. Adjust the setting by the following procedure when the CPU board battery or the XCONT-2018 board is replaced.

8.14.1 To Enter CPU Board Battery Usage Information Setting

1 **Enter the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

2 **Press in order of the Oblique-view button and the abdominal radiography button to enter the CPU board battery usage information setting mode.**

The LEDs on the following buttons are:.

- The Body thickness change button goes off.
- The Oblique-view button blinks.
- The Abdominal radiography button blinks.

8.14.2 To Check CPU Board Battery Usage Information

The usage information is displayed at kV and mAs displays.

Fig. 8-35 EXAMPLE: CPU board battery's starting date of use (Aug. 6, 2021)

8.14.3 To Reset CPU Board Battery Usage Information

It is required to reset the starting date of use when CPU board battery is replaced.

1 **While the usage information is displayed, press and hold the Registration button for 3 seconds.**

The starting date of CPU board battery usage is set to the current date.

8.14.4 To Finish CPU Board Battery Usage Information Setting

1 **Finish the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

8.15 Emergency Brake Release Battery Usage Information Setting Mode

This mode is used to check the starting date of use of the emergency brake release battery, and to set its starting date of use when the emergency brake release battery or the XCONT-2018 board is replaced. Adjust the setting by the following procedure when the emergency brake release battery or the XCONT-2018 board is replaced.

8.15.1 To Enter Emergency Brake Release Battery Usage Information Setting Mode

1 **Enter the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

2 **Press in order of the Oblique-view button and the Lumbar radiography button to enter the emergency brake release battery usage information setting mode.**

The LEDs on the following buttons are:.

- The Body thickness change button goes off.
- The Oblique-view button blinks.
- The Lumbar radiography button blinks.

8.15.2 To Check Emergency brake release battery usage information

The usage information is displayed at kV and mAs displays.

Fig. 8-36 EXAMPLE: Emergency brake release battery's starting date of use (Aug. 6, 2021)

8.15.3 To Reset Emergency Brake Release Battery Usage Information

It is required to reset the starting date of use after replacing the emergency brake release battery.

1 **While the usage information is displayed, press and hold the Registration button for 3 seconds.**

The starting date of Emergency brake release battery is set to the current date.

8.15.4 To Finish Emergency Brake Release Battery Usage Information Setting Mode

1 **Finish the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

8.16 X-ray Tube Unit Usage Information Setting Mode

This mode is used to check the X-ray tube unit's starting date of use and the count of radiography, and to set the X-ray tube unit usage information when the X-ray tube unit or the XCONT-2018 board is replaced. Adjust the setting by the following procedure when the X-ray tube unit or the XCONT-2018 board is replaced.

8.16.1 To Enter X-ray Tube Unit Usage Information Setting Mode

1 **Enter the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

2 **Press in order of the Oblique-view button and the Arm radiography button to enter the X-ray tube unit usage information setting mode.**

The LEDs on the following buttons are:.

- The Body thickness change button goes off.
- The Oblique-view button blinks.
- The Arm radiography button blinks.

8.16.2 To Check X-ray Tube Unit Usage Information

The usage information is displayed at kV and mAs displays.

Battery usage and the charge count are displayed at the first to fifth digits of mGy display. Press the Collimator Lamp key to switch the usage information on the mGy display.

Table 8-19

6th Digits of mGy	Item
0	Radiography count (by single count)
1	Accumulated mAs (by 1000 mAs unit)
2	Tube discharge count (by single count)
3	Accumulated filament heating time (by 100 sec unit)
4	Accumulated starter rotation time (by 100 sec unit)

Fig. 8-37 EXAMPLE: X-ray tube unit's starting date of use and count of radiography (Aug. 6, 2021, 56 counts)

8.16.3 To Reset X-ray Tube Unit Usage Information

It is required to reset the starting date of use after the X-ray tube unit is replaced.

1 **While the usage information is displayed, press and hold the Registration button for 3 seconds.**

The starting date of X-ray tube unit usage is set to the current date, and the count of radiography is set to 0.

8.16.4 To Finish X-ray Tube Unit Usage Information Setting Mode

1 **Finish the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

8.17 Motor Usage Information Setting Mode

This mode is used to set the usage information, such as the starting date of use of motor and the accumulated running distance, when the motor is replaced. Adjust with the following procedure.

8.17.1 To Enter Motor Usage Information Setting Mode

1 **Enter the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

2 **Press in order of the Oblique-view button and the Foot radiography button to enter the motor usage information setting mode.**

The LEDs on the following buttons are:.

- The Body thickness change button goes off.
- The Oblique-view button blinks.
- The Foot radiography button blinks.

8.17.2 To Check Motor Usage Information

The usage information is displayed at kV and mAs displays.

The accumulated running distance and the starting date of use of motor is displayed at mGy display. Press the Collimator Lamp key to switch the usage information on the mGy display.

Fig. 8-38 EXAMPLE: Starting date of use of motor and accumulated running distance of motor (Aug. 6, 2021, 55 km)

8.17.3 To Reset Motor Usage Information

It is required to reset the starting date of use after replacement of the motor.

1 **While the usage information is displayed, press and hold the Registration button for 3 seconds.**

The starting date of the motor usage is set to the current date, and the accumulated running distance is set to 0.

8.17.4 To FinishX-ray Tube Unit Usage Information Setting Mode

1 **Finish the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

8.18 Arm Usage Information Setting Mode

This mode is used set the usage information, such as wire's starting date of use and the arm moving distance, when the wire is replaced. Adjust with the following procedure.

8.18.1 To Enter Arm Usage Information Setting Mode

1 **Enter the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

2 **Press the Body thickness change button to blink, and then press in order of the Oblique-view button and the Chest radiogrqphy button to enter the arm usage information setting mode.**

The LEDs on the following buttons are:.

- The Body thickness change button blinks.
- The Oblique-view button blinks.
- The Chest radiogrqphy button blinks.

8.18.2 To Check Arm Usage Information

The usage information is displayed at kV and mAs displays.

The accumulated running distance and the starting date of use of motor. Press the Collimator Lamp key to switch the usage information on the mGy display.

Table 8-20

6th Digits of mGy	Item
0	Arm upward moving distance (by 1 km unit)
1	Arm downward moving distance (by 1 km unit)
2	Encoder usage setting

Fig. 8-39 EXAMPLE: Starting date of use of wire and accumulated moving distance of arm (Aug. 6, 2021, 3km)

8.18.3 To Reset Arm Usage Information

It is required to reset the starting date of use after replacement of the wire.

1 **While the usage information is displayed, press and hold the Registration button for 3 seconds.**

The starting date of the wire usage is set to the current date, and the arm moving distance is set to 0.

8.18.4 To Check Encoder Usage Setting

MobileArt Evolution does not have an encoder, so make sure the Encoder usage setting is set to "oFF".
If the setting is "on", follow the procedure below to set it to "oFF".

1 **With the encoder setting indicated as "oFF", and press and hold the Added filter button for three seconds.**

The Encoder usage setting is set to "oFF".

Fig. 8-40 EXAMPLE: Display of the encoder usage setting ("oFF" is set)

8.18.5 To Finish Arm Usage Information Setting Mode

1 **Finish the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

8.19 Usage Information Collective Setting Mode

This mode is used to reset the starting date of use of components collectively, such as the battery, the CPU board battery, the emergency brake release battery, the X-ray tube unit, the motor, and the charging count, radiography count and the accumulated running distance.

8.19.1 To Enter Usage Information Collective Setting Mode

1 **Enter the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

2 **Press the Body thickness change button to turn off the light, and press in order of the Oblique-view button and the Hand radiography button to enter the Usage Information Collective Setting Mode.**

The LEDs on the following buttons are:.

- The Body thickness change button goes off.
- The Oblique-view button blinks.
- The Hand radiography button blinks.

8.19.2 To Reset Usage Information of All Components

When provide the system or replace all components, it is required to reset the usage information of all components.

1 **In this mode, the current date is displayed at the kV and mAs displays.**

2 **Press and hold the Registration button for 5 seconds.**

The starting dates of use of all components are set to the current date. The count of charging, the count of radiography, and the accumulated running distance are set to 0.

Fig. 8-41 EXAMPLE: Usage information collective setting mode (When the current date is Aug. 6, 2021)

8.19.3 To Finish Usage Information Collective Setting Mode

1 **Finish the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

8.20 Dose Area Product Calculation Unit Setting Mode

> **NOTE** When the registered information is modified, that information must be saved in the memory by pressing the Registration keyon the initial screen of the control panel.
>
> The registered information in each adjustment mode is the interim registration. If the adjustment mode is finished without saving in the memory, the modified information cannot be refleted to the system.

8.20.1 To Remove Collimator Cover

1 **Remove the power plug from the outlet, and turn OFF the power to the system.**

2 **Remove the collimator cover, and remove screws at eight positions as shown below.**

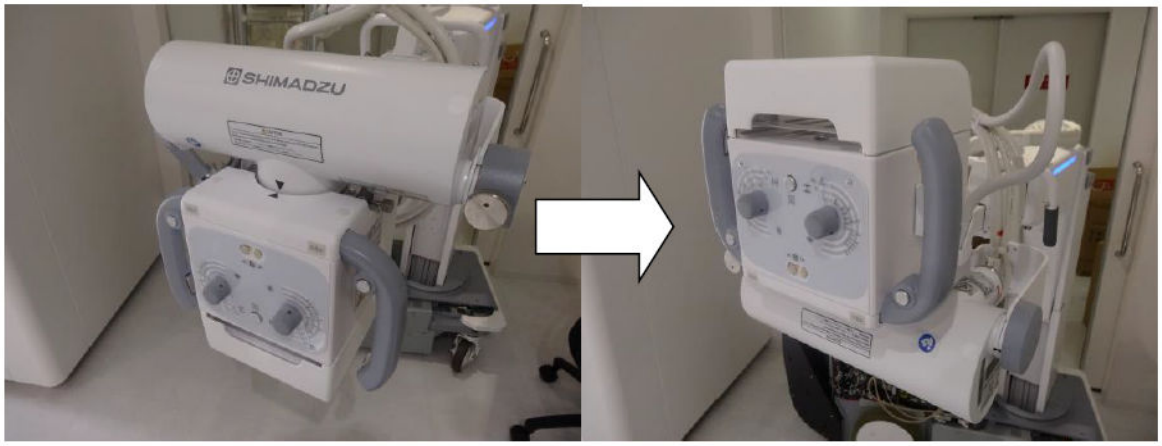


Fig. 8-42

Fig. 8-43 Front side of Collimator

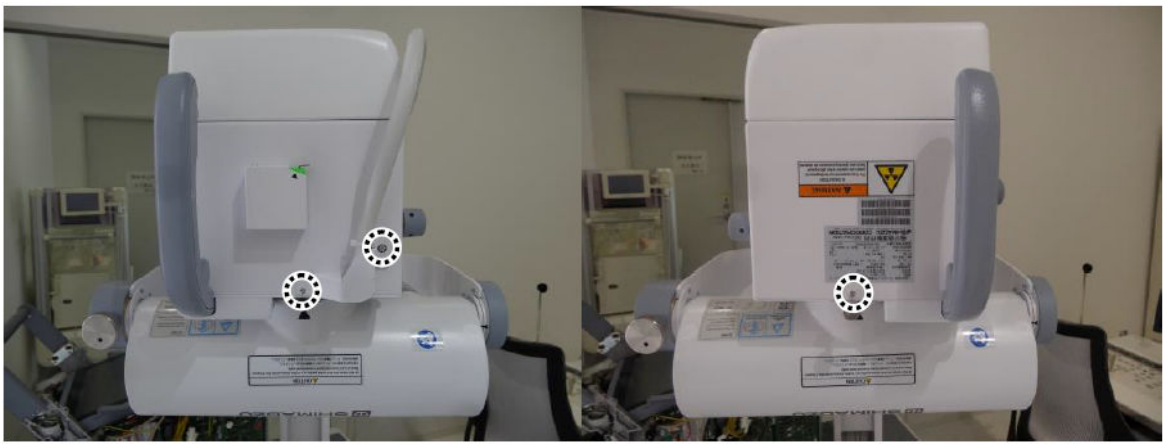


Fig. 8-44 Lateral side of Collimator

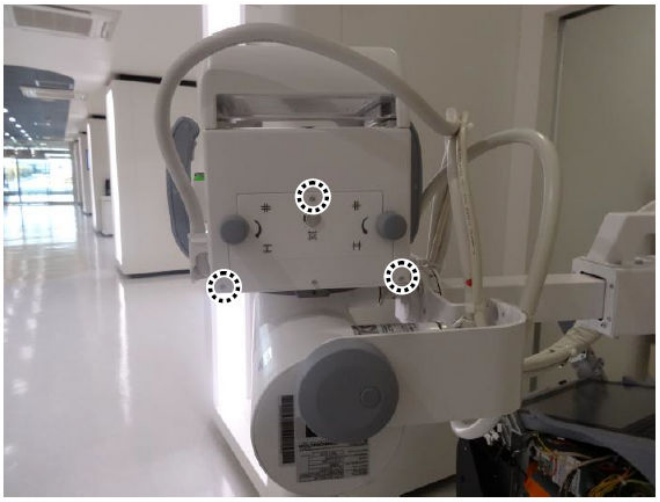


Fig. 8-45 Rear side of Collimator

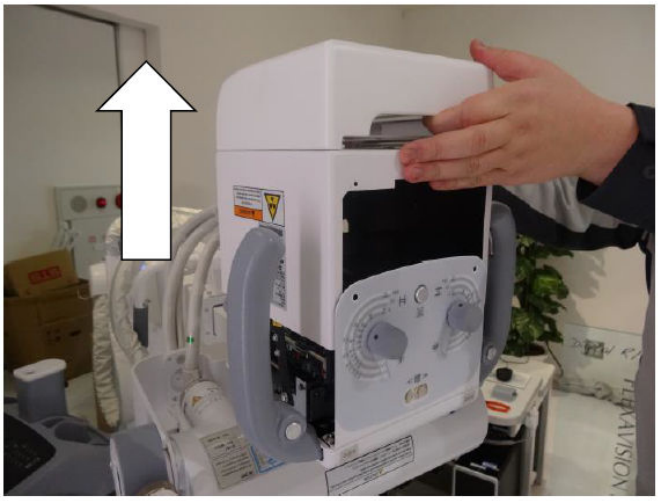


Fig. 8-46

8.20.2 To Enter Dose Aarea Product Calculation Unit Setting Mode

1 **Remove the power plug from the outlet, and turn OFF the power to the system.**

2 **Remove the right-inside cover and right-side cover.**

Reference "3.5 How to Open/Close Covers"

3 **Set each DIP SW on the XCONT-2018 board as shown below, and turn ON the power.**

- DIPSW3-5: OFF
- DIPSW3-8 (for execution of setting mode): ON

4 **Press the Body thickness change button to blink, and then press the Abdominal radiography button to enter the dose area product calculation unit setting mode.**

The LEDs on the following buttons are:.

- The Body thickness change button blinks.
- The Front-view button blinks.
- The Abdominal radiography button blinks.

5 **When enter the dose area product calculation unit setting mode, the exposure condition is displayed in the control panel.**

6 **Make sure the default condition is displayed as 40kV, 10mAs.**

7 **Press the Added filter button to switch the screen to the ON/OFF setting of the dose area product calculation unit.**

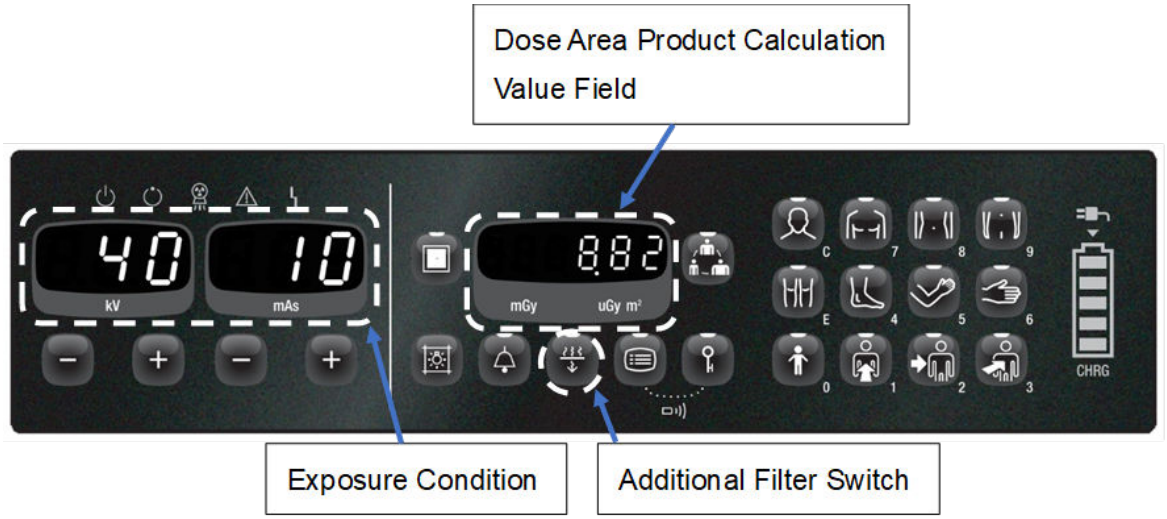


Fig. 8-47 Initial screen of the dose area product calculation unit

8.20.3 ON/OFF Setting of Dose Area Product Calculation Unit

1 **The ON/OFF setting of the dose area product calculation unit is displayed in the kV display.**

2 **Make sure the setting is ON. If OFF, press the Tube voltage button to change the setting to ON, and then the Registration key.**

3 **Press the Added filter button to switch the screen to the setting the collimator leaf opening amount.**

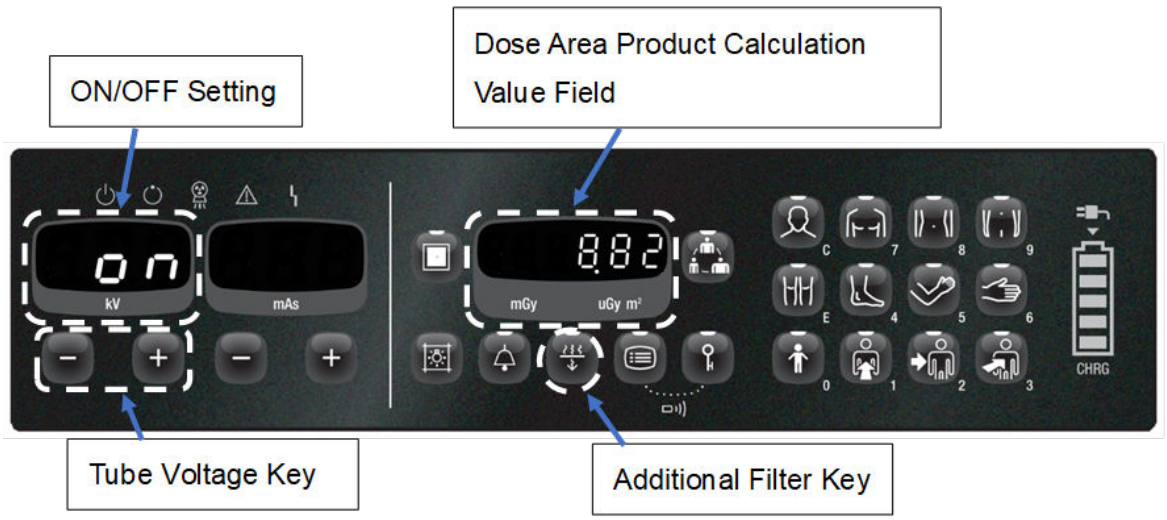


Fig. 8-48 ON/OFF setting screen of the dose area product calculation unit

8.20.4 Setting of collimator leaf opening amount

The collimator leaf opening amount has been specified at the factory. This setting is needed only for the re-adjustment of opening amount when the collimator is replaced. Press the Added filter button to enter the adjustment of whole number coefficients.

1 **The following information is displayed in the kV display field and mAs display field.**

Table 8-21

kV display field	mAs display field
The registration point from “h0”, “h35”, “v0”, “v35”	The registered detection voltage (0.00to 5.00V) of the collimator leaf opening amount at the registration point displayed in kV display field.

* *Press the tube voltage switch to switch the registration point in kV display field.

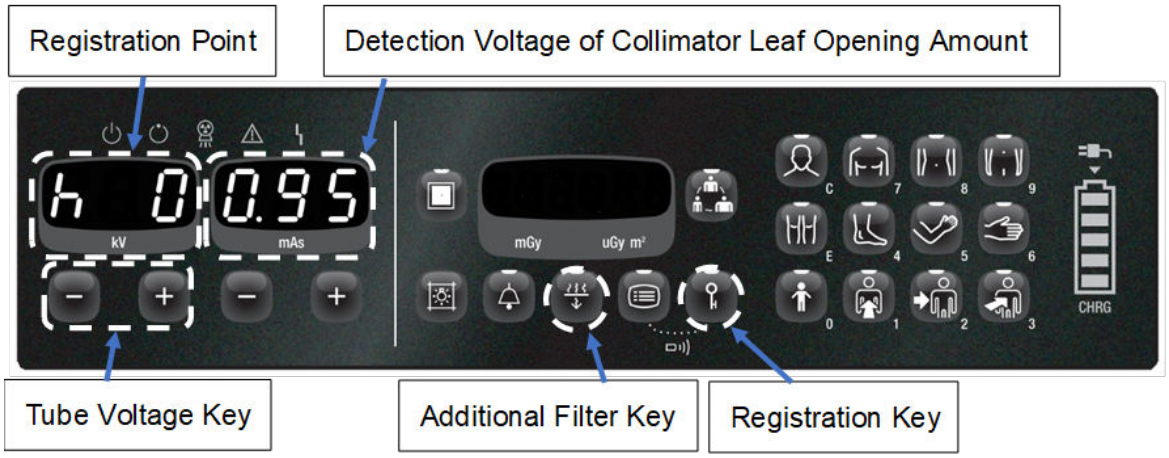


Fig. 8-49 Setting screen of the collimator leaf opening amount

2 **Remove the collimator cover.**

Register the collimator leaf opening amount at each registration point, as shown in the table below. After adjustment, the opening amount at the registration point can be updated by pressing the Registration buttonat the registration point.

Table 8-22

Registration point (kV display field)	Collimator leaf state	How to adjust the collimator
h0	The horizontal leaf level when the horizontal radiation field becomes 0cm.	Adjust by the lighting field.
h35	The horizontal leaf level when the horizontal radiation field becomes 35cm at 1m of SID.	Adjust using the adjustment jig (*).
v0	The horizontal leaf level when the vertical leaf becomes 0cm.	Adjust by the lighting field.
v35	The vertical leaf level when the vertical radiation field becomes 35cm at 1m of SID.	Adjust using the adjustment jig (*).

* *(*)Protective tool supplied with the collimator

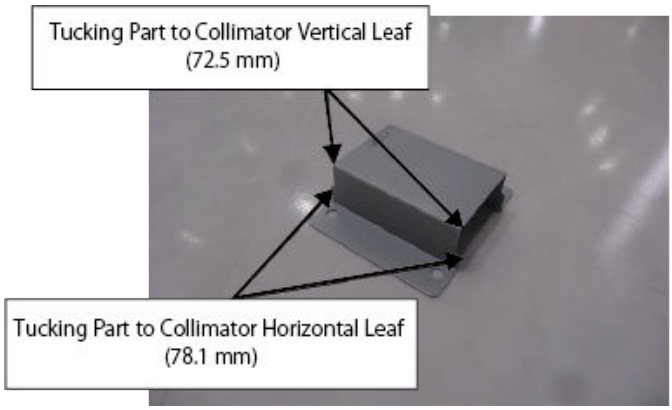


Fig. 8-50 Ajustment jig

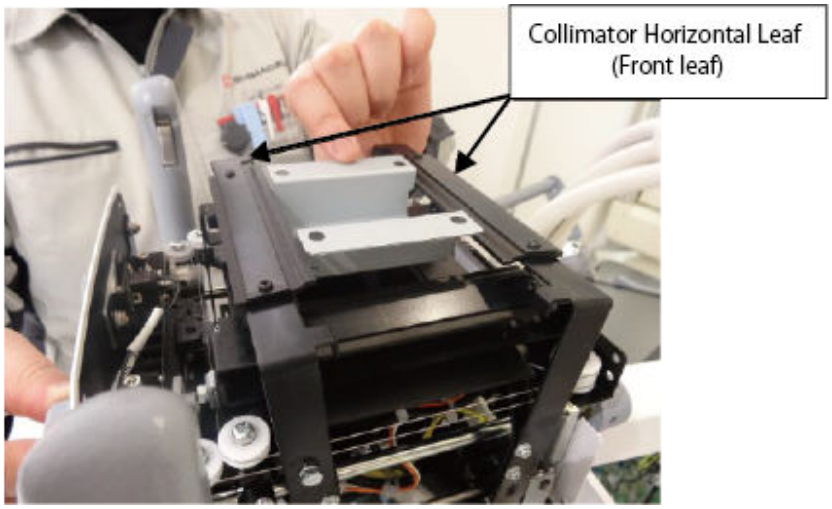


Fig. 8-51 Adjustment of horizontal leaf level

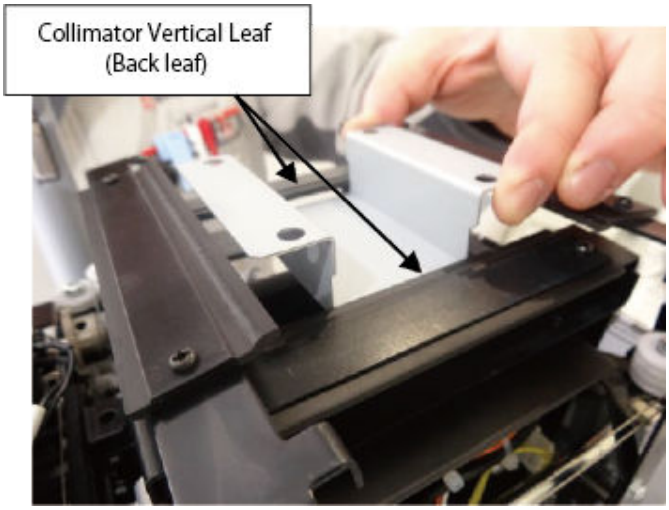


Fig. 8-52 Adjustment of vertical leaf level

3 **Press the Added filter button for two times to enter the dose area product calculation unit setting mode.**

4 **When the control panel shows the initial screen for the dose area product calculation unit setting mode, press the Registration key.**

All data registered on each adjustment screen is stored to the memory.

Fig. 8-53 Registration of all data

> **NOTE** When the registered information is modified, that information must be saved in the memory by pressing the Registration keyon the initial screen of the control panel.
>
> The registered information in each adjustment mode is the interim registration. If the adjustment mode is finished without saving in the memory, the modified information cannot be refleted to the system.

5 **Turn OFF th power to the system.**

6 **Attach the removed collimator cover.**

7 **Turn ON the power to the system.**

8 **Press the Body thickness change button to blink, and then press the Abdominal radiography button to enter the dose area product calculation unit setting mode.**

The LEDs on the following buttons are:.

- The Body thickness change button blinks.
- The Front-view button blinks.
- The Abdominal radiography button blinks.

9 **Press the Added filter button for three times to switch the screen to set the whole number coefficients and select the additional filter.**

8.20.5 To Adjust Whole Coefficient and Select Added Filter

1 **ALL Comp value (whole adjustment coefficient) of the dose area product meter is displayed in the kV display field, and the following Added filter is displayed in the mAs display field.**

Table 8-23

mAs display field	Filter information
0	No filter
1	Added filter1 (Arbitrary additional filter)
2	Added filter2 (Arbitrary additional filter)

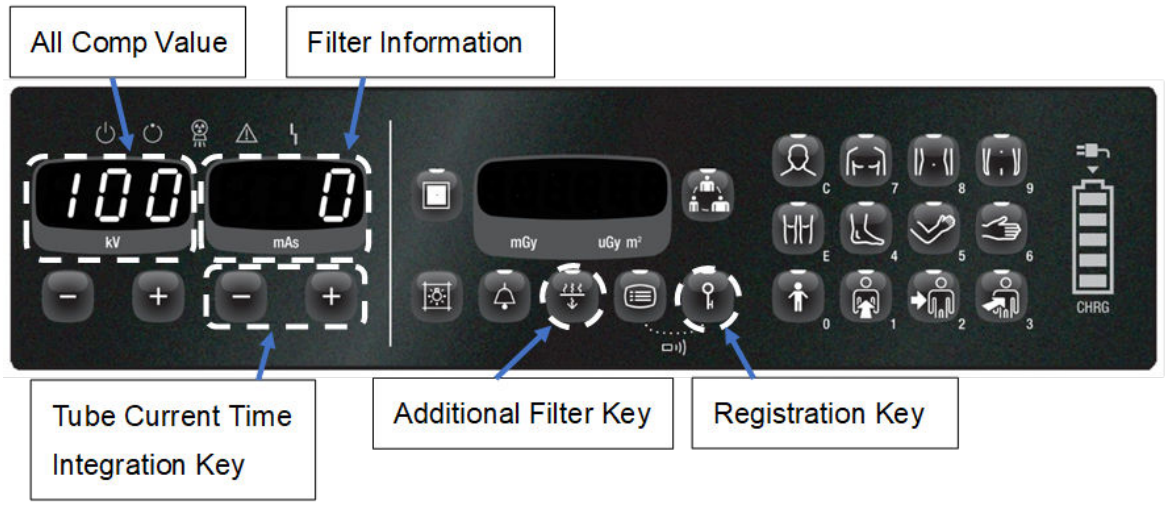


Fig. 8-54 Screen to adjust the whole coefficient and to select the additional filter

2 **Make sure the All Comp value is 100. The modification of All Comp value is not needed. If modify, change the value by the Tube voltage button , and press the Registration button to save that value.**

3 **Press the tube current time integration key to select a filter type.**

4 **Press the Added filter button to switch the screen to the calibrating the dose area product meter for the displayed filter type.**

8.20.6 To Calibrate Dose Area Product Calculation Unit

Calibrate the dose area product calculation unit for the filter type selected in the previous section, "To adjust the whole coefficients and to select the additional filter".

When "0" is selected in the filter type, this setting is not required because the calibration of the dose area product calculation unit has been made at the factory.

When Added filter1 and 2 are used, the dose area product calculation unit must be calibrated.

Use the calibrated dose area product calculation unit to set the measurement system as shown below.

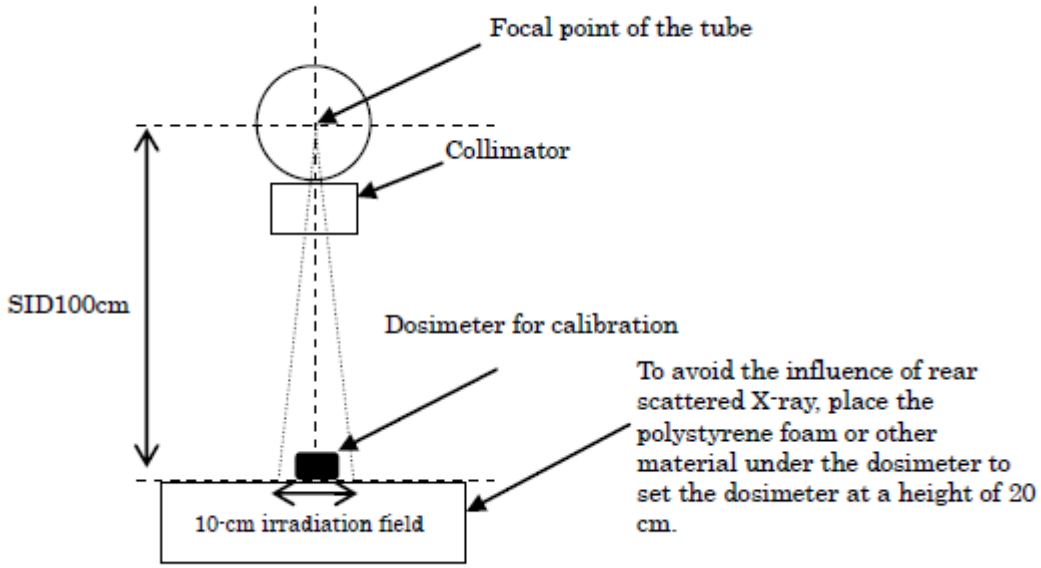


Fig. 8-55 Measurement system

1 The following information is displayed in each display field.

Table 8-24

kV display field	mAs display field	Ares dose display field
Tube voltage to be adjusted. The value is changed by step below: • 32kW type (MobileArt Evolution high-power type) 40kV, 50kV, 60kV, 70kV, 80kV, 90kV, 100kV, 110kV, 120kV, 130kV, 133kV • 12.5kW type (MobileArt Evolution standard type) 40kV, 50kV, 60kV, 70kV, 80kV, 90kV, 100kV, 110kV, 120kV, 125kV	Filter type selected in the adjustment of whole coefficients: 0: No filter 1: Added filter1 2: Added filter2	ckV at kV displayed in the kV display field. (Tube voltage total filtering coefficient basic data)

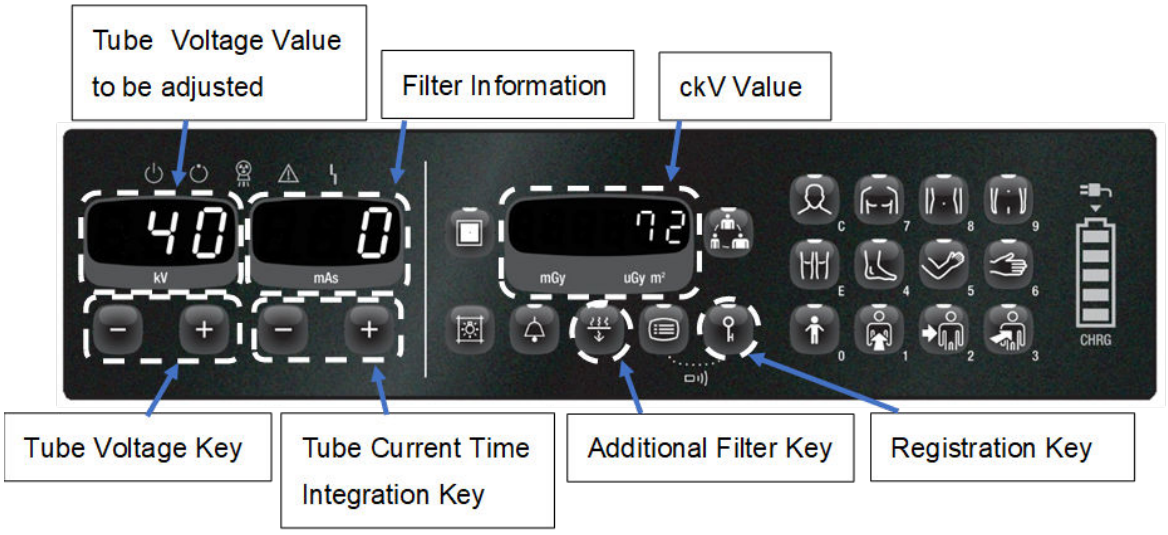


Fig. 8-56

Table 8-25 Calibration screen of dose area product calculation unit

2 Press the Tube voltage button to change kV to be set.

NOTE The mAs value is not displayed. The value (10mAs) that was set on the initial screen is displayed on the mAs display.

3 **By changing kV, set ckV at each kV in the table below. Select the filter type, and perform an exposure by the measurement system at each kV, then write down the measurement value of the dosimeter in the table below. And also, calculate ckV at each kV using the following calculation formula, and write down the value in the table below. (These values must be memorized for the later confirmation.)**

- For the dosimeter in mGy unit ckV = ESD (measured value) x 1000
- For the dosimeter in µGy unit ckV = ESD (measured value)
- ESD = Measured dose value [mGy] or [µGy]

Table 8-26 32kW type (MobileArt Evolution high-power type)

kV	Measured value (µGy)	ckV
40		
50		
60		
70		
80		
90		
100		
110		
120		
130		
133		

*

Table 8-27 12.5kW type (MobileArt Evolution standard type)

kV	Measured value (µGy)	ckV
40		
50		
60		
70		
80		
90		
100		
110		
120		
125		

* (Reference) 115 mR = 1 mGy, 1 mGy = 1000µGy

* (Reference) The software of dose area product calculation unit calculates the following information.

ESD＝A/100×C/10000×R×1/D
ESD[mGy]: Irradiation dose
A: All Comp value (whole adjustment coefficient)
C: ckV value (Tube voltage total filtering coefficient basic data)
R[mAs]: Tube current time integration
D[m]: SID (=1m)
DAP=ESD×H×V

DAP[μGym2]: Value of dose area product calculation unit (by 0..01μGym02)
H[cm]: Horizontal opening amount of the collimator leaf at 1m of SID (by 0.001 m)
V[cm]: Vertical opening amount of the collimator leaf at 1m of SID (by 0.001 m)

4 **Press the Current-time product buttons to change kV and set ckV at the displayed kV.**

5 **Press the Registration button to save ckV at each kV.**

6 **Press the Added filter button to go back to the initial screen of the dose area product calculation unit setting mode.**

7 **In the initial screen of the dose area product calculation unit setting mode, press the Registration button to save all data regitered in each adjustment mode.**

Fig. 8-57 Registration of All Data

NOTE When the registered information is modified, that information must be saved in the memory by pressing the Registration keyon the initial screen of the control panel.

The registered information in each adjustment mode is the interim registration. If the adjustment mode is finished without saving in the memory, the modified information cannot be refleted to the system.

8 **When you adjust the dose area product calculation unit for other filter types, press the Added fitler button for several times to display the screen of adjusting the whole coefficients and selecting the additional filter, and change a filter type and make adjustment.**

8.20.7 Initial Screen of Dose Area Product Calculation Unit (To Save Registered Information in Memory)

1 **Turn OFF the power to the system.**

2 **Set the DIP switch SW3-5 and SW3-8 on the XCONT-2018 board as shown below.**

- DIPSW3-5 5(Dose area product meter): OFF
- DIPSW3-8 (Adjustment mode): OFF

3 **Attach the removed covers in reverse order.**

Reference "3.5 How to Open/Close Covers"

8.21 Cord Reel Usage Information Setting Mode

This mode is used to check the starting date of the cord reel usage and the number of connection to the outlet, and to set the usage information of the Cord Reel.

Adjust the setting by the following procedure when the cord reel is replaced.

8.21.1 To Enter Cord Reel Usage Information Setting Mode

1 **Enter the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

2 **Press the Body thickness change button to blink, and then press in order of the Oblique-view button button and Abdominal radiography button to enter the cord reel usage information setting mode.**

The LEDs on the following buttons are:.

- The Body thickness change button blinks.
- The Oblique-view button blinks.
- The Abdominal radiography button blinks.

8.21.2 To Check Cord Reel Usage Information

When the cord reel usage information setting mode is entered, the starting date of the cord reel usage is displayed in the kV and mAs display fields.

The number of connection to the outlet is displayed in the dose display field. The number of connection to the outlet is displayed by single count.

Fig. 8-58 EXAMPLE: Starting date of use of cord reel and the number of connection to the outlet (Aug. 6, 2021, 62 times)

8.21.3 To Reset Cord Reel Usage Information

It is required to reset the usage information when the cord reel is replaced.

1 **Enter the cord reel usage information setting mode, and press the Registration button for 3 seconds.**

The starting date of the cord reel is reset to the current date, and the number of connection to the outlet is reset to 0.

8.21.4 To Finish Cord Reel Usage Information Setting Mode

1 **Finish the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

8.22 FSR Usage Information Setting Mode

This mode is used to check the starting date of the FSR usage and the number of FSR enhancement, and to set the usage information of the FSR. Adjust the setting by the following procedure when FSR is replaced.

8.22.1 To Enter FSR Usage Information Setting Mode

1 **Enter the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

2 **Press the Body thickness change button to blink, and then press in order of the Oblique-view button and the Lumbar radiography button to enter the FSR usage information setting mode.**

The LEDs on the following buttons are:.

- The Body thickness change button blinks.
- The Oblique-view button blinks.
- The Lumbar radiography button blinks.

8.22.2 To Check FSR Usage Information

When the FSR usage information setting mode is entered, the starting date of FSR usage is displayed in the kV and mAs display fields.

The number of FSR enhancement is displayed in the dose display field. The information in the dose display field can be switched by pressing the collimator lamp switch.

Table 8-28

6th Digits of mGy	Item
0	The number of left FSR enhancement (by single count)
1	The number of right FSR enhancement (by single count)

Fig. 8-59 EXAMPLE: Starting date of FSR usage and the number of SR enhancement (Aug. 6, 2021, 365 times of left FSR enhancement)

8.22.3 To Reset FSR Usage Information

It is required to reset the usage information when FSR is replaced.

1 **Enter the FSR usage information setting mode, and press the Registration button for 3 seconds.**

The starting date of the cord reel is reset to the current date., and the number of enhancement of FSR is reset to 0.

8.22.4 To Finish FSR Usage Information Setting Mode

1 **Finish the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

8.23 Mechanical Performance Usage Information Setting Mode

8.23.1 To Enter Mechanical Performance Usage Information Setting Mode

1 **Enter the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

2 **Press the Body thickness change button to blink, and then press in order of the Oblique-view button and the Foot radiography button to enter the mechanical performance usage information setting mode.**

The LEDs on the following buttons are:.

- The Body thickness change button blinks
- The Oblique-view button blinks
- The Foot radiography button blinks

8.23.2 To Check Mechanical Performance Usage Information

When the mechanical performance usage information setting mode is entered, the starting date of mechanical performance usage is displayed in the kV and mAs display fields.

The usage information of each item is displayed in the dose display field. The information on kV/mAs, and dose display fields can be switched by pressing the collimator lamp switch.

Table 8-29

kV/mAs display field	Description
Obstr	The number of obstacle detection (by single count)
Catch	The number of arm lock (by single count)

Fig. 8-60 EXAMPLE: The number of obstacle detection (96 times)

Fig. 8-61 EXAMPLE: The number of arm lock (158 times)

8.23.3 To Reset Mechanical Performance Usage Information

1 **Enter the mechanical performance usage information setting mode, and press the Registration button for 3 seconds.**

The number of obstacle detection and the number of arm lock are reset to 0.

8.23.4 To Finish Mechanical Performance Usage Information Setting Mode

1 **Finish the adjustment mode.**

Reference "8.1 How to Start Adjustment Mode"

9 [Appendix] Adjusting Methods

The adjustments in this chapter are not required in the standard installation procedure of the unit. Perform the adjusting methods in case such as when the circuit boards or other parts are replaced and readjustment is required.

> NOTE In this chapter, following names are used to locate certain place on the circuit board. Characters "xx" represent the ID number.
>
> - VRxx : variable resistance
> - CPxx : check pin
> - TERxx : terminal

9.1 Preparation

Before making the detailed adjustment in this section, confirm the following items:

1 **The connections of connector P1 and P2 are compliant with the supply voltage.**

Reference "4.9 Connection Change According to Power-Supply Voltage"

2 **The settings of DIP SW and jumper on the XCONT-2018 board are matched with the unit type and the options.**

Reference
- "6.1 XCONT-2018 Board"
- "6.2 XCONT-2018 Board Jumper Setting"

3 **The settings of DIP SW and jumper on the INVERTER UNIT-32K board, INVERTER UNIT-C1 board, POWER-200D board(for MobileDaRt Evolution), or MUX CHARGE-32KX board are matched with the unit type and the options.**

Reference
- "6.3 INVERTER UNIT-32K Board"
- "6.4 INVERTER UNIT-C1 Board"
- "6.5 POWER-200D Board"
- "6.9 MUX CHARGE-32KX Board"

4 **Move the arm out of parking position. (For viewing the display, radiography preparation and exposure)**

5 Check that the combinations such as X-ray tube device, high voltage transformer are correct.

⚠ CAUTION

Make sure that the combinations are as described in the following table.
If the combinations such as X-ray tube devices, high-pressure transformers are not correct, the equipment can be damaged.
DO NOT install X-ray tube device 0.7U161CS-36 to 32kW type (MobileDaRt Evolution/MobileArt Evolution high-power type). The equipment will certainly be damaged

Table 9-1

	Models	High Voltage Transformer	X-ray Tube Device	KVADJ board	XCONT board DIP Switch
32kW type	MobileDaRt Evolution	D150MH-40 (P/N：562-25900)	0.7/1.3U163C-36	KV ADJ 32K	2-2：OFF 2-6：OFF
		D150MH-40（S）(P/N：565-16863)			2-2：OFF 2-6：ON
	MobileArt Evolution high-power type	D150MH-40(S) (P/N：565-16863)			2-2：OFF 2-6：ON
12.5kW type	MobileArt Evolution standard type	D125PH-C1	0.7U161CS-36	KV ADJ COCK	2-2：ON 2-6：OFF

NOTE To initialize the XCONT-2018 board, before performing a unit adjustment, carry out the following steps:

Reference
- For MobileDaRt Evolution: "7.2 Initialization of XCONT-2018 Board"
- For MobileArt Evolution:"8.2 Initialization of XCONT-2018 Board"

1 Turn the main breaker OFF.

2 Connect the power plug.

3 Turn ON the power supply to the equipment.

9.2 Adjusting the battery-voltage detection circuit

9.2.1 Preparation

> NOTE Steps 2 and 3 are not necessary when the adjustment of the unit is confirmed and only the adjustment for the battery voltage detection is performed.

1 **Confirm that the main breaker and the key switch are turned OFF, and the power plug is not connected.**

2 **Check that the discharge procedure has been performed for the internal capacitor.**

Reference "3.4 Discharging the Internal Capacitor"

3 **Turn on the DIPSW 3-8 (adjustment mode) of the XCONT-2018 board.**

4 **Turn ON the main breaker.**

5 **Turn the key switch ON. (System power turns ON.)**

9.2.2 Adjusting the positive-side of battery-voltage detection

1 **Press 4 on the numeric keypad or the Foot radiography button.**

Reference
- For MobileDaRt Evolution: "7.6 Adjustment of Battery Voltage Detection Circuit"
- For MobileArt Evolution: "8.6 Adjustment of Battery Voltage Detection"

2 **Measure the positive-side of the actual battery voltage (between TER3 [B120+O] and TER5 [B0] on the MUX POWER 32K board).**

3 **Press the mAs+ and mAs- key so that the same value with the actual voltage is displayed. If the voltage difference adjusted is within ±0.2 V, press the Program Entry key.**

9.2.3 Adjusting the negative-side of battery-voltage detection

1 **Press the kV- key.**

The mode to adjust the negative-side of battery-voltage detection will start.

Reference
- For MobileDaRt Evolution: "7.6 Adjustment of Battery Voltage Detection Circuit"
- For MobileArt Evolution: "8.6 Adjustment of Battery Voltage Detection"

2 **Measure the minus-side of actual battery voltage (the voltage between TER5 [B0] and TER4 [B120-O] on MUX POWER 32K board).**

3 **Press the mAs+ and mAs- key so that the same value with the actual voltage is displayed. If the voltage difference adjusted is within ±0.2 V, press the Program Entry key.**

4 **After the adjustments are completed, turn the key switch OFF.**

System power turns OFF.

9.2.4 Work after adjustment is confirmed

1 **Undo the DIP switch setting performed in steps 2, 3 of the section "9.2.1 Preparation".**

9.3 Adjusting MUX CHARGE-32KX board

9.3.1 Replacing the board

1 **Set the DIP switch: S1 as "6.9 MUX CHARGE-32KX Board" and replace the board.**

9.4 Adjusting XCONT-2018 Board

9.4.1 Setting of XCONT-2018 Board

1 **Set the switches and jumpers on the XCONT-2018 board.**

Reference
- "6.1 XCONT-2018 Board"
- "6.2 XCONT-2018 Board Jumper Setting"

2 **Initialize the XCONT-2018 board.**

Reference
- MobileDaRt Evolution: "7.2 Initialization of XCONT-2018 Board"
- MobileArt Evolution: "8.2 Initialization of XCONT-2018 Board"

3 **Download and upload XCONT-2018 board data as necessary.**

Reference "11.9 PC Maintenance Function"

9.4.2 Adjusting mAs circuit of XCONT-2018

■ Preparation

1 **Check that the main breaker and the key switch are turned OFF, and that the power cable plug is disconnected.**

2 **Check that the discharge procedure has been performed for the internal capacitor.**

Reference "3.4 Discharging the Internal Capacitor"

3 **Short-circuit CP3 [CHECK1] and CP4 [CHECK2] on the XCONT-2018 board.**

4 **Turn VR1 [mAs] on the XCONT-2018 board counterclockwise to the maximum.**

Hint This step is not necessary if the XCONT-2018 board is already adjusted.
If the device has been checked and only readjustment is performed, the above steps 2 to 4 are unnecessary.

■ Adjusting mAs circuit

1 **Short-circuit CP1 [+5V] and CP30 [mA-IN] on the XCONT-2018 board.**

Hint For the XCONT-2018 board, the CP65 [+5V] can be used instead of the CP1[+5V].

2 **Turn ON the main breaker and the key switch.**
System power turns ON.

3 **Adjust VR [mAs] so that the relation between the frequency of CP28 [MASCLK] on the XCONT-2018 board and the voltage of CP30 [mA-IN] on the XCONT-2018 board is 5 kHz/V.**

Hint For example, if the voltage at CP30 [mA-IN] is 5.10 V, the frequency at CP28 [MASCLK] must be adjusted to 25.5 kHz.

4 **Turn OFF the key switch.**
System power turns OFF.

5 **Remove the short-circuit between CP1 [+5V] and CP30 [mA-IN] on the XCONT-2018 board.**

■ Procedure after adjustment

1 **Undo the procedures in steps 2,3 of sectionPreparation.**

9.4.3 Adjusting inverter control circuit of XCONT-2018 board

■ Preparation

1 **Check that the main breaker and the key switch are turned OFF, and that the power cable plug is disconnected.**

2 **Check that the discharge procedure has been performed for the internal capacitor.**

Reference "3.4 Discharging the Internal Capacitor"

3 **Disconnect the cables from TER3 [B120+O] and TER4 [B120-O] on the MUX POWER 32K board, and perform insulation.**

4 **Disconnect the JCH2 connector on the MUX CHARGE-32KX board and the JM2 connector on the INVERTER UNIT-32K board or INVERTER UNIT-C1 board.**

5 **Short-circuit CP3 [CHECK1] and CP4 [CHECK2] on the XCONT-2018 board.**

6 **Short-circuit CP17 [+15V] and CP5 [VSIN].**

■ Waveform check

NOTE For the XCONT-2018 board, the inverter pulse width has been adjusted at the time of factory shipment. Check the inverter pulse width according to the following procedure, and adjust only when the pulse width is not within the adjustment range.

1 **Turn ON the main breaker and the key switch.**

System power turns ON.

2 Check the inverter pulse width.

1 Connect oscilloscope probes between CP5 [Ga] - CP6 [Ea] and CP10 [Gb] - CP12 [Eb] on INVERTER UNIT board.

2 Perform READY and XRAY operations, and check the gate waveform of IGBT.

3 Confirm that the pulse width at +5 V is as shown in the following table.

Table 9-2

	Models	High Voltage Transformer	pulse width
32kW type	MobileDaRt Evolution	D150MH-40 (P/N：562-25900)	7.8 - 8.0 μs
		D150MH-40(S) (P/N：565-16863)	10.9 - 11.1 μs
	MobileArt Evolution high-power type	D150MH-40(S) (P/N：565-16863)	10.9 - 11.1 μs
12.5kW type	MobileArt Evolution standard type	D125PH-C1	7.8 - 8.0 μs

4 If necessary, readjust the pulse width using the procedure in "9.4.4 Adjusting inverter pulse width of XCONT-2018 board".

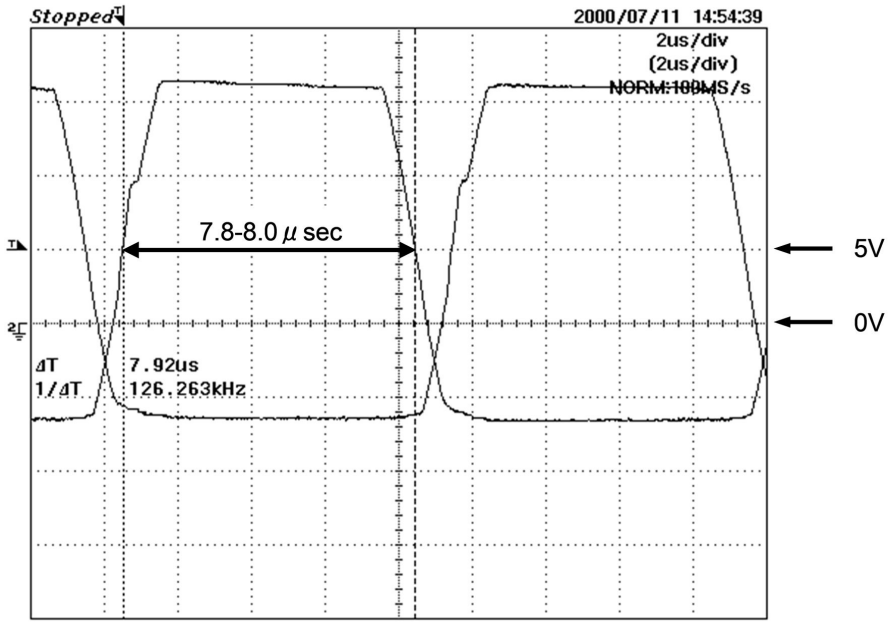


Fig. 9-1 Waveform between CP[Ga] – CP[Ea], CP[Gb] – CP[Eb] on INVERTER UNIT board (for D125PH-C1 and D150MH-40)

NOTE Check the gate waveform of the time period later than 10 ms after the start of the X-ray exposure. This is to ignore the initial peak value and enable accurate reading.

3 **Confirm the pulse width between CP3 [Gc] - CP4 [Ec], and CP8 [Gd] - CP9 [Ed] with the same procedure as Step 2.**

4 **Turn OFF the key switch.**
System power turns OFF.

5 **Remove the short-circuit between CP24 [+15V] and CP5 [VSIN].**

9.4.4 Adjusting inverter pulse width of XCONT-2018 board

> NOTE For the XCONT-2018 board, "9.4 Adjusting XCONT-2018 Board" is done by connecting the PC with the LAN cable.

■ Preparation

1 **Check that the main breaker and the key switch are turned OFF, and that the power cable plug is disconnected.**

2 **Check that the discharge procedure has been performed for the internal capacitor.**
Reference "3.4 Discharging the Internal Capacitor"

3 **Disconnect the cables from TER3 [B120+O] and TER4 [B120-O] on the MUX POWER 32K board, and perform insulation.**

4 **Disconnect the JCH2 connector on the MUX CHARGE-32KX board and the JM2 connector on the INVERTER UNIT board.**

5 **Short-circuit CP3 [CHECK1] and CP4 [CHECK2] on the XCONT-2018 board.**

6 **Short-circuit CP17 [+15V] and CP5 [VSIN].**

■ Connection/Adjustment

1 **Configure the PC as described in "11.10 Connecting the XCONT-2018 board to the LAN".**

2 **Connect the LAN cable to the LAN port on the XCONT-2018 board and connect it to the PC.**

3 **Turn ON DIP-SW3-8 on the XCONT-2018 board.**

4 **When you turn on the power of the system and open the web browser (http://192.168.1.1/index.html), the following screen appears. Click [Inverter Control Circuit].**

> **NOTE** Use Internet Explorer or Internet Explorer mode of Microsoft Edge for web browser.

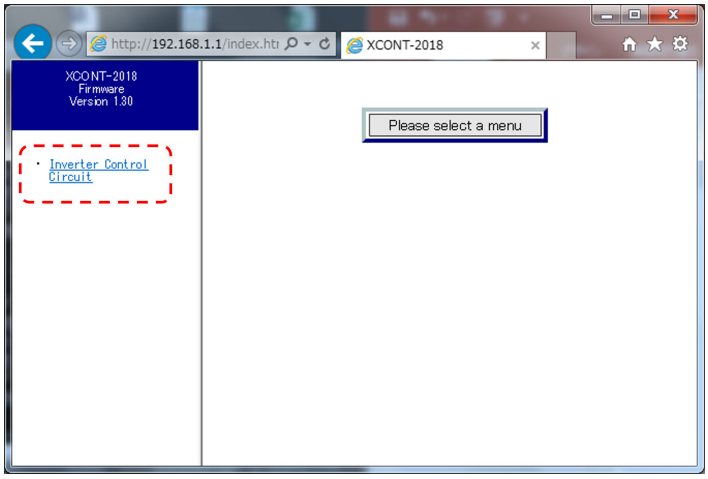


Fig. 9-2

5 Enter a value for [IGBT GateWidth] and make adjustments.

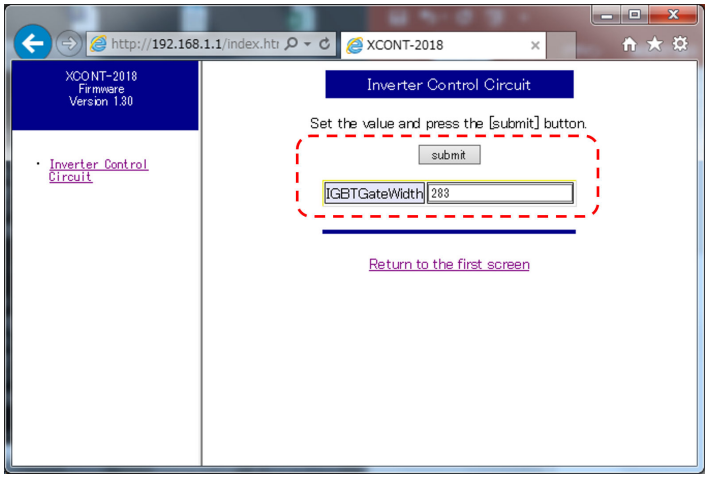


Fig. 9-3

NOTE If you change the value, the value on the XCONT-2018 board will not be updated unless you click [Submit].

Hint Changing the number by 1 changes the pulse width by 0.025 μs. For example, changing the number by 4 changes the pulse width by 0.1 μsec.

6 When you click Submit, the following screen appears and the value of the XCONT-2018 board is updated from the PC. In this state, make an exposure and check the pulse width as described in "9.4.3 Adjusting inverter control circuit of XCONT-2018 board", and make an adjustment to obtain an appropriate pulse width.

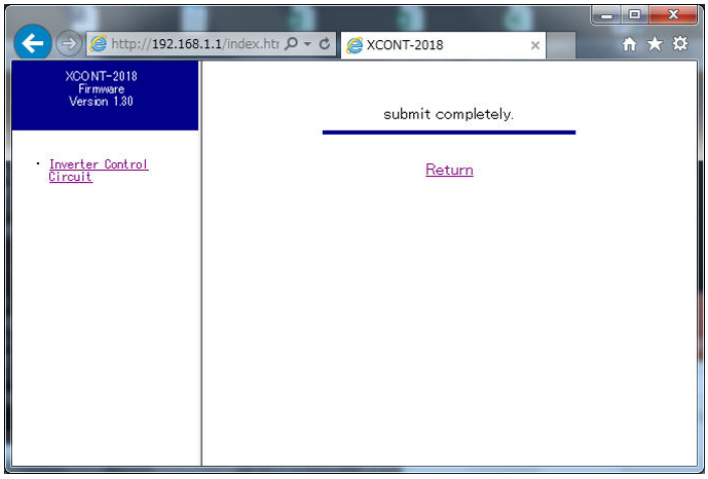


Fig. 9-4

7 When you finish making the adjustment, close the browser window.

8 Turn off the power of the system and unplug the LAN cable.

9 **Turn OFF the DIP-SW3-8 on the XCONT-2018 board.**

■ **Procedure after adjustment**

1 **Undo the procedures in steps 2, 3, 4, 5 and 6.**

9.4.5 Adjusting VR9 and VR10 on XCONT-2018 Board

■ **Preparation**

1 **Ensure that the main breaker and key switch are OFF, and power plug is not connected.**

■ **Adjustment**

1 **Using the tester, adjust the VR9 on the XCONT-2018 board as follows.**

Table 9-3

	Models	High Voltage Transformer	VR9
32kW type	MobileDaRt Evolution	D150MH-40 (P/N：562-25900)	Turn VR9 clockwise 5 increments
		D150MH-40(S) (P/N：565-16863)	Turn VR9 clockwise 6 increments
	MobileArt Evolution high-power type	D150MH-40(S) (P/N：565-16863)	Turn VR9 clockwise 6 increments
12.5kW type	MobileArt Evolution standard type	D125PH-C1	Turn VR9 clockwise 5 increments

2 **Using a tester, measure the resistance between CP61 [FB RES] and CP42 [VINCK3] on the XCONT-2018 board, and adjust VR10 as follows:**

Table 9-4

	Models	High Voltage Transformer	VR10
32kW type	MobileDaRt Evolution	D150MH-40 (P/N：562-25900)	5.8kΩ±5%
		D150MH-40(S) (P/N：565-16863)	10kΩ(Turn VR10 clockwise 5 increaments.)
	MobileArt Evolution high-power type	D150MH-40(S) (P/N：565-16863)	10kΩ(Turn VR10 clockwise 5 increments.)

	Models	High Voltage Transformer	VR10
12.5kW type	MobileArt Evolution standard type	D125PH-C1	10kΩ(Turn VR10 clockwise 5 increments.)

9.5 Operation Check of Capacitor Charging

> NOTE This operation check is required only for 32kW type (MobileDaRt Evolution/MobileArt Evolution high-power type)

9.5.1 Preparation

1 **Check that the main breaker and the key switch are turned OFF, and that the power cable plug is disconnected.**

2 **Connect cables to TER3 [B120+O] and TER4 [B120-O] on the MUX POWER 32K board.**

3 **Undo the discharge of the internal capacitor and switch to normally mode.**

Reference "3.4 Discharging the Internal Capacitor"

1. Set the discharge switch to OFF (left side).
2. Change the connection of the CN1/CN2 cable on the MUX CAPACITOR board from connector CN1 to CN2.

4 **Disconnect the JCH2 connector on the MUX CHARGE-32KX board and the JM2 connector on the INVERTER UNIT 32K board.**

5 **Disconnect the JINV connector on the INVERTER UNIT-32K board.**

6 **Short-circuit CP3 [CHECK1] and CP4 [CHECK2] on the XCONT-2018 board.**

7 **Connect a tester to CP1 [VC+] - CP2 [VC0] on the MUX CAPACITOR board.**

9.5.2 Operation Check

1 **Turn the main breaker ON.**

2 **Turn ON the key switch.**

System power turns ON.

3 Approximately 10 seconds after power ON, capacitor charging will start, and the voltage between CP1 [VC+] - CP2 [VC0] on the MUX CAPACITOR board will rise.

4 Approximately 30 to 60 seconds after the start of the charging process, voltage increases to approximately 170 V and the charge suppression resistance value decreases, speeding up the charging process.

5 After approximately 10 to 30 seconds, voltage increases to 220 to 240 V and the system preparation indicator lamp turns ON.

6 When performing READY operation, the voltage becomes the same as the battery voltage.

7 Turn the key switch OFF.

9.5.3 Procedure after adjustment

1 Undo the procedures in steps 4, 5, 6, 7 of the section "9.5.1 Preparation".

9.6 Checking the X-ray exposure

9.6.1 Preparation

1 **Check that the main breaker and the key switch are turned OFF, and that the power cable plug is disconnected.**
For MobileDaRt Evolution, check that the DR system is turned OFF.

2 **Check that the connection type for the internal capacitor is for normal charging.**

Reference "3.4 Discharging the Internal Capacitor"

1 Set the discharge switch to OFF (left side).

2 The CN1/CN2 cable on the MUX CAPACITOR board is connected to CN2.

3 **Turn ON the DIP switch SW3-8 on the XCONT-2018 board. (Adjustment mode)**

4 **Confirm there is no short-wire between check pins on the XCONT-2018 board.**

5 **Turn VR3 [WF-ADJ] and VR4 [kV-ADJ] on KVADJ board to center position in case that KVADJ board has not been adjusted.**

Hint This step is not necessary if the KVADJ board is already adjusted.

6 **Turn VR9 and VR10 on the XCONT-2018 board to center position if they have not been adjusted.**

Hint Normally, advanced adjustment is not necessary for VR9 and VR10.

9.6.2 Performance check of READY operation

1 **Turn ON the main breaker and the key switch.**
System power turns ON.

2 **Confirm that the collimator lamp turns on.**

3 **Perform the READY operation. Confirm that the anode of X-ray tube is rotating and the exposure-ready lamp turns ON (green) within approximately 3 seconds.**

4 **During READY operation, check the waveform of filament heating voltage between following check points on INVERTER UNIT-32K board or INVERTER UNIT-C1 board. The waveform should look like the waveform shown below.**

- For 32kW type (MobileDaRt Evolution/MobileArt Evolution high-power type)
  - CP15[CS1]-CP2[C0] for small focal spot.
  - CP14[CL1]-CP2[C0] for large focal spot.
- For 12.5kW type (MobileArt Evolution standard type)
  - CP1[CL] - CP2[C0}

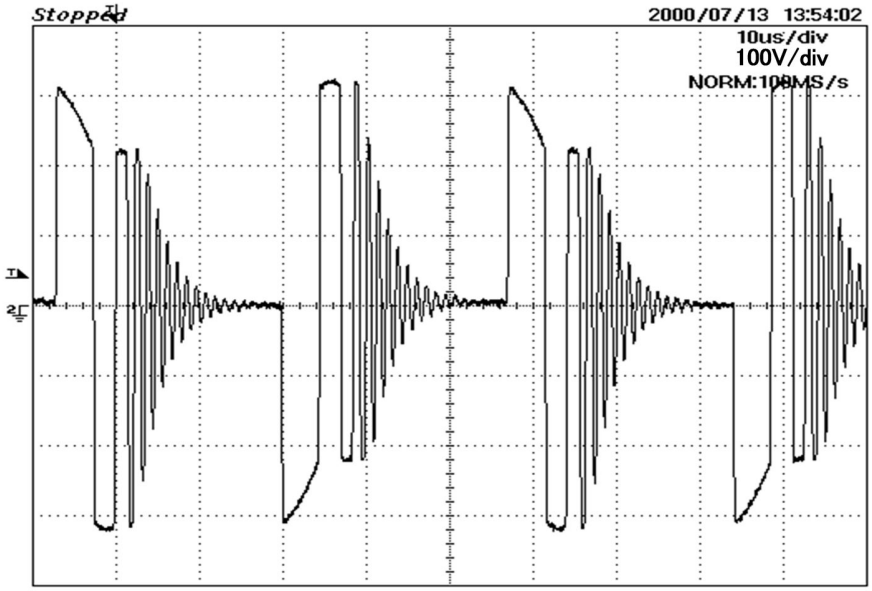


Fig. 9-5 Waveform between CP14[CL1] - CP2[C0] on the INVERTER UNIT-32K board

9.6.3 Checking the waveform for tube voltage, and adjustment of tube current

1 **Enter the adjustment mode.**

Reference
- For MobileDaRt Evolution: "7.1 How to Start Adjustment Mode"
- For MobileArt Evolution: "8.1 How to Start Adjustment Mode"

2 **Press 1 on the numeric key pad or the Front-view button, and enter the Tube Current 2-points Adjustment Mode and run the tube current two-point adjustment for each focus.**

Reference
- For MobileDaRt Evolution: "7.4.4 Tube Current Two-point Adjustment Mode"
- For MobileArt Evolution: "8.4.4 Tube Current Two-point Adjustment Mode"

3 **If necessary, enter the Tube Current Each-point Adjustment Mode and then check that measurement value is with error within -15 to +5 % range of the setting value.**

Reference
- For MobileDaRt Evolution: "7.4.5 Tube Current Each Point Adjustment Mode"
- For MobileArt Evolution: "8.4.5 Tube Current Each Point Adjustment Mode"

NOTE
- Use a neutral point tube current meter for measuring the tube current. The neutral point tube current meter should be connected to [mA+] and [mA-] on the terminal board X1 on the or XCONT-2018 board. Ensure the short-circuit plate between mA+ and mA- is removed before any measurements are taken.
- Tube current measurement may be adversely affected by any voltmeter connected to the unit. All tube current measurements / adjustments must therefore be performed with no voltmeter connected.

9.6.4 Fine adjustment of tube voltage waveform

NOTE Adjustment of the tube voltage waveform will be required if the high voltage transformer and / or kVADJ board are replaced.

Hint In case of the XCONT-2018 board replacement, the following adjustment is not required if the old kVADJ board is not replaced..

1 **Enter the Tube Current Two-point Adjustment Mode.**

Reference
- For MobileDaRt Evolution: "7.1 How to Start Adjustment Mode" and "7.4 Adjustment of Tube Current"
- For MobileArt Evolution: "8.1 How to Start Adjustment Mode" and "8.4 Adjustment of Tube Current"

2 **Set the adjustment point as shown below.**

Table 9-5

	Models	High Voltage Transformer	Adjustment point
32kW type	MobileDaRt Evolution	D150MH-40 (P/N：562-25900)	80kV 50mA
		D150MH-40(S) (P/N：565-16863)	80kV 200mA
	MobileArt Evolution high-power type	D150MH-40(S) (P/N：565-16863)	80kV 200mA
12.5kW type	MobileArt Evolution standard type	D125PH-C1	80kV 50mA

3 **Adjust VR4 [kV-ADJ] on KVADJ board so that a tube voltmeter displays 80kV±1kV.**

4 **Adjust VR3 [WF-ADJ] on kVADJ board so that waveform of the tube voltmeter becomes flat.**

Hint Rotate VR3 clockwise, and the waveform changes toward overshoot.
Rotate VR3 counterclockwise, and the waveform changes toward under shoot.

9.6.5 Procedure after adjustment

1 **Set the DIP switch SW3-8 to OFF.**

9.7 Checking battery charging

9.7.1 Preparation

1 **Turn ON the main breaker.**

2 **Turn the key switch OFF.**

3 **Turn ON the DIP switch SW3-8 on the XCONT-2018 board (Adjustment mode).**

4 **Confirm that there is no short-wire between check pins on the XCONT-2018 board.**

9.7.2 Procedure for checking the battery charge

1 **Turn ON the key switch.**

The system powers ON and enters the adjustment mode.

2 **Enter the battery charging test mode.**

For MobileDaRt Evolution, press the INFO switch and then press numeric keypad 4.
For MobileArt Evolution, press body thickness change button and then chest radiography button.

A preset value of the charging current is displayed in the kV display field and an actual measurement of the charging current is displayed in 0.01 A unit on the mA display field.

3 **Connect the power plug.**

4 **Turn OFF the key switch.**

The battery charge starts after 10 seconds and charges for about 1 minute.

5 **Confirm that the waveform of the voltage between CP5 [IC-] and CP7 [GND] on MUX CHARGE-04B board looks like the waveform shown below.**

6 **Unplug the unit.**

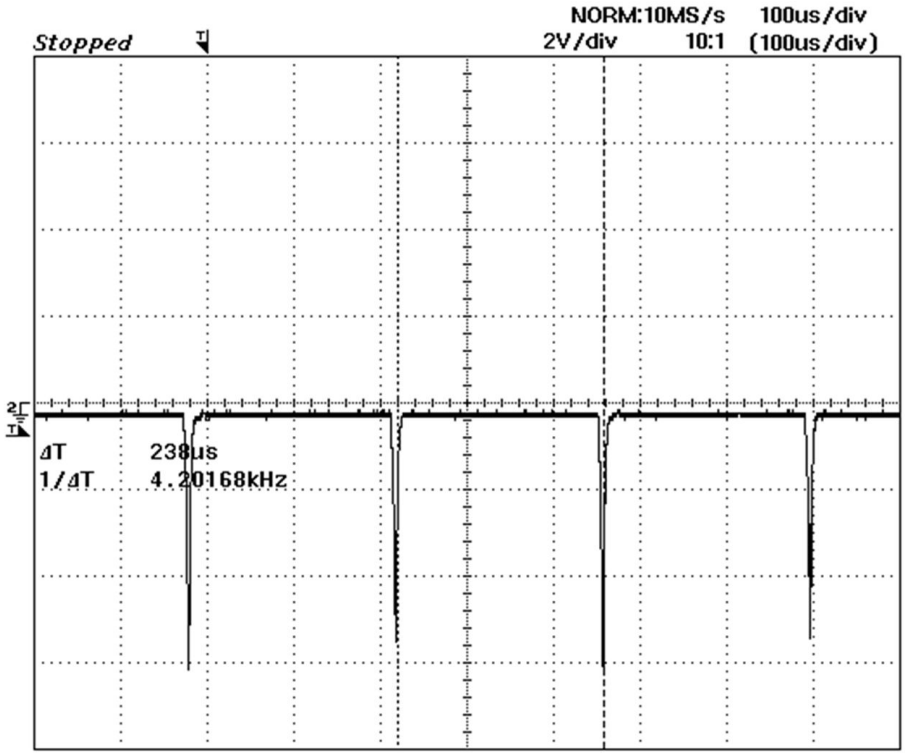


Fig. 9-6 Waveform between CP18 - GND

NOTE The waveform shown above is an example. The pulse width and wave height value may differ.

If the charge confirmation mode is entered when the battery is fully charged, the error (F5b, F51, and F53, etc.) may appear.

9.7.3 Procedure after the adjustment

1 **Set the DIP switch SW3-8 to OFF.**

9.8 Procedure after the adjustment

- Return the DIP SW on the XCONT-2018 board to its normal use condition.
- Confirm that no short-circuits are on the boards.
- Confirm that all connectors and terminals are connected on the boards.
- Confirm that the XCONT-2018 board terminal X11[mA+] and X12[mA-] is short-circuited.
- Confirm that the discharge function is released and the system is in the normal state of charge. (Cable CN1/CN2 is connected to connector CN2 (NORMAL), and the discharge switch is NORMAL (left side).)

10 [Appendix] Error Message List

10.1 Error message list

The Fault Indication lamp will light during error and radiography process will be disallowed. The content of the error is indicated by the code on the mAs display.

Table 10-1 Fault error message list

Error code	Content of the error	Treatment
F01	Main supply voltage is higher than the standard voltage by 20% or more.	Check the power supply voltage of outlet. Check the signal [LV] on XCONT board.
F02	Main supply voltage is lower than the standard voltage by 30% or more.	Check the power supply voltage of outlet. Check the signal [LV] on XCONT board.
F03	A failure in –15V supply. ("-15V OK signal" not supplied)	Check the connection of DC power supply. Check the signal [-15V] on XCONT board.
F04	The main power relay or the inrush prevention relay is slightly adhered.	Check the voltage between [TER3] and [TER4] during key switch OFF.
F11	X-ray tube voltage during irradiation is detected to be higher than the set value by 30kV or more.	Check the signal [TKV] [KV+] [KV-] [VP1] [VP2] [VIN] on XCONT board during an exposure.
F12	Measured X-ray tube voltage during irradiation is lower than the set value by 15% or more.	Check the signal [TKV] [KV+] [KV-] [VP1] [VP2] [VIN] on XCONT board during an exposure.
F13	Measured X-ray tube voltage during irradiation is lower than 10kV or more.	Check the signal [TKV] [KV+] [KV-] [VP1] [VP2] [VIN] on XCONT board during an exposure.
F14	The minute discharge of the X-ray tube occurred three times. Exposure is stopped.	Perform the tube aging. Replace the X-ray tube if not improved.
F15	Measured X-ray tube voltage during irradiation is lower than 90% of the set value. (Irradiation is not aborted.) Battery performance may have deteriorated.	Check the battery voltage. Check the FVR adjustment. Check the signal [TKV] [KV+] [KV-] [VP1] [VP2] [VIN] on XCONT board during an exposure.
F16	The kV feedback start signal during irradiation is not detected. (Irradiation is not aborted.) Battery performance may have deteriorated.	Check the battery voltage. Check the FVR adjustment. Check the signal [TKV] [KV+] [KV-] [VP1] [VP2] [VIN] on XCONT board during an exposure.
F17	The minute discharge of the X-ray tube occurred once.	Perform the tube aging.
F18	The minute discharge of the X-ray tube occurred twice.	Perform the tube aging.

Error code	Content of the error	Treatment
F19	Measured tube voltage exceeds 144kV.	Perform the tube aging. Replace the X-ray tube if not improved.
F1a	The KVADJ Board is not mounted.	Check the mount of KVADJ board on the XCONT board
F1b	Mounted KVADJ Board is not conformed to the mounted High voltage transformer.	Check the type of KVADJ board on the XCONT board type 32kW : KVADJ 32K type 12.5kW : KVADJ COCK
F1c	The FPGA revision of the XCONT-2018 board does not support the high-voltage transformer D150MH-40(S) (P/N: 565-16863).	Check the revision seal of FPGA and M35 on the XCONT-2018 board
F1d	The FPGA revision of the XCONT-2018 board does not support Built-in AEC Assistance. Or, the FPGA revision of the XCONT-2018 board does not support serial radiography.	Check the revision seal of FPGA and M35 on the XCONT-2108 board
F1e	Dongle is disconnected.	Check the dongle connection.
F1F	Authentication of dongle is Failed.	Check the part number of the dongle.
F21	Measured X-ray tube current during irradiation is larger than the set value by 100mA or more. →Detection by software	Check the FVR adjustment. Check the signal [TKV] [TMA] [IFV] [VIN] on XCONT board during an exposure.
F22	Measured X-ray tube current during irradiation is smaller than the set value by 50% or more.	Check the FVR adjustment. Check the signal [TKV] [TMA] [IFV] [VIN] on XCONT board during an exposure.
F23	Measured X-ray tube current during irradiation is smaller than 10mA.	Check the FVR adjustment. Check the signal [TKV] [TMA] [IFV] [VIN] on XCONT board during an exposure.
F24	Measured X-ray tube current during irradiation is larger than the set value by 100mA or more. →Detection by hardware	Check the FVR adjustment. Check the signal [TKV] [TMA] [IFV] [VIN] on XCONT board during an exposure.
F31	Measured filament current is higher than the set value by 50% or more. →Detection by software	Check the FVR adjustment. Check the signal [TKV] [TMA] [IFV] [VIN] on XCONT board during an exposure.
F32	Measured filament current is lower than the set value by 50% or more.	Check the FVR adjustment. Check the signal [TKV] [TMA] [IFV] [VIN] on XCONT board during an exposure.
F33	Measured filament current is lower than 1A.	Check the connection of the filament current line. Check the FVR adjustment. Check the signal [TKV] [TMA] [IFV] [VIN] on XCONT board during an exposure.

Error code	Content of the error	Treatment
F34	Measured filament current is higher than the set value by 50% or more. →Detection by hardware	Check the connection of the filament current line. Check the FVR adjustment. Check the signal [TKV] [TMA] [IFV] [VIN] on XCONT board during an exposure.
F35	Measured filament current is higher than the set value by 50% or more. →Detection by hardware	Check the connection of the filament current line. Check the FVR adjustment. Check the signal [TKV] [TMA] [IFV] [VIN] on XCONT board during an exposure.
F41	Capacitor voltage is higher than 300V.	Check the voltage between [VC+] and [VC0] on MUX CAPACITOR board.
F42	Capacitor voltage is lower than 170V after 60 seconds since the start of capacitor charging.	Check the voltage between [VC+] and [VC0] on MUX CAPACITOR board.
F43	Capacitor voltage is lower than 30V after 60 seconds since the start of capacitor charging.	Check the discharge processing of the internal capacitor. Check the voltage between [VC+] and [VC0] on MUX CAPACITOR board.
F44	Signal “VCOK” is not returned but the capacitor voltage is normal.	Check the signal [VC] [VP1] on MUX CHARGE-32KX board.
F45	System preparation is not finished 30 seconds after the initiation of capacitor changing relay "K2".	Check the voltage between [VC+] and [VC0] on MUX CAPACITOR board.
F46	Relay K3 for charging capacitor is not turned on immediately before the start of exposure.	Check the voltage between [VC+] and [VC0] on MUX CAPACITOR board.
F47	Capacitor voltage is lower than system preparation voltage for more than 30 seconds after system preparation is completed.	Check the voltage between [VC+] and [VC0] on MUX CAPACITOR board.
F51	Battery voltage of plus-side during charging is higher than 155V.	Check the signal [VP1] [VP2] on XCONT board during a charge. Check the signal [120+] [120-] [VP1] [VP2] on MUX CHARGE-32KX board during a charge.
F52	Battery voltage of plus-side during charging is lower than 50V.	Check the signal [VP1] [VP2] on XCONT board during a charge. Check the signal [120+] [120-] [VP1] [VP2] on MUX CHARGE-32KX board during a charge.
F53	Battery voltage of minus-side during charging is higher than 155V.	Check the signal [VP1] [VP2] on XCONT board during a charge. Check the signal [120+] [120-] [VP1] [VP2] on MUX CHARGE-32KX board during a charge.

Error code	Content of the error	Treatment
F54	Battery voltage of minus-side during charging is lower than 50V.	Check the signal [VP1] [VP2] on XCONT board during a charge. Check the signal [120+] [120-] [VP1] [VP2] on MUX CHARGE-32KX board during a charge.
F55	Battery voltage of the plus-side is higher than the minus-side by more than 12V during battery driving.	Check the signal [VP1] [VP2] on XCONT board during battery driving. Check the signal [120+] [120-] [VP1] [VP2] on MUX CHARGE-32KX board during battery driving.
F56	Battery voltage of the minus-side is higher than the plus-side by more than 12V during battery driving.	Check the signal [VP1] [VP2] on XCONT board during battery driving. Check the signal [120+] [120-] [VP1] [VP2] on MUX CHARGE-32KX board during battery driving.
F57	Battery voltage of the plus-side is higher than the minus-side by more than 12V during charging.	Check the signal [VP1] [VP2] on XCONT board during a charge. Check the signal [120+] [120-] [VP1] [VP2] on MUX CHARGE-32KX board during a charge.
F58	Battery voltage of the minus-side is higher than the plus-side by more than 12V during charging.	Check the signal [VP1] [VP2] on XCONT board during a charge. Check the signal [120+] [120-] [VP1] [VP2] on MUX CHARGE-32KX board during a charge.
F5a	Charging current is higher than 3.8A. (Signal CHBR1)	Check the signal [IC] on XCONT board during a charge. Check the signal [IC1] on MUX CHARGE-32KX board during a charge. Check the signal [IC-] on MUX CHARGE-04B board during a charge.
F5b	Battery voltage + or - side during charging is higher than 155 V. (Signal CHBR2)	Check the signal [120+] [120-] on MUX CHARGE-32KX board.
F5c	The power plug is connected when the main breaker is turned off.	Check the main breaker. Check the battery voltage.
F5E	Measured charging current is larger than the set value by 50% or more for 10 seconds.	Check the signal [IC] on XCONT board during a charge. Check the signal [IC1] on MUX CHARGE-32KX board during a charge. Check the signal [IC-] on MUX CHARGE-04B board during a charge.

Error code	Content of the error	Treatment
F5F	Measured charging current is smaller than the set value by 30% or more for 10 seconds.	Check the signal [IC] on XCONT board during a charge. Check the signal [IC1] on MUX CHARGE-32KX board during a charge. Check the fuse F1 on MUX CHARGE-04B board.
F61	STOK signal is not returned even after the starter drive time is over.	Check the connection of starter drive line. Check the signal [YC] [ZC] on INVERTER UNIT-32K board. (Use [GND] on INVERTER UNIT-32K board for a grounding.)
F63	FPD does not start image store when the frame cycle is reached.	Reboot the equipment and confirm that there is nothing wrong with the DR system.
F66	During serial radiography, the exposure time of one frame is longer than the maximum image store time.	Reboot the equipment and check the FVR adjustment.
F67	FPD stops store images during radiography.	Reboot the equipment and confirm that there is nothing wrong with the DR system.
F6A	The generator can not recieve "Image store end notice"command from the DR system.	Reboot the DR system and the generator.
F6b	Serial radiography does not end when the accumulated collection time reaches the maximum collection time.	Reboot the equipment and confirm that there is nothing wrong with the DR system.
F6E	Irradiation does not finish by backup time / mAs.	Check the signal [TKV] [TMA] [KVT] on XCONT board during an exposure.
F6F	KVT does not input after 250 ms has passed since EXRR output. Or, KVT does not input after 1000 ms has passed since EXRR output in case of serial radiography.	Check the signal [TKV] [TMA] [KVT] [EXRR] on XCONT board during an exposure.
F71	Communication timeout of Dose Area Product Meter	Check the connection and setting of the dose area product meter.
F72	Communication error of Dose Area Product Meter	Restart the system and check the acquisition operation.
F73	Failure of Dose Area Product Meter	Replace the dose area product meter.
F81	The setting of the DIP switch 2 -2 on the XCONT board does not match the maximum output of the device (32 kW/12.5 kW).	Check the DIP switch settings on the XCONT board.
F85	The setting of the DIP switch 2 -6 on the XCONT board does not match the type of the 32 kW type high-voltage transformer mounted on the device.	Check the DIP switch settings on the XCONT board.

Error code	Content of the error	Treatment
F86	The setting of the DIP switch 2 -1 on the XCONT board does not match the type of the device (MobileDaRt Evolution or MobileArt Evolution).	Check the DIP switch settings on the XCONT board.
F90	When serial radiography is selected, for static radiography APR is set. Or when static radiography is selected, for sereal radiography APR is set.	Review of radiography conditions
F91	Serial radiography setting protocol is set other than serial radiography ON/OFF.	Re-select the imaging protocol on the DR console. The light turns off when a command is sent fromthe DR system.
F92	A frame rate outside the configurable range is set in the serial radiography setup protocol.	Re-select the imaging protocol on the DR console. The light turns off when a command is sent fromthe DR system.
F93	The maximum store time outside the configurable range is set in the serial radiography setup protocol.	Re-select the imaging protocol on the DR console. The light turns off when a command is sent fromthe DR system.
F94	The maximum number of frames outside the configurable range is set in the serial radiography setup protocol.	Re-select the imaging protocol on the DR console. The light turns off when a command is sent fromthe DR system.
d15	Welding of the Relay of a motor driving section is detected.	Move the system to a safe place. In this case, use the function of emergency brake release without the power supply to move the system. Check the electrical connection between Pin1 and 2 of the CN17 with the CN17 connected.
d16	Short mode failure of the right FSR is detected.	Move the system to a safe place. In this case, use the function of emergency brake release without the power supply to move the system. Check the cable between the FSR R ASSY and the MUDRIVER board for coating damage due to the cable pinch. Check the ground fault to FSR R ASSY.

Error code	Content of the error	Treatment
d17	Short mode failure of the left FSR is detected.	Move the system to a safe place. In this case, use the function of emergency brake release without the power supply to move the system. Check the cable between the FSR L ASSY and the MUDRIVER board for coating damage due to the cable pinch. Check the ground fault to FSR L ASSY.
d20	Abnormality of the FET for motor of the right motor drive circuit is detected.	Move the system to a safe place. In this case, use the function of emergency brake release without the power supply to move the system. Check the electrical connection of the FET (Q11, 12, 14, 15) on the MU DRIVER board with the system power turned OFF.
d21	Abnormality of the FET for motor of the left motor drive circuit is detected.	Move the system to a safe place. In this case, use the function of emergency brake release without the power supply to move the system. Check the electrical connection of the FET (Q1, 2, 8, 9) on the MU DRIVER board with the system power turned OFF.
d22	Short circuit of the right motor drive circuit is detected.	Move the system to a safe place. In this case, use the function of emergency brake release without the power supply to move the system. Check the electrical connection of the FET (Q11, 12, 14, 15) on the MU DRIVER board with the system power turned OFF.
d23	Short circuit of the left motor drive circuit is detected.	Move the system to a safe place. In this case, use the function of emergency brake release without the power supply to move the system. Check the electrical connection of the FET (Q1, 2, 8, 9) on the MU DRIVER board with the system power turned OFF.

Error code	Content of the error	Treatment
d24	Abnormal rotation of the right motor is detected.	Move the system to a safe place. In this case, use the function of emergency brake release without the power supply to move the system. Check whether the connections of CN6 and 7 on the MU DRIVER board are correct or not. Check whether the CN7 on the MU DRIVER board is connected firmly. Check whether the GND cable of the encoder cable (connector label CN7) of the right motor is firmly fixed to the device casing with the screw.
d25	Abnormal rotation of the left motor is detected.	Move the system to a safe place. In this case, use the function of emergency brake release without the power supply to move the system. Check whether the connections of CN6 and 7 on the MU DRIVER board are correct or not. Check whether the CN6 on the MU DRIVER board is connected firmly. Check whether the GND cable of the encoder cable (connector label CN6) of the left motor is firmly fixed to the device casing with the screw.
d26	Excess current of the right motor is detected.	Move the system to a safe place. In this case, use the function of emergency brake release without the power supply to move the system. Check whether or not the wheel on the right side is locked or it contacts the system due to deformation.
d27	Excess current of the left motor is detected.	Move the system to a safe place. In this case, use the function of emergency brake release without the power supply to move the system. Check whether or not the wheel on the left side is locked or it contacts the system due to deformation.

Error code	Content of the error	Treatment
d28	Abnormality of the encoder for the right motor is detected.	Move the system to a safe place. Check whether the CN7 on the MU DRIVER board is connected firmly. Check whether the GND cable of the encoder cable (connector label CN7) of the right motor is firmly fixed to the device casing with the screw. Check whether or not a value of right motor is appropriate (or an abnormal value like “0” is not displayed) in the displaying running speed mode.
d29	Abnormality of the encoder for the left motor is detected.	Move the system to a safe place. Check whether the CN6 on the MU DRIVER board is connected firmly. Check whether the GND cable of the encoder cable (connector label CN6) of the left motor is firmly fixed to the device casing with the screw. Check whether or not a value of left motor is appropriate (or an abnormal value like "0" is not displayed) in the displaying running speed mode.
d30	Temperature rise of FET for motor of the right motor drive circuit is detected.	Move the system to a safe place. In this case, use the function of emergency brake release without the power supply to move the system. Check whether the CN19 on the MU DRIVER board is connected firmly. Turn OFF the main breaker of the system, and check the temperature of FET (Q11, 12, 14, 15) on the MU DRIVER board.
d31	Temperature rise of FET for motor of the left motor drive circuit is detected.	Move the system to a safe place. this case, use the function ofemergency brake release without the power supply to move thesystem. Check whether the CN18 on the MU DRIVER board is connected firmly. Turn OFF the main breaker of the system, and check the temperature of FET (Q1, 2, 8, 9) on the MU DRIVER board.

Error code	Content of the error	Treatment
d40	Fuse Melting is detected.	Move the system to a safe place. In this case, use the function of emergency brake release without the power supply to move the system. - Check the electrical connection of Fuses F4 and F5 mounted on the left side of the system.
d42	24V power is not supplied.	Move the system to a safe place. In this case, use the function of emergency brake release without the power supply to move the system. Check the connection of CN9 on the MU DRIVER board. Check whether the connector of DC power PS2 mounted in the front side of the system is connected firmly.
d44	+5/+15V power for the right motor is not supplied.	Move the system to a safe place. this case, use the function ofemergency brake release without the power supply to move thesystem.Check whether the connector of DC power PS22 mounted in the right side of the system is connected firmly. Check whether the CN22 and 23 on the MU DRIVER board are connected firmly.
d45	+5/+15V power for the left motor is not supplied.	Move the system to a safe place. In this case, use the function of emergency brake release without the power supply to move the system. Check whether the connector of DC power PS21 mounted in the right side of the system is connected firmly. Check whether the CN22 and 23 on the MU DRIVER board are connected firmly.
d60	Breaking of a wire is detected.	Move the system to a safe place. In this case, use the function of emergency brake release without the power supply to move the system. Check the wire of the support stand for disconnection.
d61	Abnormal connection with MU DRIVER	Check whether the CN21 on the MU DRIVER board and the CNUM on the XCONT board are connected firmly.

The Warning Indication lamp ⚠ will light during warning. The content of the error is indicated by the code on the mAs display. The code is displayed for approx. 3 seconds.

Table 10-2 Warning error message list

Error code	Content of the error	Treatment
F64	Invalid HX signal input. (EX. Hand switch was pushed just after turning ON the system power.)	The lamp turns off by releasing the hand switch.
F65	Invalid AR signal input. (EX. Hand switch was pushed just after turning ON the system power.)	The lamp turns off by releasing the hand switch.
F68	The X-ray exposure is done before the ready preparation of the FPD device is completed. Disappears by pressing the FPD Radiography Mode button.	Exposure should be performed after the completion of READY on DR system. The lamp turns off by pressing the switch for FPD / Cassette selection.
F69	Command for setting Built-in AEC Assistance was not received. Or, command for setting serial radiography was not received	Reselect the protocol on the DR console. The lamp turns off when the protocol is transmitted from DR system.
F6c	Radiography was ended before reaching the target pixel value of Built-in AEC Assistance.	Modify radiographic condition.
F6d	Emergency stop switch has being pressed. (EMG signal) Disappears by releasing the Emergency stop switch.	Eliminate any abnormality on the device. The lamp turns off by turning the emergency switch in an arrow direction and pulling it back.
F76	An illegal value was transmitted from DR-ID 800. (EX. Out of range APR No.) Make the anatomical code correspond with the DR-ID 800.	Modify the exposure protocol on DR console. The lamp turns off when the correct value is transmitted from DR system.
H1	The heat accumulated on the anode of the X-ray tube has reached its tolerance limit. (calculated value). Radiography is not allowed unless the radiography parameters are reduced, or until the warning indicator lamp turns off. Disappears when X-ray tube assembly cools sufficiently.	Modify the exposure conditions to lower, or stop radiography until the warning indication lamp turns off. The lamp turns off when the body temperature goes down by cooling the X-ray tube unit.
H2	The heat accumulated on the anode of the X-ray tube has reached its tolerance limit. (measured value) Radiography is not allowed until the warning indicator lamp turns off. Disappears when X-ray tube assembly cools sufficiently.	Modify the exposure conditions to lower, or stop radiography until the warning indication lamp turns off. The lamp turns off when the body temperature goes down by cooling the X-ray tube unit.

Error code	Content of the error	Treatment
H4	THERMAL signal is not supplied. Radiography is not allowed until the warning indicator lamp turns off. Disappears when X-ray tube assembly cools sufficiently.	Modify the exposure conditions to lower, or stop radiography until the warning indication lamp turns off. The lamp turns off when the body temperature goes down by cooling the X-ray tube unit.
E	CPU battery voltage has dropped. Change the CPU battery.	Although the usual traveling and radiography are available, the replacement of CPU board battery for backup is needed.
EEE	CPU battery is completely empty. Change the CPU battery.	Although the usual traveling and radiography are available, the replacement of CPU board battery for backup is needed.
E5	Voltage of the battery for emergency brake release has dropped. Change the battery.	Although the usual traveling and radiography are available, the brake cannot be released in an emergency. The replacement of the emergency brake release battery is needed.
bbb	Main battery voltage has dropped. Radiography is not allowed. Only the movement of the system is possible.	Charge the battery.
b1	The specified radiography conditions set an exposure duration exceeding the upper limit for the current image accumulation time setting, since the system is currently set in low battery mode.	Charge the battery. Warning clears when the battery is charged sufficiently.
bb0	Main battery voltage has dropped. Radiography is not allowed and the system is automatically turned OFF. The system may have been left with the key switch ON.	Charge the battery if it is low.
bb1	Main battery voltage has dropped. Radiography is not allowed and the system is automatically turned OFF. The system may have been left with the power of DR turned ON.	Charge the battery if it is low. If the battery is still low after the charge, the battery may be deteriorated.
A2	The cable between the short connector (CN38A, CN38B) of the handle unit and MU DRIVER board is disconnected.	Check between the short connector (CN38A, CN38B) of the handle unit and MU DRIVER board for disconnection. Check the short connector (CN38A, CN38B) of the handle unit is securely connected.
A4	Brake release switch is ON when the system power is turned ON.	Release the brake release bar. Although the radiography is available, the system cannot be moved until the warning is released.

Error code	Content of the error	Treatment
A5	Small movement switch (forward) is ON when the system power is turned ON.	Release the small movement switch. Although the radiography is available, the system cannot be moved until the warning is released.
A6	Small movement switch (backward) is ON when the system power is turned ON.	Release the small movement switch. Although the radiography is available, the system cannot be moved until the warning is released.
A7	Drive handle input (left) is detected when the system power is turned ON.	Release the drive handle. Although the radiography is available, the system cannot be moved until the warning is released.
A8	Drive handle input (right) is detectedwhen the system power is turned ON.	Release the drive handle. Although the radiography is available, the system cannot be moved until the warning is released.
A9	Bumper switch is ON when the system power is turned ON.	Release the bumper switch. Although the radiography is available, the system cannot be moved until the warning is released.
A10	Arm lock release switch is ON when the system power is turned ON.	Release the arm lock release switch. Although the radiography is available, the system cannot be moved until the warning is released.
A11	Arm lock release lever is ON when the system power is turned ON.	Release the arm lock release lever. Although the radiography is available, the system cannot be moved until the warning is released.
d62	Abnormality in the arm travel distance detection encoder is detected.	Turn on the power of the system again, and move the arm in full stroke to check whether or not the encoder of the support stand spins around.

10.2 Error Message on DR-ID800

NOTE This sction is only applicable to MobileDaRt Evolution for DR-ID800 type.

If an error message is displayed on DR-ID800 PC, the appropriate instruction is given in the service manual below.

Reference [MT: Machine Troubleshooting] in DR-ID800CL Service Manual or DR-ID800 Service Manual

To reinstall DR-ID800CL software, perform the following steps.

1 **Uninstall DR-ID800CL software.**

Reference [Appx IN: Installation – Appendix, Uninstalling the AP] in DR-ID800CL Service Manual

2 **Install DR-ID800CL software again.**

Reference [MC: Inspection, Replacement and Adjustment, Installing Database Management System] in DR-ID800CL Service Manual

11 [Appendix] Maintenance and Inspection

Basically, it is not necessary to make the replacement of any parts described in this Chapter during installation of the unit.

11.1 Batteries Replacement

11.1.1 Precautions for batteries replacement

NOTE 20 pieces are needed when replacing batteries.

⚠ WARNING

- The unit contains a total of 240 VDC battery.
- If the terminals of the battery are short-circuited, electric shock or fire may occur, resulting in malfunction or accident. In the worst case, equipment and hospital facilities cannot be used.
- During work, insulate the terminals by covering them with an insulating tape, etc., so that the terminals do not contact the housing unintentionally.
- When replacing the batteries, make sure to place the unit on the flat floor. Otherwise, the unit may fall down.
- Do not wear metallic accessories such as the bracelet, hanging jewelries, and clocks during the battery exchange work. It is very dangerous when coming in contact between terminals of the battery.
- Pay attention to polarity of +/- when arranging the battery in the battery unit. Do not short-circuit and contact between terminals.

⚠ CAUTION

Do not use different types of batteries (such as usage status, size, quality, and brand) together. This may cause system malfunction.
The following requirements are needed for any battery replacement.

- All 20 batteries should be replaced with new ones at the same time.
- All 10 batteries on the positive pole side (Battery ASSY BAT1, BAT2) must have the same manufacturing date.
- All 10 batteries on the negative pole side (Battery ASSY BAT3, BAT4) must have the same manufacturing date.
- The difference of manufacturing date between batteries for the positive pole and negative pole must be less than one month.

The manufacturing date is stamped on the upper surface of the battery, and written in the following form:
050217## …, i.e. manufactured on Feb.17, 2005 (## indicates any alphanumeric characters).

11.1.2 Procedure of the battery replacement

1 **Turn OFF both the key switch and the main breaker.**

2 **Remove the inside cover and both side covers.**

Reference "3.5 How to Open/Close Covers"

3 **Discharge the internal capacistor.**

Reference "3.4 Discharging the Internal Capacitor"

⚠ WARNING

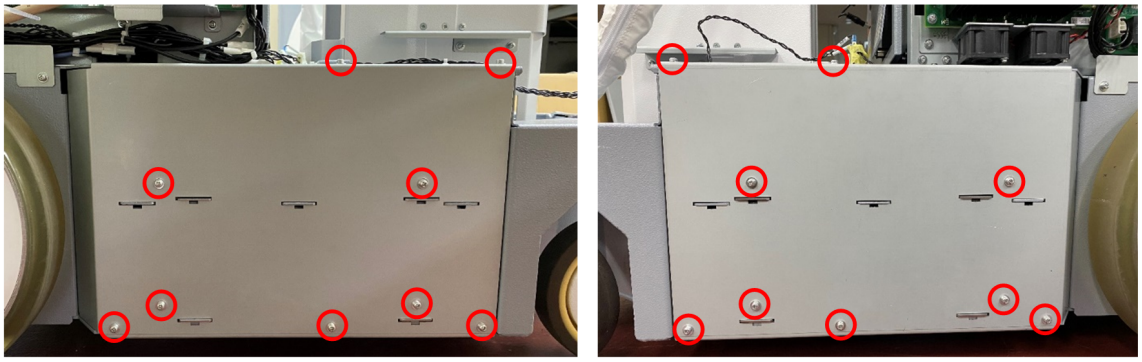
Prior to replacing the battery, be sure to discharge the capacitor.

Reference "3.4 Discharging the Internal Capacitor"

4 **Remove the battery cover as shown below.**

Right Side Left Side

Fig. 11-1



Right Side **Left Side**

Fig. 11-2

5 **Disconnect the wires from the terminal blocks in order of the numbers shown in the figure below. The terminal (-) on BAT 2 must be disconnected first.**

Right Side **Left Side**

Fig. 11-3 Battery Units (Both sides)

⚠ WARNING

Insulate the terminals of the wires removed from the terminal block by covering them with insulating vinyl tape, etc., so that the terminals do not contact the housing unintentionally.

⚠ CAUTION

Removing the wires in the wrong order increases the risk of electric shock or fire. Be sure to remove them in the order shown in the figure (①→②→...→⑧).

6 **Draw the battery ASSY.**

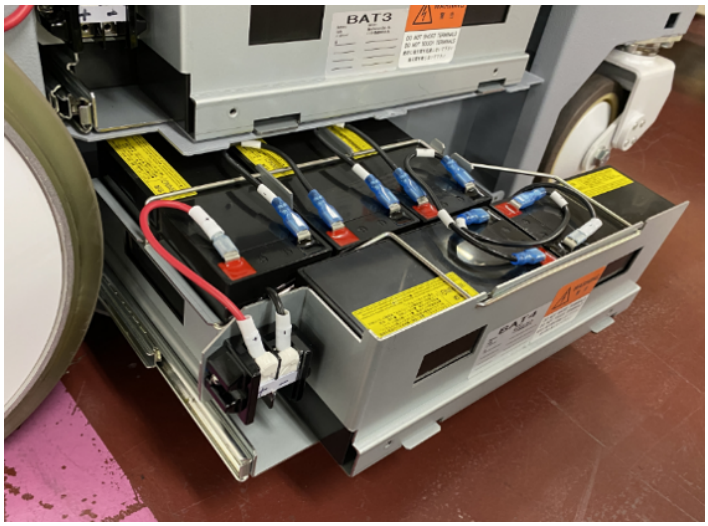


Fig. 11-4 Removing the Battery ASSY (Right Side)

7 Press the arrow position on the presser bar to release the presser bar from the tab.

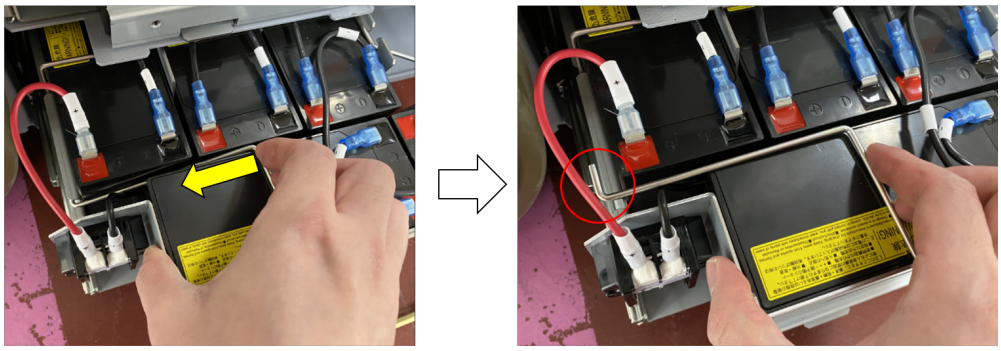


Fig. 11-5 Battery ASSY

8 Remove the presser bar from the tab to remove the bar in the order shown in the figure.

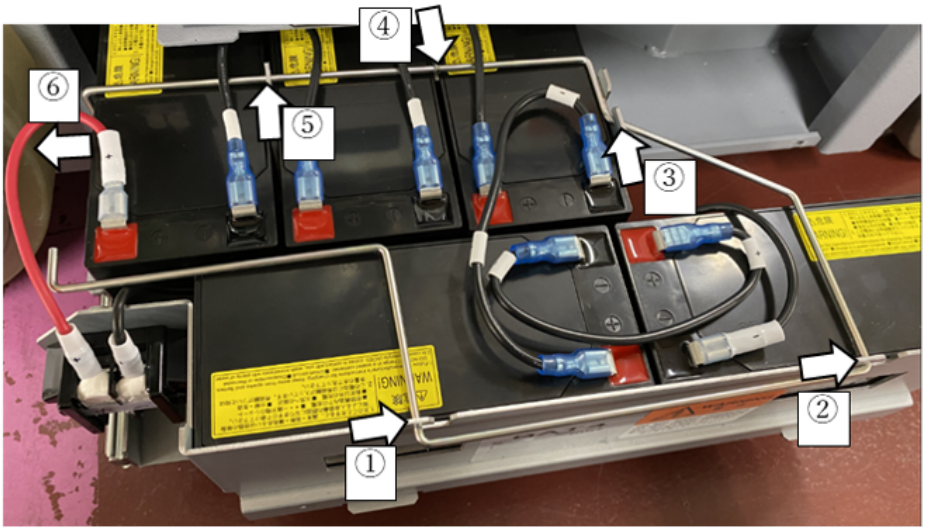


Fig. 11-6 Battery ASSY

9 Disconnect the battery wires and remove all the batteries.

10 Write the battery code and manufacture date of each new battery on the label of each battery ASSY.

11 **Insert the new battery, connect the battery ASSY terminal block and the wiring, and restore it to its original state.**

Connection order should be in reverse of the order shown in step 5.
(Connect them in the opposite order (⑧→⑦→⑥→⑤→④→③→②→①) in "Fig. 11-3 Battery Units (Both sides) ". Be sure to start from the (-) terminal of BAT 4.

⚠ CAUTION

Connection in the wrong order increases the risk of electric shock or fire.
Be sure to connect them in the opposite order shown in the figure (⑧→⑦→⑥→⑤→④→③→②→①).

12 **Attach the presser bar by hooking it to the tab in the order of the numbers in the figure.**

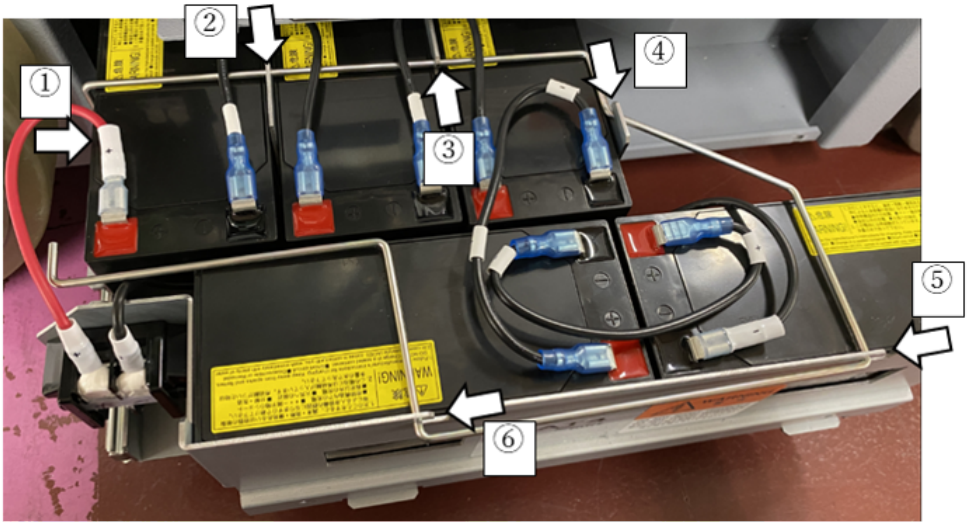


Fig. 11-7

13 **Press the arrow position on the presser bar to fix the bar to the tab.**
If you push the Battery ASSY slightly to the back, it will be easier to fix the bar.

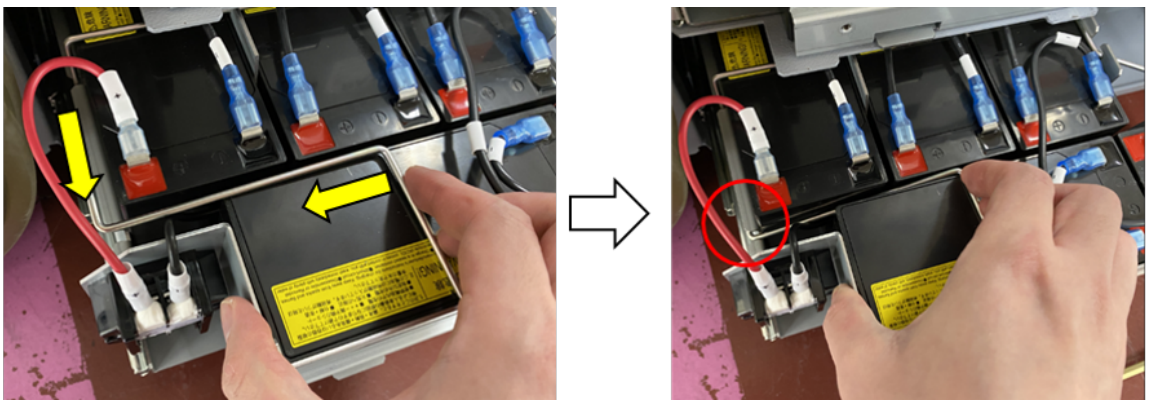


Fig. 11-8

14 Gently push the Battery ASSY to the back, taking care not to pinch the cable. Replace the other three locations in the same manner.

15 Re-install the battery cover.

16 Reset the internal capacity discharge to "NORMAL". (Both discharge switch and connector should be reset.)

17 Replace the inside cover and the side covers.

18 Turn ON both the main breaker and the key switch. Confirm that power is supplied to the system.

19 Clear the usage information of the Battery.

Reference
- For MobileDaRt Evolution: "7.13 Battery Usage Setting Mode"
- For MobileArt Evolution: "8.13 Battery Usage Setting Mode"

11.2 Replacement of the Motor

11.2.1 Procedure of the Motor Replacement

■ Detaching of the motor

1 **Turn OFF the main breaker.**

2 **Remove the inside cover and side cover.**

Reference "3.5 How to Open/Close Covers"

3 **Remove 3 caps on the wheel cover and loose screws. Turn the wheel cover to clockwise to remove it.**

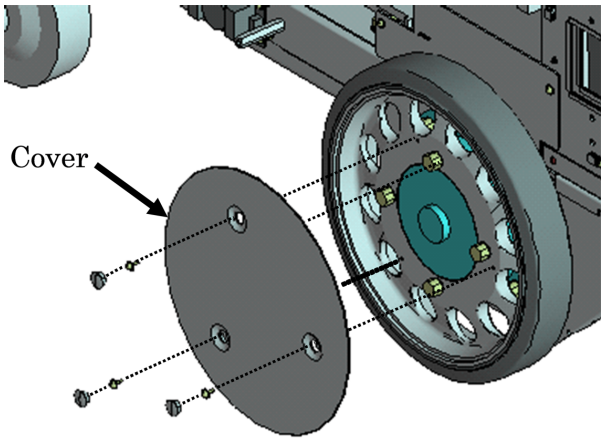


Fig. 11-9

4 Loose lightly 4 bolts of the wheel.

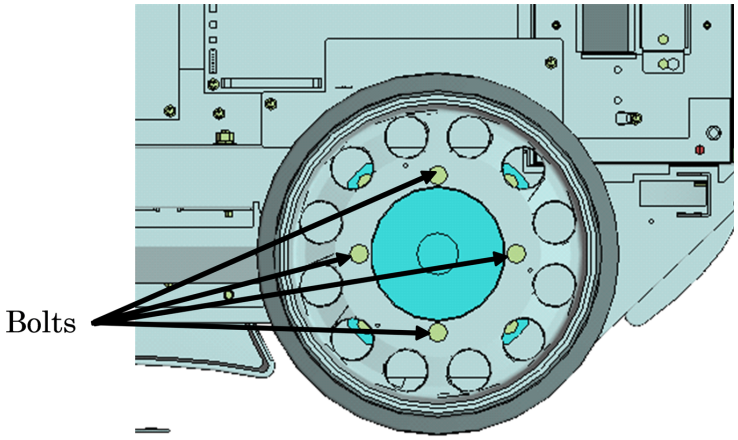


Fig. 11-10

5 Lift the wheel up by a jack etc. and keep that to set woods under the system.

6 Disconnect connectors on MU DRIVER 07 Board corresponding to the replaced motor. CN6 and CN14 are connectors for the left motor. CN7 and CN16 are connectors for right motor. Grounding earth is also disconnected.

7 Remove the wheel.

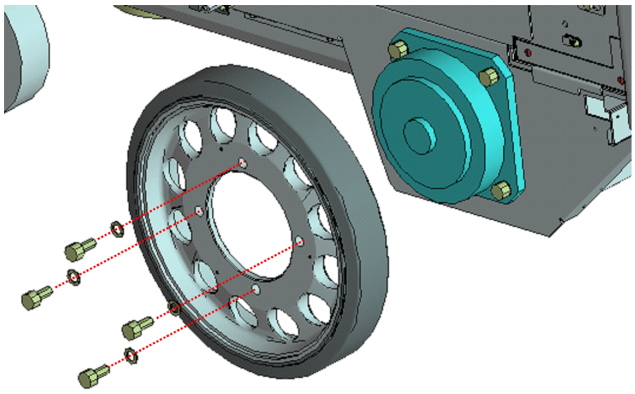


Fig. 11-11

8 **Remove the motor.**

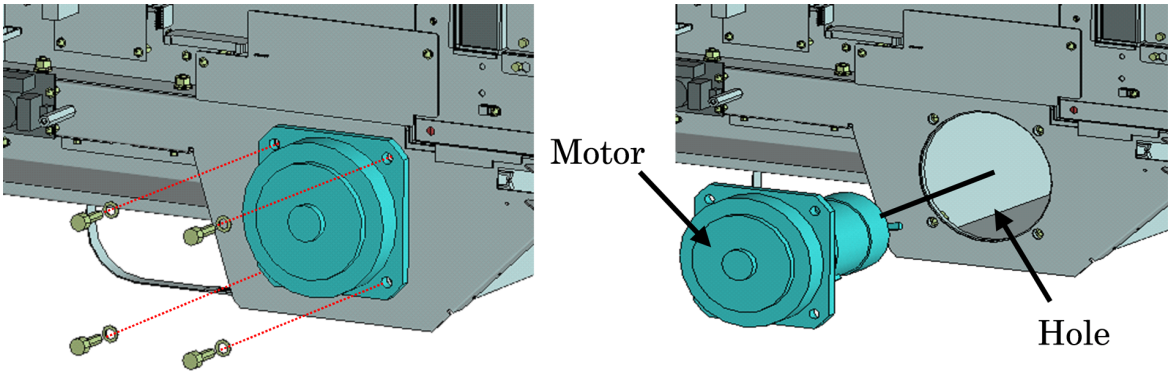


Fig. 11-12

9 **Pull motor cables out.**

■ Mounting the motor

1 **Wire the motor cable to rearward of the system.**

2 **Mount the motor.**

> NOTE The gear reducer of the motor is eccentric from the output shaft. Mount the motors as shown below.

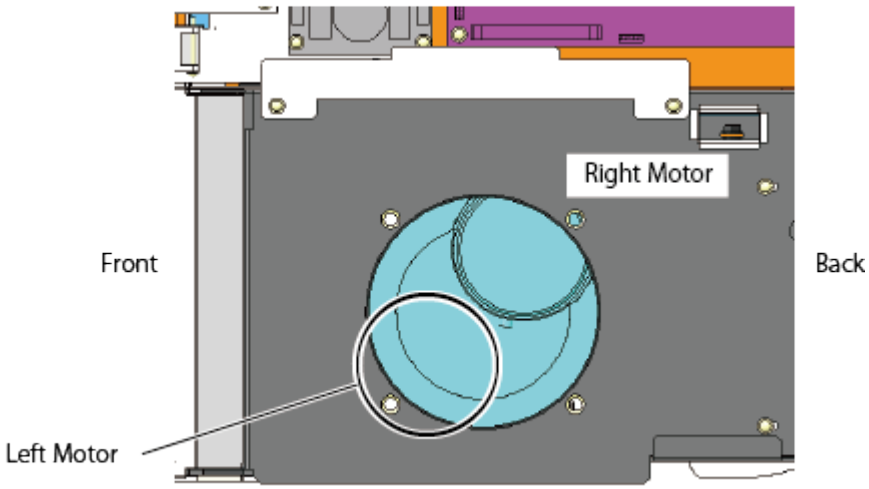


Fig. 11-13

3 **Connect CN6, CN14, CN7, CN16, and the grounding cable.**

4 Turn ON both the main breaker and the key switch, and check the performance without the wheel.

5 Make sure the performance is normal. Mount the wheels loosely.

6 Remove the woods under the system.

7 Tighten the bolts of the wheels.

8 Replace the wheel covers.

9 Attach the removed covers again.

10 Make sure the performance is normal.

11 Clear the usage information of the motor.

Reference
- For MobileDaRt Evolution: "7.17 Motor Usage Information Setting Mode"
- For MobileArt Evolution: "8.17 Motor Usage Information Setting Mode"

11.3 CPU Backup Battery Replacement

11.3.1 CPU Backup Battery replacement

■ **Procedure of replacement for the CPU backup battery.**

1 **Remove the right inside cover and the right-side cover.**

Reference "3.5 How to Open/Close Covers"

2 **Turn on the key switch to apply the power.**

NOTE The key switch must be turned ON whenever the CPU backup battery requires replacing. Failure to do this will cause all stored configuration information to be lost.

3 **Replace the CPU backup battery where is on the XCONT-2018 board.**

4 **Clear the usage information of the CPU Backup Battery.**

Reference
- For MobileDaRt Evolution: "7.14 CPU Board Battery Usage Information Setting Mode"
- For MobileArt Evolution: "8.14 CPU Board Battery Usage Information Setting Mode"

11.4 Maintenance of rotary X-ray tube section

If the rotation of the X-ray tube section is found to be abnormal or short of lubrication, grease part ① and ② as shown below ("Fig. 11-15 Detail of Part A", "Fig. 11-16 Detail of Part B", and "Fig. 11-17 Detail of C" for the detail of part A, B, and C).

* Recommended grease: Albania grease.

⚠ CAUTION

Before maintenance, make sure the lift operation of the column is locked by the column lift-lock brackets and pin.

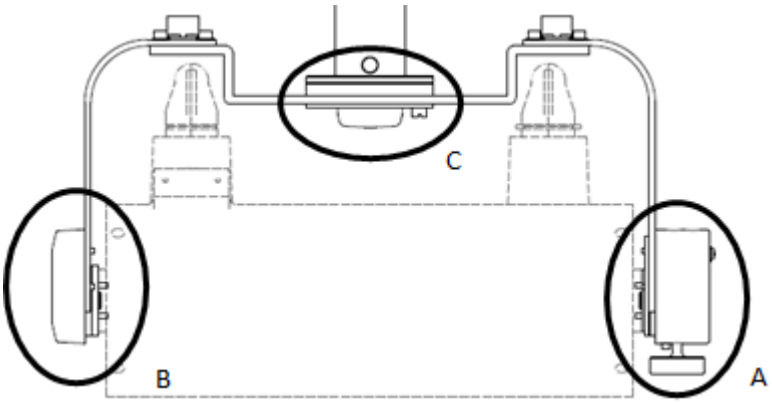


Fig. 11-14 X-ray tube section

⚠ CAUTION

Before removing the X-ray tube (approximately 13 kg), grasp the detail of the fixing structure. Be careful not to drop the tube by loosening other bolts incautiously. Dropping the tube may cause the break of the tube and injury.

11.4.1 Greasing Part [A] of the X-ray tube section

The detailed drawing of Part [A] of "Fig. 11-14 X-ray tube section" is shown below.

■ **Greasing steps**

1 **Unscrew M4 bind screw (a) and remove cover A.**

2 **Unscrew three locking screws (b), and make two plungers (c) loose.**

3 **Unscrew two M3 flat countersunk head screws (d), and remove cover C.**

4 **Unscrew four M6 hexagon socket head cap screws (e), and remove bearing A.**

5 **Grease part ①.**

6 **Bearing A and X-ray tube are separated each other by unscrewing four M6 hexagon socket head cap screws (f).**

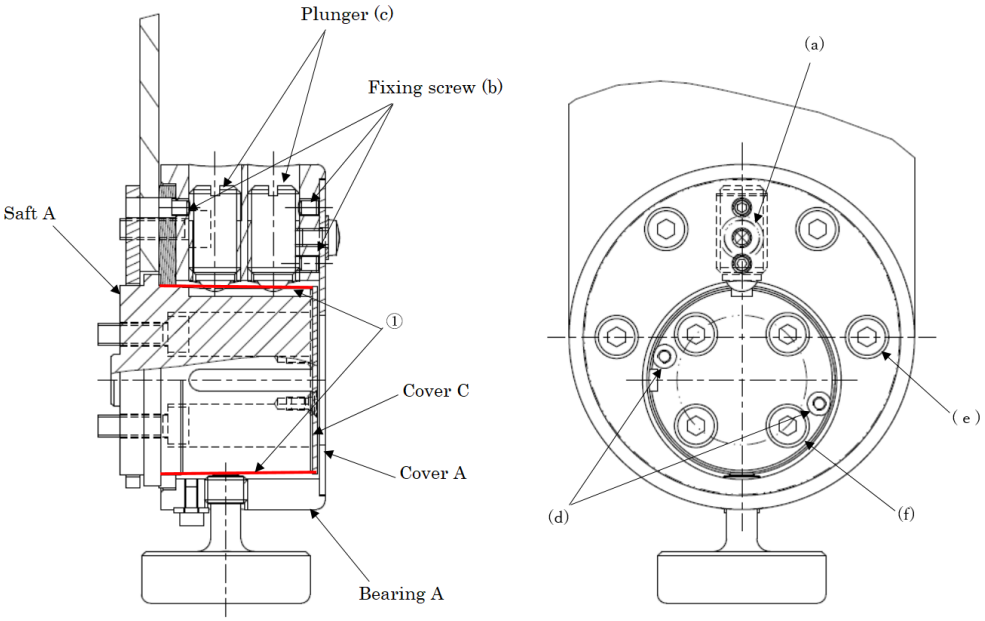


Fig. 11-15 Detail of Part A

11.4.2 Greasing Part [B] of the X-ray tube section

The detailed drawing of Part [B] of "Fig. 11-14 X-ray tube section" is shown below.

■ **Greasing steps.**

1 **Unscrew M4 bind screw (g) and remove cover B.**

2 **Unscrew four M5 hexagon socket head cap screws (h), and remove the X-ray tube from the holding frame.**

3 **Put the X-ray tube removed from the holding frame on a table or the like. (Use a cushion under the tube.)**

4 **Unscrew locking nut (i), disc spring (j), and four M5 hexagon socket head cap screws (k), and remove bearing B. In this state, bearing B and X-ray tube can be separated each other.**

5 **Grease Part ①.**

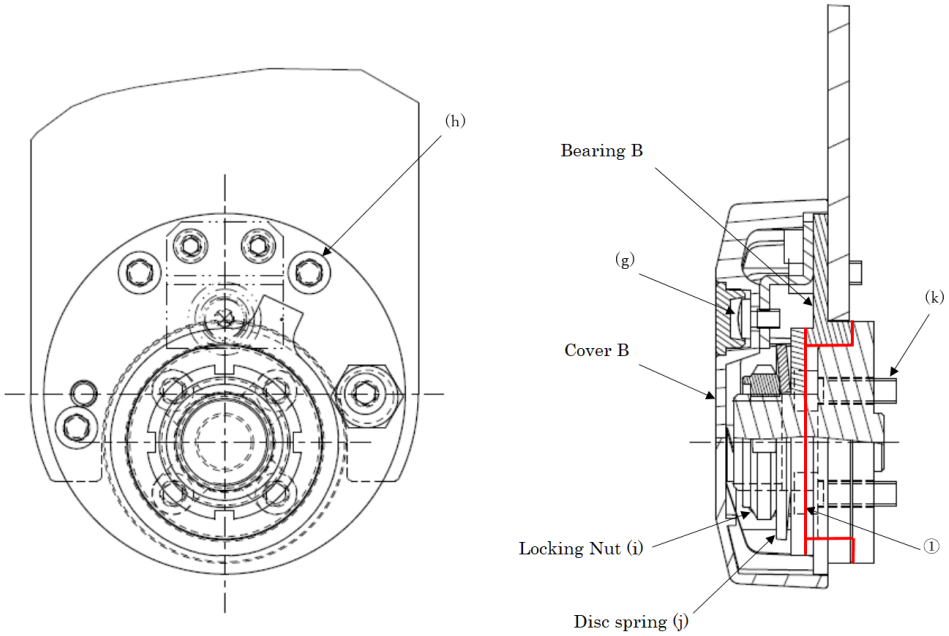


Fig. 11-16 Detail of Part B

11.4.3 Greasing Part [C] of the X-ray tube section

The detailed drawing of Part [C] of "Fig. 11-14 X-ray tube section" is shown below. After assembling the X-ray tube section, make it rotate 10 to 20 times to be thoroughly lubricated by the grease.

■ Greasing steps

1 **Unscrew hexagon socket flat countersunk head screw (l) and remove the end cover.**

2 **Unscrew locking screw (m), and make plunger (n) loose.**

3 **Unscrew locking nut (o), disc spring (p), detent washer (q), and thrust washer (r).**

4 **Pull out the bearing toward the front, and remove it.**

5 **After making sure that there is nothing abnormal with Part ②, apply grease there.**

6 **After applying grease to Part ① and ②, assemble the X-ray tube section in the reverse steps.**

NOTE If the fluted edge of Axis A is rounded off by the plunger in the Part A, fix Axis A at 90 degrees to this position.

⚠ CAUTION

Check the direction of stopper at B, and thrust DU washer at C.

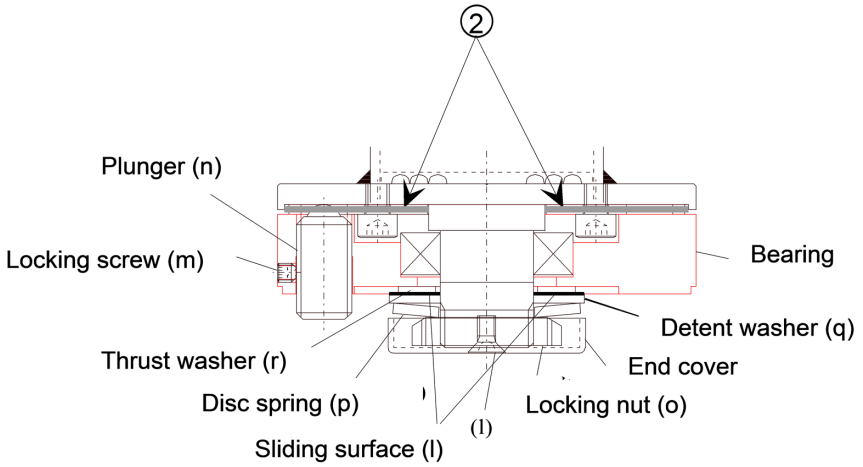


Fig. 11-17 Detail of C

⚠ CAUTION

Assemble the X-ray tube section so that the plunger in Part A, locking nut in Part B, and the plunger and locking nut in Part C are fixed with the following pressure and torque.

11.4.4 Adjustment of Click Feeling

■ Back-and-forth direction rotation

1 **Remove the cover.**

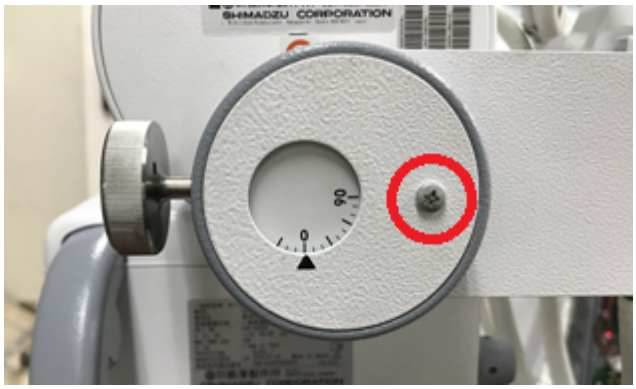


Fig. 11-18

2 **Loosen the set screw fixing the plunger (at 3 places).**

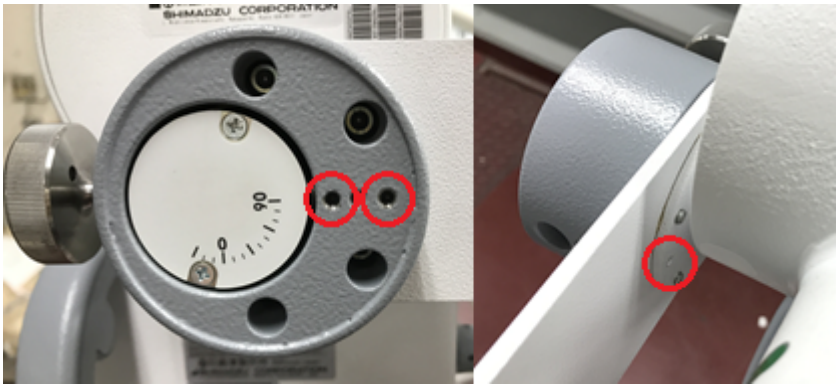


Fig. 11-19

3 **Turn the ball plunger clockwise, and push it in until the ball enters the plunger and the plunger body hits the shaft.**

> **⚠ CAUTION**
>
> Be careful not to push the ball plunger too much. The rotating shaft may deform and cause unusual sound and malfunction.

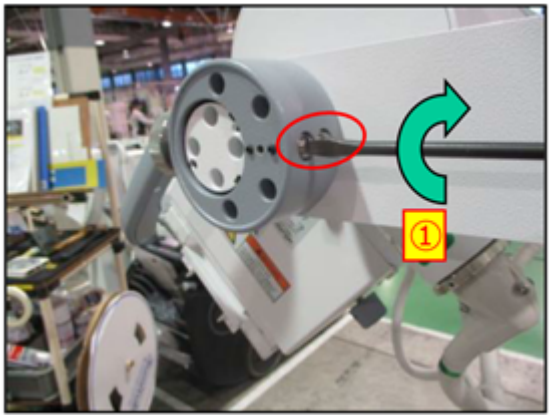


Fig. 11-20

4 **Turn back 45 degrees counterclock.**

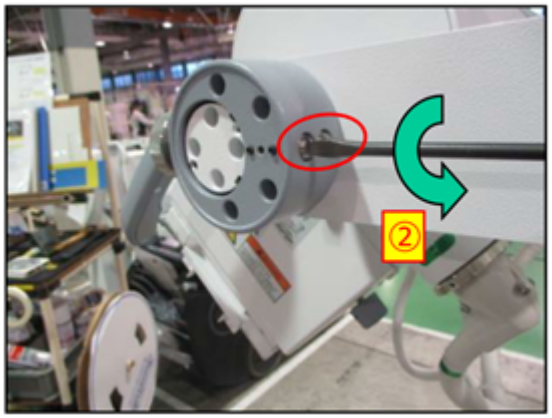


Fig. 11-21

5 **If the click feeling is not enough, turn the plunger clockwise little by little to check the click feeling.**

6 **Tighten the set screw loosened in step 2 (at 3 places).**

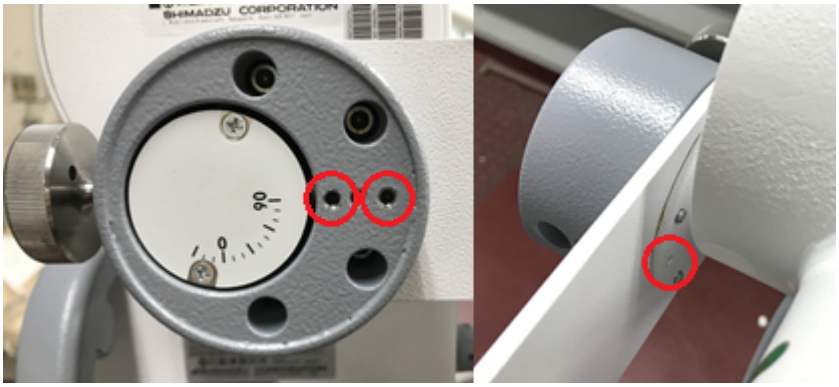


Fig. 11-22

7 **Attach the cover removed in step 1.**

■ Tube horizontal-shaft rotation

1 **Loosen a set screw (at 1 place).**

Fig. 11-23

2 **Turn the ball plunger clockwise, and push it in until the ball enters the plunger and the plunger body hits the shaft.**

> **⚠ CAUTION**
>
> Be careful not to push the ball plunger too much. The rotating shaft may deform and cause unusual sound and malfunction.

Fig. 11-24

3 **Turn back 45 degrees counterclock.**

Fig. 11-25

4 **If the click feeling is not enough, turn the plunger clockwise little by little to check the click feeling.**

5 **Tighten the set screw loosened in step 1 (at 1 place).**

Fig. 11-26

11.4.5 Adjustment of Tube Rotation Operating Force

■ Back-and-forth direction rotation

1 **Remove the cover.**

Fig. 11-27

2 **Adjust the lock nut so that the force starting to move by pressing the position of the arrow in the state of 80 degrees rotated foreground may be 30±3N.**

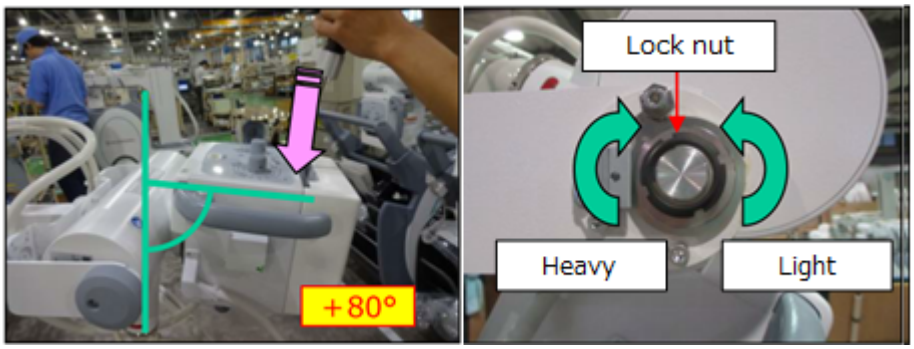


Fig. 11-28

3 **Attach the cover removed in step 1.**

■ Tube horizontal-shaft rotation

1 **Remove the cover.**

Fig. 11-29

2 **Raise the click of the rotation stop washer.**

Fig. 11-30

3 **Adjust the lock nut so that the force starting to move by pressing the position of the arrow in the state of 80 degrees rotated rightward may be 20±3N.**

Fig. 11-31

4 **Fold the click of the rotation stop washer to fix the lock nut.**

5 **Attach the cover removed in step 1.**

11.5 Centering the Collimator and X-Ray Focus

There are following two kinds of discrepancy between the collimator illumination field and X-ray irradiation field. When installing the apparatus, make sure which discrepancy is occurring and adjust the discrepancy:

- Discrepancy between the collimator illumination field center and X-ray irradiation field center.
- Discrepancy between the collimator illumination field size and X-ray irradiation field size.

11.5.1 Discrepancy between the collimator illumination field center and X-ray irradiation field center

■ How to adjust the collimator illumination field center and X-ray irradiation field center.

If the center axis of the collimator blades and the axis of conical X-ray beam are discrepant when the collimator is mounted on the X-ray tube, the collimator illumination field center shifts from the X-ray irradiation field center.

In this case, align the center axis of the collimator blades with the axis of conical X-ray beam by moving the collimator itself without displacing the lamp mounting board as follows:

1 **Show the illumination field on a cassette placed on the exposure top panel (10-cm square).**

2 **Put a piece of line solder along the four sides of the illumination field and operate exposure.**

3 **Then, widen the illumination field further (15-cm square), leaving the line solder as it is.**

4 **Operate another exposure (double exposure).**

5 **After developing the radiograph, compare the illumination field center (center of the image of the line solder) with the X-ray irradiation center, and measure the discrepancy between them.**

6 **Attach the screw and loosen the bolt a little, and then correct the discrepancy measured in the step 5 by adjusting the position where the screw is tightened. After adjustment, tighten the bolt firmly, and then remove the screw.**

⚠ CAUTION

Because the center of the collimator blades and the illumination field center are aligned each other using a dedicated device in factory assembling, never move the lamp mounting board.

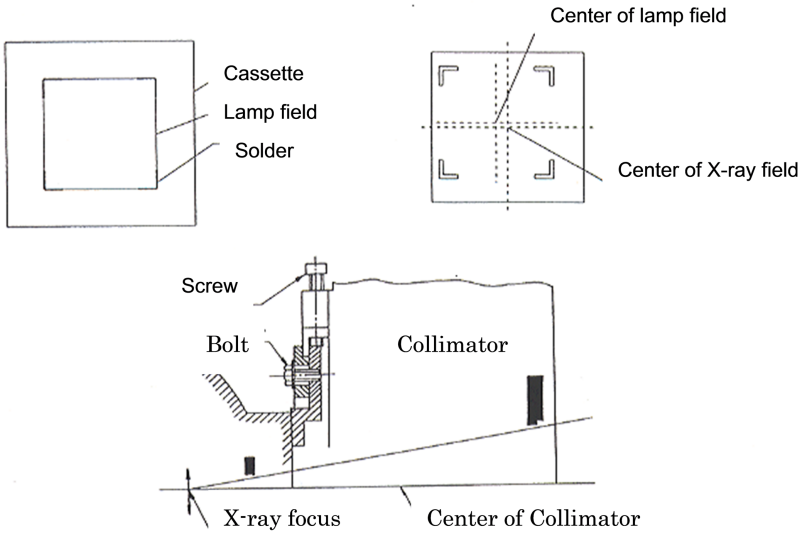


Fig. 11-32 Adjust field center

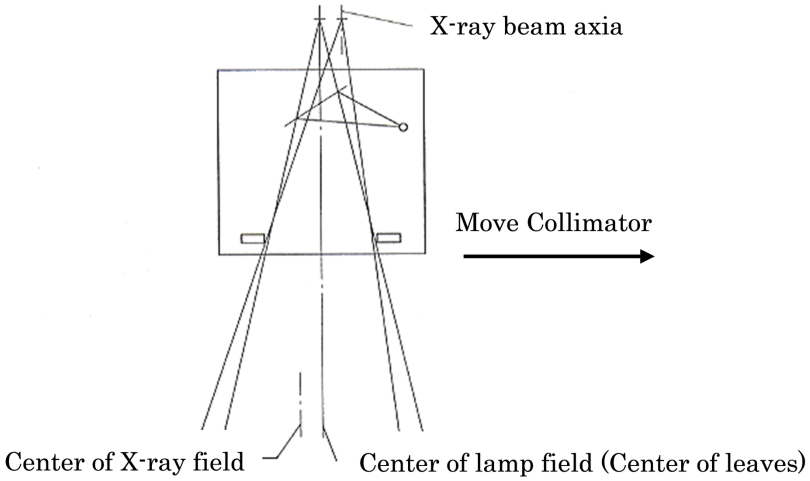


Fig. 11-33 How to adjust center

11.5.2 Discrepancy between the collimator illumination field size and X-ray irradiation field size

■ Why the discrepancy between collimator illumination field size and X-ray irradiation field size occur.

The illumination field size is always different from the X-ray irradiation field size. Because the filament size of illumination lamp is larger than 3 mm in comparison with the X-ray focus size (less than 1 mm), a portion called half-shadow is physically occurs on the illumination field, and due to the half-shadow the illumination field becomes larger than the X-ray irradiation field.

Discrepancy between the illumination field and X-ray irradiation field is adjusted minimum discrepancy at S.I.D 1m. Because the half-shadow is a physical phenomenon, it is impossible to eliminate it.

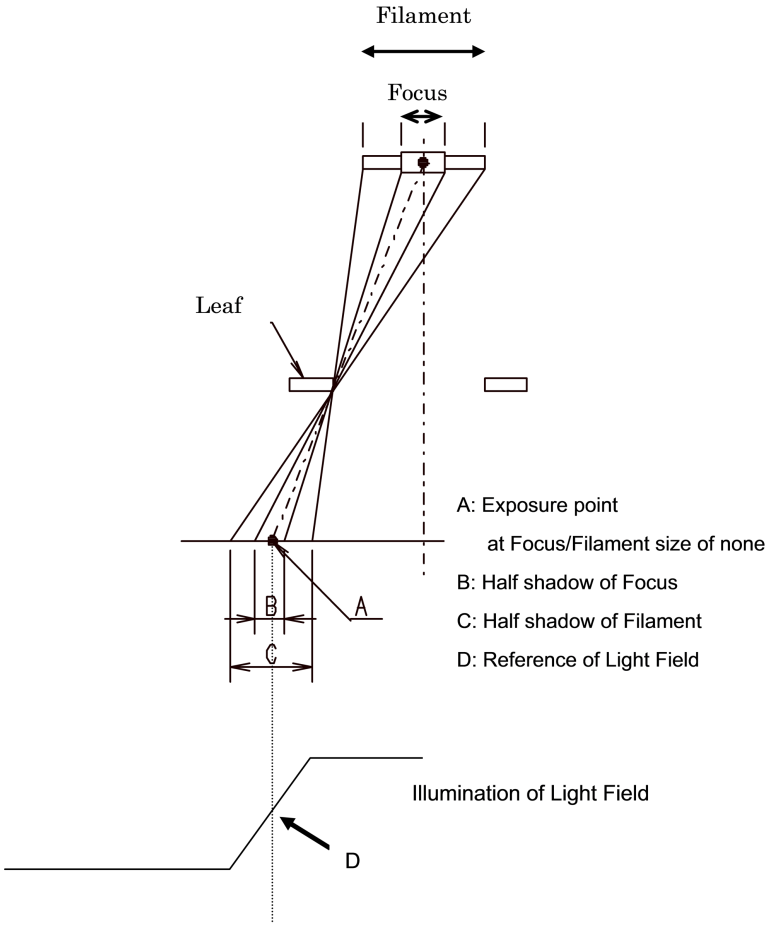


Fig. 11-34 Discrepancy between the collimator illumination field and X-ray irradiation field.

■ Technique to minimize the discrepancy.

Adjust the X-ray irradiation field size so that the edge of the X-ray irradiation field is set on the point D of the edge of the illumination field, after determining the illumination field size in accordance with the exposing portion of the subject.

In this way discrepancy between the collimator illumination field center and X-ray irradiation field can be minimized.

Discrepancy between the Point D and X-ray irradiation field is conformed to IEC regulation, adjusted within 2cm (within 2% of S.I.D) at S.I.D 1m prior to shipment.

11.6 Cleaning and disinfections

11.6.1 Cleaning and disinfections

⚠ CAUTION

Cleaning and disinfections work should work, after turning off equipment power.

If equipment is turned on the power, there is a possibility of causing an electric shock and a short circuit.

Do not splash antiseptic on this equipment. Do not spray of the antiseptic on this equipment directly.

If an antiseptic goes into the inside of equipment, it will become failure and the cause of the accident.

When you switch on a power supply after cleaning and disinfections work, do fully ventilate the inside of a cleaning and disinfections working space.

If inflammable gas remains cleaning and disinfections working space, when a power supply is switched on, there is danger of ignition, emitting smoke, explosion, and an electric shock.

NOTE This equipment is not protected against any form of liquid infiltration.

Ensure care is taken to prevent liquid infiltration when cleaning or disinfecting the unit.

There is no guarantee for the system performance and safety when the following disinfectants such as follows:

- A chlorinated disinfectant.
- A disinfectant that corrodes metal, plastic, rubber, or paint.
- A disinfectant that is unsuited to metal, plastic rubber, or paint.
- A spray disinfectant.
- A volatile disinfectant.
- A disinfectant with a possibility of infiltration into the product.

1 **Check the equipment is turned off power.**

2 **Cleaning and disinfections work are done.**

Clean with the cloth moistened with alcohol or disinfectant.
Do cleaning and disinfections work certainly also to what has unevenness in the surface using a swab etc.

3 **After a cleaning and disinfections work end, should be checked the following item, before switching on a power supply of equipment.**

- Neither water nor the antiseptic has adhered to equipment.
- The tool used for cleaning and disinfections work is tidied up.

11.6.2 Notes in the case of taking over, return of apparatus etc.

⚠ WARNING

In the medical institution where the various microbes which affect a human body exist, a certain incubation period may be set by infection, and it may suffer from the "infection" which appears and is attacked with condition.

In connection with carrying out of the product in the case of recession from a medical institution, a tool, etc., there is the possibility of secondary infection out of a medical institution.
In order to prevent this, disinfections and positive implementation of sterilization are important with a prevention measure and healthy management.

■ General notes

In order to carry out positive disinfections and to prevent secondary infection, demand implementation of disinfections in a medical institution as much as possible.

For the sake of appearance, the existence of contamination is unclear.
Should be sure to perform a listening comprehension check to medical institution staffs.

■ Handling cautions of disinfections

Wipe external surface with the cloth dipped with glutaraldehyde (GA) liquid 2%.

Parts (especially machine part) are decomposing in the range, which can be decomposed. Parts wipe with the cloth dipped with glutaraldehyde (GA) liquid 2%, or disinfect by ethylene oxide gas.

11.7 Replacing the Emergency Brake Release Battery

11.7.1 Replacing the Emergency Brake Release Battery

■ Emergency Brake Release Battery Replacement Procedure

1 **Turn OFF the system power and the main breaker.**

2 **Remove the right inside cover and the right-side cover.**

Reference "3.5 How to Open/Close Covers"

3 **Remove the plate from the battery box.**

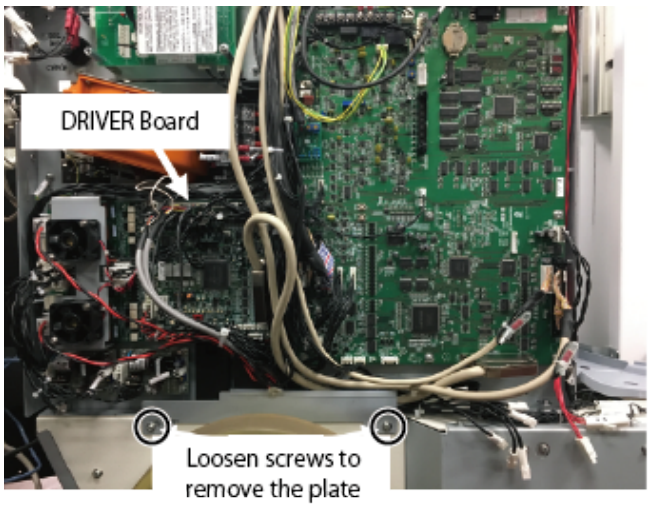


Fig. 11-35

4 Replace Battery (Specifications vary depending on the production date.)

- For the specification on the left figure, open the cover of the battery box, and replace the batteries (2 pcs).
- For the specification on the right figure, remove the screws indicated by arrows, and remove the metal cover, then replace the batteries (3 pcs).

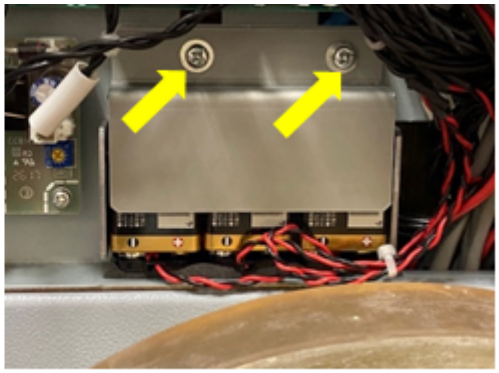


Fig. 11-36

5 Clear the usage information of the Emergency Break Release Battery.

Reference
- For MobileDaRt Evolution "7.15 Emergency Brake Release Battery Usage Information Setting Mode"
- For MobileArt Evolution "8.15 Emergency Brake Release Battery Usage Information Setting Mode"

11.8 Updating Software

This section describes how to upgrade the software via communication between a PC and the XCONT-2018 board.

■ Required Tools and Software

- PC equipped with Windows 7 or later, and with RS-232C serial port.
  If RS-232C serial port is not equipped, USB/RS232-C conversion cable should be separately prepared.
- Serial cable for RS-232C (9pin female - 9pin female, Cross type)
- Software
  →XCONT-2018 board: MXSH7085_SW_xxx （XXX indicates the version number.）

■ Preparation for the XCONT-2018 board

1 **Turn off the power.**

2 **Remove the right inside cover and the right-side cover.**

Reference "3.5 How to Open/Close Covers"

3 **Preparation for the version up S/W**

1. Turn off Main breaker NFB.
2. Unplug the unit.
3. Turn on Emergency switch.

4 **Set the DIP switch SW4 on the XCONT-2018 board. Set only SW4-2 to ON side and set other switch of SW4 to OFF side.**

5 **Connect the J4 connector of the XCONT-2018 and serial port of the PC with the serial cable.**

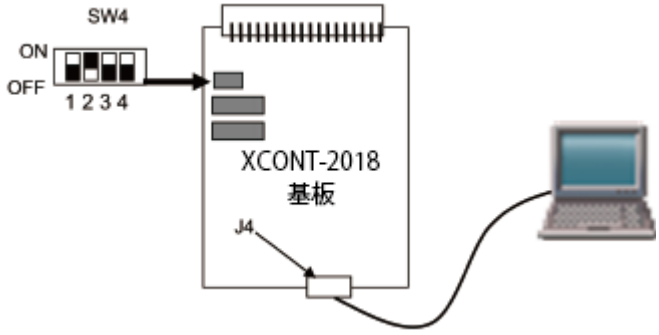


Fig. 11-37 Preparation of XCONT-2018

6 **Connect power plug. The power is supplied to the system.**

■ Setting of PC

1 **Turn on the PC.**

- For the media distribution software
  Insert the media storing the distributed software into the PC drive, and copy software "MXSH7085SW_XXX.exe"(XXX indicates the version number) to the hard disk of PC.
  For later use, copy this file to the folder easy to find.
- For the digital distribution software
  Decompress MXSHXXX.sdfx (XXX indicates the version number), the distributed software with a SDF extension, by using Sdf_Extract, the extraction software. MXSHXXX folder will be created in the location to be decompressed.
  For later use, decompress this file in the location easy to find.

2 **Make sure there are following files in the copied or decompressed folder.**

- •SoftUD.exe : Software update executable file.
- setup.ini : Software update setting file
- MXSH7085SW_XXX（XXX indicates the version number.） : Software data.

■ Backup of S/W

Before installing the S/W, backup the S/W already installed in the board for backup purpose.

> NOTE Be sure to terminate the serial communication software (e.g. Tera-Term) before starting the setup program.
> Otherwise, the backup process may fail.

1 **Double-click the SoftUD.exe. The following screen is displayed.**

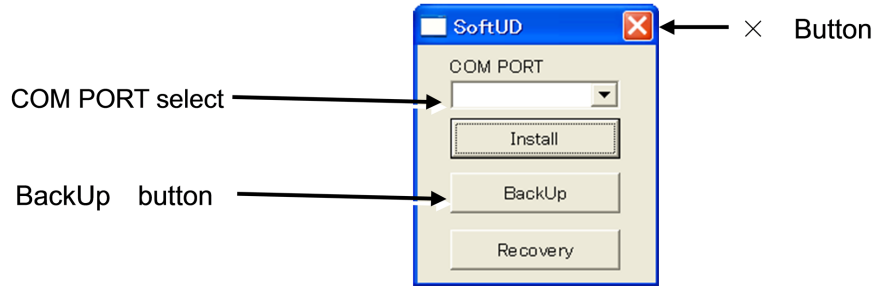


Fig. 11-38 SoftUD screen

2 **Select the COM PORT of PC that is connected with the serial cable and click the Backup button.**

3 **Confirmation screen is displayed. If proceeding backup of S/W, click the OK button.**

4 **Backup is started. The status screen is displayed. After completing the backup, screen of completion is displayed.**

5 **Return to the SoftUD screen in 1. Click the × button and the screen will be closed.**

■ Install of S/W

When version up of S/W, execute install of S/W.

> NOTE Be sure to terminate the serial communication software (e.g. Tera-Term) before starting the setup program.
> Otherwise, the backup process may fail.

1 **Double-click the SoftUD.exe. The following screen is displayed.**

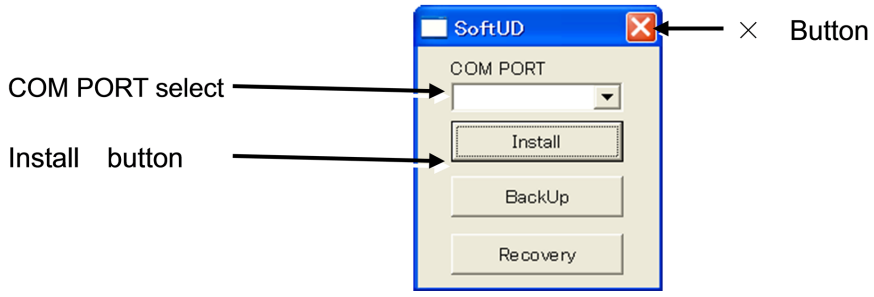


Fig. 11-39 SoftUD screen

2 **Select the COM PORT of PC that is connected with the serial cable and click the Install button.**

3 **Confirmation screen is displayed. To proceed to the installation of S/W, click the OK button.**

4 **Install is started. The status screen is displayed. After completing the install, screen of completion is displayed.**

5 **Return to the SoftUD screen in 1. Click the × button and the screen will be closed.**

6 **Set the current date and time.**

Reference
- For MobileDaRt Evolution: "7.7 Setting Date and Time"
- For MobileArt Evolution: "8.7 Setting Date and Time"

■ Recovery of S/W

When restoring the S/W that was back up, execute the recovery.

> NOTE Be sure to terminate the serial communication software (e.g. Tera-Term) before starting the setup program.
> Otherwise, the backup process may fail.

1 **Double-click the SoftUD.exe. The following screen is displayed.**

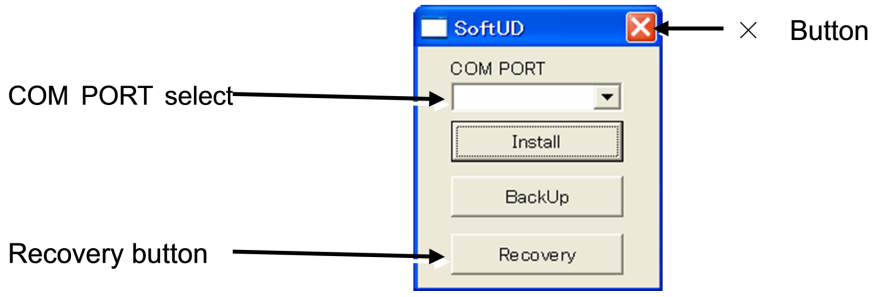


Fig. 11-40 SoftUD screen

2 **Select the COM PORT of PC that is connected with the serial cable and click the Recovery button.**

3 **Confirmation screen is displayed. If proceeding the recovery of S/W, click the OK button.**

4 **Following procedures is the same as install of S/W.**

■ Operation Check

After the install or the recovery of S/W, confirm the operation of XCONT-2018 board.

Follow the opposite procedure in "Preparation for the XCONT-2018 board" to put the system back to the original state, and then, turn the power ON. Enter the adjustment mode according to the procedure below, and check the software version.

If the version of installed software does not correspond to the displayed version of software in mAs display, try the installation or recovery again.

> NOTE When the XCONT-2018 board is replaced in the installation or recovery of the S/W, the new board must be initialized.
> Reference: "7.2 Initialization of XCONT-2018 Board".
> After the initialization, ensure the XCONT-2018 board operates correctly.

1 **Prepare the followings for operation check.**

1. Disconnect the power plug from the outlet.
2. Turn ON the main breaker NFB.
3. Release the emergency stop switch. (Set to OFF.)

2 **Disconnect the serial cable between J4 connector on the XCONT-2018 board and the serial port on PC.**

3 **Set DIP SW4 on the XCONT-2018 board. Return SW4-2 to OFF.**

4 **Turn ON the system and make sure the system starts normally.**

5 **Turn OFF the power.**

6 **Attach the inside cover and the side cover.**

7 **Turn ON the power.**

8 **Press and hold the user settings button for about 3 seconds to enter the change setting mode.**

9 When the display of is changed to (personal ID setting), press 7 on the numeric keypad or the Chest radiography button, and then the collimator lamp button within 3 seconds.

10 is displayed for about 1 second, and then is displayed in the kV display. In this case, the current version of software is displayed in the mAs display. By pressing the collimator lamp button, "8" indicating a software type is displayed in mAs display.

11 Turn OFF then ON the power to restart the system.

11.9 PC Maintenance Function

11.9.1 Introduction

It is possible to perform maintenance work to show below by connecting PC and the XCONT-2018 board with a serial cable.

- Back up (download) of the data below memorized in the XCONT-2018 board.
- Restoration (upload) of the data which has been back up into the XCONT-2018 board.

The data which can be communicated with a PC are the following data memorizing with EPROM in the XCONT-2018 board.

- APR data
- Tube data (Tube type data, Tube setting data)
- Rating data (Setting of apparatus ratings)
- Technique data (Technique keys data)
- FVR data
- Other setting data (motor control setting data etc.)
- Option setting data (Dose calculation, Dose Area Product Meter etc.)
- Status data

11.9.2 Necessary tools and software

- PC equipped with Windows 7 or later, and with RS-232C serial port. If RS-232C serial port is not equipped, USB/RS232-C conversion cable should be separately prepared.
- Serial cable for RS-232C (9pin female - 9pin female, cross type)
- Serial Communication software.
  It has a binary file Transfer Protocol XMODEM (CRC).
  Terminal program Tera-Term Ver4.95 or later is recommended. Tera-Term is freeware which can be downloaded from the following Internet site.To download Tera Term, use the version 4.95 or higher installer (the installer 4.94 and earlier is vulnerable). If the program is already installed, updating is not required.
  https://ja.osdn.net/projects/ttssh2/releases/

11.9.3 Serial cable connection

> NOTE Set the Auto Power Off function to OFF to prevent the system power from being turned off during operation. When you finish the operation, reset the value to the original.
> Reference: Reference Guide "2.1.6 Auto Power OFF [APO]"

1 **Turn off the power and open the right side cover.**

2 **Check and make a note of the setting of DIPSW2-4 and DIPSW2-7 on the XCONT-2018 board.**

3 **Turn the DIPSW2-4 to ON side and SW2-7 to OFF side.**
Turn the DIPSW3-8 to ON.

> NOTE By turning DIPSW3-8 ON, the system's baud rate is set to 9600. If DIPSW3-8 is OFF, the baud rate will be 38400, which will prevent connection with the PC even if set according to the instruction manual.

4 **Connect the serial cable to the J4 connector of the XCONT-2018 board.**

5 **Connect the serial cable to the serial connector of PC.**

11.9.4 Setup of Tera Term

1 **Turn on the PC and start Terminal program Tera Term after Starting Windows.**

2 **The setting of a new connection is displayed. Select the COM port of PC connecting to the serial cable, and click the OK button.**

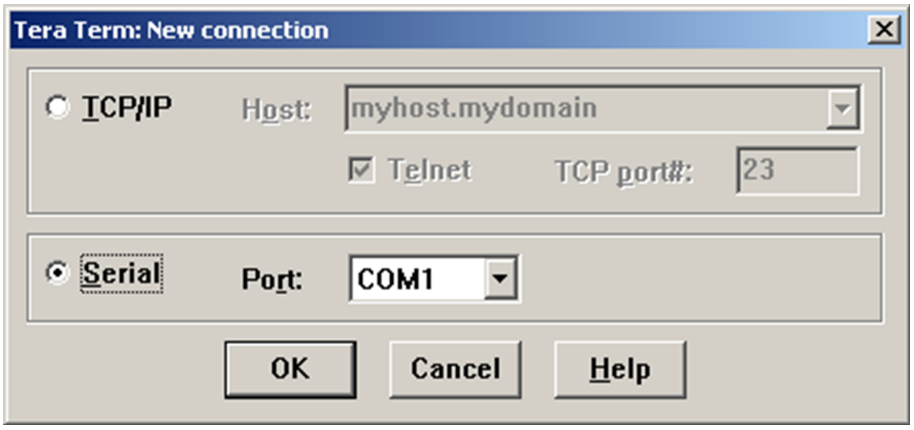


Fig. 11-41

3 **The window of Tera Term is displayed.**

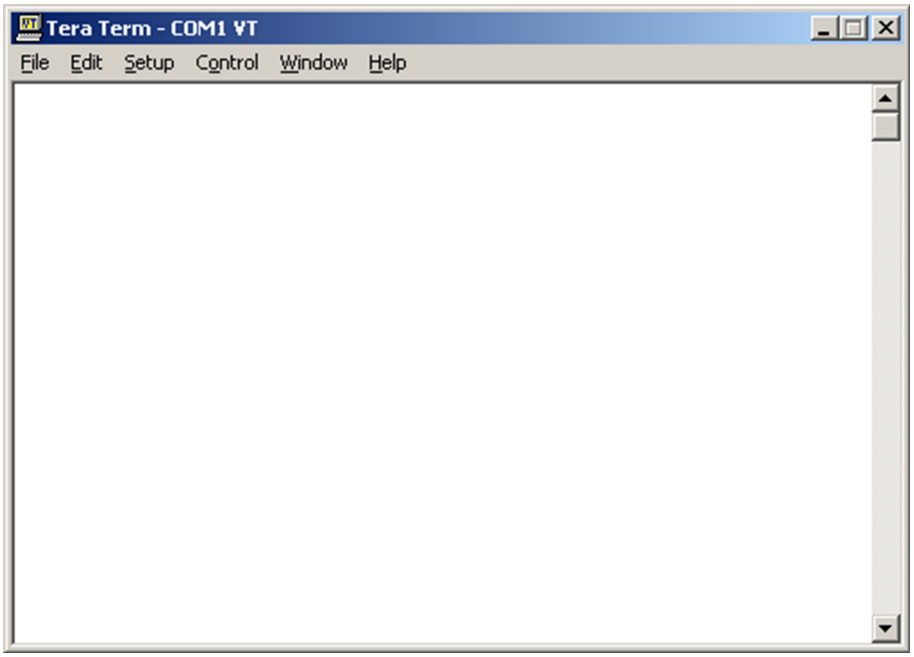


Fig. 11-42

4 **Set terminal setting. Select "Setup - Terminal".**

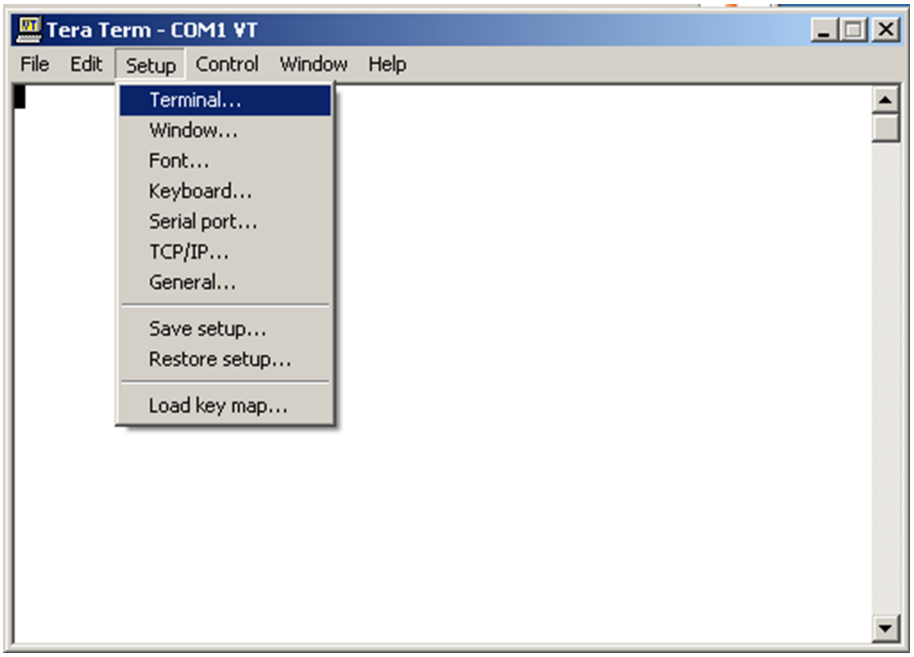


Fig. 11-43

5 **Set CR+LF on New-line - Transmit. Check the setting and click OK.**

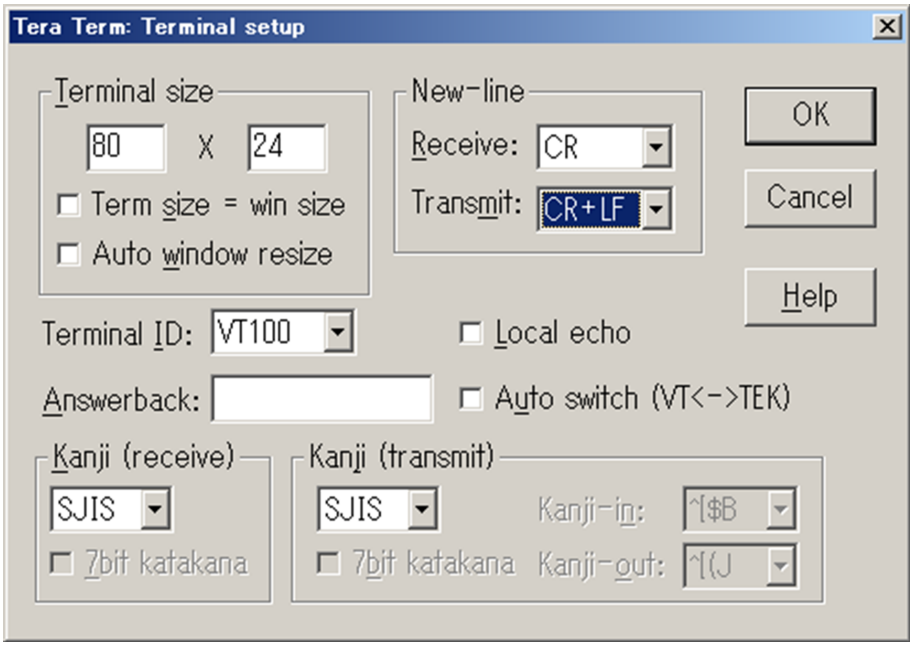


Fig. 11-44

6 Set serial port setting. Select "Setup - Serial port".

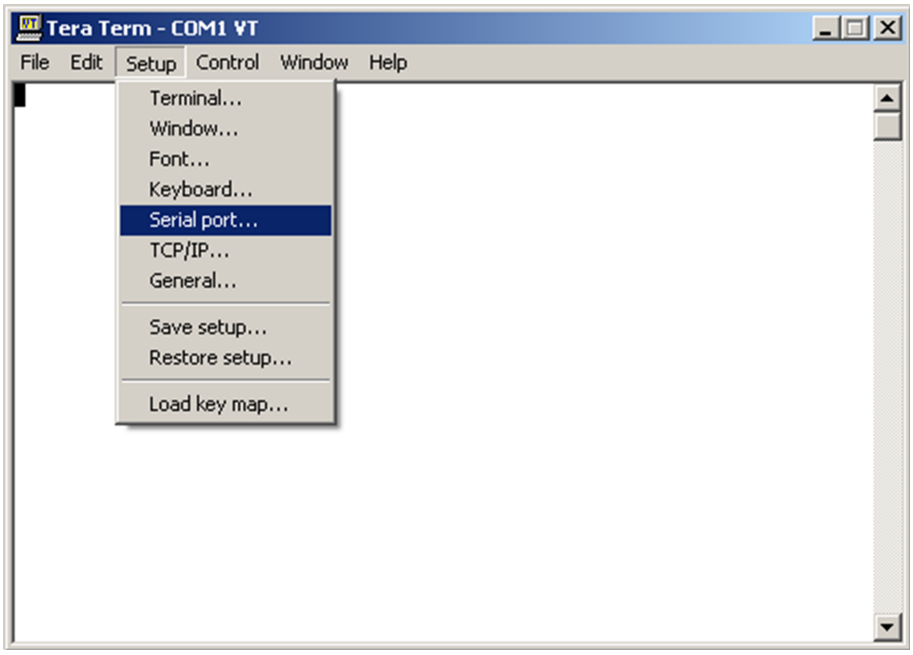


Fig. 11-45

7 Set serial port set up as shown below. Check the setting and click OK.

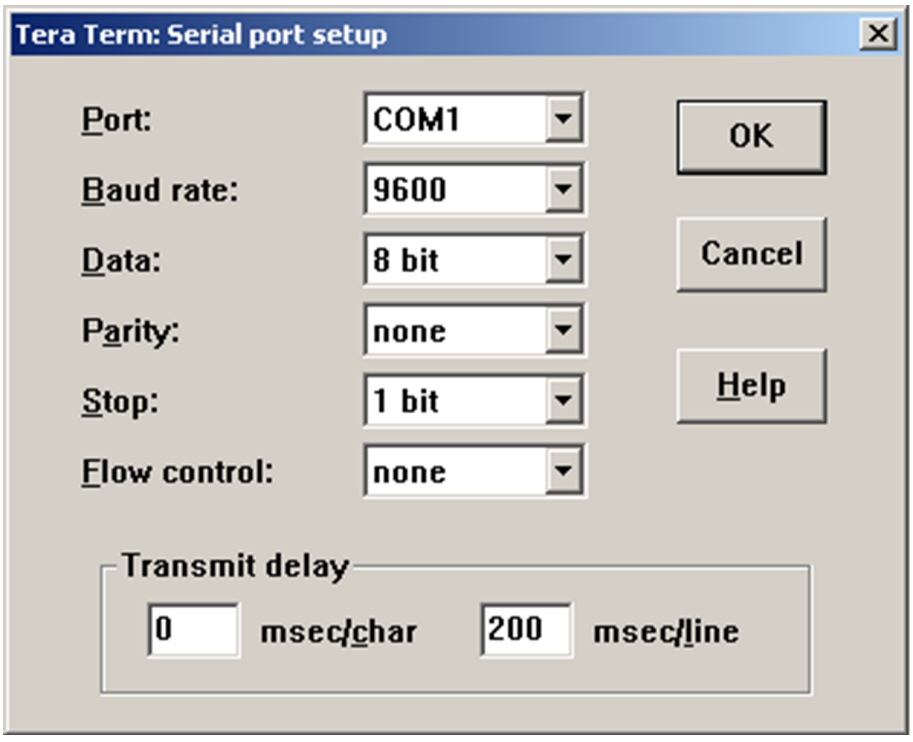


Fig. 11-46

8 **Setup is finished.**

NOTE Do not perform any PC maintenance function on the system during the communication between PC and the XCONT-2018 board.
Failure to follow this instruction may cause a data communication error.

11.9.5 Backup of data (download)

Transfer the memory of the XCONT-2018 board to the computer, and create a backup log file and save it.

1 **Connection of serial cable**

Connect the serial cable before the system turned on.

Reference "11.9.3 Serial cable connection"

2 **Start of Tera Term**

Start Tera Term after PC turned on, and Windows starts.

Reference "11.9.4 Setup of Tera Term"

■ Beginning of download processing

1 Turn on the system.

2 When Tera Term is started, the following Tera Term screen is displayed.

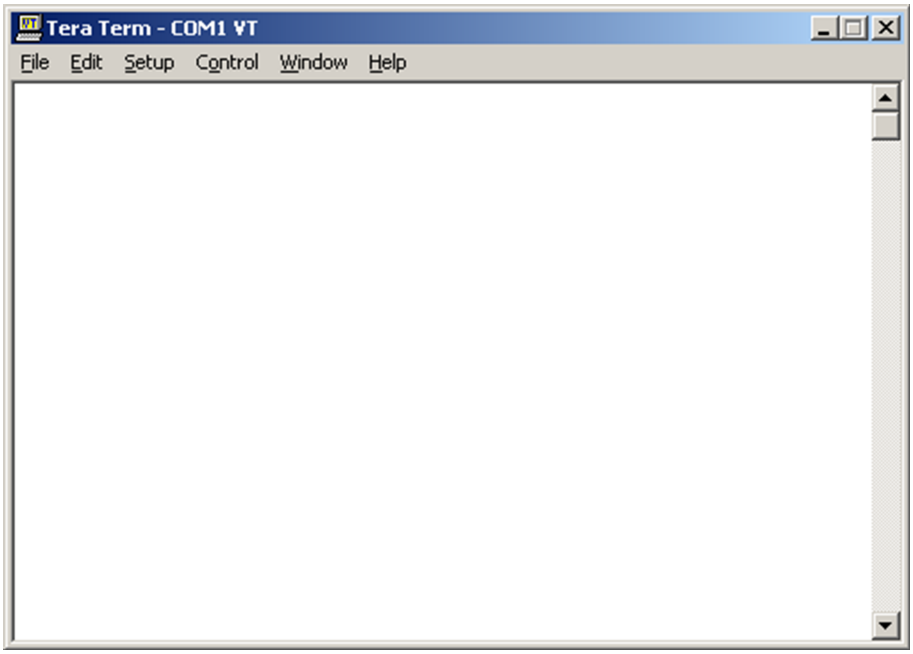


Fig. 11-47

3 **Press the Enter (↓) in the window of Tera Term.**

MAIN> is displayed.

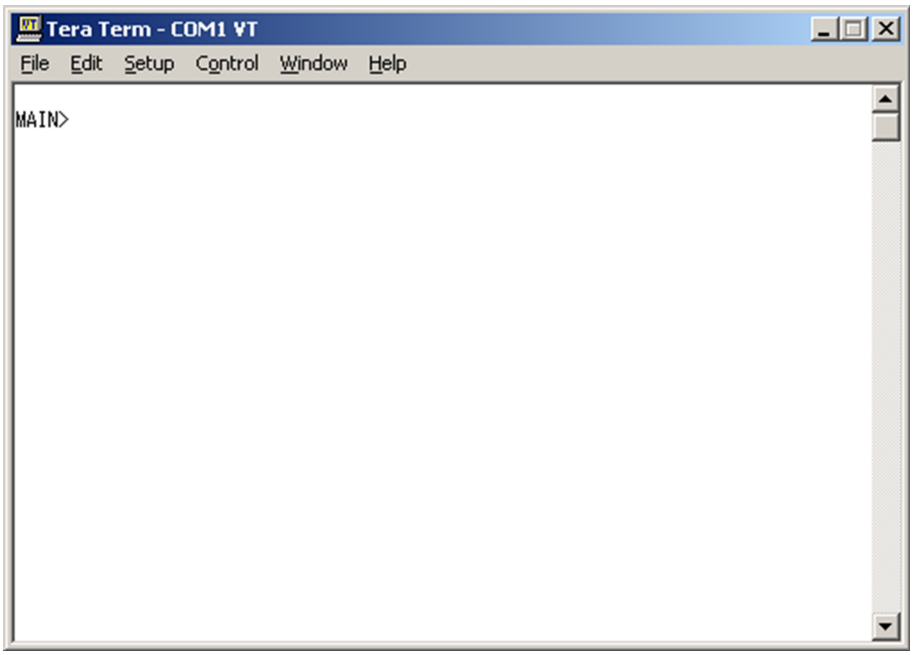


Fig. 11-48

NOTE If characters are garbled or not shown on the Tera Term display, check the network status or settings as follows:

- Make sure a serial cable connection between the XCONT-2018 board and PC is normal.
- Make sure the network configurations for Tera Term is correct.

4 **Input "maintenance" and press Enter (↓).**

- Maintenance Mode -

* is displayed.

Therefore, it comes to be able to exchange data between PC and the XCONT-2018 board.

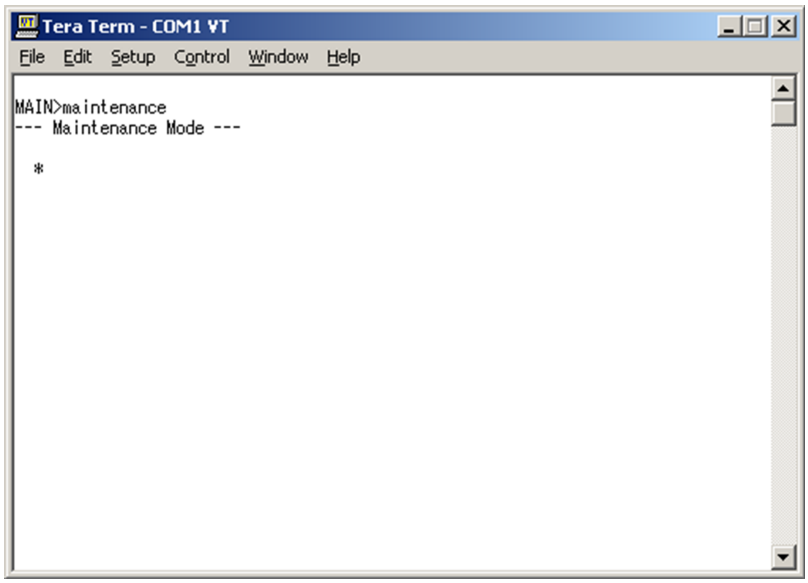


Fig. 11-49

5 **Input "help" and press the Enter (↓).**

The list of the use commands is displayed.
The list of the use command is shown as follows.
Input command for necessary data to need to transferred.
---- <Maintenance Mode Help> ----

Command	Content of command
all[r/w]	All EPROM stored data.
heaprtec[r/w]	APR data
heratio[r/w]	Ratio data
hetubedt[r/w]	Tube data
hetechinit[r/w]	Technique data
hefvrkvdata[r/w]	FVR data
hec[r/w]	Other setting data
hdose1[r/w]	Option setting data1
hdose2[r/w]	Option setting data2
hide1[r/w]	Personal ID setting data1
hide2[r/w]	Personal ID setting data2
hrep[r/w]	Usage information data
sstatus[r/w]	Status data1
sctrl[r/w]	Status data2
help	The list of commands is displayed.
exit	Return to the main menu.

all180[w]	Upload all EEPROM storage data of the device that does not support serial radiography to the EEPROM storage data of the device that does support serial radiography.
heaprtec180[w]	Upload the APR data of a device that does not support serial radiography to the APR data of a device that does support serial radiography.

NOTE APR data for serial radiography does not correspond to APR data for non-serial radiography.

6 **Input the specific command for necessary data to be transferred and a blank and "r". And press Enter (↓).**

(Example) In case of APR data, input "heaprtec r" and (↓)

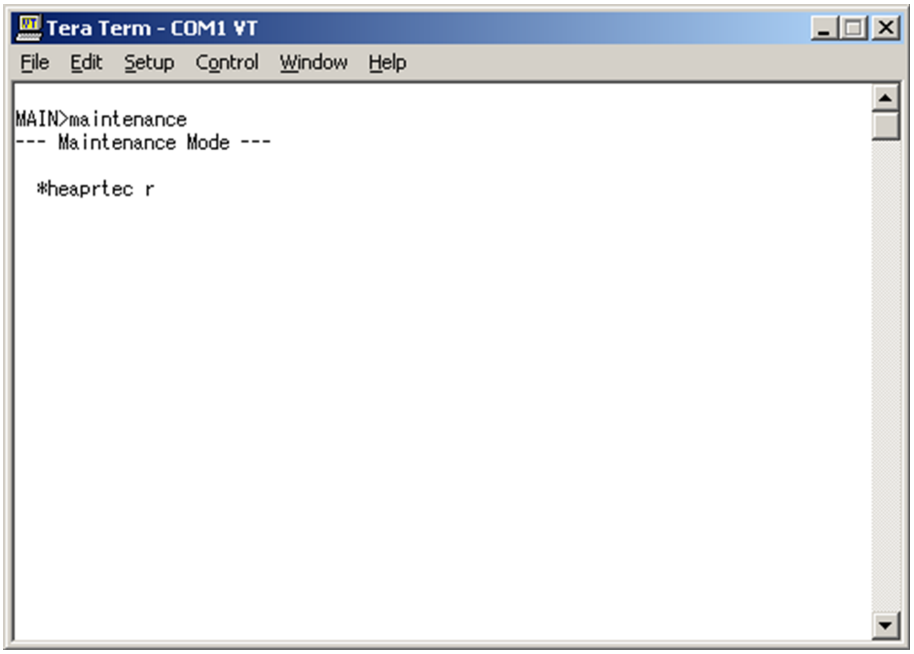


Fig. 11-50

7 **Confirmation screen is displayed as follows.**

Data Download to Disk.
Prepare log file.
Hit [Enter] key to start Download.

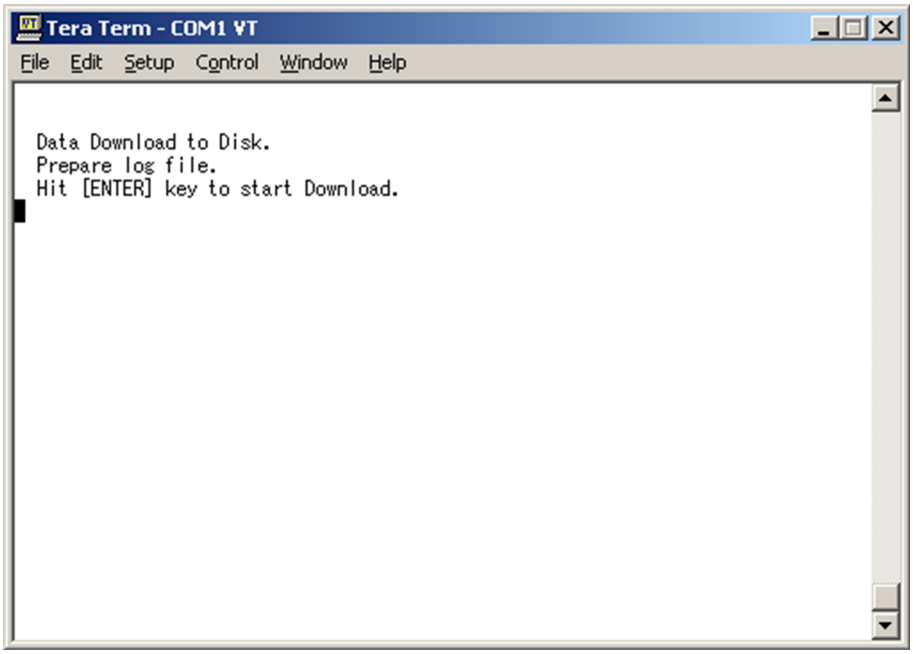


Fig. 11-51

8 **A log file for storing download data (Log file will be saved in PC.) is specified. Select "File - Log" from the menu bar.**

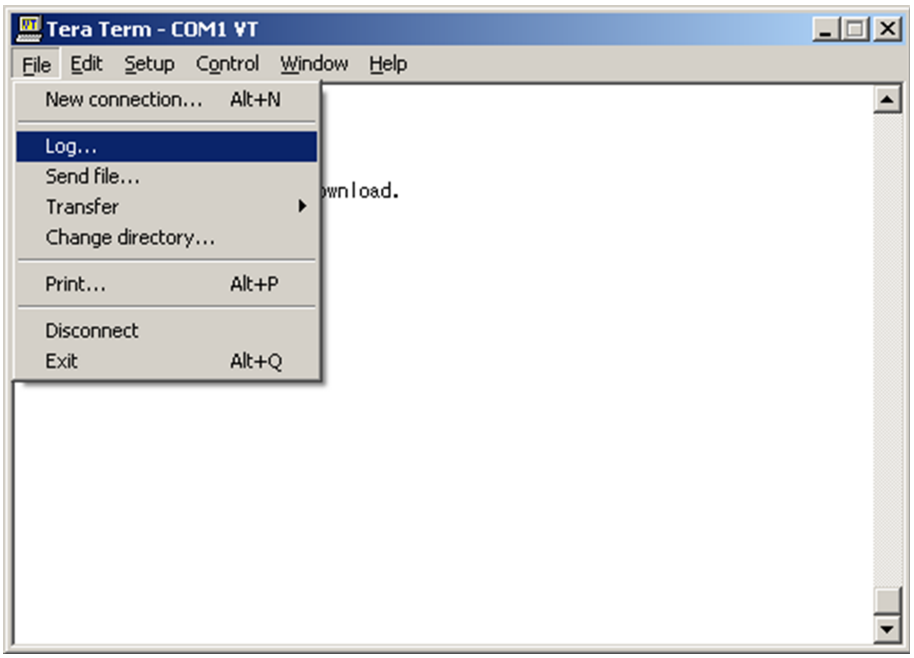


Fig. 11-52

9 **In next, a log file that stores download data is specified.**

Determine the file name of the log file that stores download data, click Open. The log file is opened.

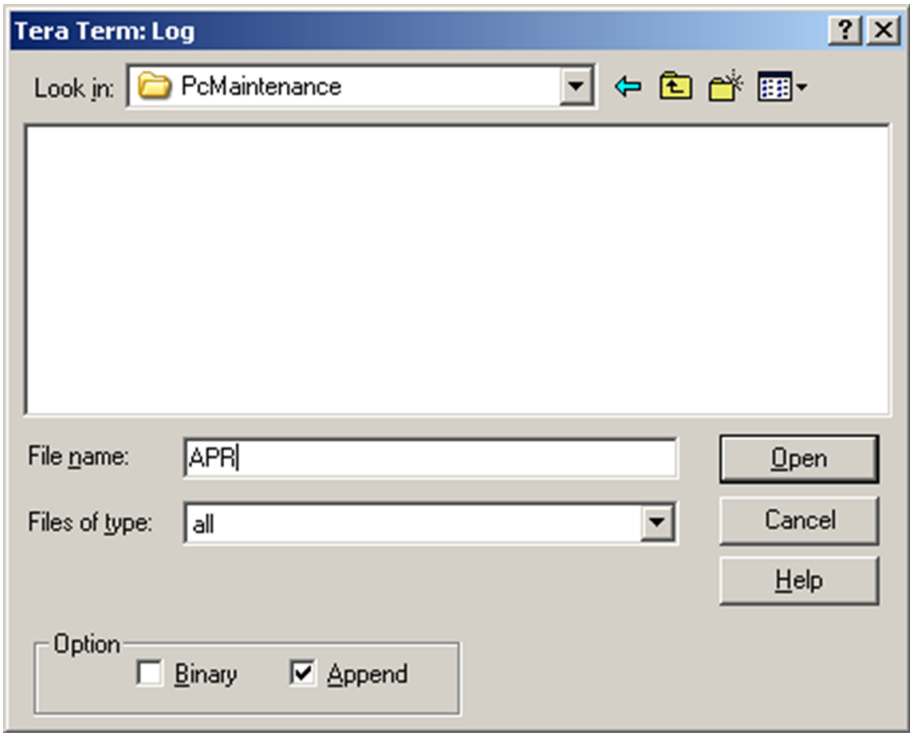


Fig. 11-53

10 **When Enter (↓) is pressed, download starts.**

The data that currently is processed for download appear in the screen. When download is completed, the following messages are displayed.
Data Download completed.
Close Log File.
Hit [Enter] key to Return menu.

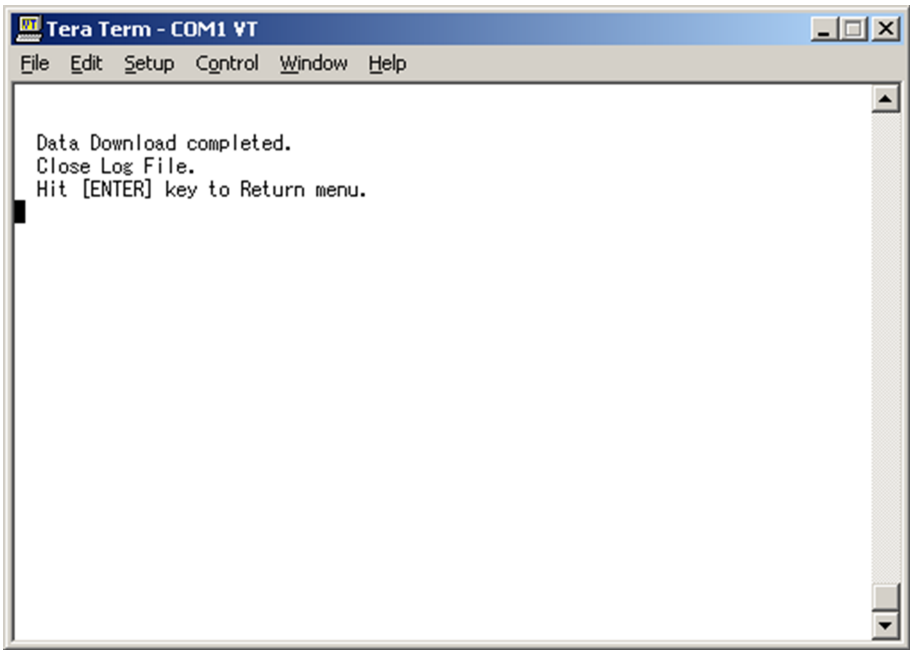


Fig. 11-54

11 **The window of Tera Term:Log is displayed. Click Close. The log file that stores the downloaded data is closed.**

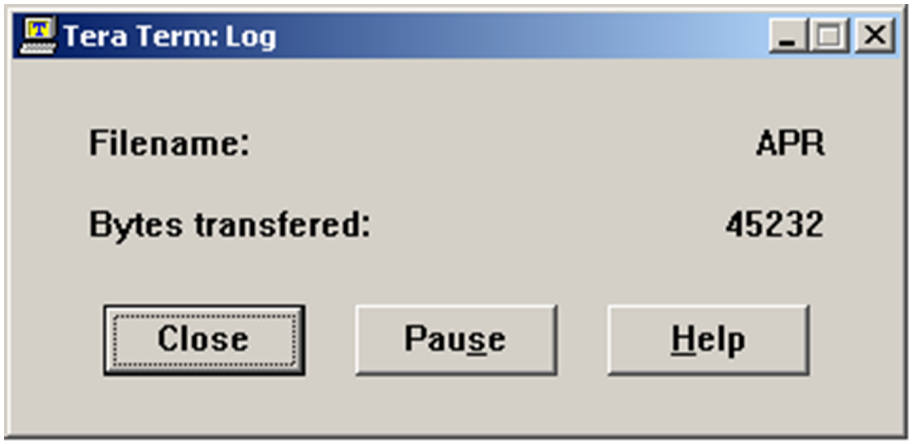


Fig. 11-55

NOTE Be sure to close the "Log" function after completing the download process. This will prevent downloading unnecessary data.

12 **If Enter (↓) is pressed, it becomes the waiting mode for inputting command. It is displayed as follows.**

Data Download/Upload
1. Download Data to Disk
2. Upload Data form Disk
3. Verify Data
E. Exit Data Download/Upload

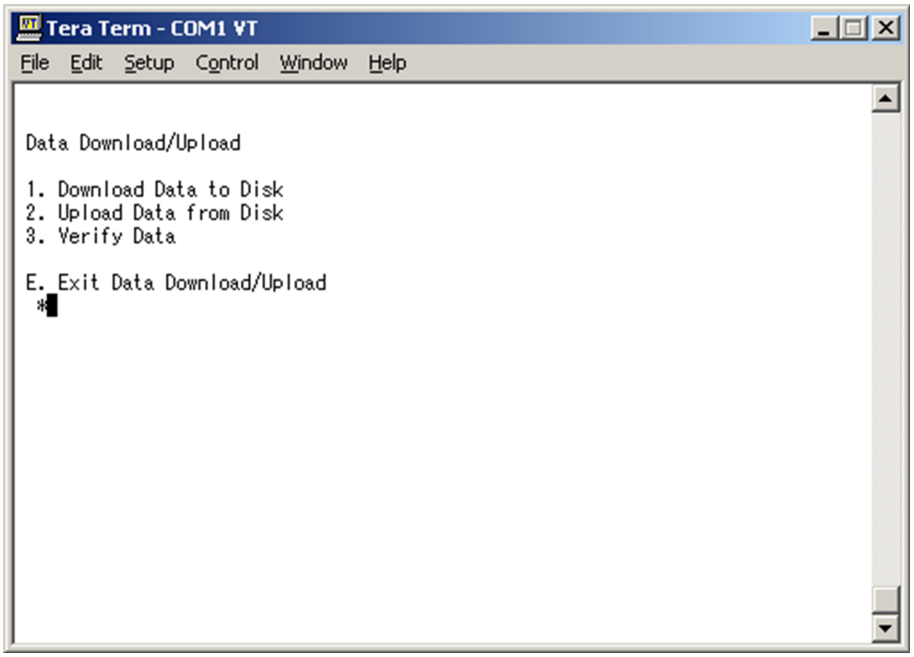


Fig. 11-56

Execute the same procedure from item 6 when continuing download for other data.

13 **Input "exit" and press the Enter (↓) when finishing download. "MAIN >" is displayed.**

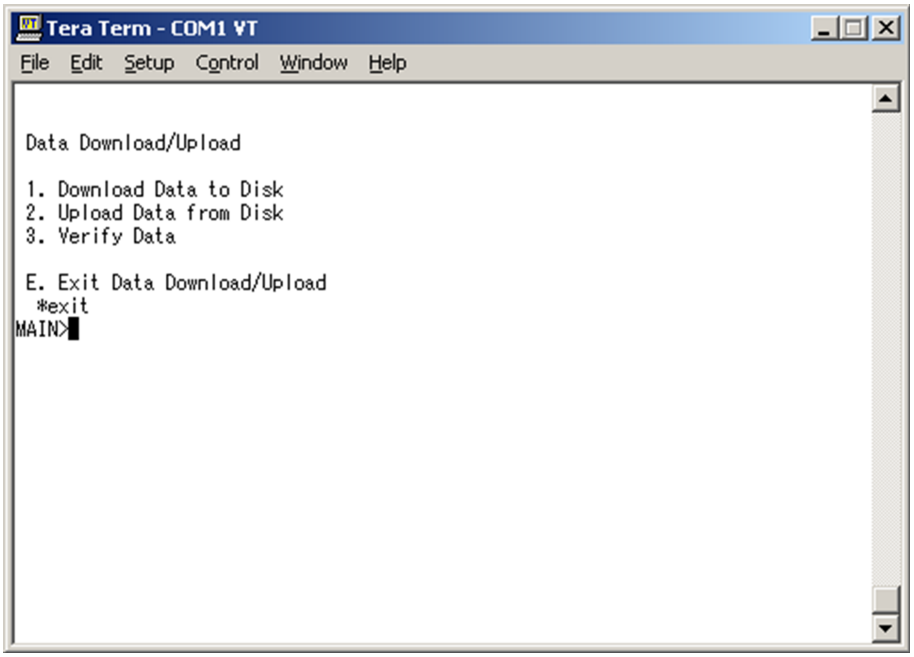


Fig. 11-57

14 **Exit Tera Term. Select "File, Exit" in menu bar of Tera Term.**

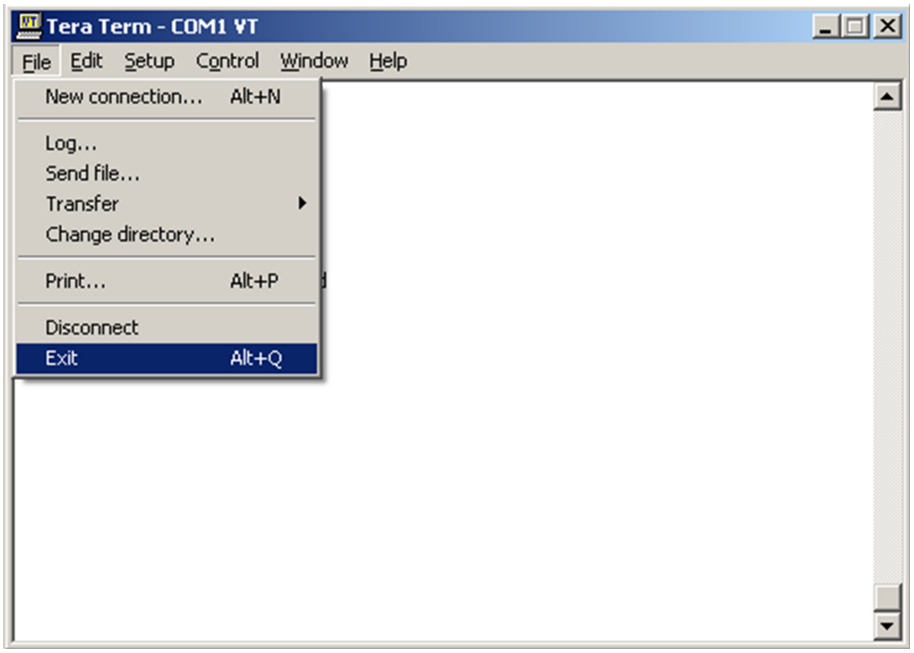


Fig. 11-58

15 **Tera Term is ended.**

16 **Turn off the PC and the system. And disconnect the serial cable.**

17 **Set back SW2-4 and SW2-7 on the XCONT-2018 board to the original state. Turn the DIPSW3-8 to OFF.**

11.9.6 Restoration of data (upload)

The RESTORATION of DATA includes the following process.
Uploading the file that was saved in PC to the XCONT-2018 board.
Restoring the data in the XCONT-2018 board.

1 **Connection of serial cable**

Connect the serial cable before turning on the system.

Reference "11.9.3 Serial cable connection"

2 **Start of Tera Term**

Start Tera Term refer to after PC turned on, and Windows starts.

Reference "11.9.4 Setup of Tera Term"

■ Beginning of upload processing

1 Turn on the system.

2 When Tera Term is started, the following Tera Term screen is displayed.

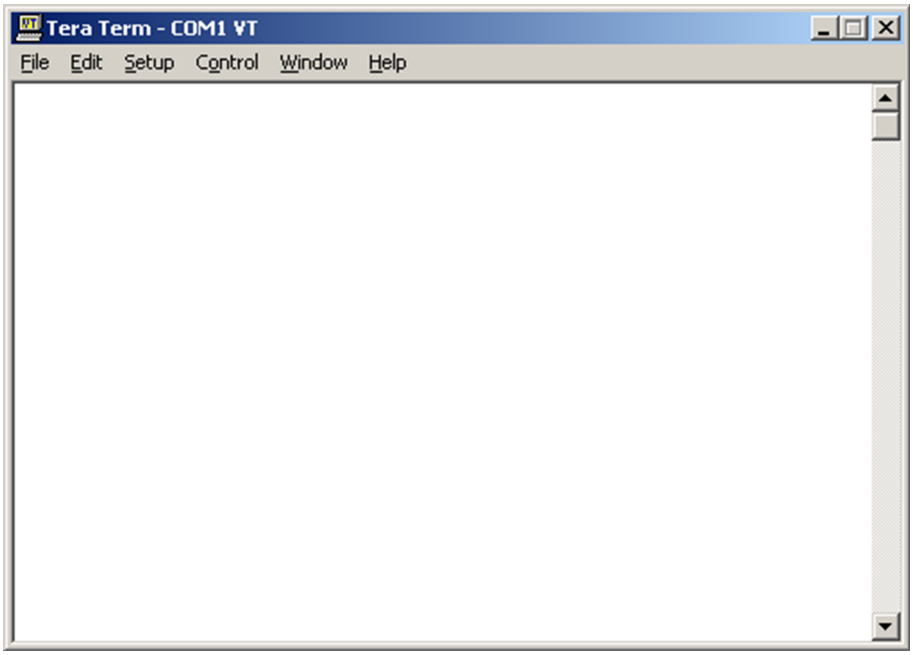


Fig. 11-59

3 **Press the Enter (↓) in the window of Tera Term.**

"MAIN>" is displayed.

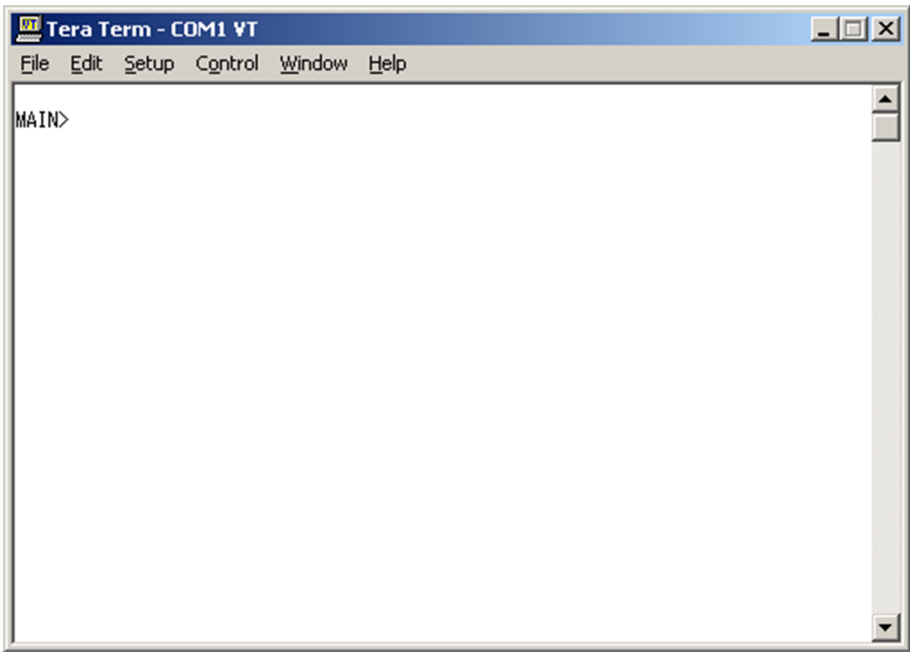


Fig. 11-60

NOTE If characters are garbled or not shown on the Tera Term window, check the network status or settings as follows:

- Make sure a serial cable connection between the XCONT-2018 board and PC is normal.
- Make sure the network configurations for Tera-Term is correct.

4 Input "maintenance" and press the Enter (↓).

- Maintenance Mode -
* is displayed.
Therefore, it comes to be able to exchange data between PC and the XCONT-2018 board.

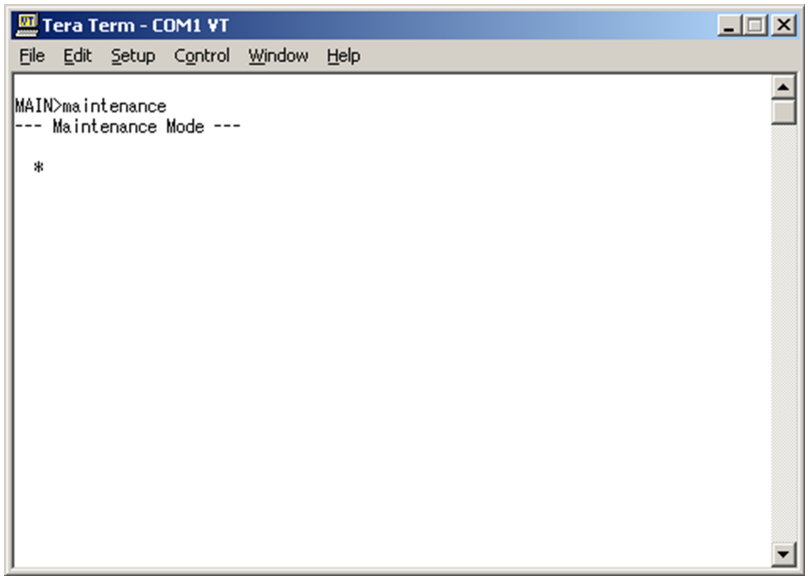


Fig. 11-61

5 Input "help" and press the Enter (↓).

The list of the use command is displayed.
The list of the use command is shown as follows.
Input command for necessary data to need to transfer.
---- <Maintenance Mode Help> ----

Command	Content of command
all[r/w]	All EPROM stored data.
heaprtec[r/w]	APR data
heratio[r/w]	Ratio data
hetubedt[r/w]	Tube data
hetechinit[r/w]	Technique data
hefvrkvdata[r/w]	FVR data
hec[r/w]	Other setting data
hdose1[r/w]	Option setting data1
hdose2[r/w]	Option setting data2
hide1[r/w]	Personal ID setting data1
hide2[r/w]	Personal ID setting data2
hrep[r/w]	Usage information data
sstatus[r/w]	Status data1
sctrl[r/w]	Status data2
help	The list of commands is displayed.
exit	Return to the main menu.

6 **Input the specific command for necessary data to be transferred and a blank and "w". And press Enter (↓).**

(Example) In case of APR data, input "heaprtec w" and (↓).

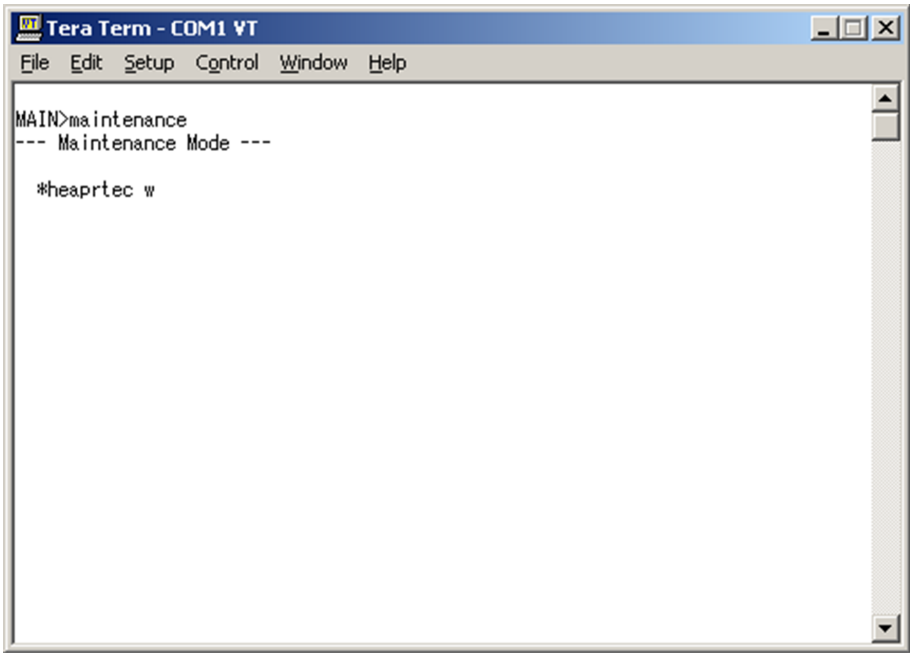


Fig. 11-62

7 **Confirmation screen is displayed as follows,**

Data Upload from Disk.
Send log file to Serial Port.
Type "END" to exit.

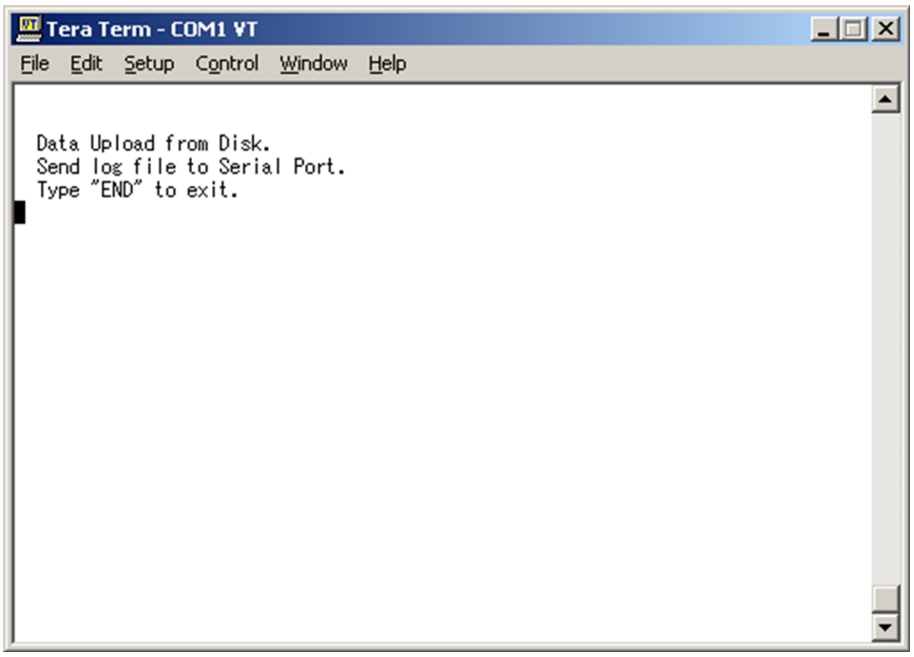


Fig. 11-63

8 **The log file for upload log is specified. Select "File ?Send file" from the menu bar.**

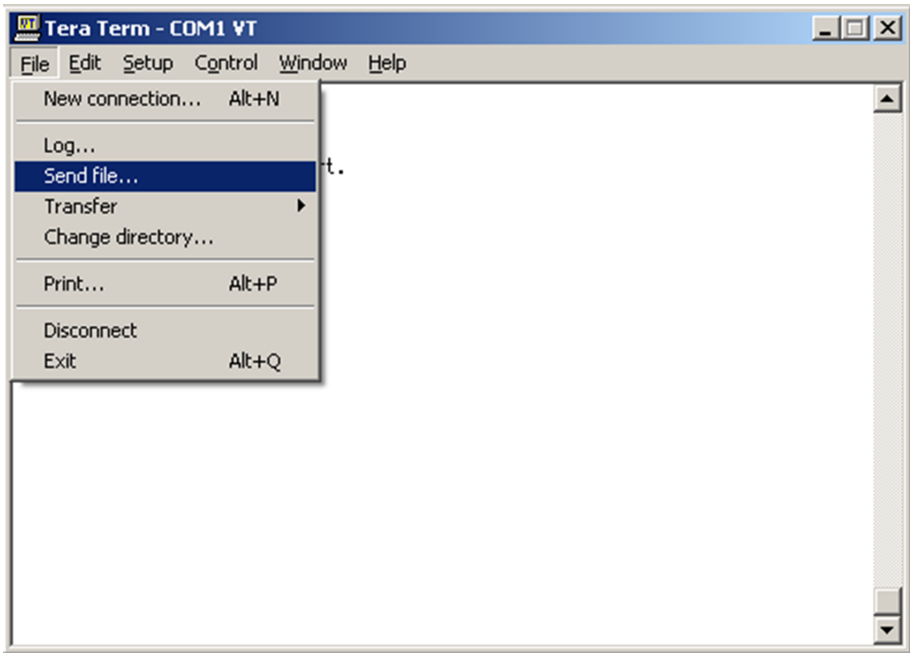


Fig. 11-64

9 **Select the log file for upload, and click "Open". Upload is started. It takes several minutes for upload. Do not operate anything of tera term until the screen for completion of upload is displayed.**

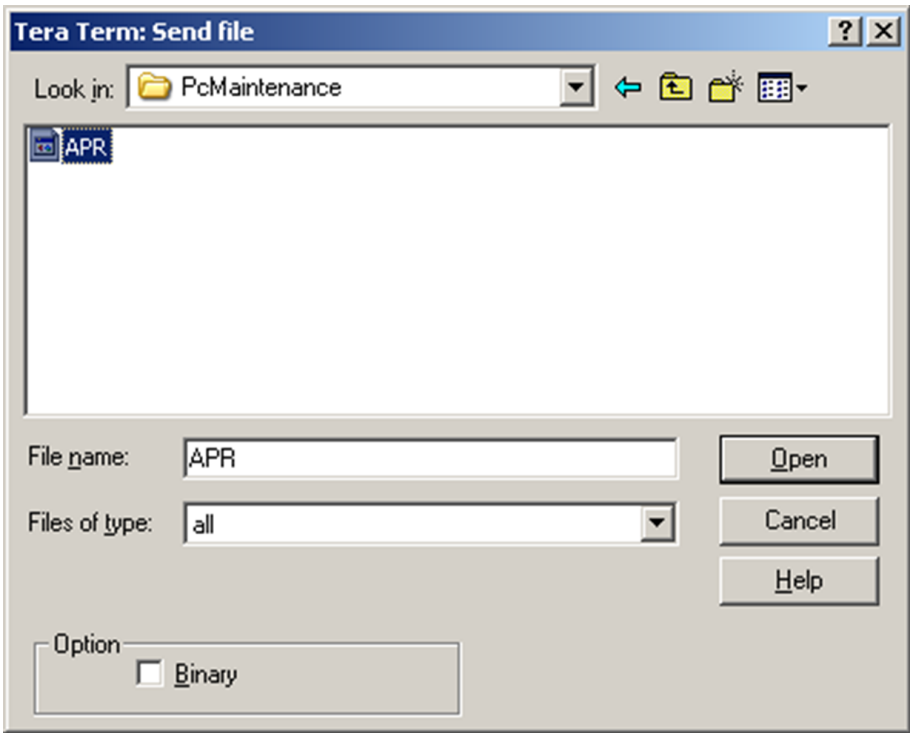


Fig. 11-65

10 When upload is completed, the screen is displayed as follows,

Data Upload completed.
Hit [Enter] key to Return menu.

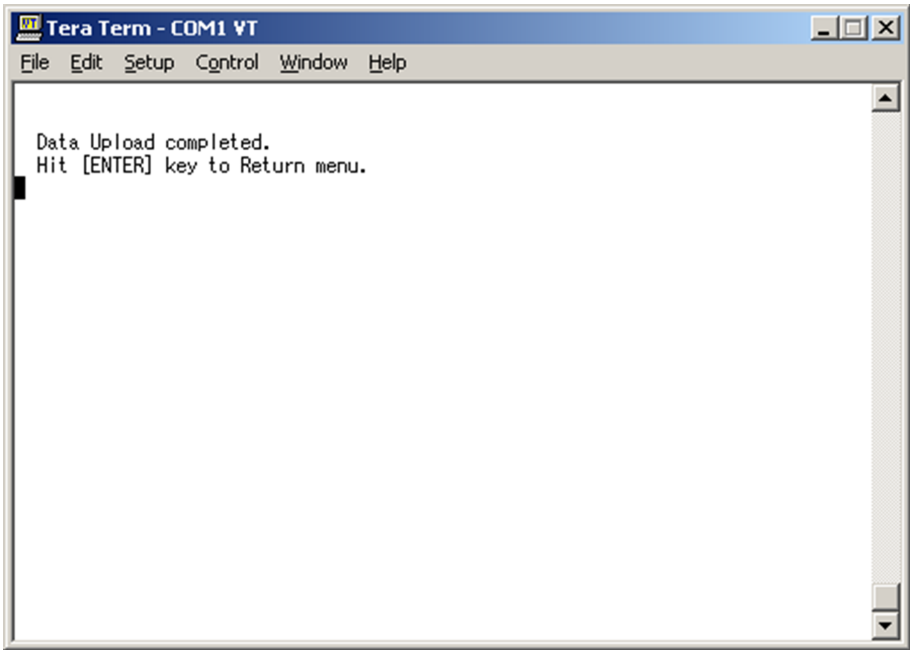


Fig. 11-66

NOTE If data cannot be uploaded, an error message will be displayed. In this case, set Line delay to more than 200 milliseconds in ASCII Setup dialog shown below, and then, try it again.

Reference "11.9.4 Setup of Tera Term"

11 If Enter (↓) is pressed it becomes the waiting mode for inputting command. It is displayed as follows,

Data Download/Upload
1. Download Data to Disk
2. Upload Data form Disk
3. Verify Data
E. Exit Data Download/Upload

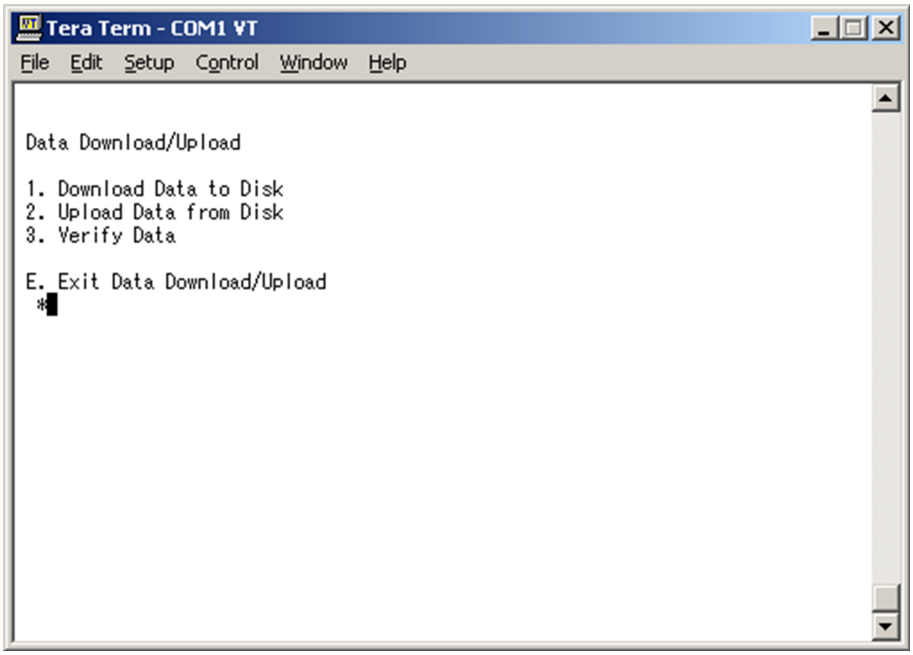


Fig. 11-67

Execute the same procedure from item 6 when continuing upload for other data.

12 **Input "exit" and press the Enter (↓) when finishing upload. "MAIN >" is displayed.**

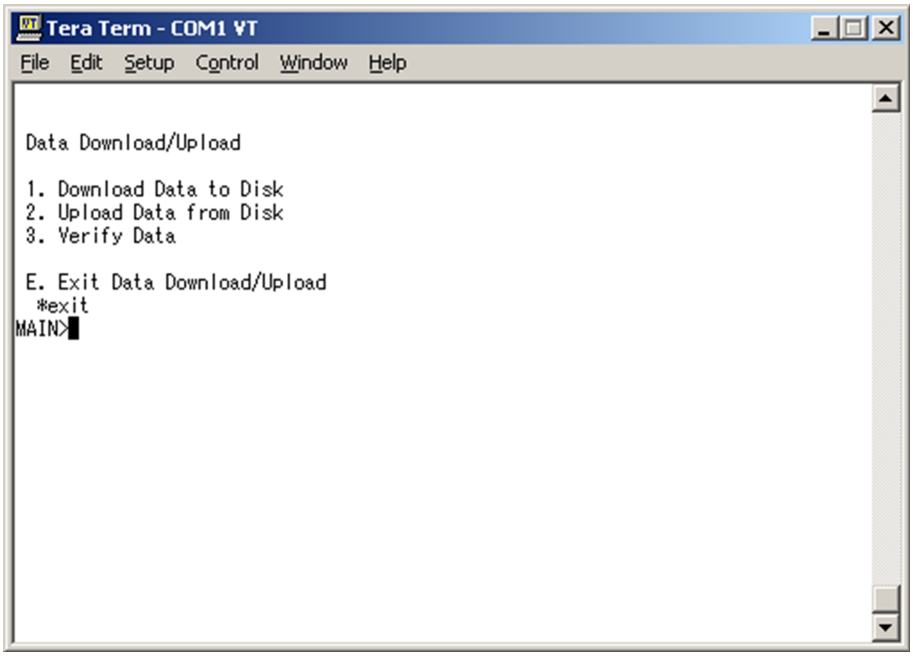


Fig. 11-68

13 **Exit Tera Term. Select "File, Exit" in menu bar of Tera Term.**

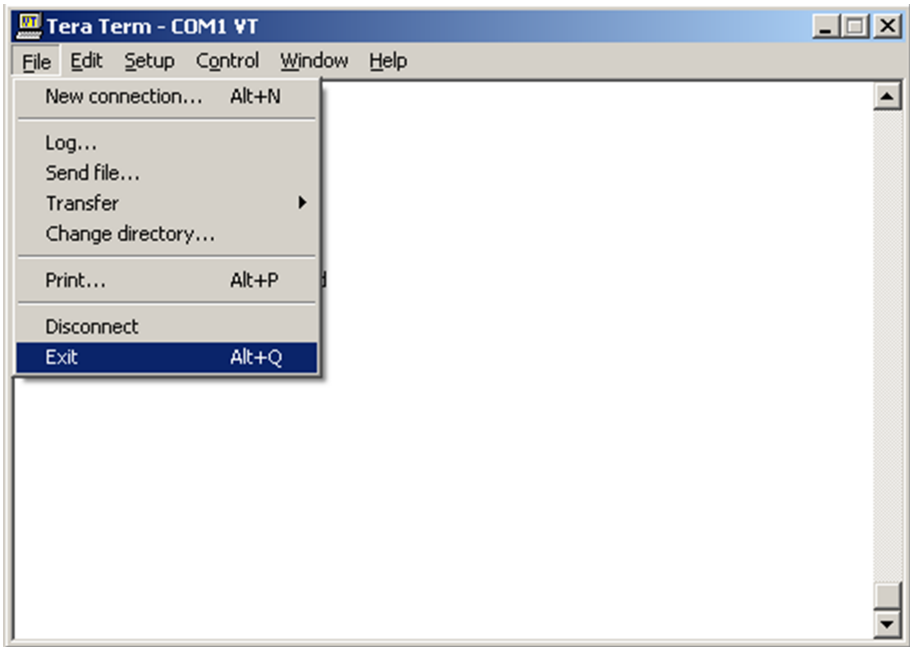


Fig. 11-69

14 **Tera Term is ended.**

15 **Turn off the PC and the system, and disconnect the serial cable.**

16 **Set back SW2-4 and SW2-7 on the XCONT-2018 board to the original state. Turn the DIPSW3-8 to OFF.**

11.10 Connecting the XCONT-2018 board to the LAN

The XCONT-2018 board can be adjusted using a web browser by connecting it to a PC with a LAN cable.
The following IP address setting is required for LAN communication.

> NOTE Use Internet Explorer or Internet Explorer mode of Microsoft Edge for web browser.

1 **On the Windows Start menu, click "Windows System" and then click "Control Panel" to start.**

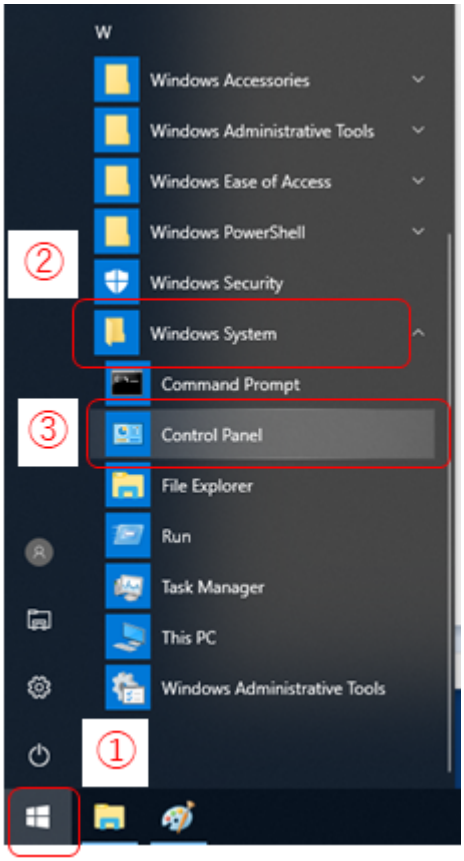


Fig. 11-70

2 **Click "View network status and tasks".**

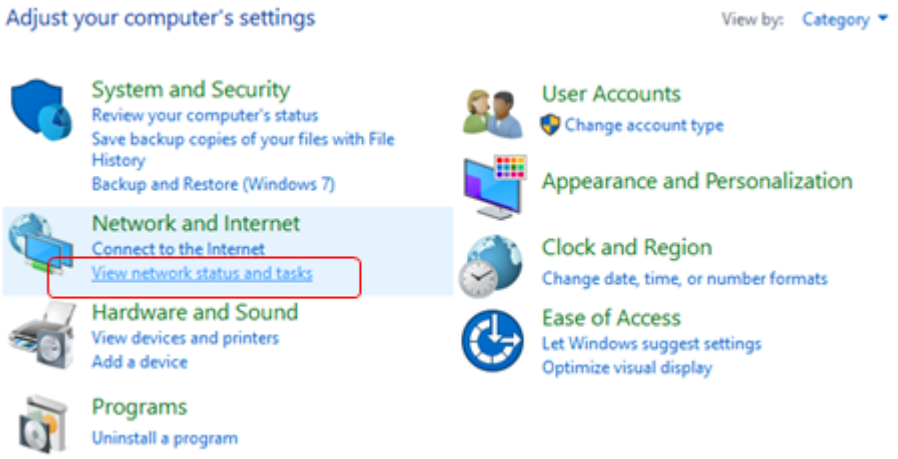


Fig. 11-71

3 **Click "Change adapter settings".**

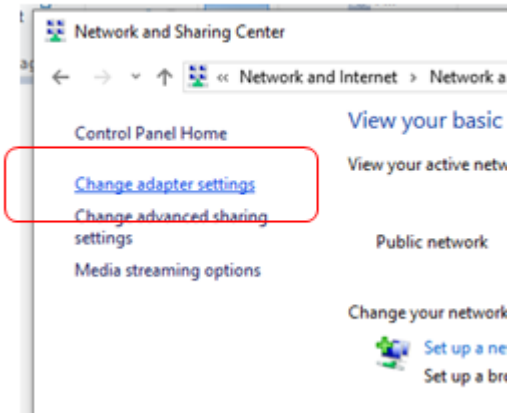


Fig. 11-72

4 **Right-click the network that you are currently using, and then click Properties.**

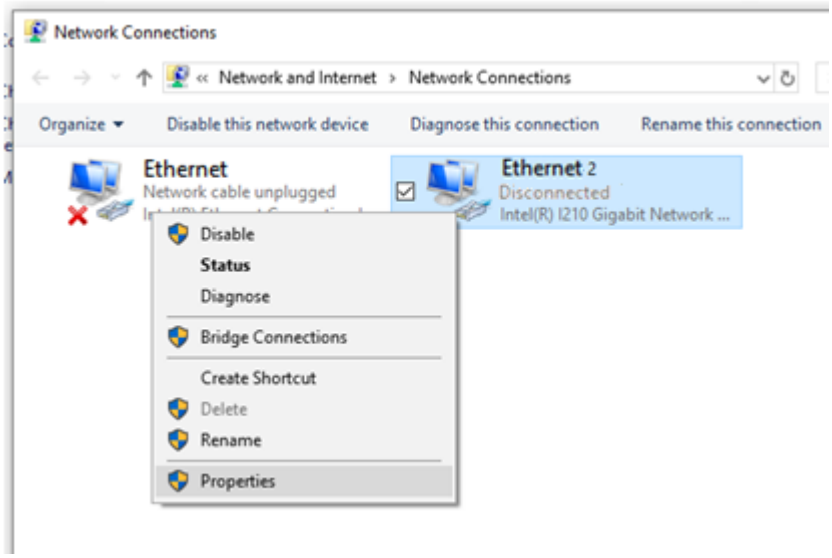


Fig. 11-73

5 **Select "Internet Protocol version 4 (TCP/IPv4)" and click "Properties".**

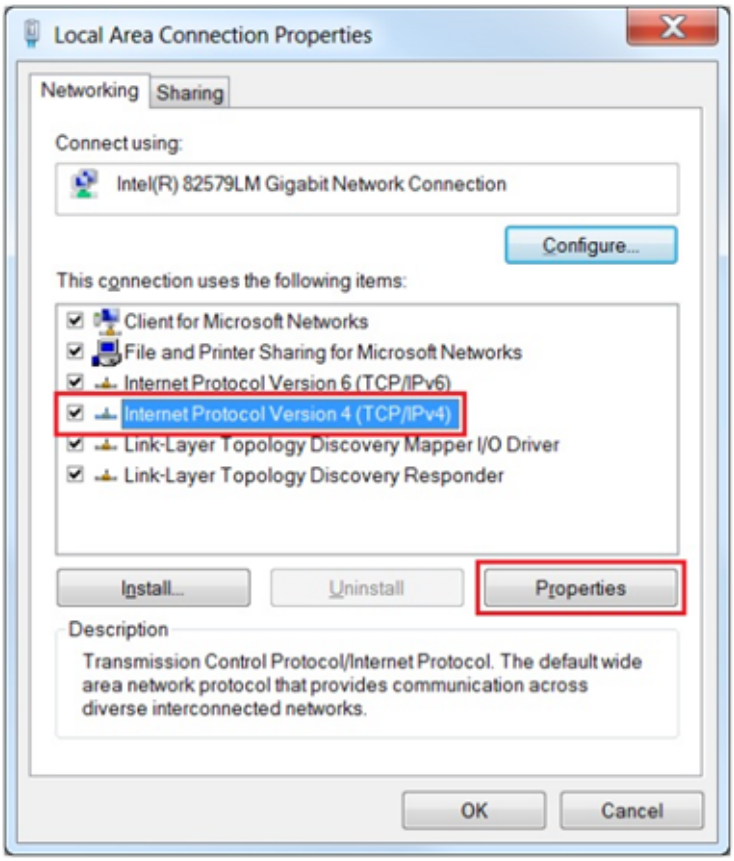


Fig. 11-74

6 Set the IP address as shonw below:.

NOTE Make sure you remember the original IP address of the PC.

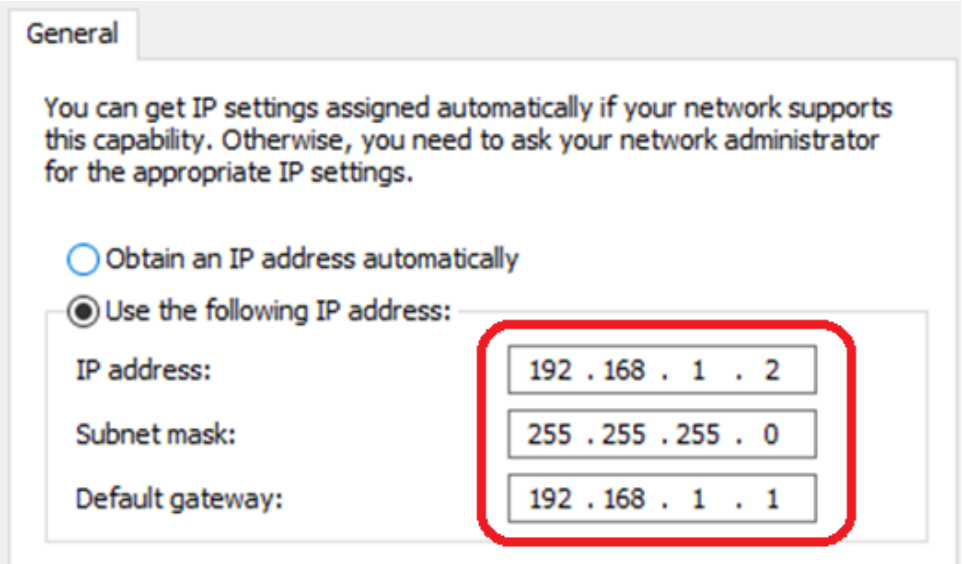


Fig. 11-75

7 Close the window. This completes the IP address setup.

12 [Appendix] Exposure Conditions

This chaper describes conditions for each exposure.

12.1 For 32kW type (MobileDaRt Evolution / MobileArt Evolution high-power type)

12.1.1 Large Focus (at normal condition)

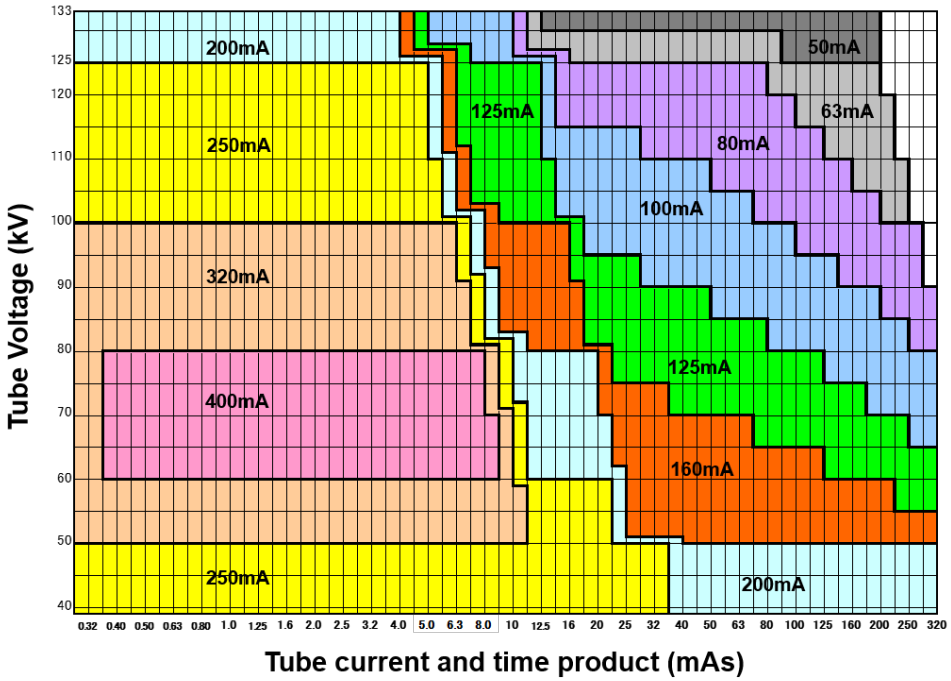


Fig. 12-1

Table Example: Conditions that can be set up at 90kV

- 90kV, a setup of a maximum of 320 mAs is possible.
- Since the tube current at 90kV and 100mAs is 100mA, time is 1.0sec.
- A setup of 320mA of tube current is possible to take an exposure with 0.32-7.1mAs.

12.1.2 Small Focus (at normal condition)

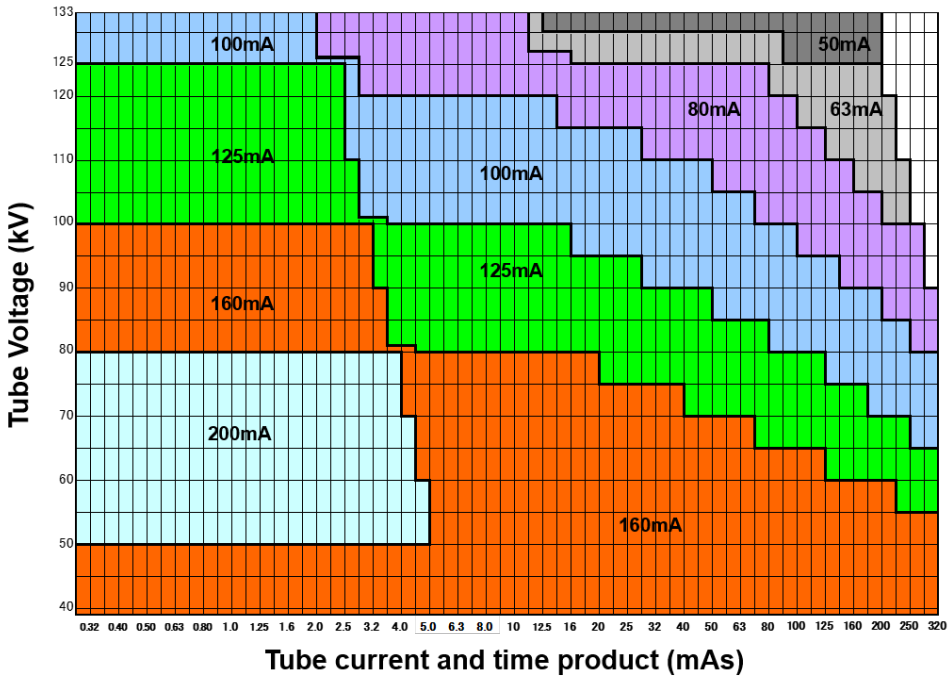


Fig. 12-2

Table Example: Conditions that can be set up at 90kV

- At 90kV, a setup of a maximum of 320 mAs is possible.
- Since the tube current at 90kV and 100mAs is 100mA, time is 1.0sec.
- A setup of 160mA of tube current is possible to take an exposure with 0.32-3.6mAs.

12.1.3 Large focus (at low level battery)

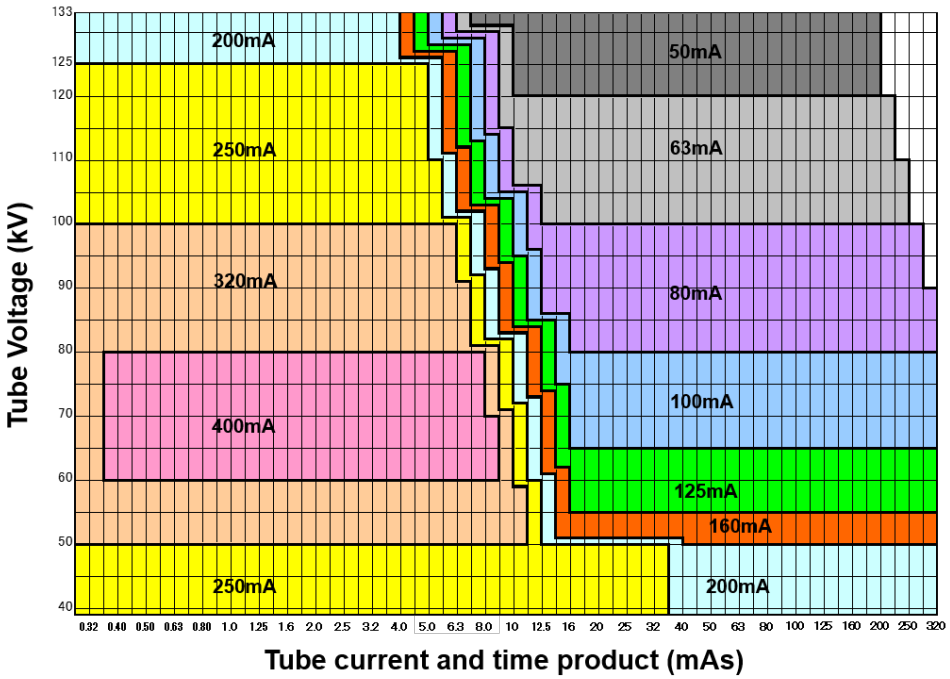


Fig. 12-3

Table Example: Conditions that can be set up at 90kV.

- At 90kV, a setup of a maximum of 320 mAs is possible.
- Since the tube current at 90kV and 100mAs is 80mA, time is 1.25sec.
- A setup of 320mA of tube current is possible to take an exposure with 0.32-7.1mAs.

12.1.4 Small Focus (at low level battery)

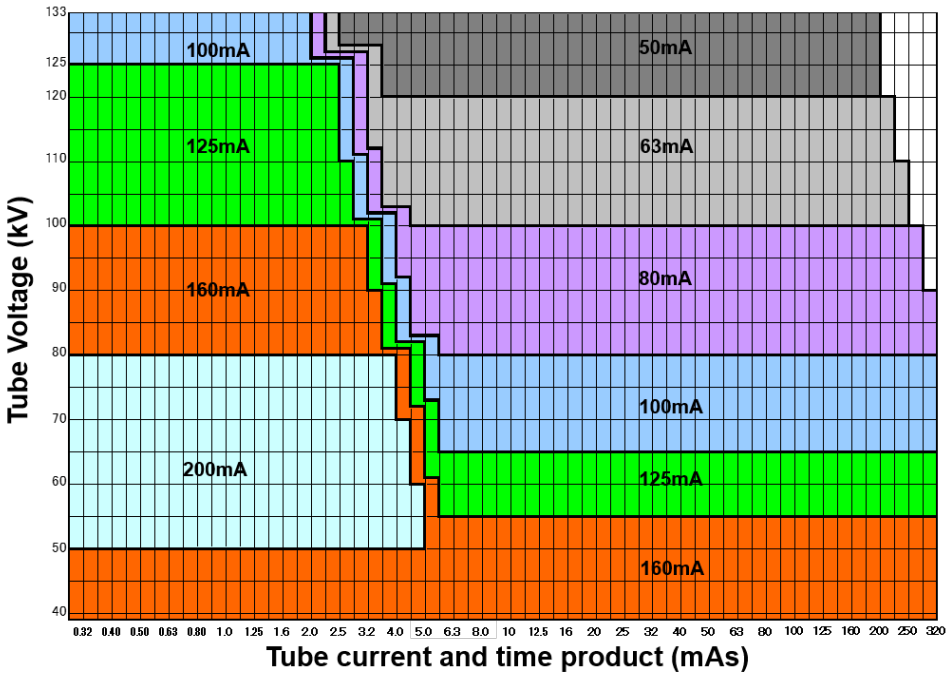


Fig. 12-4

Table Example: Conditions that can be set up at 90kV

- At 90kV, a setup of a maximum of 320 mAs is possible.
- Since the tube current at 90kV and 80mAs is 80mA, time is 1.0sec.
- A setup of 160mA of tube current is possible to take an exposure with 0.32-3.6mAs.

12.2 For 12.5kW type (MobileArt Evolution standard type)

12.2.1 At normal condition

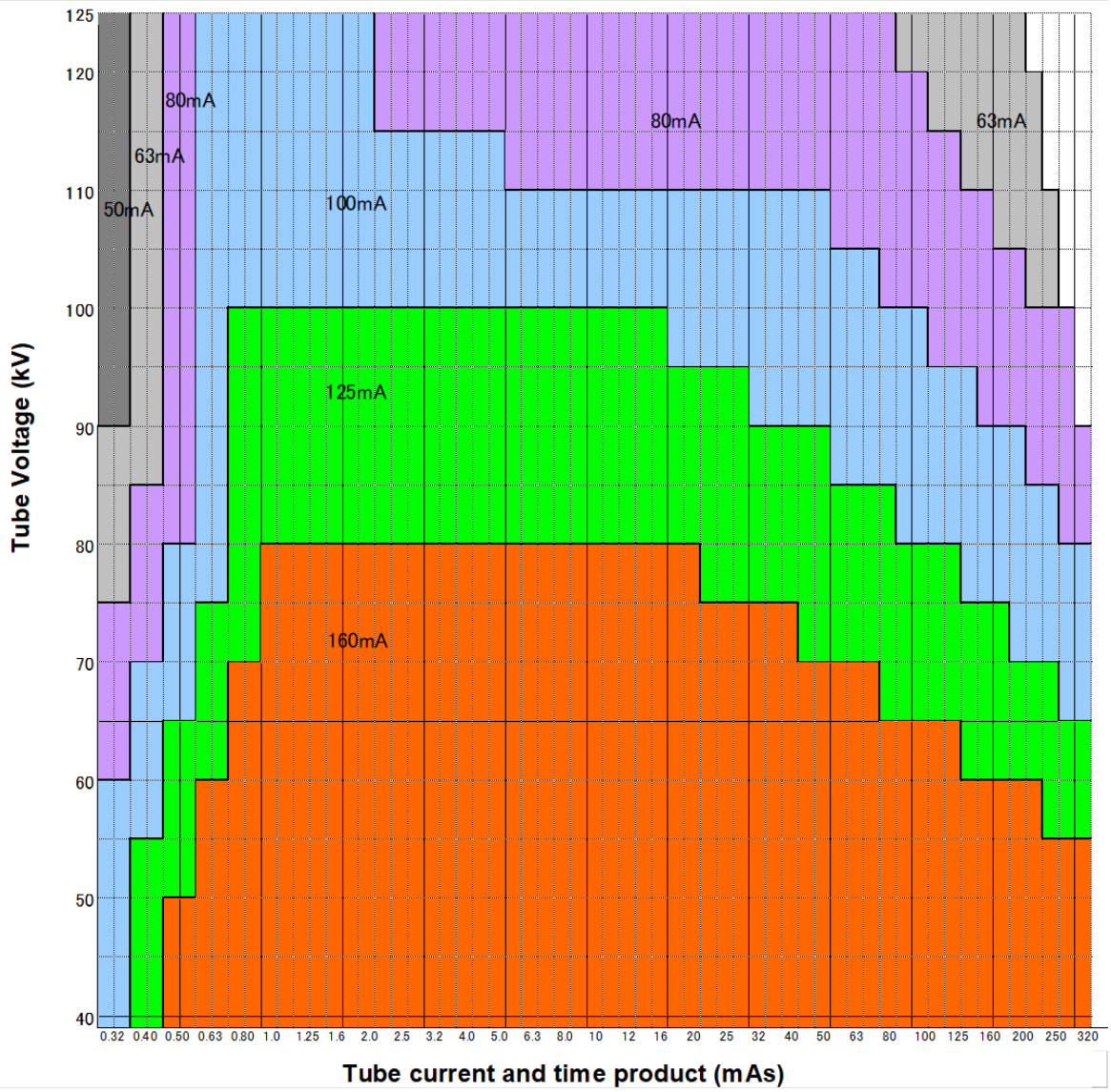


Fig. 12-5

Table Example: Conditions that can be set up at 90kV

- At 90kV, a setup of a maximum of 320 mAs is possible.
- Since the tube current at 90kV and 80mAs is 100mA, time is 0.8sec.
- A setup of 125mA of tube current is possible to take an exposure with 0.80-50mAs.

12.2.2 At low level battery

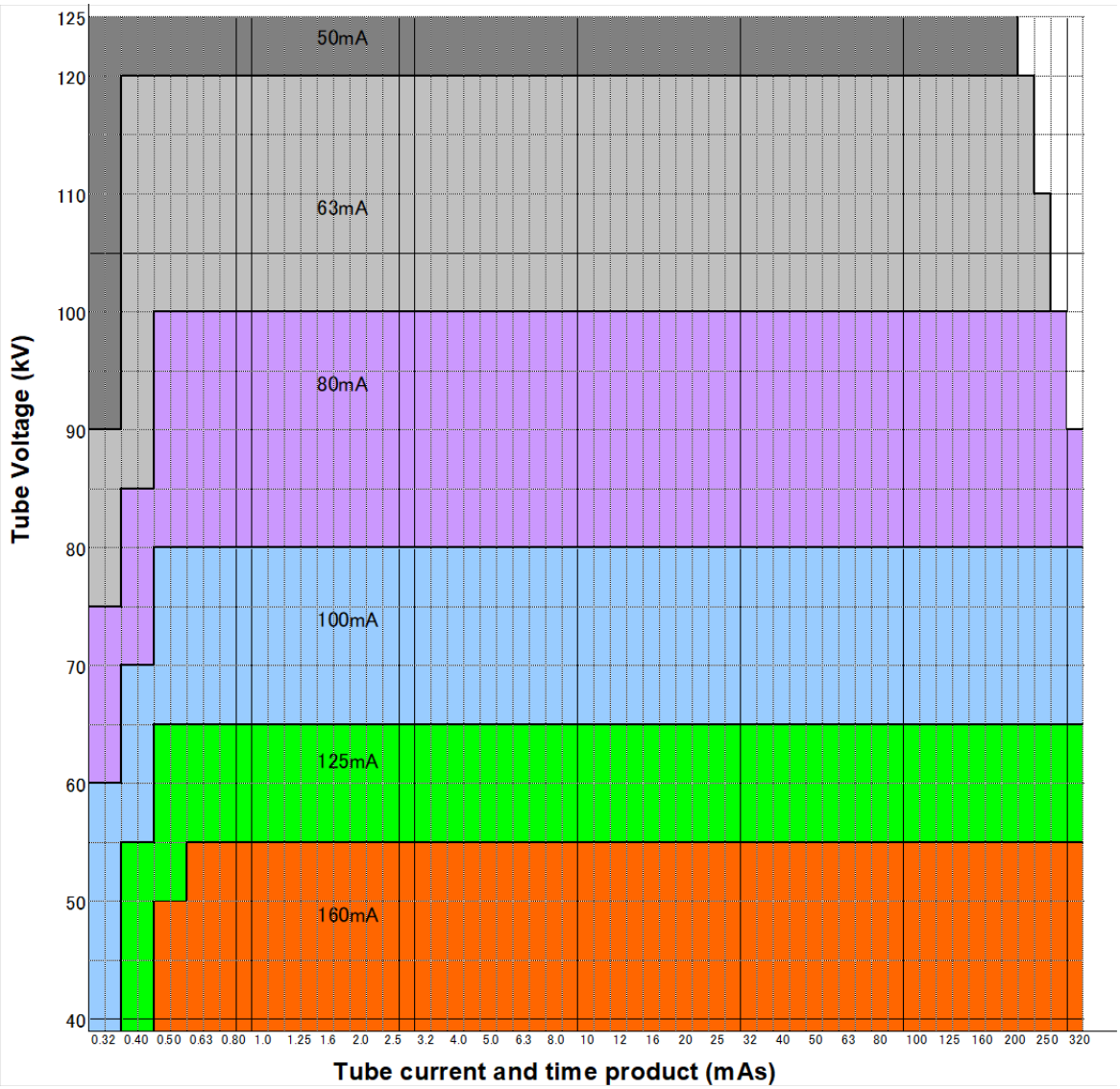


Fig. 12-6

Table Example: Conditions that can be set up at 90kV

- At 90kV, a setup of a maximum of 320 mAs is possible.
- Since the tube current at 90kV and 80mAs is 80mA, time is 1.0sec.
- A setup of 80mA of tube current is possible to take an exposure with 0.50-320mAs.

12.3 For 32kW type (CS-7 with serial radiography)

12.3.1 Large Focus (at normal condition)

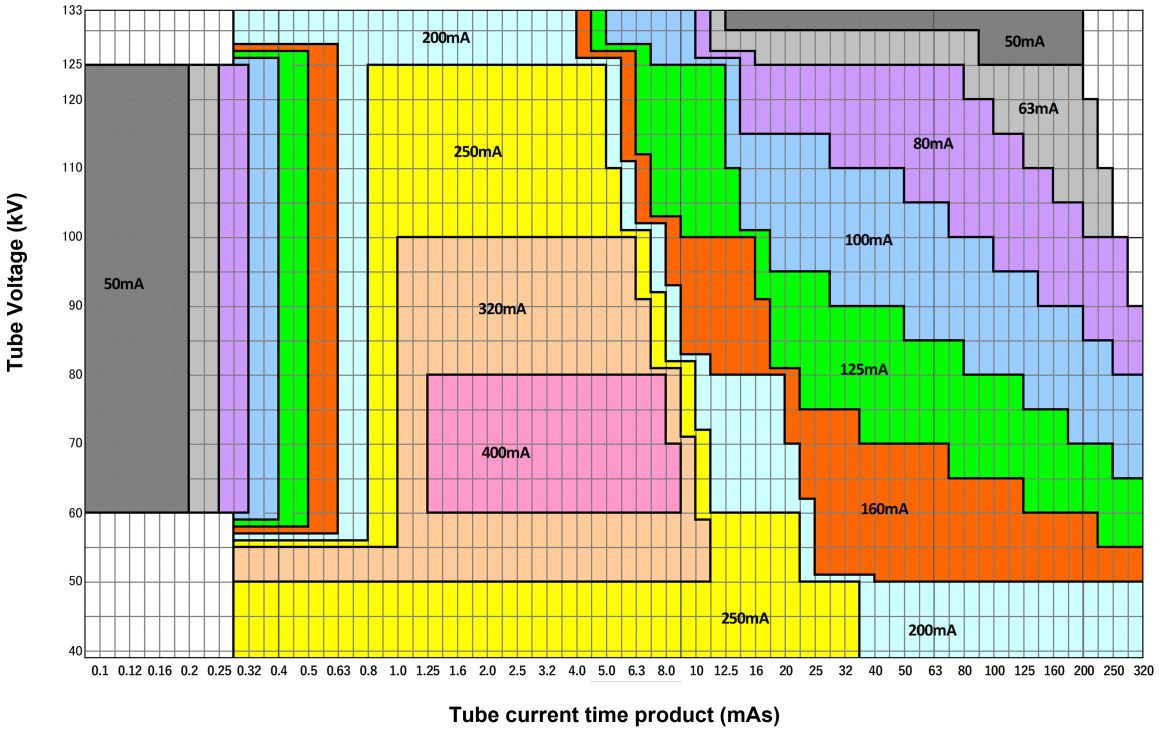


Fig. 12-7

Table Example: Conditions that can be set up at 90kV

- 90kV, a setup of a maximum of 320 mAs is possible.
- Since the tube current at 90kV and 100mAs is 100mA, time is 1.0sec.
- A setup of 320mA of tube current is possible to take an exposure with 1.1-7.1mAs.

12.3.2 Small Focus (at normal condition)

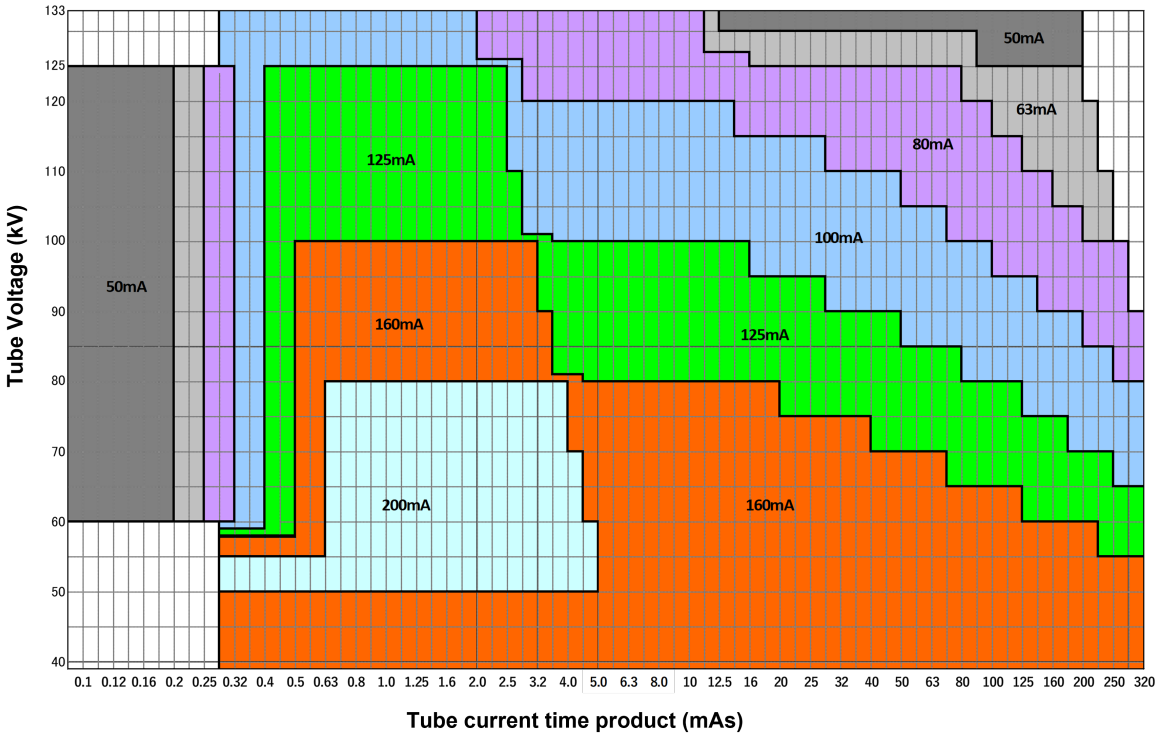


Fig. 12-8

Table Example: Conditions that can be set up at 90kV

- 90kV, a setup of a maximum of 320 mAs is possible.
- Since the tube current at 90kV and 100mAs is 100mA, time is 1.0sec.
- A setup of 160mA of tube current is possible to take an exposure with 0.56-3.6mAs.

12.3.3 Large focus (at low level battery)

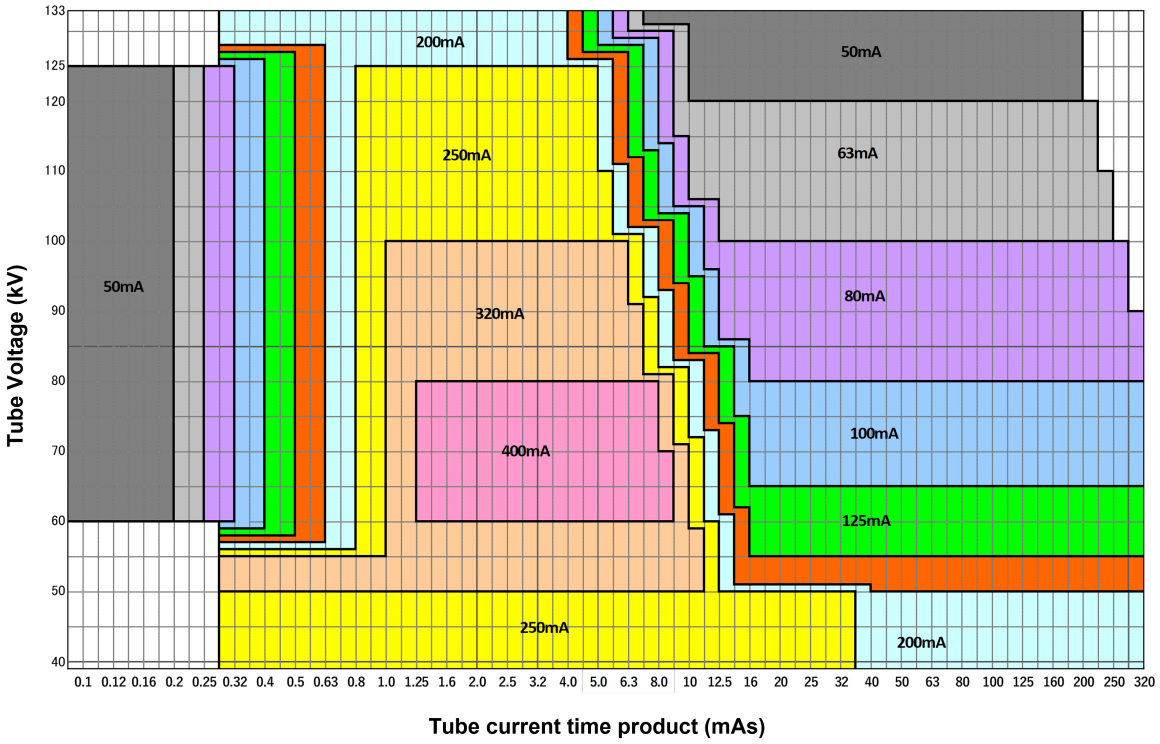


Fig. 12-9

Table Example: Conditions that can be set up at 90kV

- 90kV, a setup of a maximum of 320 mAs is possible.
- Since the tube current at 90kV and 100mAs is 80mA, time is 1.25sec.
- A setup of 320mA of tube current is possible to take an exposure with 1.1-7.1mAs.

12.3.4 Small Focus (at low level battery)

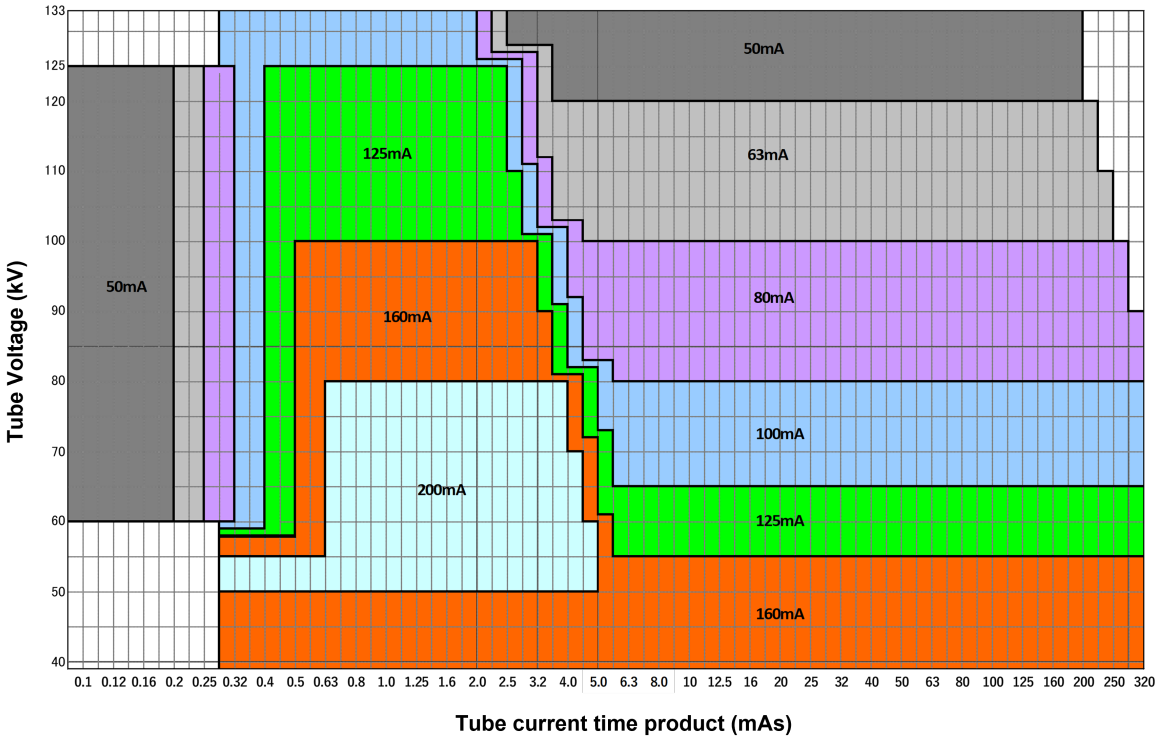


Fig. 12-10

Table Example: Conditions that can be set up at 90kV

- 90kV, a setup of a maximum of 320mAs is possible.
- Since the tube current at 90kV and 80mAs is 80mA, time is 1.0sec.
- A setup of 160mA of tube current is possible to take an exposure with 0.56-3.6mAs.

12.3.5 Serial radiography

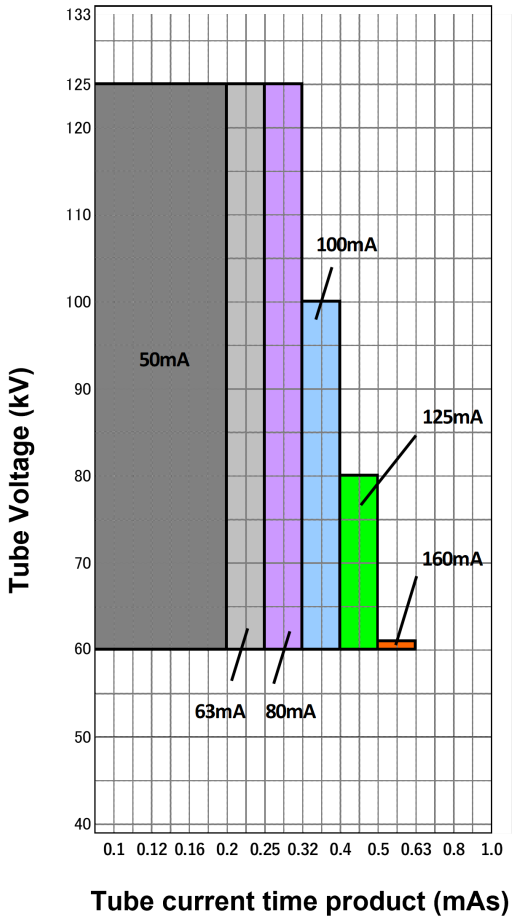


Fig. 12-11

Table Example: Conditions that can be set up at 90kV

- 90kV, a setup of a maximum of 0.4mAs is possible.
- Since the tube current at 90kV and 0.32mAs is 80mA, time is 0.004sec.
- A setup of 80mA of tube current is possible to take an exposure with 0.28-0.32mAs.

13 [Appendix] Operation of High-Voltage Cable

13.1 Spare parts for high-voltage cable

13.1.1 Spare parts of the X-ray tube assembly side mini-plug

Following parts are included in the "Mini-plug spare parts kit" (P/N 571-15061-13).

Silicon Rubber jacket 5pc

Silicon washer 5pc

Silicon Oil 5pc

13.1.2 Spare parts of the high-voltage generator side plug

Contact pin P/N 571-15061-11 (Contact pin + tool)

Silicon packing P/N 582-23029 (3mm thickness)

P/N 582-23061 (2mm thickness)

Use the 2mm thick silicon packing when using the tube voltage meter.

⚠ CAUTION

- Please exchange the silicon packing, the silicon rubber jacket and the silicon washer once every year. Also exchange these silicon parts when the cable is removed for repair, check etc.
- Please use only specified silicone oil for the X-ray tube assembly side. Do NOT use silicon grease.

13.2 Installation procedure of the high-voltage generator side plug

1 **Make sure that the MINI receptacle socket assembly is clean.**

2 **Make sure that termination plug and the mounting accessories are clean.**

3 **Screw out the Grub Screw and slip the Ring Nut over the Ring Flange.**

4 **Wet the Silicone Rubber Jacket and Washer with the Silicone Oil.**

5 **Fit the Silicone Washer over the Contact Pins.**

6 **Insert the plug into the receptacle socket until the tip meets the bottom. Turn the plug until matches the key groove and further inset the plug.**

> NOTE While inserting the plug, push the plug aside to let the air captured at the tip of the plug escape.

7 **Hand-tighten the Ring Nut.**

8 **Tighten the Grub Screw.**

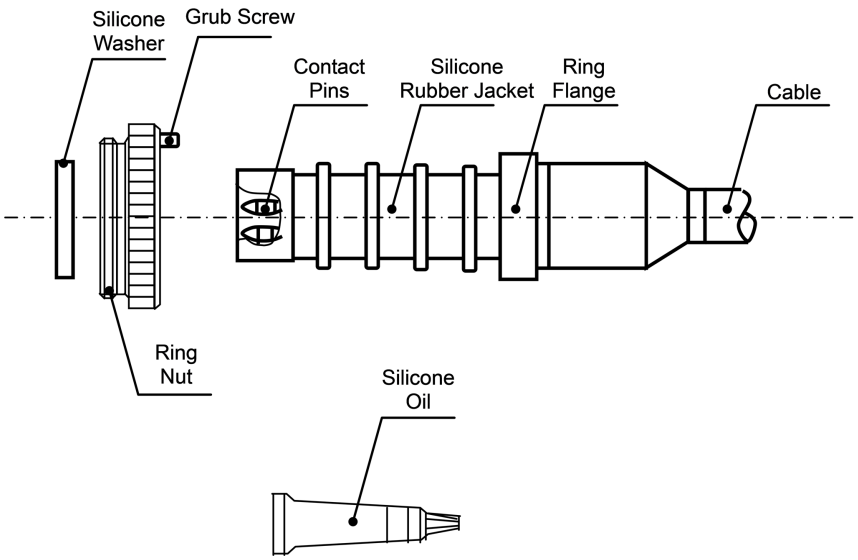


Fig. 13-1 High-voltage cable (High-voltage generator side)

13.3 Installation procedure of the X-ray tube assembly side mini-plug

1. Make sure that the receptacle socket assembly is correctly mounted on generator is clean.
2. Make sure that termination plug and the mounting accessories are clean.
3. Screw out the Grub Screw and slip the Ring Nut over the Ring Flange.
4. Place the Divided Flange onto the Ring Flange.
5. Wet the Silicone Washer (both side) and the Rubber Gasket with the Silicon Oil.
6. Fit the Rubber Gasket onto the Plug, against the Ring Flange.
7. Push the Silicone Washer over the Contact Pins.
8. Insert the Termination Plug into the Receptacle Socket (watch the Nipple on the Plug to ensure correct positioning of the Contact Pins).
9. Hang tighten the Ring Nut.
10. Fasten the grub screw to prevent the rotation of the ring nut.

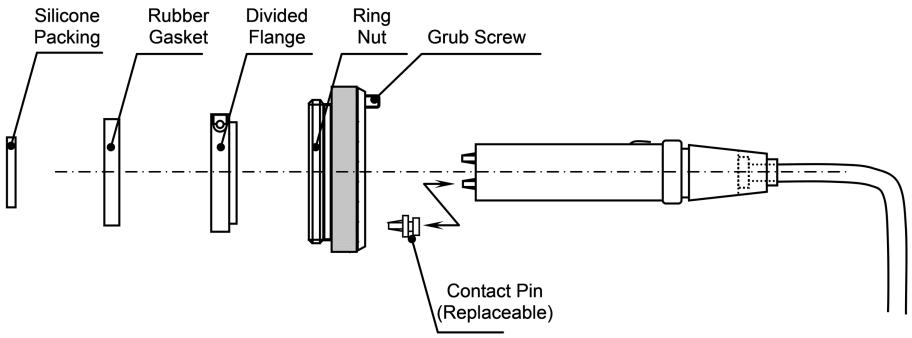


Fig. 13-2 High-voltage cable (X-ray tube assembly side)